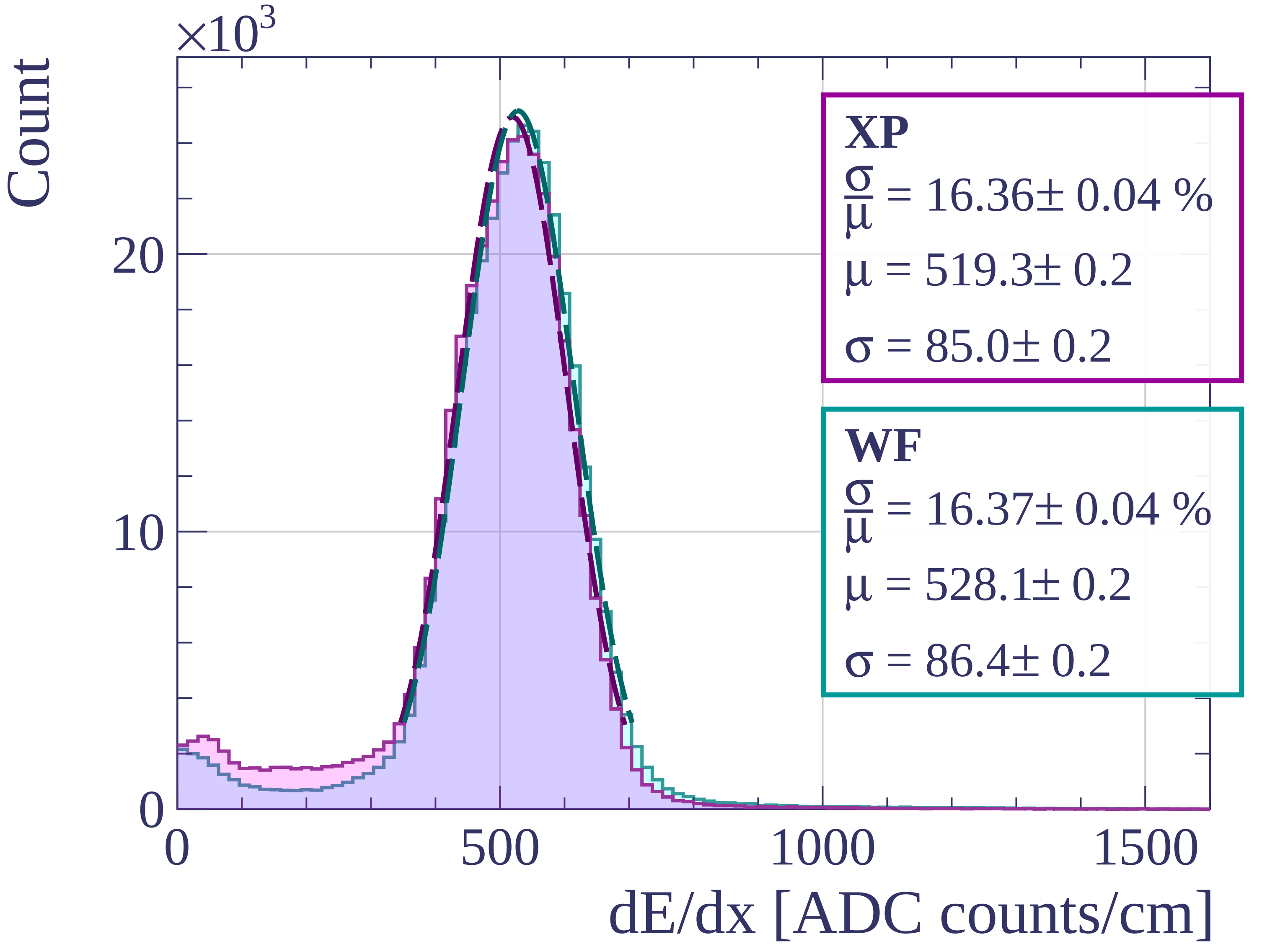
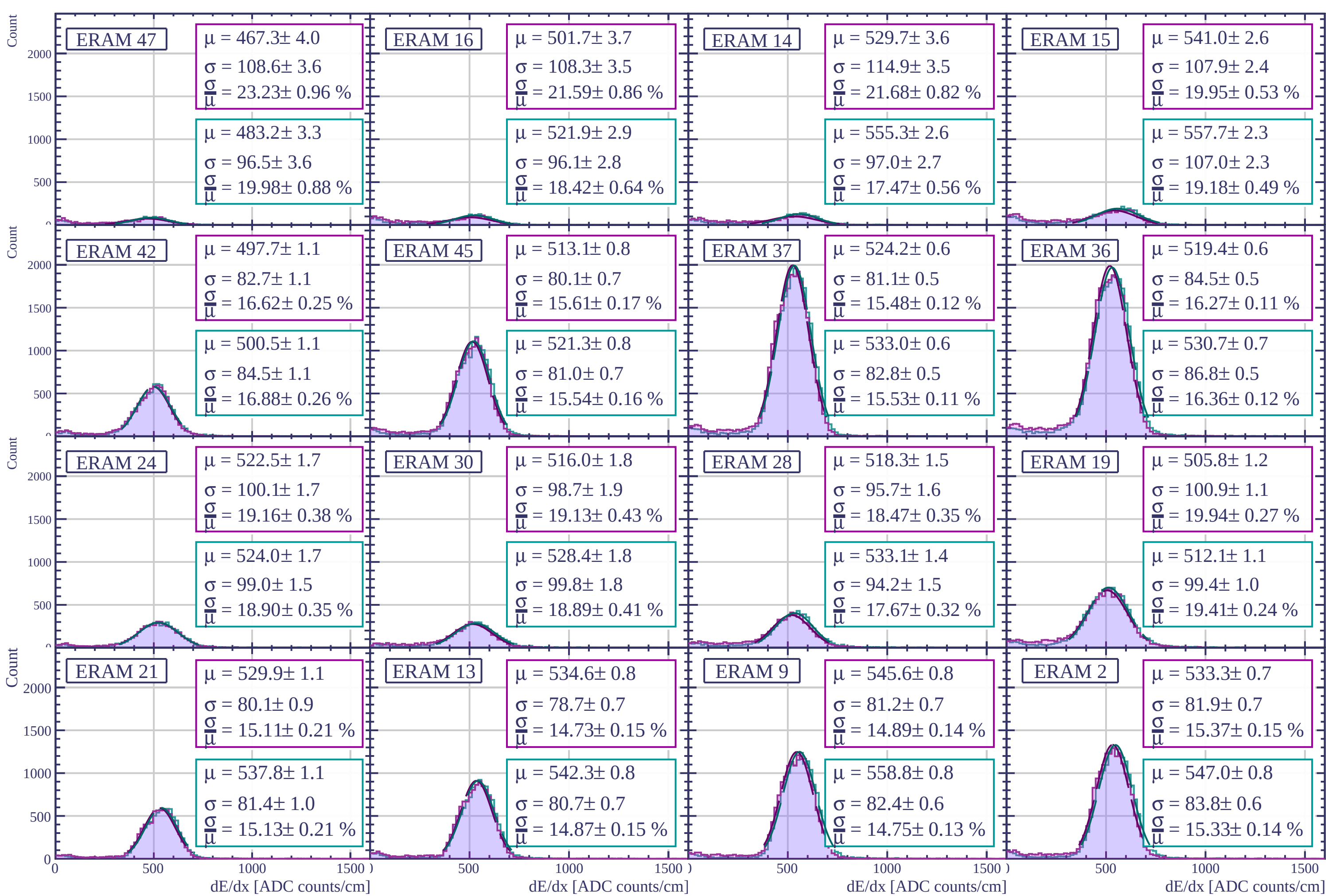
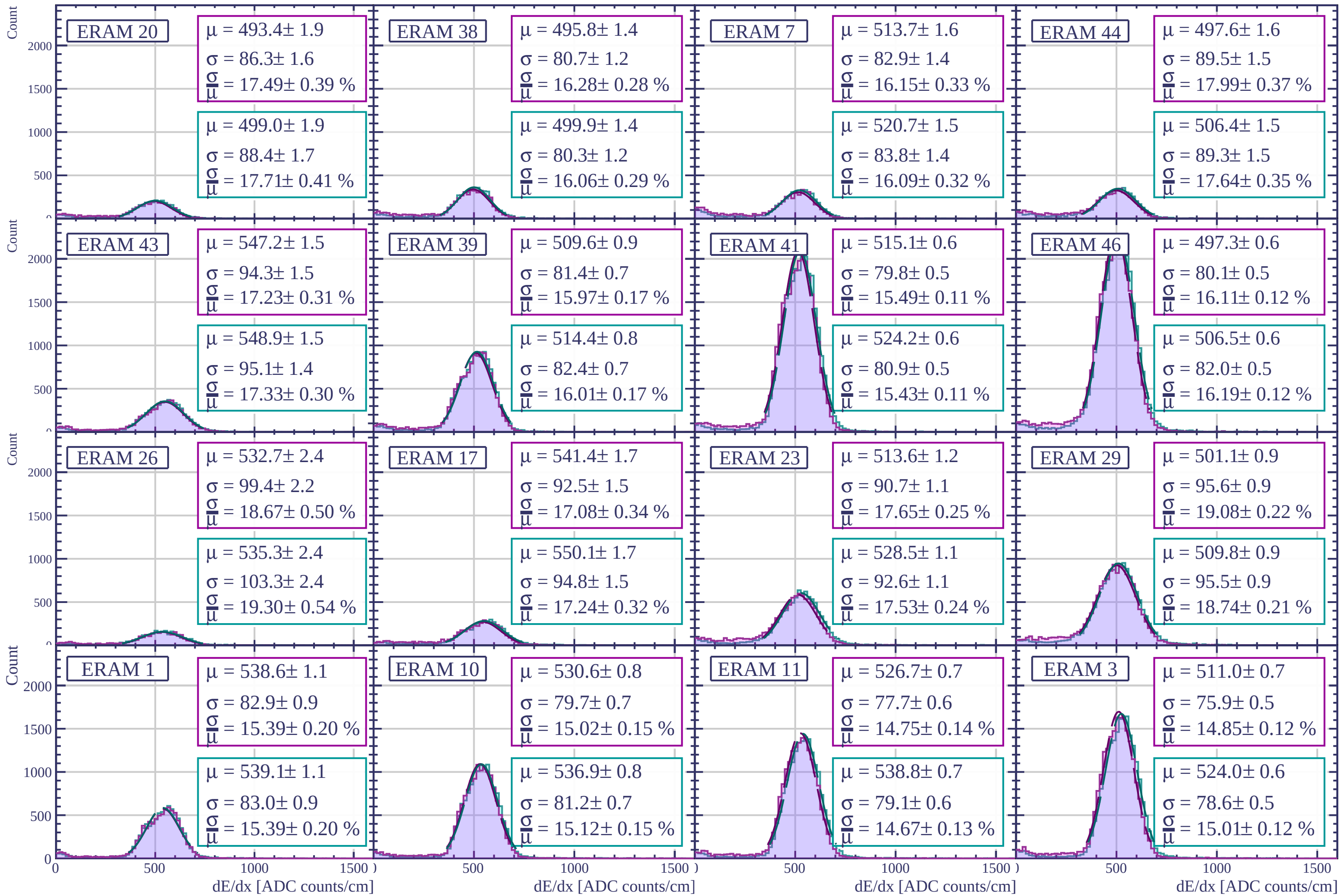
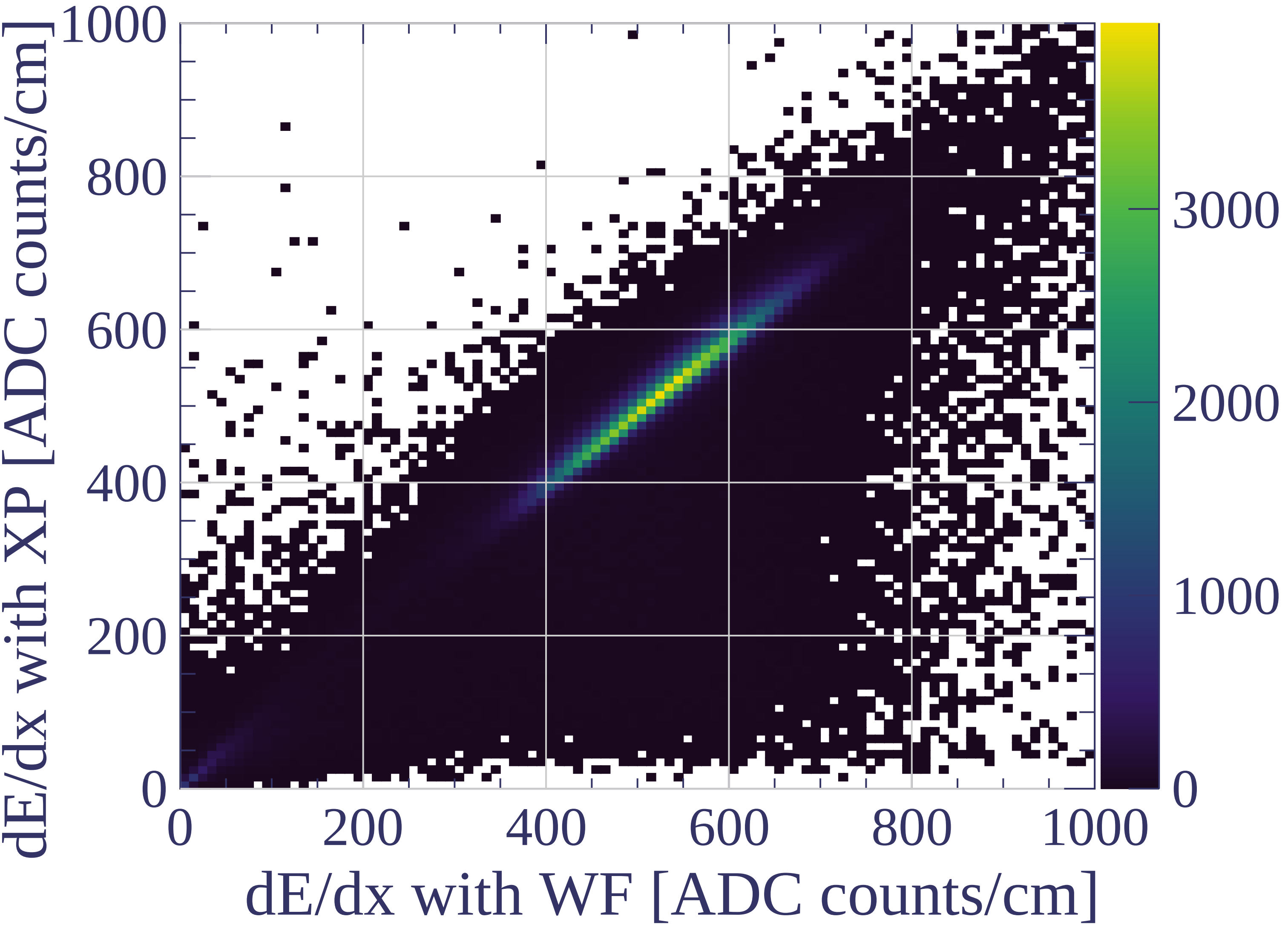

×10^3
Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 16.36± 0.04 %
μ = 519.3± 0.2
σ = 85.0± 0.2
WF
σ/μ = 16.37± 0.04 %
μ = 528.1± 0.2
σ = 86.4± 0.2

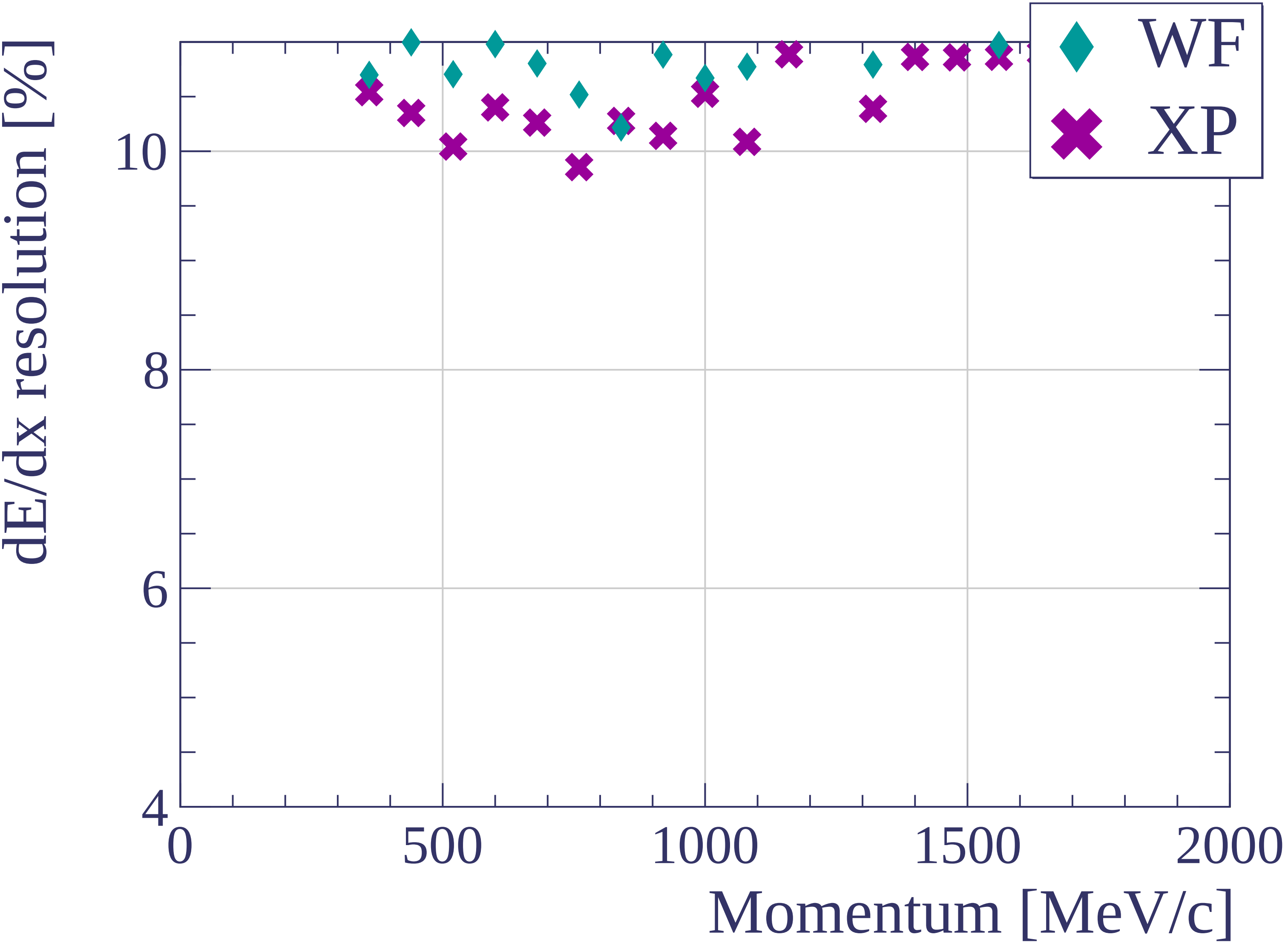


ERAM 47
μ = 467.3± 4.0
σ = 108.6± 3.6
σ/μ = 23.23± 0.96 %
μ = 483.2± 3.3
σ = 96.5± 3.6
σ/μ = 19.98± 0.88 %
ERAM 16
μ = 501.7± 3.7
σ = 108.3± 3.5
σ/μ = 21.59± 0.86 %
μ = 521.9± 2.9
σ = 96.1± 2.8
σ/μ = 18.42± 0.64 %
ERAM 14
μ = 529.7± 3.6
σ = 114.9± 3.5
σ/μ = 21.68± 0.82 %
μ = 555.3± 2.6
σ = 97.0± 2.7
σ/μ = 17.47± 0.56 %
ERAM 15
μ = 541.0± 2.6
σ = 107.9± 2.4
σ/μ = 19.95± 0.53 %
μ = 557.7± 2.3
σ = 107.0± 2.3
σ/μ = 19.18± 0.49 %
ERAM 42
μ = 497.7± 1.1
σ = 82.7± 1.1
σ/μ = 16.62± 0.25 %
μ = 500.5± 1.1
σ = 84.5± 1.1
σ/μ = 16.88± 0.26 %
ERAM 45
μ = 513.1± 0.8
σ = 80.1± 0.7
σ/μ = 15.61± 0.17 %
μ = 521.3± 0.8
σ = 81.0± 0.7
σ/μ = 15.54± 0.16 %
ERAM 37
μ = 524.2± 0.6
σ = 81.1± 0.5
σ/μ = 15.48± 0.12 %
μ = 533.0± 0.6
σ = 82.8± 0.5
σ/μ = 15.53± 0.11 %
ERAM 36
μ = 519.4± 0.6
σ = 84.5± 0.5
σ/μ = 16.27± 0.11 %
μ = 530.7± 0.7
σ = 86.8± 0.5
σ/μ = 16.36± 0.12 %
ERAM 24
μ = 522.5± 1.7
σ = 100.1± 1.7
σ/μ = 19.16± 0.38 %
μ = 524.0± 1.7
σ = 99.0± 1.5
σ/μ = 18.90± 0.35 %
ERAM 30
μ = 516.0± 1.8
σ = 98.7± 1.9
σ/μ = 19.13± 0.43 %
μ = 528.4± 1.8
σ = 99.8± 1.8
σ/μ = 18.89± 0.41 %
ERAM 28
μ = 518.3± 1.5
σ = 95.7± 1.6
σ/μ = 18.47± 0.35 %
μ = 533.1± 1.4
σ = 94.2± 1.5
σ/μ = 17.67± 0.32 %
ERAM 19
μ = 505.8± 1.2
σ = 100.9± 1.1
σ/μ = 19.94± 0.27 %
μ = 512.1± 1.1
σ = 99.4± 1.0
σ/μ = 19.41± 0.24 %
ERAM 21
μ = 529.9± 1.1
σ = 80.1± 0.9
σ/μ = 15.11± 0.21 %
μ = 537.8± 1.1
σ = 81.4± 1.0
σ/μ = 15.13± 0.21 %
ERAM 13
μ = 534.6± 0.8
σ = 78.7± 0.7
σ/μ = 14.73± 0.15 %
μ = 542.3± 0.8
σ = 80.7± 0.7
σ/μ = 14.87± 0.15 %
ERAM 9
μ = 545.6± 0.8
σ = 81.2± 0.7
σ/μ = 14.89± 0.14 %
μ = 558.8± 0.8
σ = 82.4± 0.6
σ/μ = 14.75± 0.13 %
ERAM 2
μ = 533.3± 0.7
σ = 81.9± 0.7
σ/μ = 15.37± 0.15 %
μ = 547.0± 0.8
σ = 83.8± 0.6
σ/μ = 15.33± 0.14 %
Count
0
500
1000
1500
2000
dE/dx [ADC counts/cm]

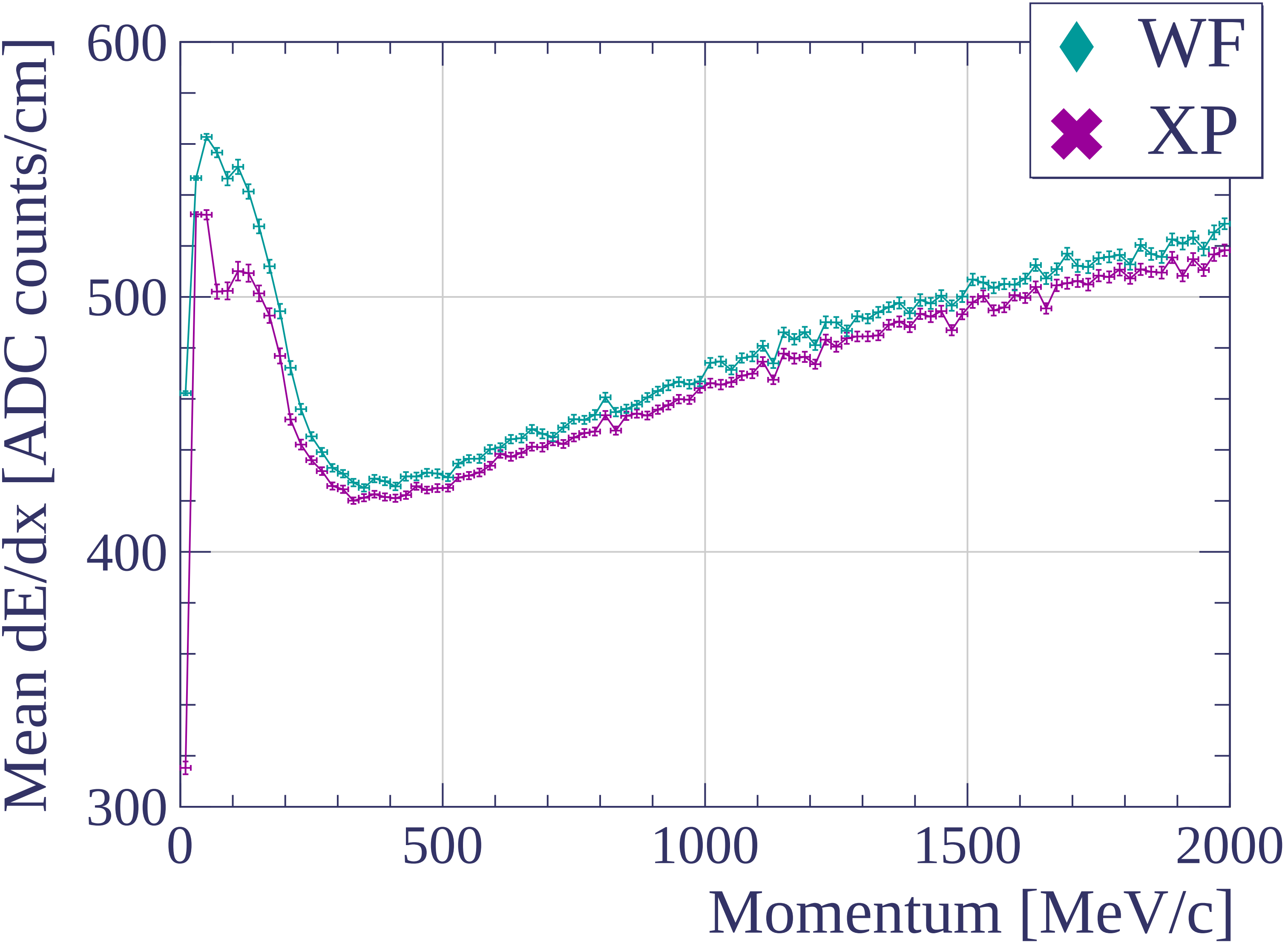

ERAM 20
μ = 493.4± 1.9
σ = 86.3± 1.6
σ/μ = 17.49± 0.39 %
μ = 499.0± 1.9
σ = 88.4± 1.7
σ/μ = 17.71± 0.41 %
ERAM 38
μ = 495.8± 1.4
σ = 80.7± 1.2
σ/μ = 16.28± 0.28 %
μ = 499.9± 1.4
σ = 80.3± 1.2
σ/μ = 16.06± 0.29 %
ERAM 7
μ = 513.7± 1.6
σ = 82.9± 1.4
σ/μ = 16.15± 0.33 %
μ = 520.7± 1.5
σ = 83.8± 1.4
σ/μ = 16.09± 0.32 %
ERAM 44
μ = 497.6± 1.6
σ = 89.5± 1.5
σ/μ = 17.99± 0.37 %
μ = 506.4± 1.5
σ = 89.3± 1.5
σ/μ = 17.64± 0.35 %
ERAM 43
μ = 547.2± 1.5
σ = 94.3± 1.5
σ/μ = 17.23± 0.31 %
μ = 548.9± 1.5
σ = 95.1± 1.4
σ/μ = 17.33± 0.30 %
ERAM 39
μ = 509.6± 0.9
σ = 81.4± 0.7
σ/μ = 15.97± 0.17 %
μ = 514.4± 0.8
σ = 82.4± 0.7
σ/μ = 16.01± 0.17 %
ERAM 41
μ = 515.1± 0.6
σ = 79.8± 0.5
σ/μ = 15.49± 0.11 %
μ = 524.2± 0.6
σ = 80.9± 0.5
σ/μ = 15.43± 0.11 %
ERAM 46
μ = 497.3± 0.6
σ = 80.1± 0.5
σ/μ = 16.11± 0.12 %
μ = 506.5± 0.6
σ = 82.0± 0.5
σ/μ = 16.19± 0.12 %
ERAM 26
μ = 532.7± 2.4
σ = 99.4± 2.2
σ/μ = 18.67± 0.50 %
μ = 535.3± 2.4
σ = 103.3± 2.4
σ/μ = 19.30± 0.54 %
ERAM 17
μ = 541.4± 1.7
σ = 92.5± 1.5
σ/μ = 17.08± 0.34 %
μ = 550.1± 1.7
σ = 94.8± 1.5
σ/μ = 17.24± 0.32 %
ERAM 23
μ = 513.6± 1.2
σ = 90.7± 1.1
σ/μ = 17.65± 0.25 %
μ = 528.5± 1.1
σ = 92.6± 1.1
σ/μ = 17.53± 0.24 %
ERAM 29
μ = 501.1± 0.9
σ = 95.6± 0.9
σ/μ = 19.08± 0.22 %
μ = 509.8± 0.9
σ = 95.5± 0.9
σ/μ = 18.74± 0.21 %
ERAM 1
μ = 538.6± 1.1
σ = 82.9± 0.9
σ/μ = 15.39± 0.20 %
μ = 539.1± 1.1
σ = 83.0± 0.9
σ/μ = 15.39± 0.20 %
ERAM 10
μ = 530.6± 0.8
σ = 79.7± 0.7
σ/μ = 15.02± 0.15 %
μ = 536.9± 0.8
σ = 81.2± 0.7
σ/μ = 15.12± 0.15 %
ERAM 11
μ = 526.7± 0.7
σ = 77.7± 0.6
σ/μ = 14.75± 0.14 %
μ = 538.8± 0.7
σ = 79.1± 0.6
σ/μ = 14.67± 0.13 %
ERAM 3
μ = 511.0± 0.7
σ = 75.9± 0.5
σ/μ = 14.85± 0.12 %
μ = 524.0± 0.6
σ = 78.6± 0.5
σ/μ = 15.01± 0.12 %
Count
dE/dx [ADC counts/cm]



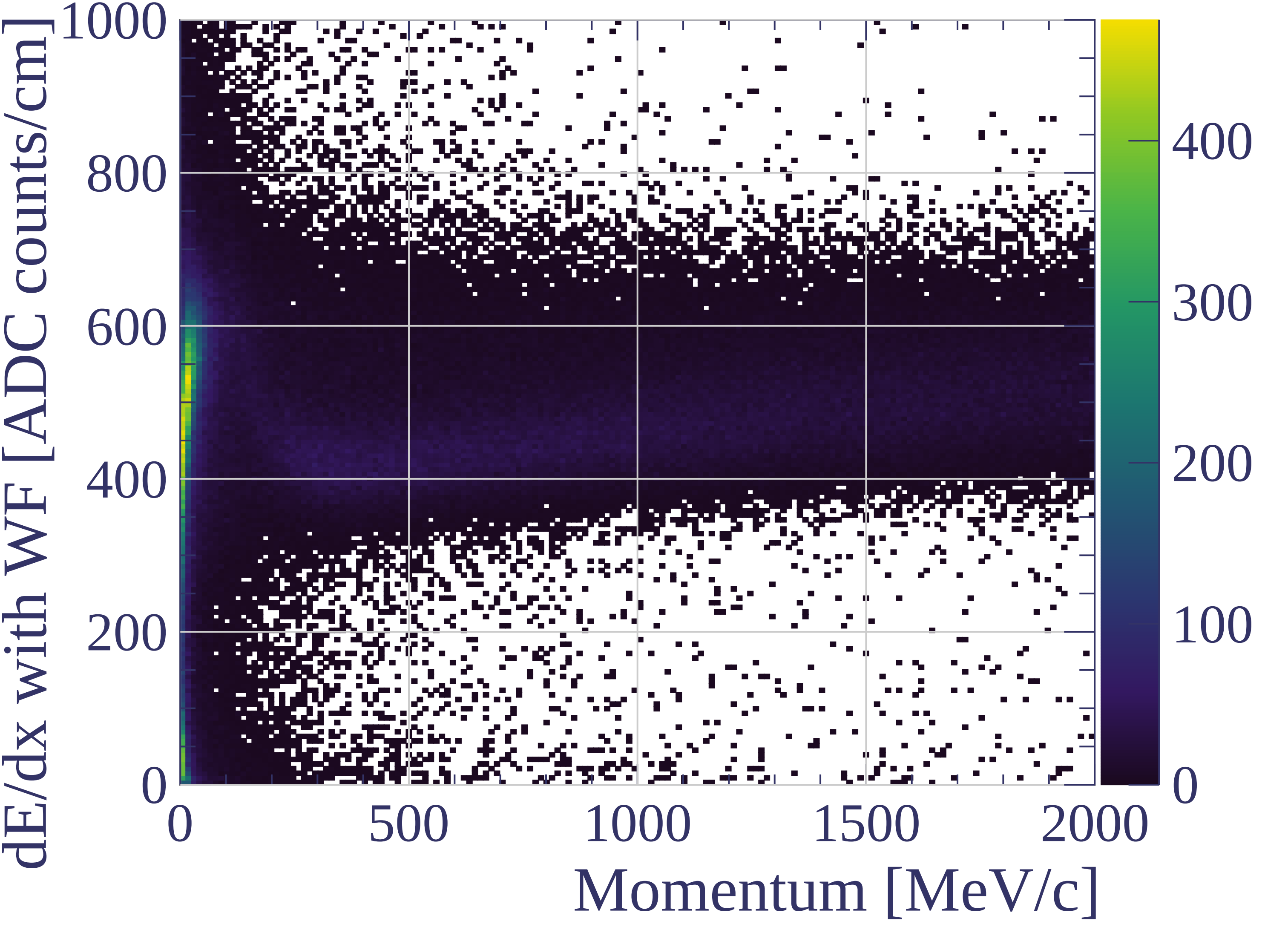
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
0
1000
2000
3000



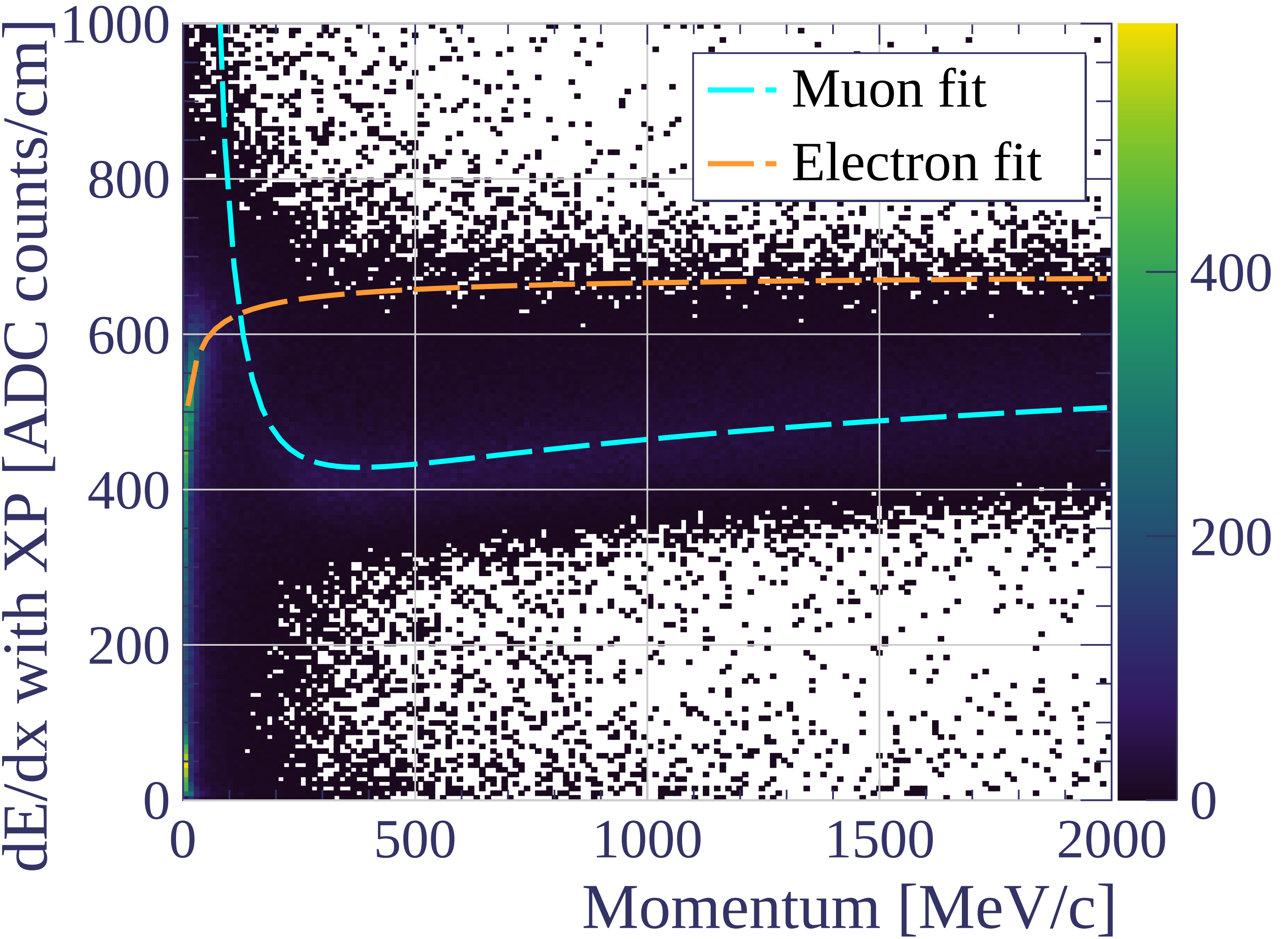
dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP

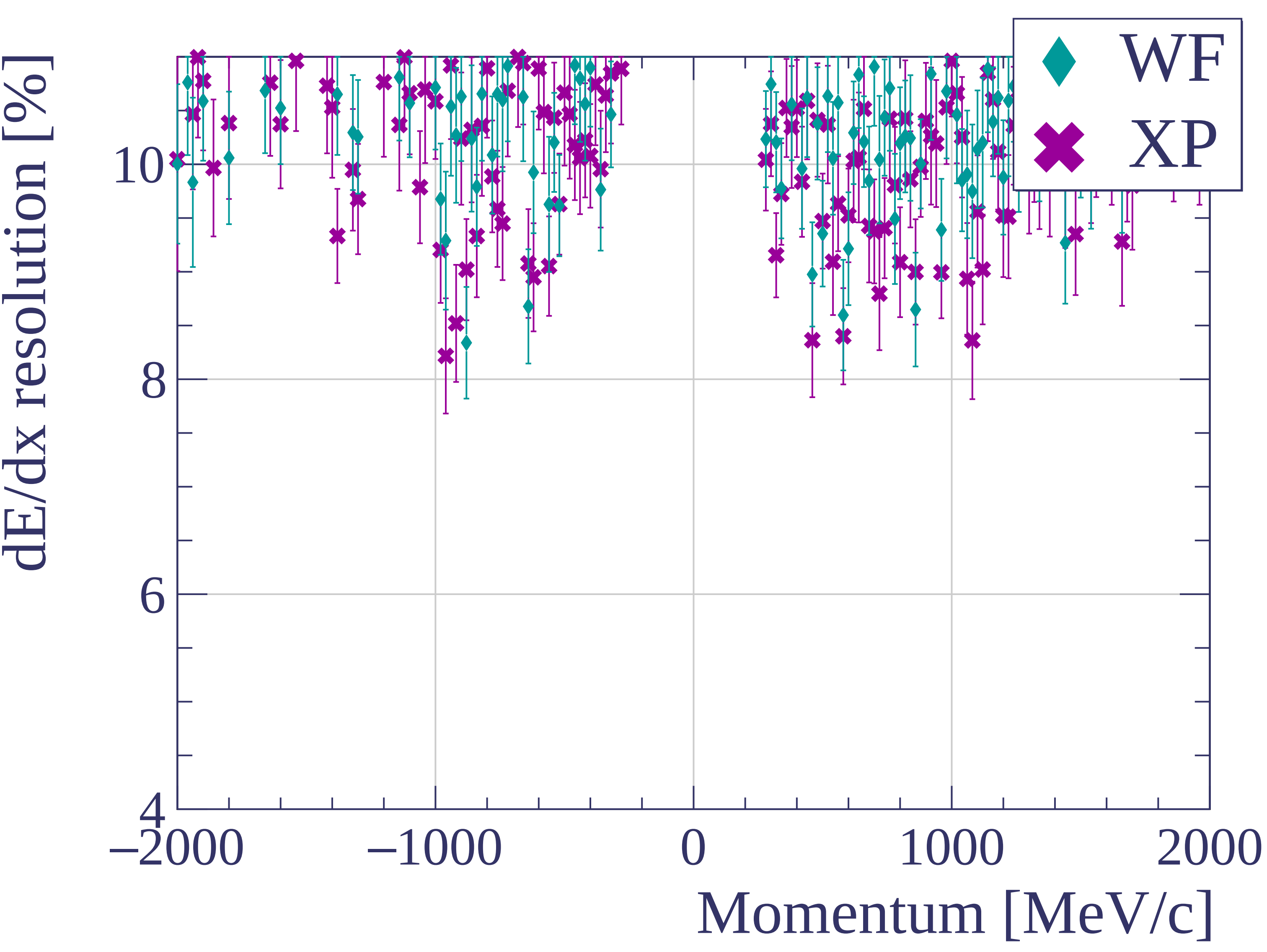


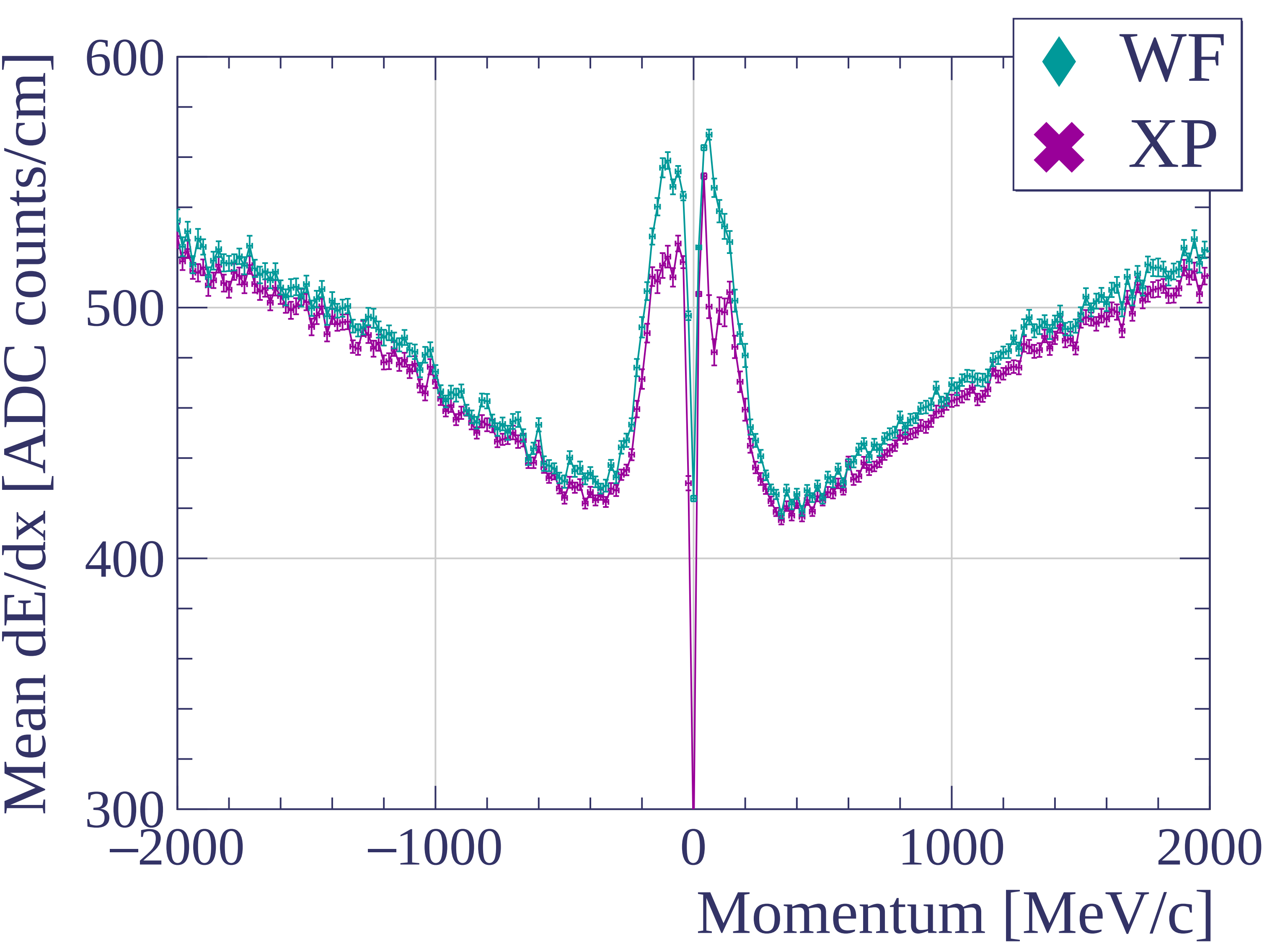
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP



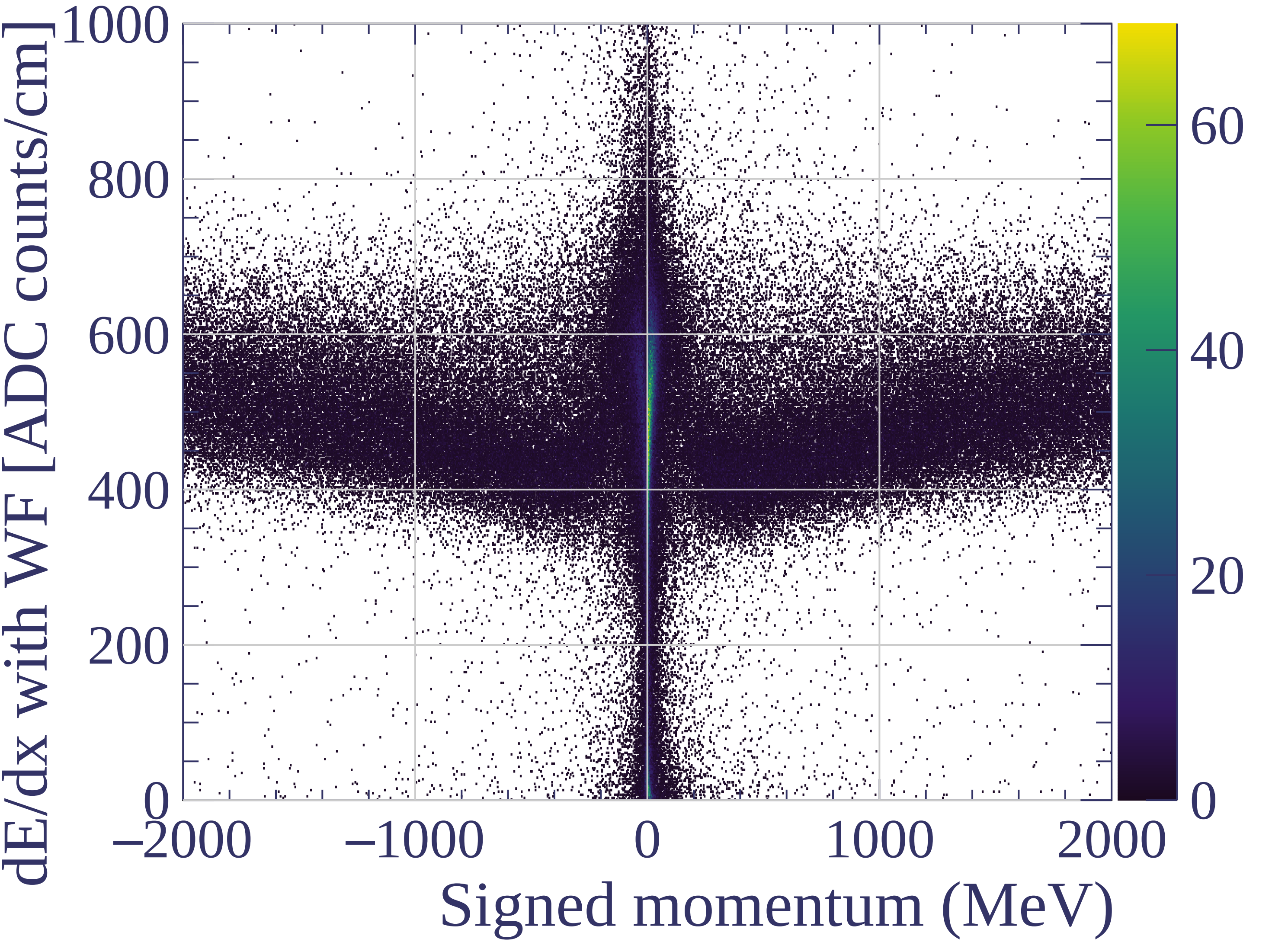
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
400
300
200
100
0


Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
0
500
1000
1500
2000
0
200
400

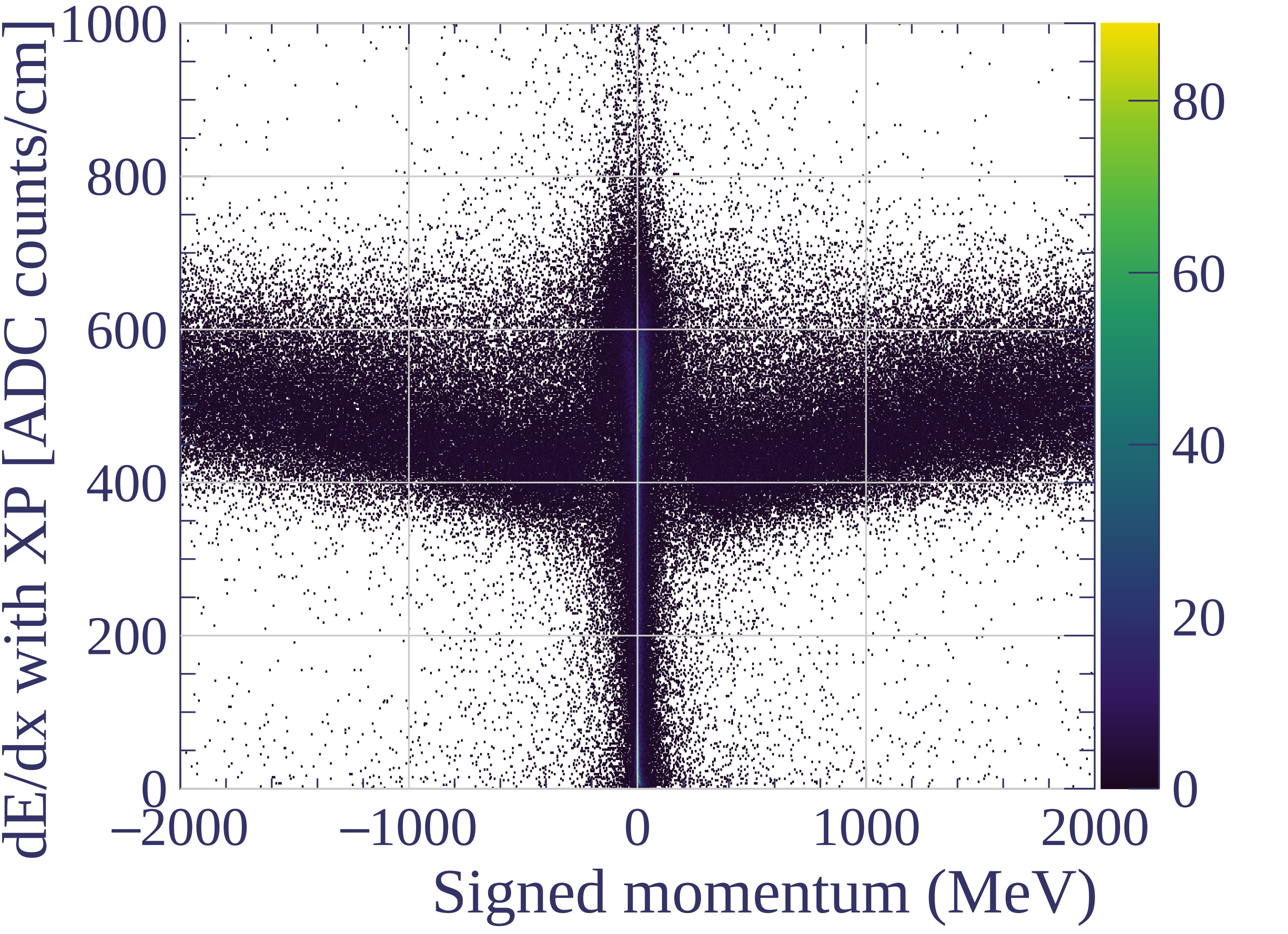

dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]
WF
XP



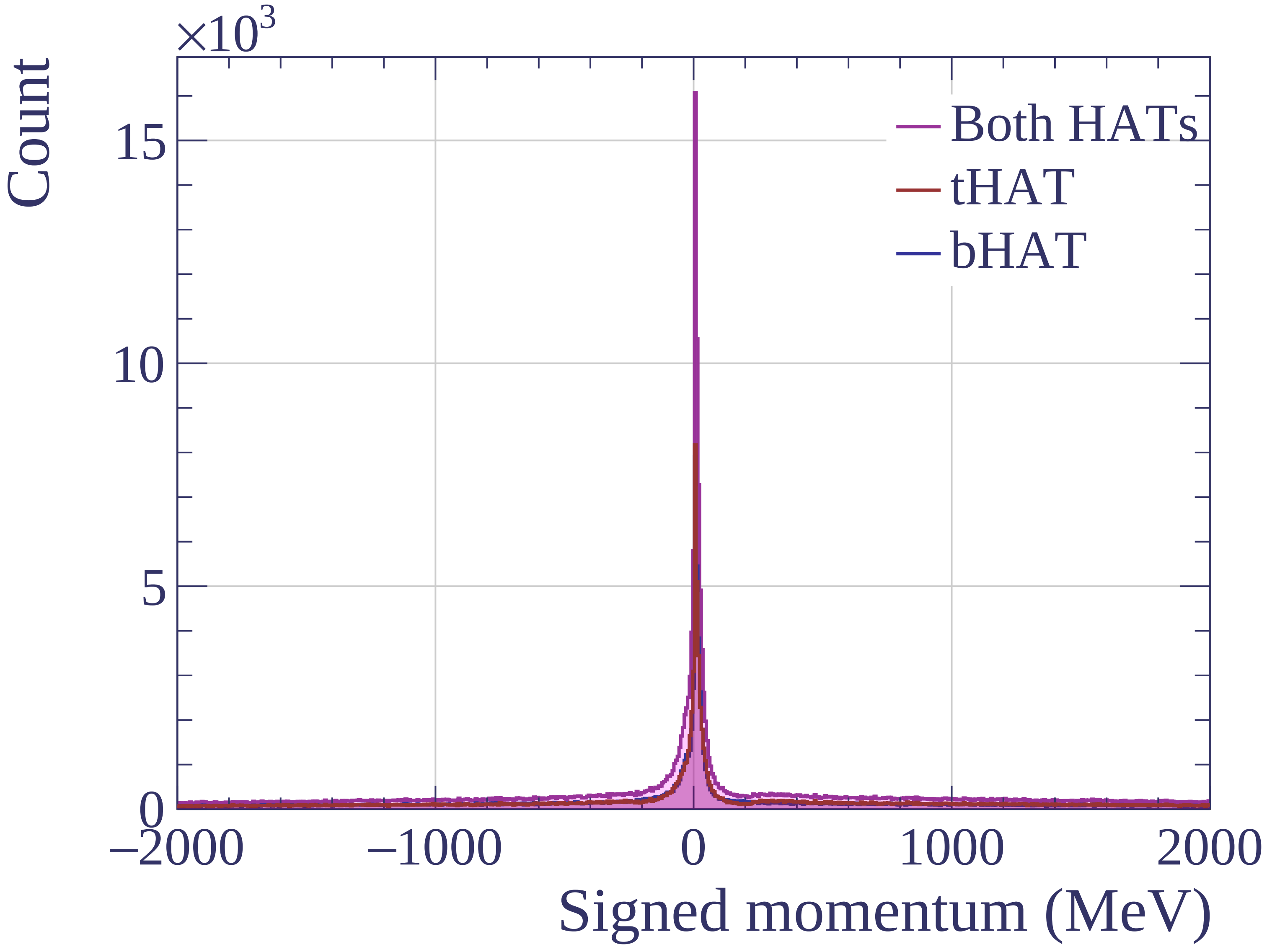
WF
XP
600
500
400
300
−2000
−1000
0
1000
2000
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]

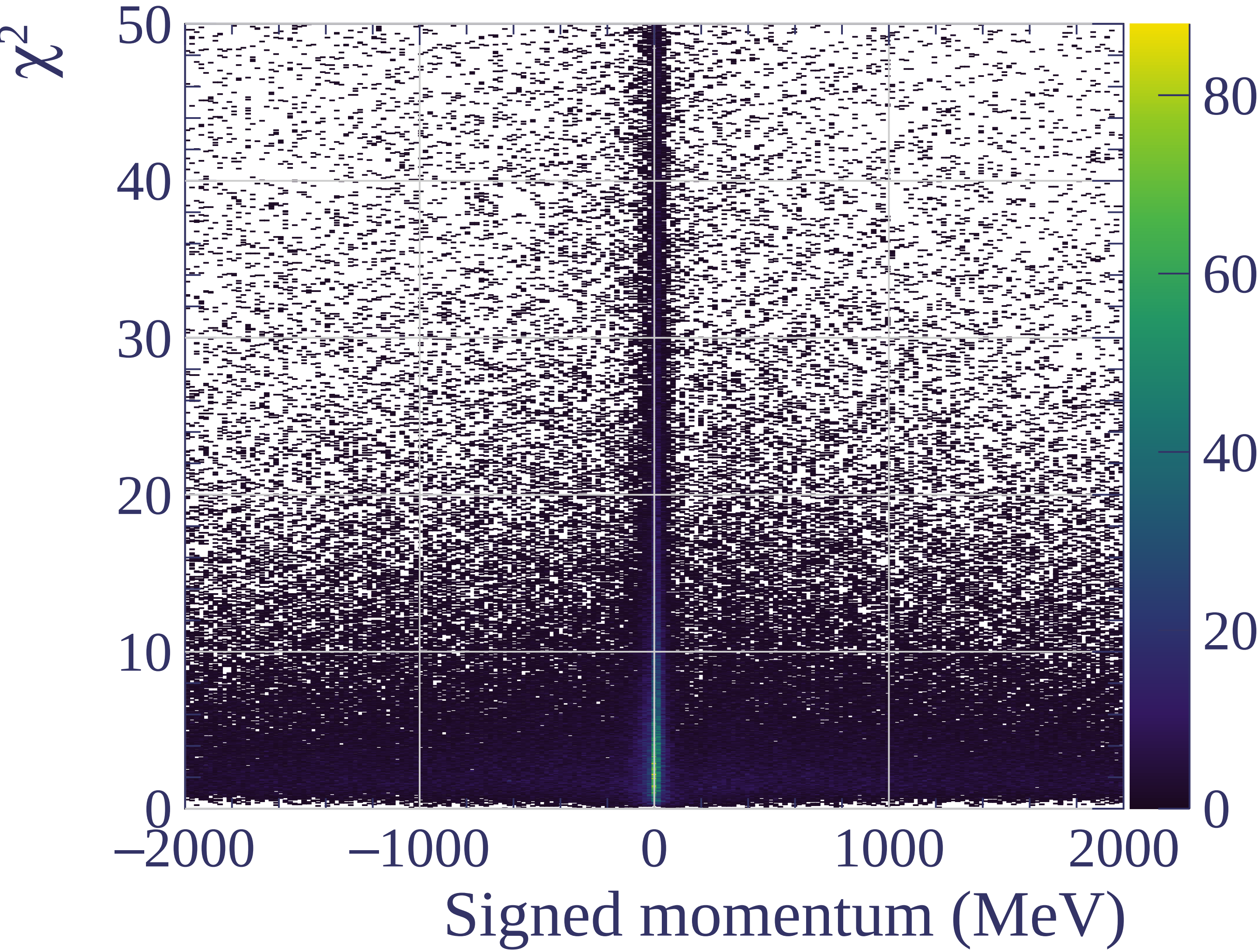


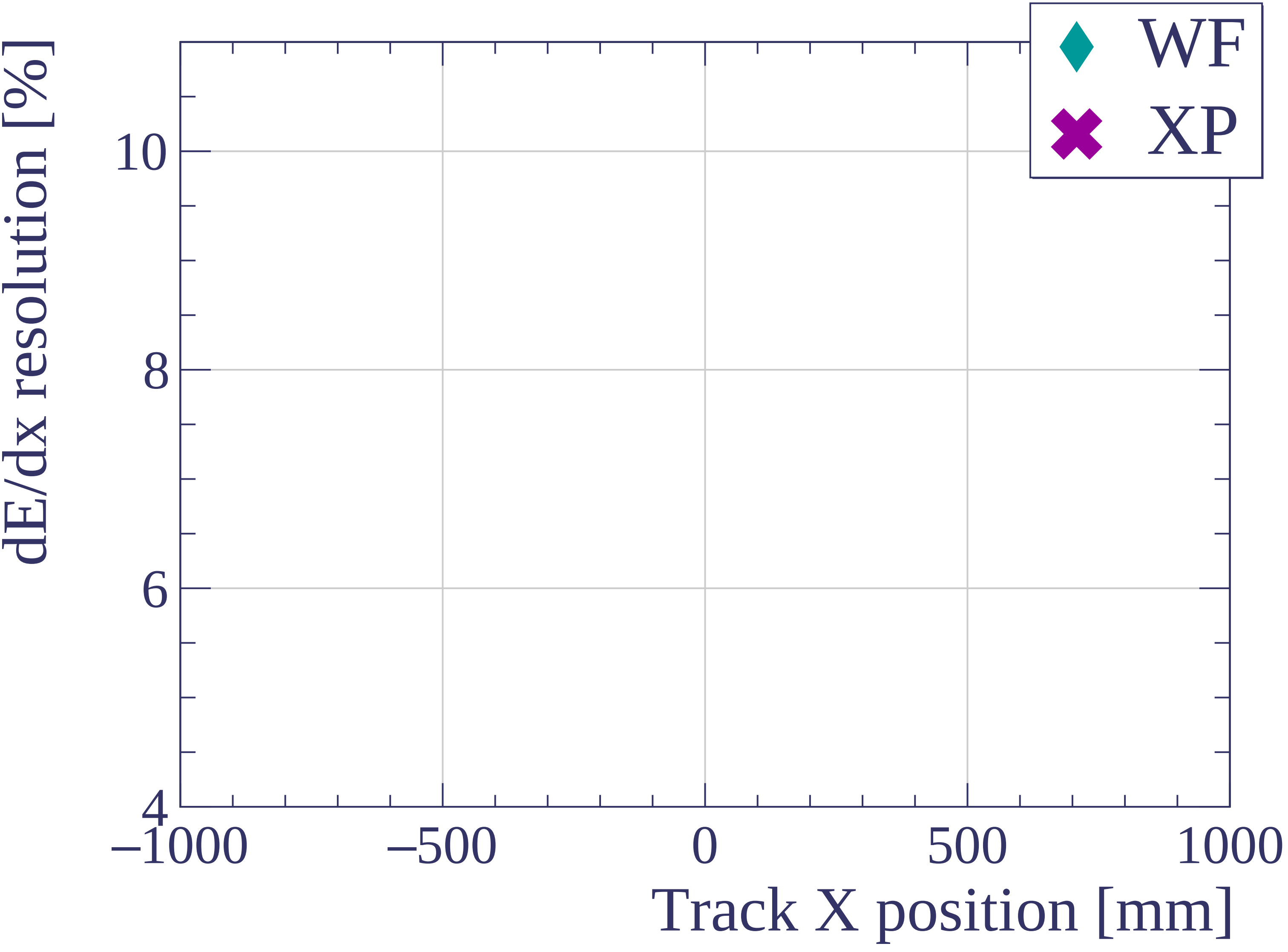
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
60
40
20
0

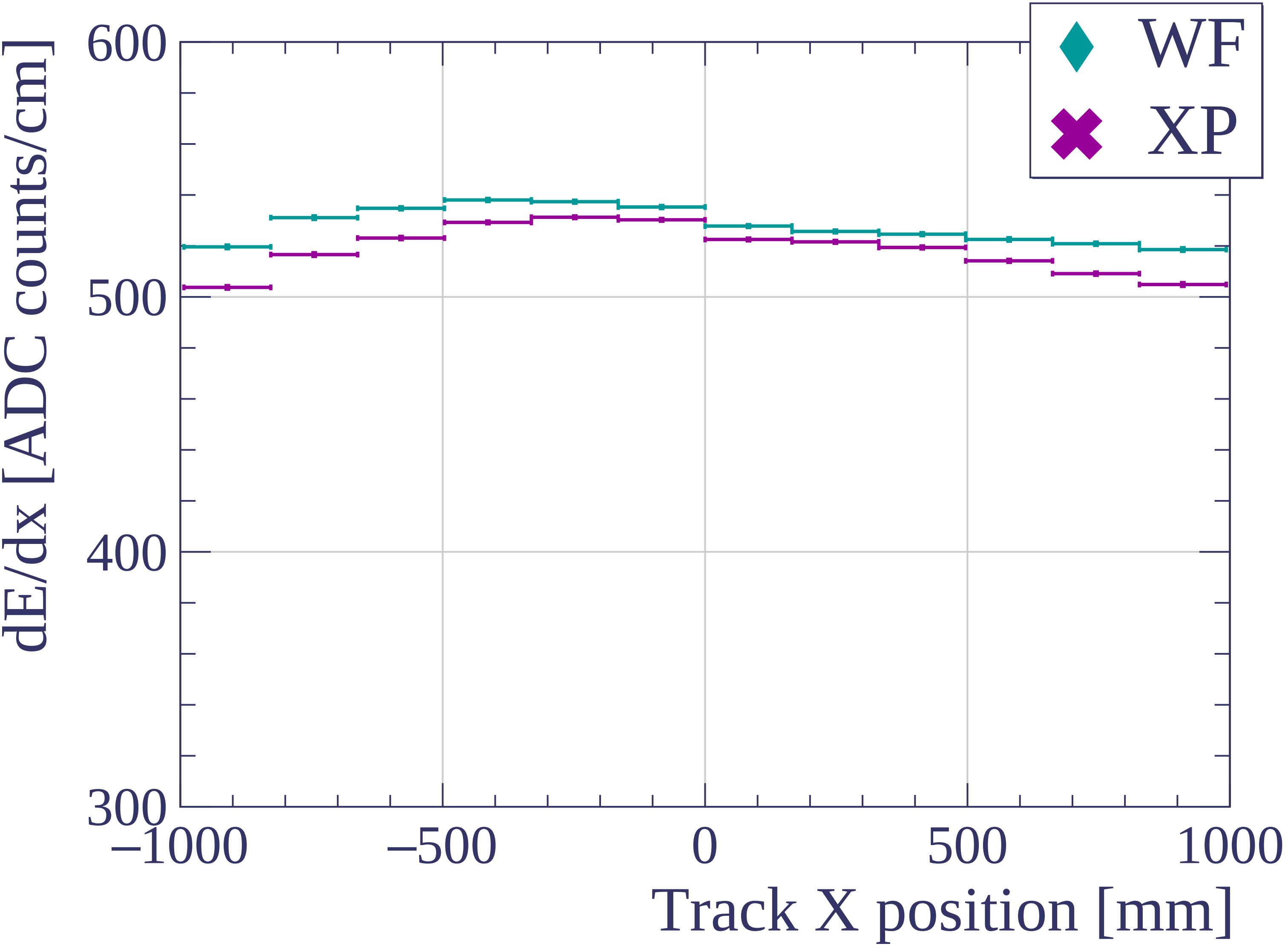


1000
800
600
400
200
0
−2000
−1000
0
1000
2000
dE/dx with XP [ADC counts/cm]
Signed momentum (MeV)
80
60
40
20
0

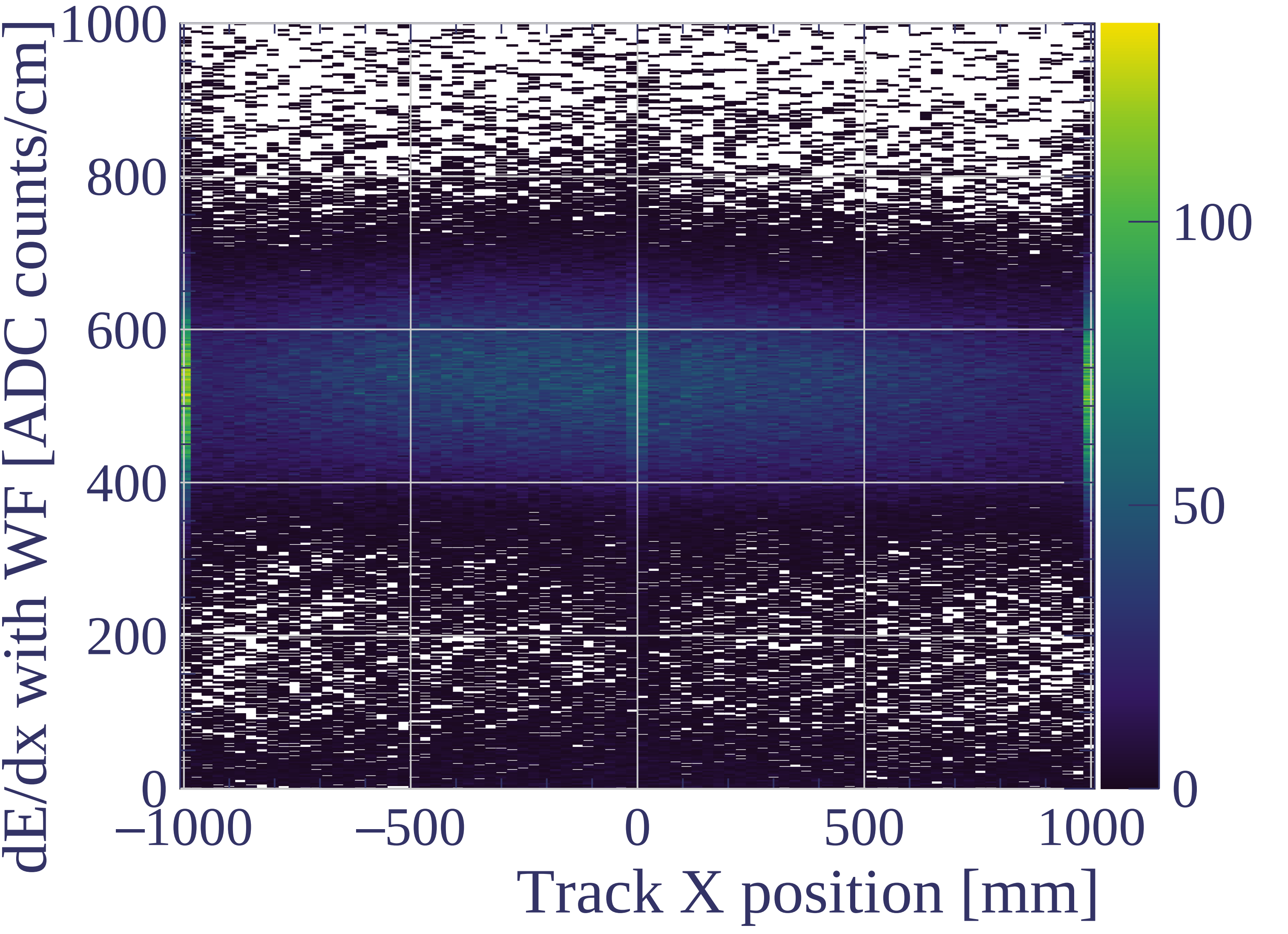

×10^3
Count
Both HATs
tHAT
bHAT
15
10
5
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)


χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
80
60
40
20
0

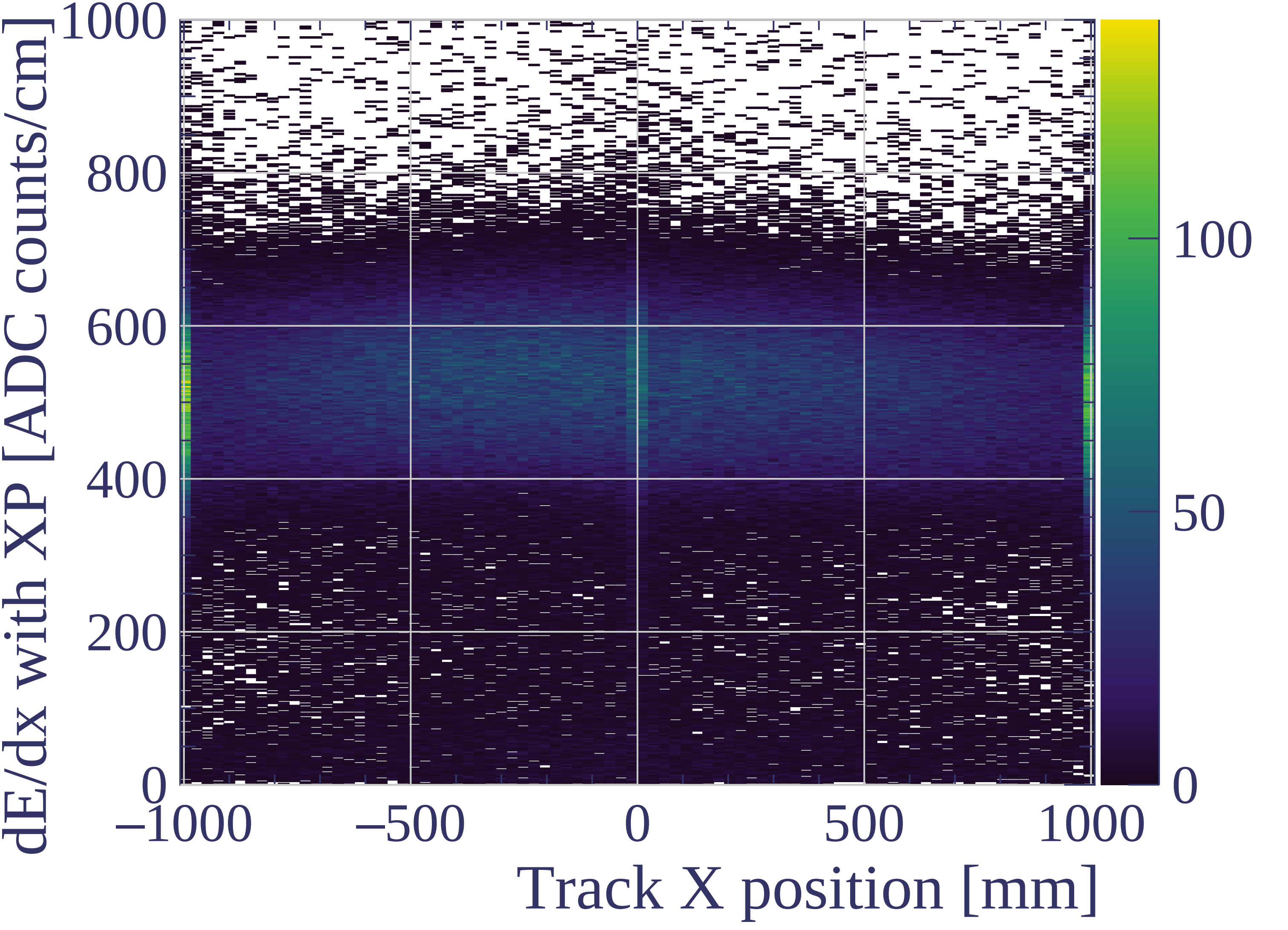

WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]



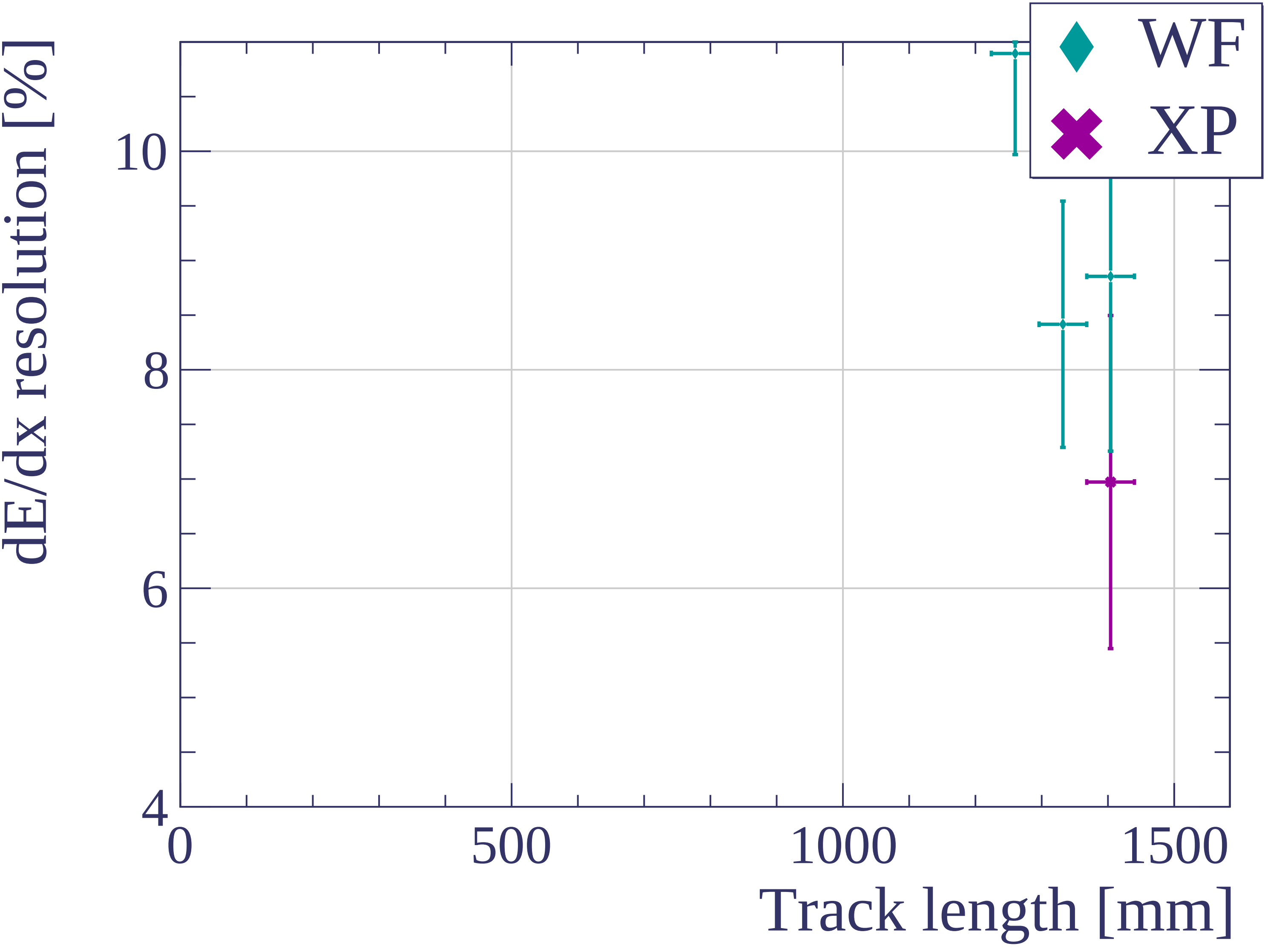
WF
XP
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]



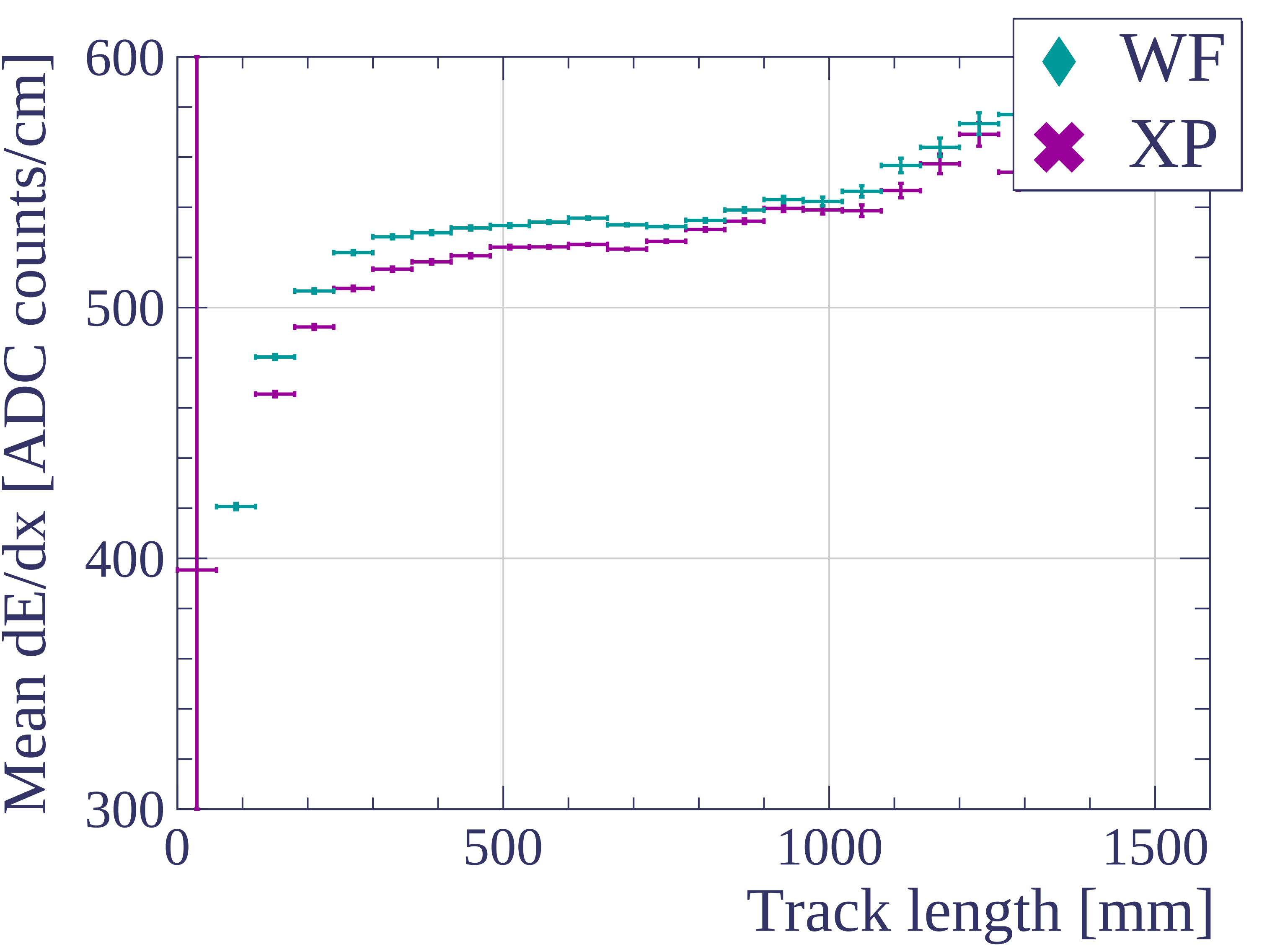
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
100
50
0



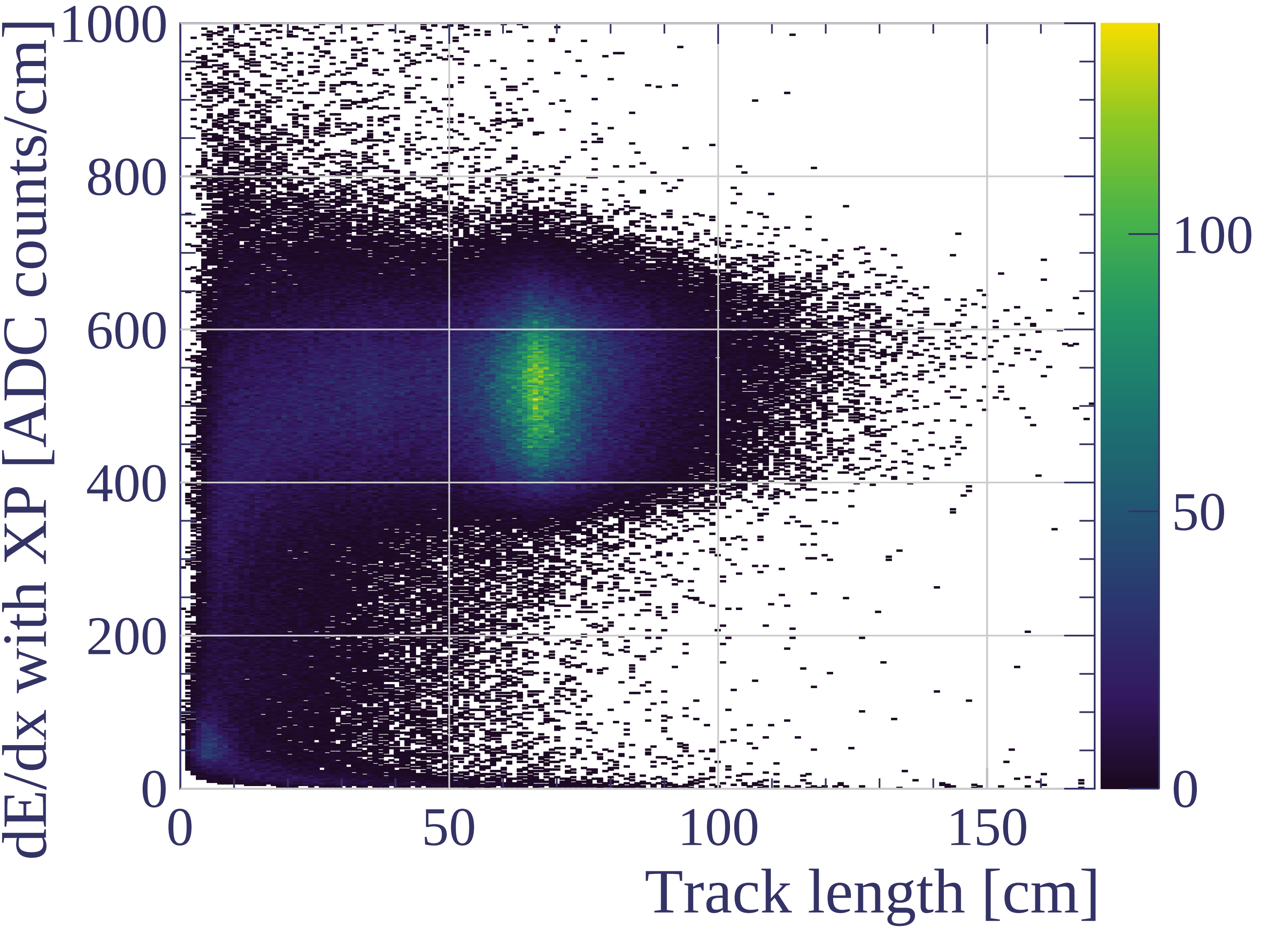
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
100
50
0



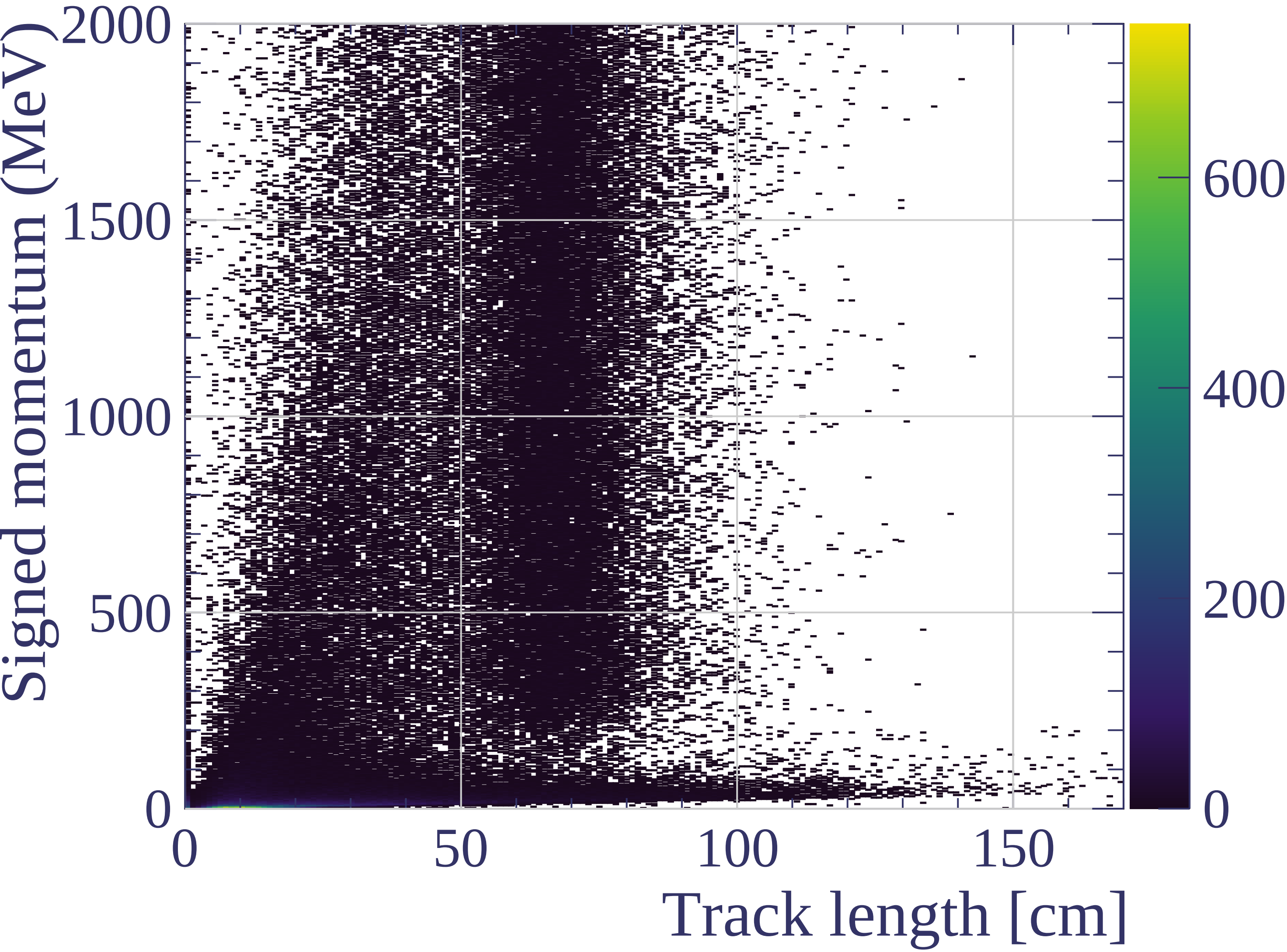
WF
XP
dE/dx resolution [%]
4
6
8
10
0
500
1000
1500
Track length [mm]



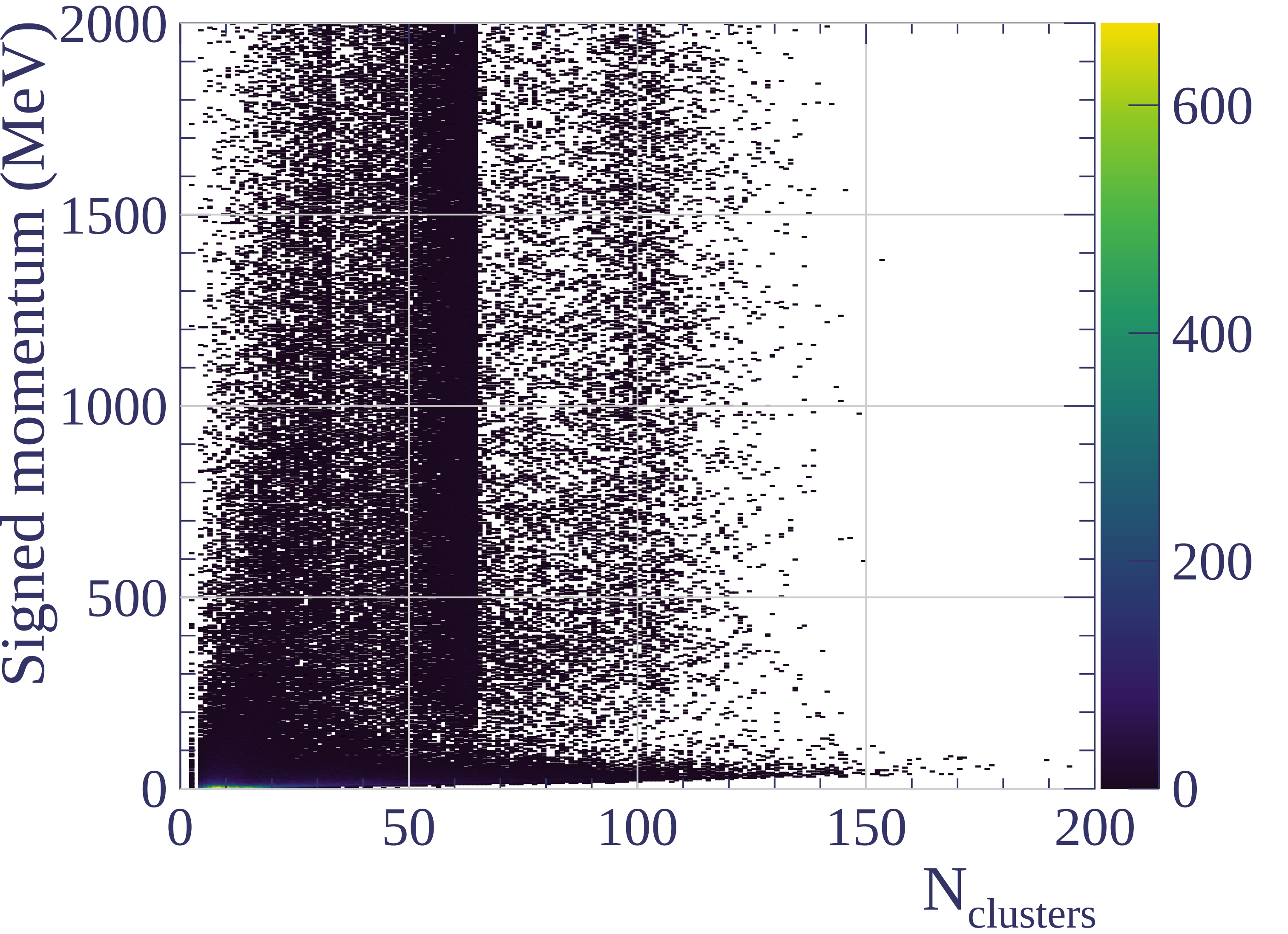
WF
XP
Mean dE/dx [ADC counts/cm]
Track length [mm]
300
400
500
600
0
500
1000
1500



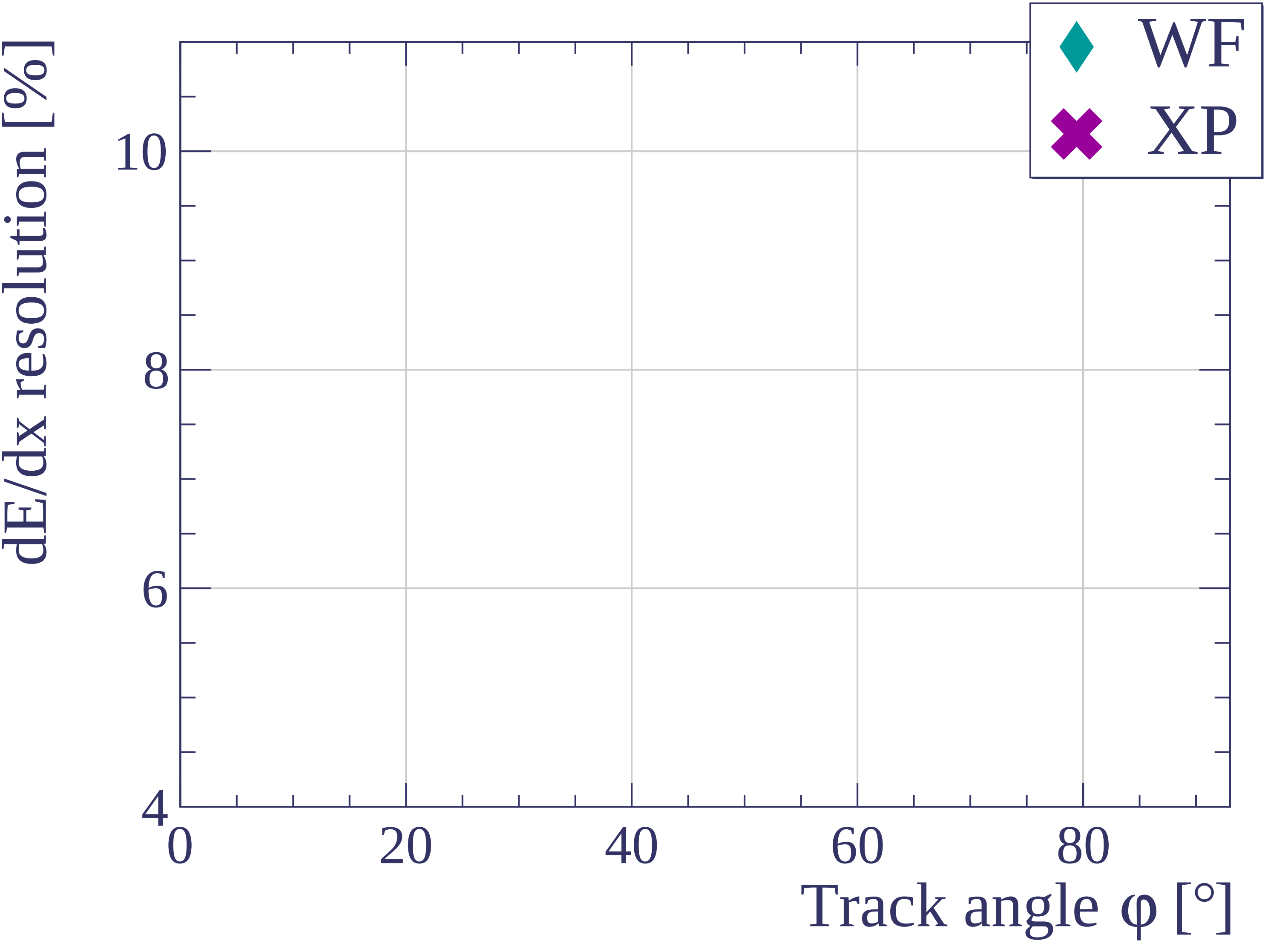
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
100
50
0



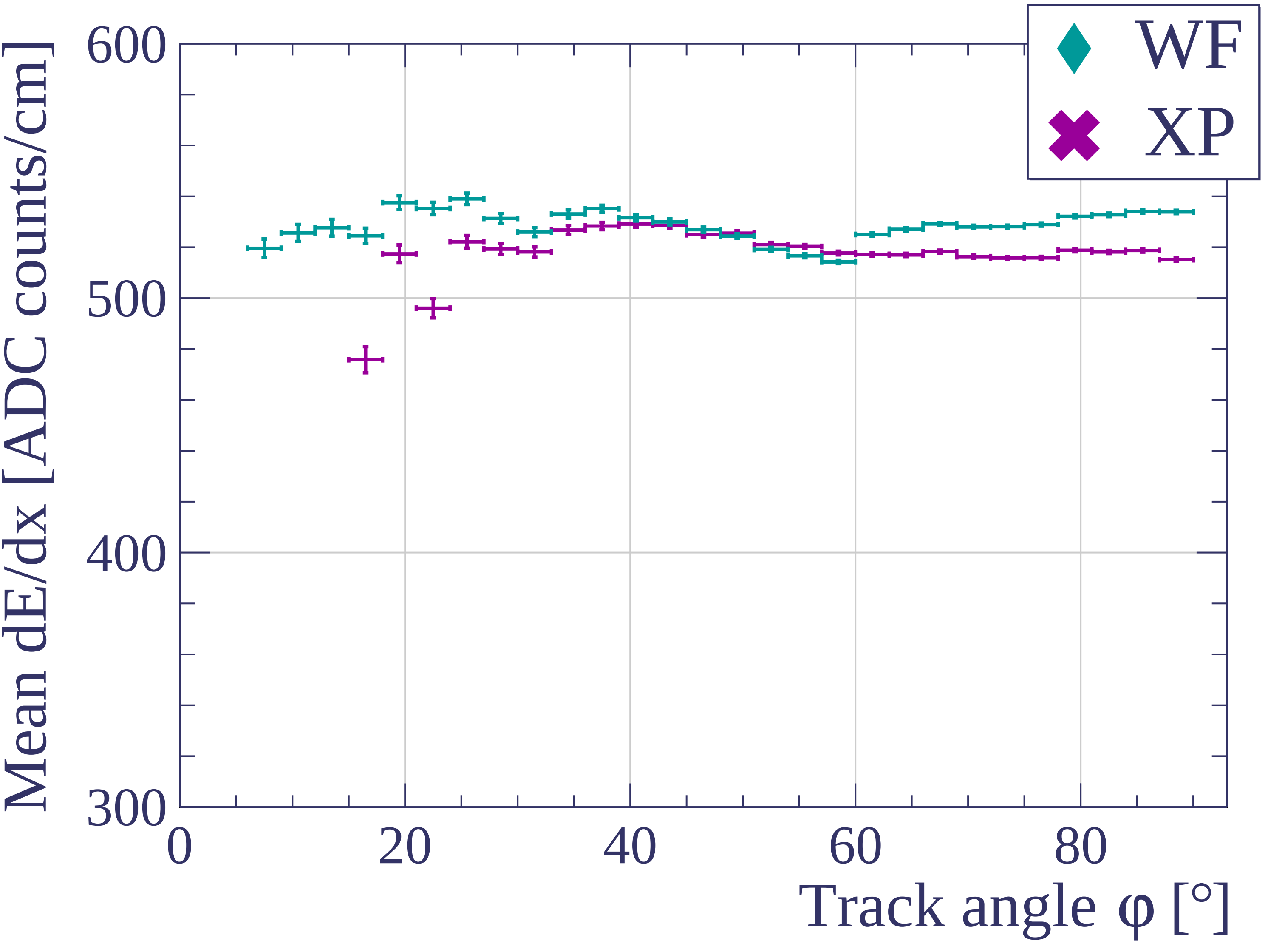
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
600
400
200
0

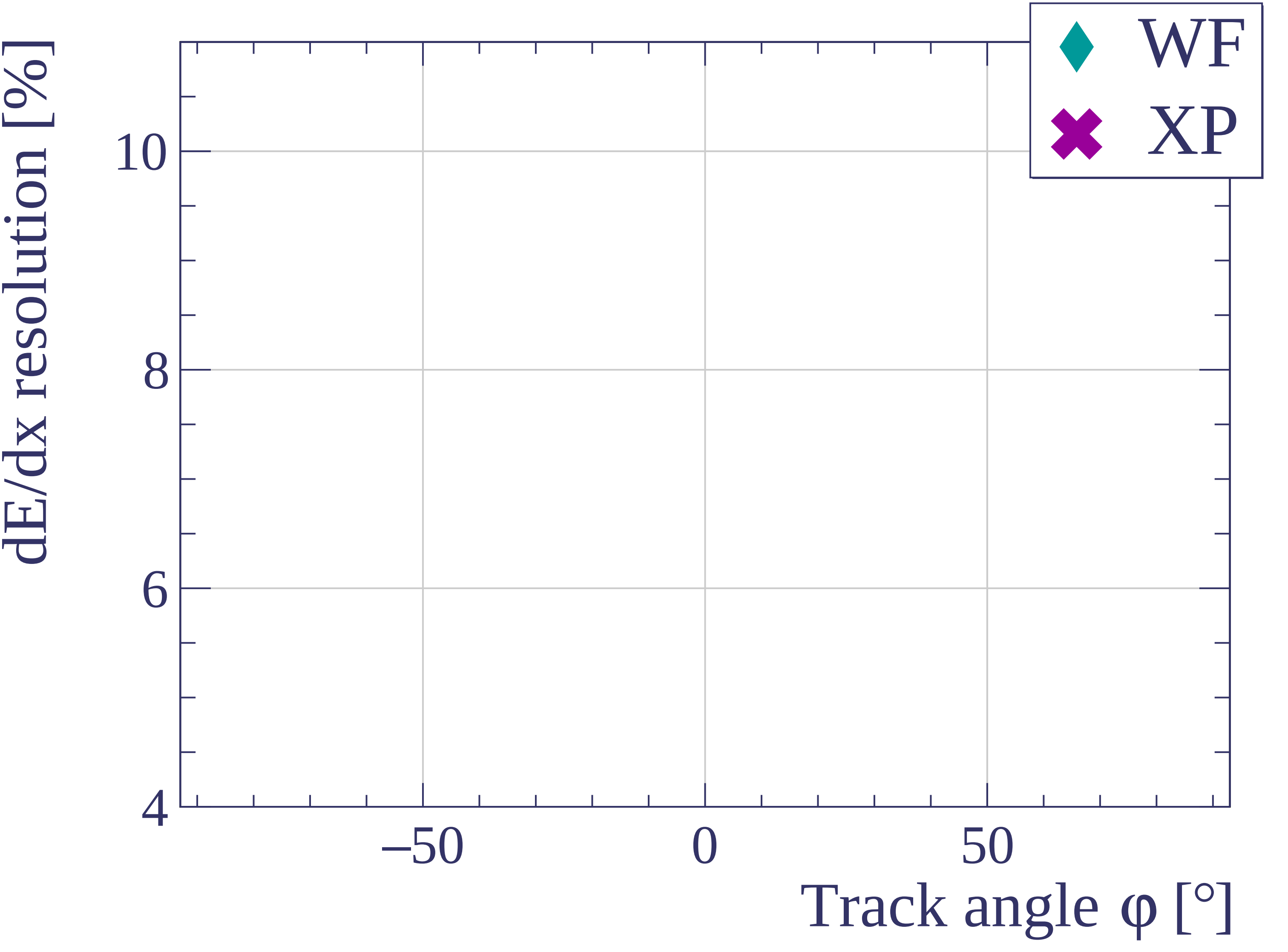


Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
600
400
200
0

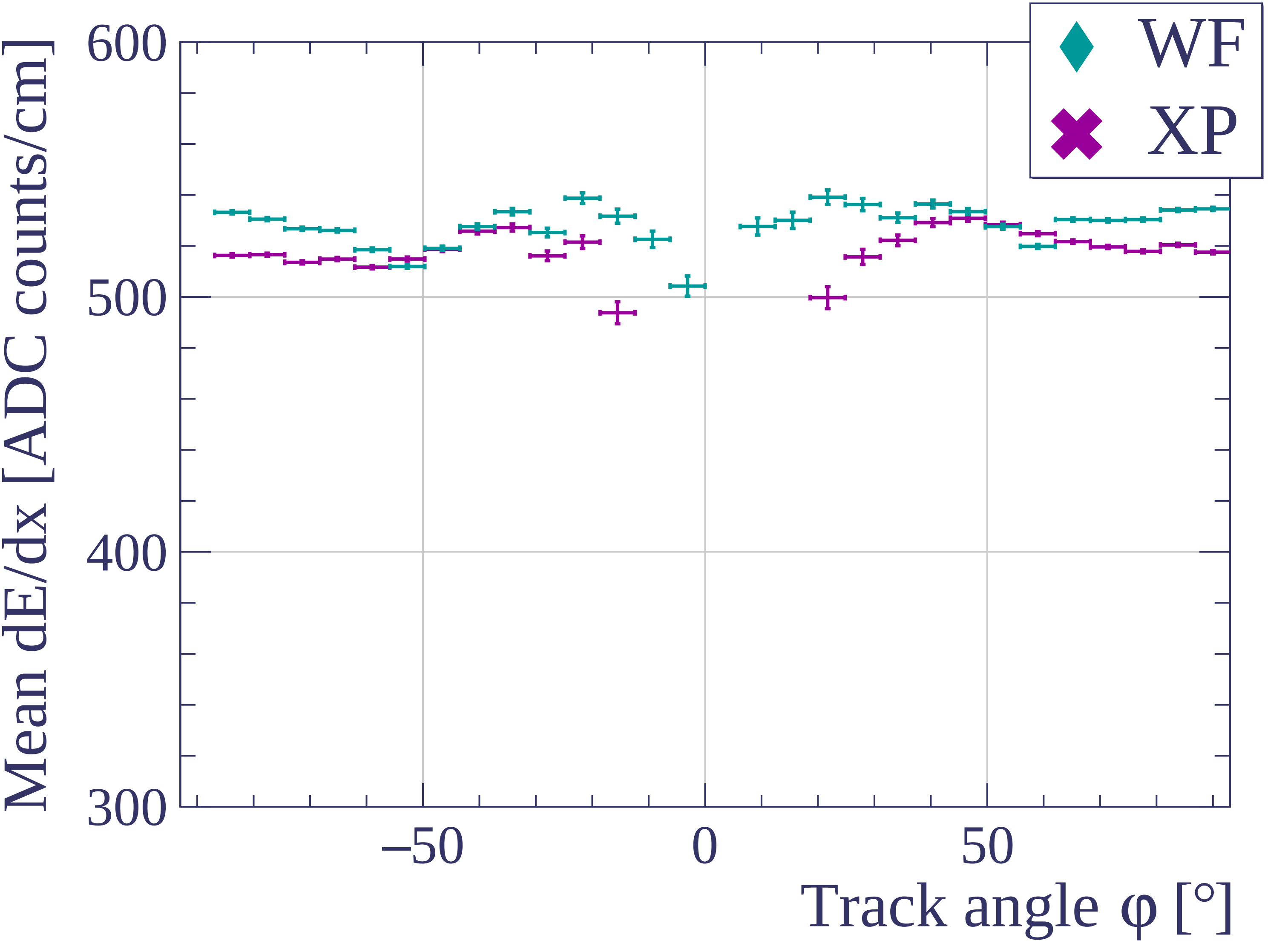


WF
XP
dE/dx resolution [%]
4
6
8
10
0
20
40
60
80
Track angle φ [°]

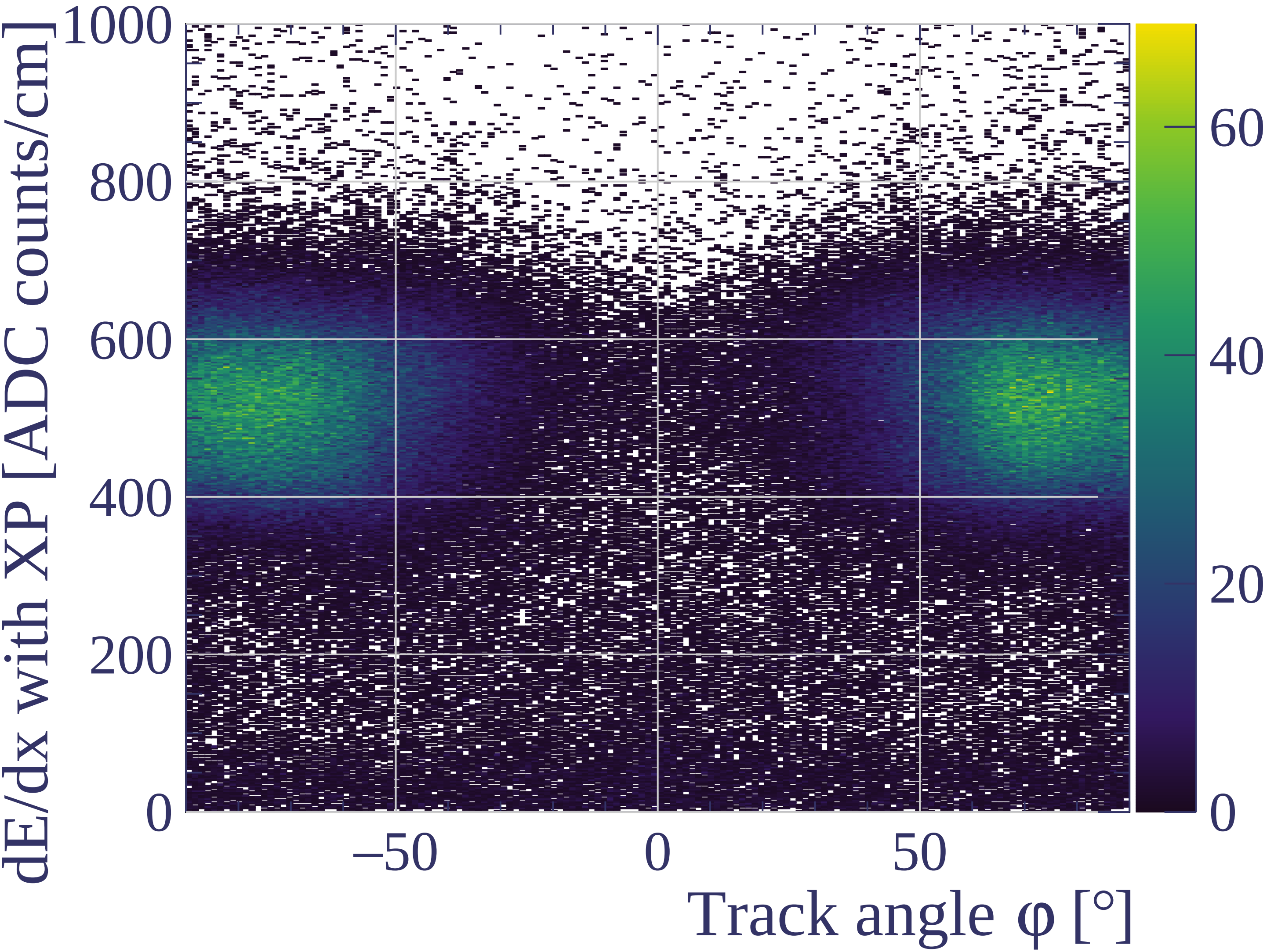

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle φ [°]

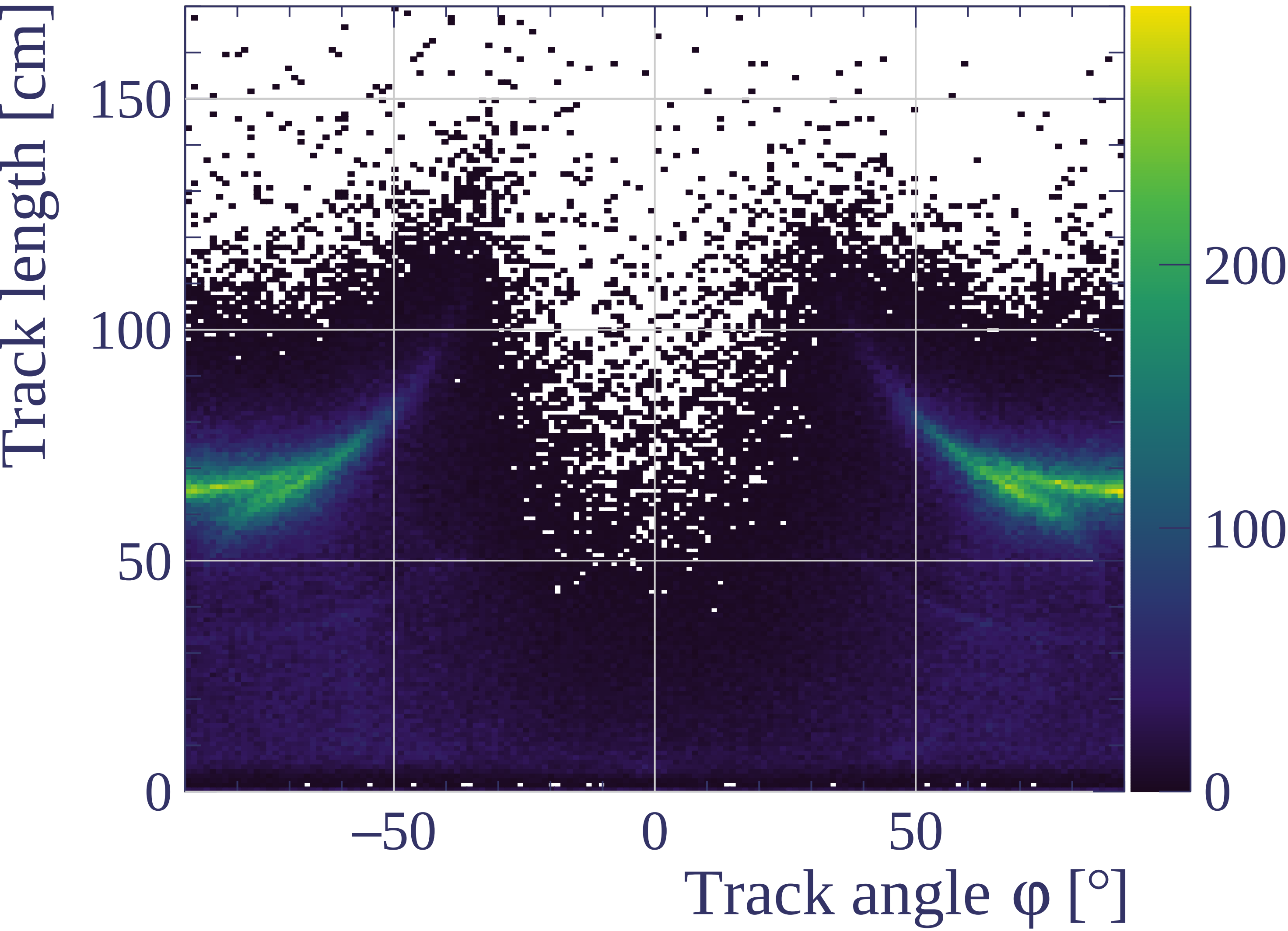


WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

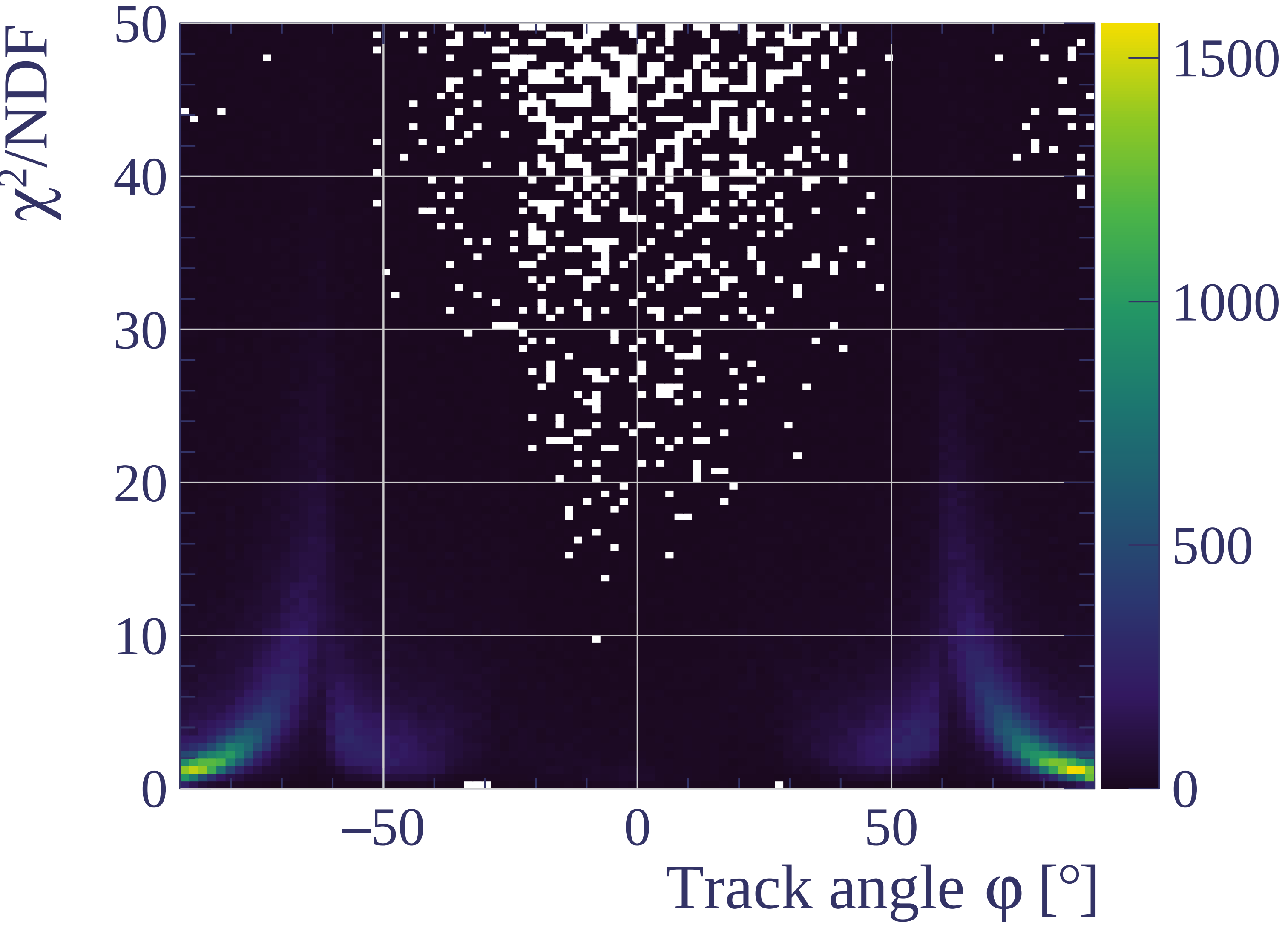


Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle φ [°]
WF
XP

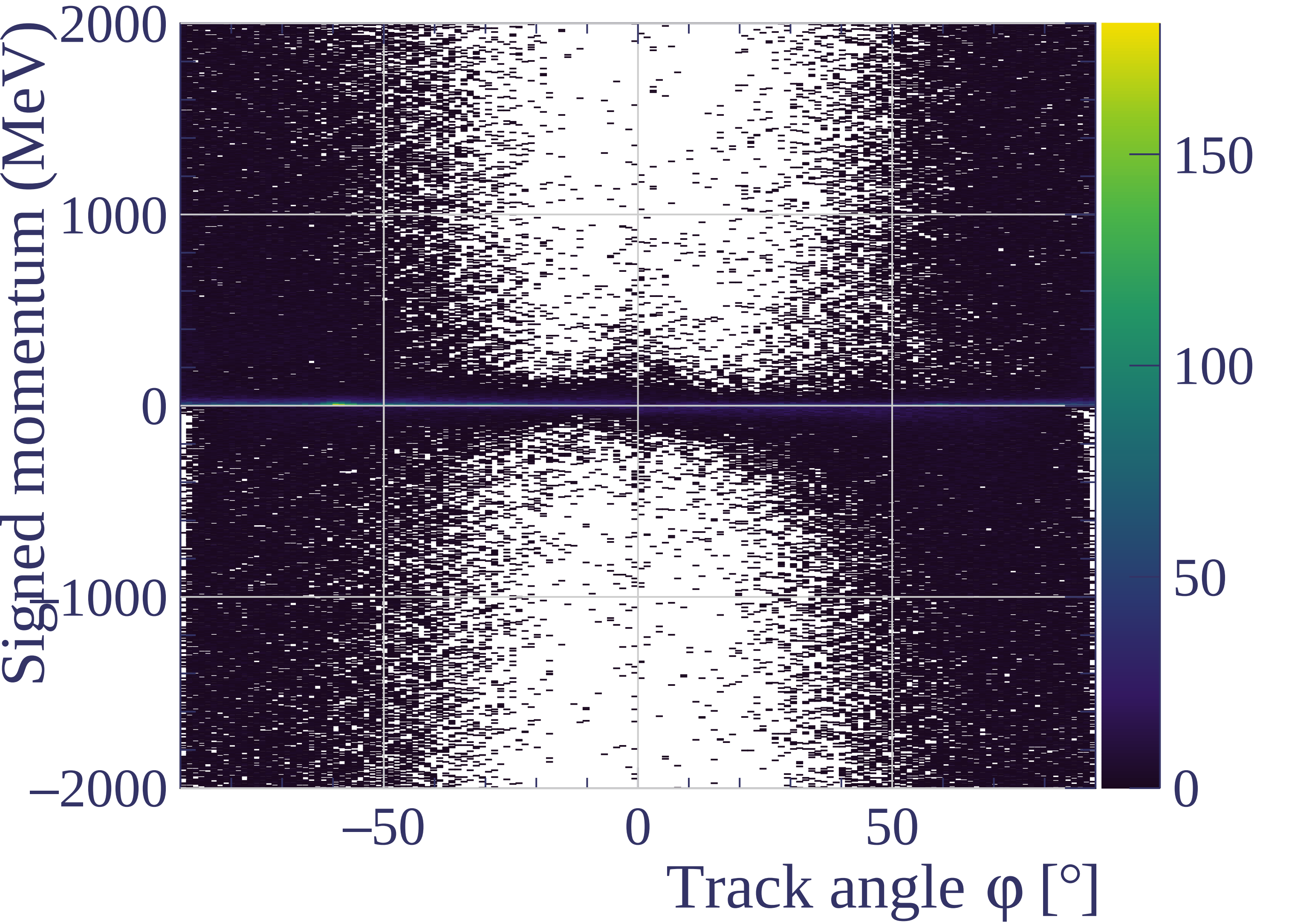

dE/dx with XP [ADC counts/cm]
Track angle φ [°]
1000
800
600
400
200
0
−50
0
50
60
40
20
0



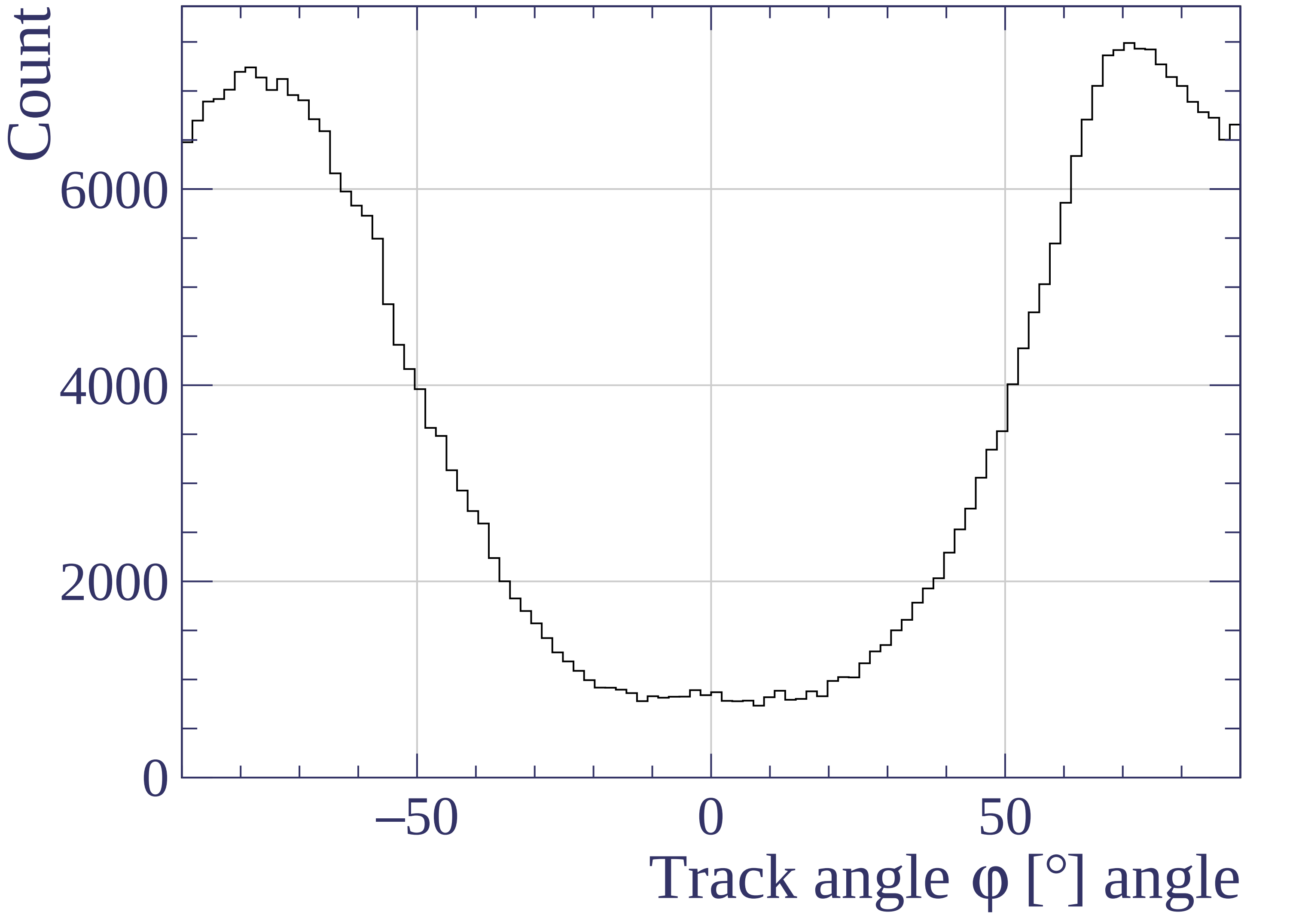
Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
200
100
0

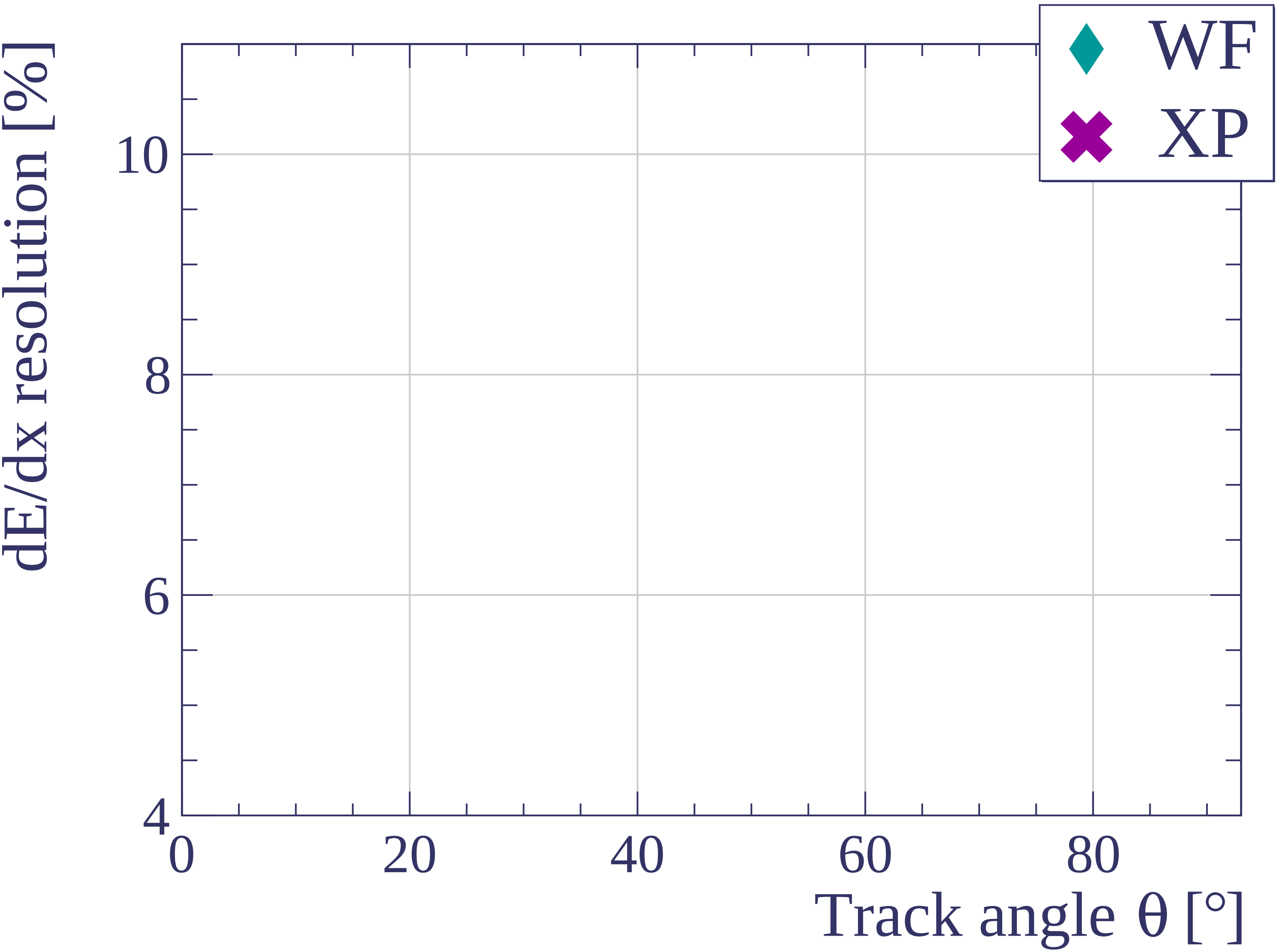


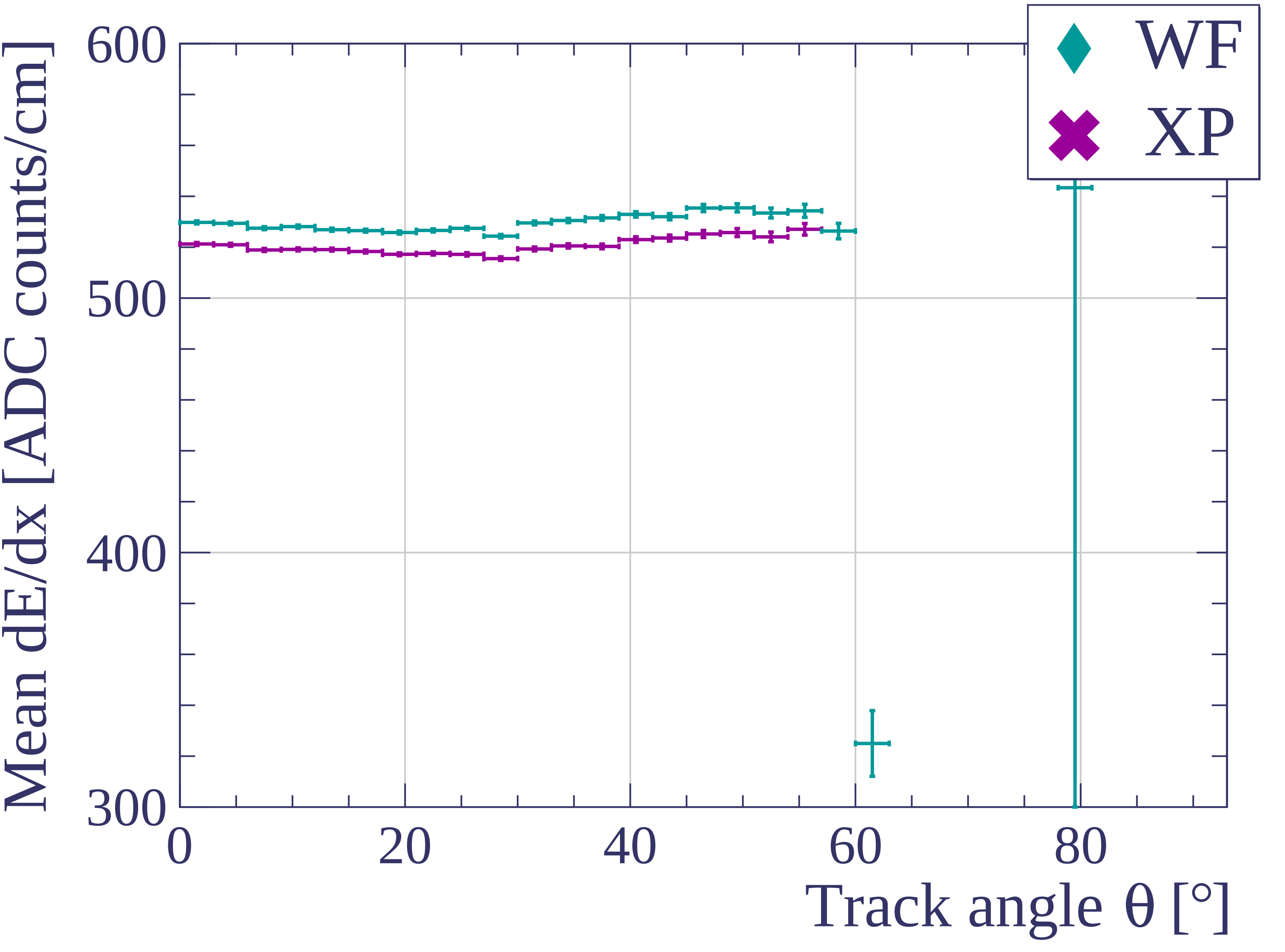
χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
1500
1000
500
0



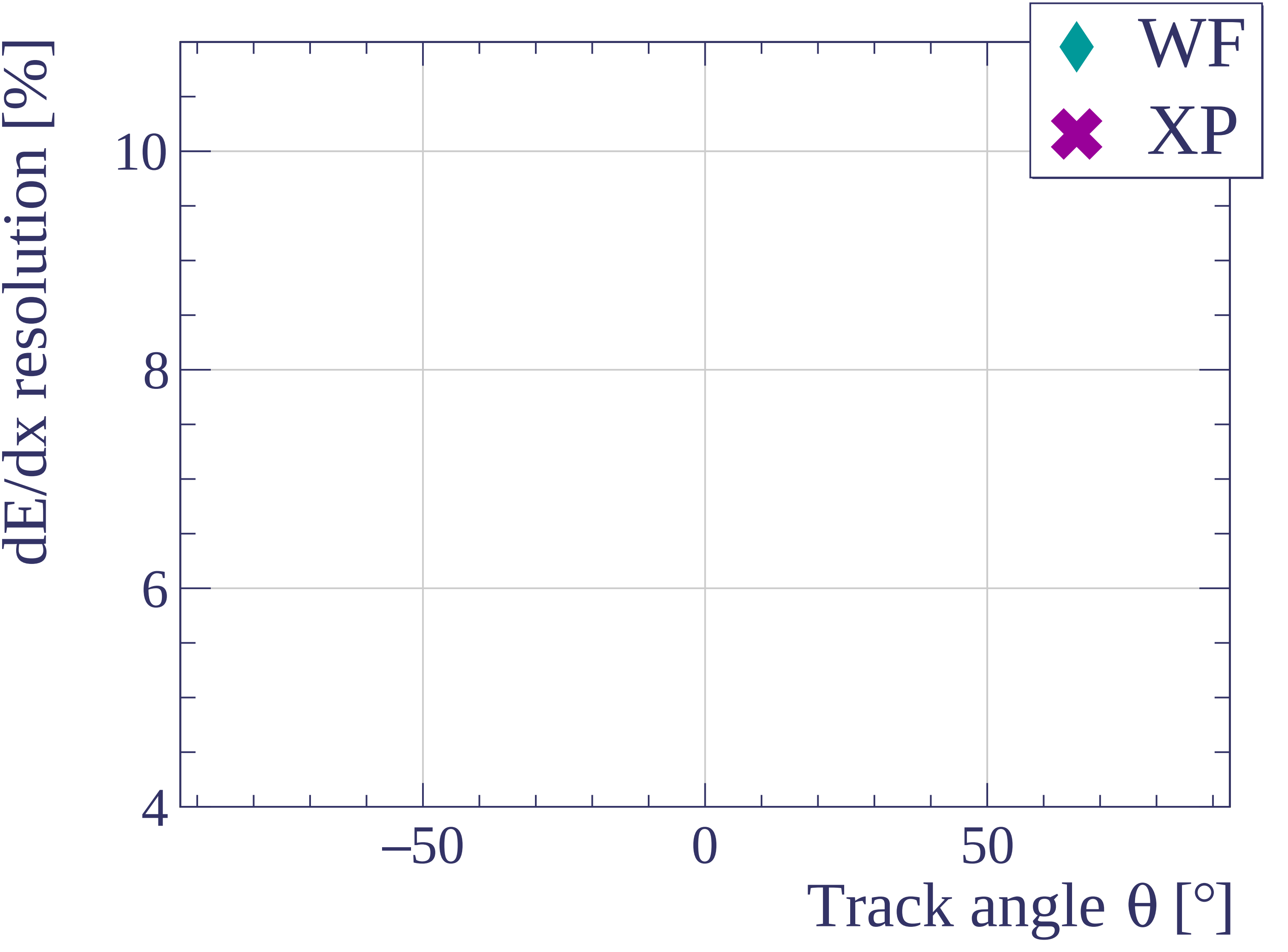
2000
1000
0
−1000
−2000
Signed momentum (MeV)
−50
0
50
Track angle φ [°]
150
100
50
0


Count
6000
4000
2000
0
−50
0
50
Track angle φ [°] angle

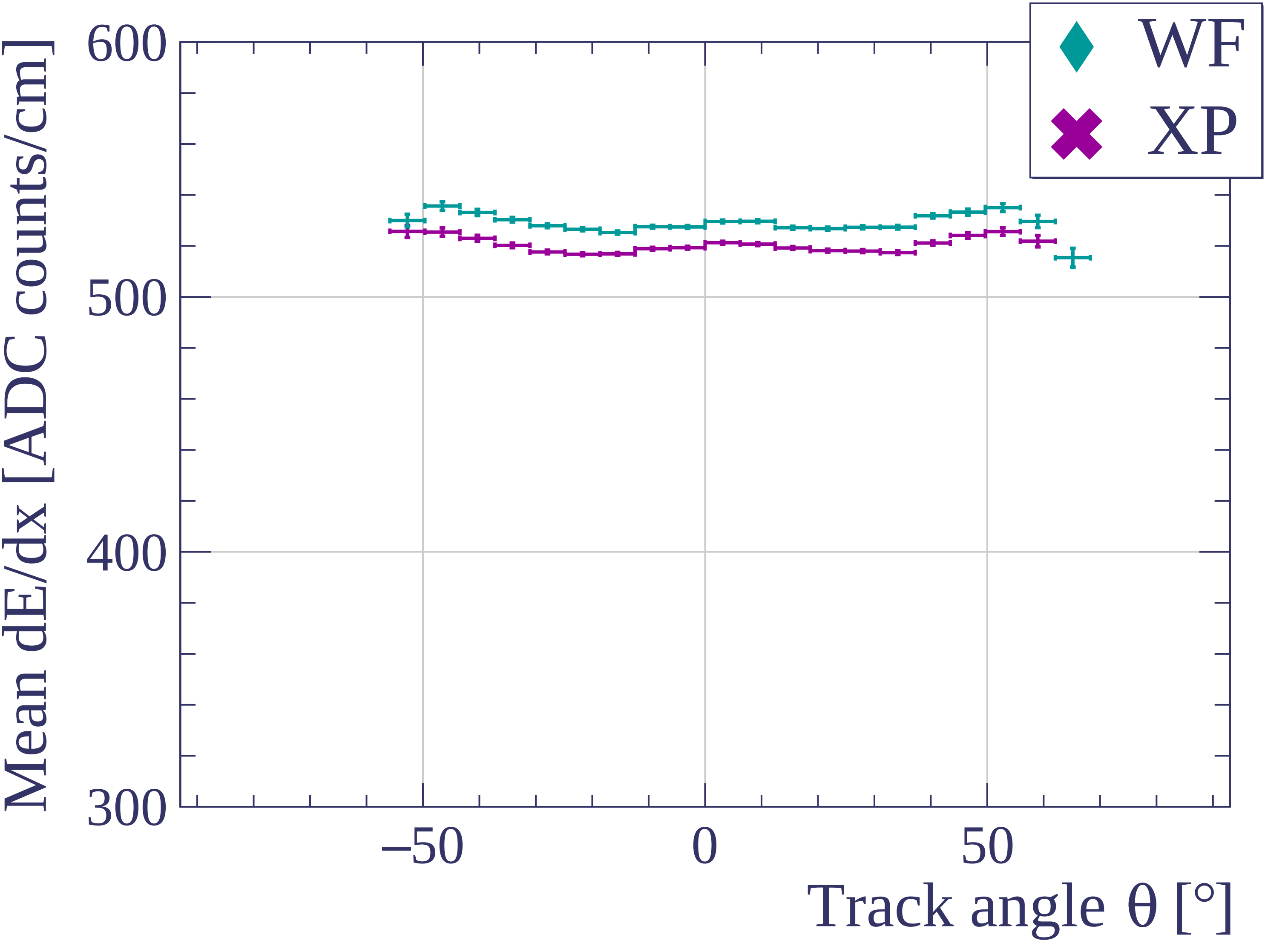

WF
XP
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle θ [°]

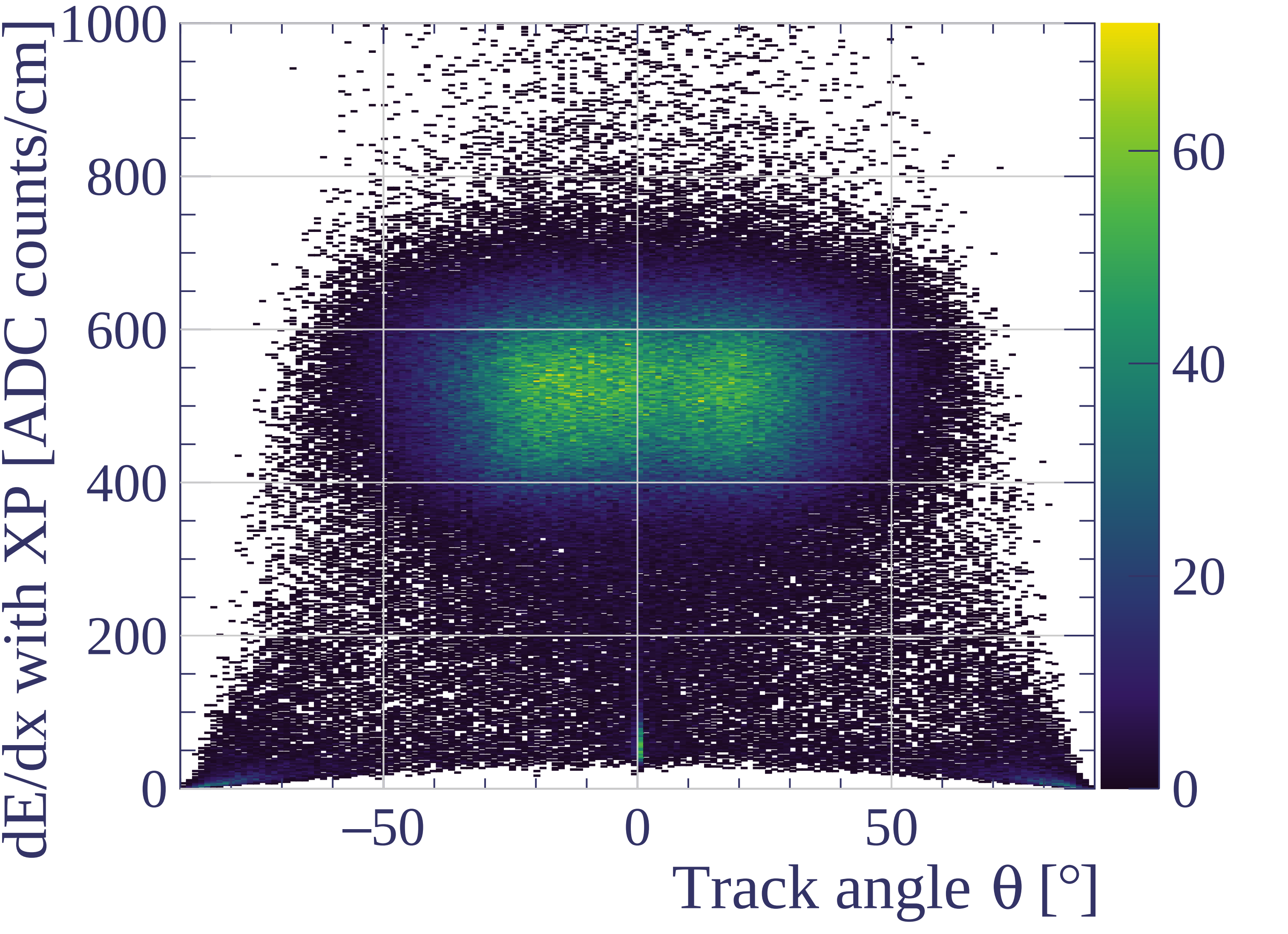


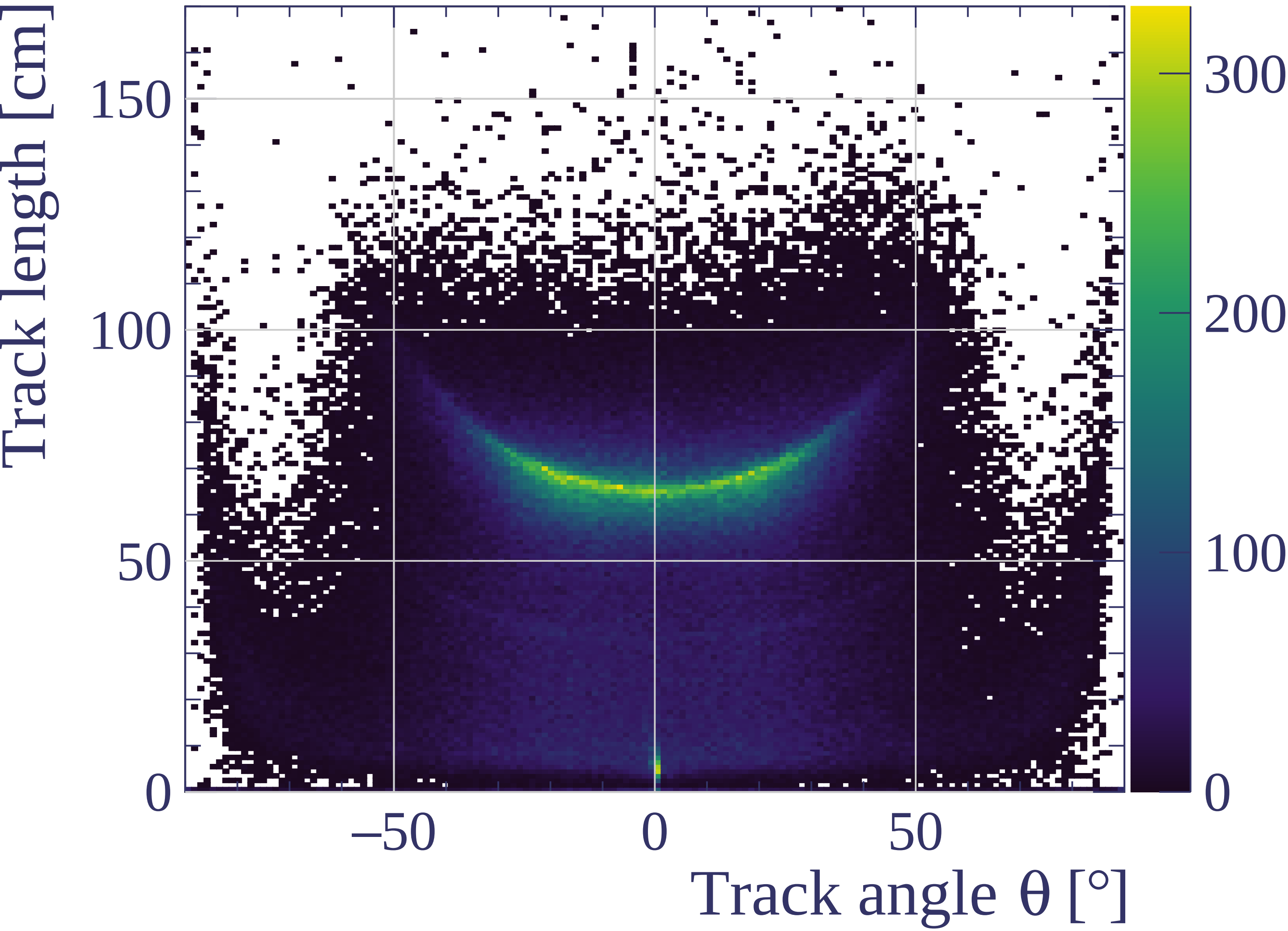
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle θ [°]



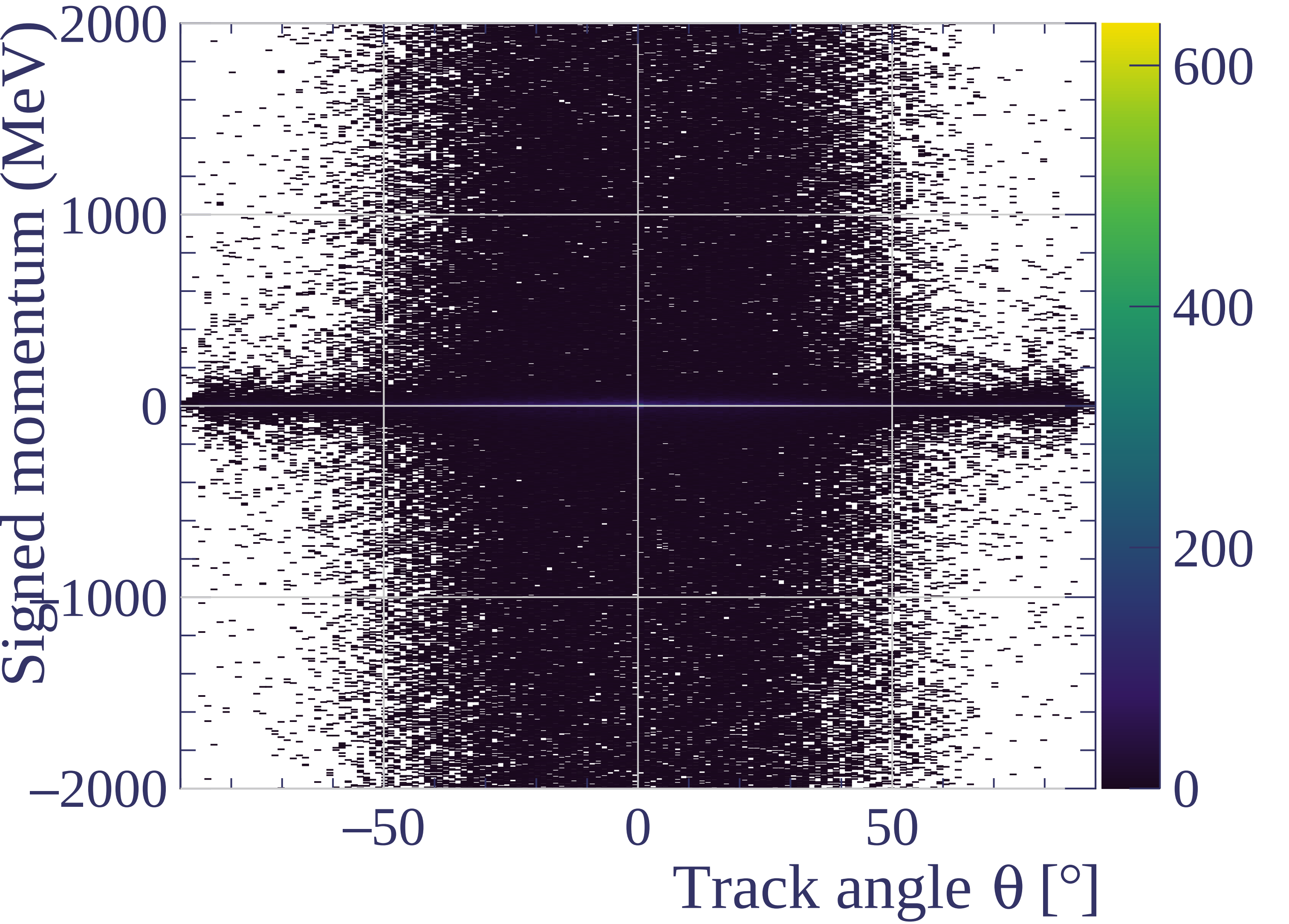
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle θ [°]

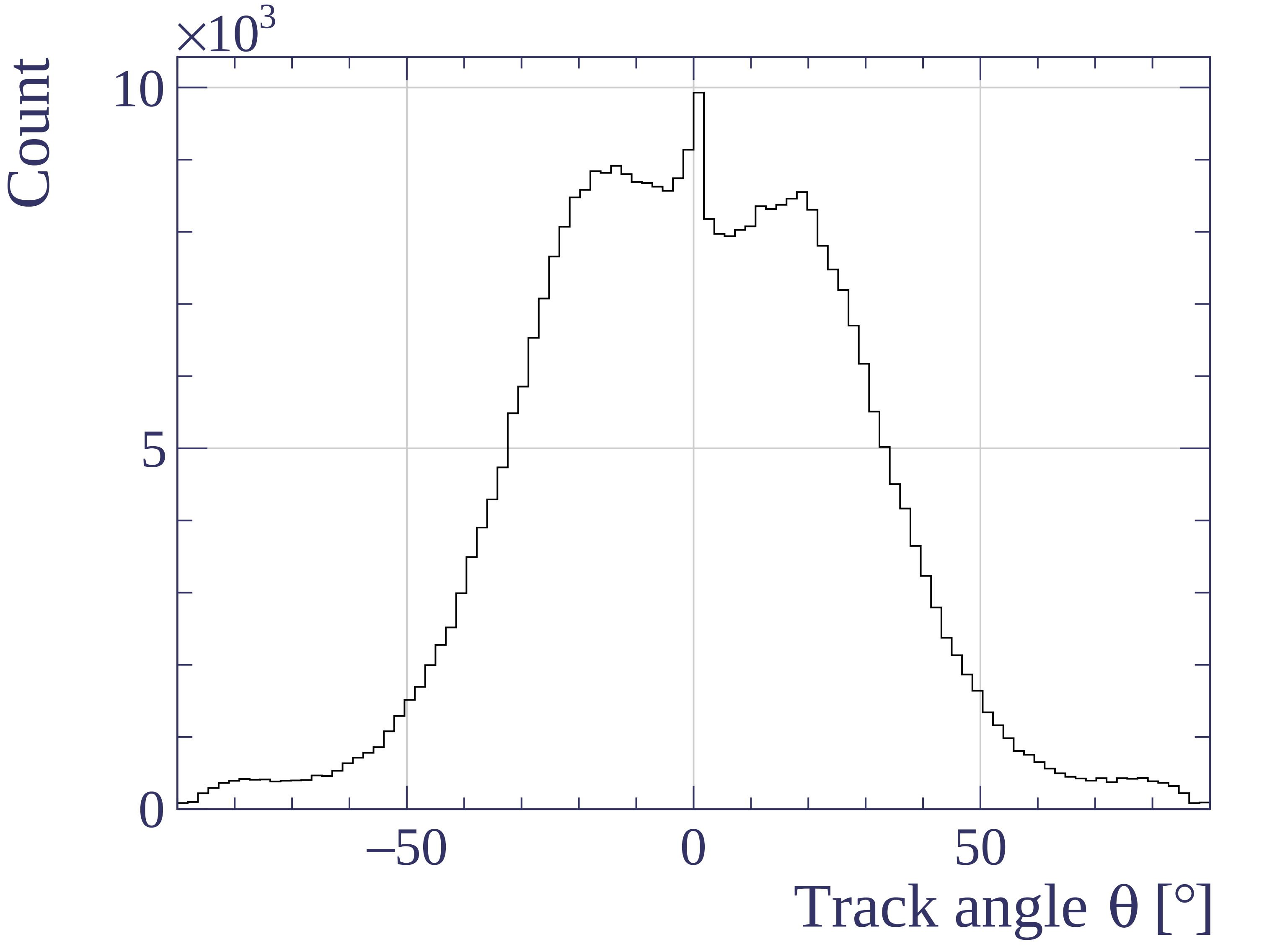

1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
dE/dx with XP [ADC counts/cm]
60
40
20
0



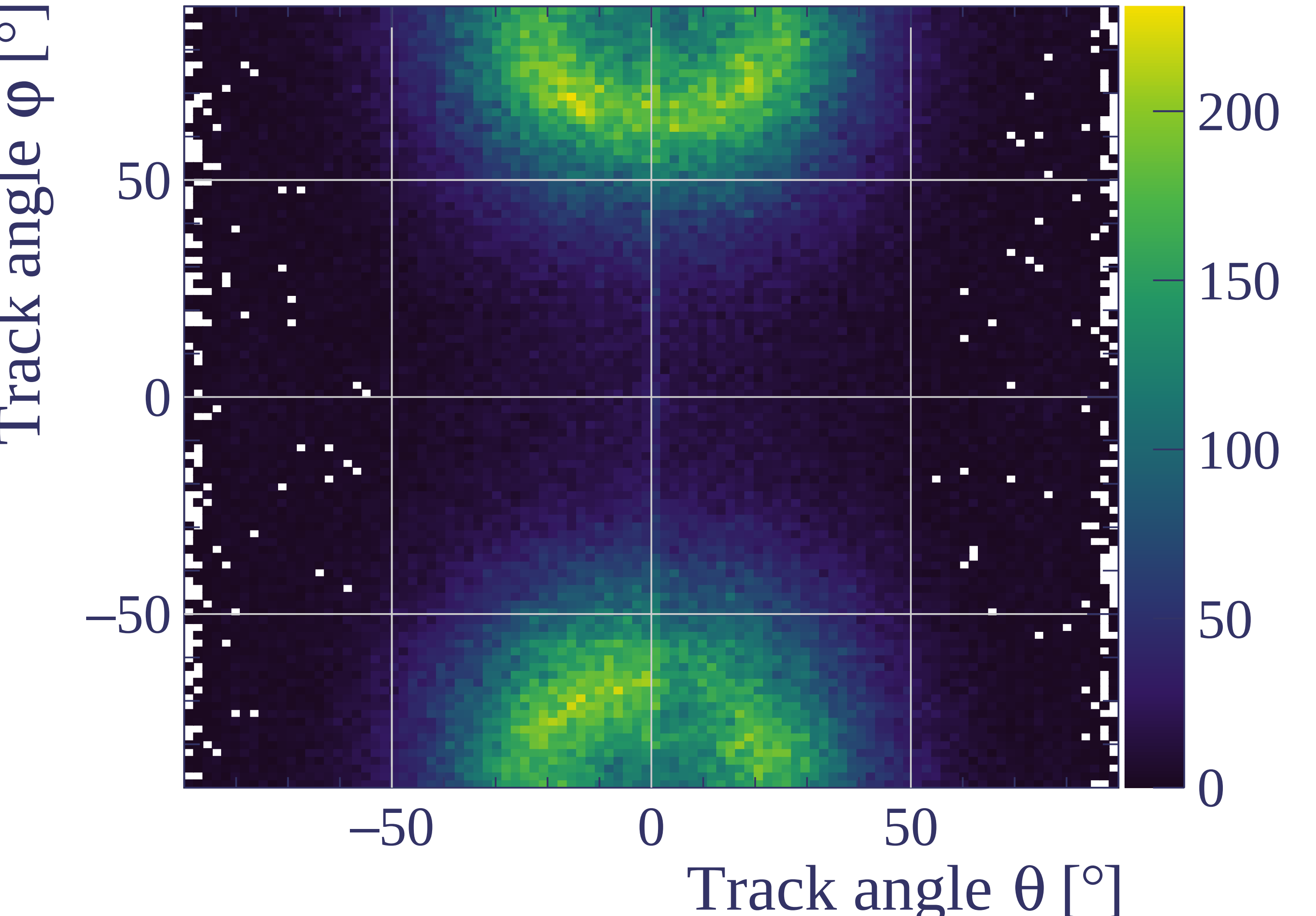
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
300
200
100
0

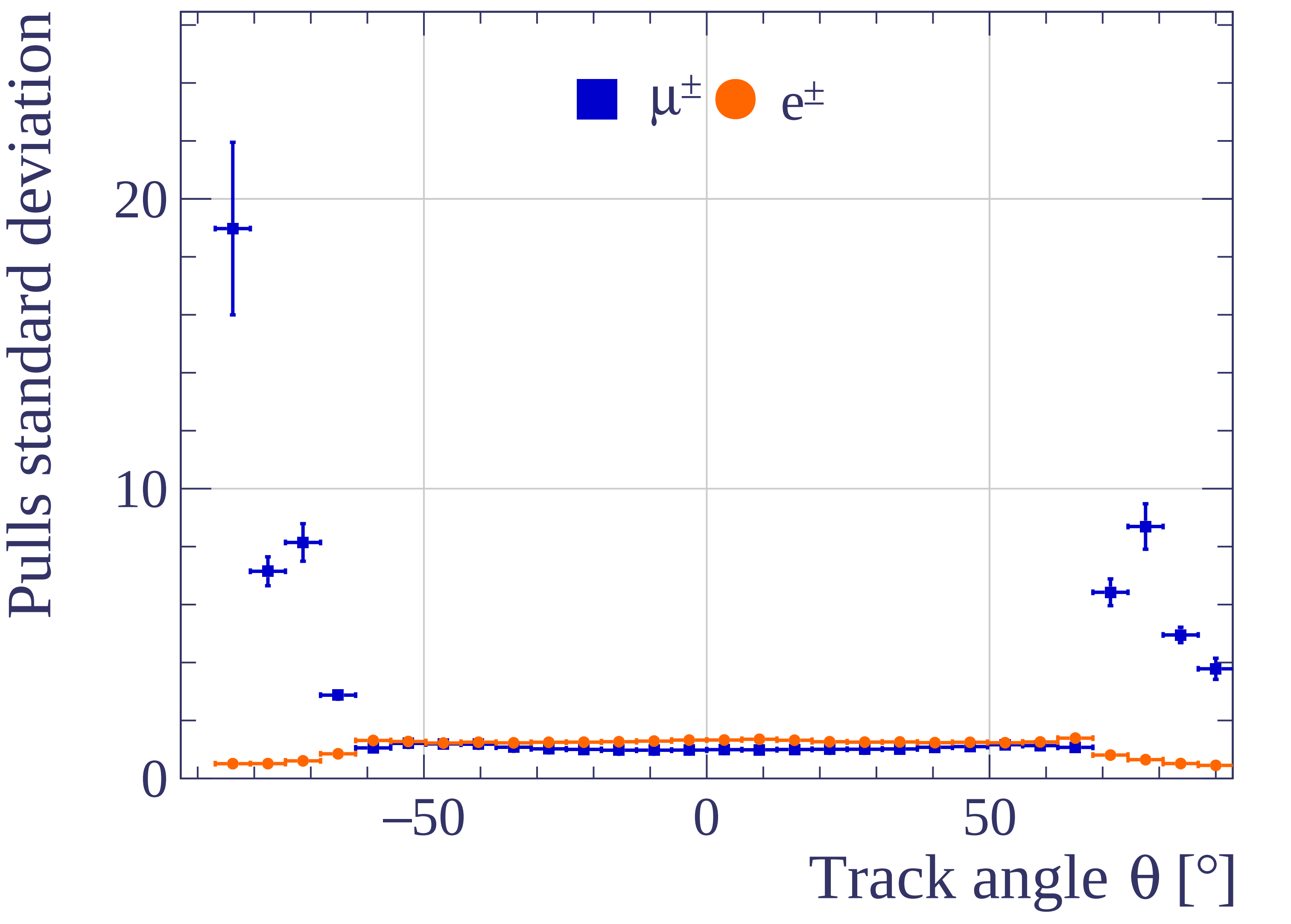


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
600
400
200
0

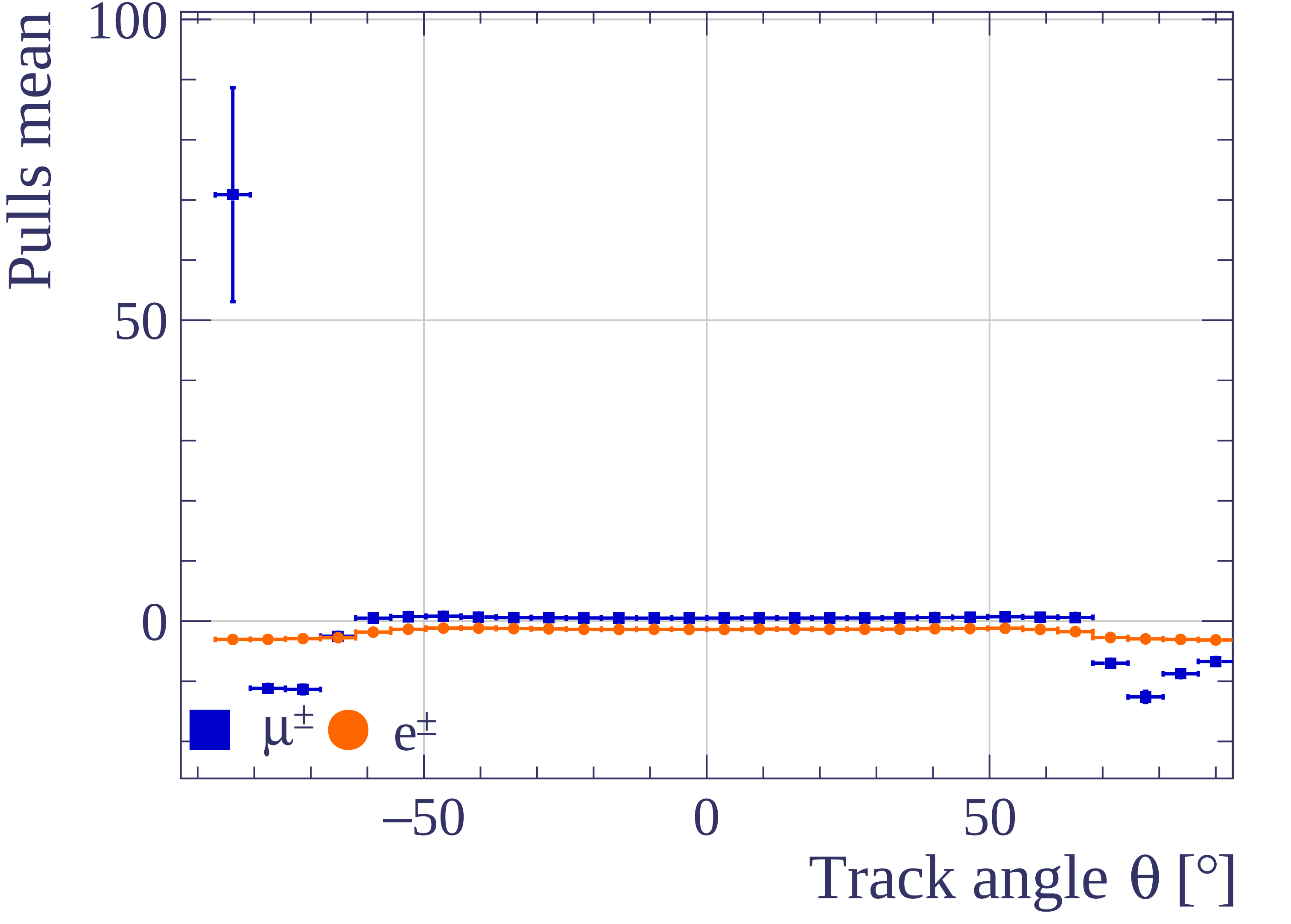


×10^3
Count
10
5
0
−50
0
50
Track angle θ [°]

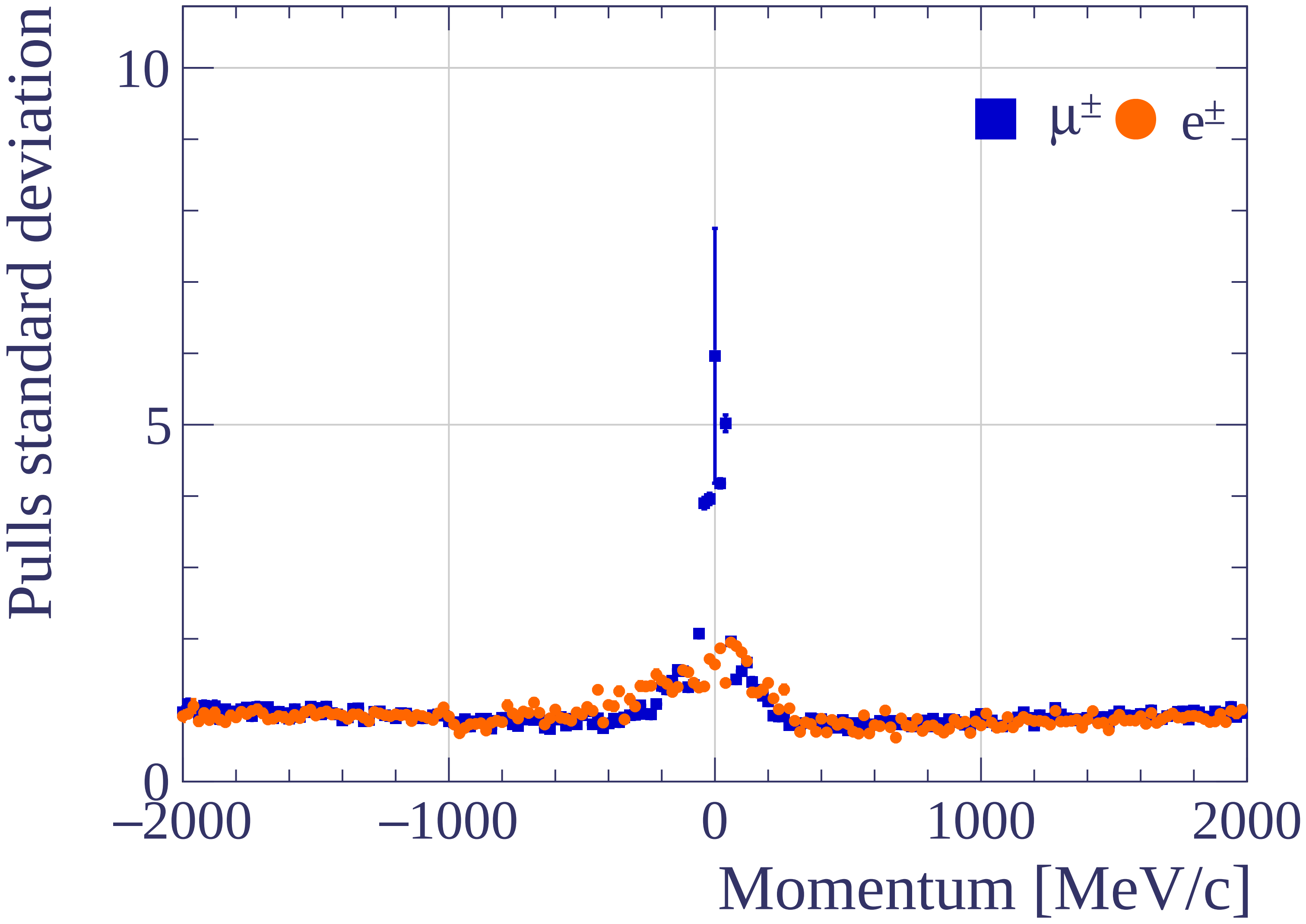

Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
200
150
100
50
0



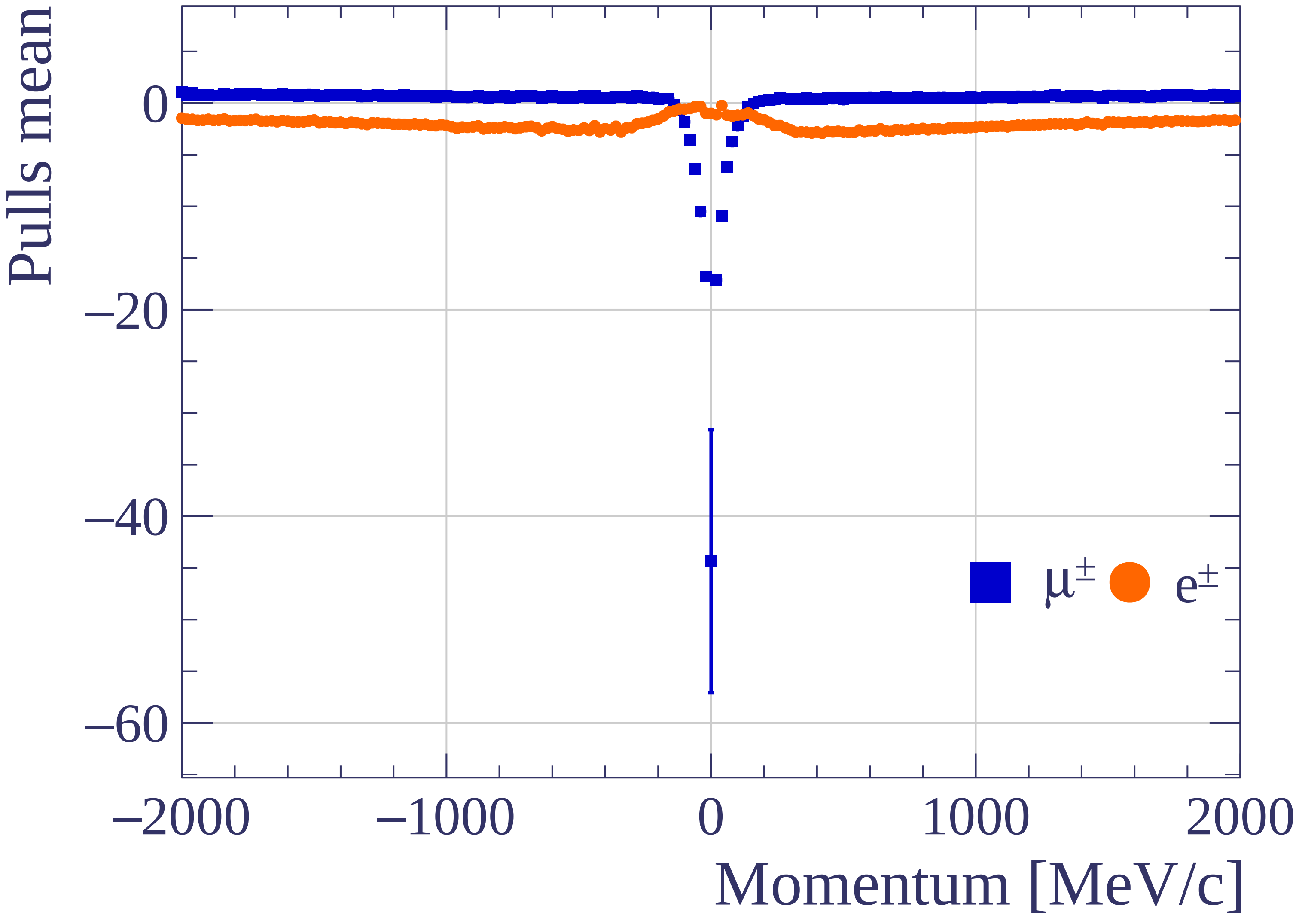
Pulls standard deviation
20
10
0
$\mu^{\pm}$
$e^{\pm}$
−50
0
50
Track angle θ [°]



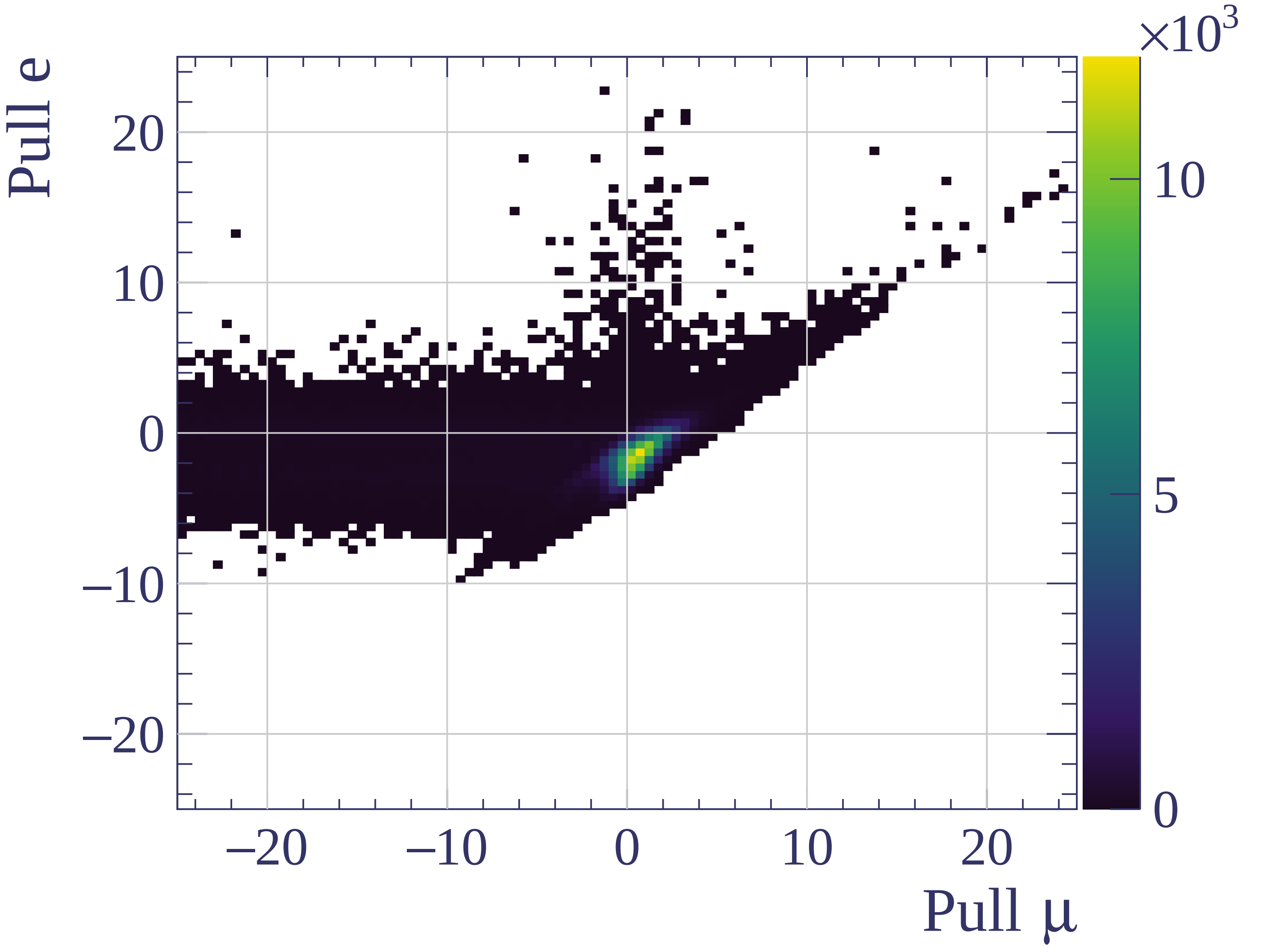
Pulls mean
100
50
0
−50
0
50
Track angle θ [°]
μ±
e±



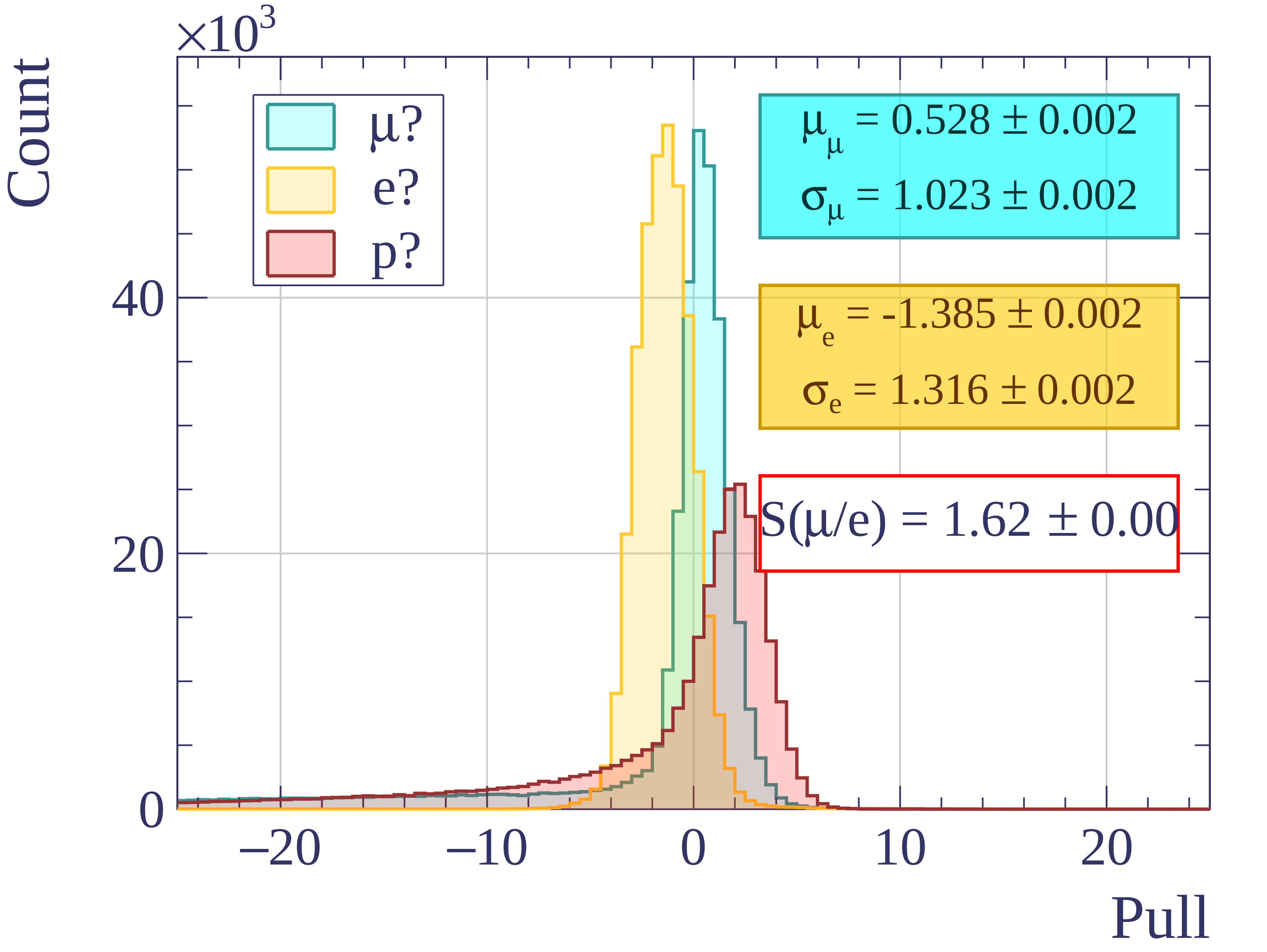
Pulls standard deviation
Momentum [MeV/c]
10
5
0
−2000
−1000
0
1000
2000
$\mu^{\pm}$
$e^{\pm}$



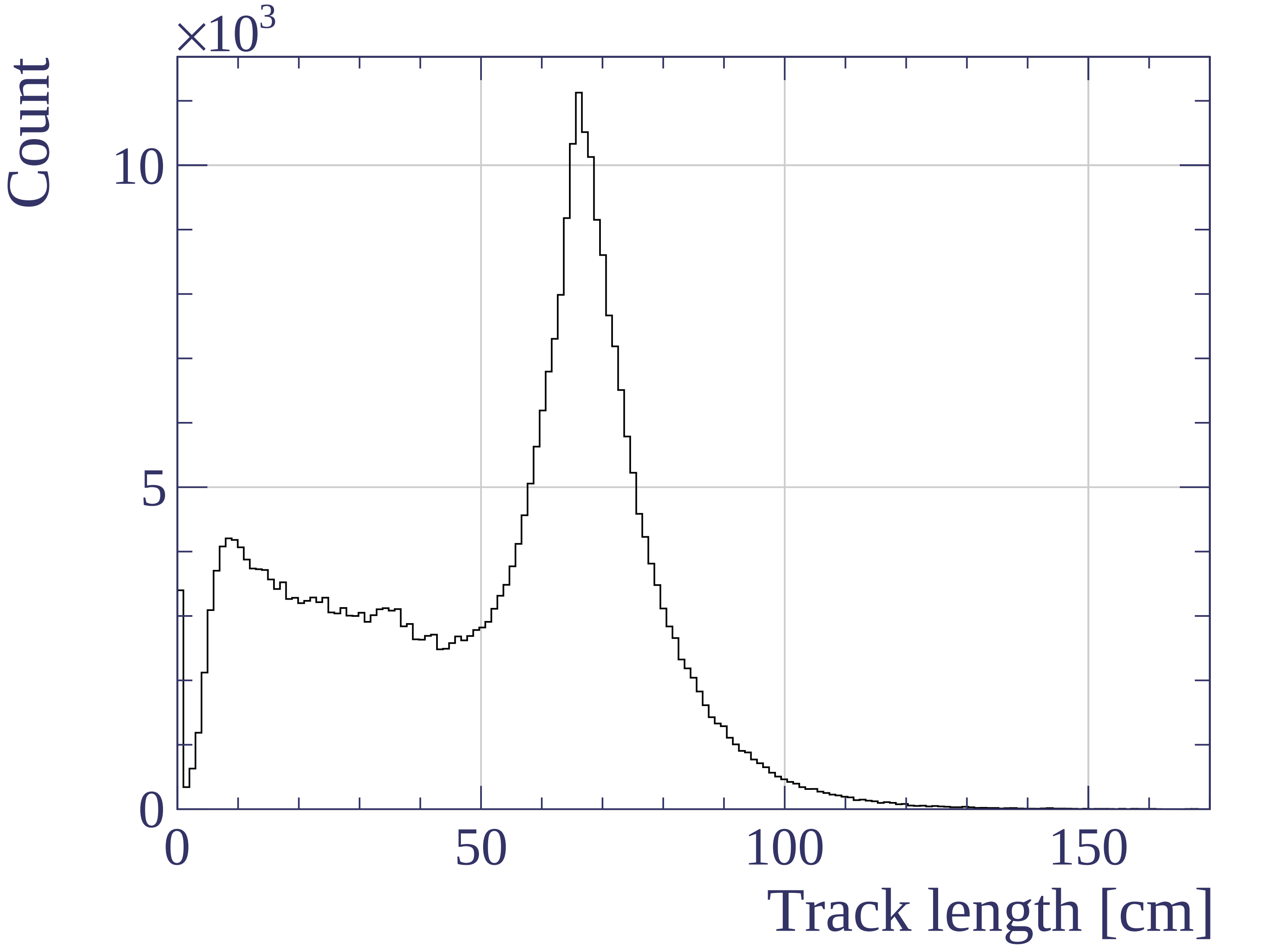
Pulls mean
0
−20
−40
−60
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



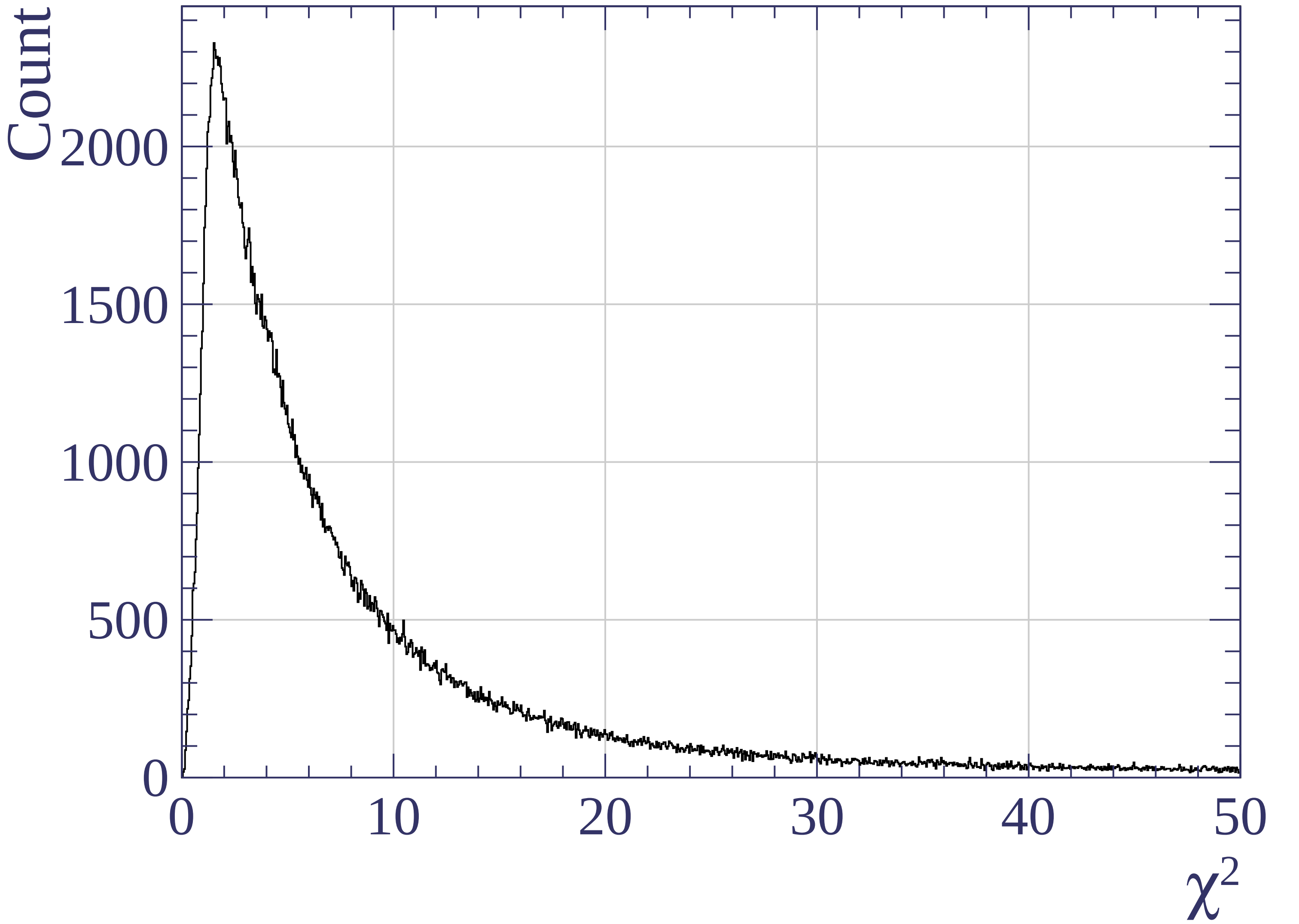
×10^3
Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
10
5
0



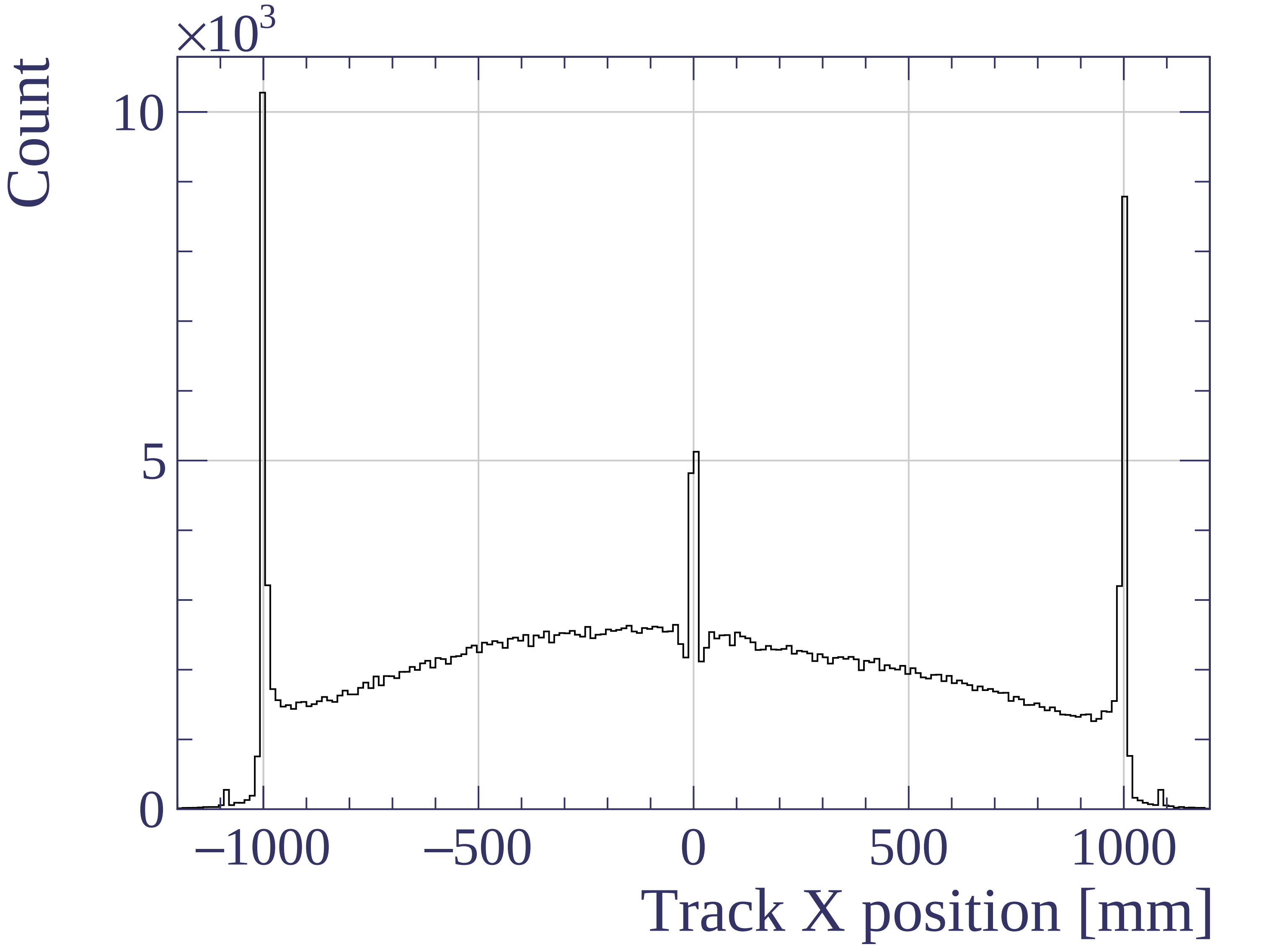
×10^3
Count
μ?
e?
p?
μ_μ = 0.528 ± 0.002
σ_μ = 1.023 ± 0.002
μ_e = -1.385 ± 0.002
σ_e = 1.316 ± 0.002
S(μ/e) = 1.62 ± 0.00
40
20
0
−20
−10
0
10
20
Pull

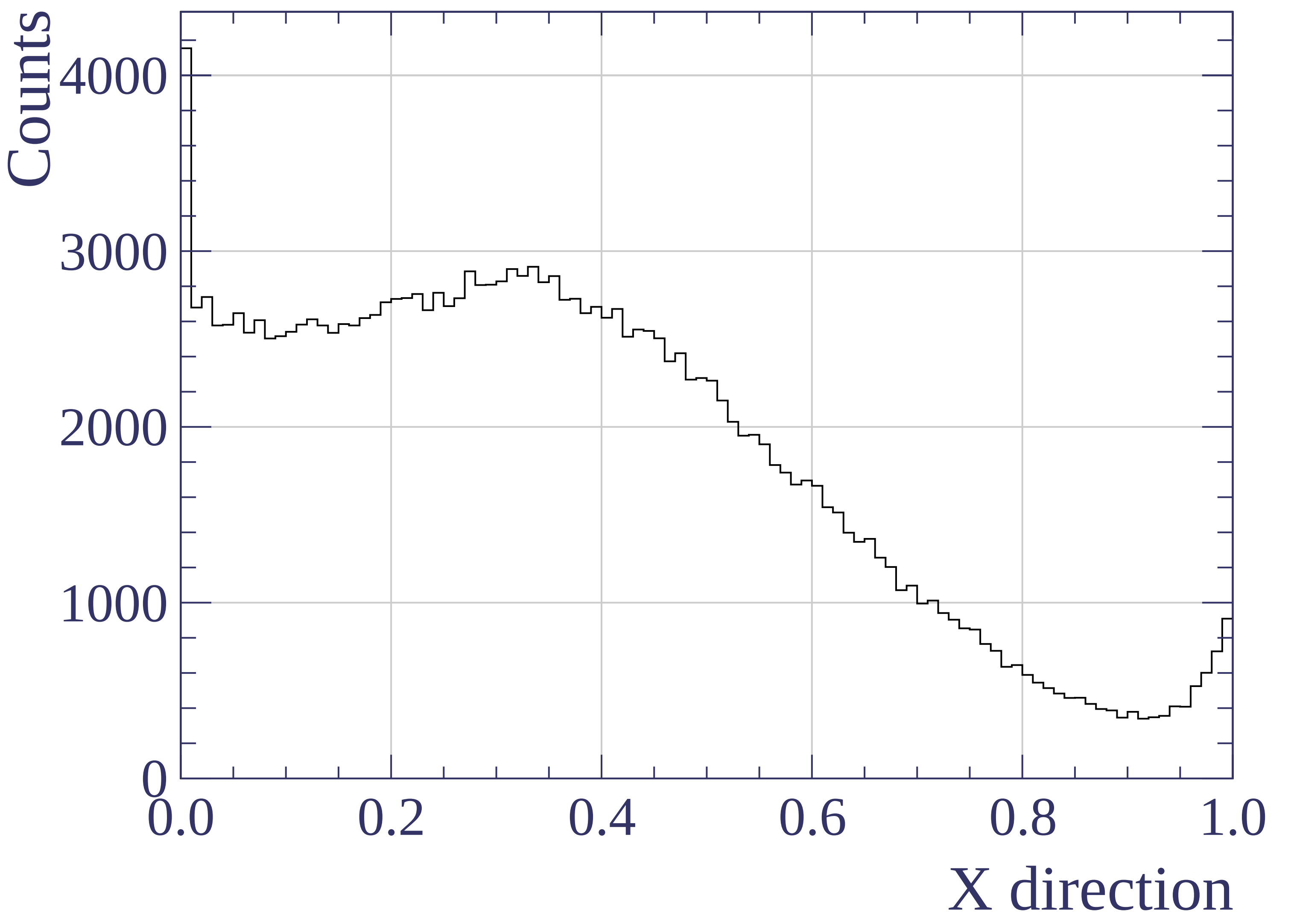


Count
×10^3
10
5
0
0
50
100
150
Track length [cm]

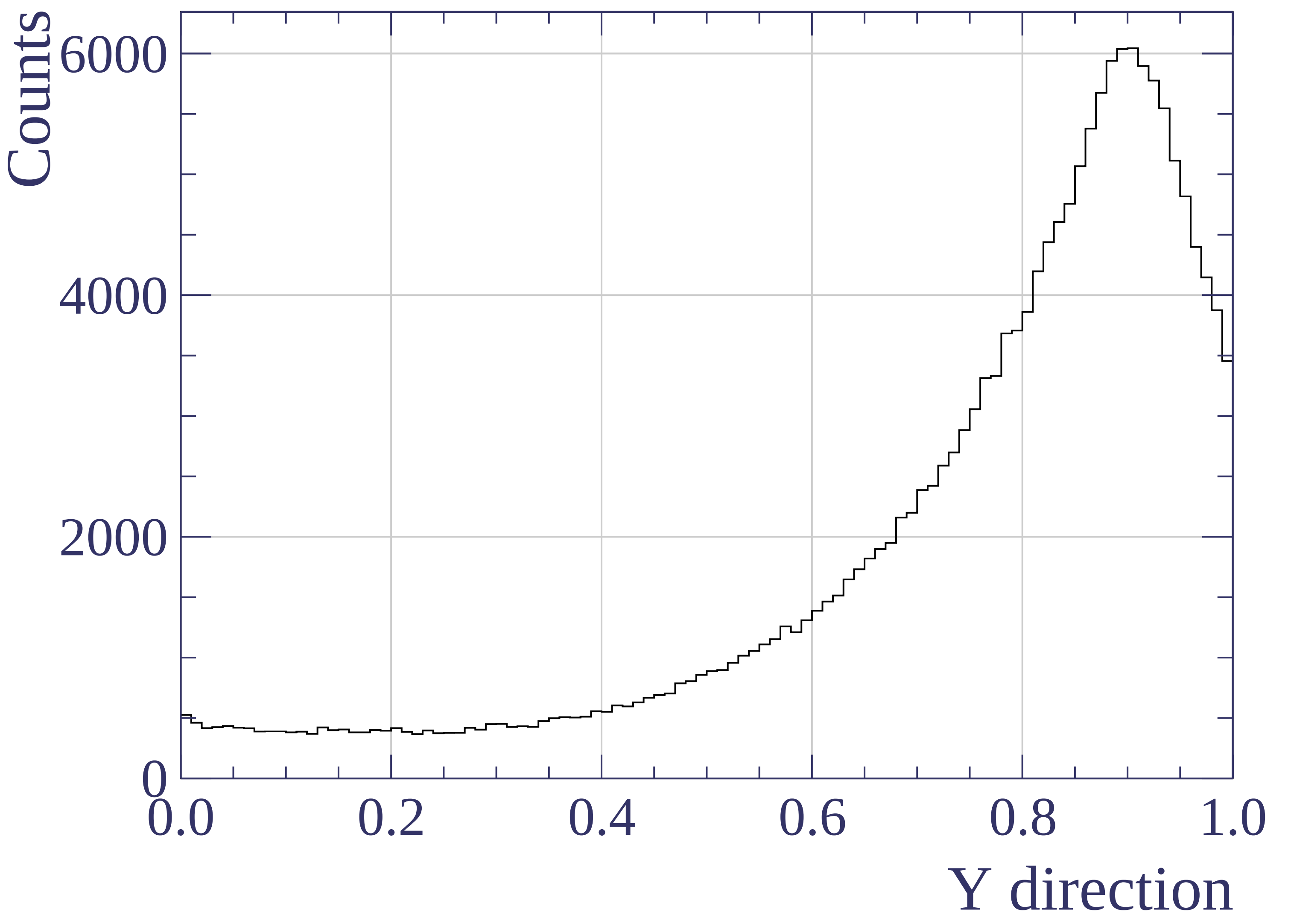


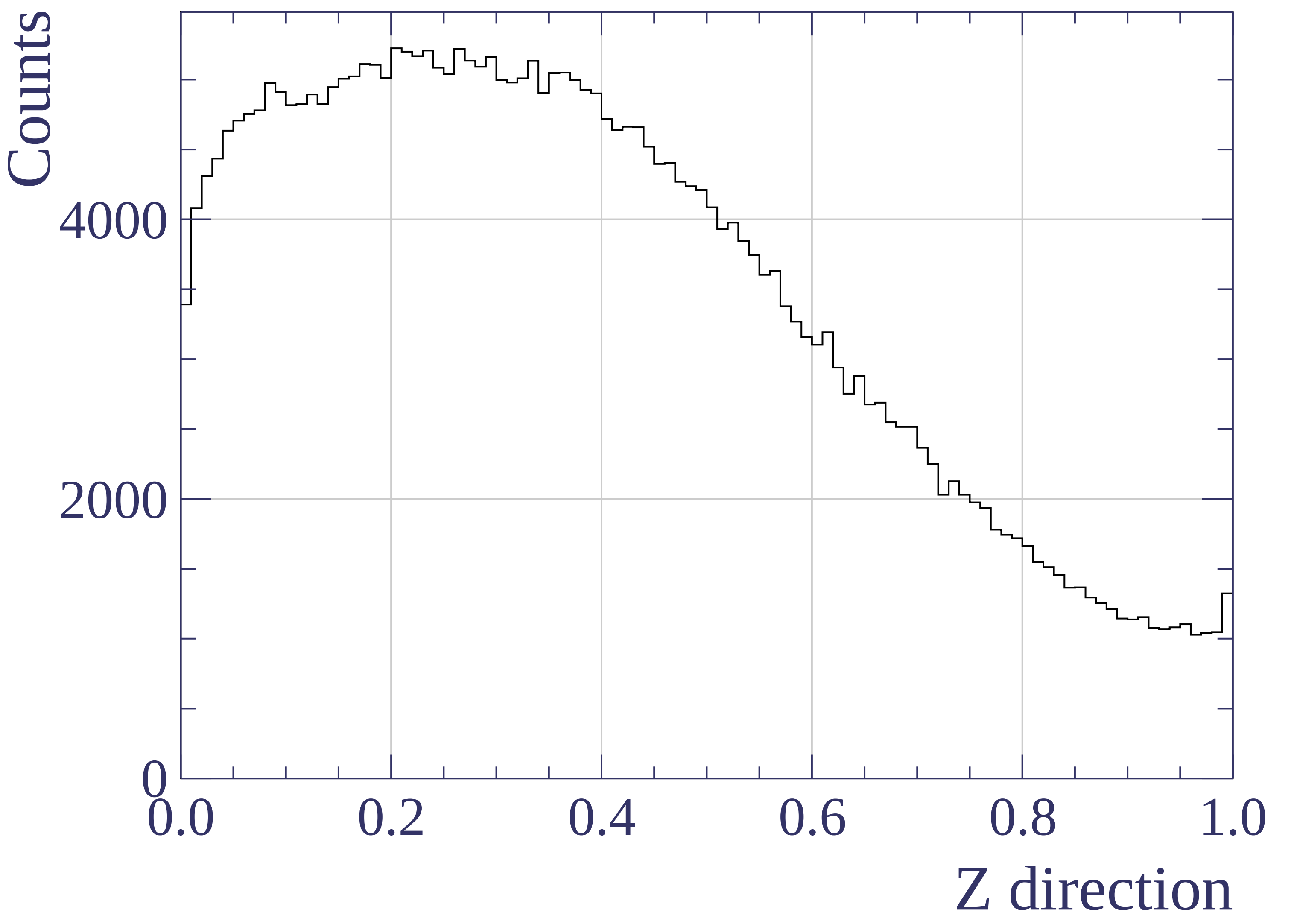
Count
2000
1500
1000
500
0
0
10
20
30
40
50
χ2


×10^3
Count
10
5
0
−1000
−500
0
500
1000
Track X position [mm]

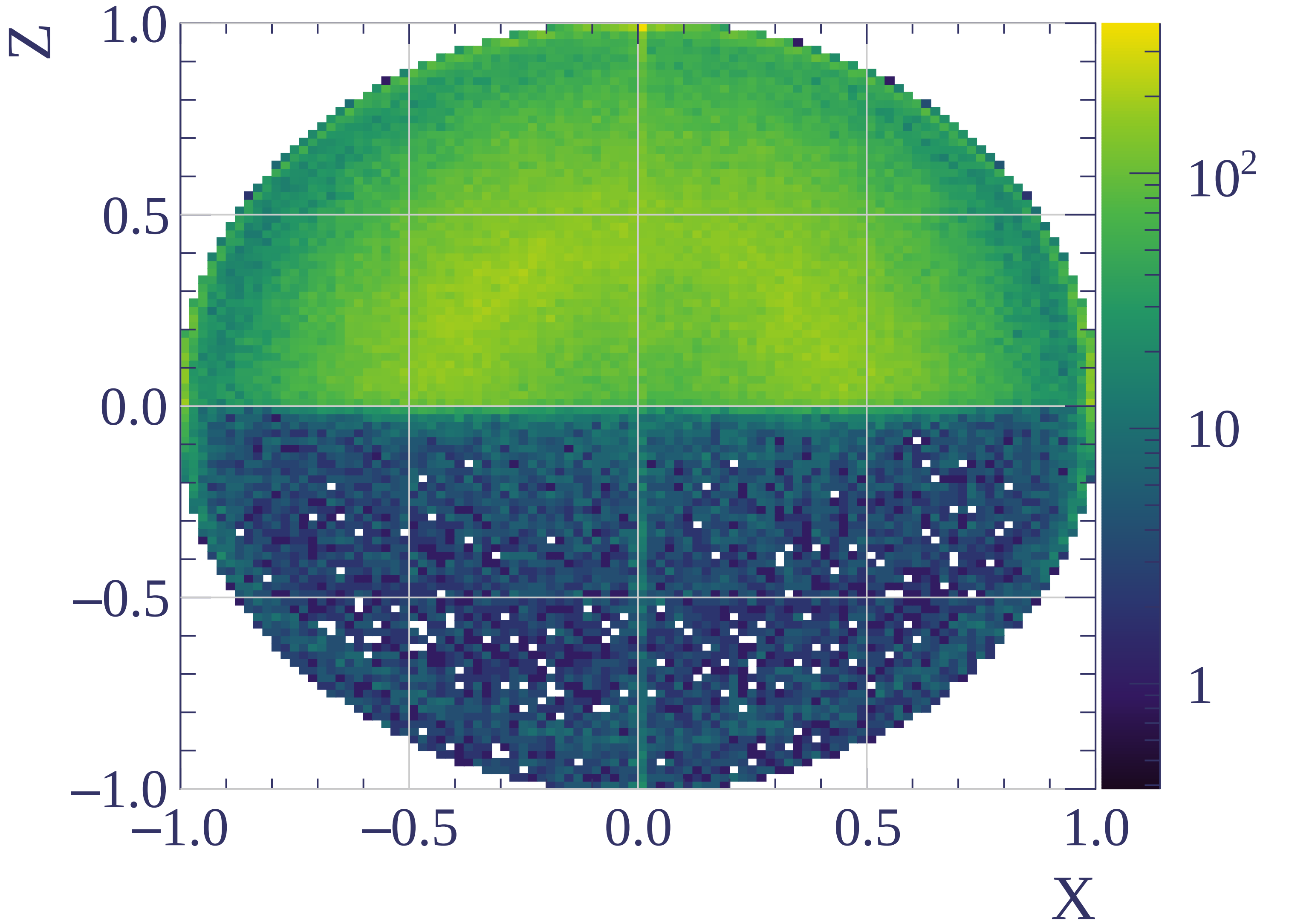


Counts
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

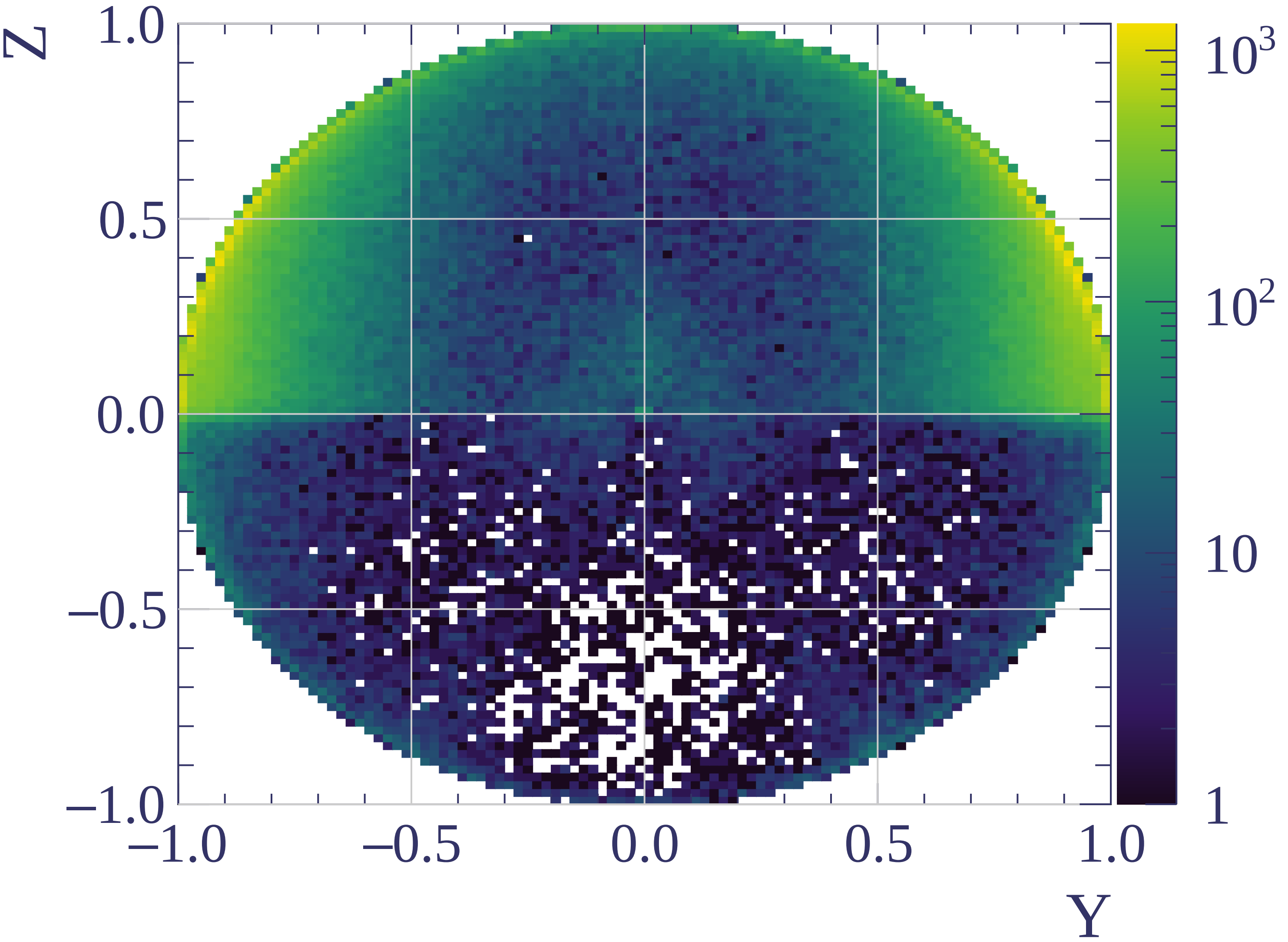

Counts
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



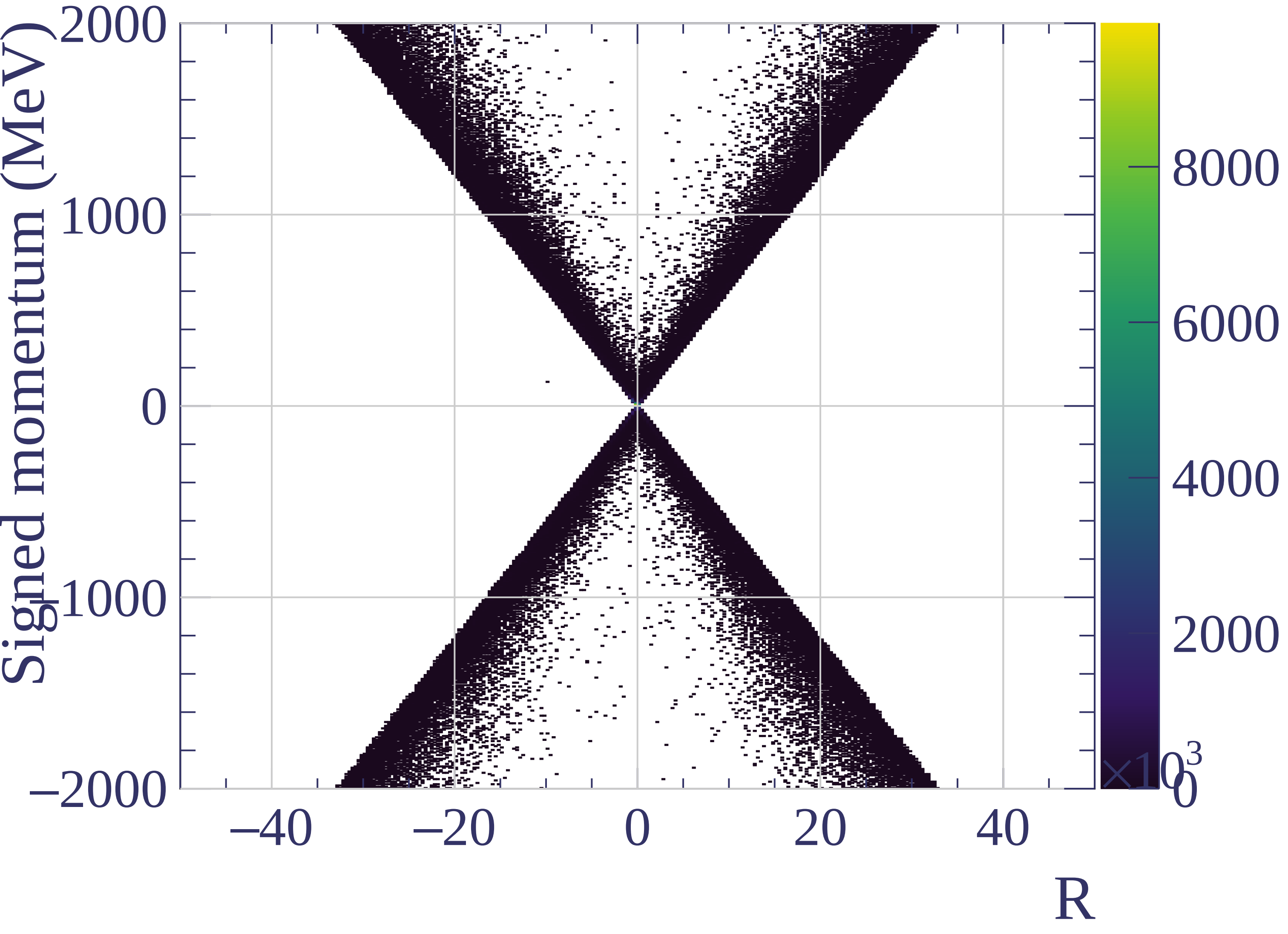
Counts
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



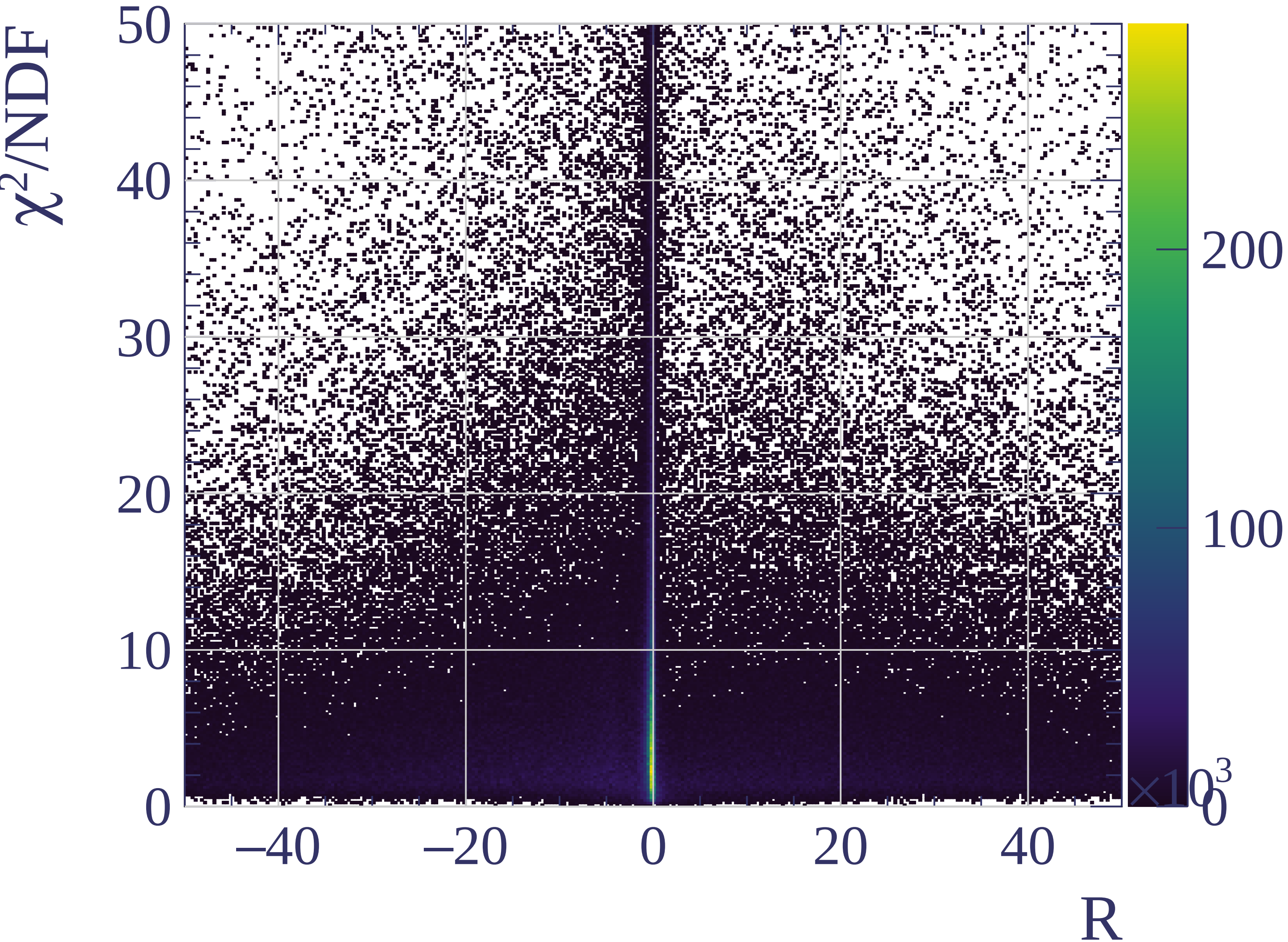
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^2
10
1



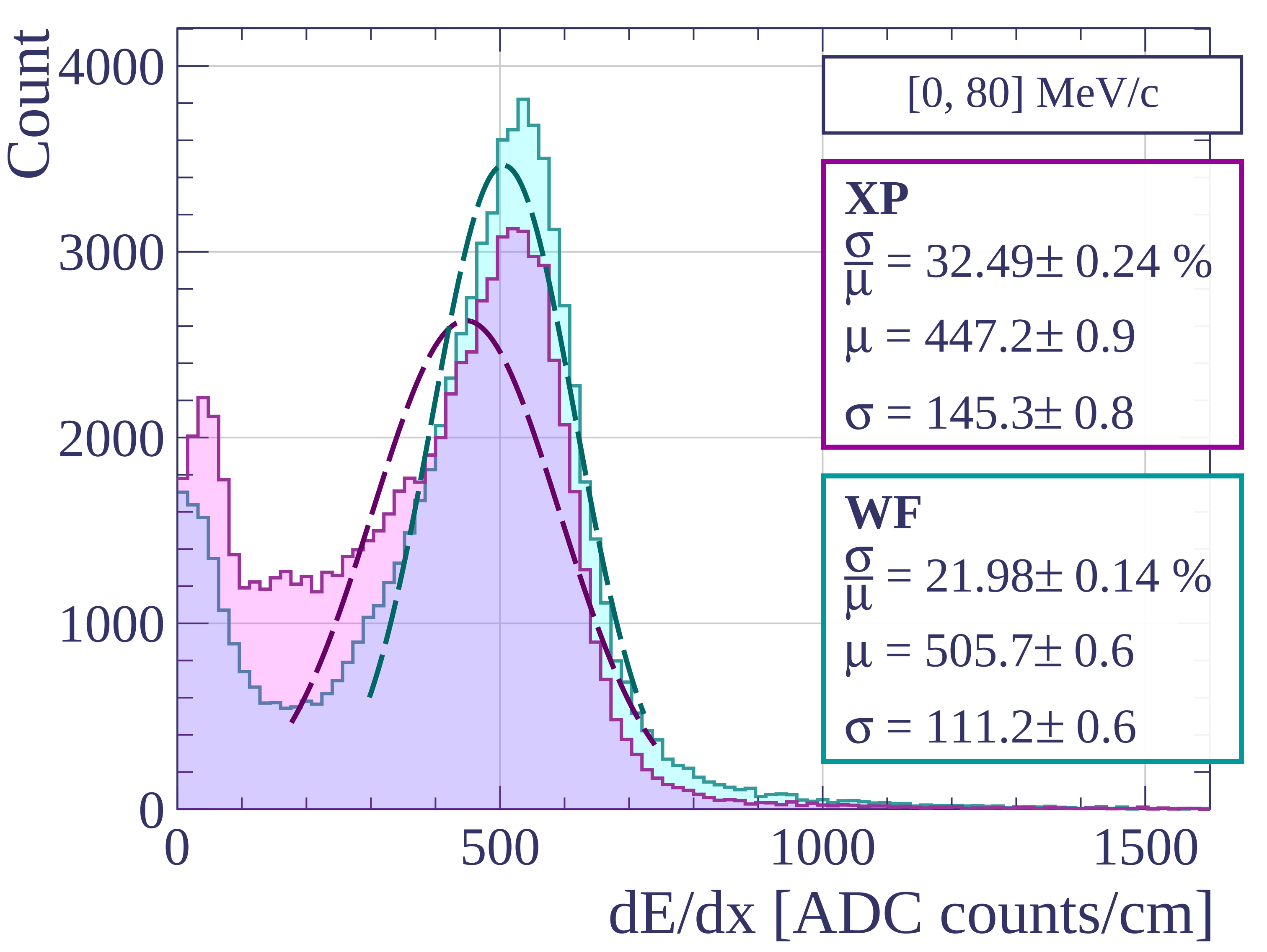
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^3
10^2
10
1



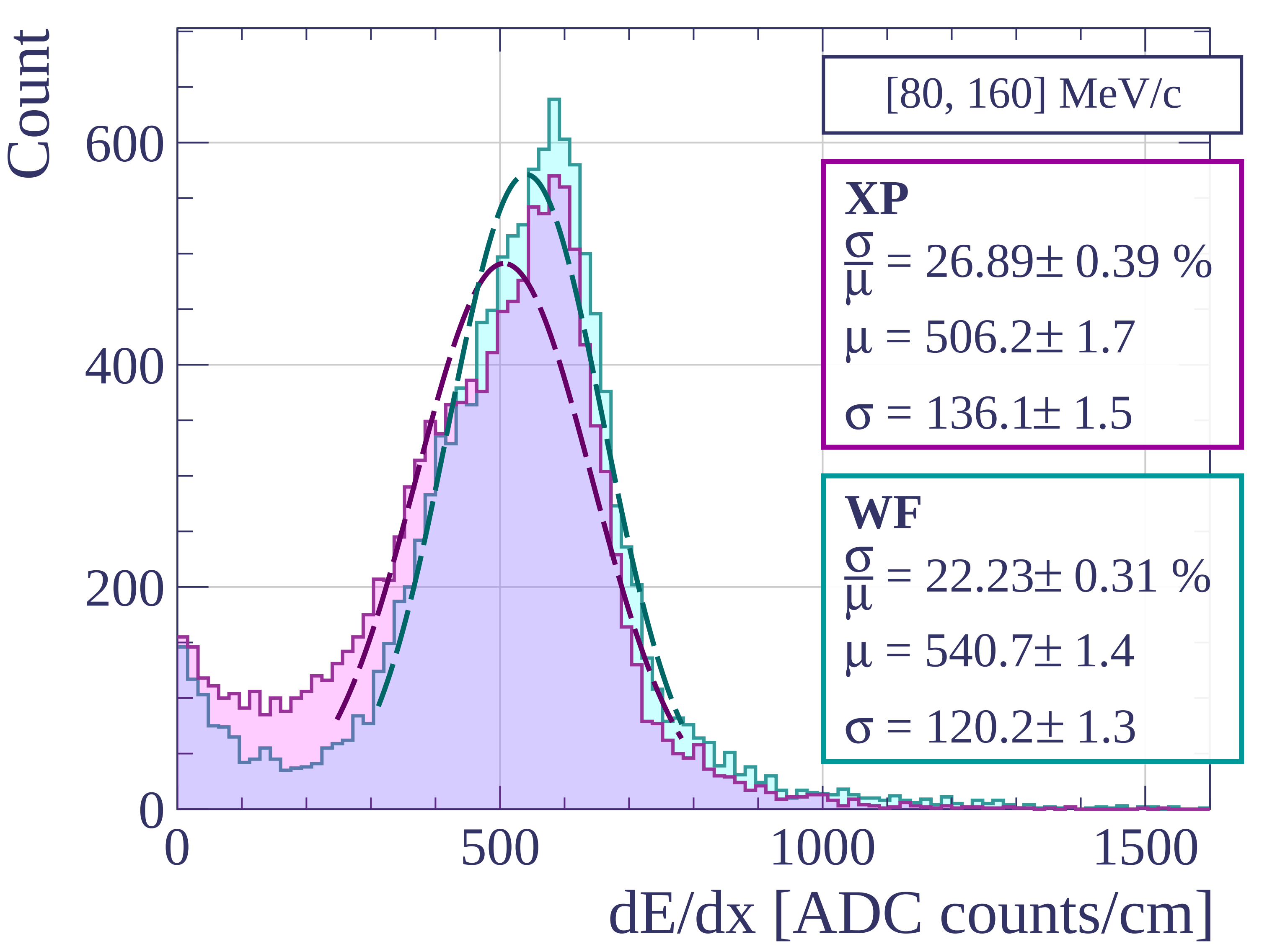
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
8000
6000
4000
2000
0
×10^3



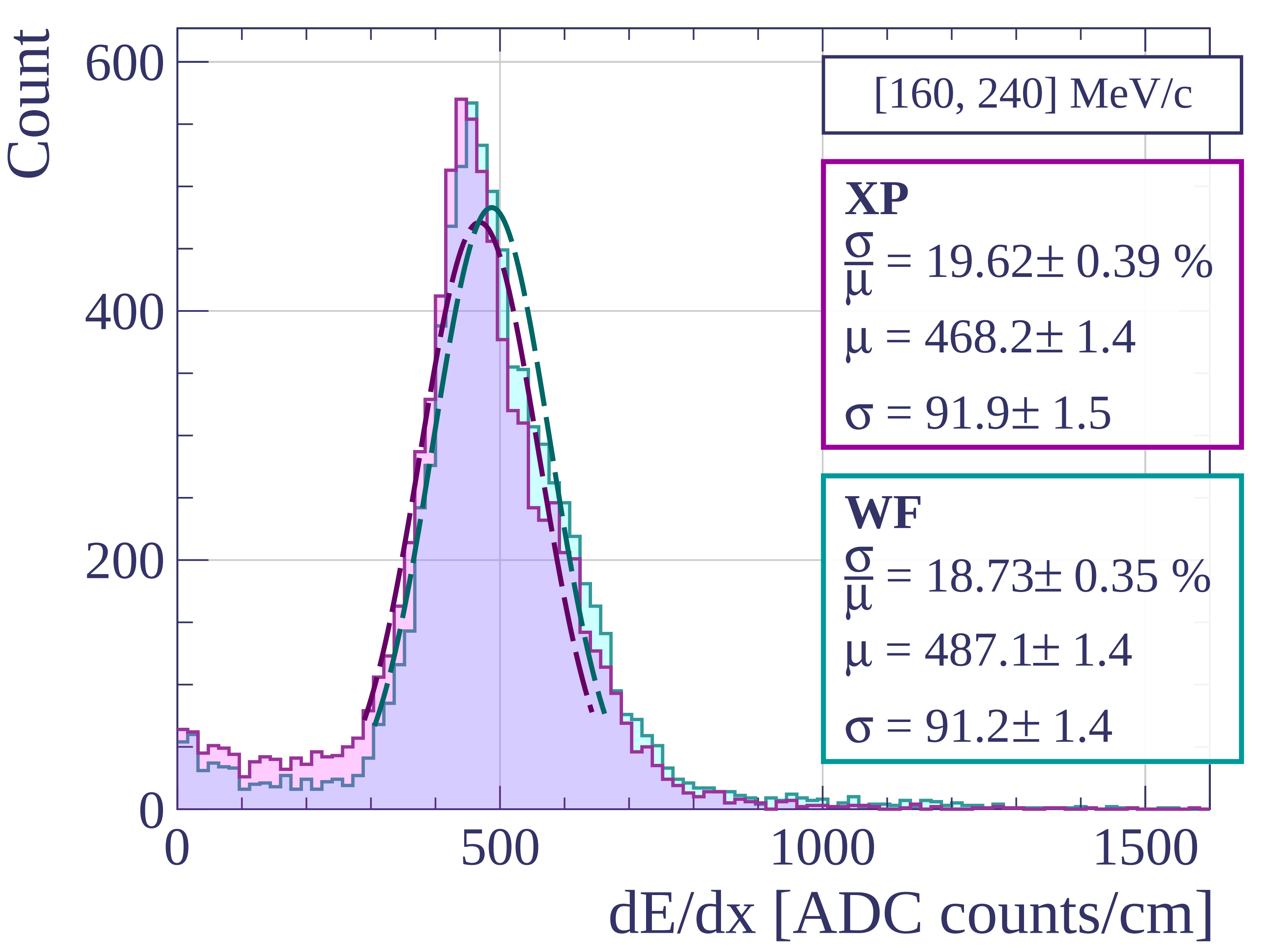
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
200
100
0
$\times10^3$

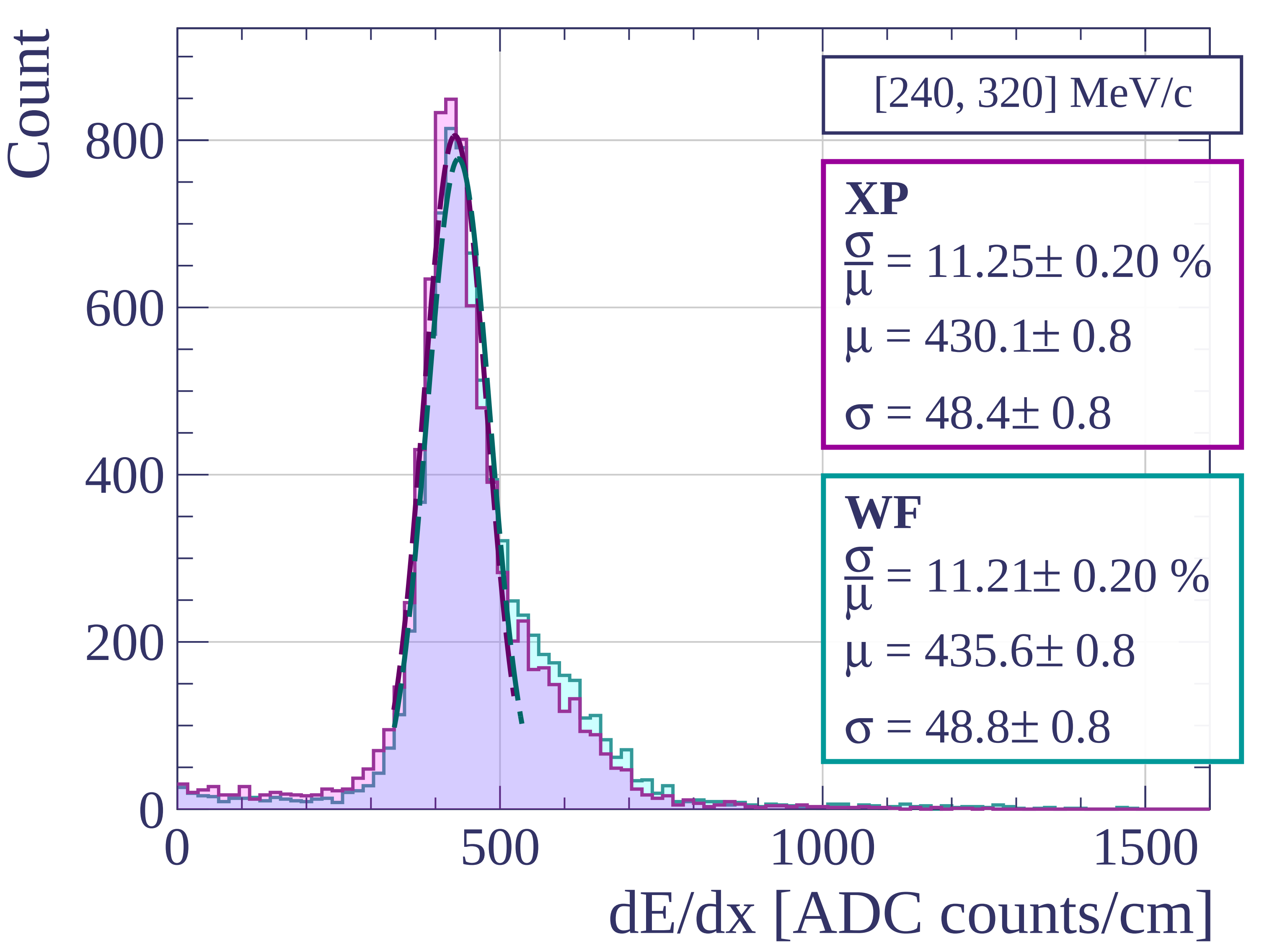


Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 32.49± 0.24 %
μ = 447.2± 0.9
σ = 145.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 21.98± 0.14 %
μ = 505.7± 0.6
σ = 111.2± 0.6

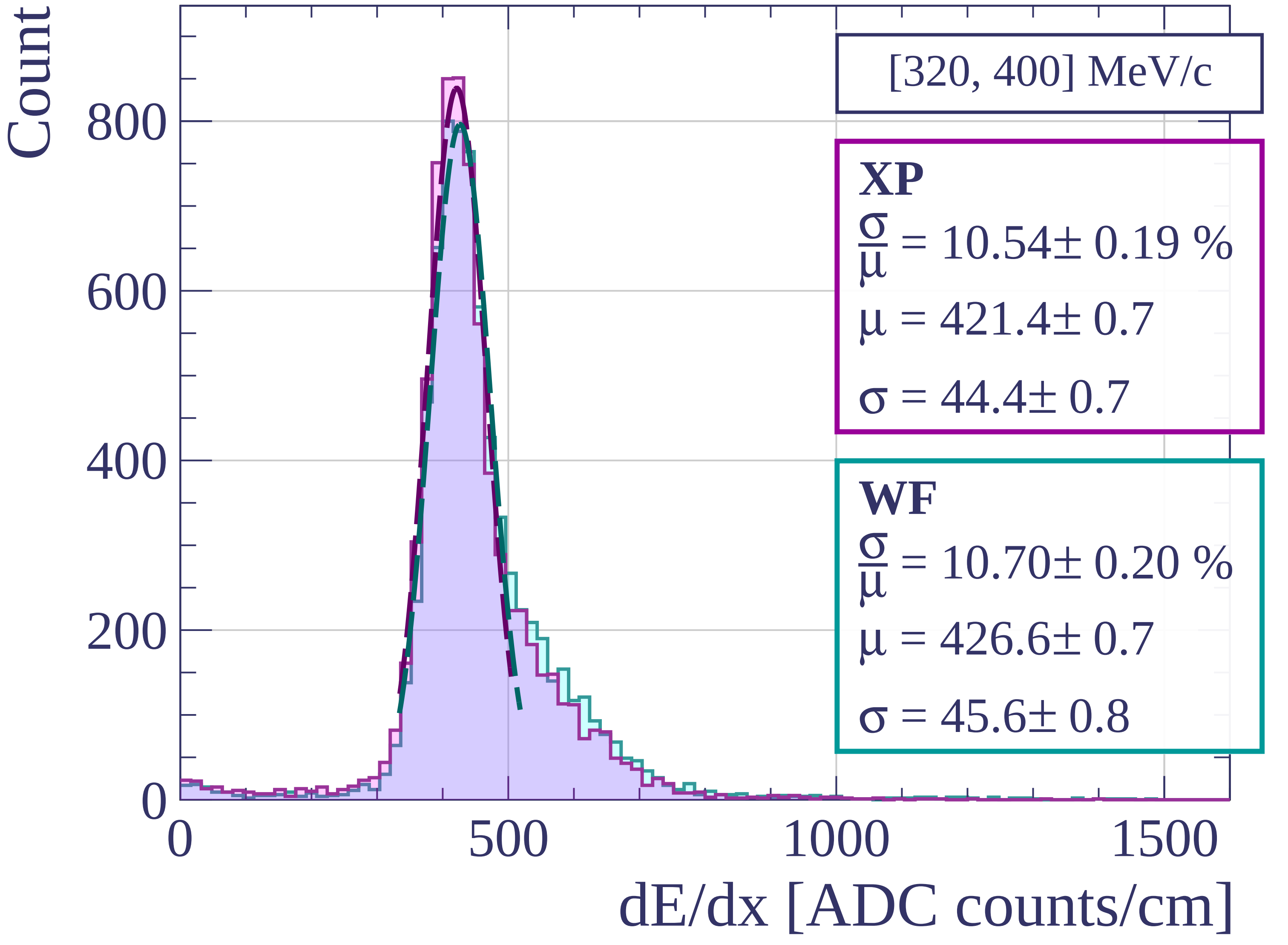


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
σ/μ = 26.89± 0.39 %
μ = 506.2± 1.7
σ = 136.1± 1.5
WF
σ/μ = 22.23± 0.31 %
μ = 540.7± 1.4
σ = 120.2± 1.3

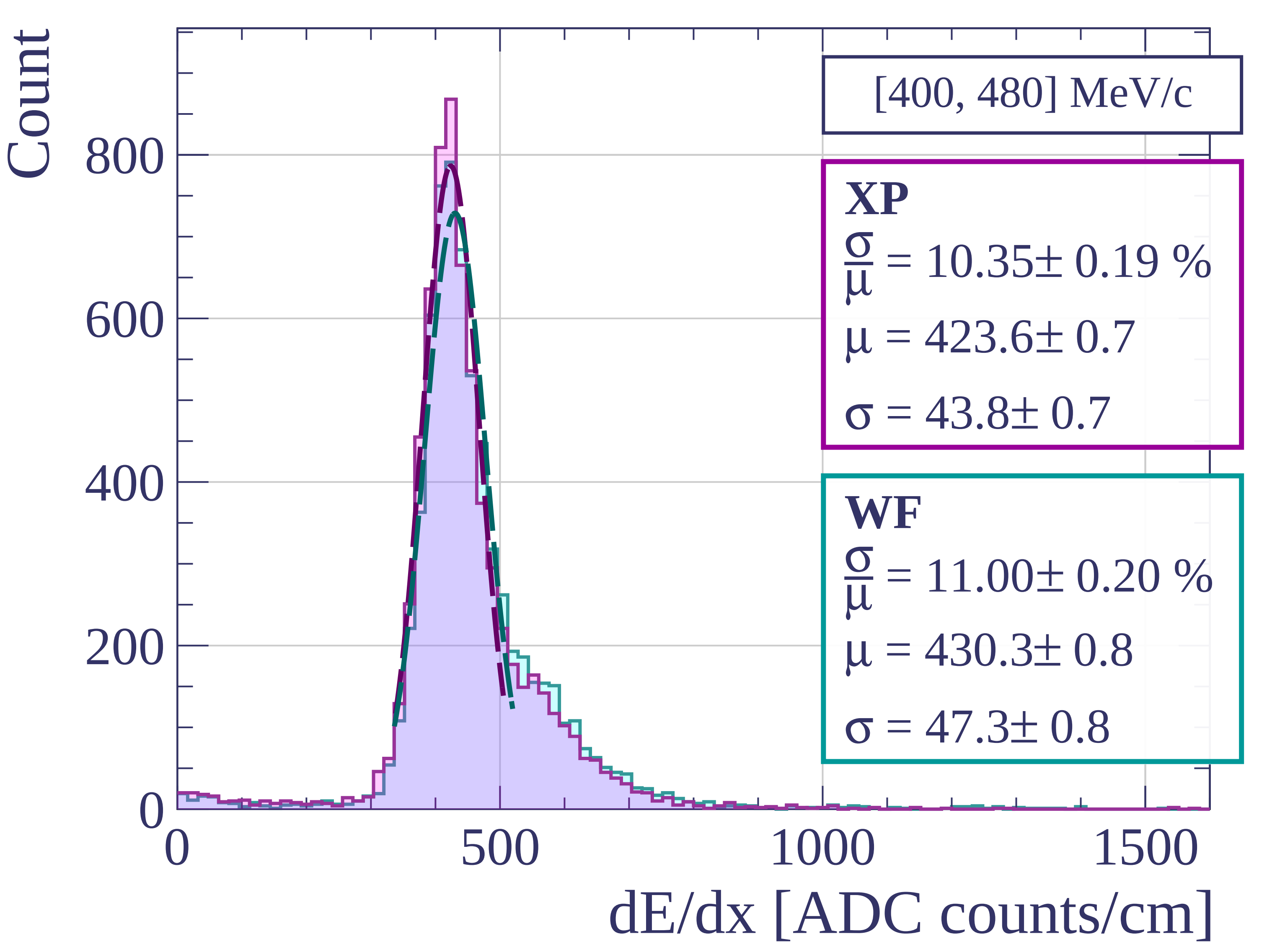

Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 19.62± 0.39 %
μ = 468.2± 1.4
σ = 91.9± 1.5
WF
$\frac{\sigma}{\mu}$ = 18.73± 0.35 %
μ = 487.1± 1.4
σ = 91.2± 1.4

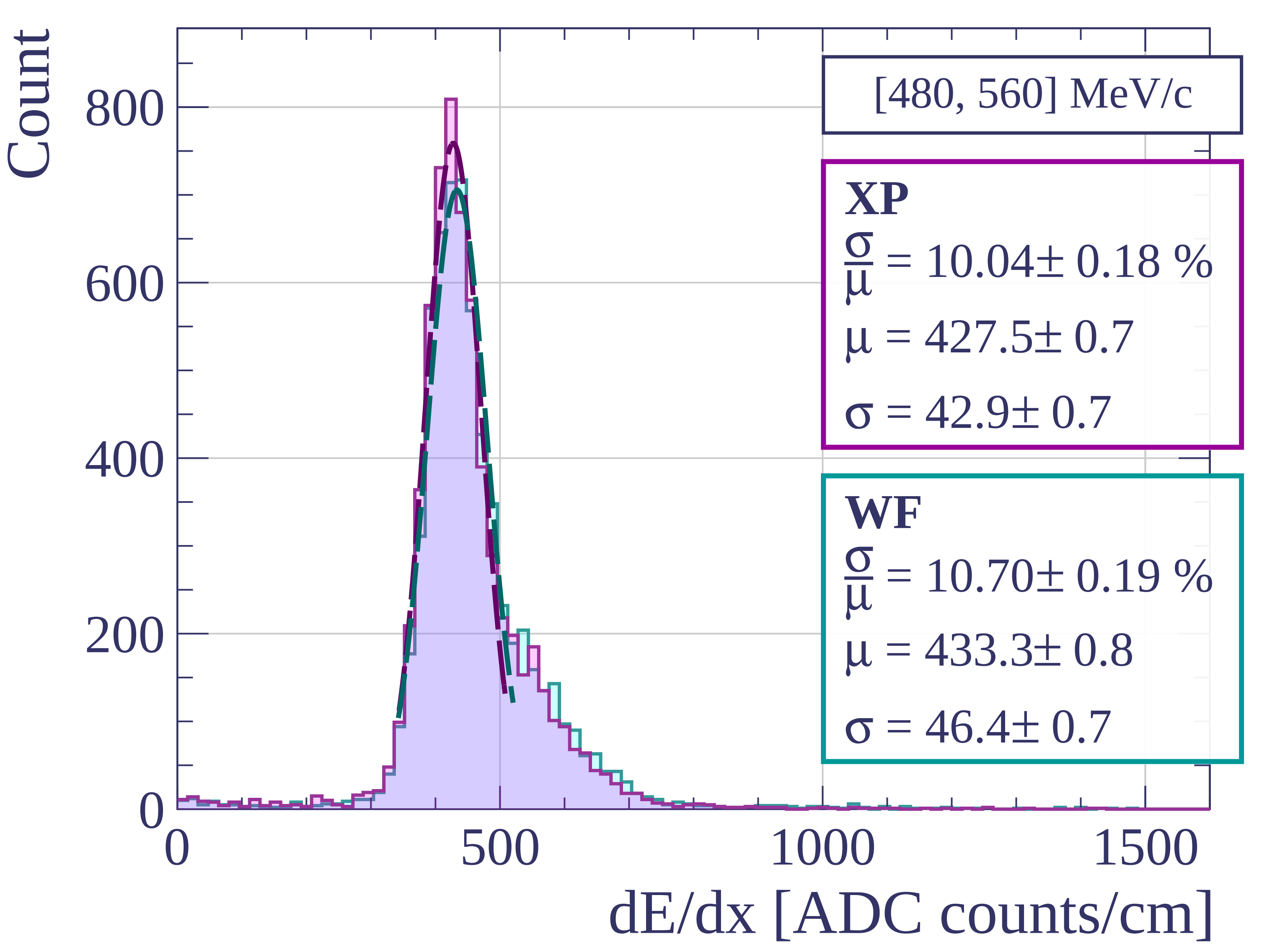


Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.25± 0.20 %
μ = 430.1± 0.8
σ = 48.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 11.21± 0.20 %
μ = 435.6± 0.8
σ = 48.8± 0.8

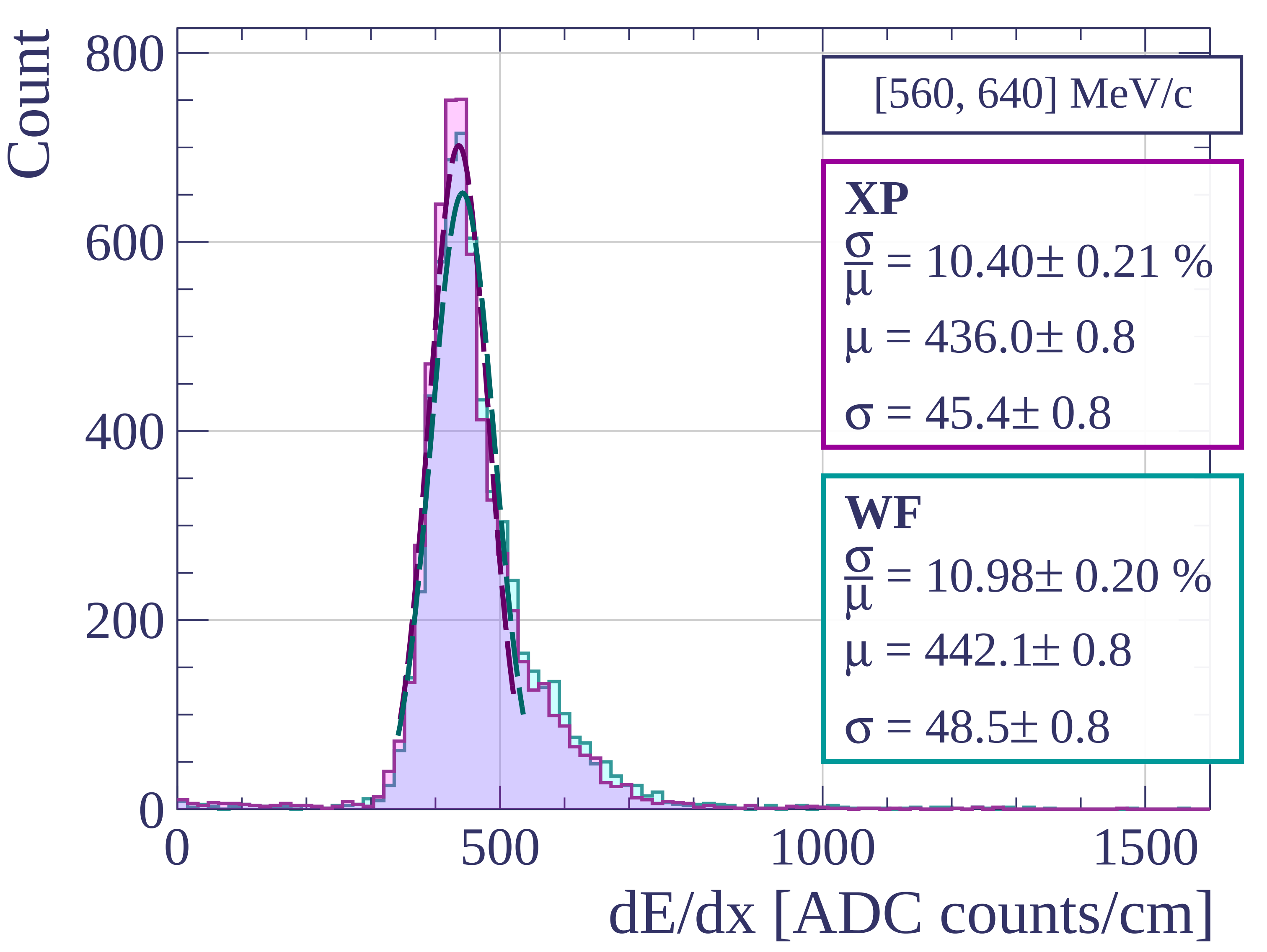


Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.54± 0.19 %
μ = 421.4± 0.7
σ = 44.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 10.70± 0.20 %
μ = 426.6± 0.7
σ = 45.6± 0.8

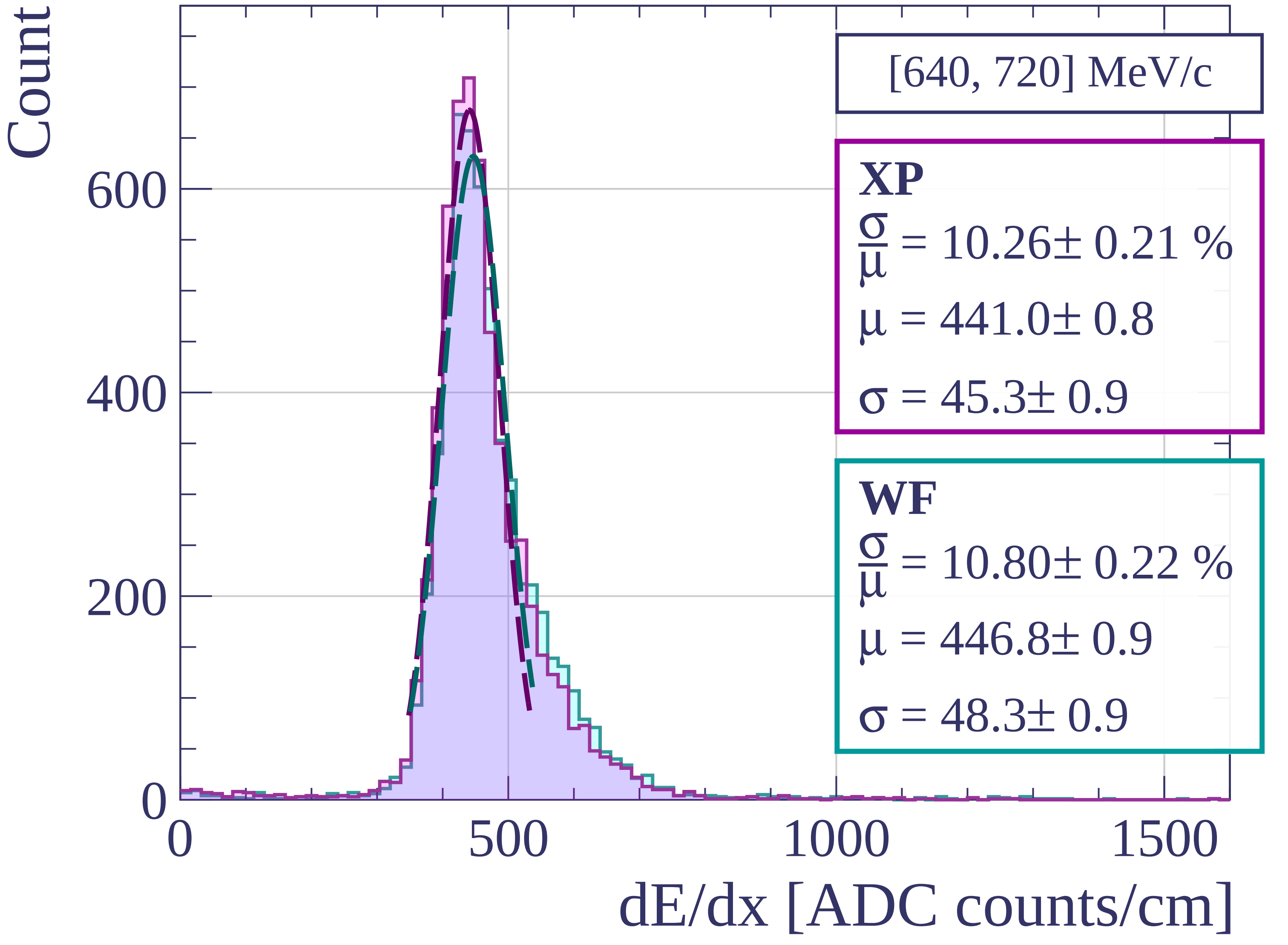

Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
σ/μ = 10.35± 0.19 %
μ = 423.6± 0.7
σ = 43.8± 0.7
WF
σ/μ = 11.00± 0.20 %
μ = 430.3± 0.8
σ = 47.3± 0.8

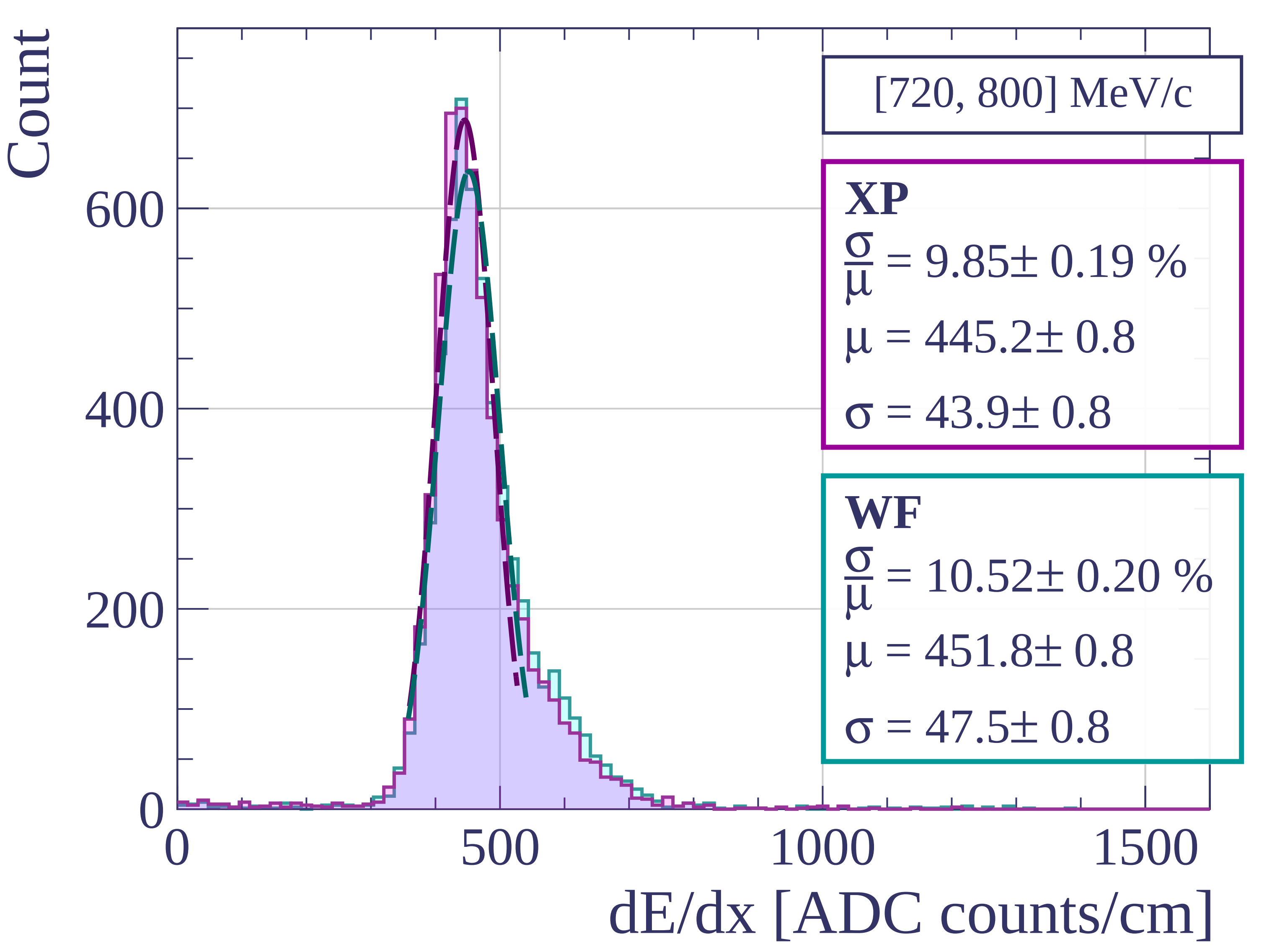


Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.04± 0.18 %
μ = 427.5± 0.7
σ = 42.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 10.70± 0.19 %
μ = 433.3± 0.8
σ = 46.4± 0.7

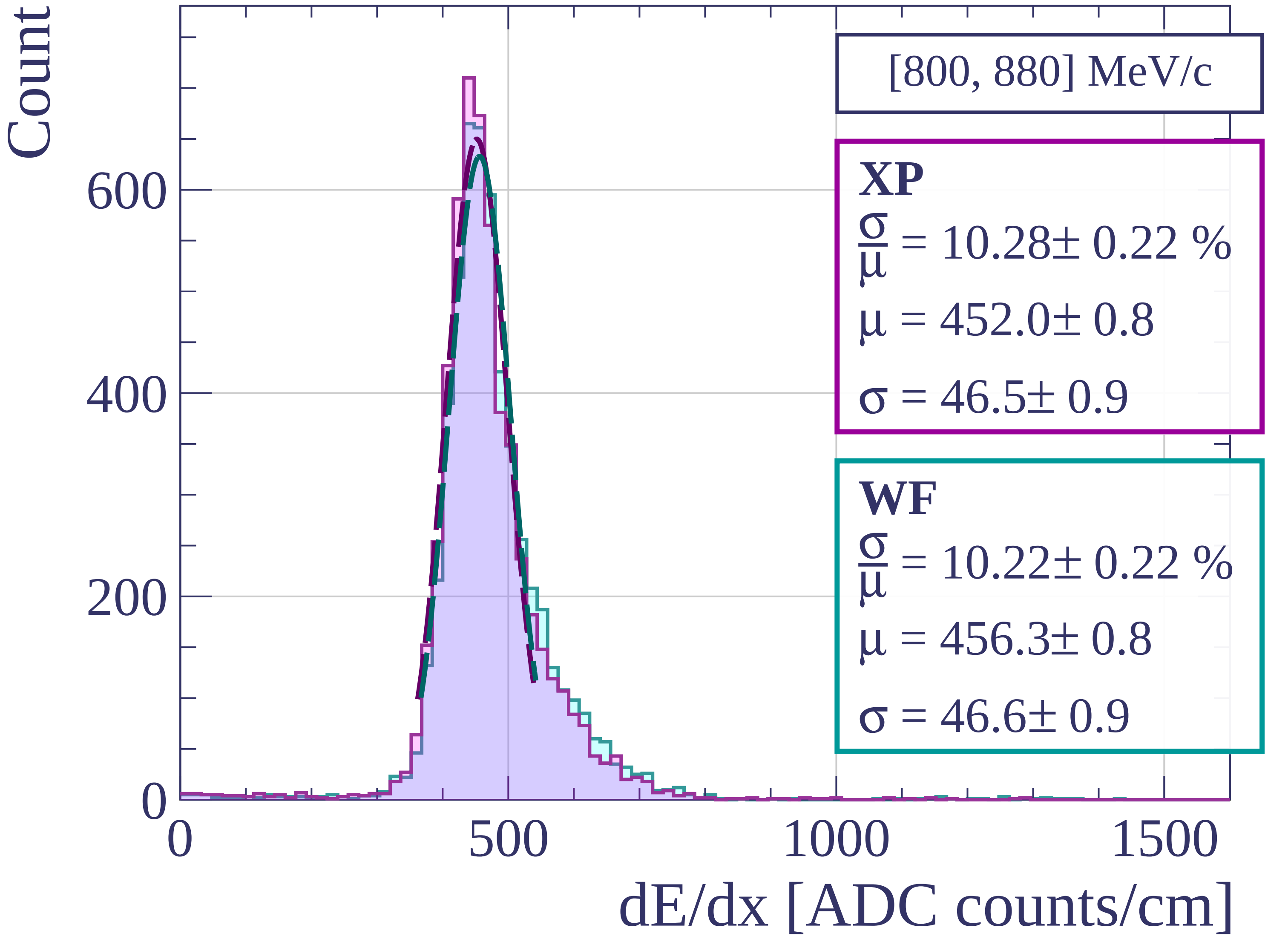


Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.40± 0.21 %
μ = 436.0± 0.8
σ = 45.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.98± 0.20 %
μ = 442.1± 0.8
σ = 48.5± 0.8

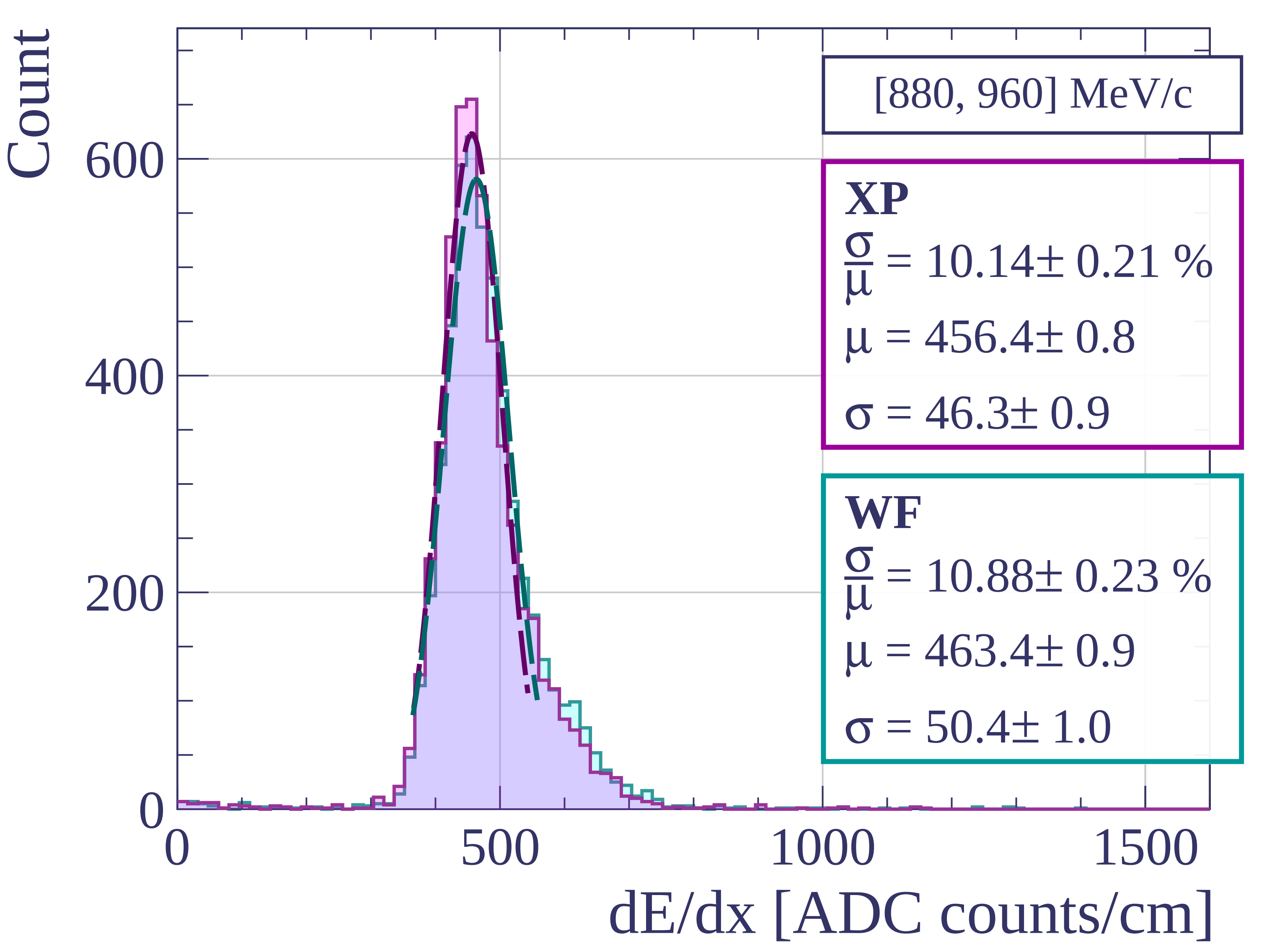

Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 720] MeV/c
XP
σ/μ = 10.26± 0.21 %
μ = 441.0± 0.8
σ = 45.3± 0.9
WF
σ/μ = 10.80± 0.22 %
μ = 446.8± 0.9
σ = 48.3± 0.9



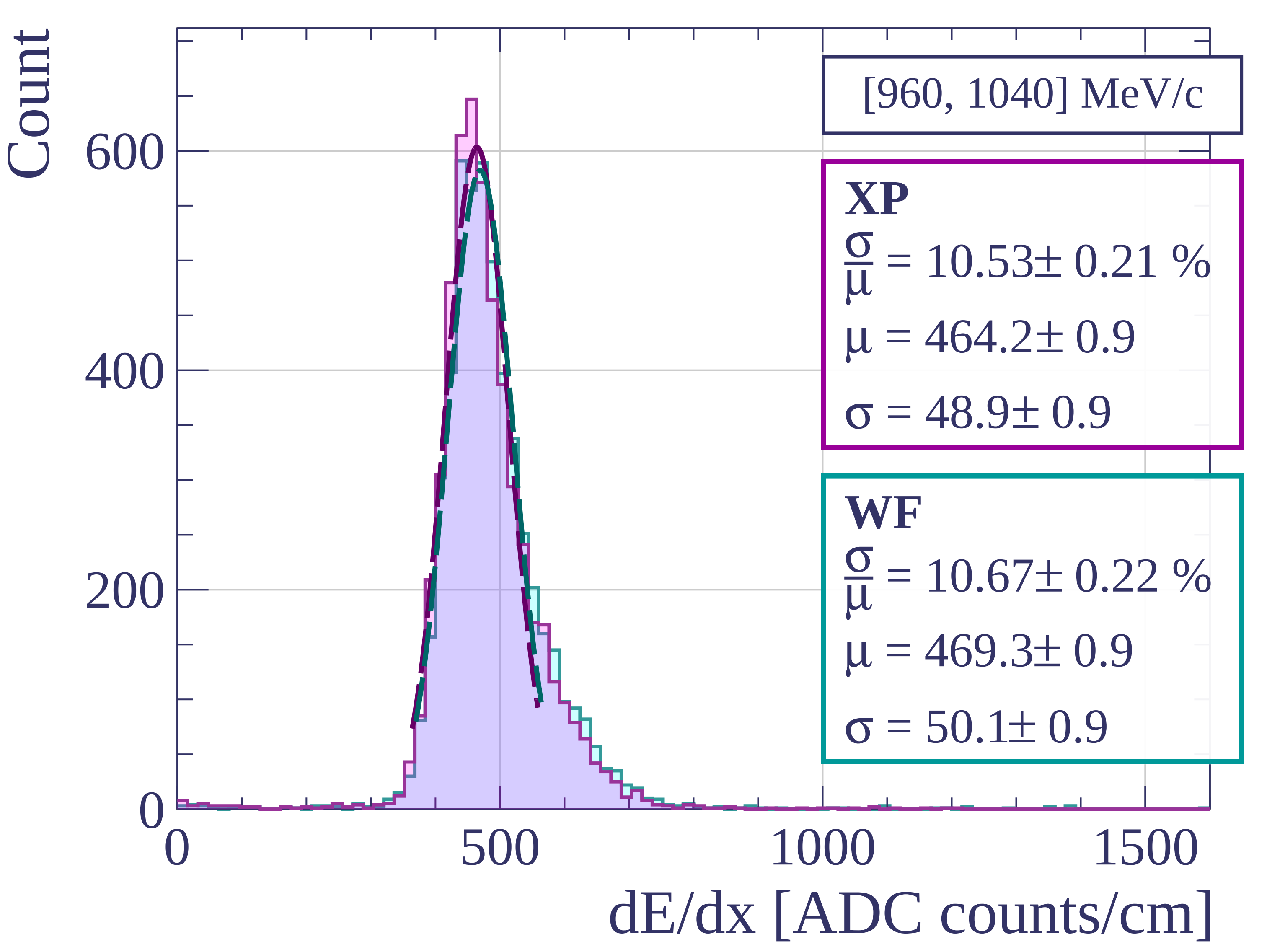
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 800] MeV/c
XP
σ/μ = 9.85± 0.19 %
μ = 445.2± 0.8
σ = 43.9± 0.8
WF
σ/μ = 10.52± 0.20 %
μ = 451.8± 0.8
σ = 47.5± 0.8



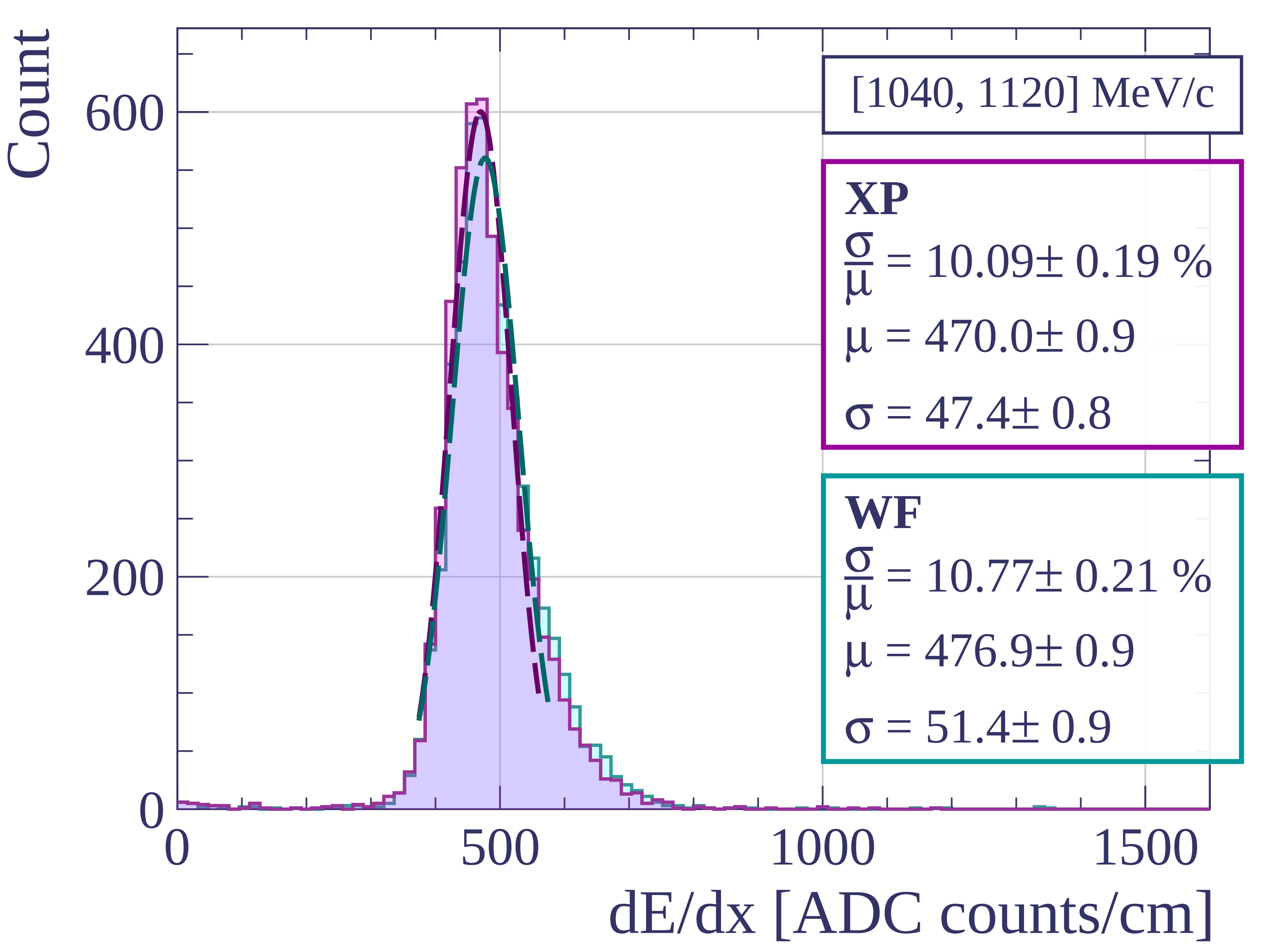
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.28± 0.22 %
μ = 452.0± 0.8
σ = 46.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.22± 0.22 %
μ = 456.3± 0.8
σ = 46.6± 0.9

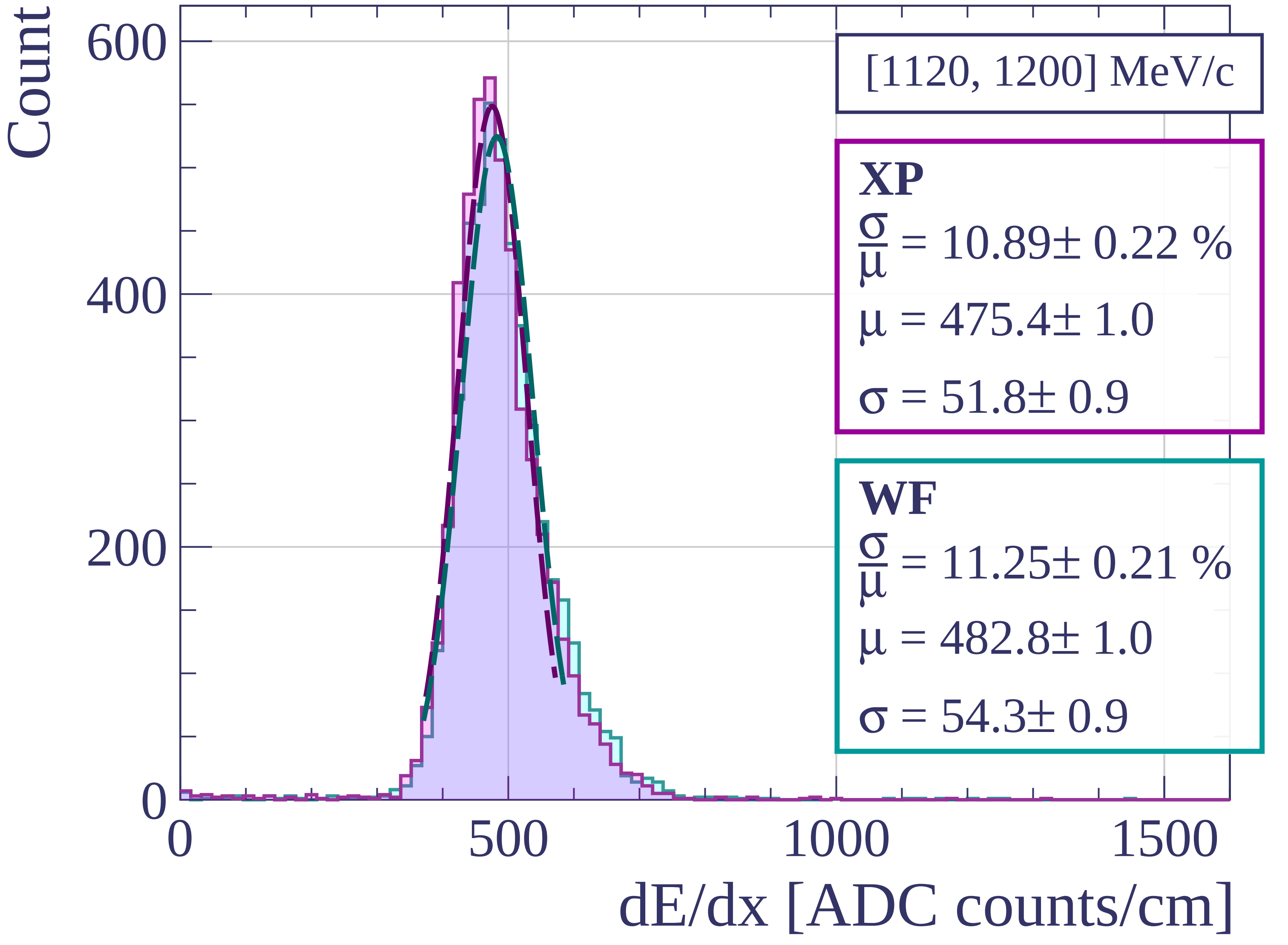


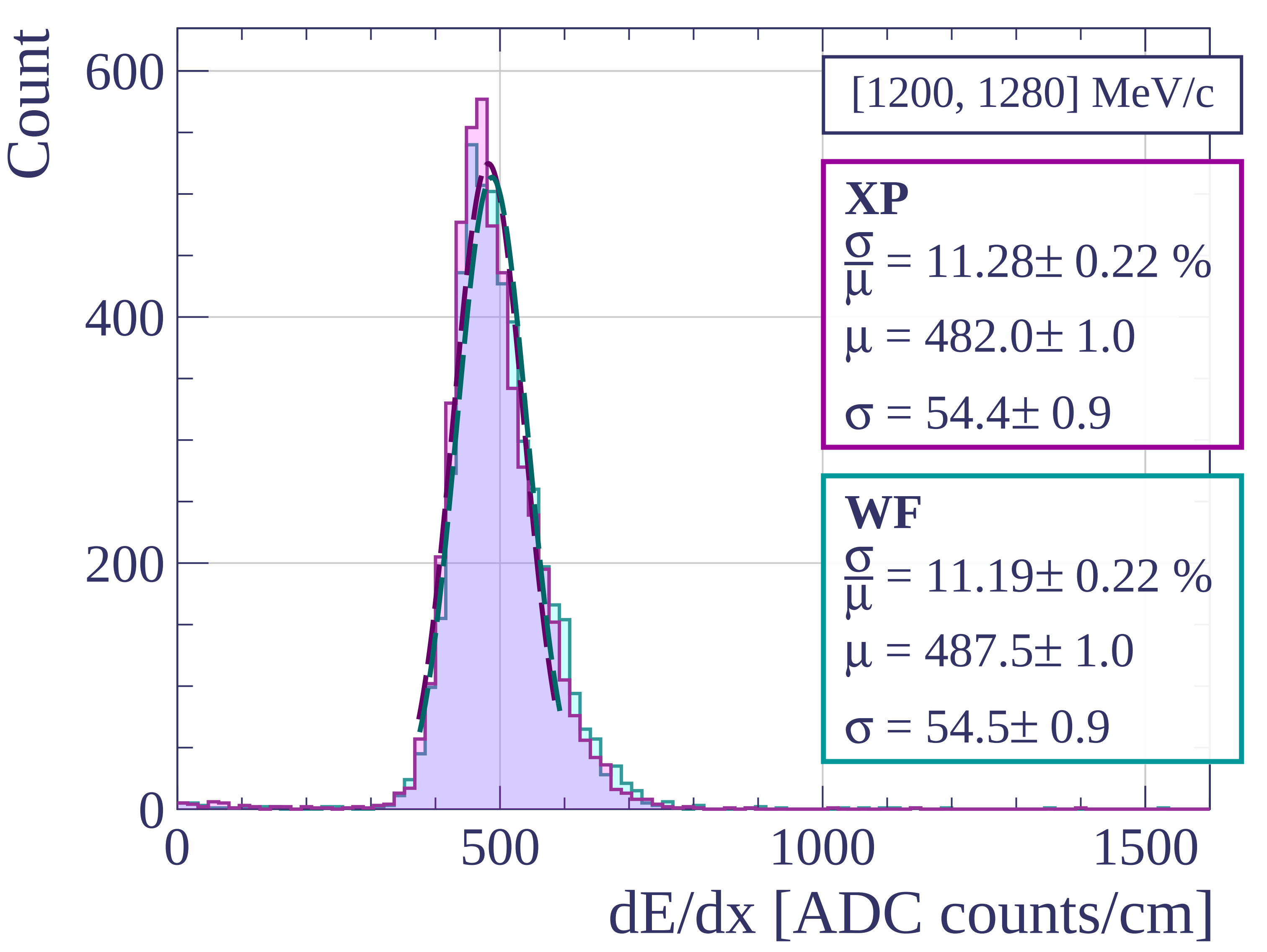
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.14± 0.21 %
μ = 456.4± 0.8
σ = 46.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.88± 0.23 %
μ = 463.4± 0.9
σ = 50.4± 1.0



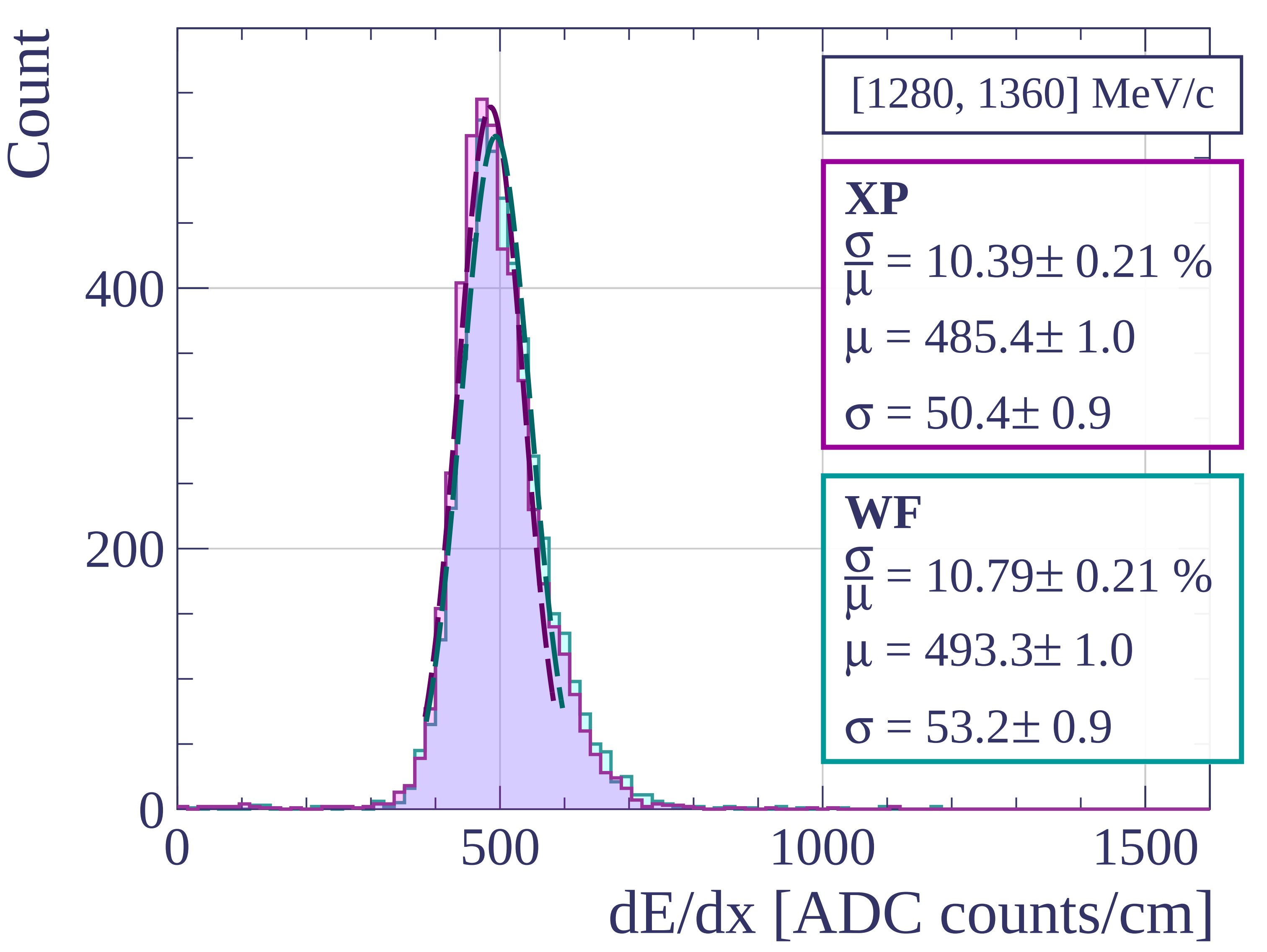
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.53± 0.21 %
μ = 464.2± 0.9
σ = 48.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.67± 0.22 %
μ = 469.3± 0.9
σ = 50.1± 0.9

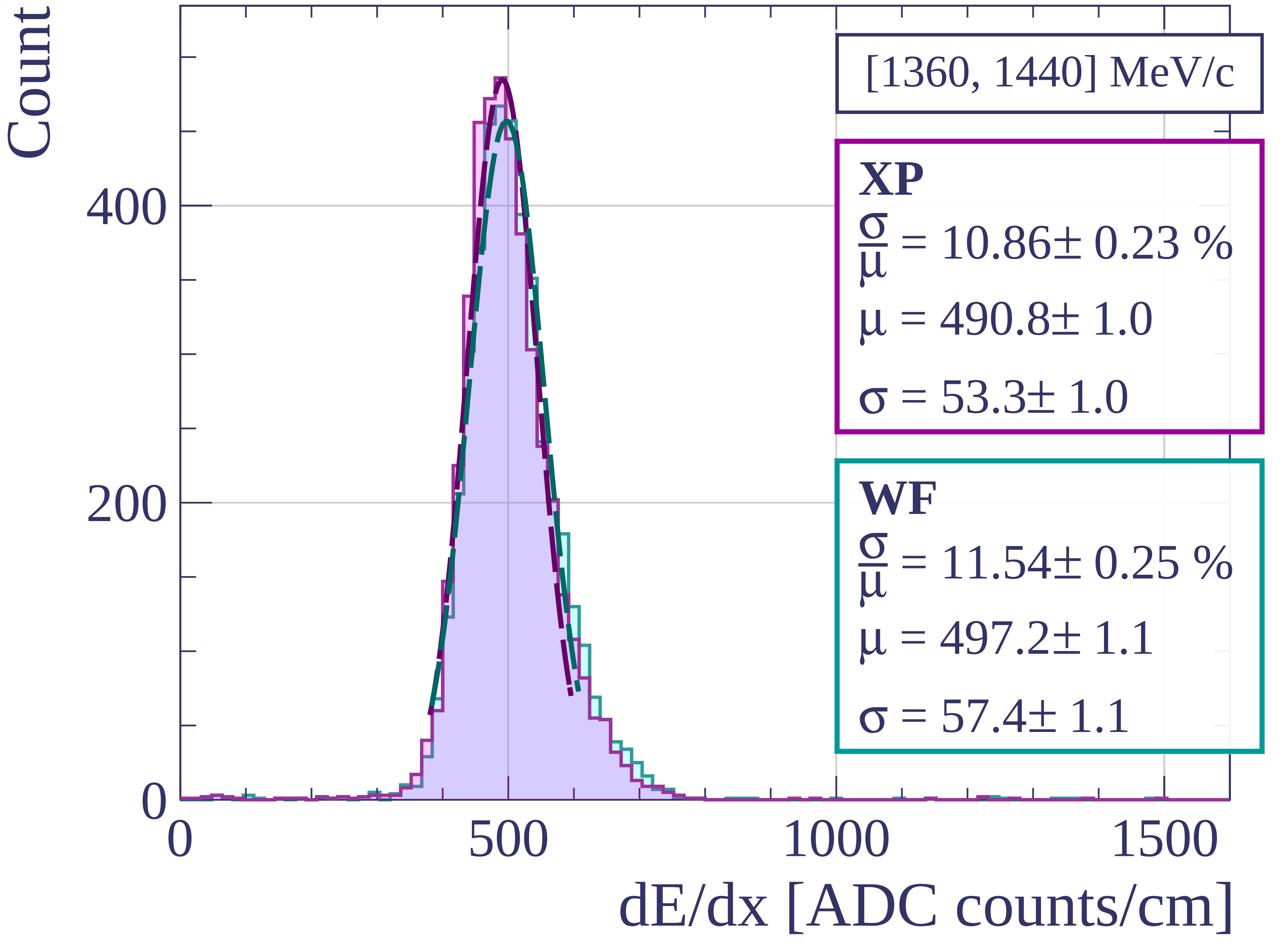

Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
[1040, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.09± 0.19 %
μ = 470.0± 0.9
σ = 47.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.77± 0.21 %
μ = 476.9± 0.9
σ = 51.4± 0.9

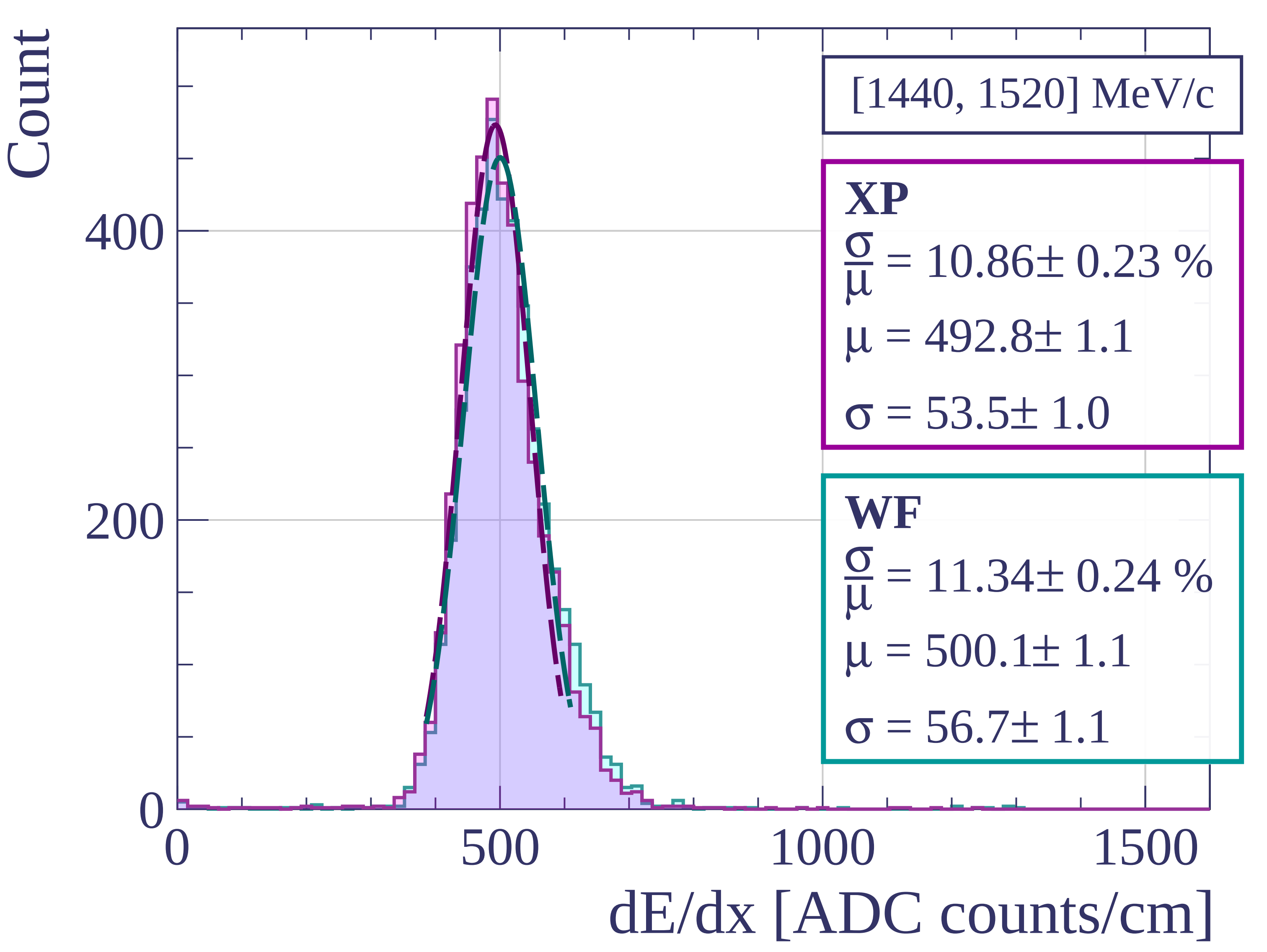

Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
σ/μ = 10.89± 0.22 %
μ = 475.4± 1.0
σ = 51.8± 0.9
WF
σ/μ = 11.25± 0.21 %
μ = 482.8± 1.0
σ = 54.3± 0.9

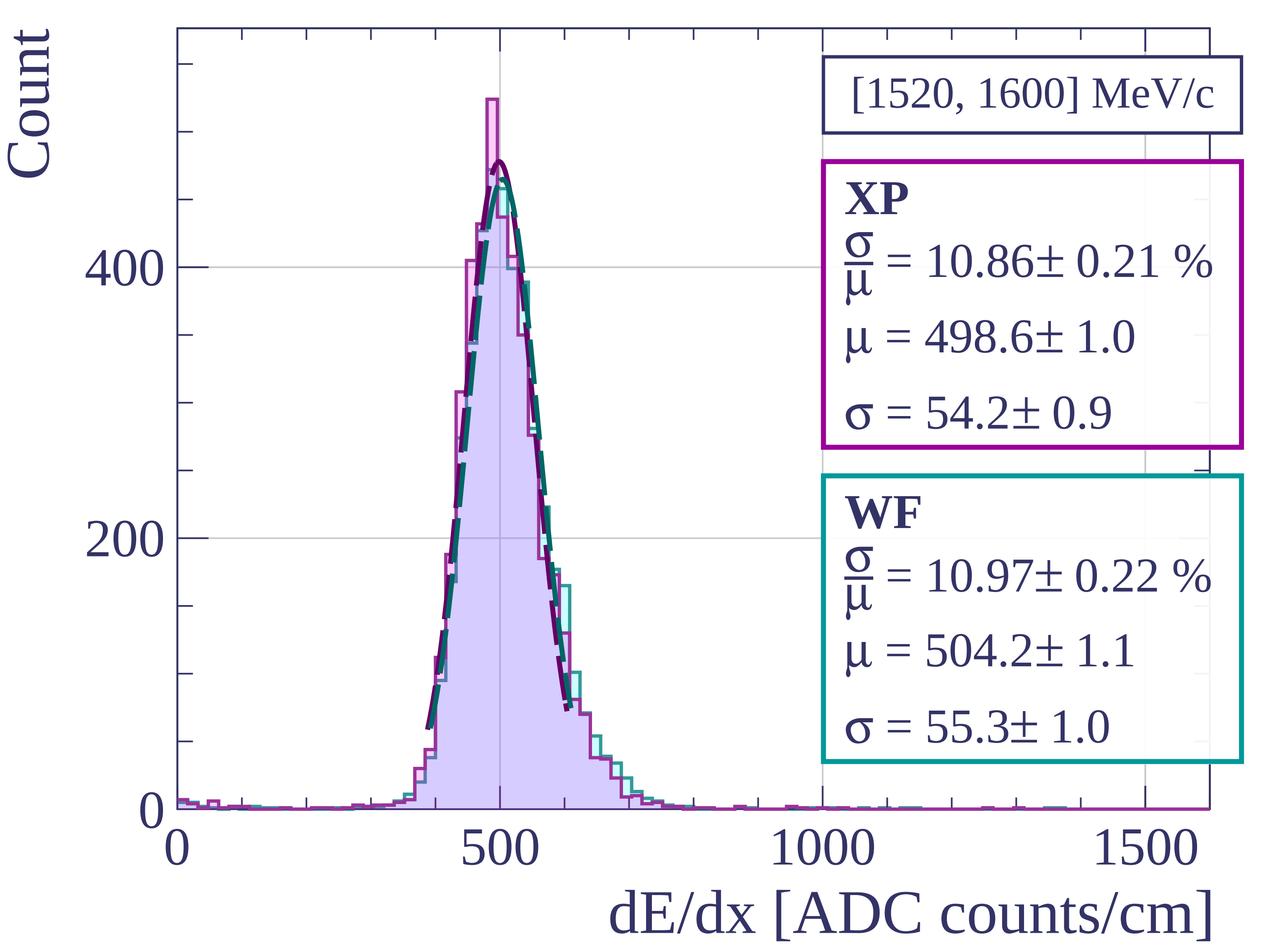


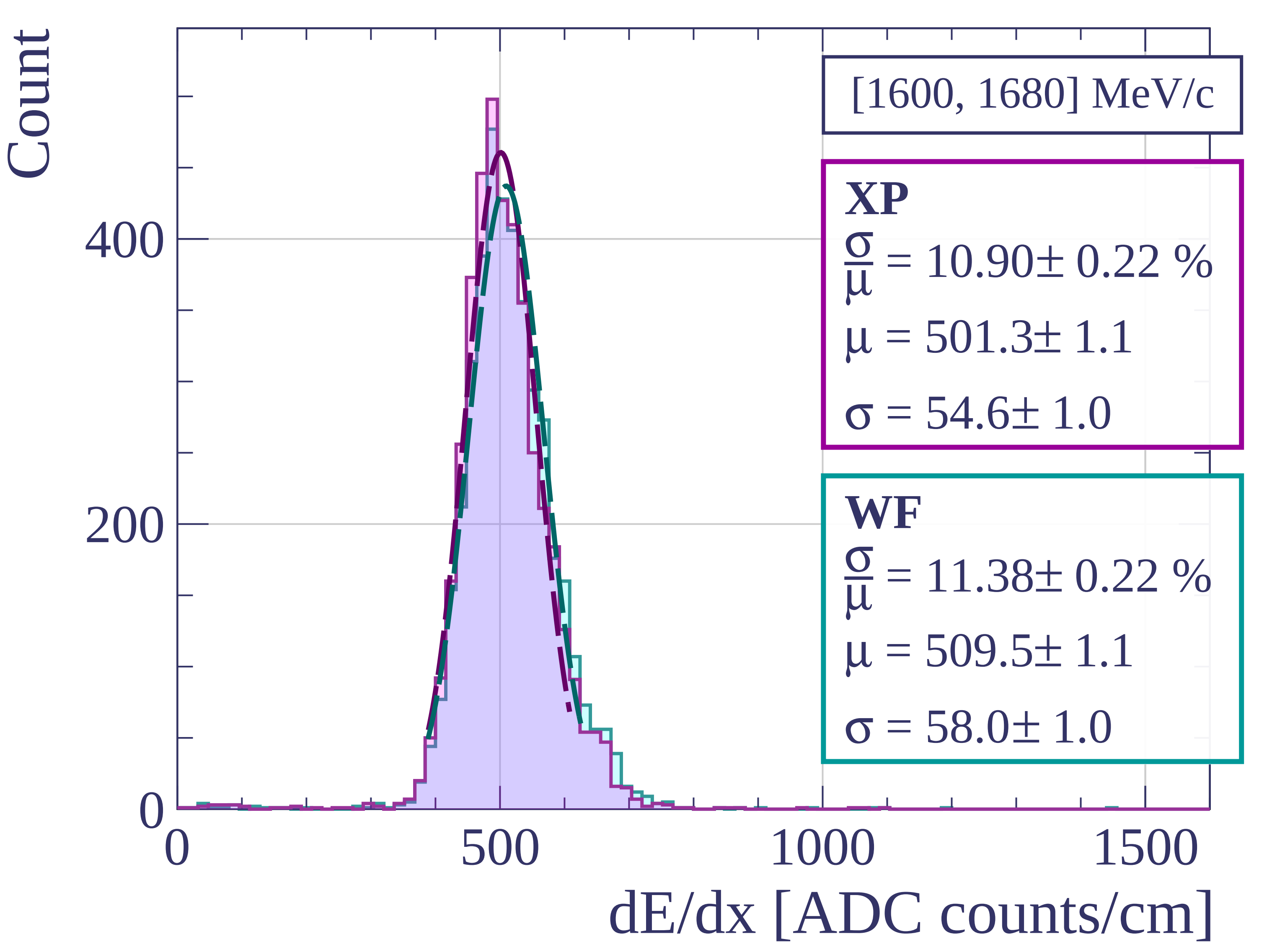
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
σ/μ = 11.28± 0.22 %
μ = 482.0± 1.0
σ = 54.4± 0.9
WF
σ/μ = 11.19± 0.22 %
μ = 487.5± 1.0
σ = 54.5± 0.9


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.39± 0.21 %
μ = 485.4± 1.0
σ = 50.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.21 %
μ = 493.3± 1.0
σ = 53.2± 0.9

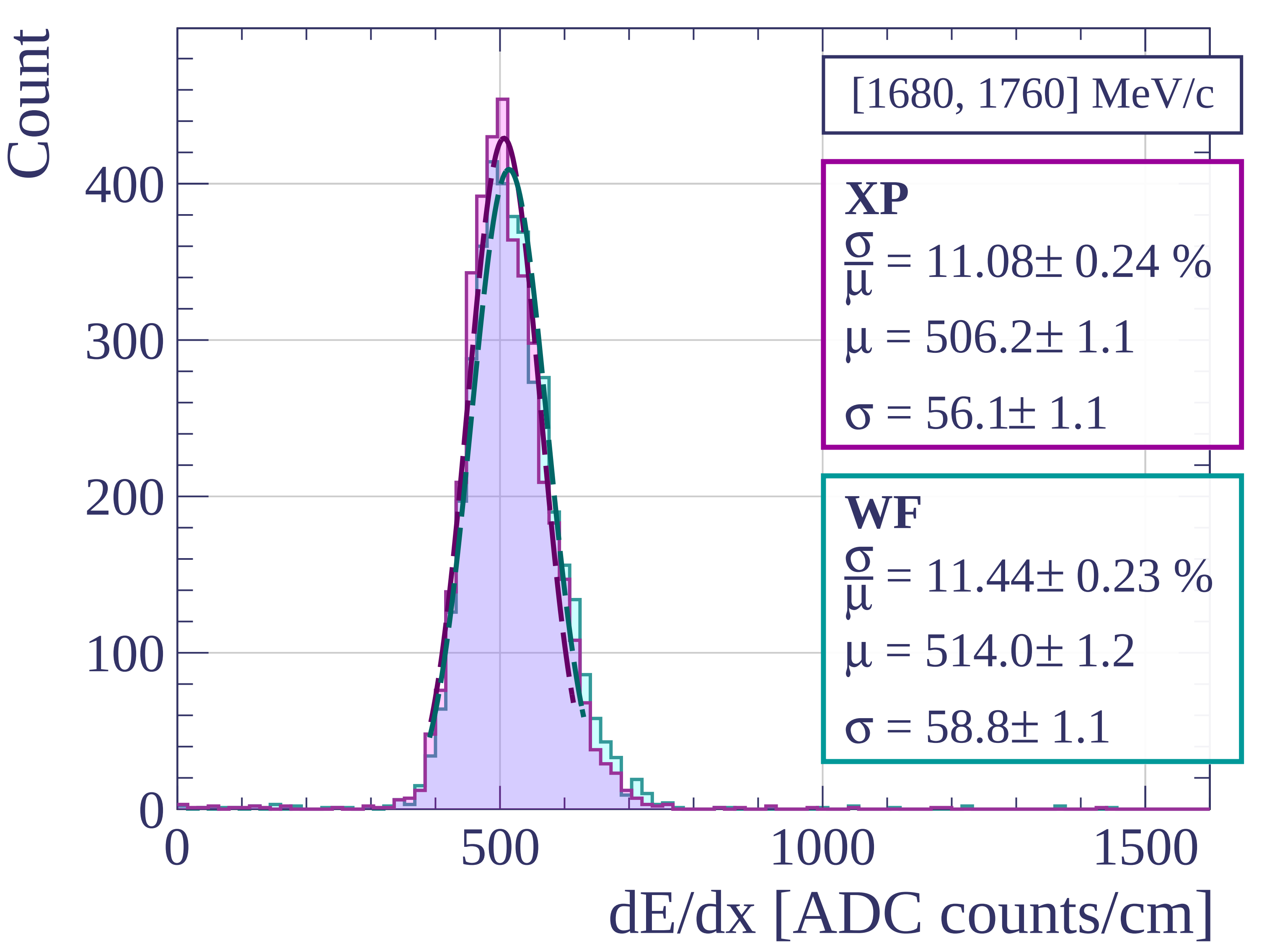

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.86± 0.23 %
μ = 490.8± 1.0
σ = 53.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.54± 0.25 %
μ = 497.2± 1.1
σ = 57.4± 1.1

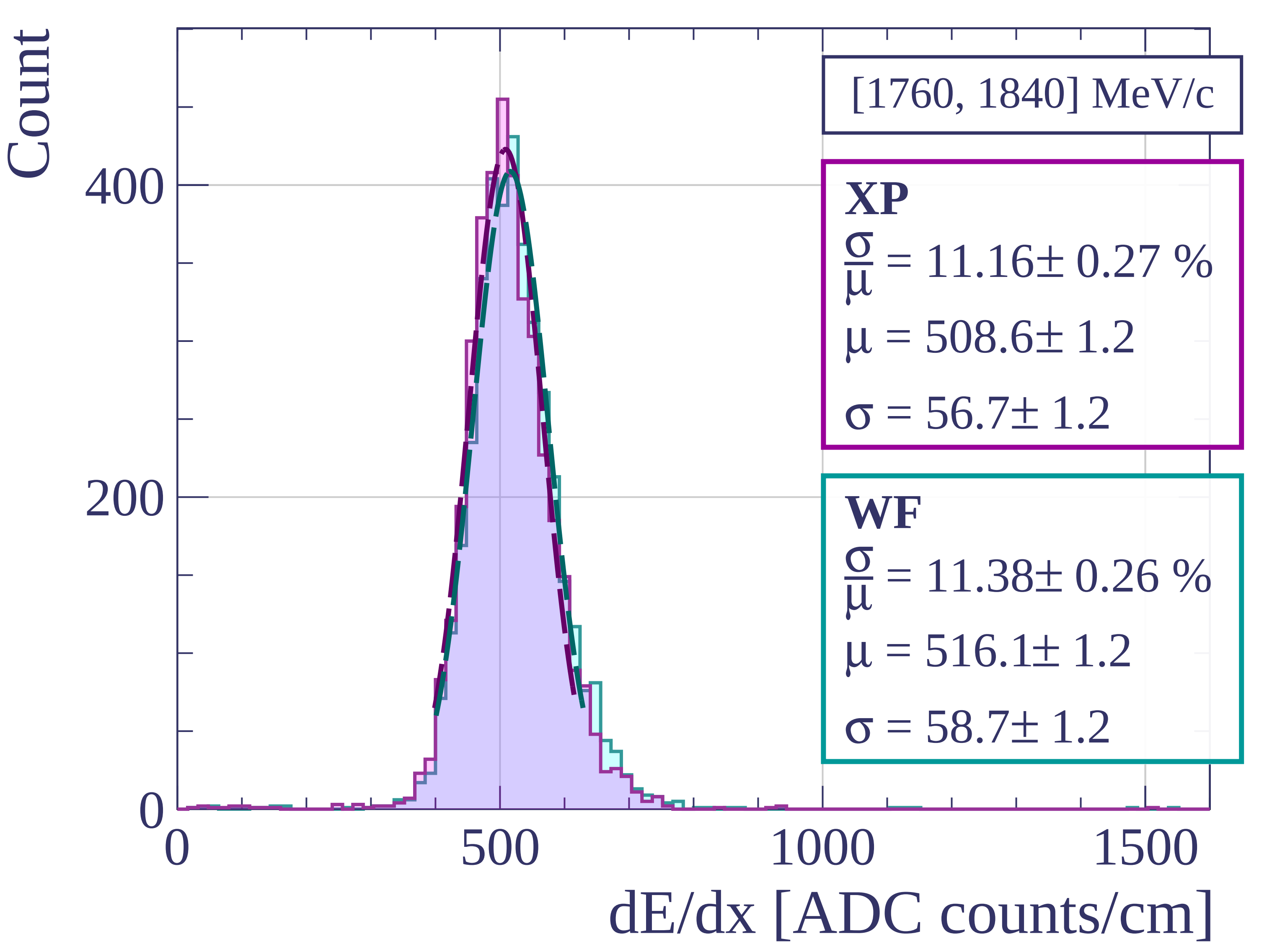

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.86± 0.23 %
μ = 492.8± 1.1
σ = 53.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.34± 0.24 %
μ = 500.1± 1.1
σ = 56.7± 1.1


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.86± 0.21 %
μ = 498.6± 1.0
σ = 54.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.97± 0.22 %
μ = 504.2± 1.1
σ = 55.3± 1.0

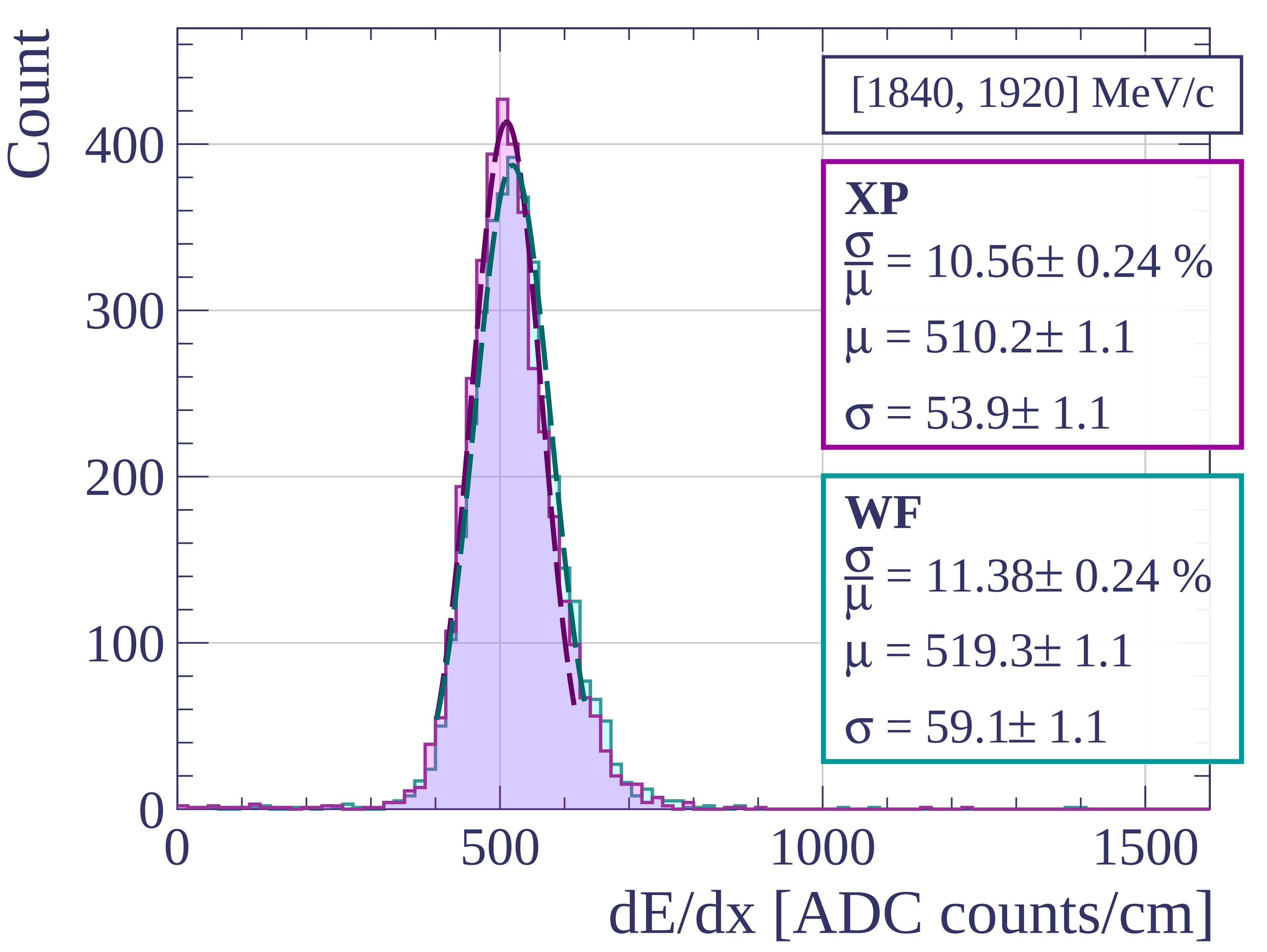


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.90± 0.22 %
μ = 501.3± 1.1
σ = 54.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.38± 0.22 %
μ = 509.5± 1.1
σ = 58.0± 1.0

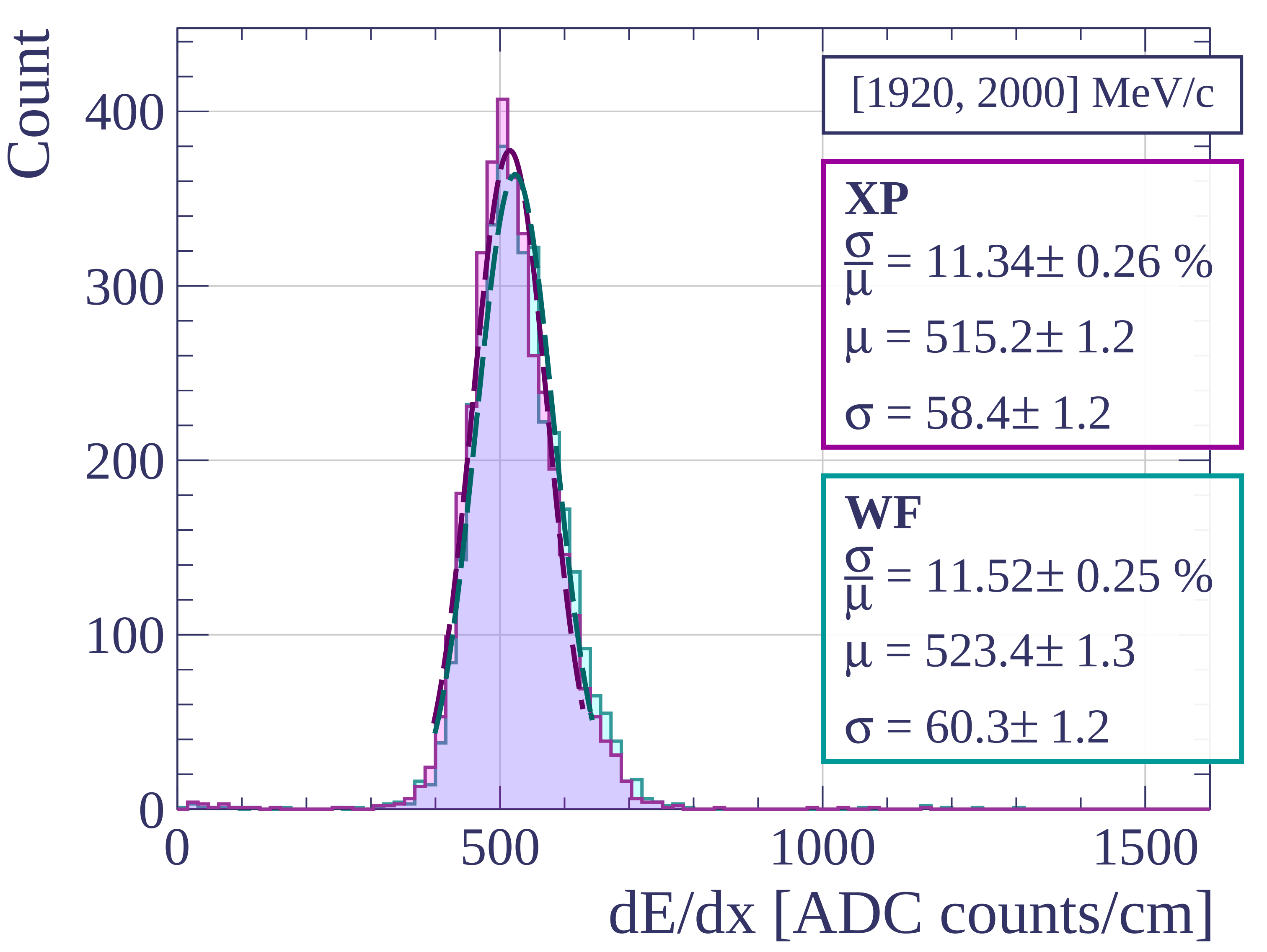

Count
dE/dx [ADC counts/cm]
[1680, 1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.08± 0.24 %
μ = 506.2± 1.1
σ = 56.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.44± 0.23 %
μ = 514.0± 1.2
σ = 58.8± 1.1
0
100
200
300
400
0
500
1000
1500



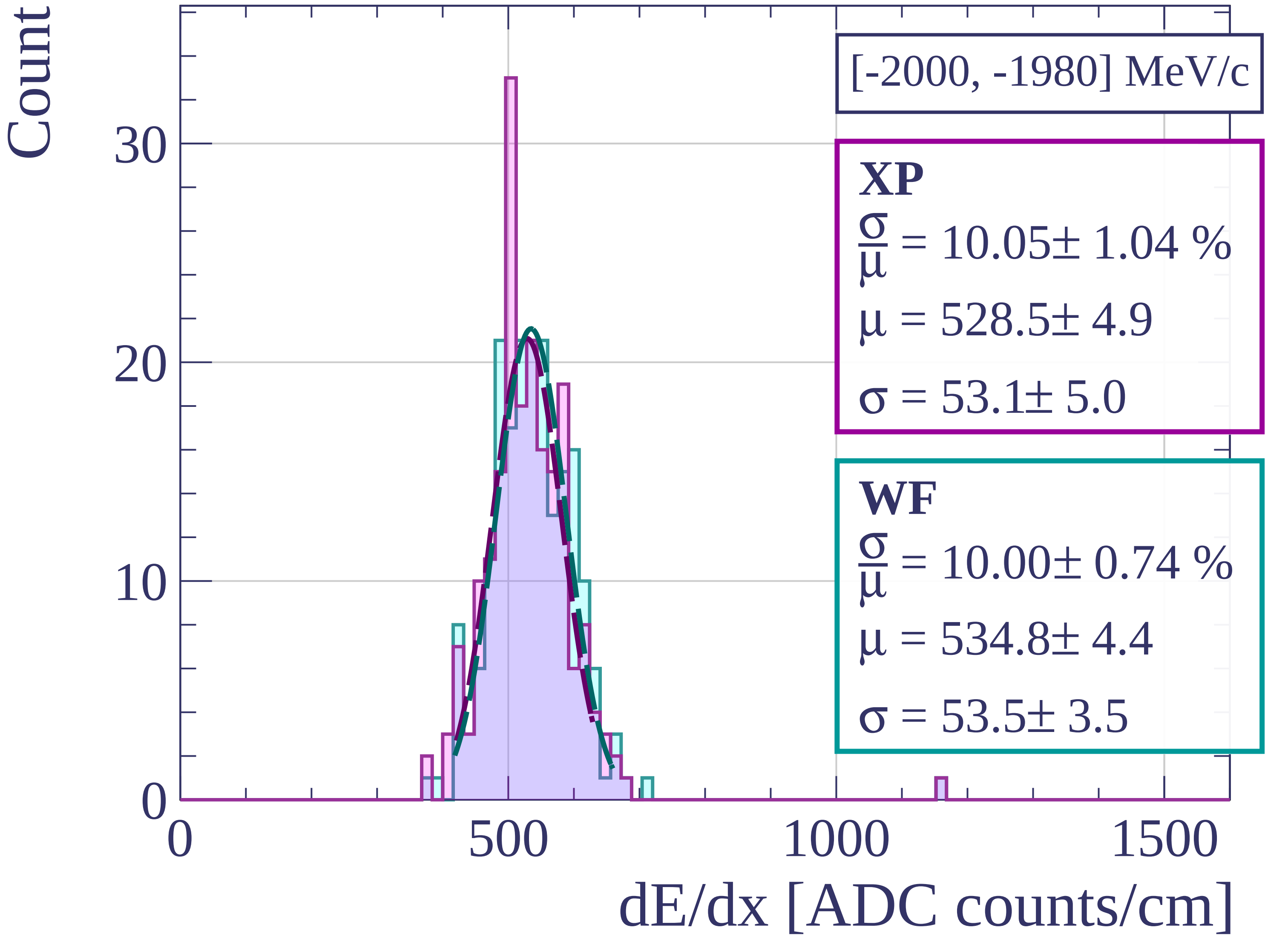
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1840] MeV/c
XP
σ/μ = 11.16± 0.27 %
μ = 508.6± 1.2
σ = 56.7± 1.2
WF
σ/μ = 11.38± 0.26 %
μ = 516.1± 1.2
σ = 58.7± 1.2



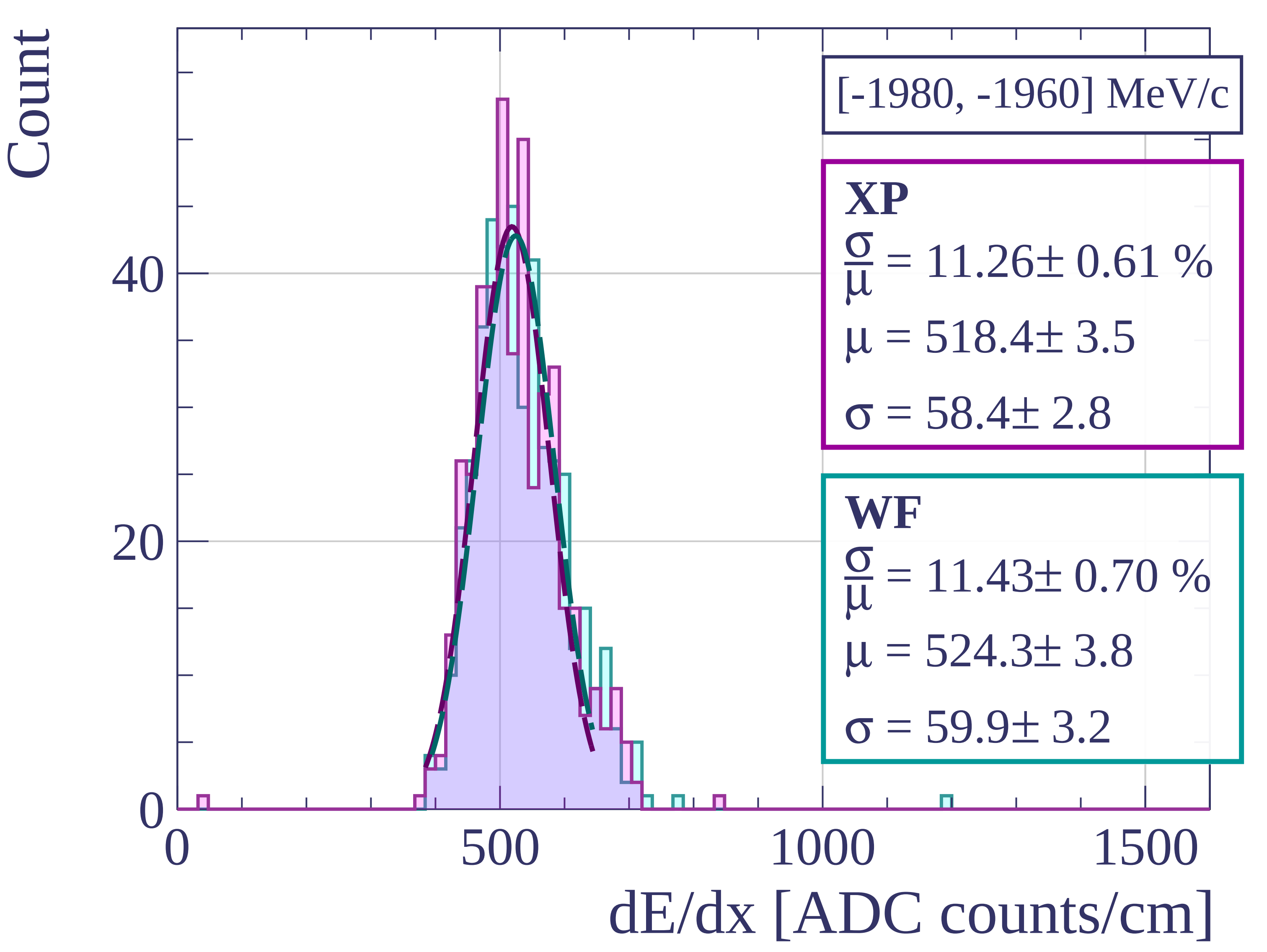
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.56± 0.24 %
μ = 510.2± 1.1
σ = 53.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.38± 0.24 %
μ = 519.3± 1.1
σ = 59.1± 1.1



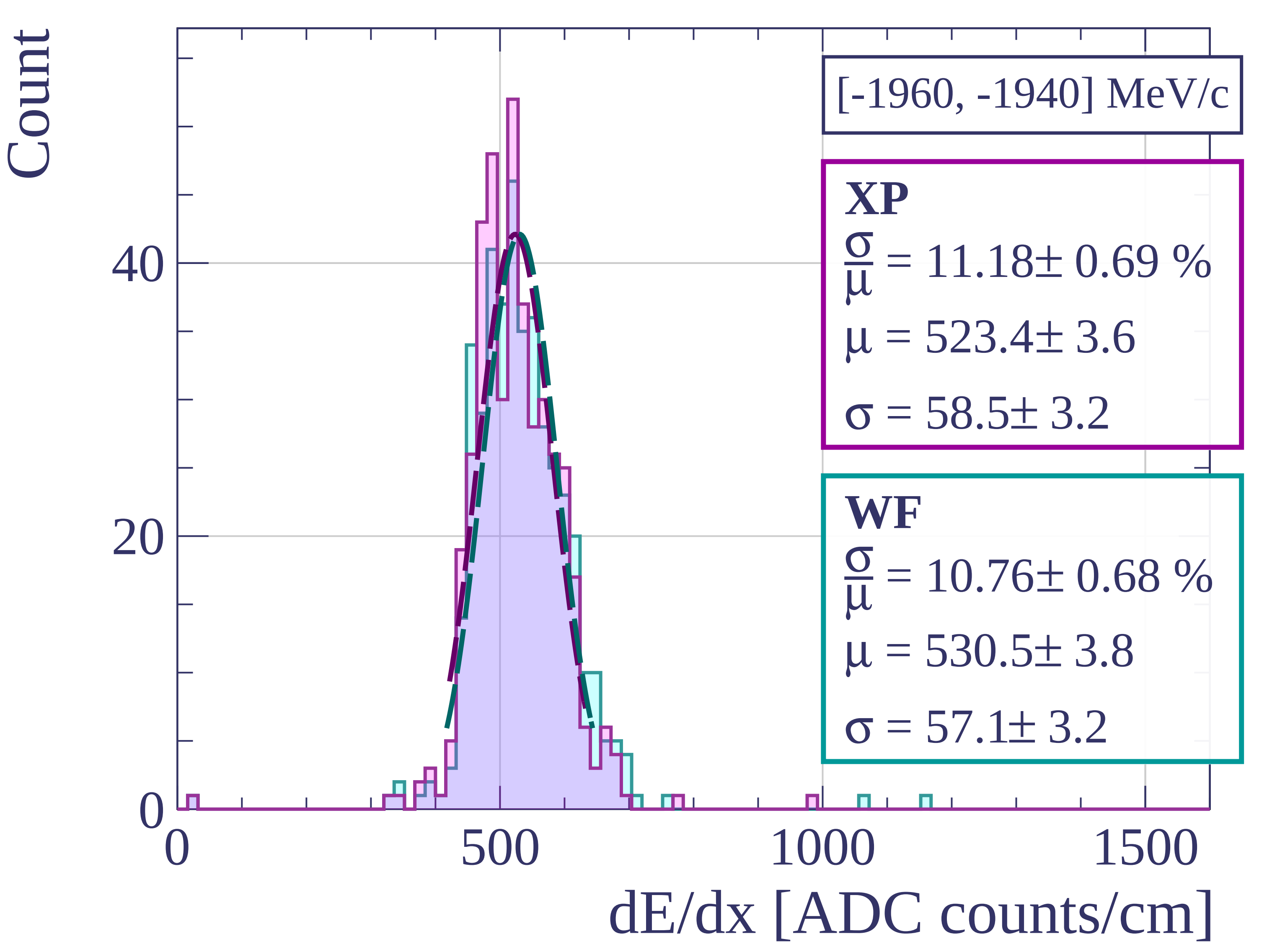
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.34± 0.26 %
μ = 515.2± 1.2
σ = 58.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.52± 0.25 %
μ = 523.4± 1.3
σ = 60.3± 1.2



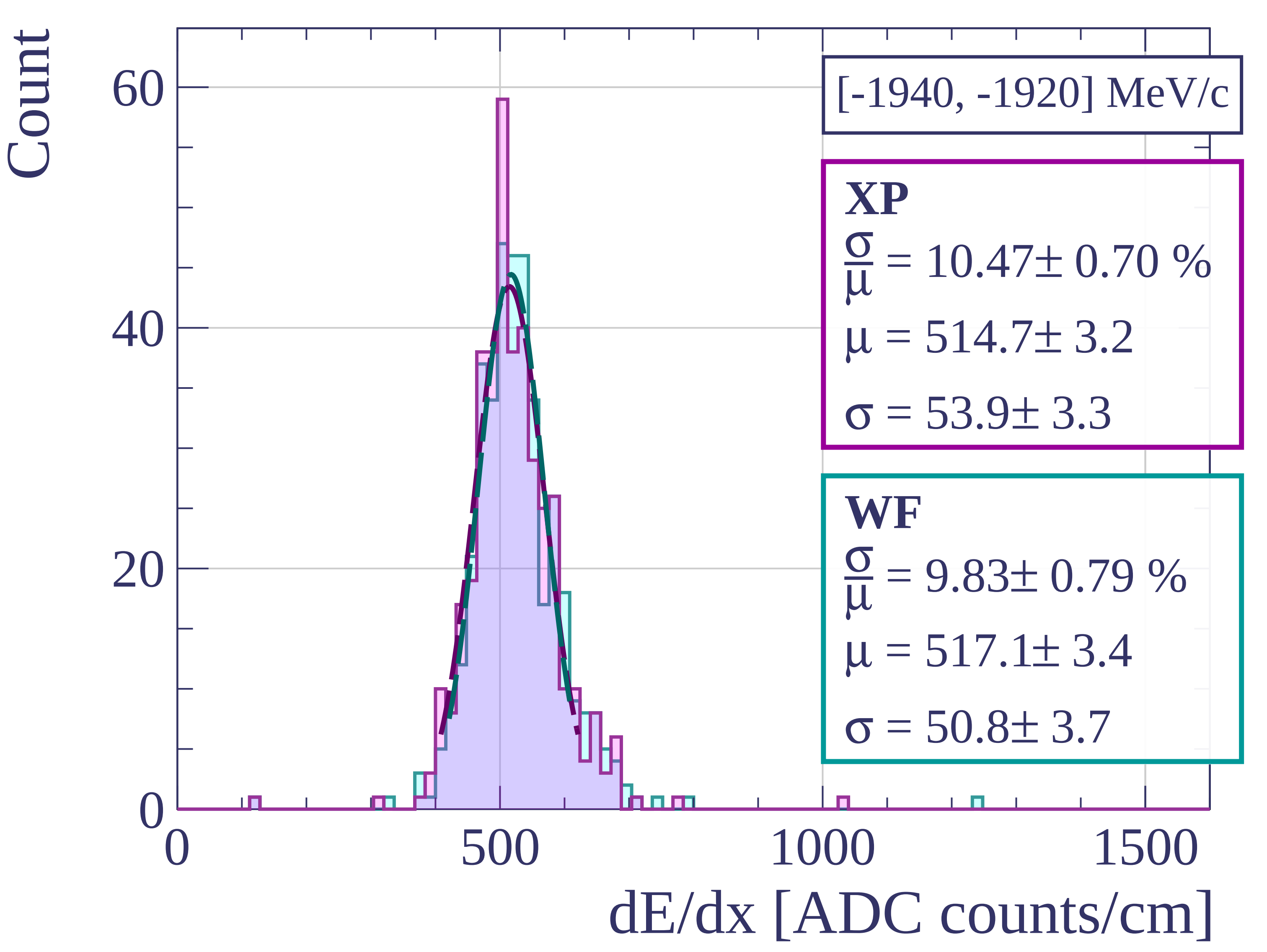
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
[-2000, -1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.05± 1.04 %
μ = 528.5± 4.9
σ = 53.1± 5.0
WF
$\frac{\sigma}{\mu}$ = 10.00± 0.74 %
μ = 534.8± 4.4
σ = 53.5± 3.5



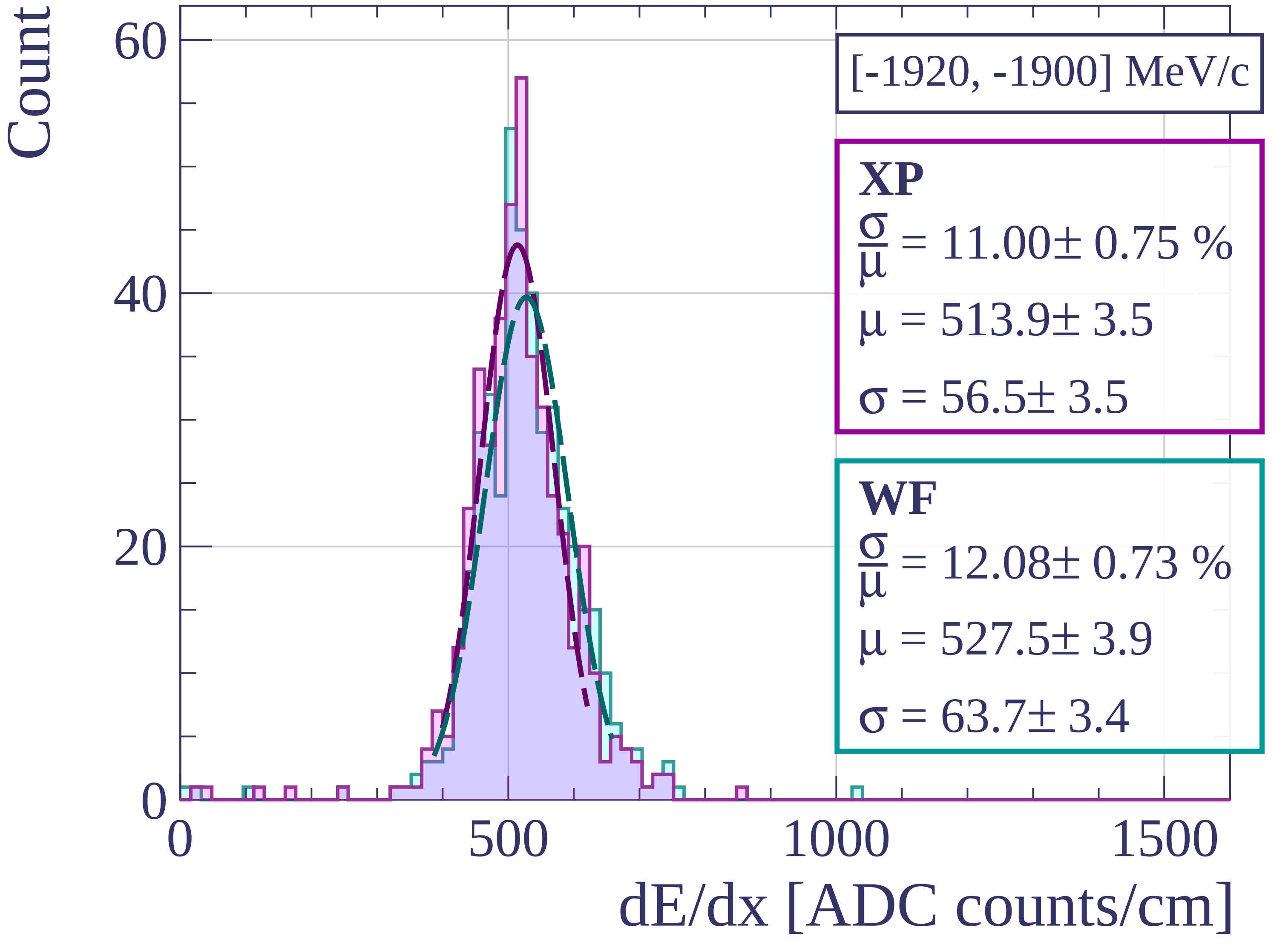
Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
σ/μ = 11.26± 0.61 %
μ = 518.4± 3.5
σ = 58.4± 2.8
WF
σ/μ = 11.43± 0.70 %
μ = 524.3± 3.8
σ = 59.9± 3.2



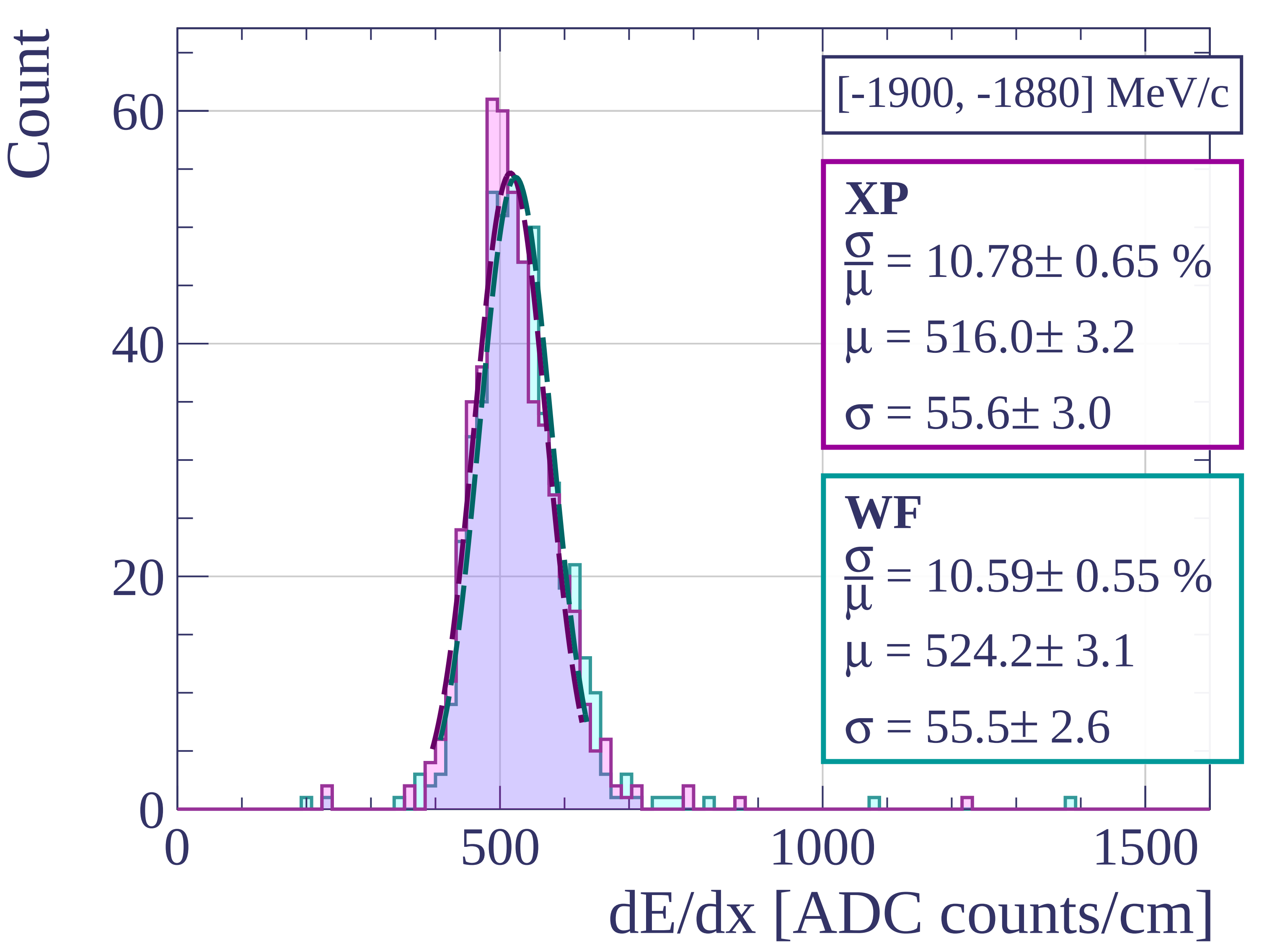
Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1960, -1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.18± 0.69 %
μ = 523.4± 3.6
σ = 58.5± 3.2
WF
$\frac{\sigma}{\mu}$ = 10.76± 0.68 %
μ = 530.5± 3.8
σ = 57.1± 3.2



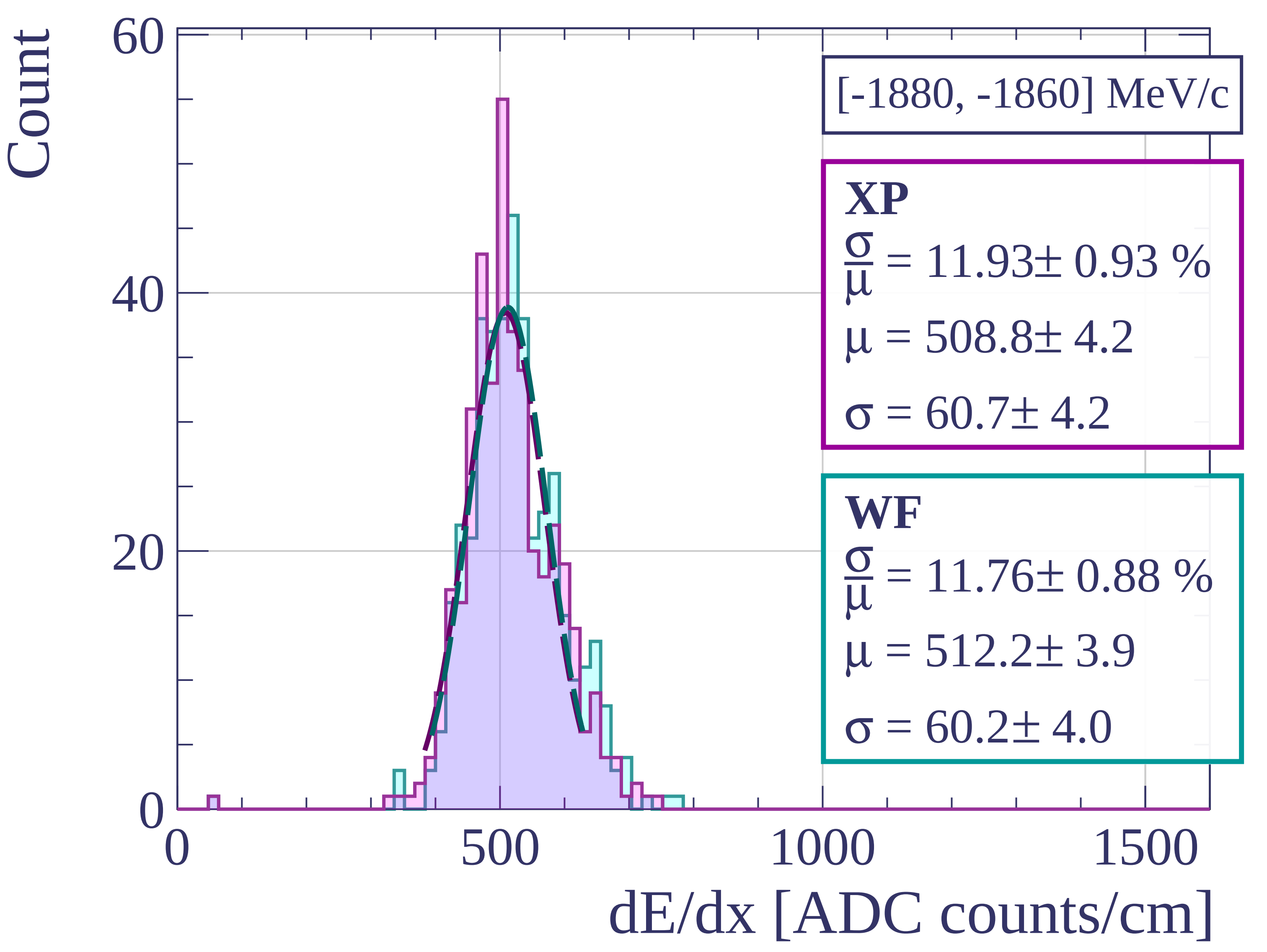
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1940, -1920] MeV/c
XP
σ/μ = 10.47± 0.70 %
μ = 514.7± 3.2
σ = 53.9± 3.3
WF
σ/μ = 9.83± 0.79 %
μ = 517.1± 3.4
σ = 50.8± 3.7

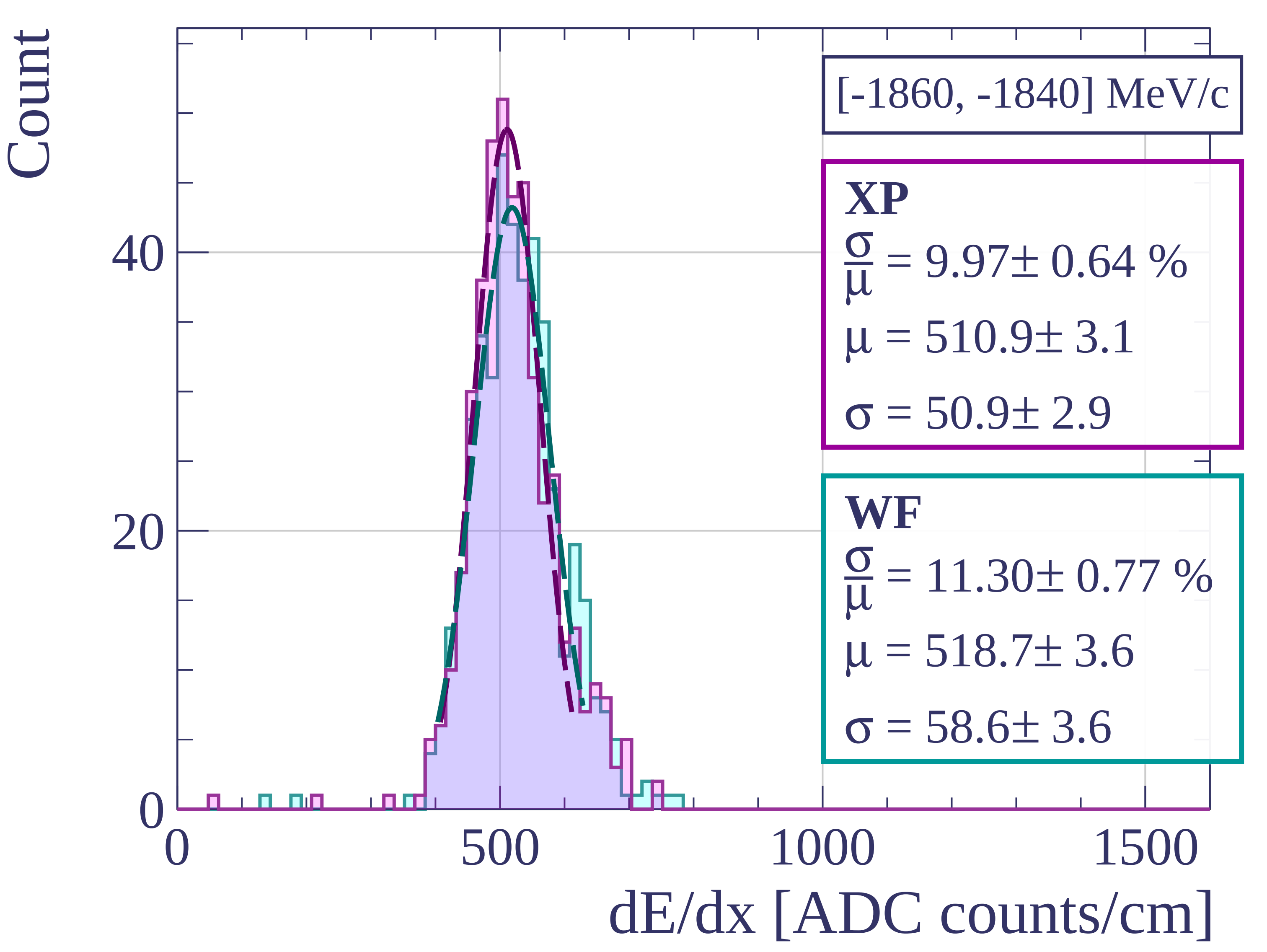


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1920, -1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.00± 0.75 %
μ = 513.9± 3.5
σ = 56.5± 3.5
WF
$\frac{\sigma}{\mu}$ = 12.08± 0.73 %
μ = 527.5± 3.9
σ = 63.7± 3.4

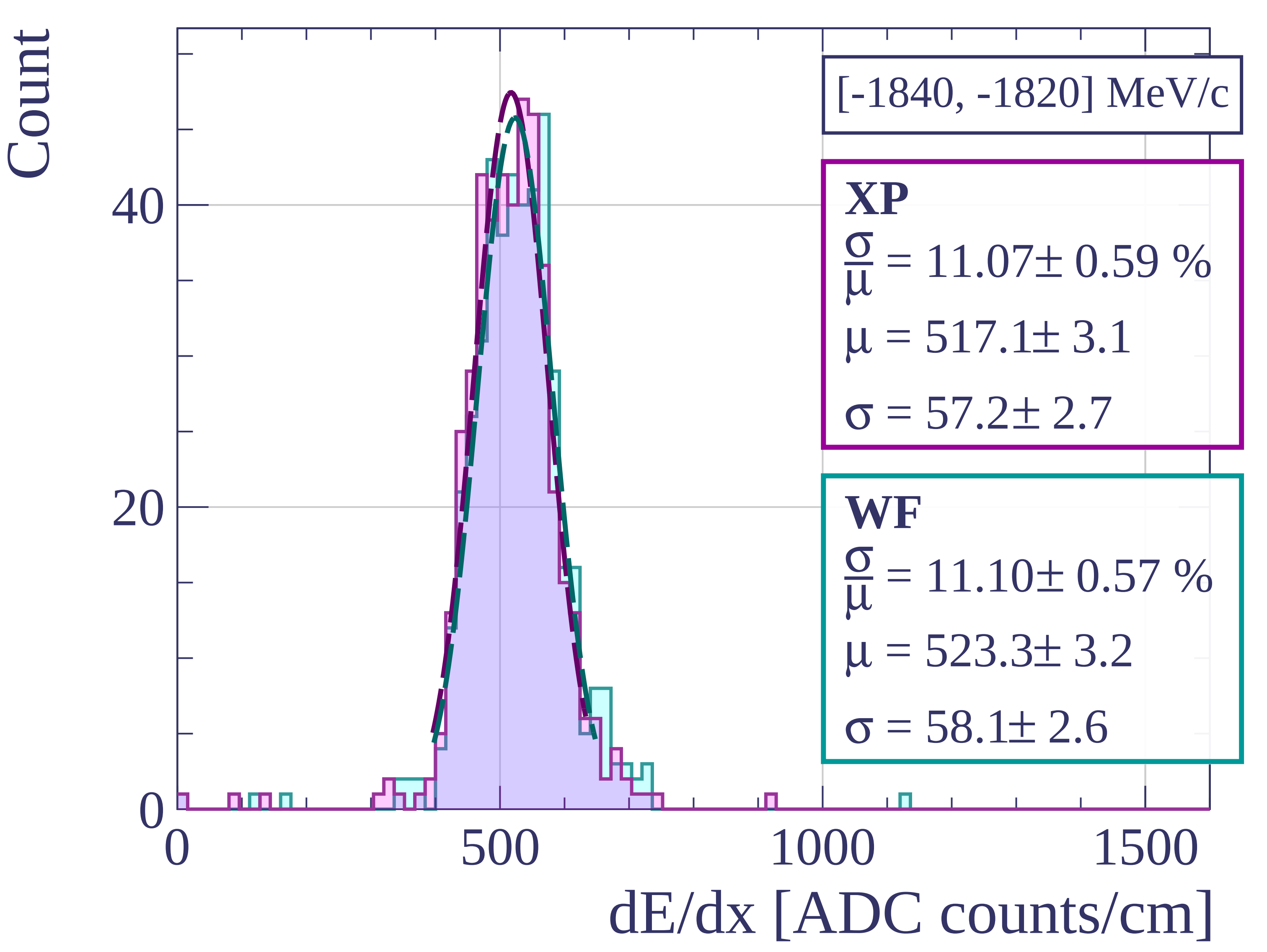


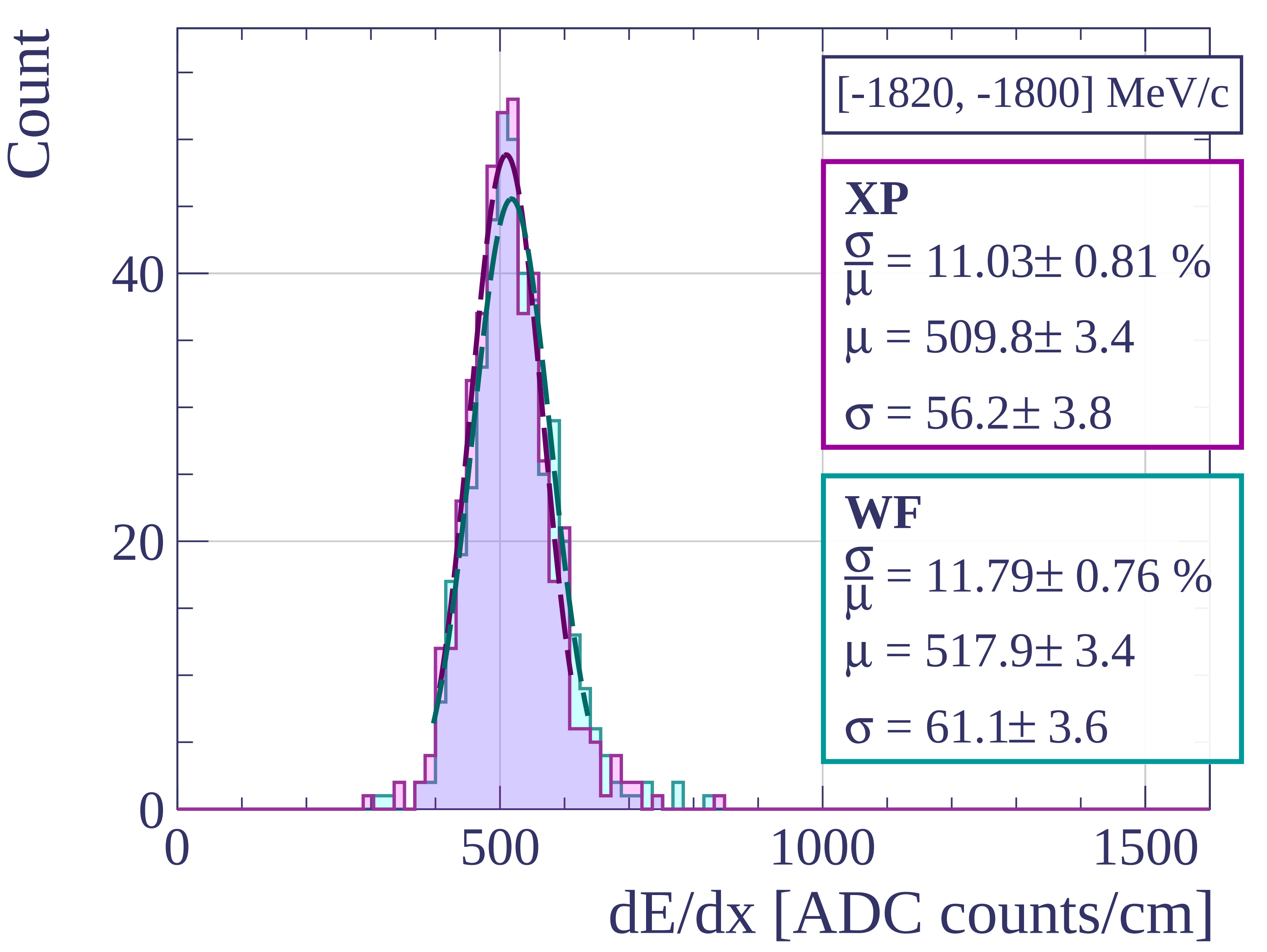
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1900, -1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.78± 0.65 %
μ = 516.0± 3.2
σ = 55.6± 3.0
WF
$\frac{\sigma}{\mu}$ = 10.59± 0.55 %
μ = 524.2± 3.1
σ = 55.5± 2.6


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1880, -1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.93± 0.93 %
μ = 508.8± 4.2
σ = 60.7± 4.2
WF
$\frac{\sigma}{\mu}$ = 11.76± 0.88 %
μ = 512.2± 3.9
σ = 60.2± 4.0

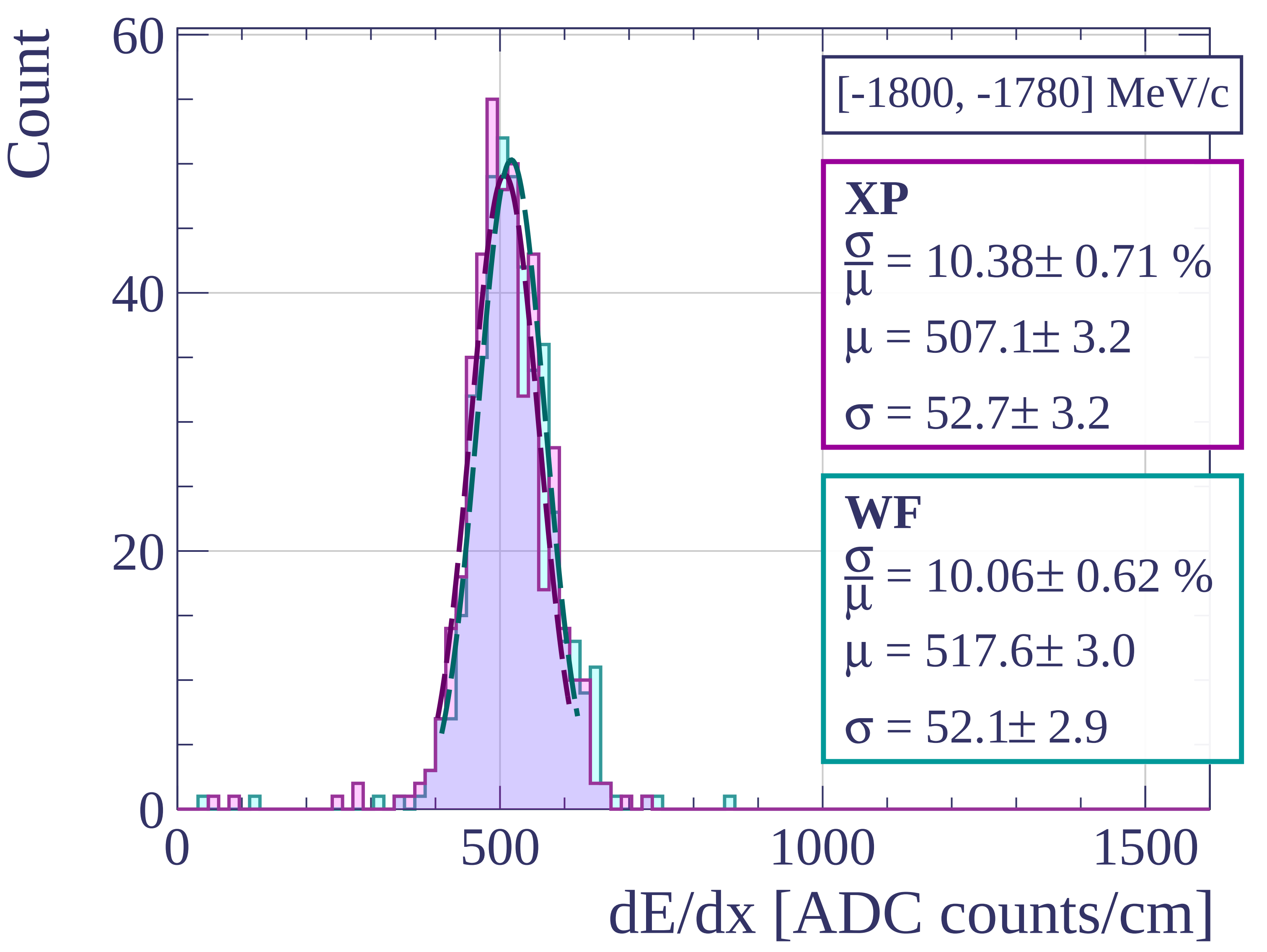


Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[-1860, -1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.97± 0.64 %
μ = 510.9± 3.1
σ = 50.9± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.30± 0.77 %
μ = 518.7± 3.6
σ = 58.6± 3.6

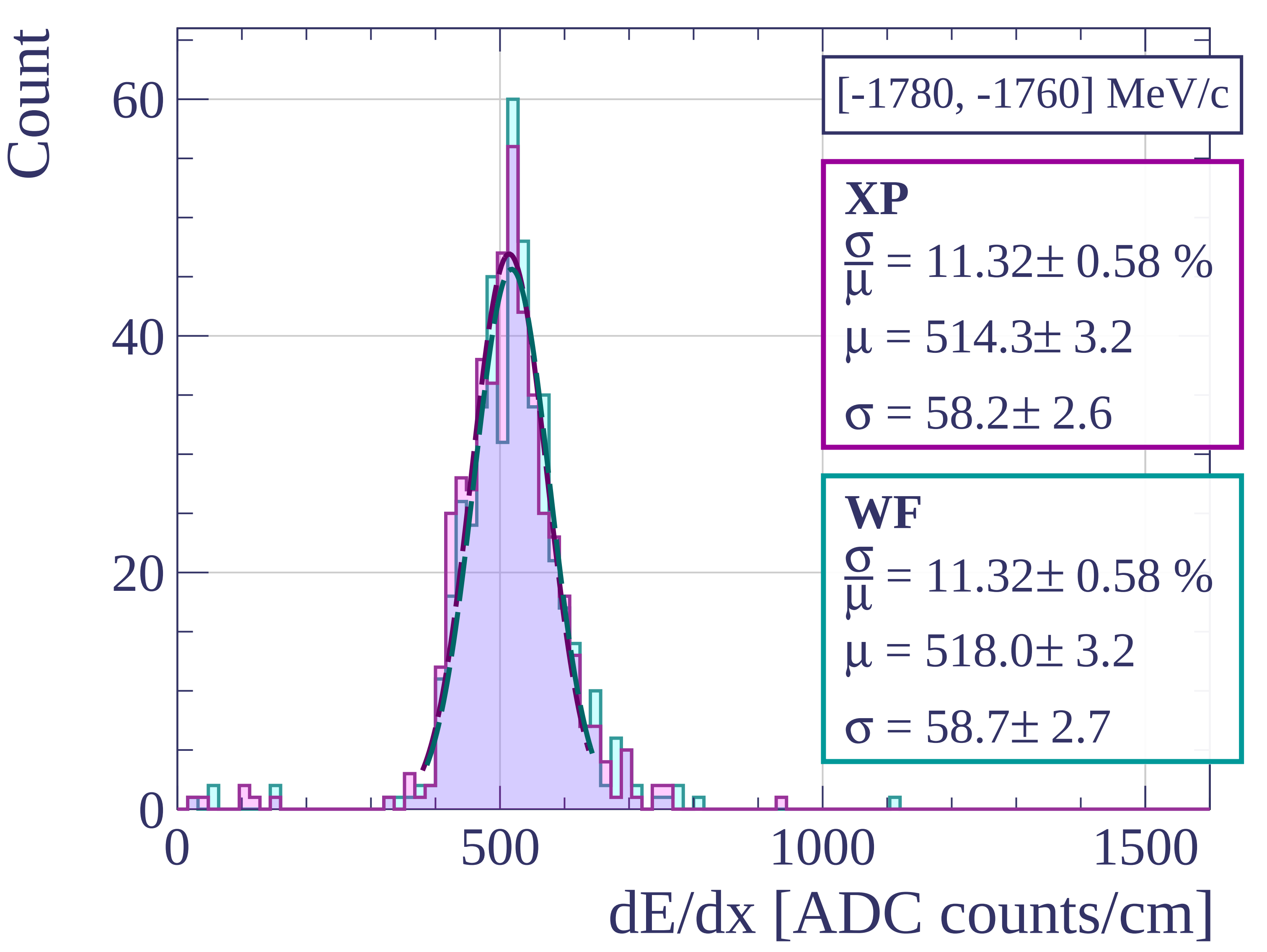

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.07± 0.59 %
μ = 517.1± 3.1
σ = 57.2± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.10± 0.57 %
μ = 523.3± 3.2
σ = 58.1± 2.6

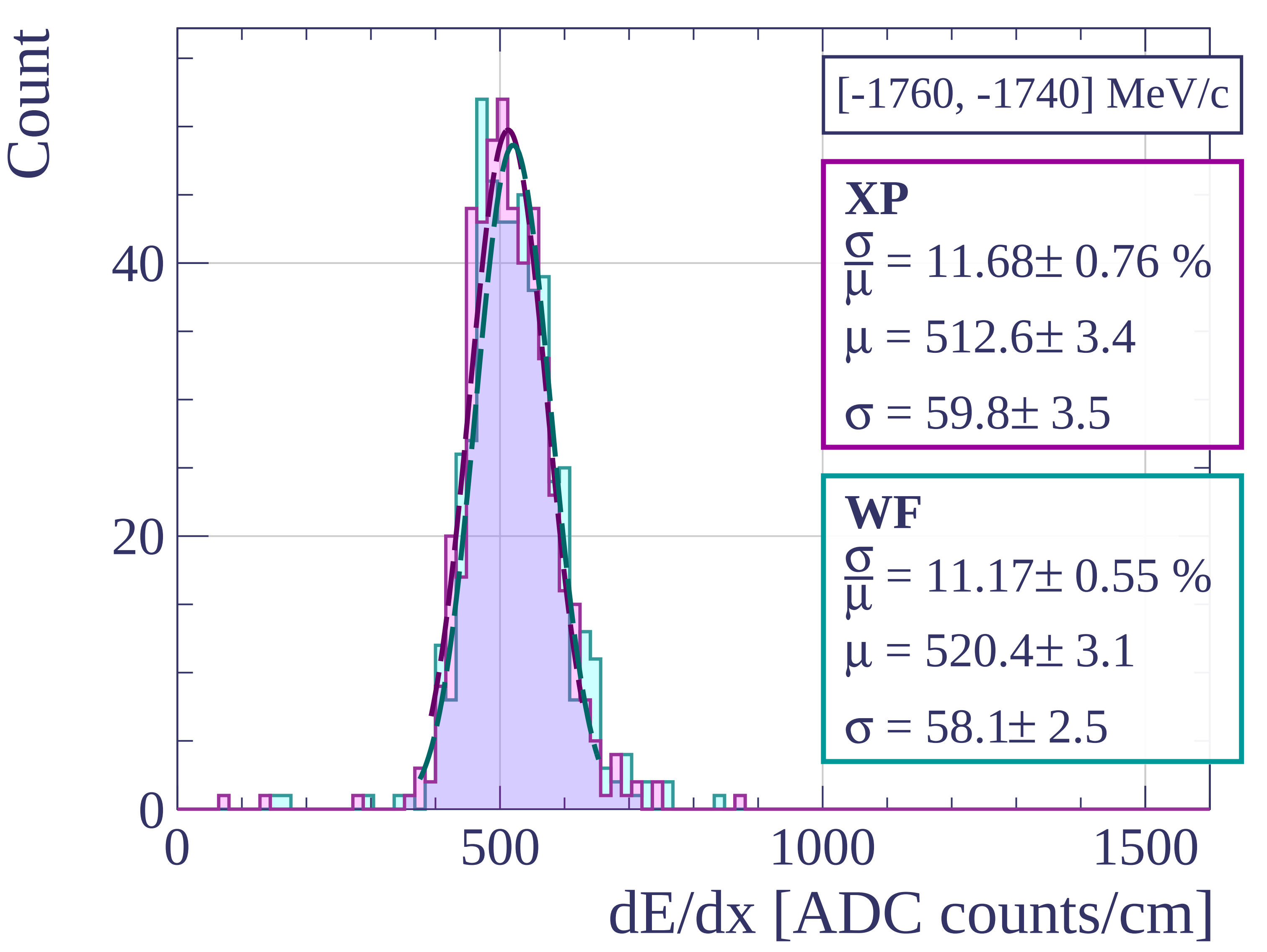


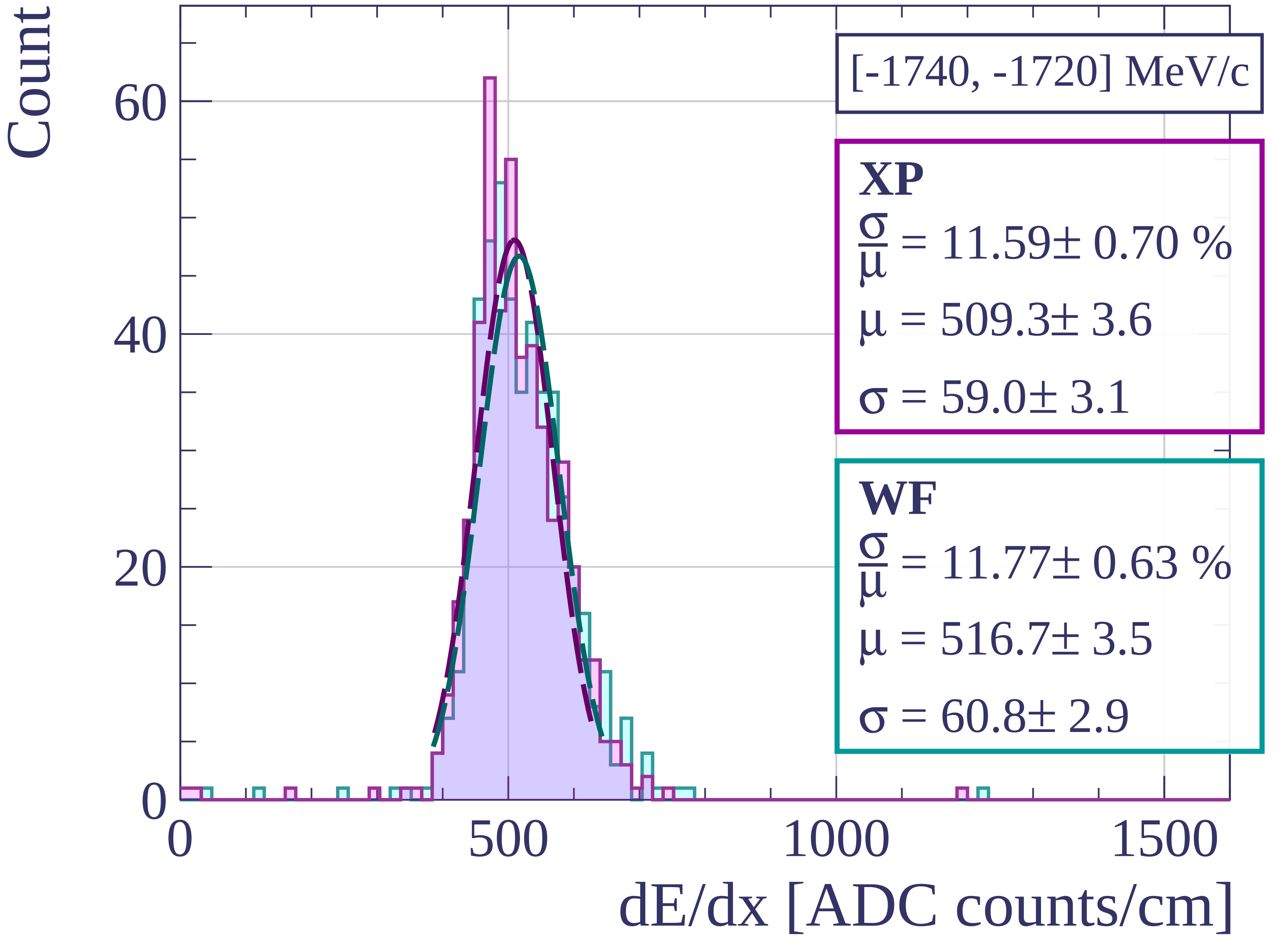
Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[-1820, -1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.03± 0.81 %
μ = 509.8± 3.4
σ = 56.2± 3.8
WF
$\frac{\sigma}{\mu}$ = 11.79± 0.76 %
μ = 517.9± 3.4
σ = 61.1± 3.6



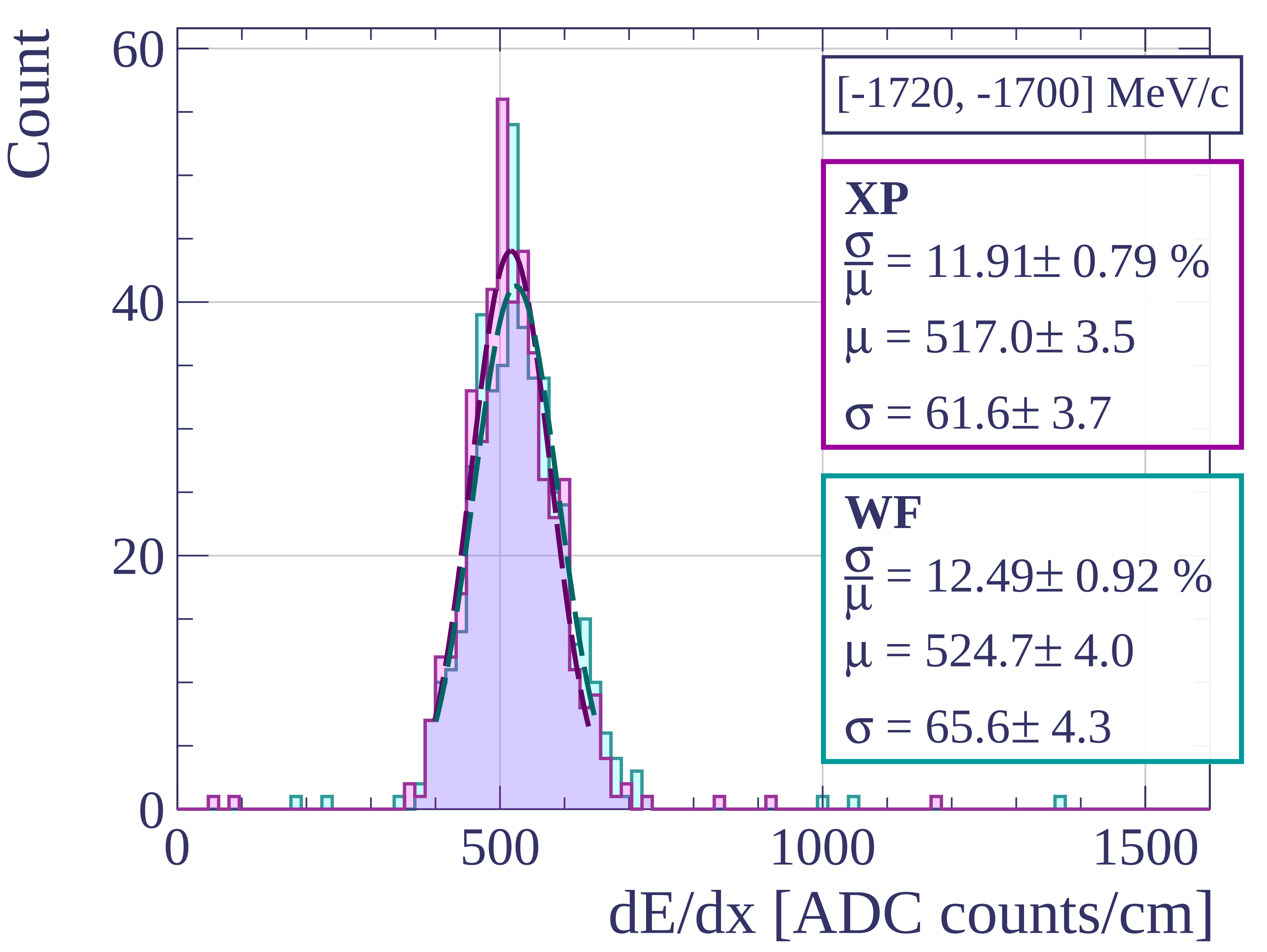
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1800, -1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.38± 0.71 %
μ = 507.1± 3.2
σ = 52.7± 3.2
WF
$\frac{\sigma}{\mu}$ = 10.06± 0.62 %
μ = 517.6± 3.0
σ = 52.1± 2.9

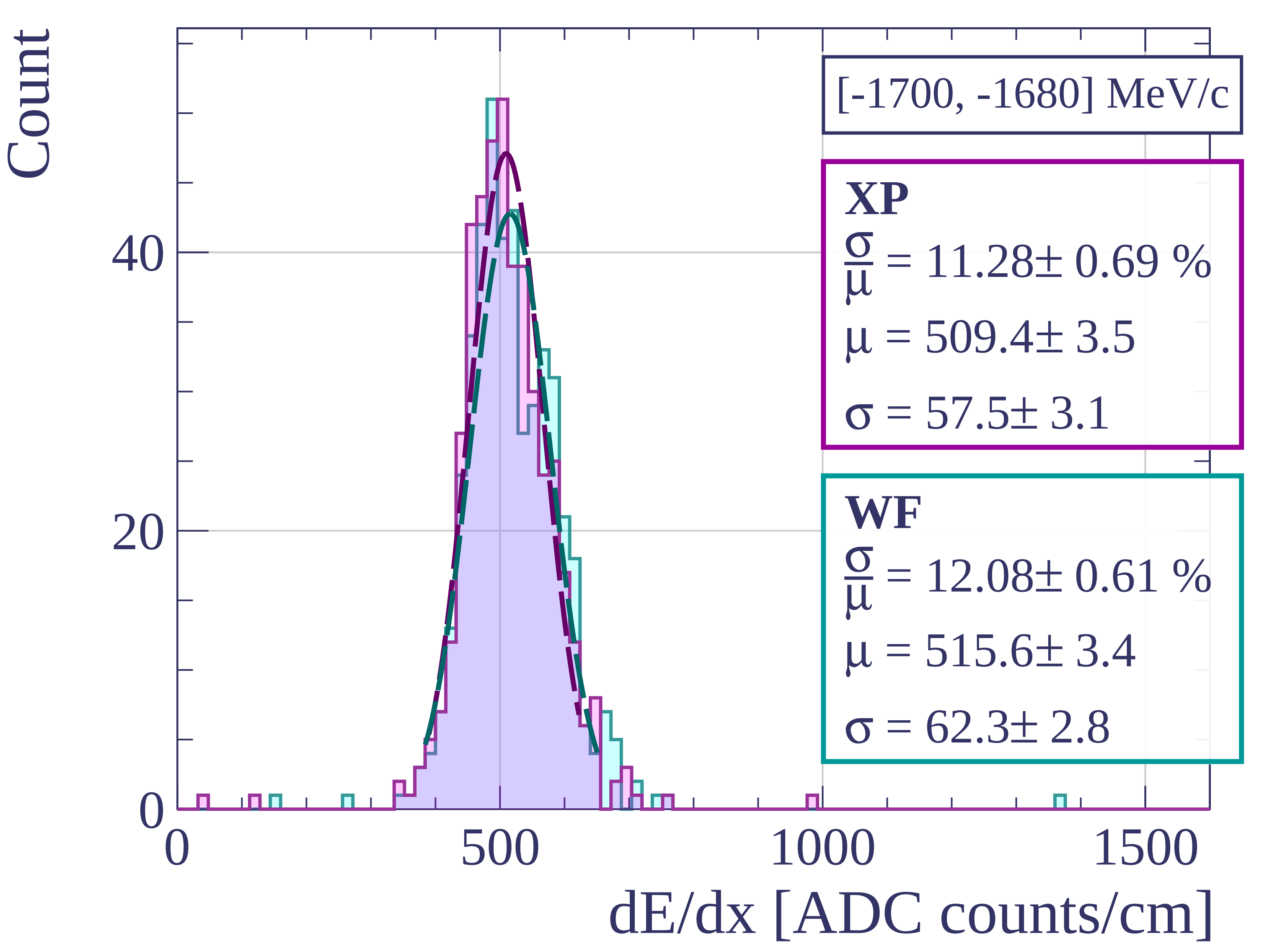

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1780, -1760] MeV/c
XP
σ/μ = 11.32± 0.58 %
μ = 514.3± 3.2
σ = 58.2± 2.6
WF
σ/μ = 11.32± 0.58 %
μ = 518.0± 3.2
σ = 58.7± 2.7

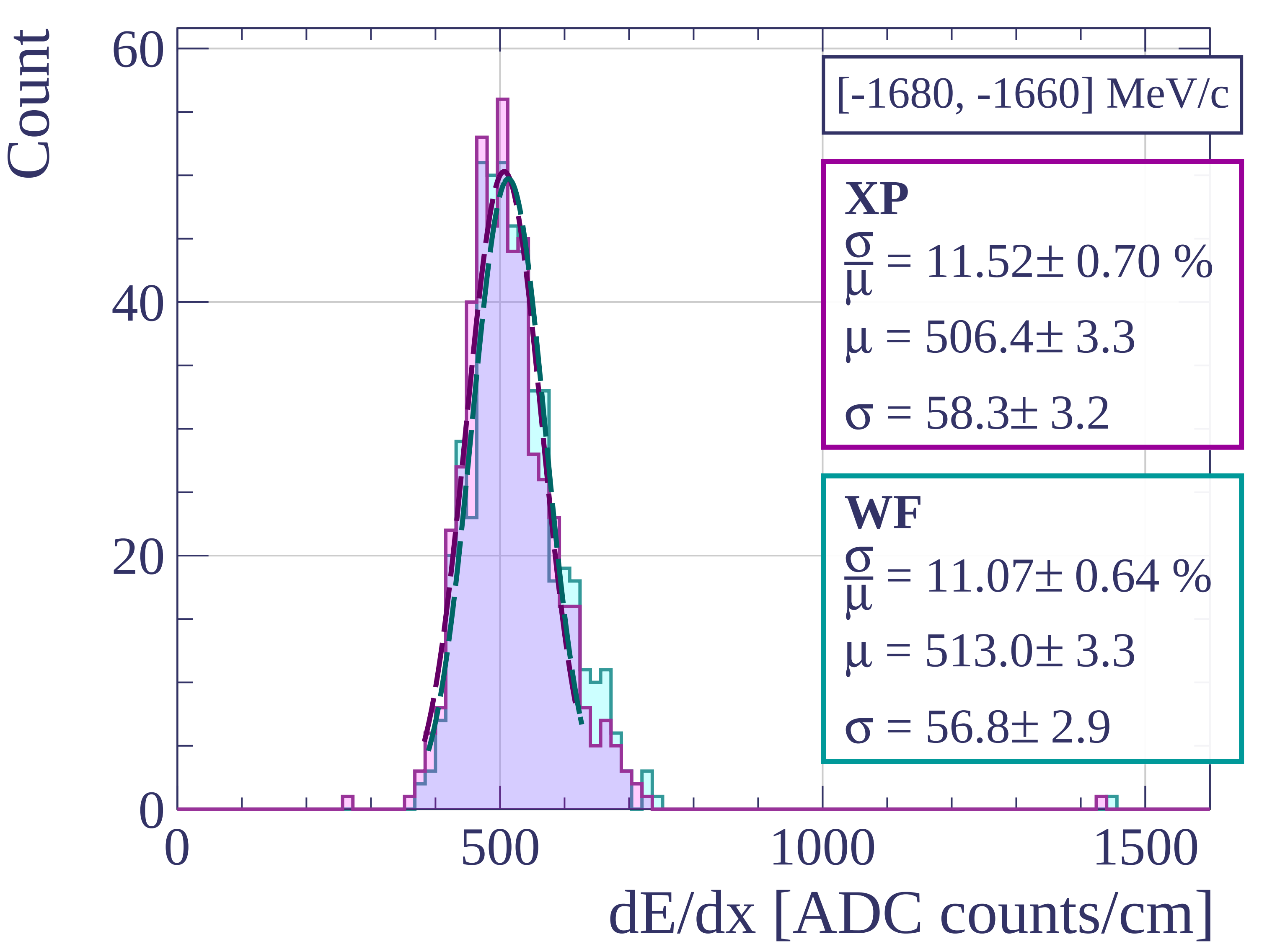

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1760, -1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.68± 0.76 %
μ = 512.6± 3.4
σ = 59.8± 3.5
WF
$\frac{\sigma}{\mu}$ = 11.17± 0.55 %
μ = 520.4± 3.1
σ = 58.1± 2.5



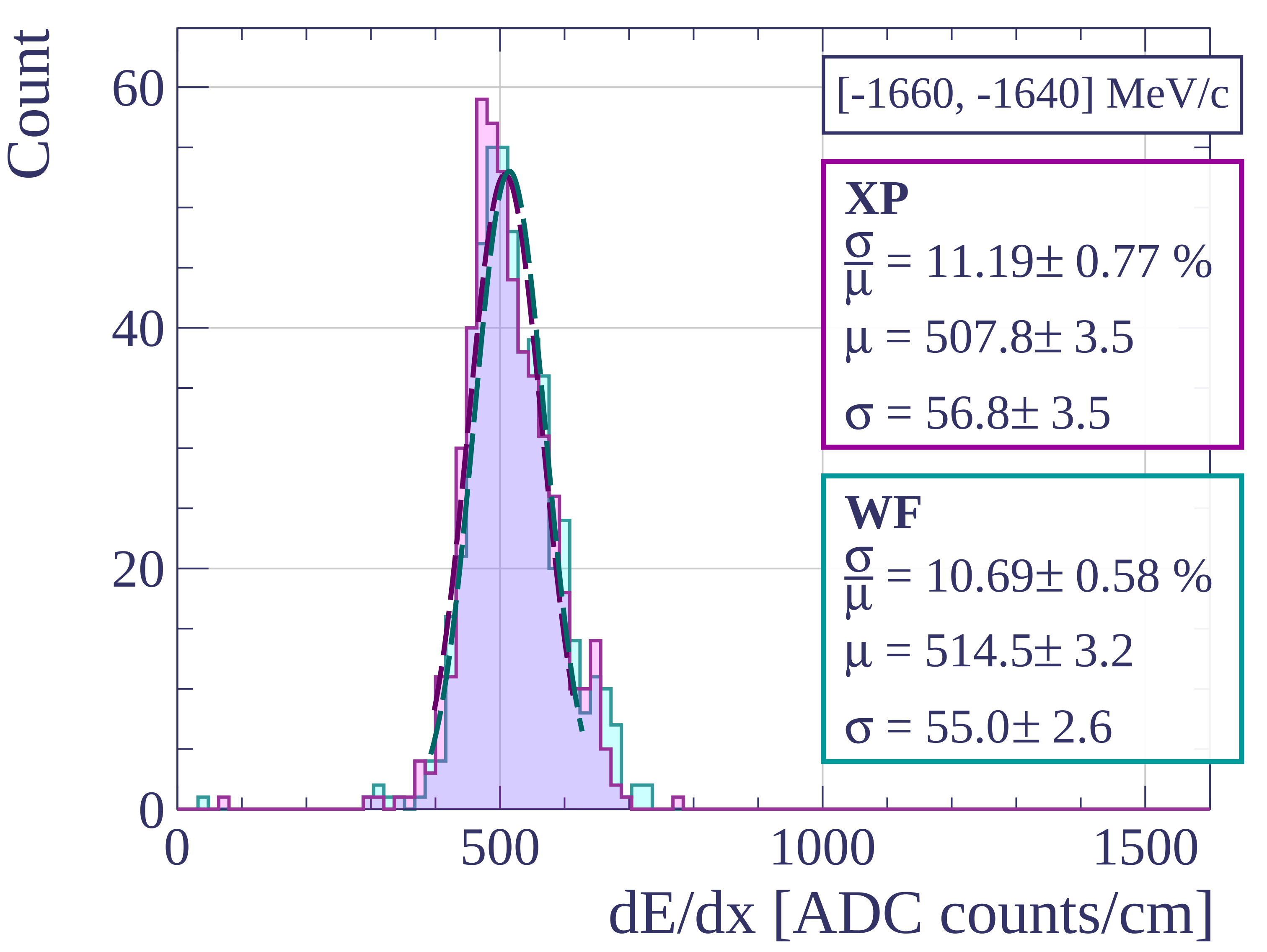
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1740, -1720] MeV/c
XP
σ/μ = 11.59± 0.70 %
μ = 509.3± 3.6
σ = 59.0± 3.1
WF
σ/μ = 11.77± 0.63 %
μ = 516.7± 3.5
σ = 60.8± 2.9

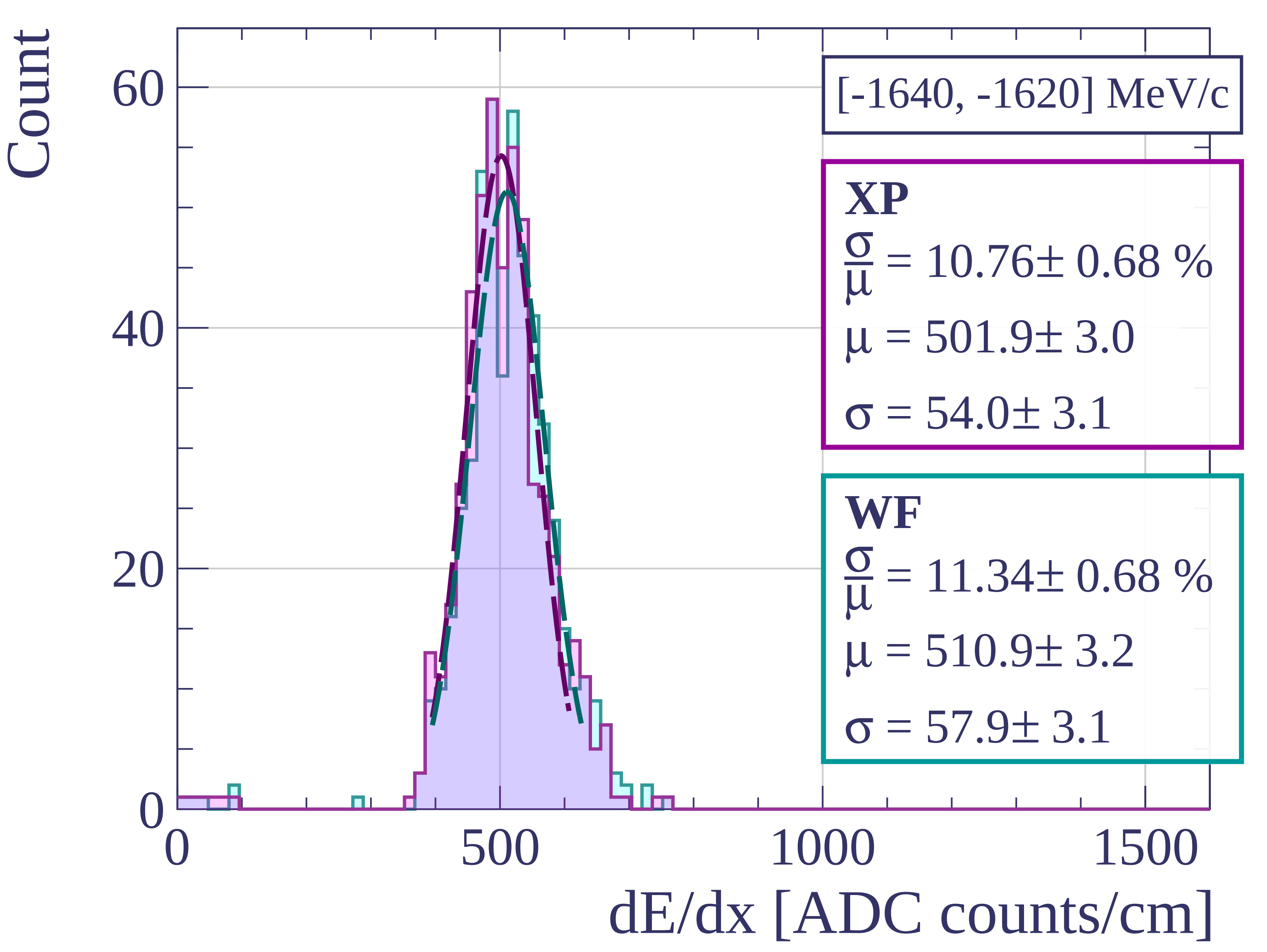

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1720, -1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.91± 0.79 %
μ = 517.0± 3.5
σ = 61.6± 3.7
WF
$\frac{\sigma}{\mu}$ = 12.49± 0.92 %
μ = 524.7± 4.0
σ = 65.6± 4.3

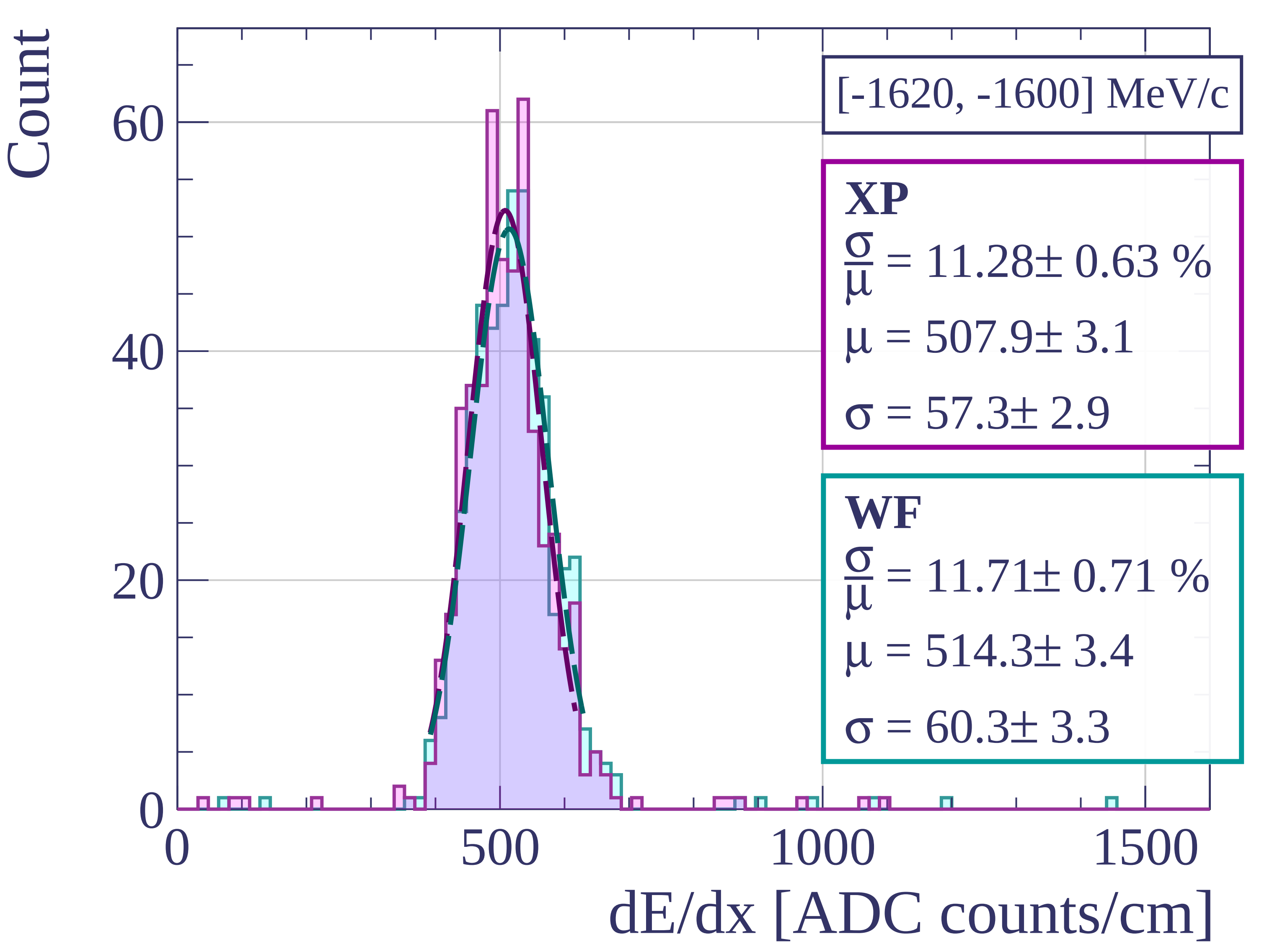

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1700, -1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.28± 0.69 %
μ = 509.4± 3.5
σ = 57.5± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.08± 0.61 %
μ = 515.6± 3.4
σ = 62.3± 2.8

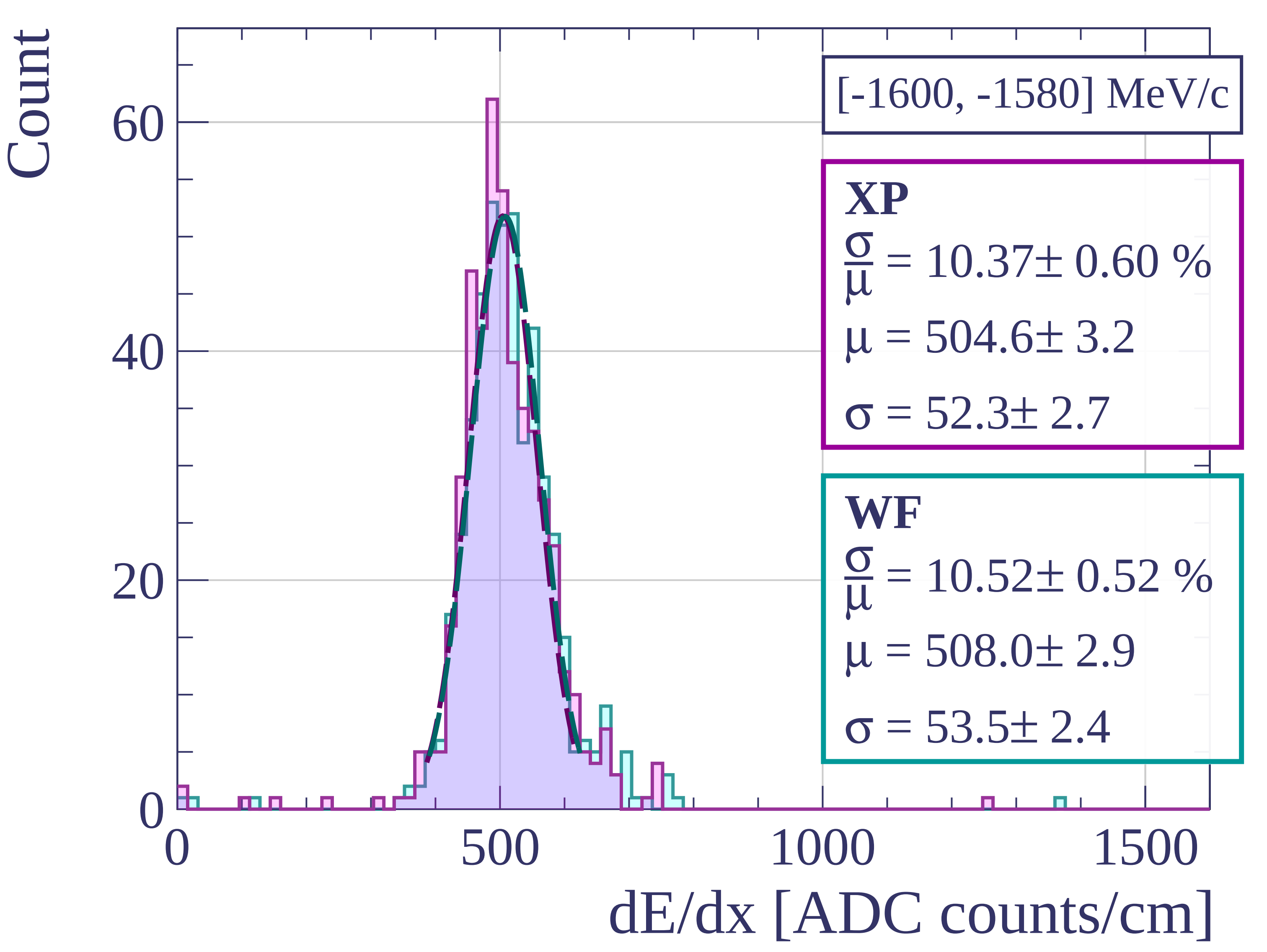


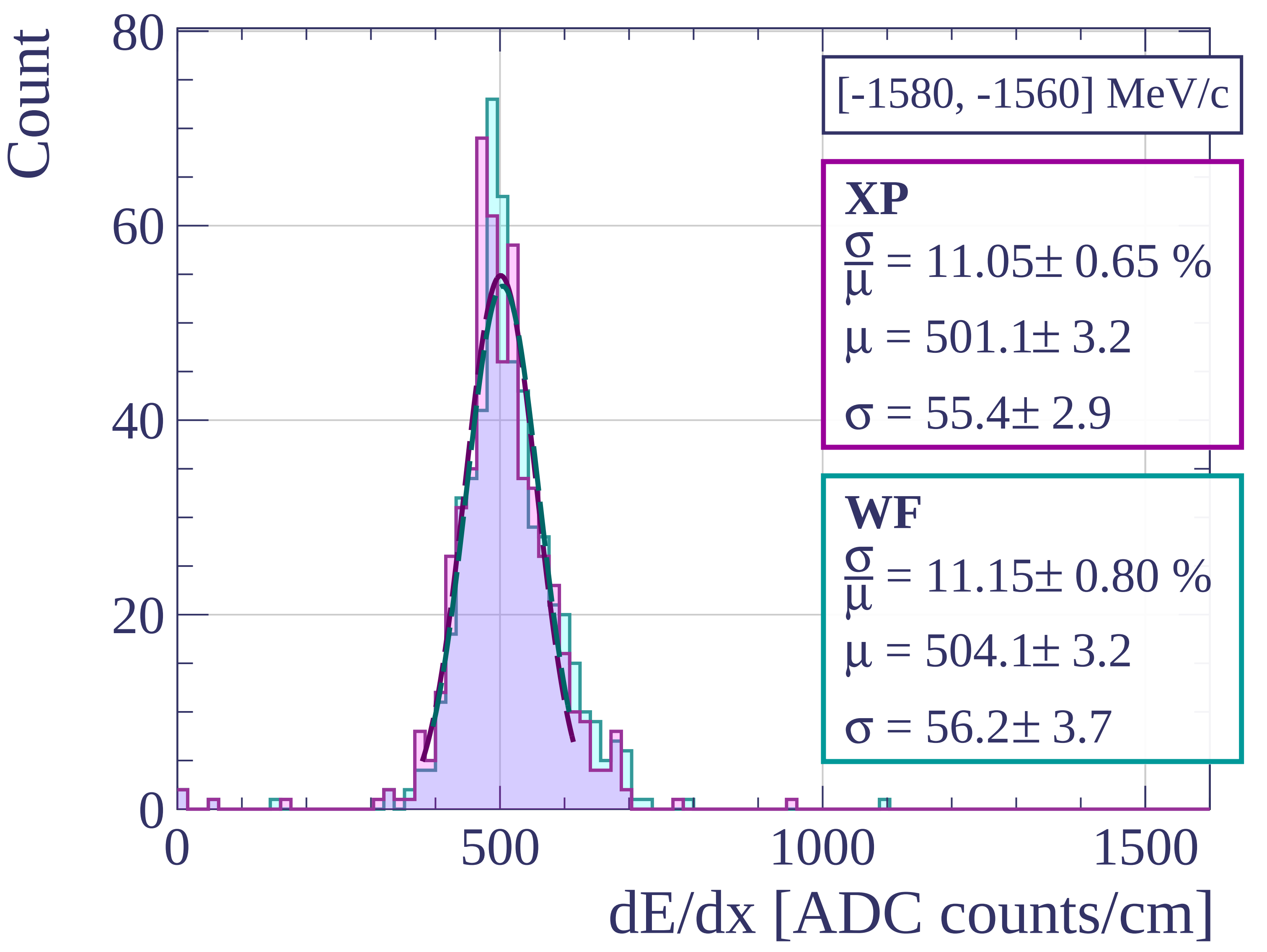
Count
dE/dx [ADC counts/cm]
[-1680, -1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.52± 0.70 %
μ = 506.4± 3.3
σ = 58.3± 3.2
WF
$\frac{\sigma}{\mu}$ = 11.07± 0.64 %
μ = 513.0± 3.3
σ = 56.8± 2.9
0
20
40
60
0
500
1000
1500


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1660, -1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.19± 0.77 %
μ = 507.8± 3.5
σ = 56.8± 3.5
WF
$\frac{\sigma}{\mu}$ = 10.69± 0.58 %
μ = 514.5± 3.2
σ = 55.0± 2.6

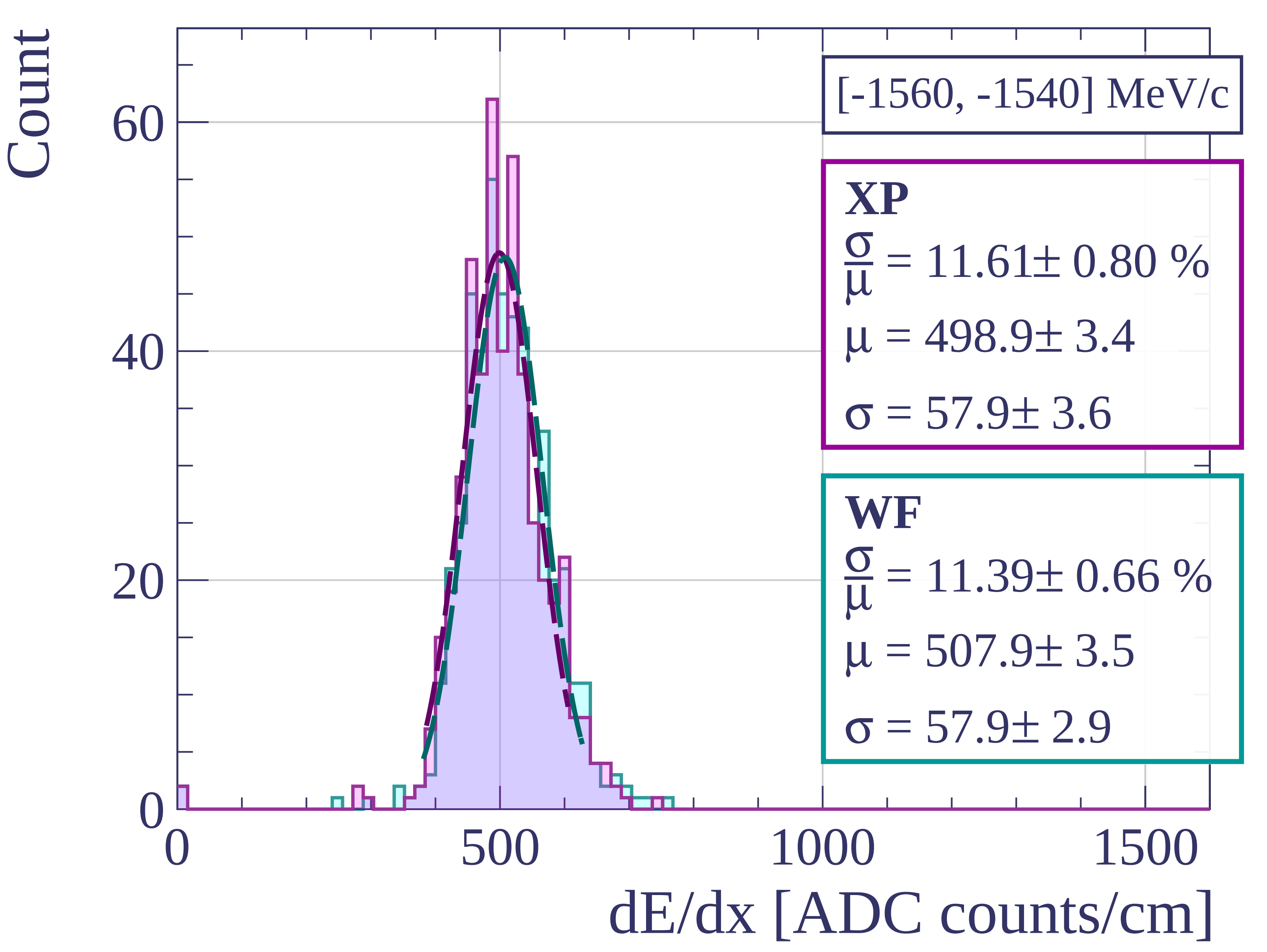

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1640, -1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.76± 0.68 %
μ = 501.9± 3.0
σ = 54.0± 3.1
WF
$\frac{\sigma}{\mu}$ = 11.34± 0.68 %
μ = 510.9± 3.2
σ = 57.9± 3.1


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1620, -1600] MeV/c
XP
σ/μ = 11.28± 0.63 %
μ = 507.9± 3.1
σ = 57.3± 2.9
WF
σ/μ = 11.71± 0.71 %
μ = 514.3± 3.4
σ = 60.3± 3.3

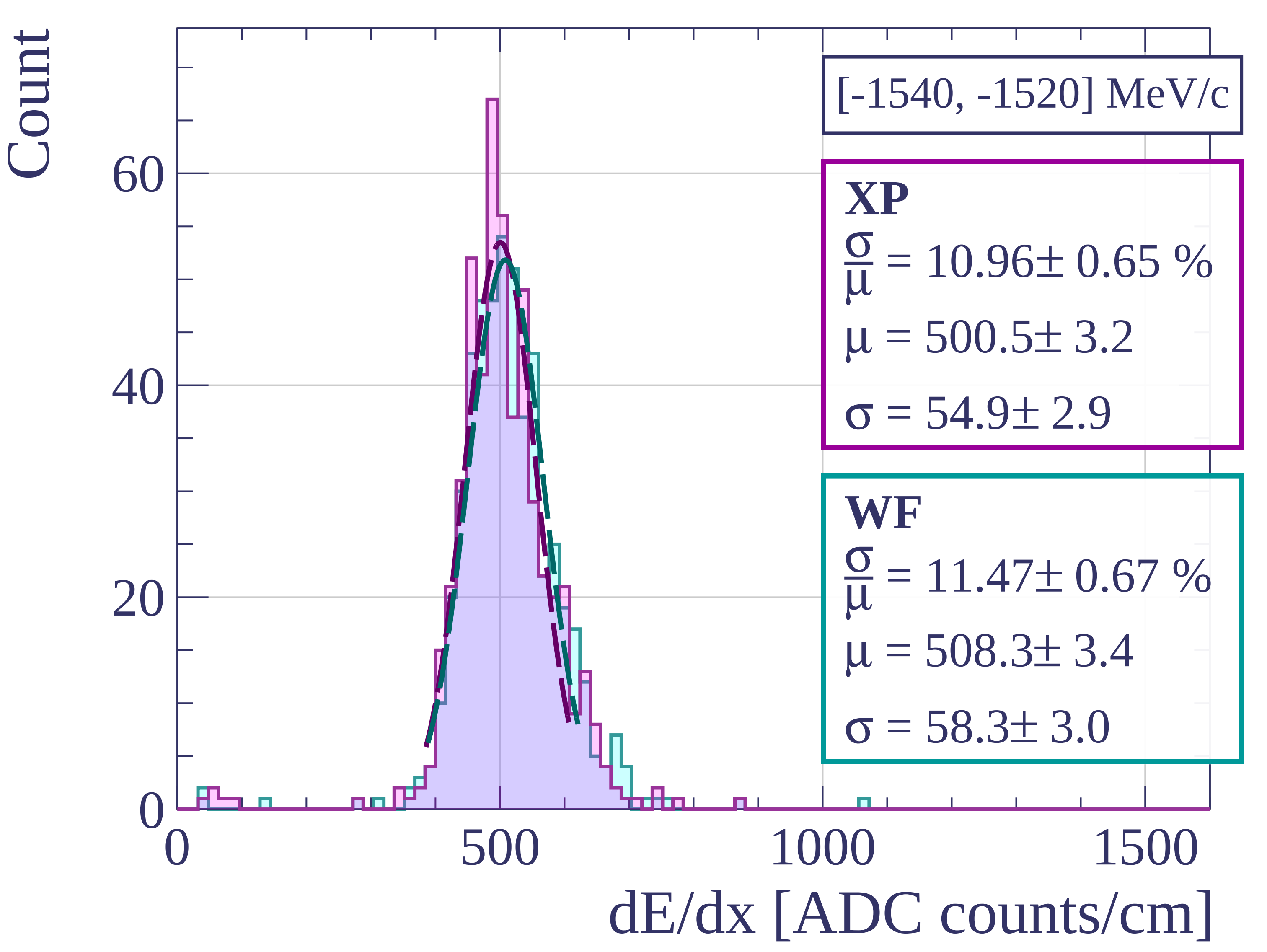

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1600, -1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.37± 0.60 %
μ = 504.6± 3.2
σ = 52.3± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.52± 0.52 %
μ = 508.0± 2.9
σ = 53.5± 2.4

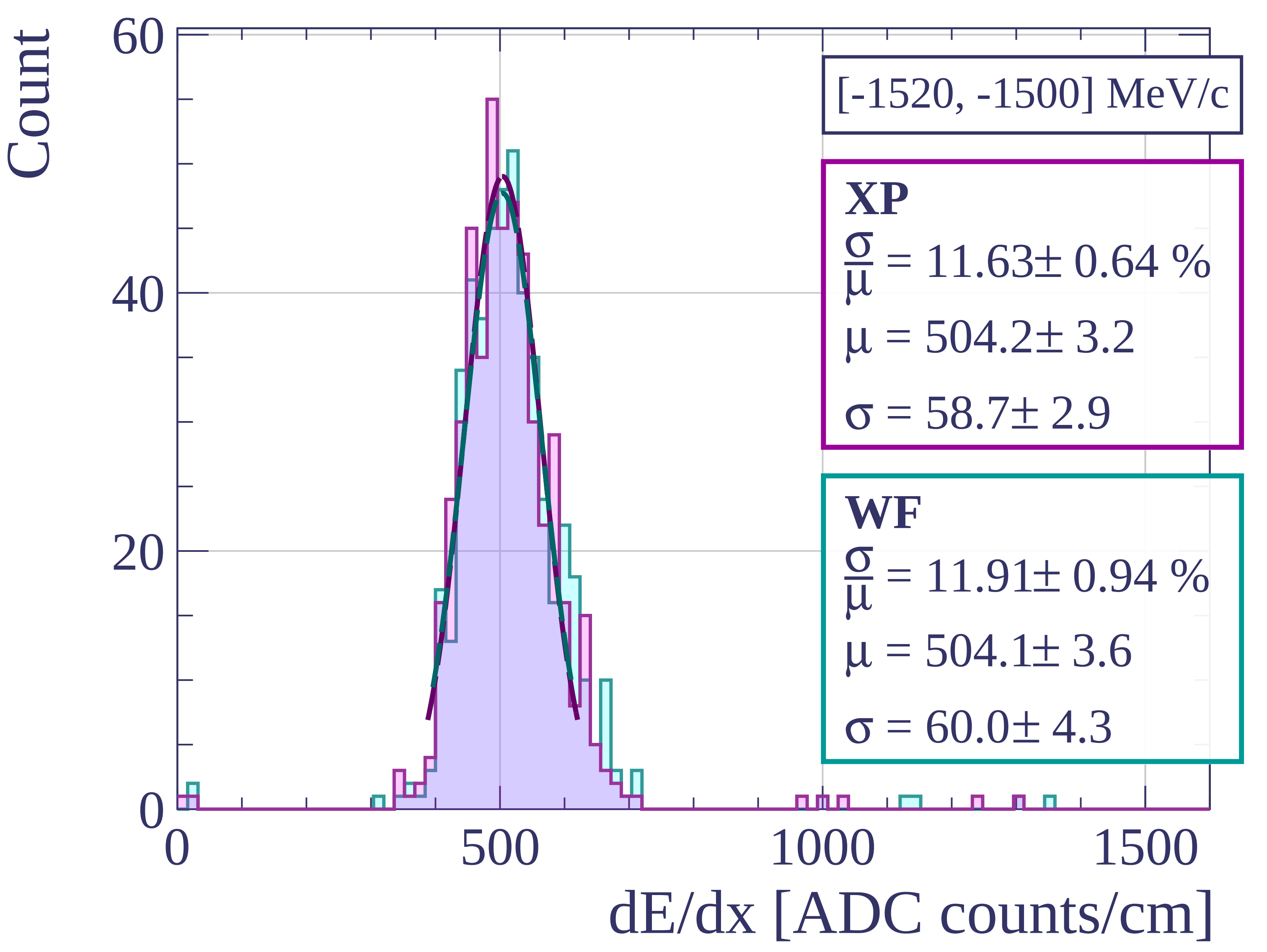


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-1580, -1560] MeV/c
XP
σ/μ = 11.05± 0.65 %
μ = 501.1± 3.2
σ = 55.4± 2.9
WF
σ/μ = 11.15± 0.80 %
μ = 504.1± 3.2
σ = 56.2± 3.7

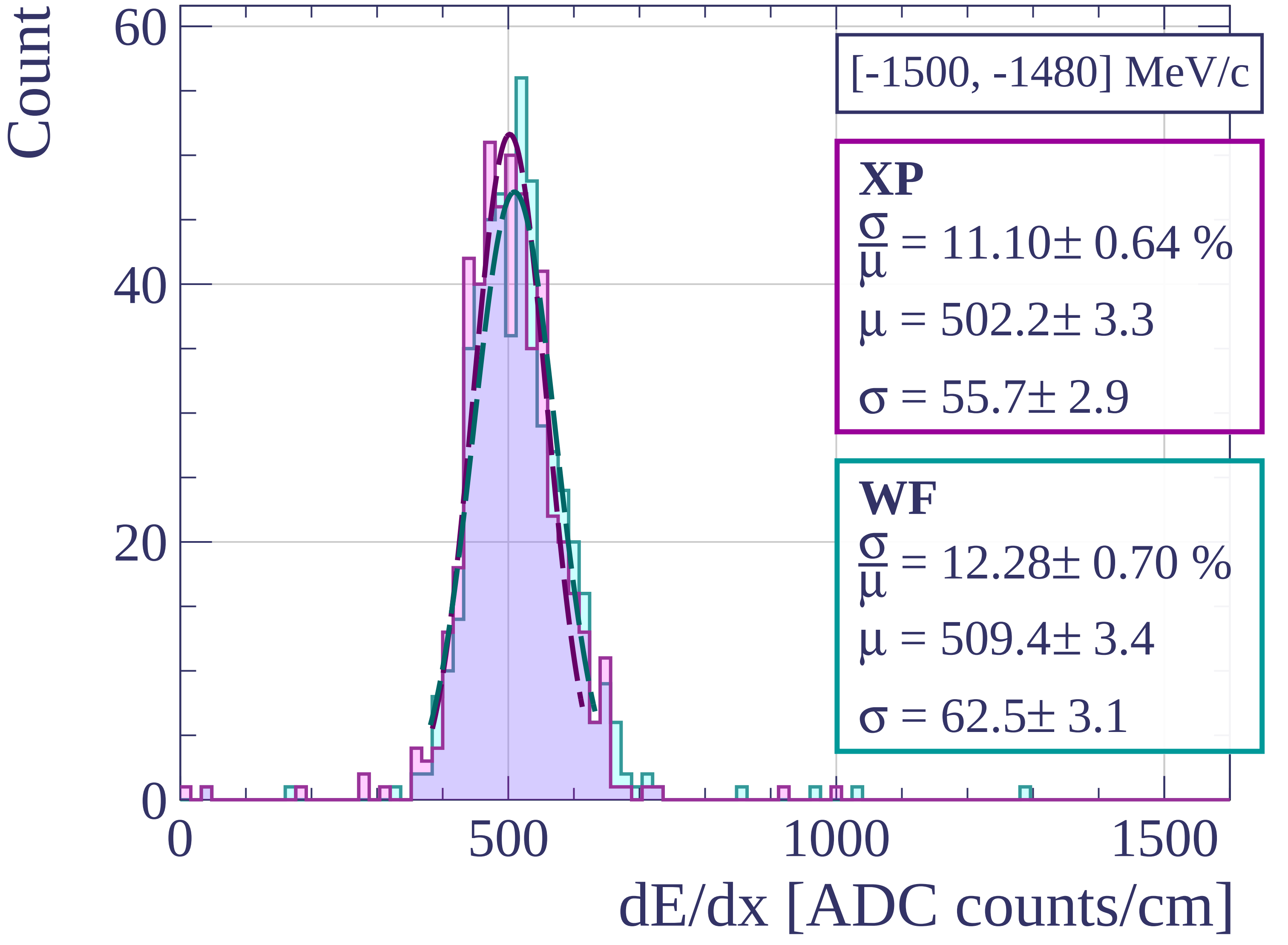


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1560, -1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.61± 0.80 %
μ = 498.9± 3.4
σ = 57.9± 3.6
WF
$\frac{\sigma}{\mu}$ = 11.39± 0.66 %
μ = 507.9± 3.5
σ = 57.9± 2.9

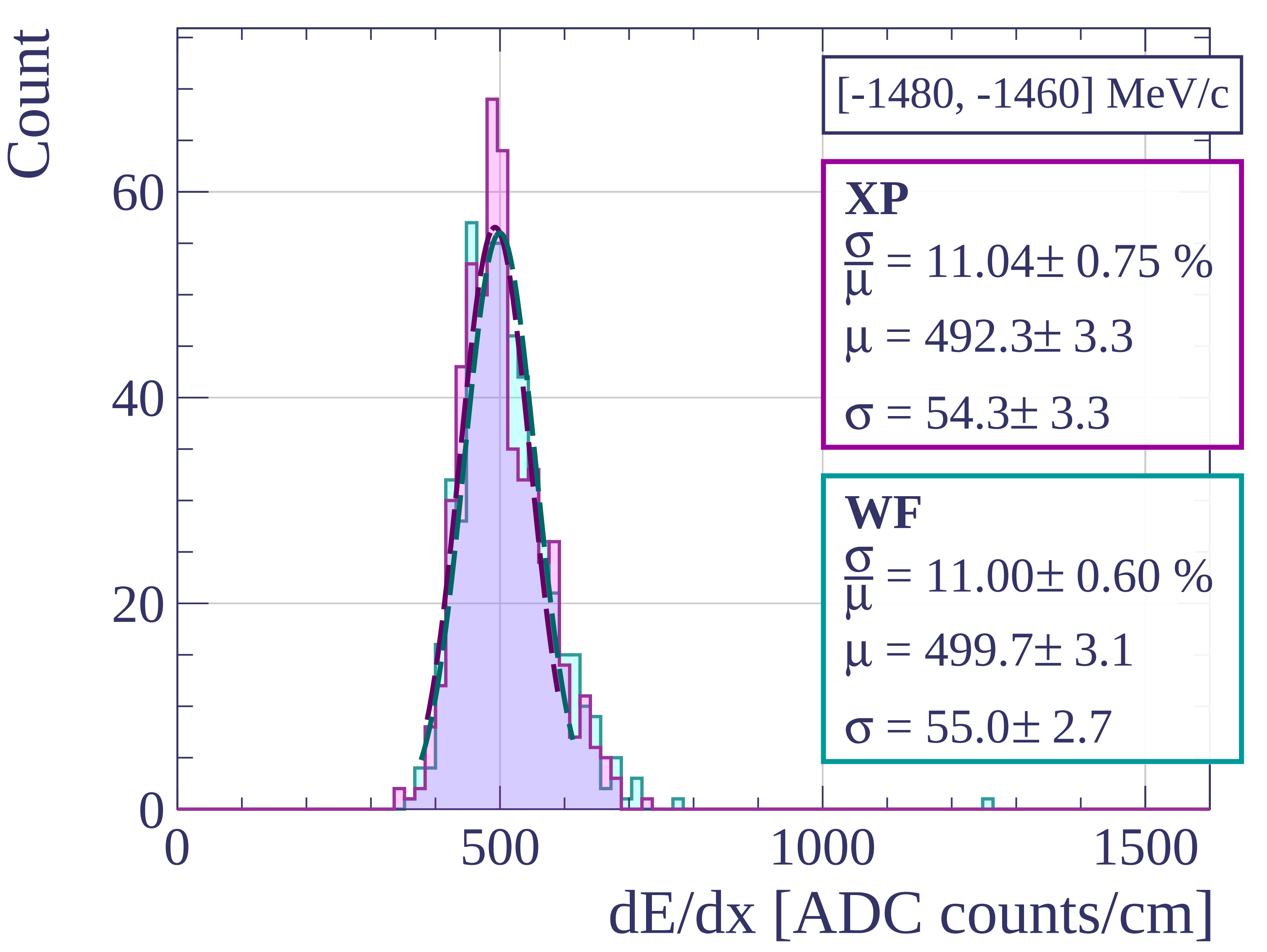

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.96± 0.65 %
μ = 500.5± 3.2
σ = 54.9± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.47± 0.67 %
μ = 508.3± 3.4
σ = 58.3± 3.0



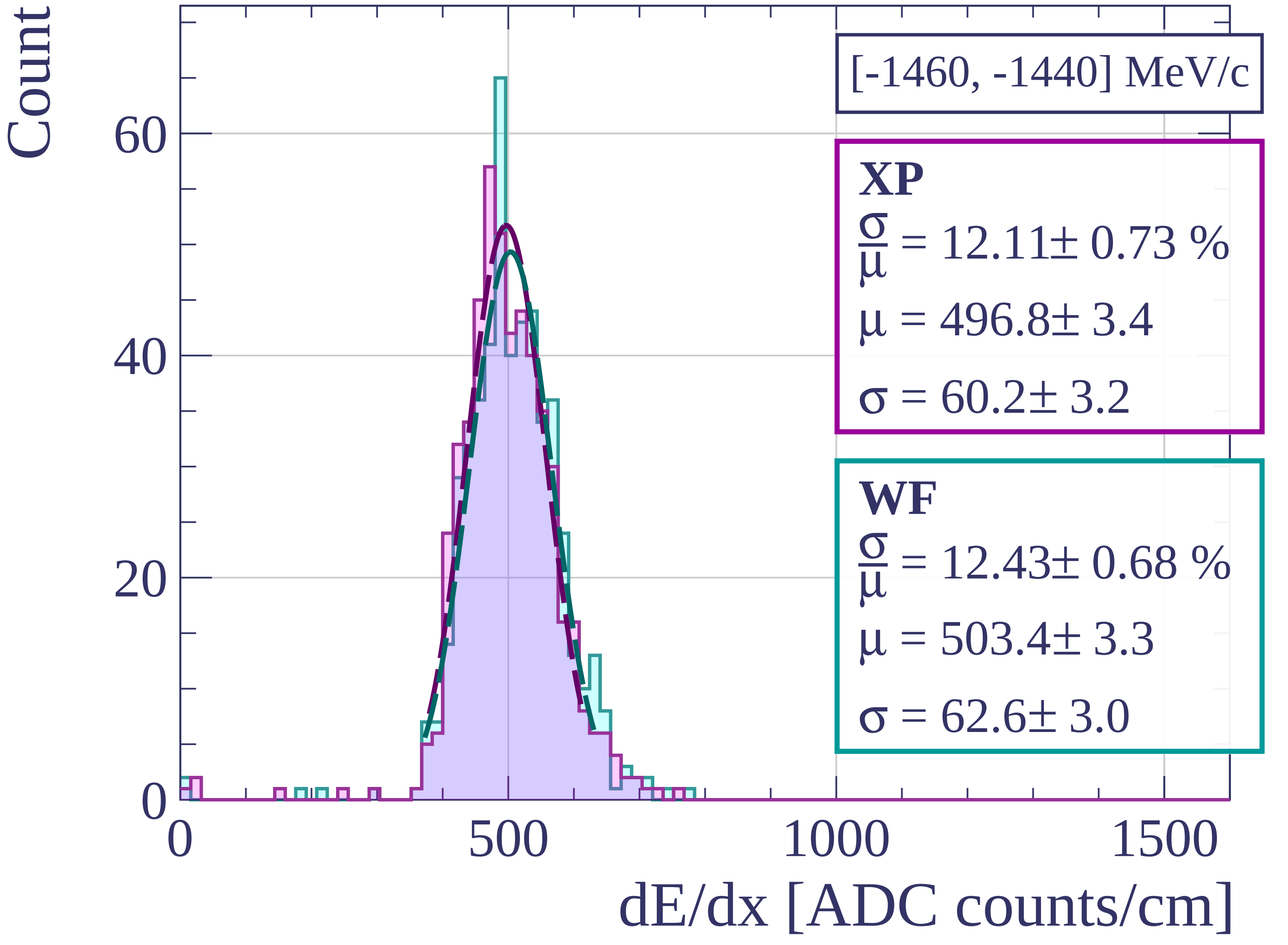
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.63± 0.64 %
μ = 504.2± 3.2
σ = 58.7± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.91± 0.94 %
μ = 504.1± 3.6
σ = 60.0± 4.3



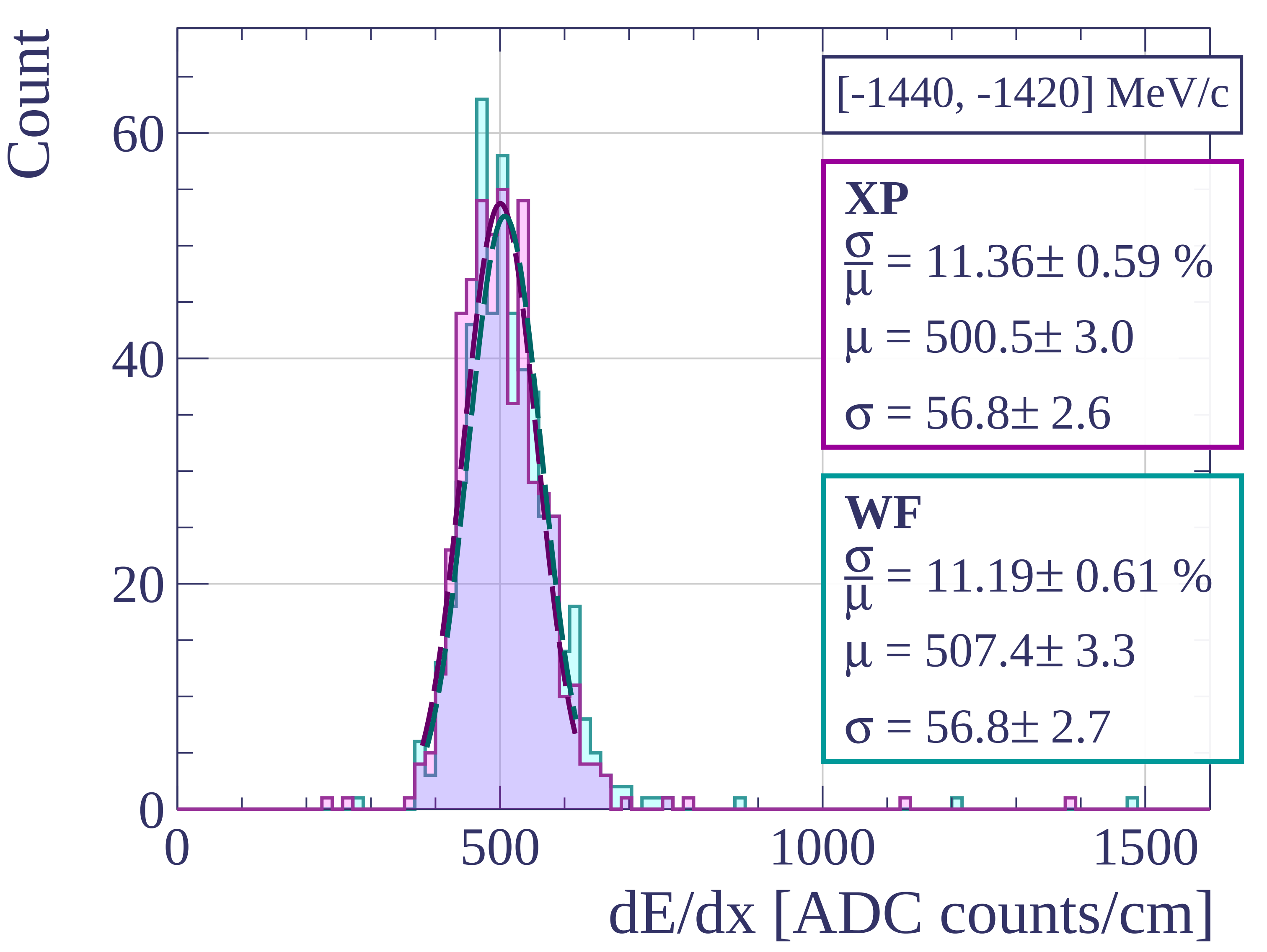
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
σ/μ = 11.10± 0.64 %
μ = 502.2± 3.3
σ = 55.7± 2.9
WF
σ/μ = 12.28± 0.70 %
μ = 509.4± 3.4
σ = 62.5± 3.1



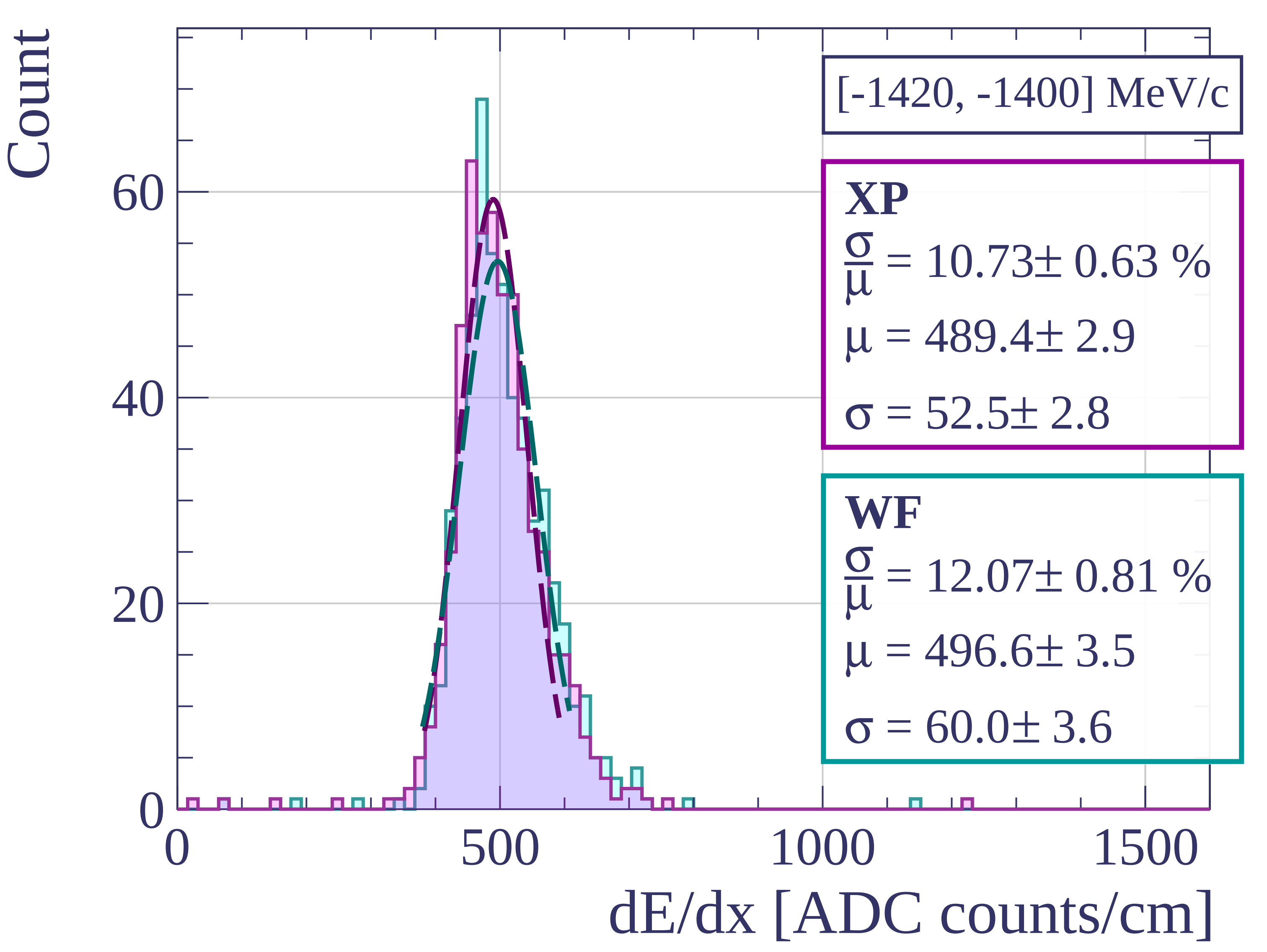
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1480, -1460] MeV/c
XP
σ/μ = 11.04± 0.75 %
μ = 492.3± 3.3
σ = 54.3± 3.3
WF
σ/μ = 11.00± 0.60 %
μ = 499.7± 3.1
σ = 55.0± 2.7



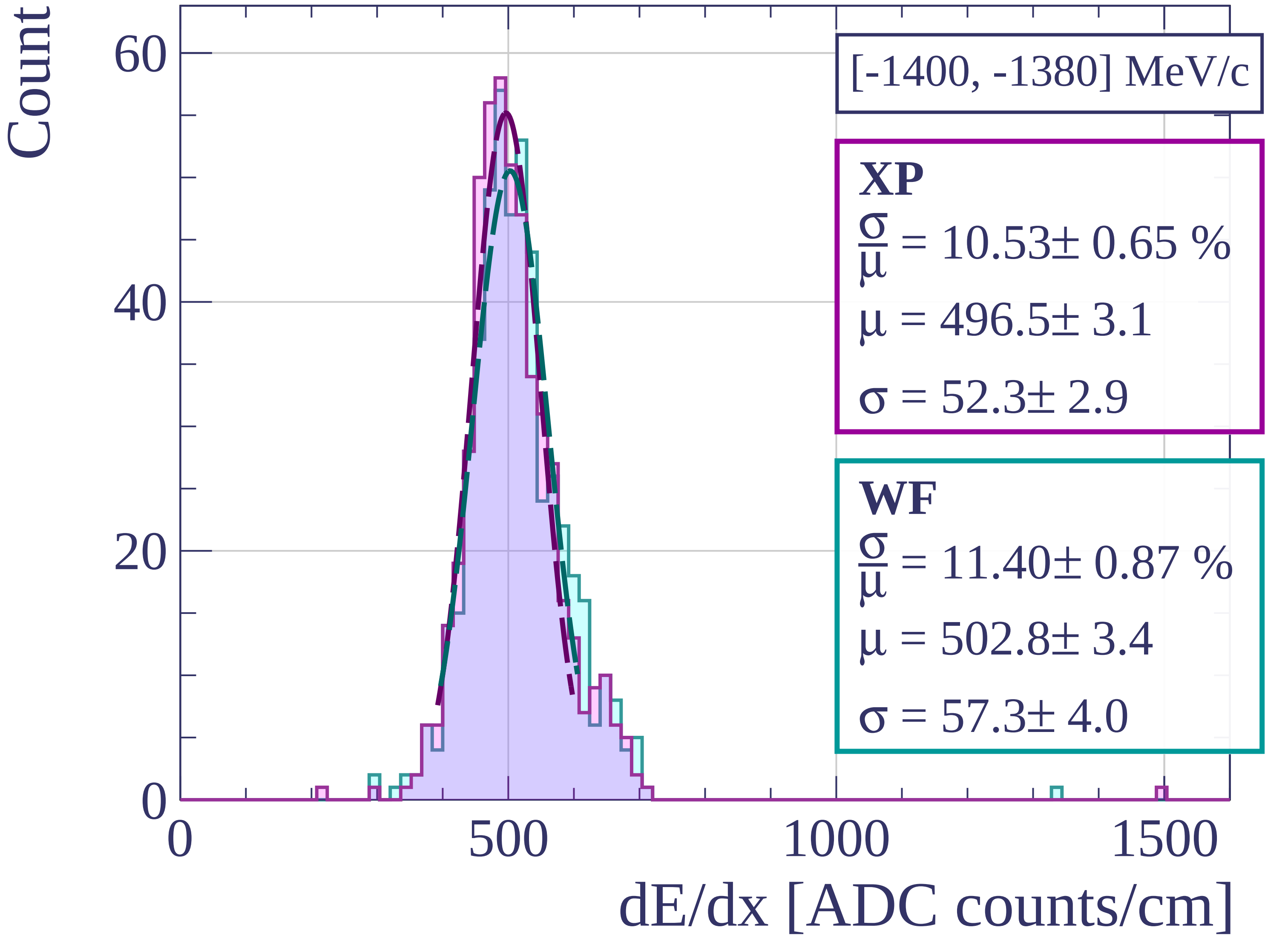
Count
dE/dx [ADC counts/cm]
[-1460, -1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.11± 0.73 %
μ = 496.8± 3.4
σ = 60.2± 3.2
WF
$\frac{\sigma}{\mu}$ = 12.43± 0.68 %
μ = 503.4± 3.3
σ = 62.6± 3.0
0
20
40
60
0
500
1000
1500



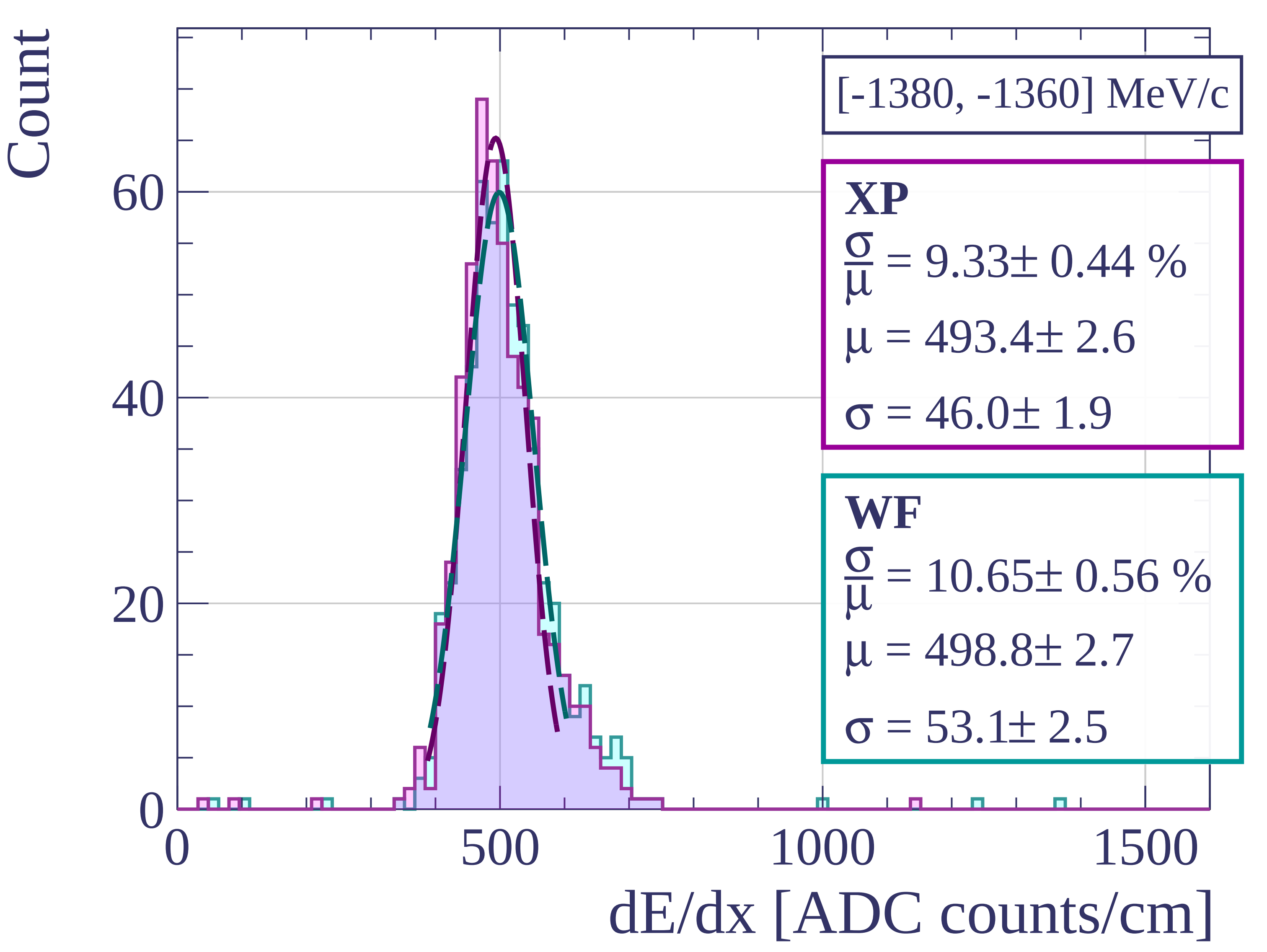
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1440, -1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.36± 0.59 %
μ = 500.5± 3.0
σ = 56.8± 2.6
WF
$\frac{\sigma}{\mu}$ = 11.19± 0.61 %
μ = 507.4± 3.3
σ = 56.8± 2.7



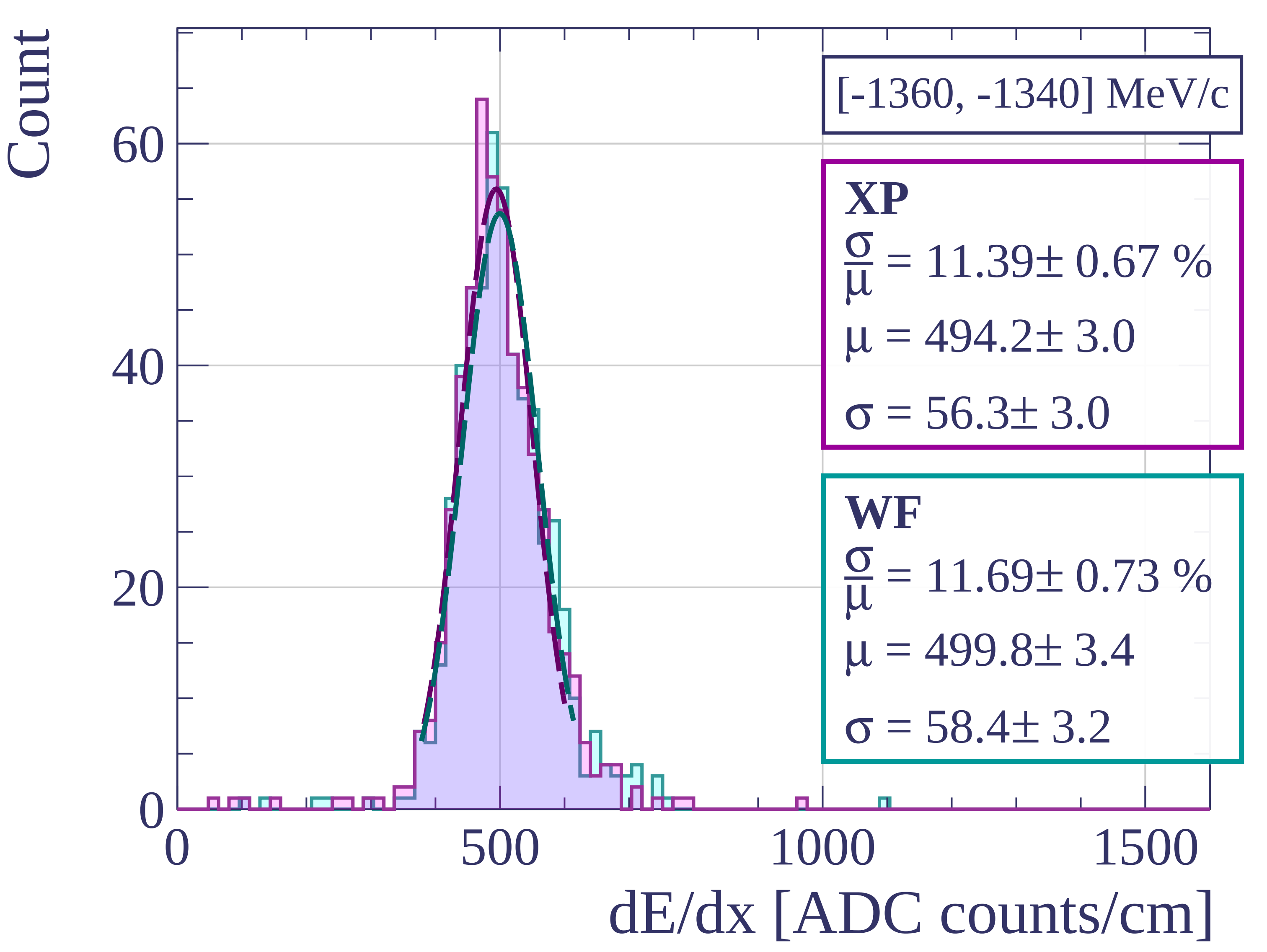
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1420, -1400] MeV/c
XP
σ/μ = 10.73± 0.63 %
μ = 489.4± 2.9
σ = 52.5± 2.8
WF
σ/μ = 12.07± 0.81 %
μ = 496.6± 3.5
σ = 60.0± 3.6



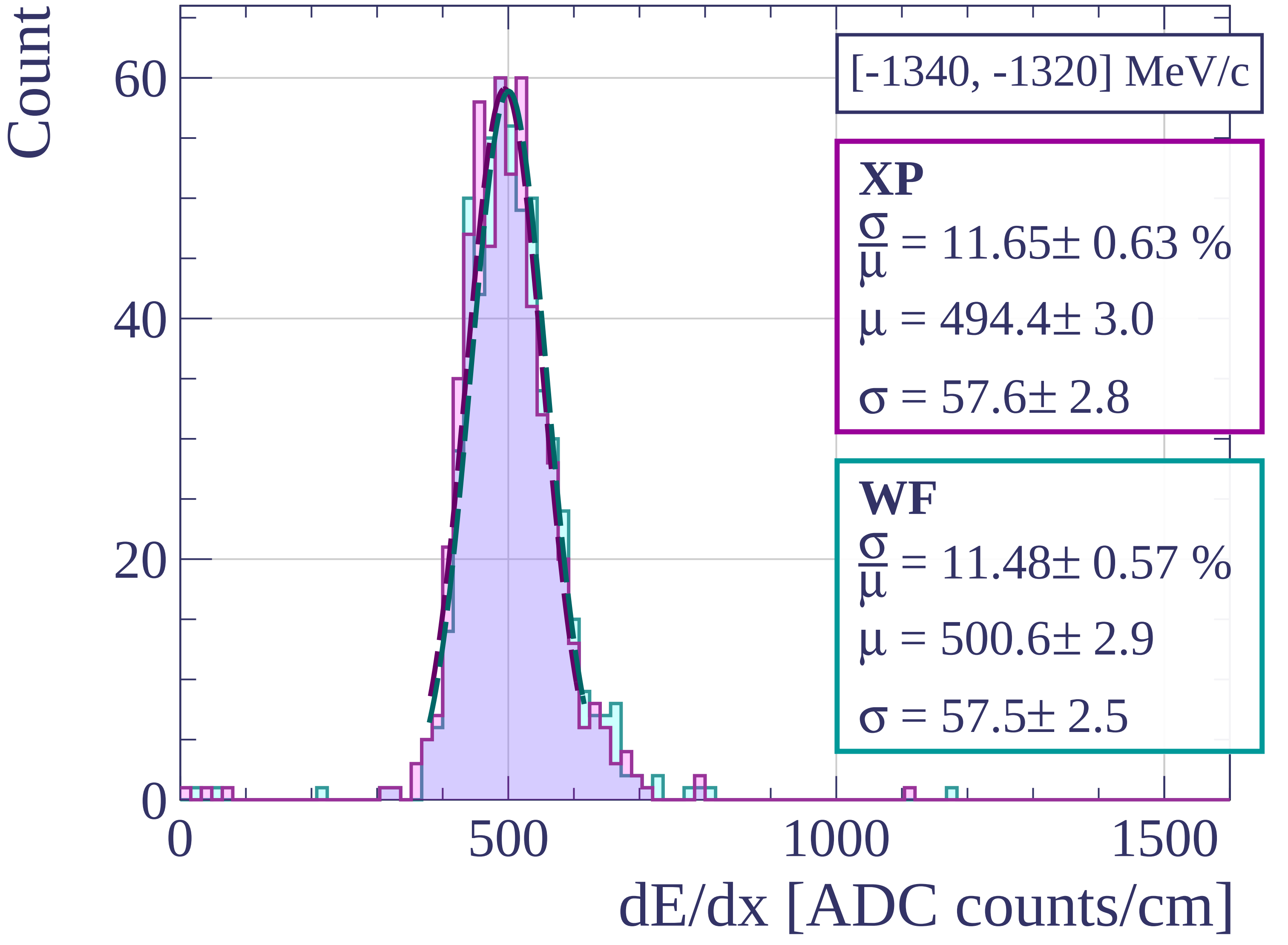
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.53± 0.65 %
μ = 496.5± 3.1
σ = 52.3± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.40± 0.87 %
μ = 502.8± 3.4
σ = 57.3± 4.0



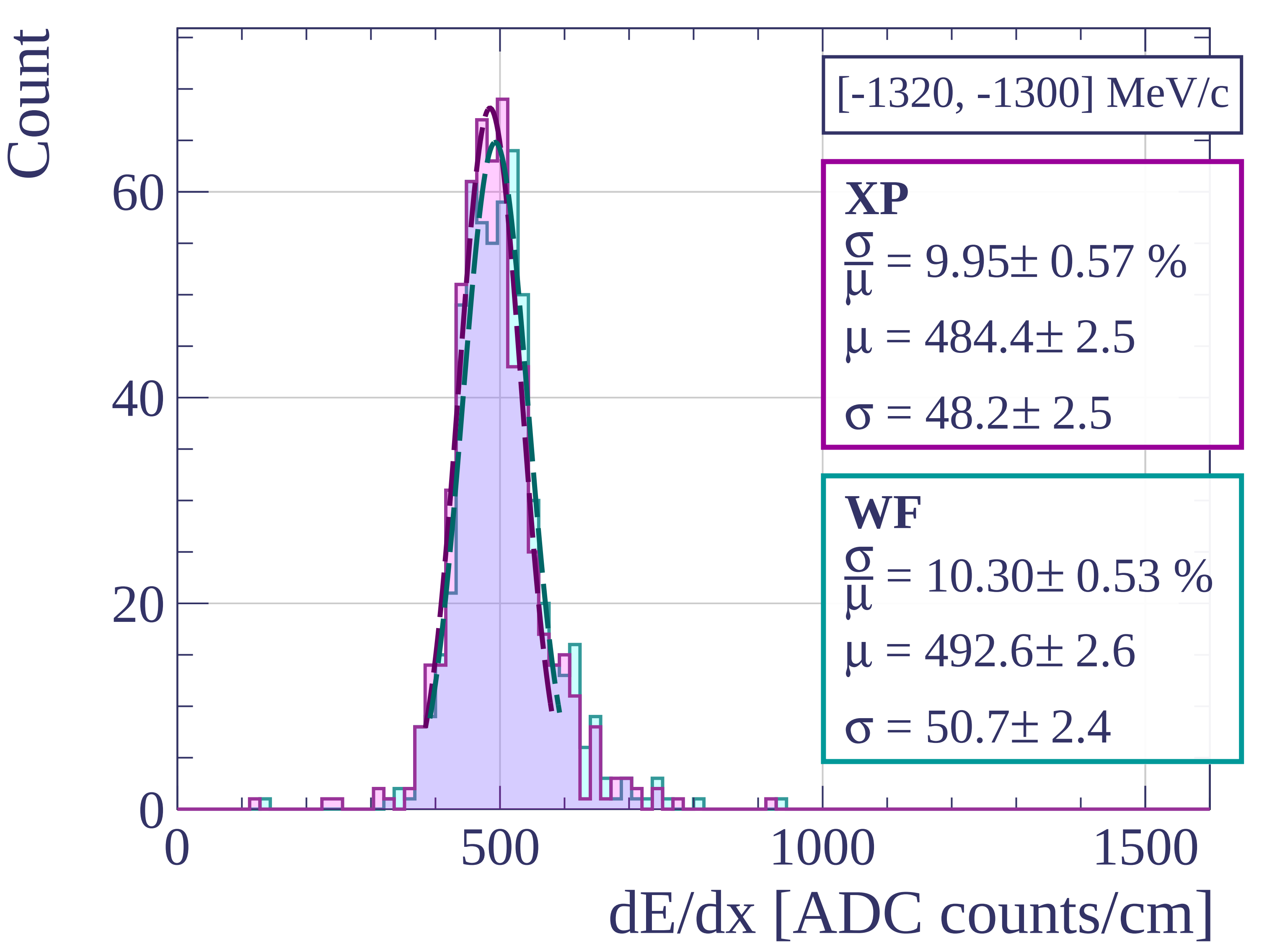
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1380, -1360] MeV/c
XP
σ/μ = 9.33± 0.44 %
μ = 493.4± 2.6
σ = 46.0± 1.9
WF
σ/μ = 10.65± 0.56 %
μ = 498.8± 2.7
σ = 53.1± 2.5

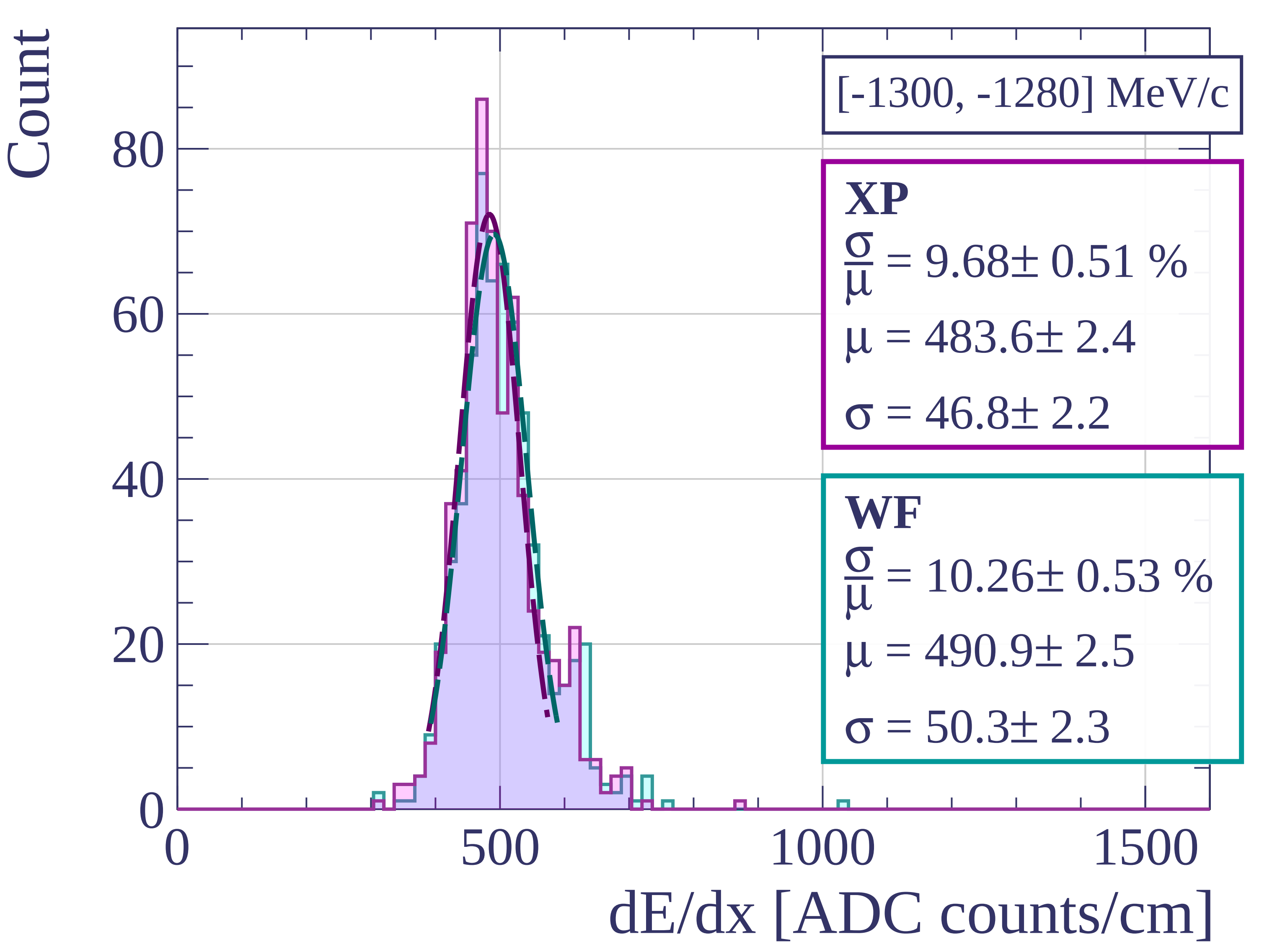


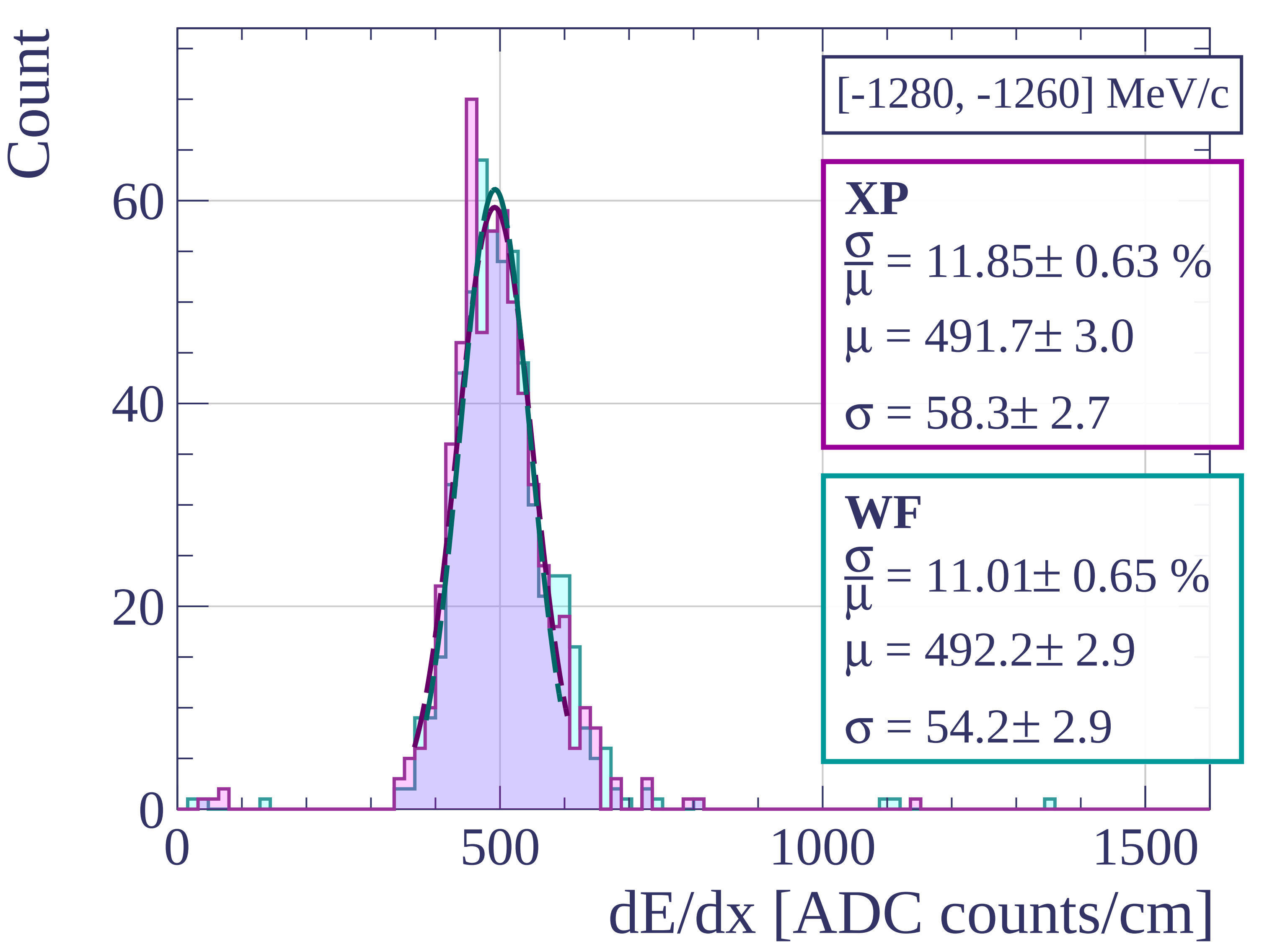
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1360, -1340] MeV/c
XP
σ/μ = 11.39± 0.67 %
μ = 494.2± 3.0
σ = 56.3± 3.0
WF
σ/μ = 11.69± 0.73 %
μ = 499.8± 3.4
σ = 58.4± 3.2



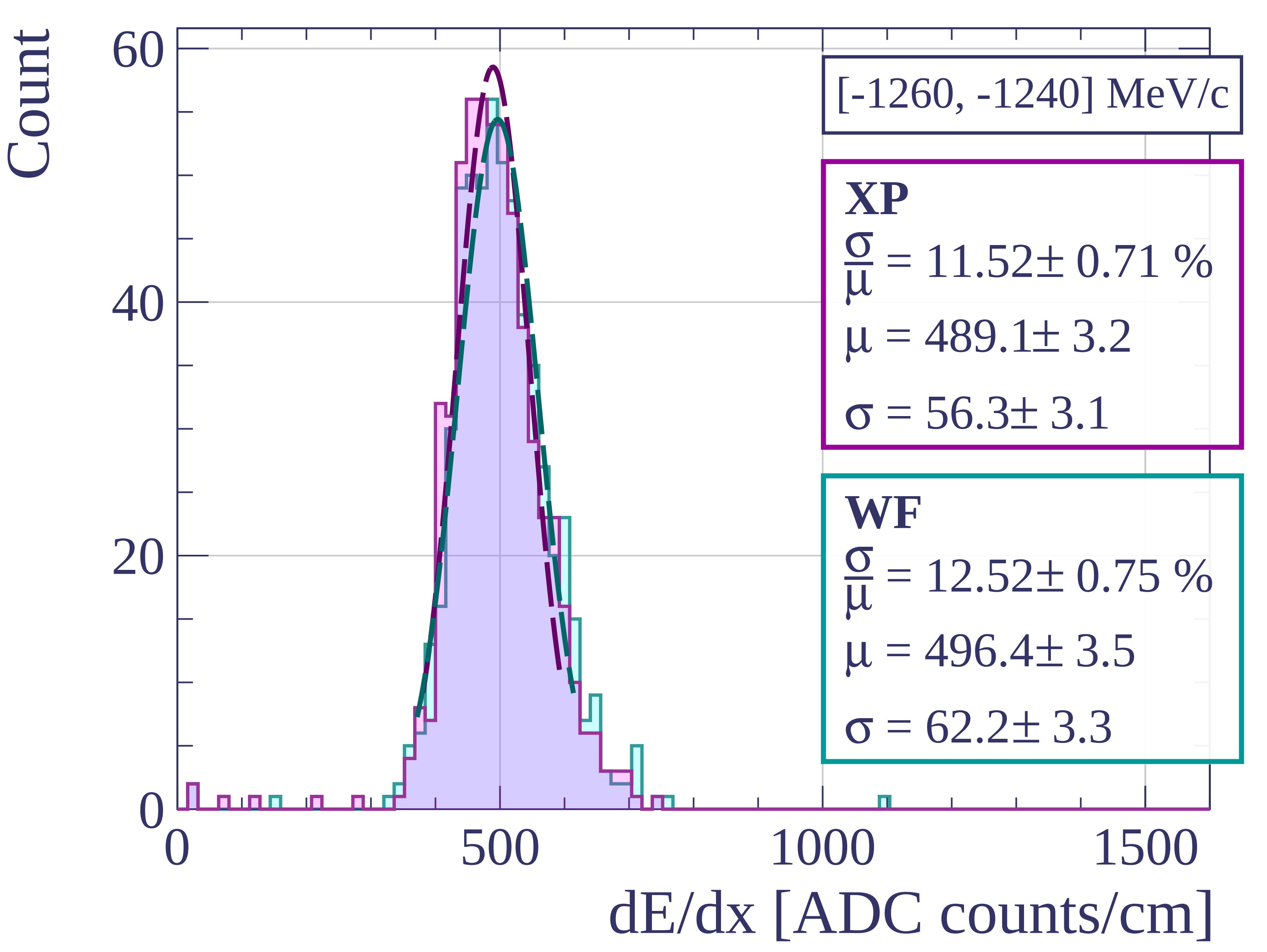
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.65± 0.63 %
μ = 494.4± 3.0
σ = 57.6± 2.8
WF
$\frac{\sigma}{\mu}$ = 11.48± 0.57 %
μ = 500.6± 2.9
σ = 57.5± 2.5


Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.95± 0.57 %
μ = 484.4± 2.5
σ = 48.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 10.30± 0.53 %
μ = 492.6± 2.6
σ = 50.7± 2.4

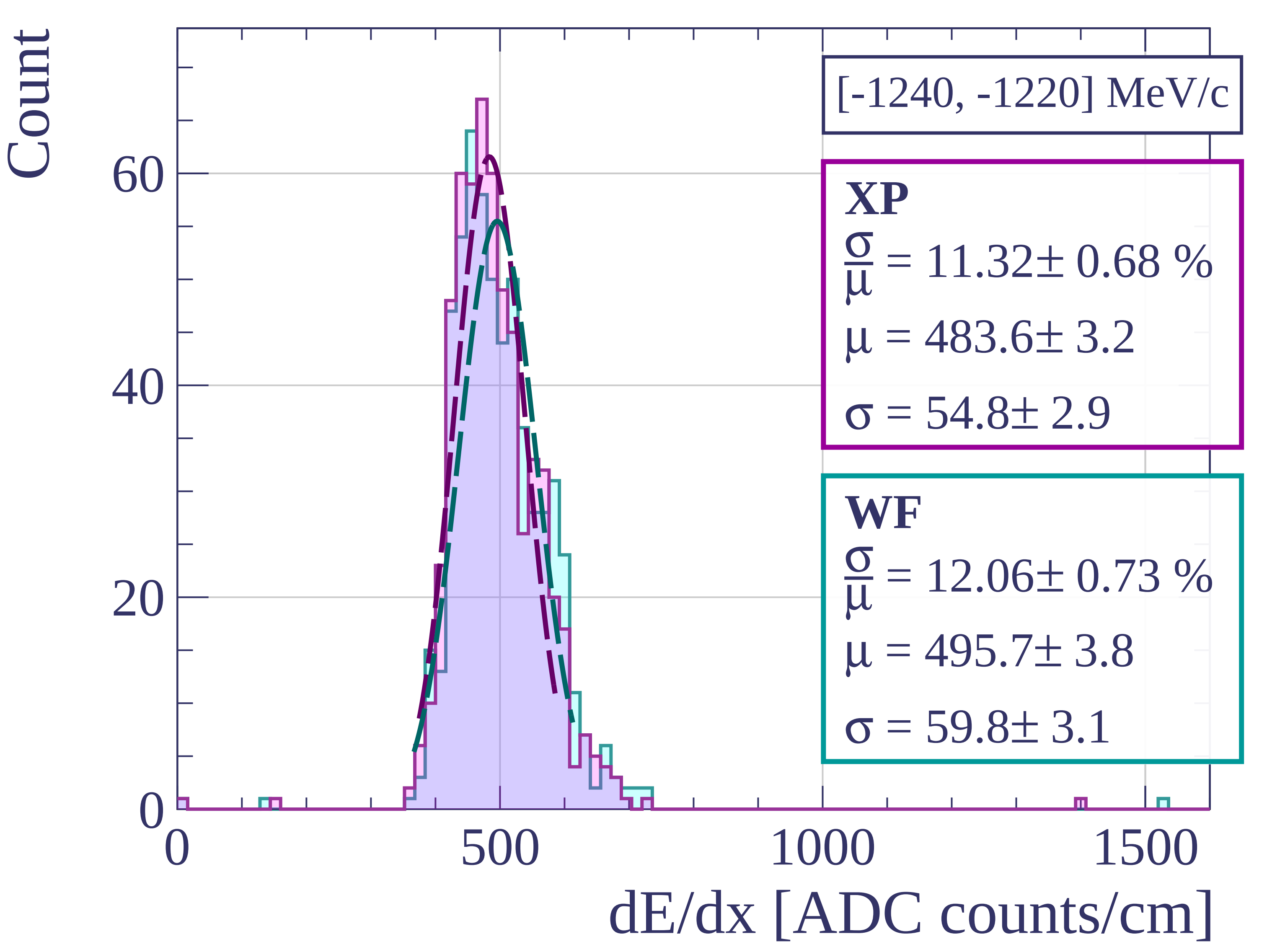

Count
[-1300, -1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.68± 0.51 %
μ = 483.6± 2.4
σ = 46.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.26± 0.53 %
μ = 490.9± 2.5
σ = 50.3± 2.3
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]



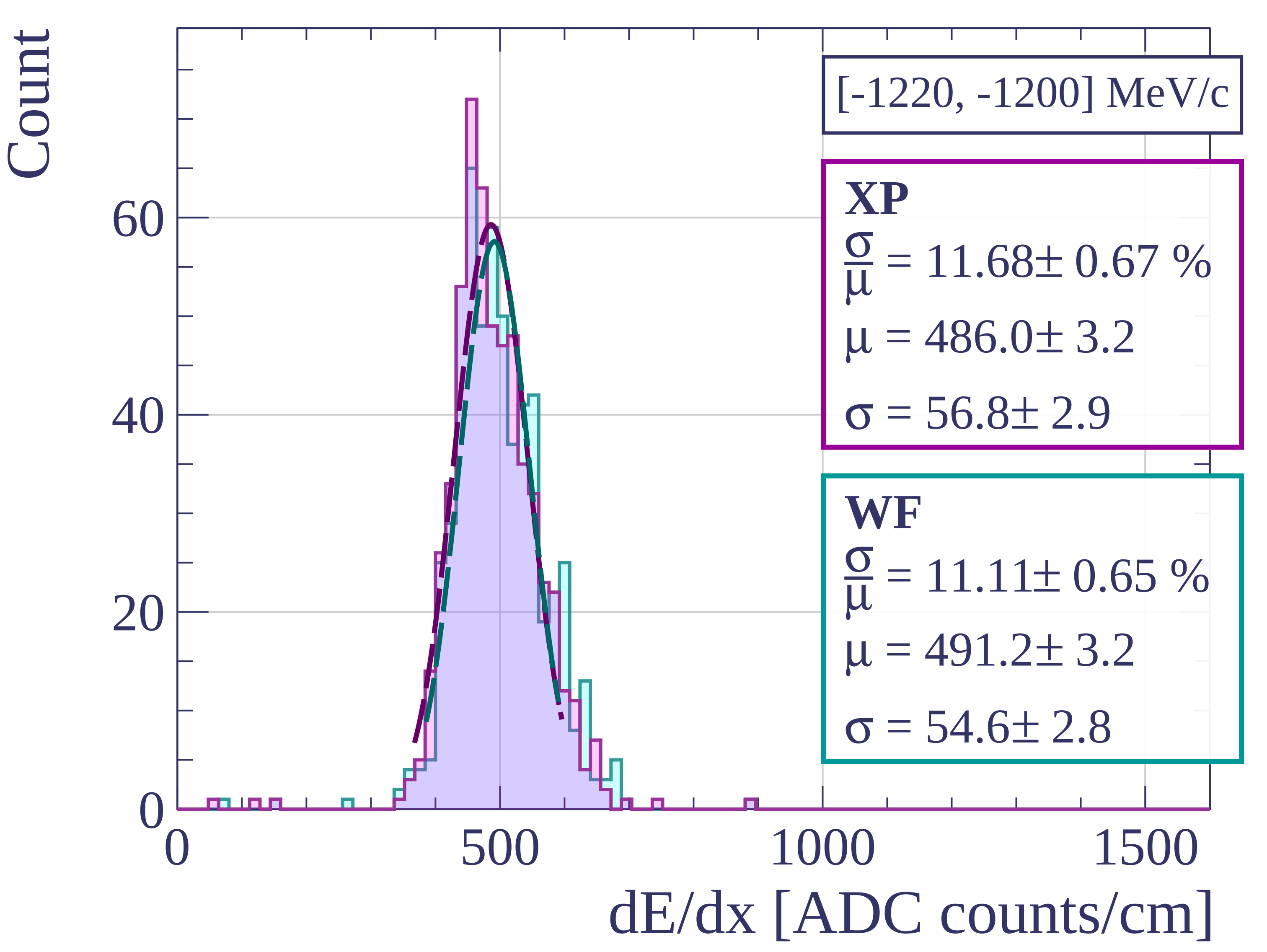
Count
dE/dx [ADC counts/cm]
[-1280, -1260] MeV/c
XP
σ/μ = 11.85± 0.63 %
μ = 491.7± 3.0
σ = 58.3± 2.7
WF
σ/μ = 11.01± 0.65 %
μ = 492.2± 2.9
σ = 54.2± 2.9
0
20
40
60
0
500
1000
1500



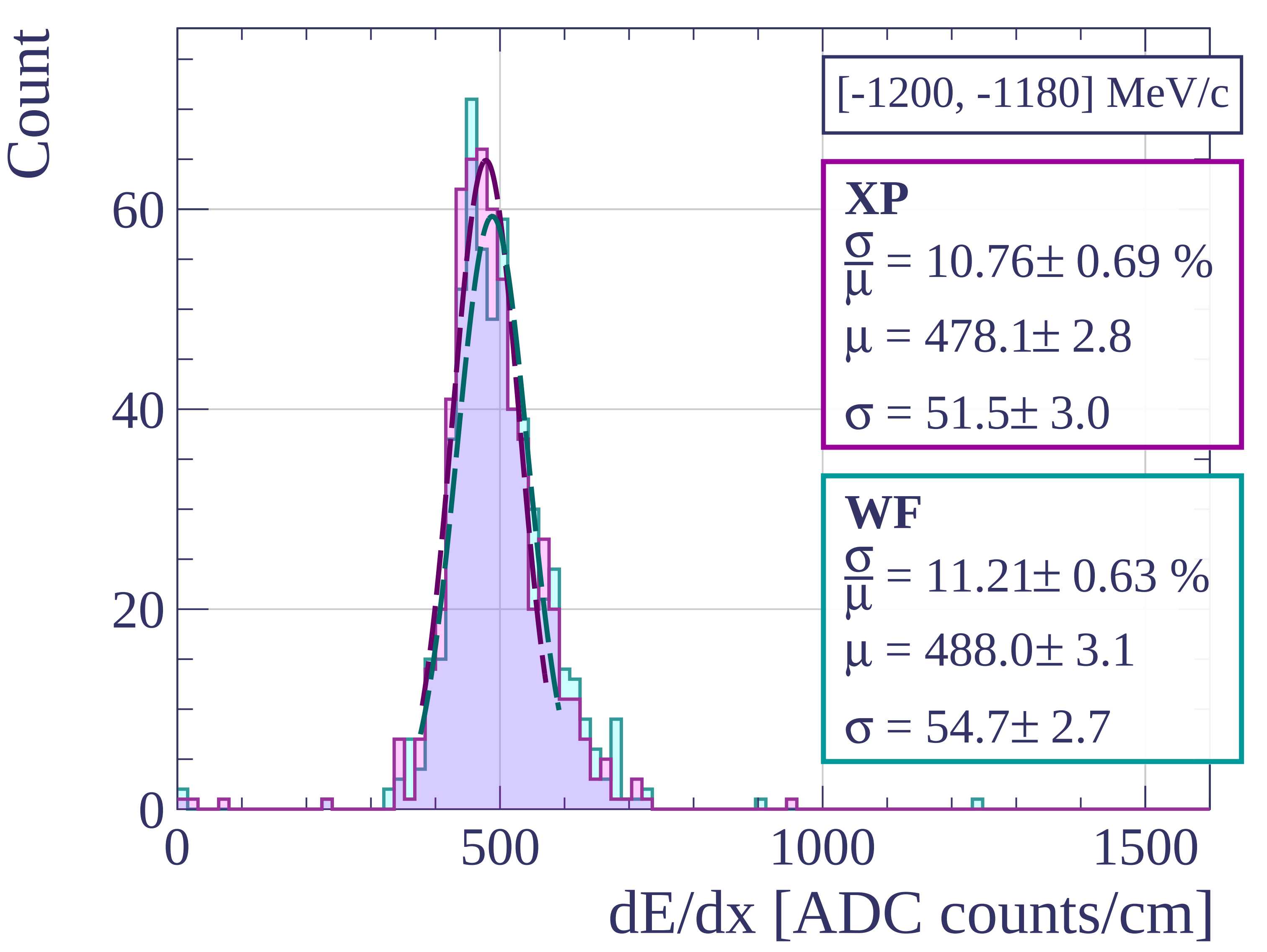
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.52± 0.71 %
μ = 489.1± 3.2
σ = 56.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.52± 0.75 %
μ = 496.4± 3.5
σ = 62.2± 3.3



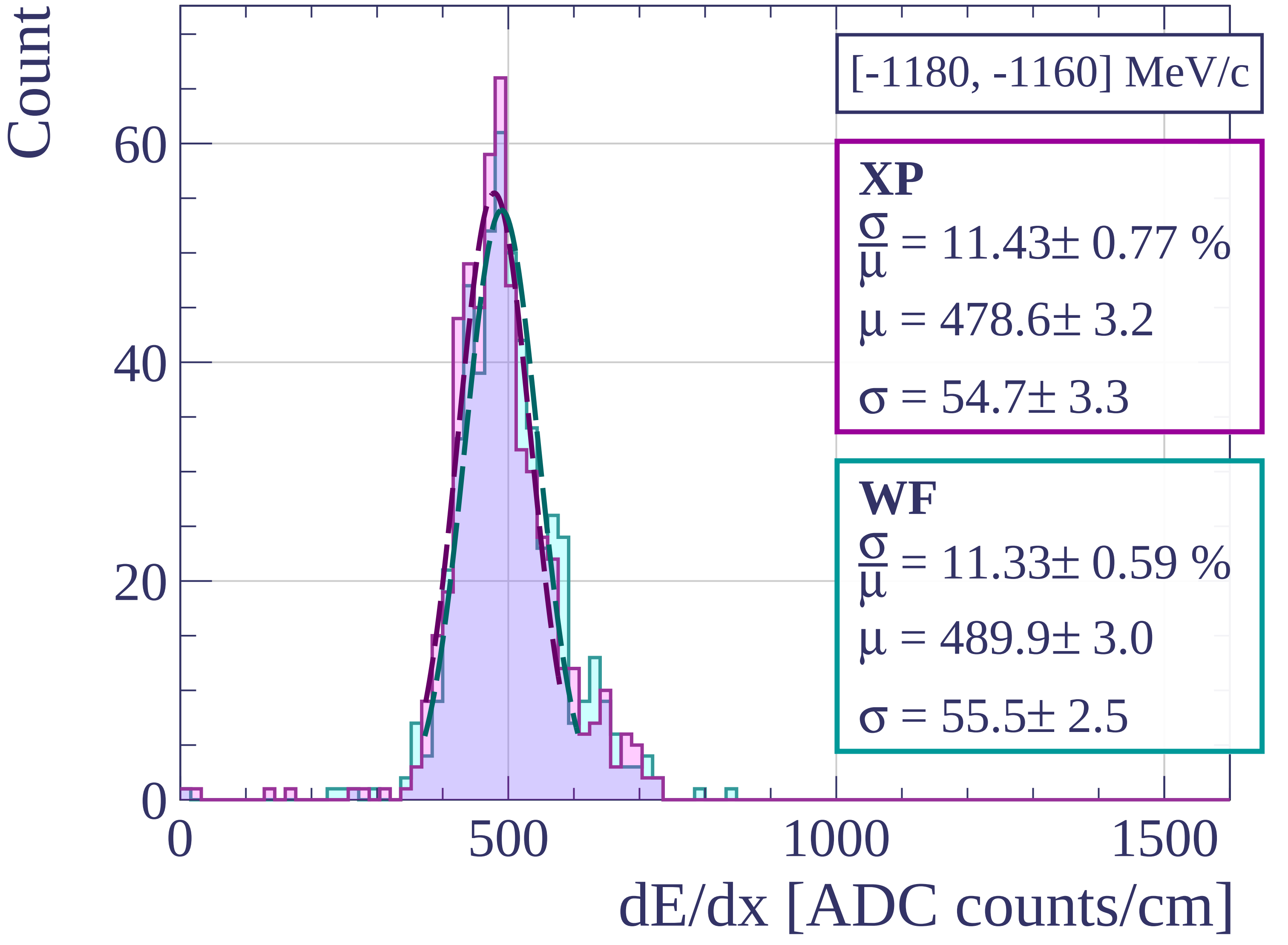
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.32± 0.68 %
μ = 483.6± 3.2
σ = 54.8± 2.9
WF
$\frac{\sigma}{\mu}$ = 12.06± 0.73 %
μ = 495.7± 3.8
σ = 59.8± 3.1



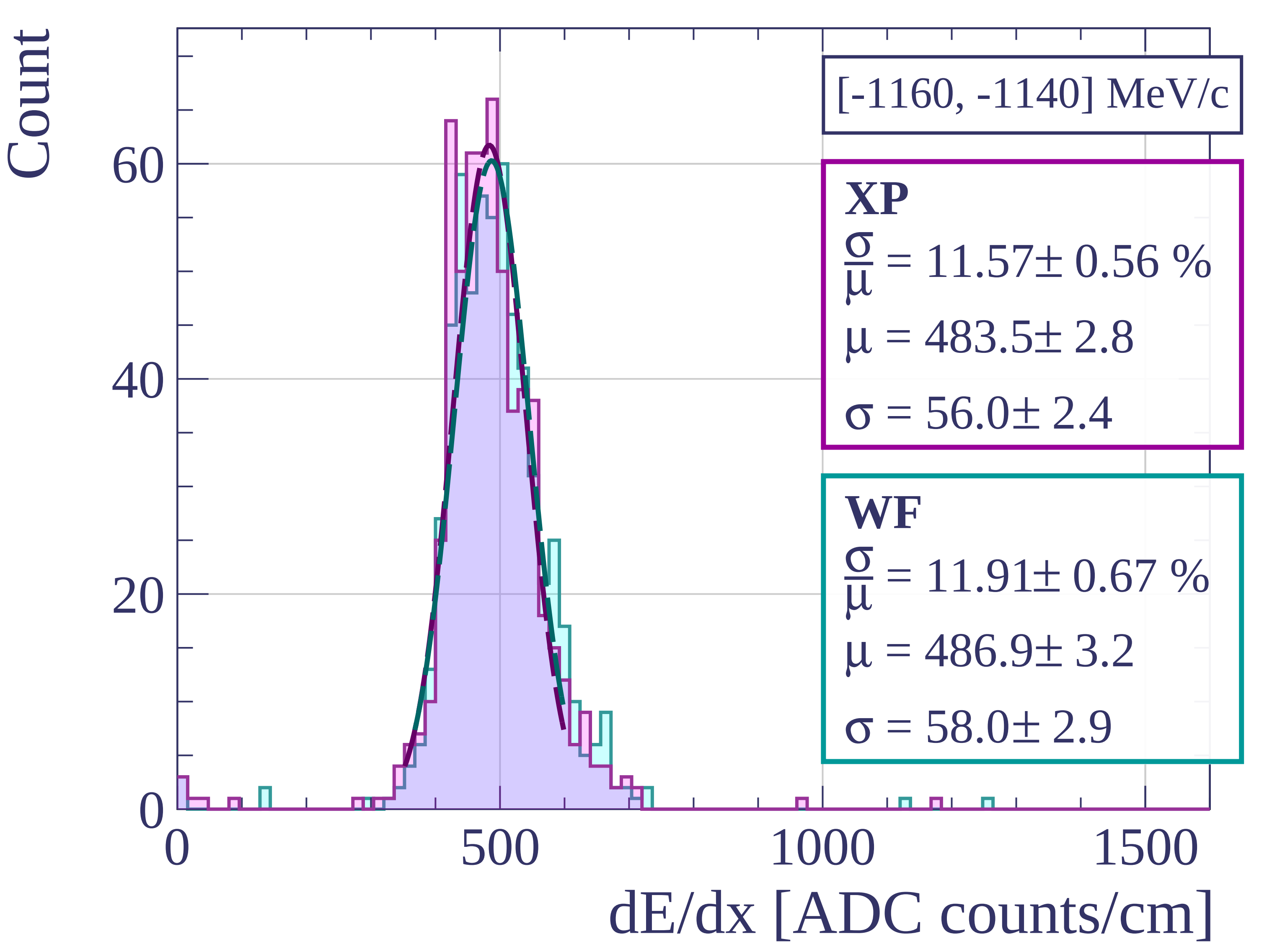
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1220, -1200] MeV/c
XP
σ/μ = 11.68± 0.67 %
μ = 486.0± 3.2
σ = 56.8± 2.9
WF
σ/μ = 11.11± 0.65 %
μ = 491.2± 3.2
σ = 54.6± 2.8



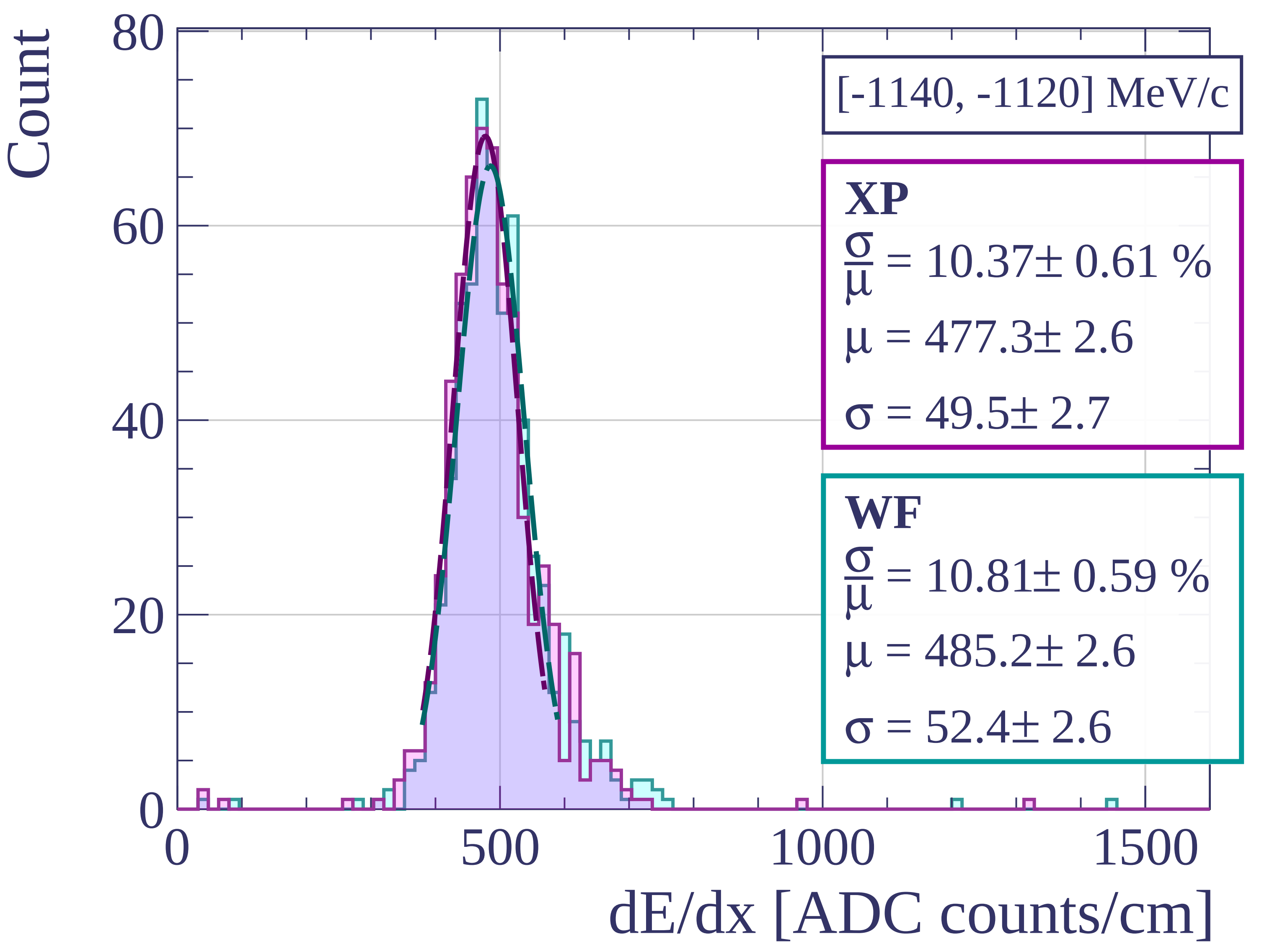
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.76± 0.69 %
μ = 478.1± 2.8
σ = 51.5± 3.0
WF
$\frac{\sigma}{\mu}$ = 11.21± 0.63 %
μ = 488.0± 3.1
σ = 54.7± 2.7



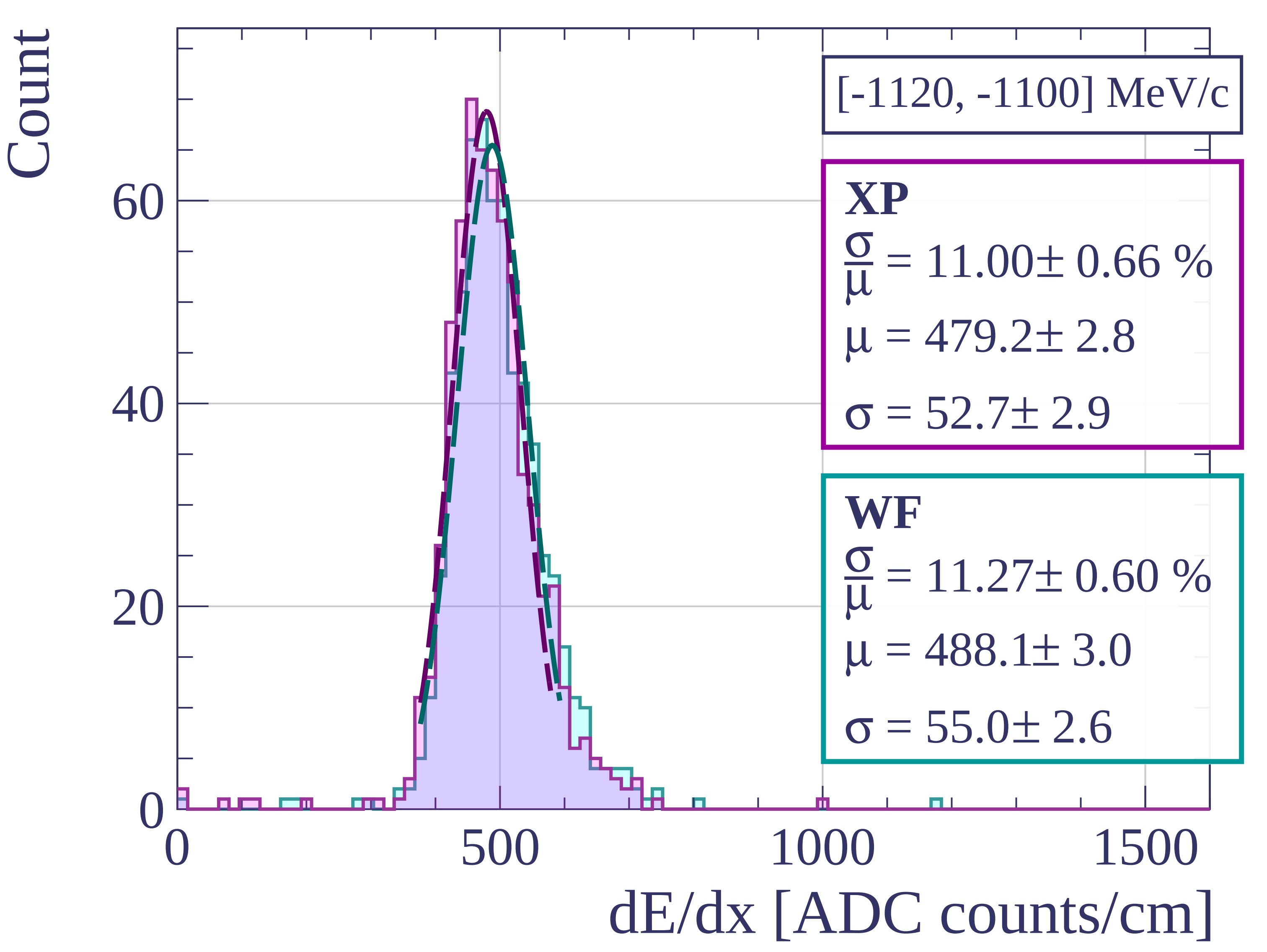
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1180, -1160] MeV/c
XP
σ/μ = 11.43± 0.77 %
μ = 478.6± 3.2
σ = 54.7± 3.3
WF
σ/μ = 11.33± 0.59 %
μ = 489.9± 3.0
σ = 55.5± 2.5



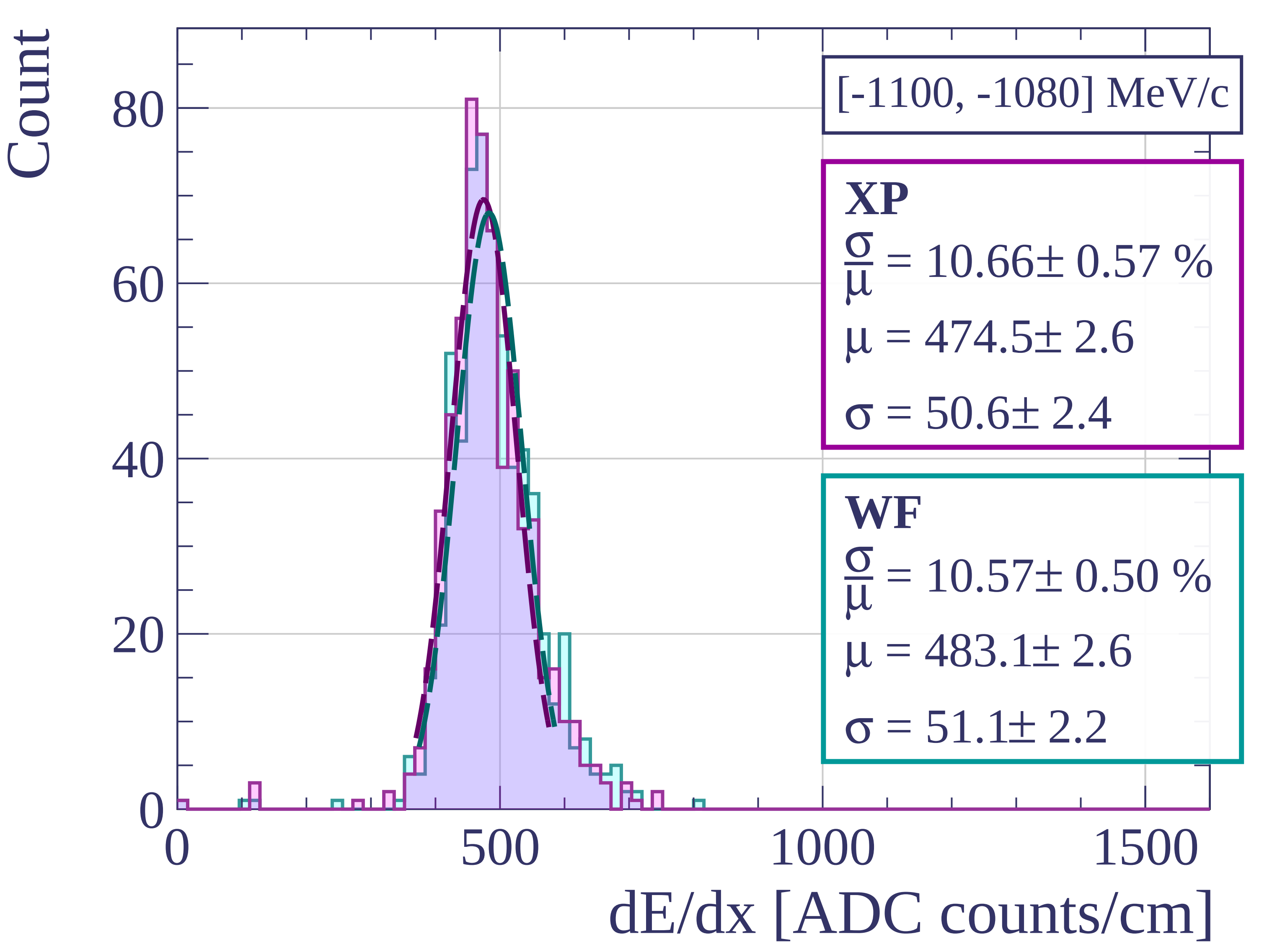
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.57± 0.56 %
μ = 483.5± 2.8
σ = 56.0± 2.4
WF
$\frac{\sigma}{\mu}$ = 11.91± 0.67 %
μ = 486.9± 3.2
σ = 58.0± 2.9



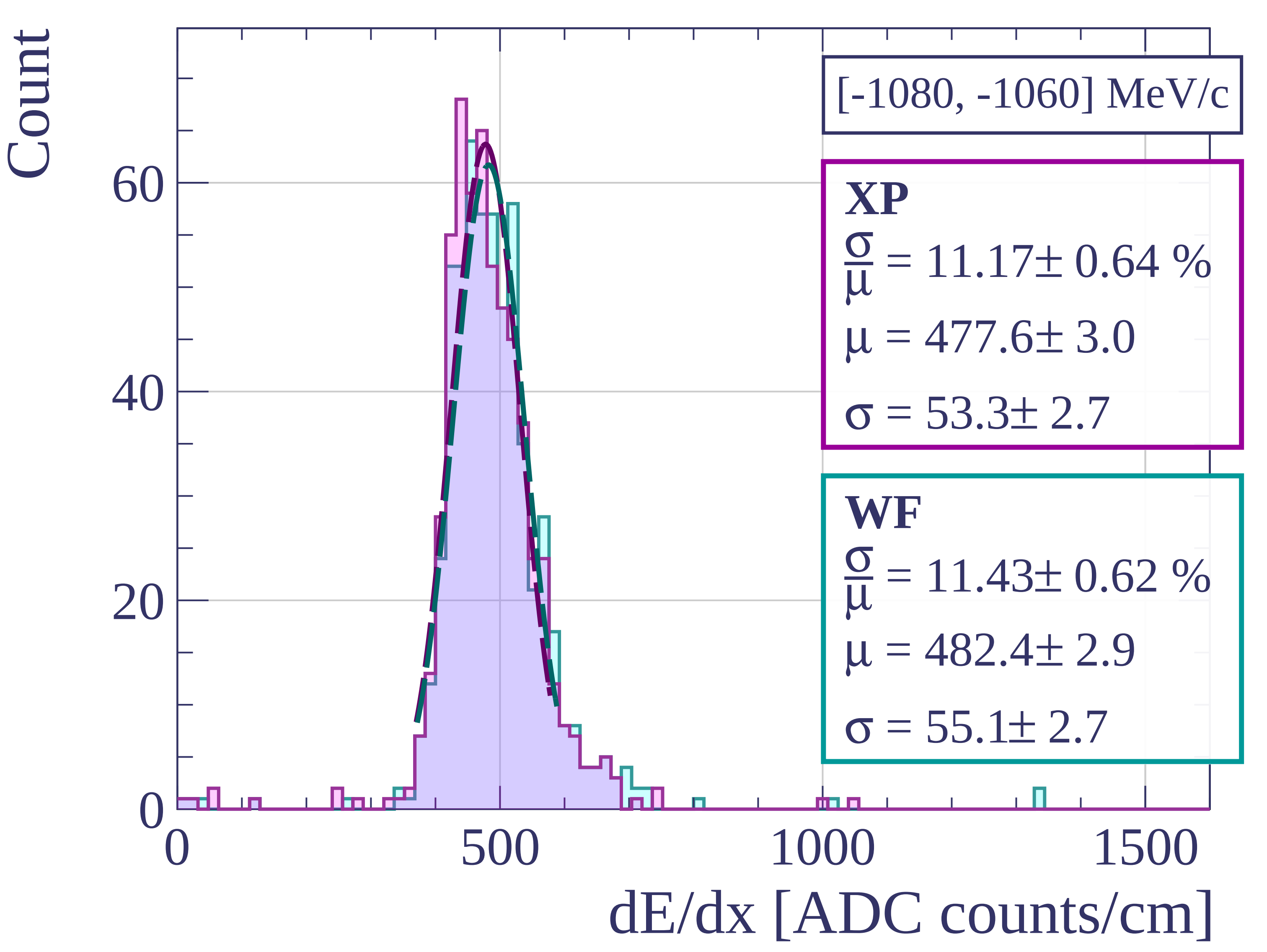
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1140, -1120] MeV/c
XP
σ/μ = 10.37± 0.61 %
μ = 477.3± 2.6
σ = 49.5± 2.7
WF
σ/μ = 10.81± 0.59 %
μ = 485.2± 2.6
σ = 52.4± 2.6

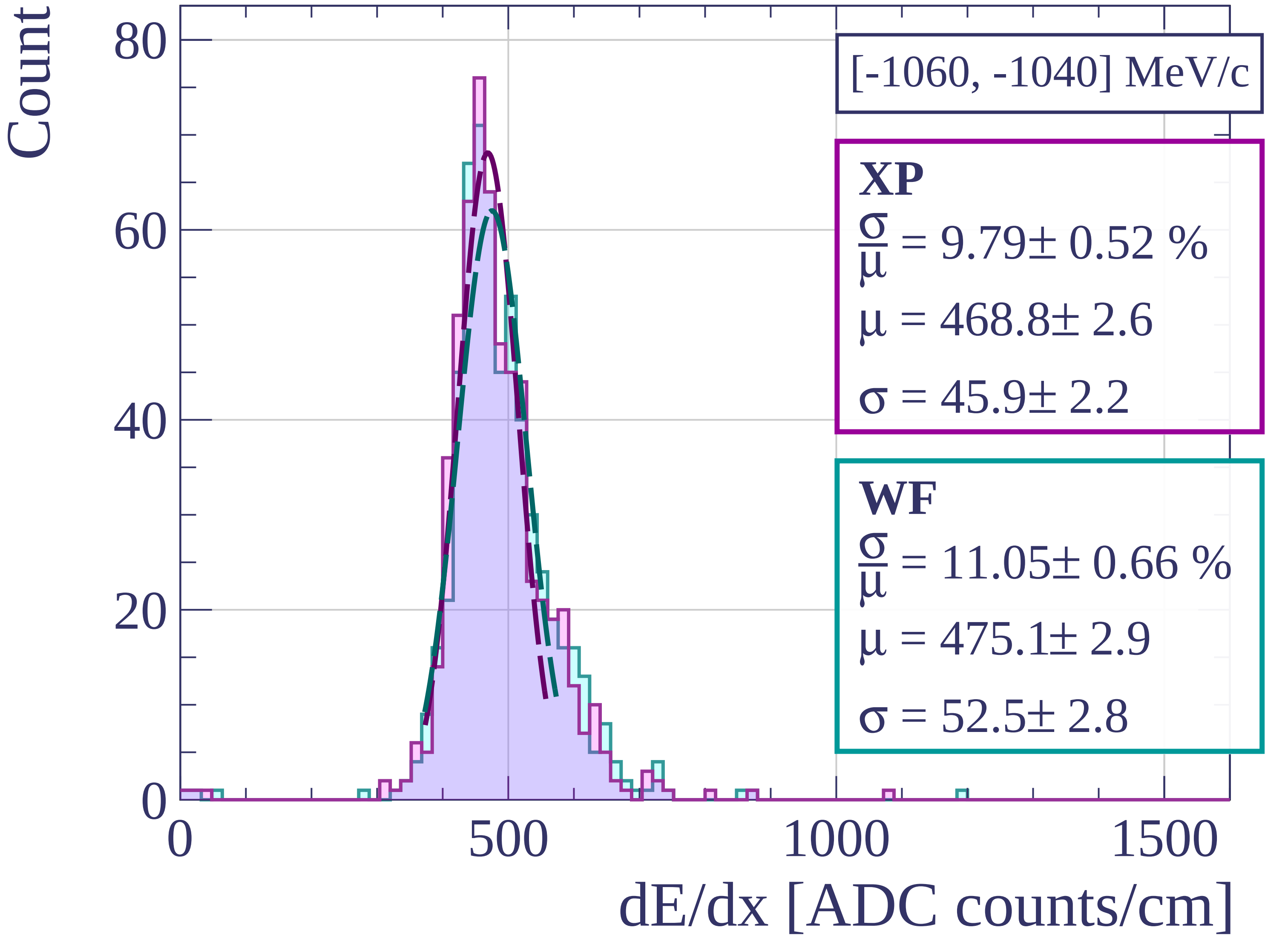


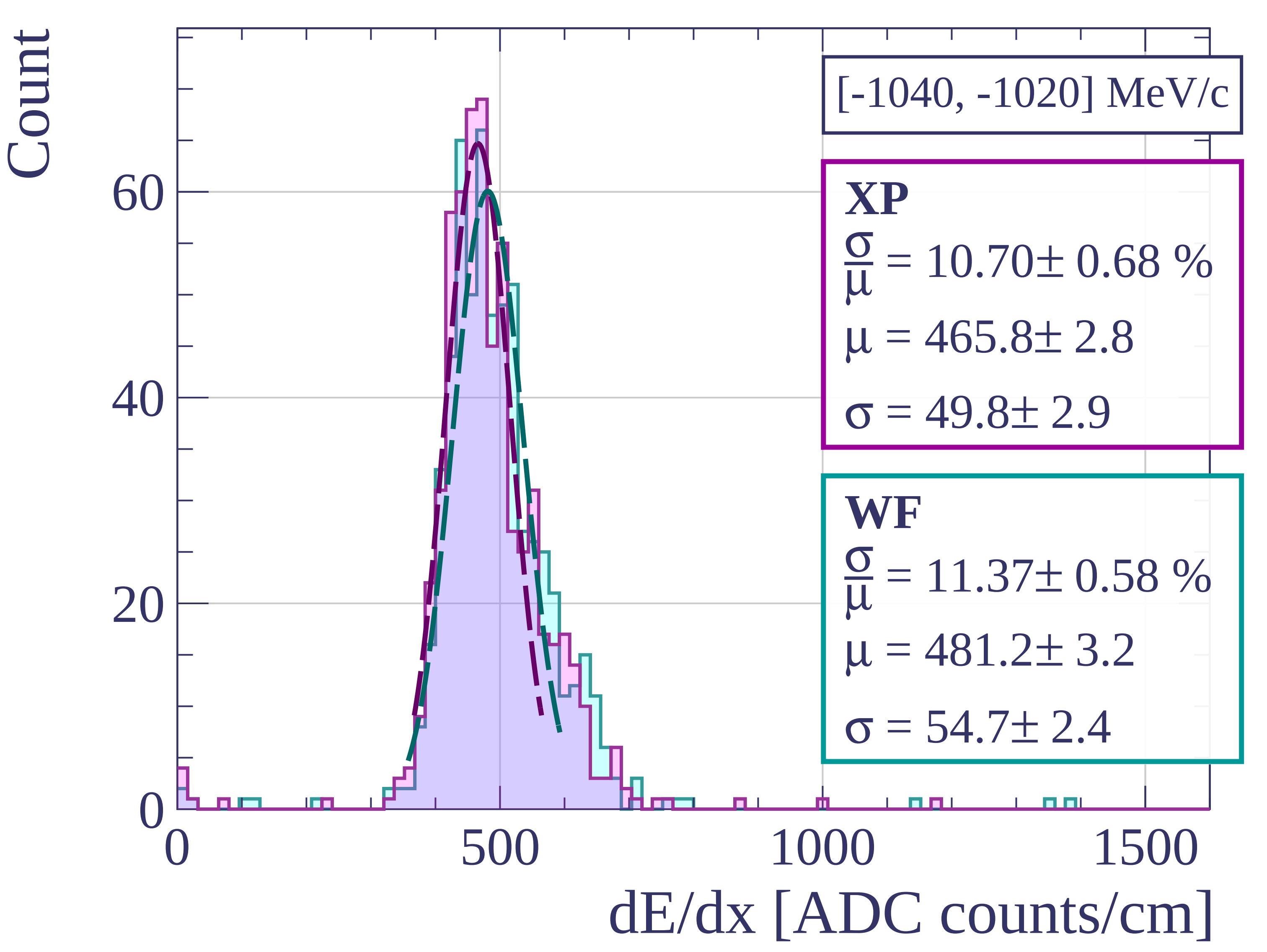
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1120, -1100] MeV/c
XP
σ/μ = 11.00± 0.66 %
μ = 479.2± 2.8
σ = 52.7± 2.9
WF
σ/μ = 11.27± 0.60 %
μ = 488.1± 3.0
σ = 55.0± 2.6

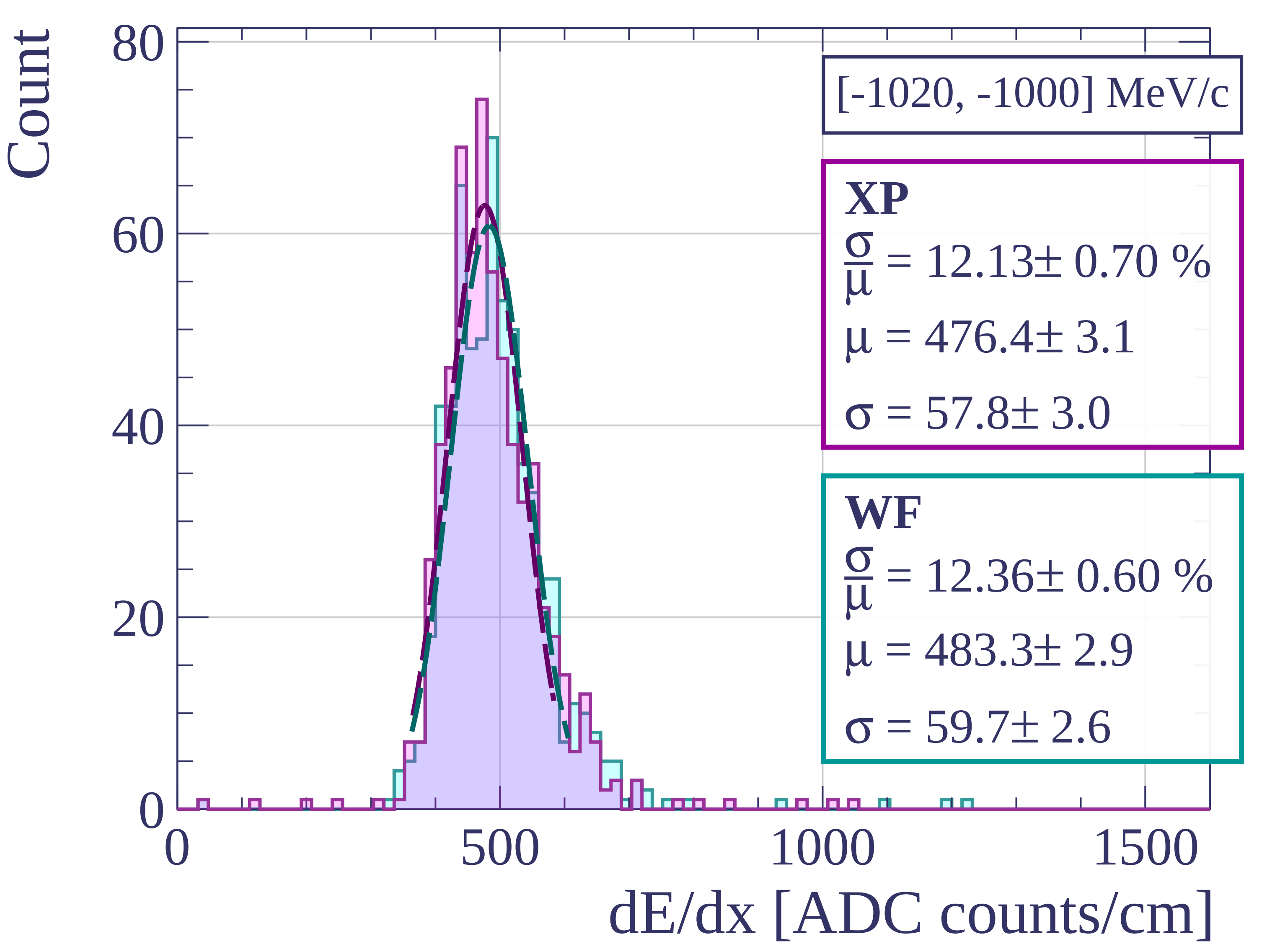


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
σ/μ = 10.66± 0.57 %
μ = 474.5± 2.6
σ = 50.6± 2.4
WF
σ/μ = 10.57± 0.50 %
μ = 483.1± 2.6
σ = 51.1± 2.2

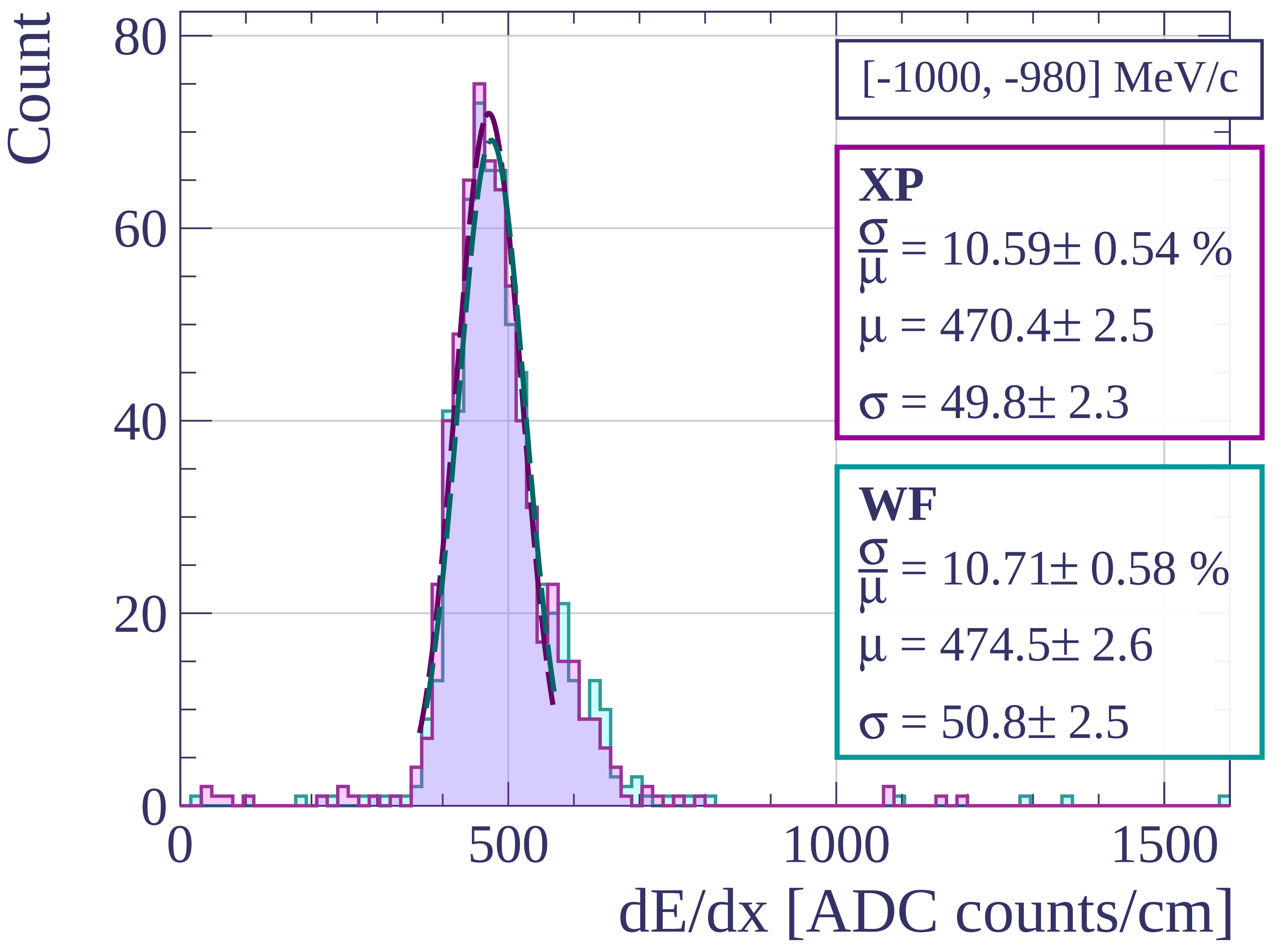

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1080, -1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.17± 0.64 %
μ = 477.6± 3.0
σ = 53.3± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.43± 0.62 %
μ = 482.4± 2.9
σ = 55.1± 2.7

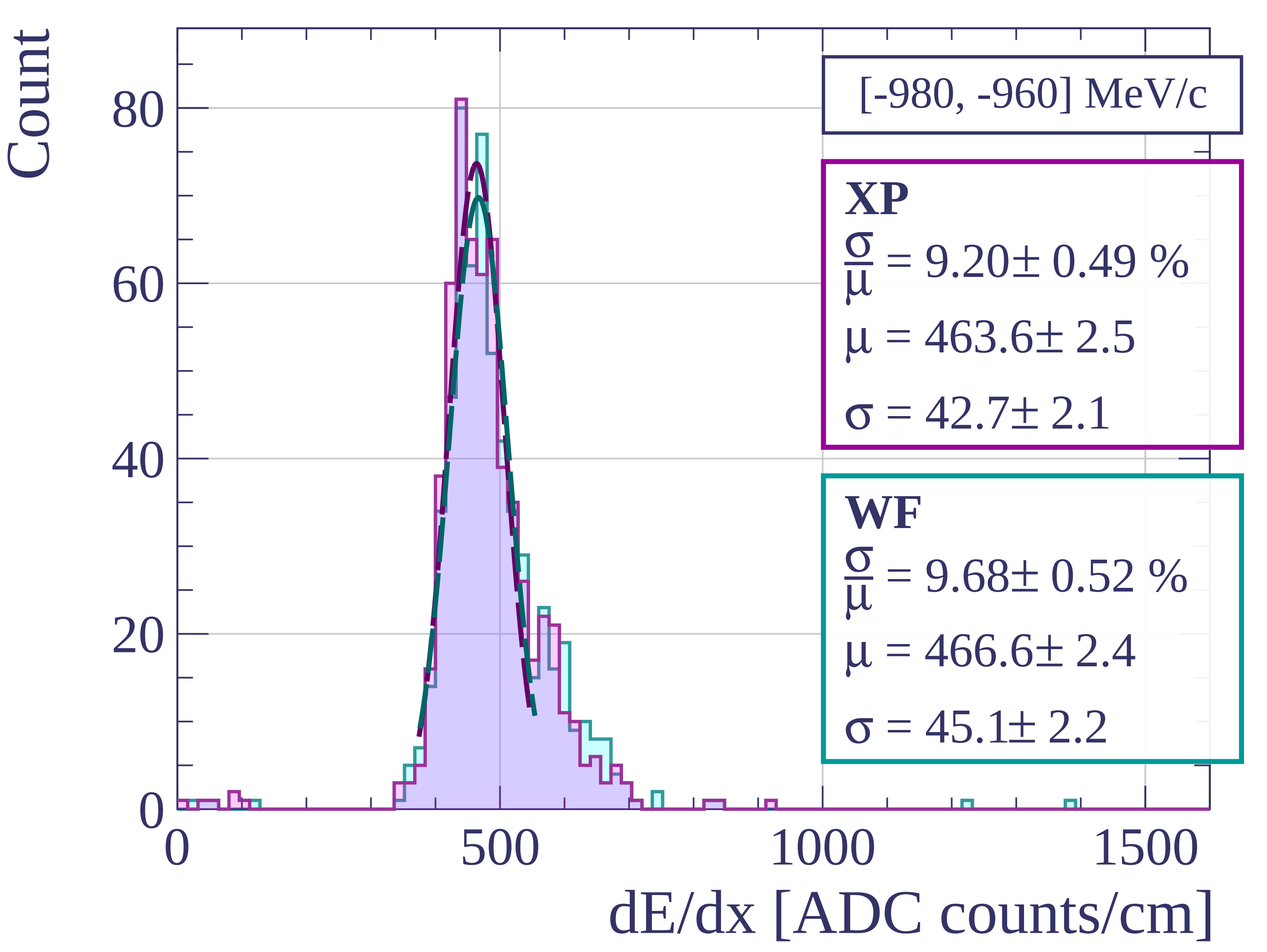

Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-1060, -1040] MeV/c
XP
σ/μ = 9.79± 0.52 %
μ = 468.8± 2.6
σ = 45.9± 2.2
WF
σ/μ = 11.05± 0.66 %
μ = 475.1± 2.9
σ = 52.5± 2.8


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1040, -1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.70± 0.68 %
μ = 465.8± 2.8
σ = 49.8± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.37± 0.58 %
μ = 481.2± 3.2
σ = 54.7± 2.4

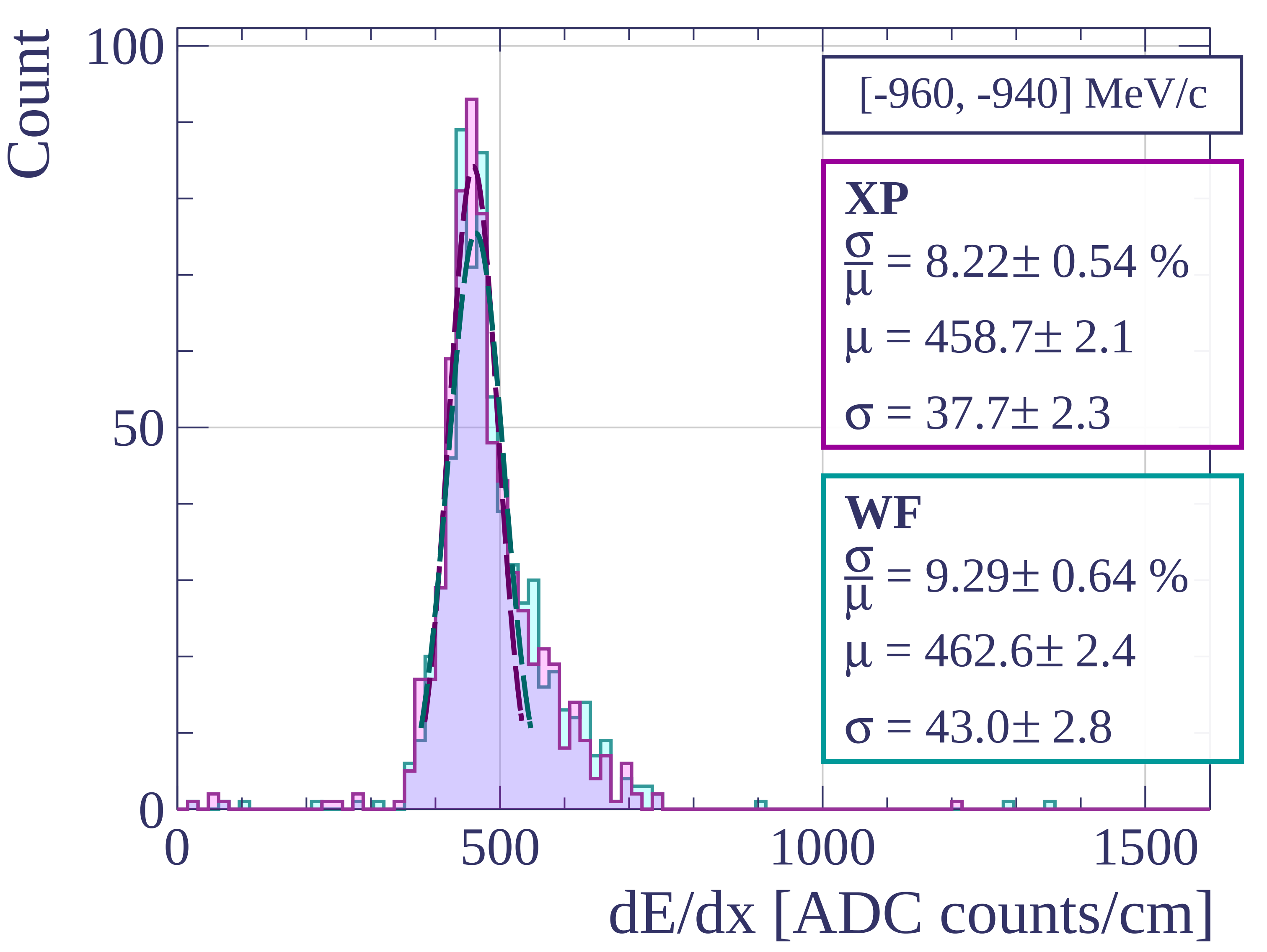


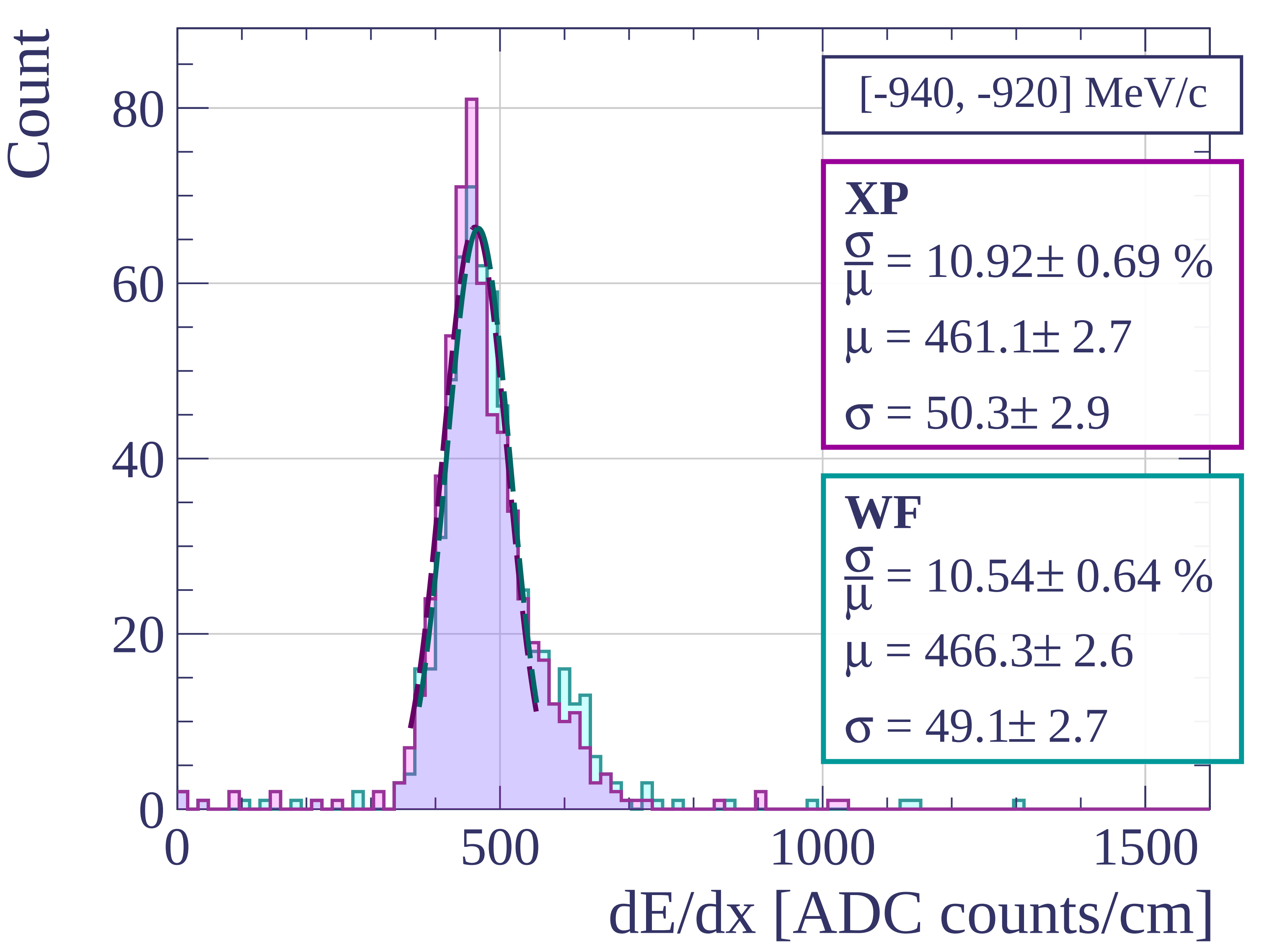
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
σ/μ = 12.13± 0.70 %
μ = 476.4± 3.1
σ = 57.8± 3.0
WF
σ/μ = 12.36± 0.60 %
μ = 483.3± 2.9
σ = 59.7± 2.6


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1000, -980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.59± 0.54 %
μ = 470.4± 2.5
σ = 49.8± 2.3
WF
$\frac{\sigma}{\mu}$ = 10.71± 0.58 %
μ = 474.5± 2.6
σ = 50.8± 2.5

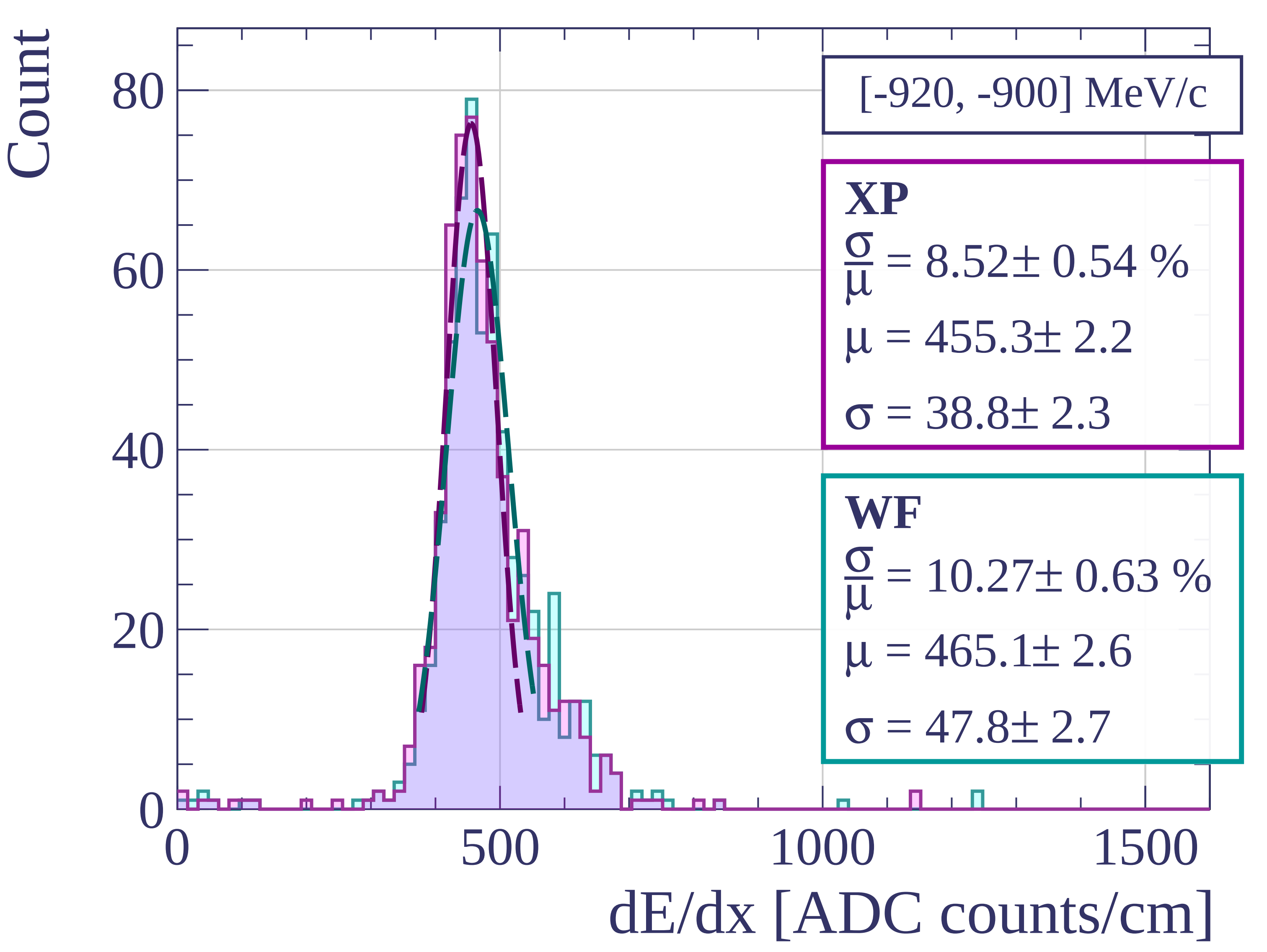


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-980, -960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.20± 0.49 %
μ = 463.6± 2.5
σ = 42.7± 2.1
WF
$\frac{\sigma}{\mu}$ = 9.68± 0.52 %
μ = 466.6± 2.4
σ = 45.1± 2.2

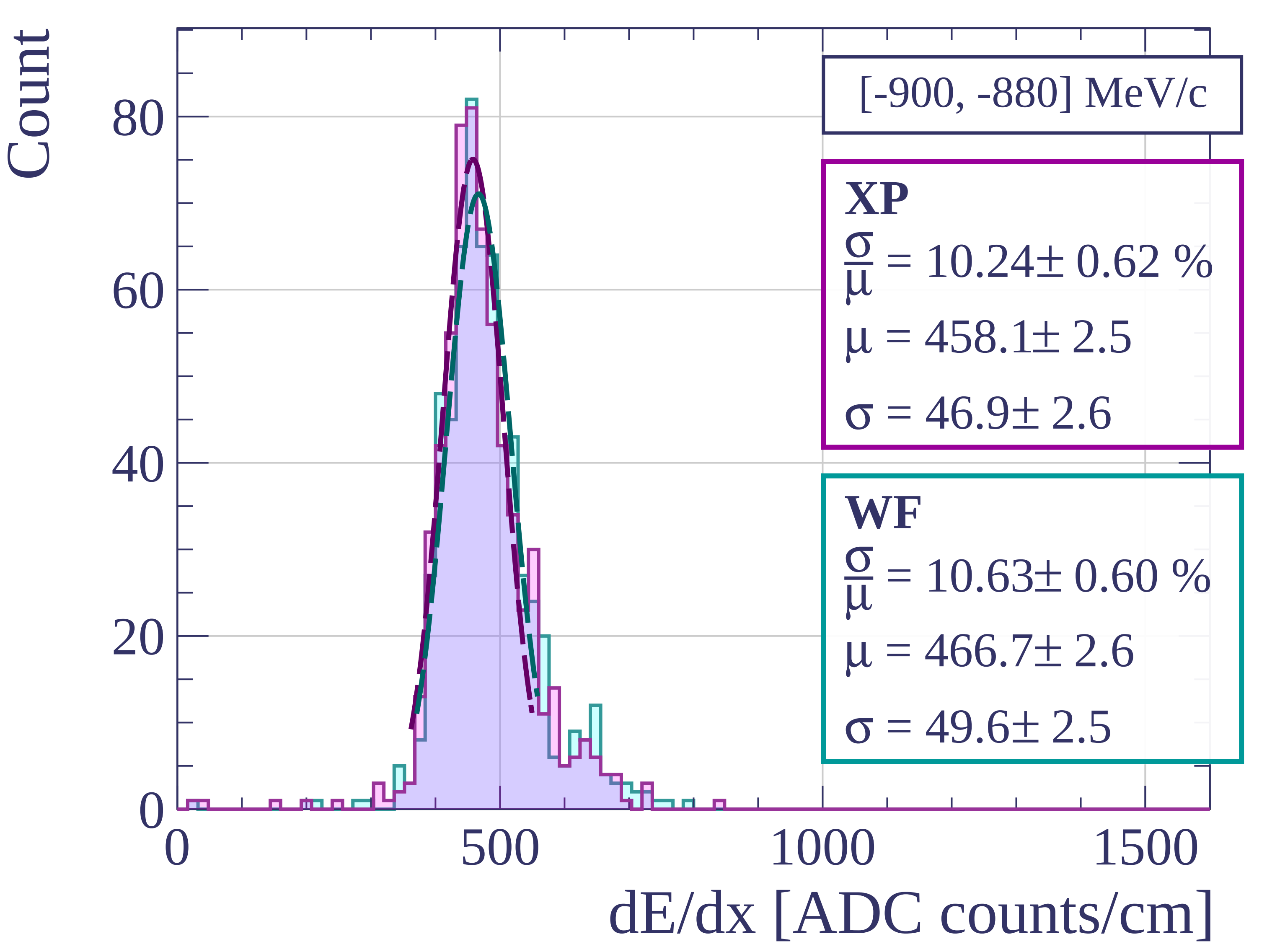

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-960, -940] MeV/c
XP
σ/μ = 8.22± 0.54 %
μ = 458.7± 2.1
σ = 37.7± 2.3
WF
σ/μ = 9.29± 0.64 %
μ = 462.6± 2.4
σ = 43.0± 2.8



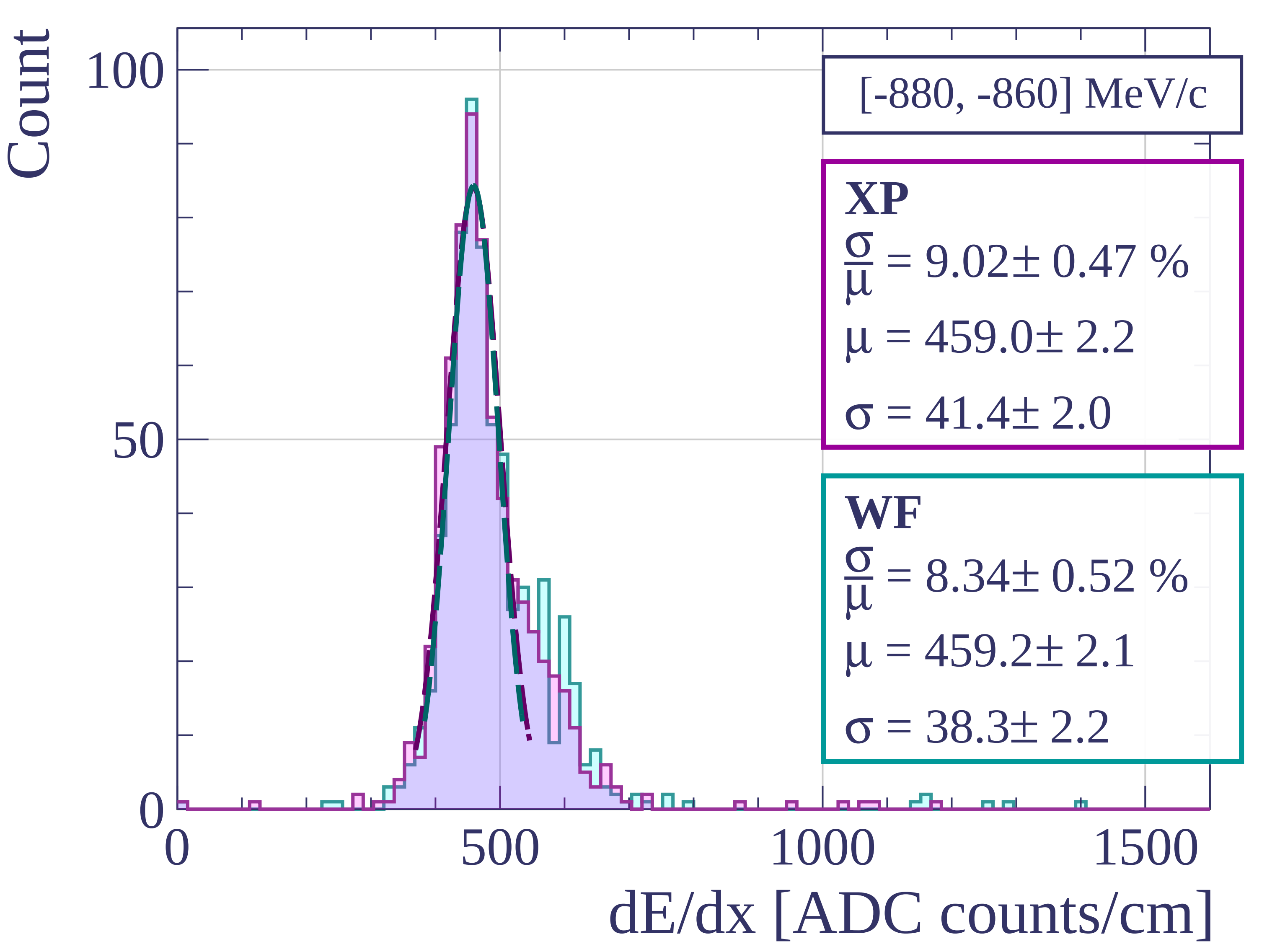
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-940, -920] MeV/c
XP
σ/μ = 10.92± 0.69 %
μ = 461.1± 2.7
σ = 50.3± 2.9
WF
σ/μ = 10.54± 0.64 %
μ = 466.3± 2.6
σ = 49.1± 2.7



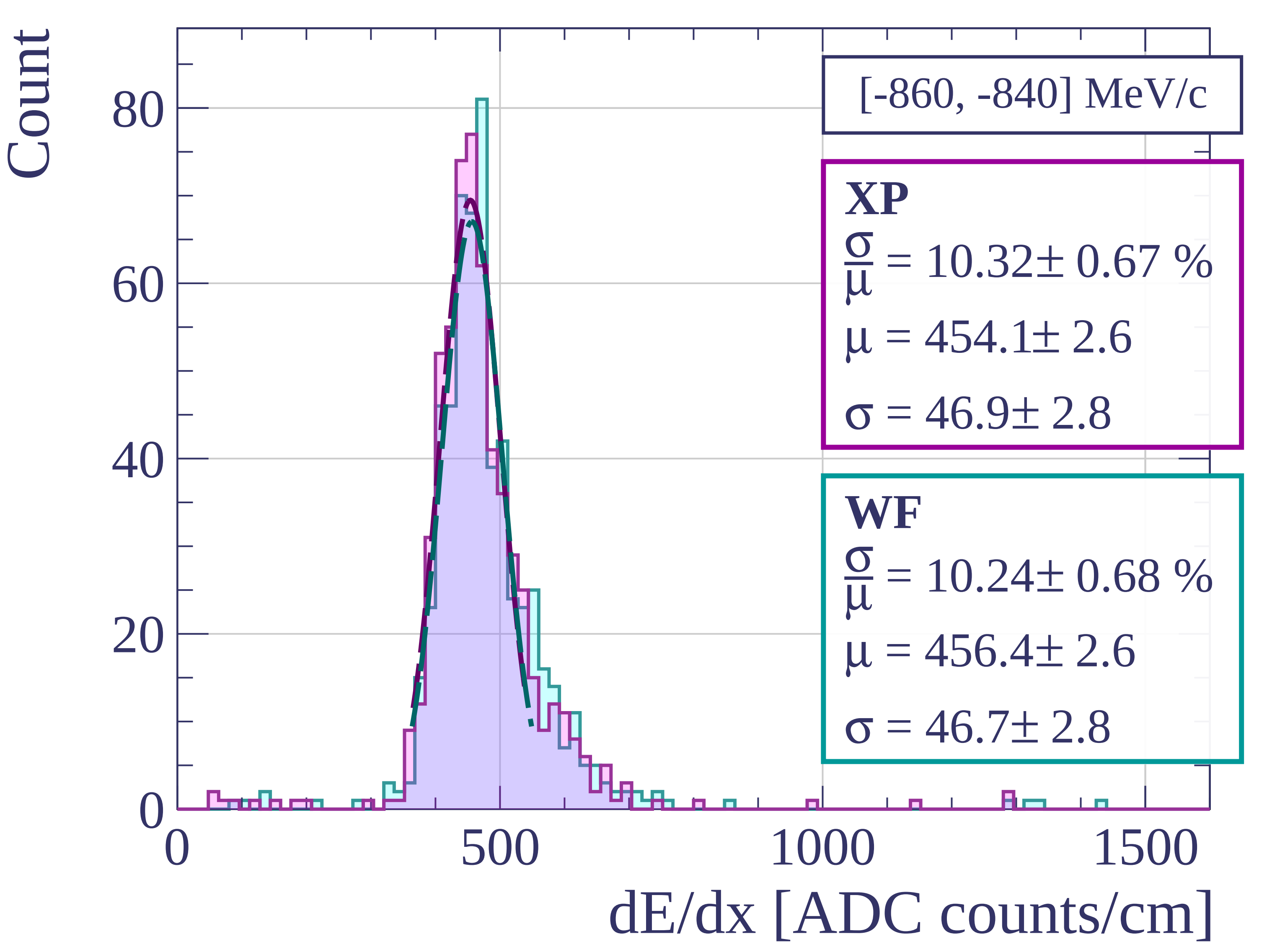
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-920, -900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.52± 0.54 %
μ = 455.3± 2.2
σ = 38.8± 2.3
WF
$\frac{\sigma}{\mu}$ = 10.27± 0.63 %
μ = 465.1± 2.6
σ = 47.8± 2.7



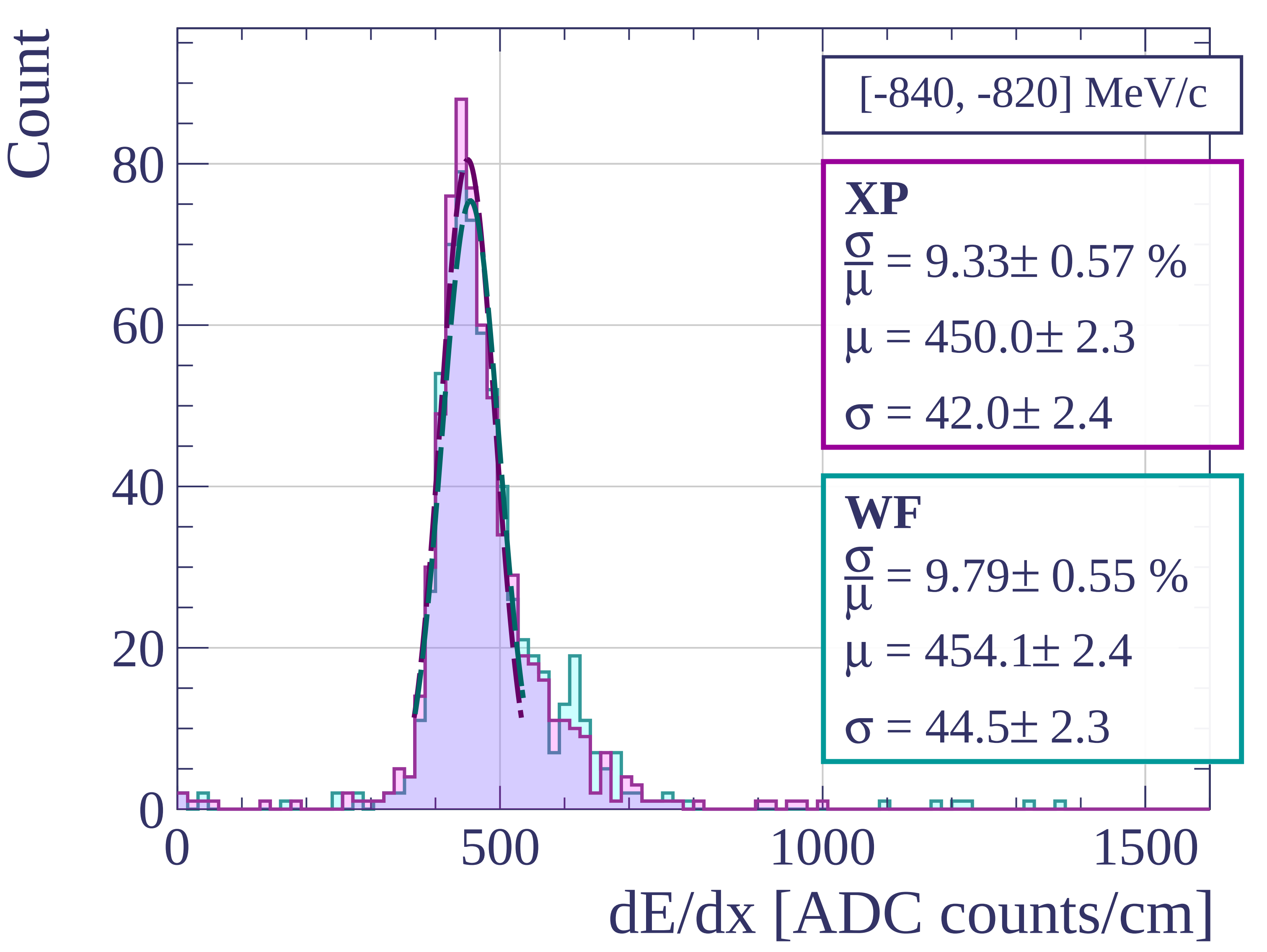
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-900, -880] MeV/c
XP
σ/μ = 10.24± 0.62 %
μ = 458.1± 2.5
σ = 46.9± 2.6
WF
σ/μ = 10.63± 0.60 %
μ = 466.7± 2.6
σ = 49.6± 2.5



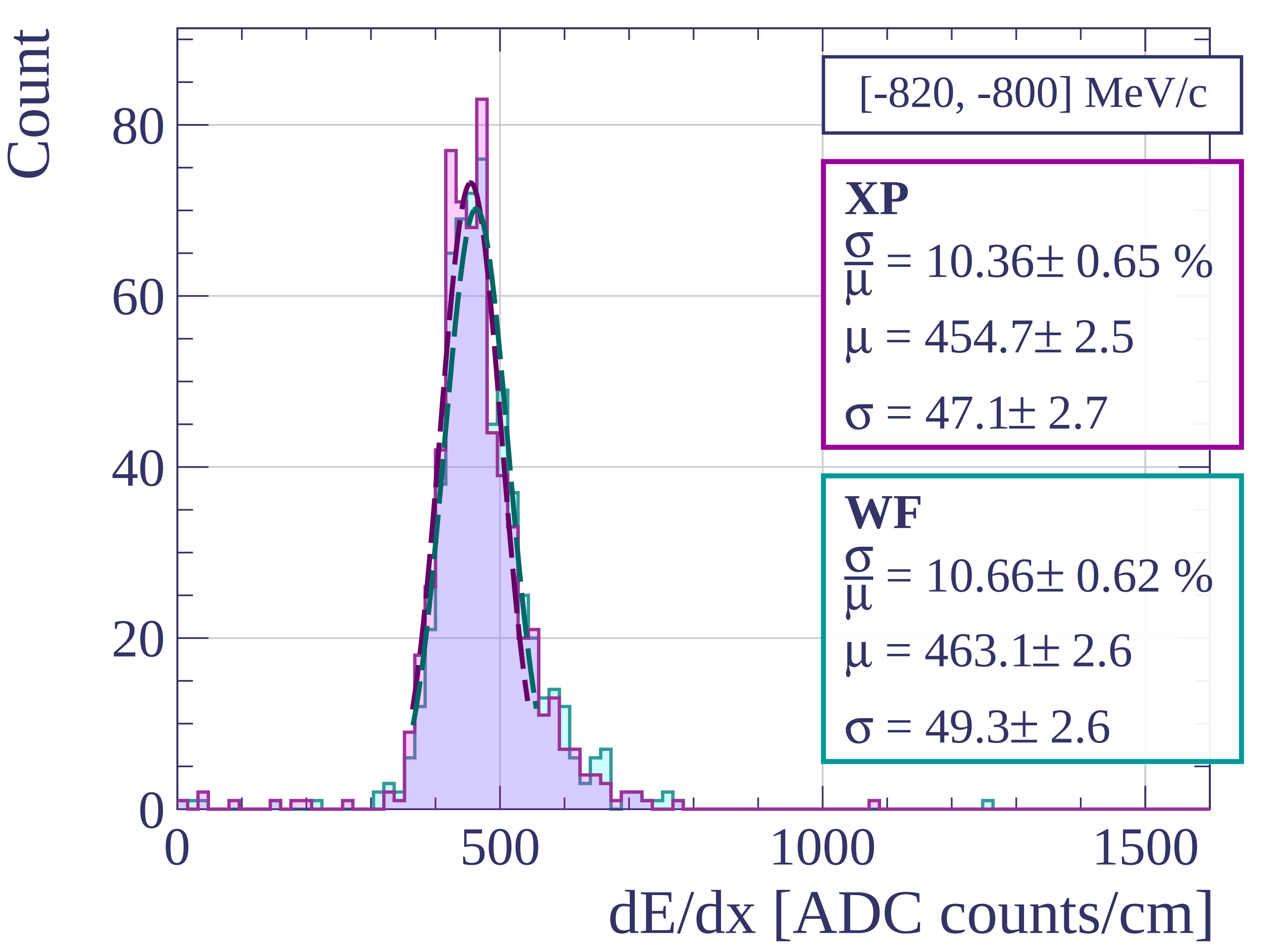
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.02± 0.47 %
μ = 459.0± 2.2
σ = 41.4± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.34± 0.52 %
μ = 459.2± 2.1
σ = 38.3± 2.2



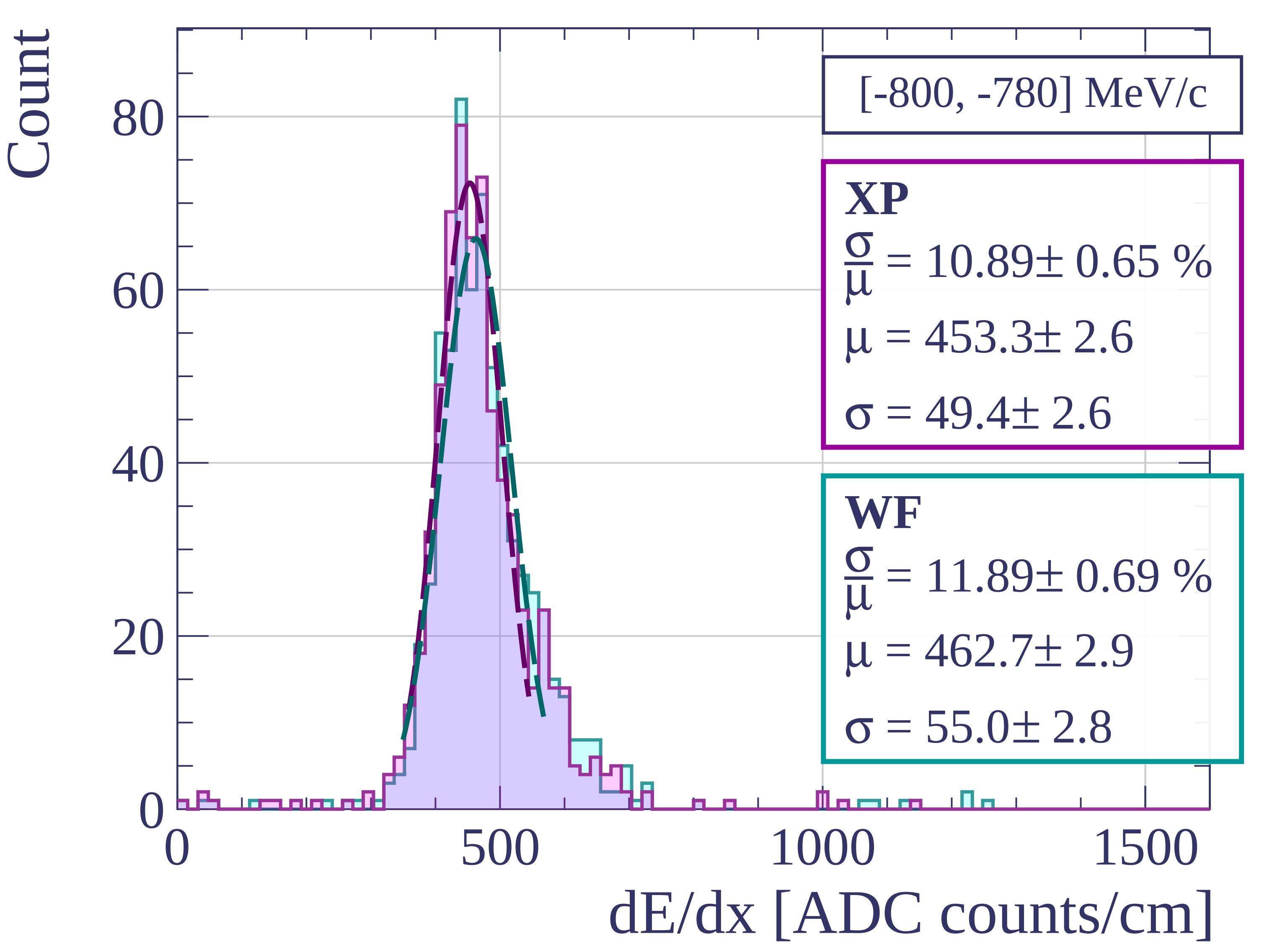
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-860, -840] MeV/c
XP
σ/μ = 10.32± 0.67 %
μ = 454.1± 2.6
σ = 46.9± 2.8
WF
σ/μ = 10.24± 0.68 %
μ = 456.4± 2.6
σ = 46.7± 2.8



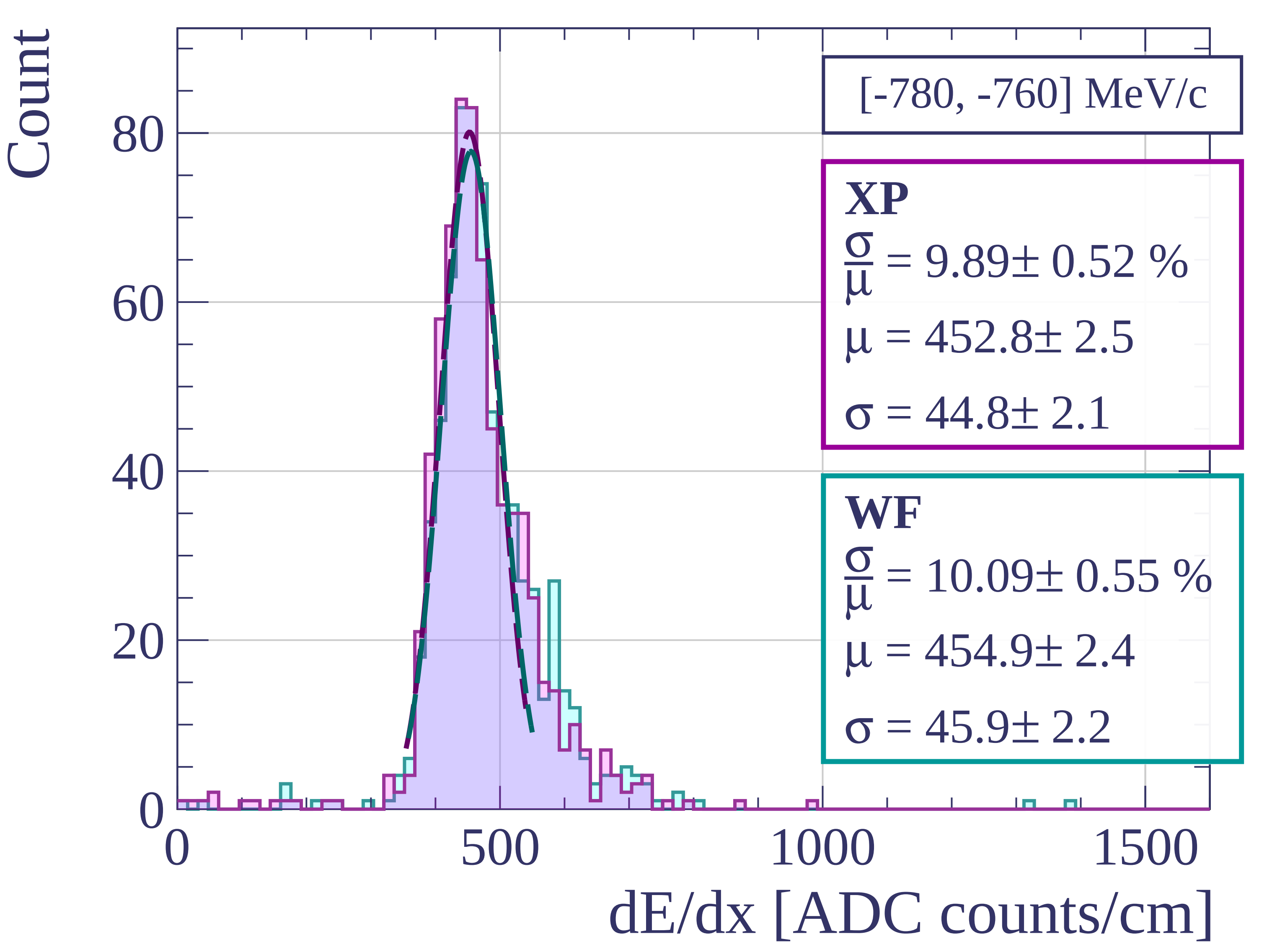
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-840, -820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.33± 0.57 %
μ = 450.0± 2.3
σ = 42.0± 2.4
WF
$\frac{\sigma}{\mu}$ = 9.79± 0.55 %
μ = 454.1± 2.4
σ = 44.5± 2.3



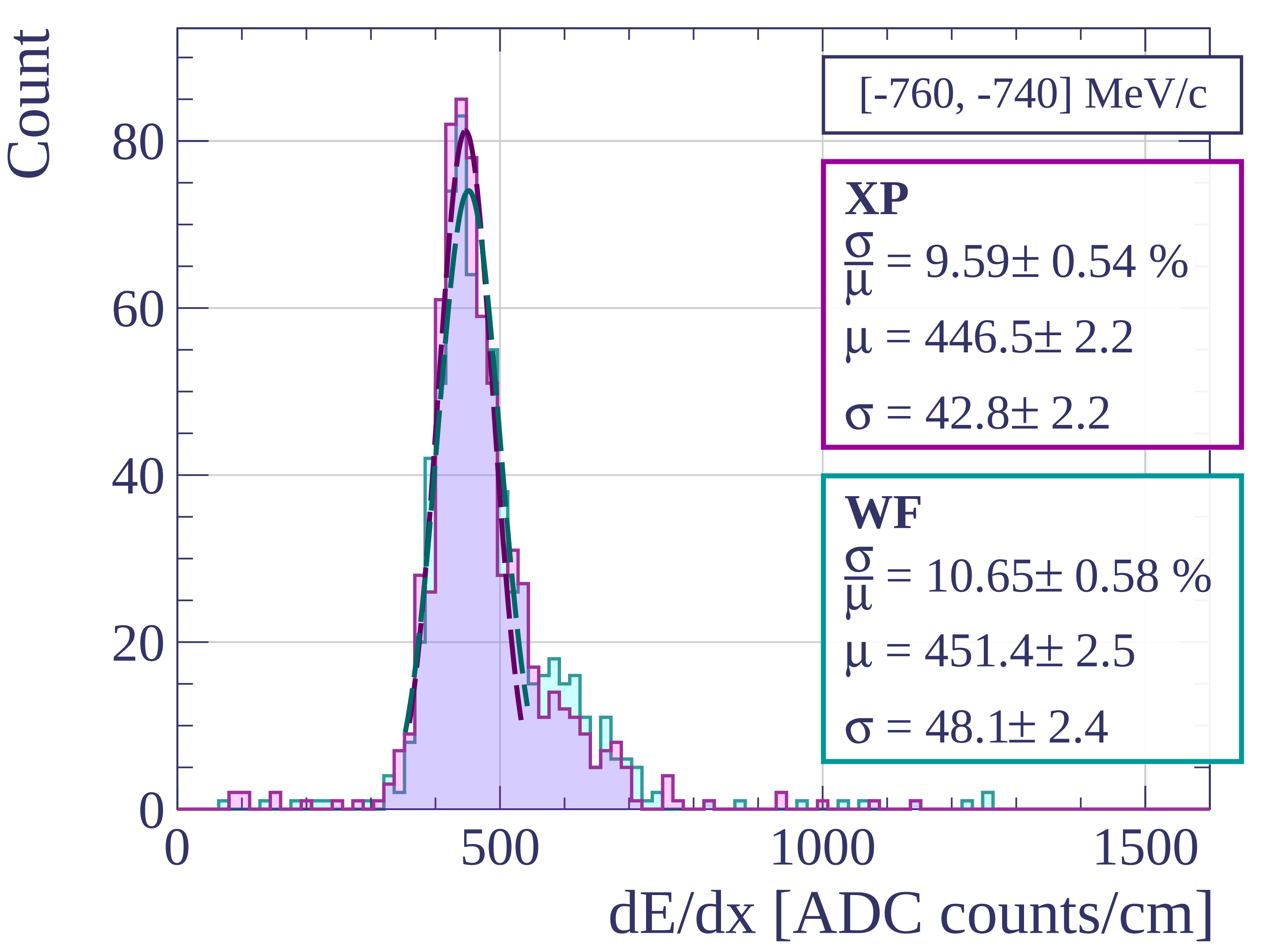
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.36± 0.65 %
μ = 454.7± 2.5
σ = 47.1± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.66± 0.62 %
μ = 463.1± 2.6
σ = 49.3± 2.6

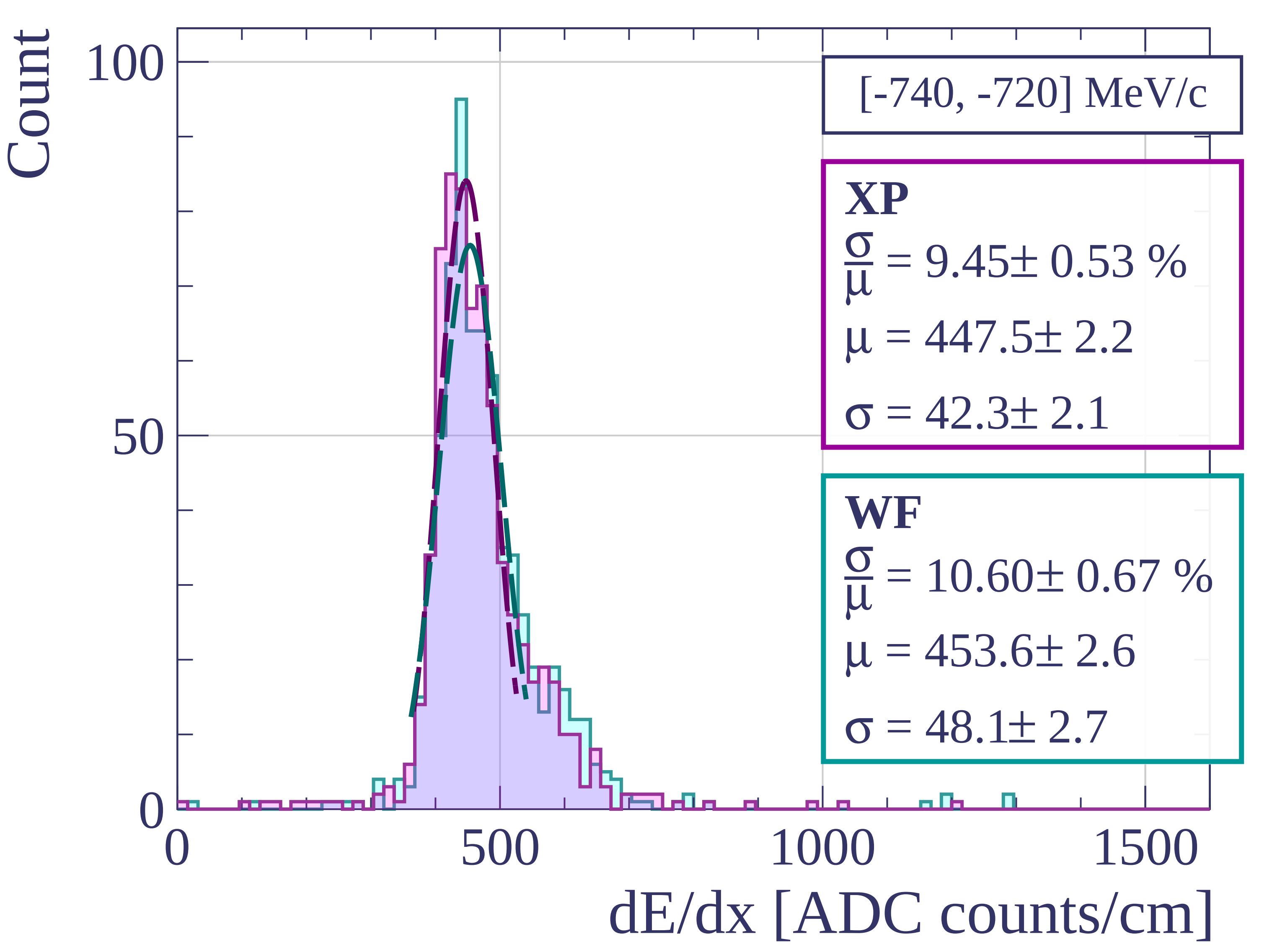


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-800, -780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.89± 0.65 %
μ = 453.3± 2.6
σ = 49.4± 2.6
WF
$\frac{\sigma}{\mu}$ = 11.89± 0.69 %
μ = 462.7± 2.9
σ = 55.0± 2.8

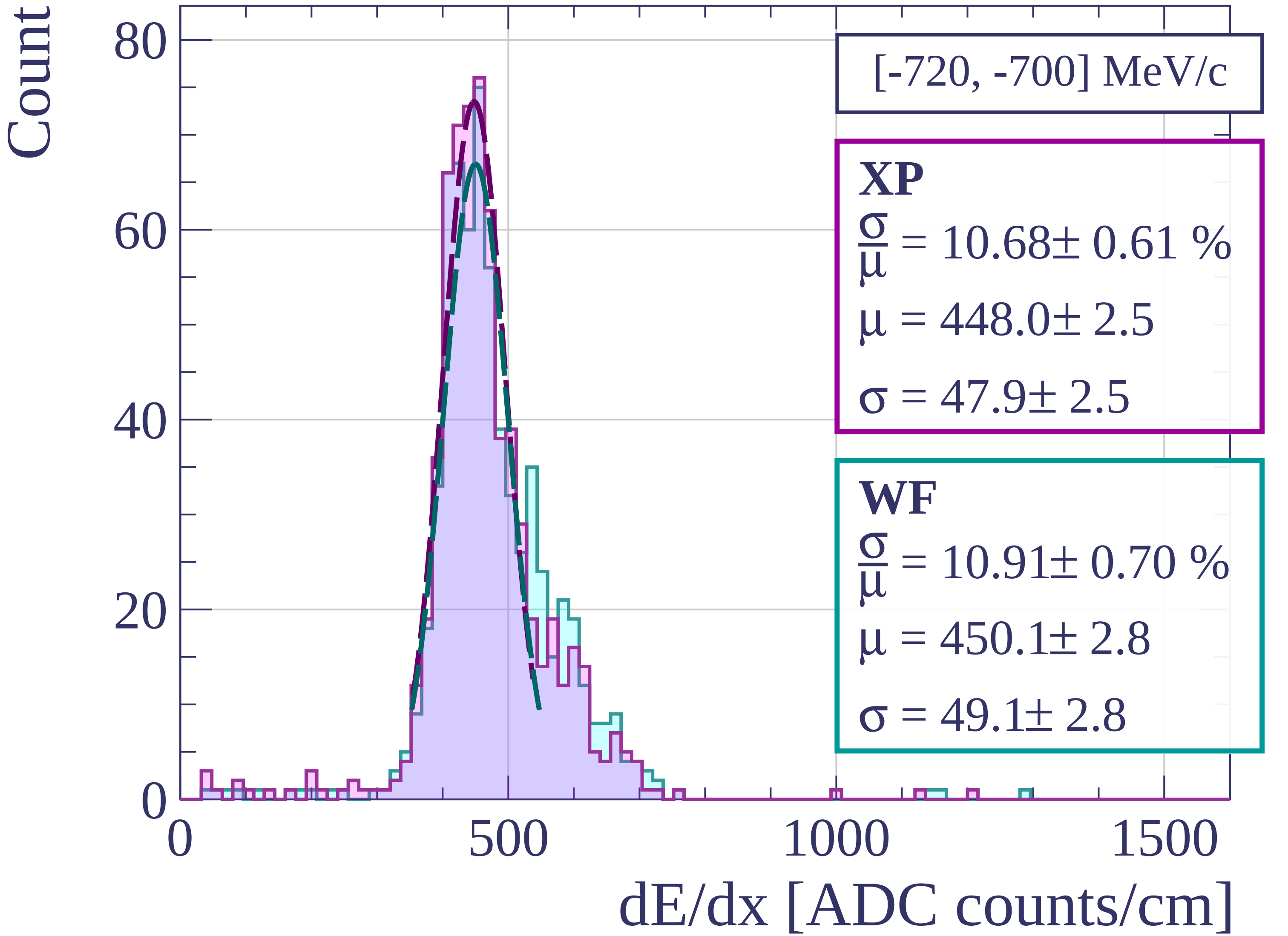


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
20
40
60
80
[-780, -760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.89± 0.52 %
μ = 452.8± 2.5
σ = 44.8± 2.1
WF
$\frac{\sigma}{\mu}$ = 10.09± 0.55 %
μ = 454.9± 2.4
σ = 45.9± 2.2

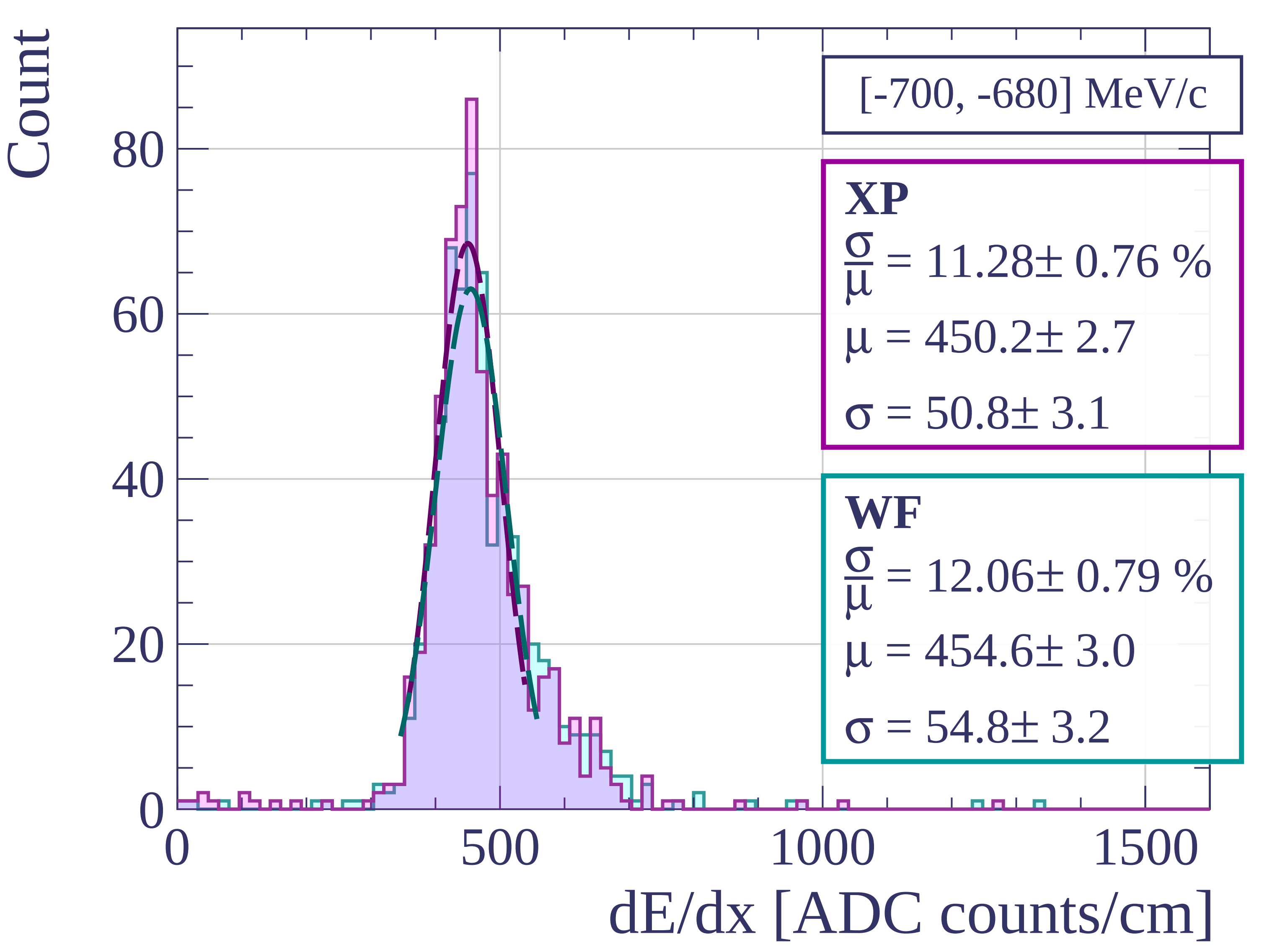

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.59± 0.54 %
μ = 446.5± 2.2
σ = 42.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.65± 0.58 %
μ = 451.4± 2.5
σ = 48.1± 2.4



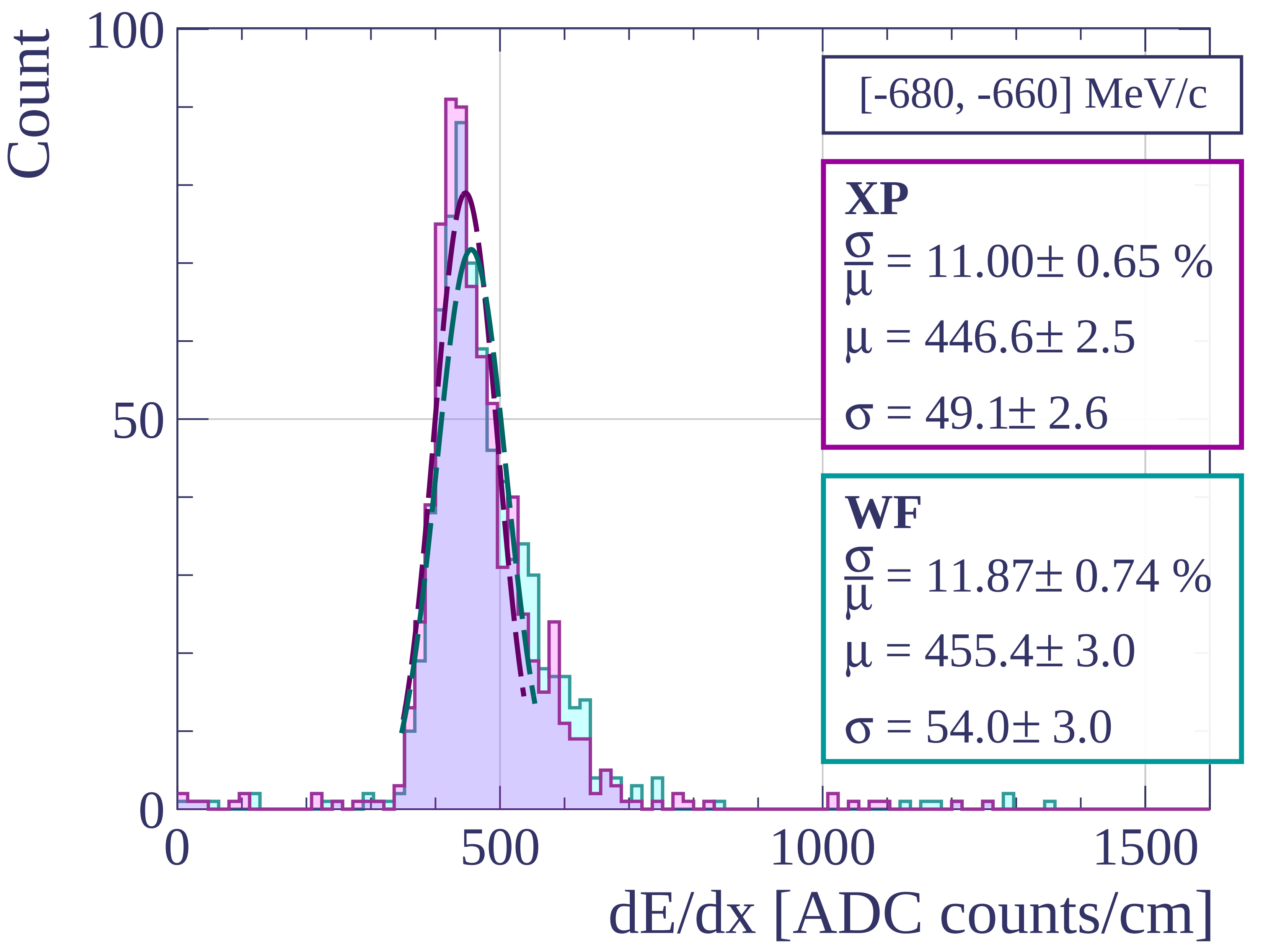
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.45± 0.53 %
μ = 447.5± 2.2
σ = 42.3± 2.1
WF
$\frac{\sigma}{\mu}$ = 10.60± 0.67 %
μ = 453.6± 2.6
σ = 48.1± 2.7



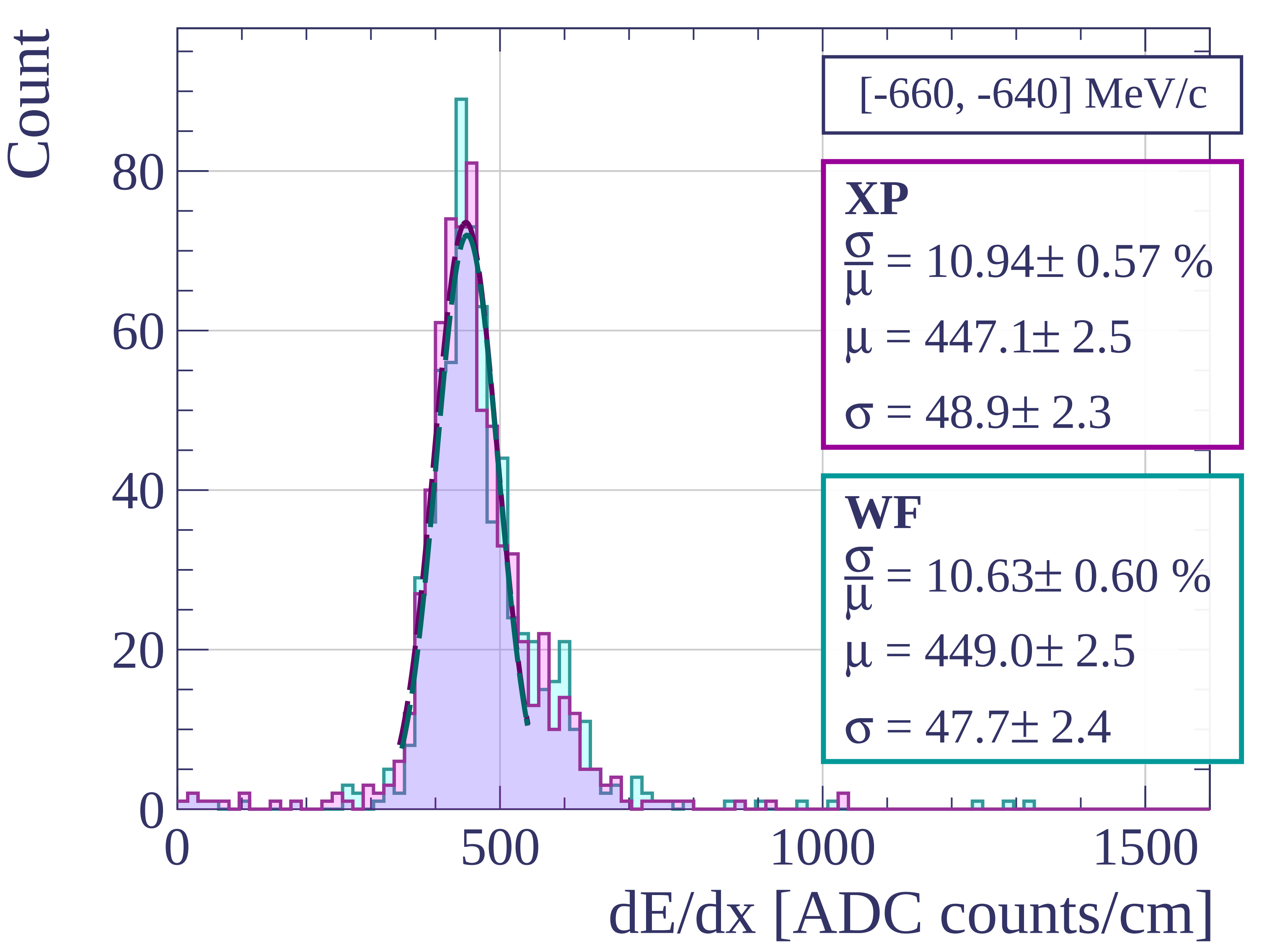
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-720, -700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.68± 0.61 %
μ = 448.0± 2.5
σ = 47.9± 2.5
WF
$\frac{\sigma}{\mu}$ = 10.91± 0.70 %
μ = 450.1± 2.8
σ = 49.1± 2.8



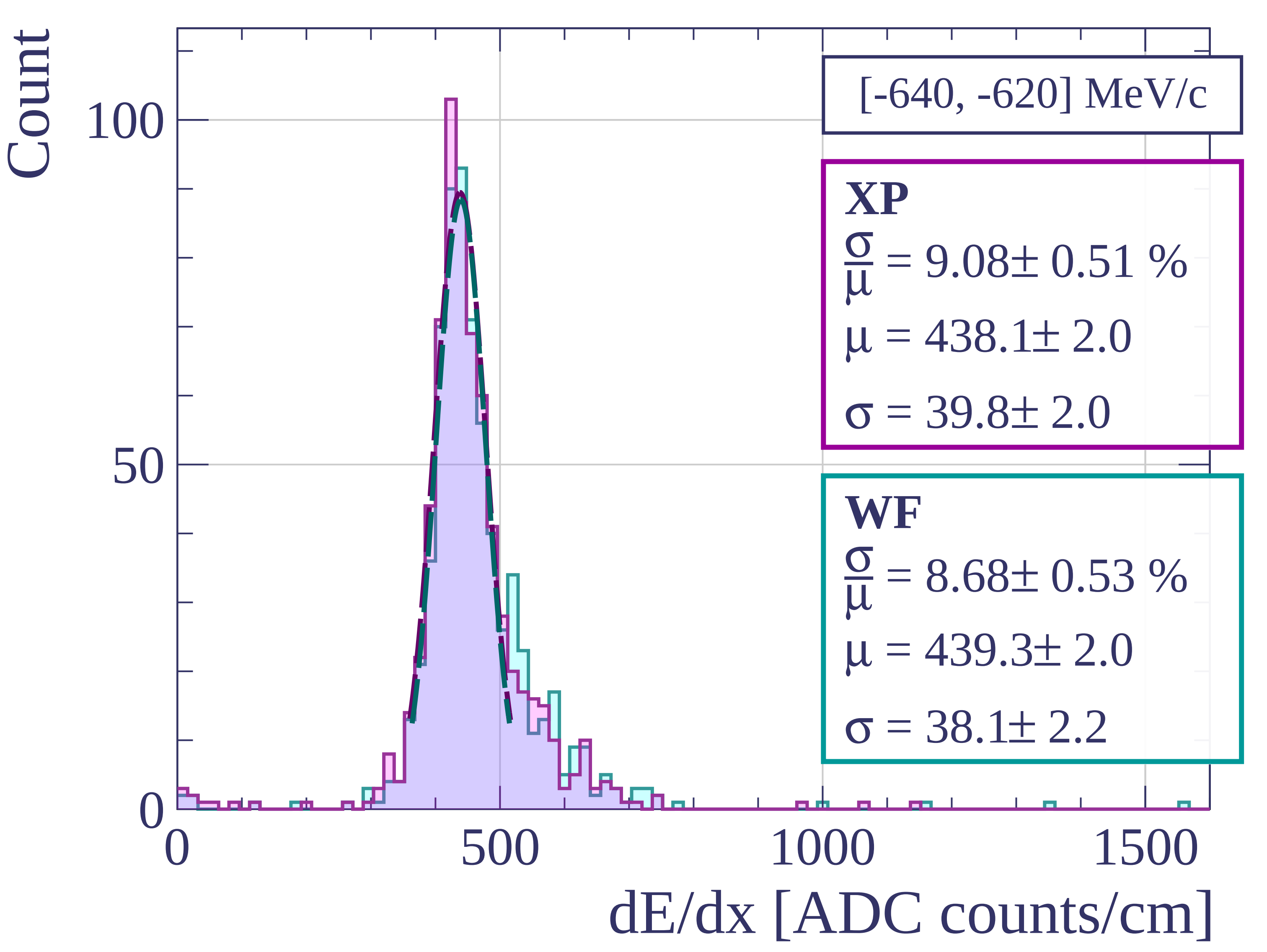
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-700, -680] MeV/c
XP
σ/μ = 11.28± 0.76 %
μ = 450.2± 2.7
σ = 50.8± 3.1
WF
σ/μ = 12.06± 0.79 %
μ = 454.6± 3.0
σ = 54.8± 3.2



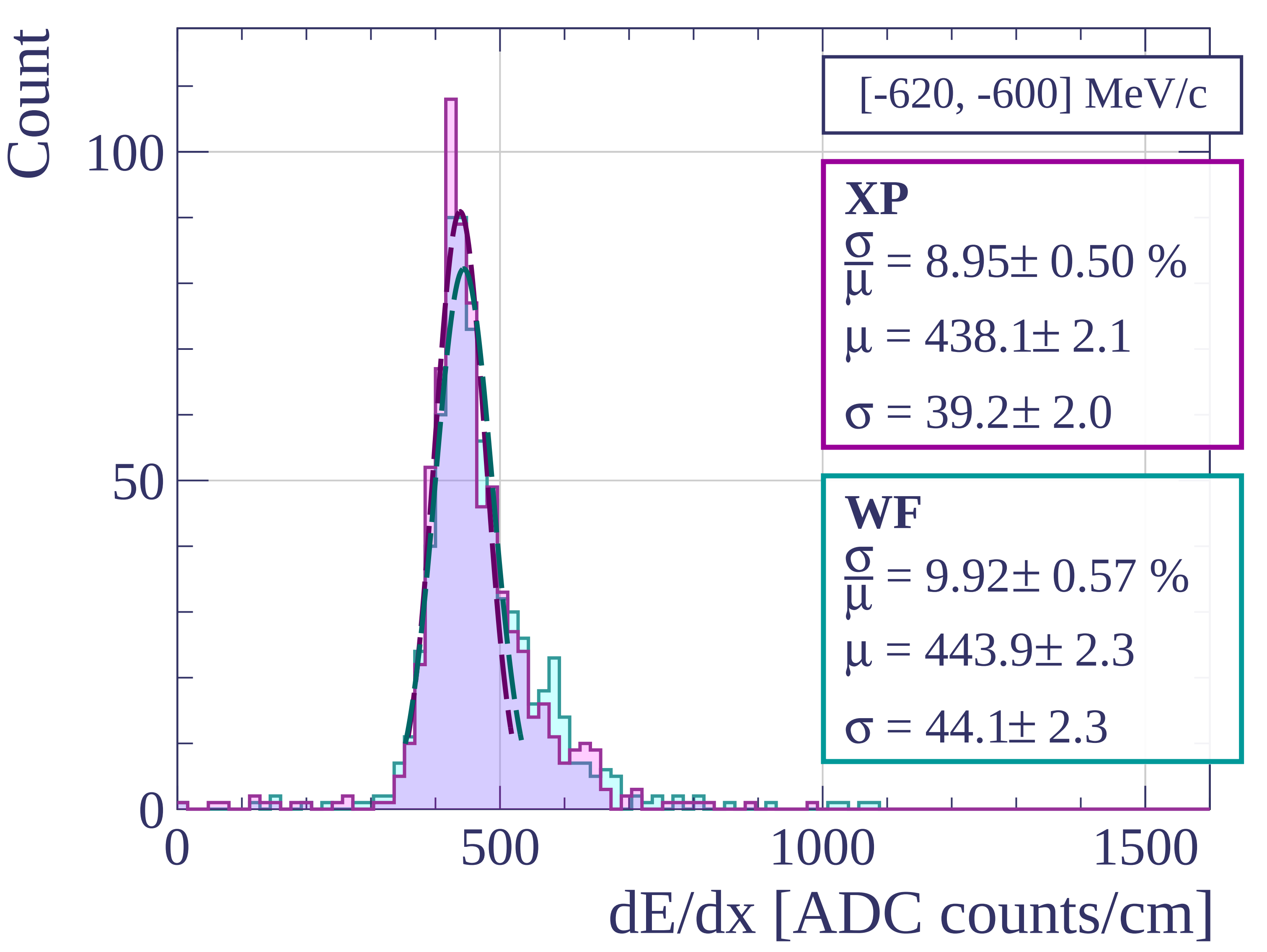
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-680, -660] MeV/c
XP
σ/μ = 11.00± 0.65 %
μ = 446.6± 2.5
σ = 49.1± 2.6
WF
σ/μ = 11.87± 0.74 %
μ = 455.4± 3.0
σ = 54.0± 3.0



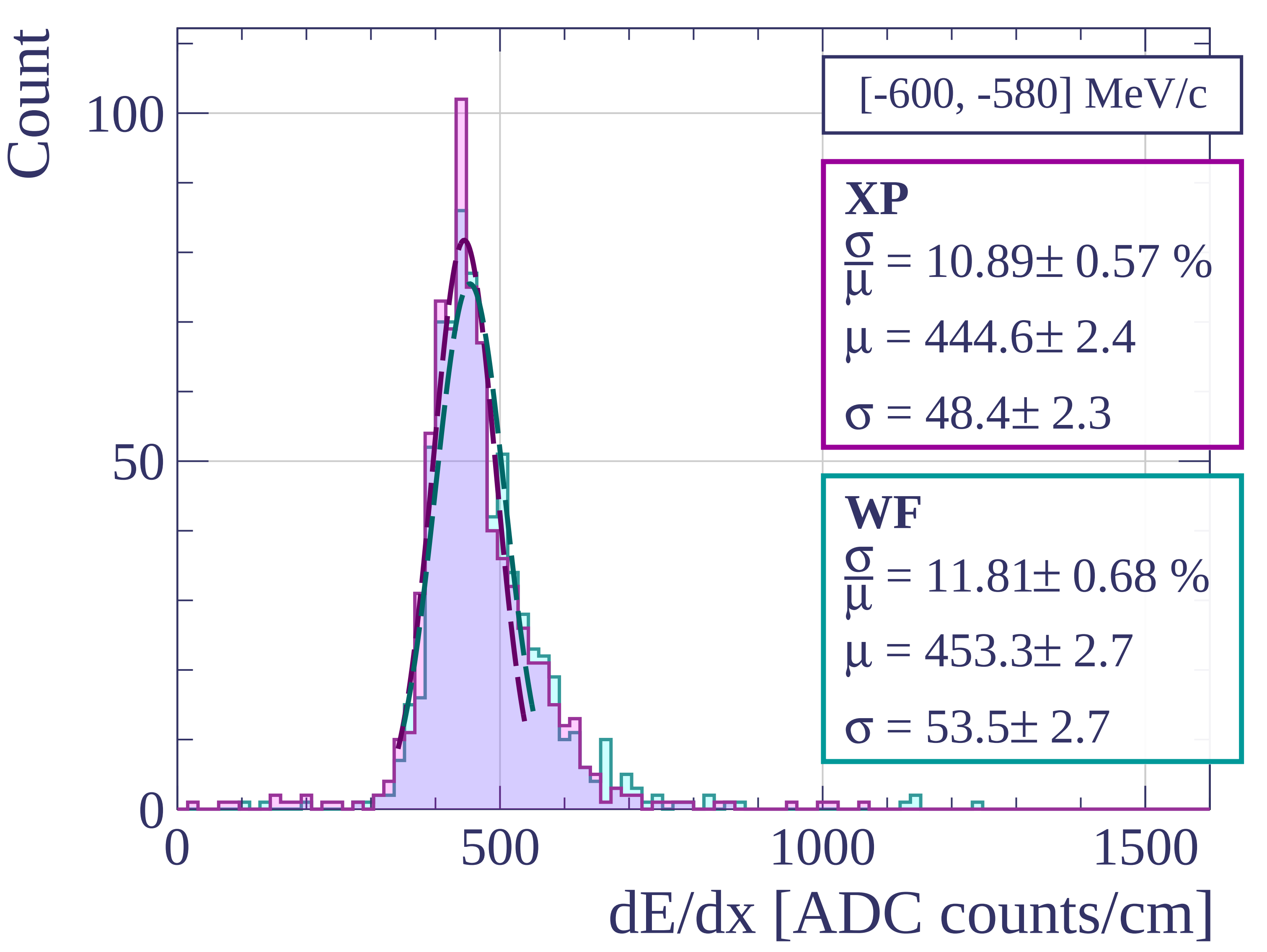
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.94± 0.57 %
μ = 447.1± 2.5
σ = 48.9± 2.3
WF
$\frac{\sigma}{\mu}$ = 10.63± 0.60 %
μ = 449.0± 2.5
σ = 47.7± 2.4

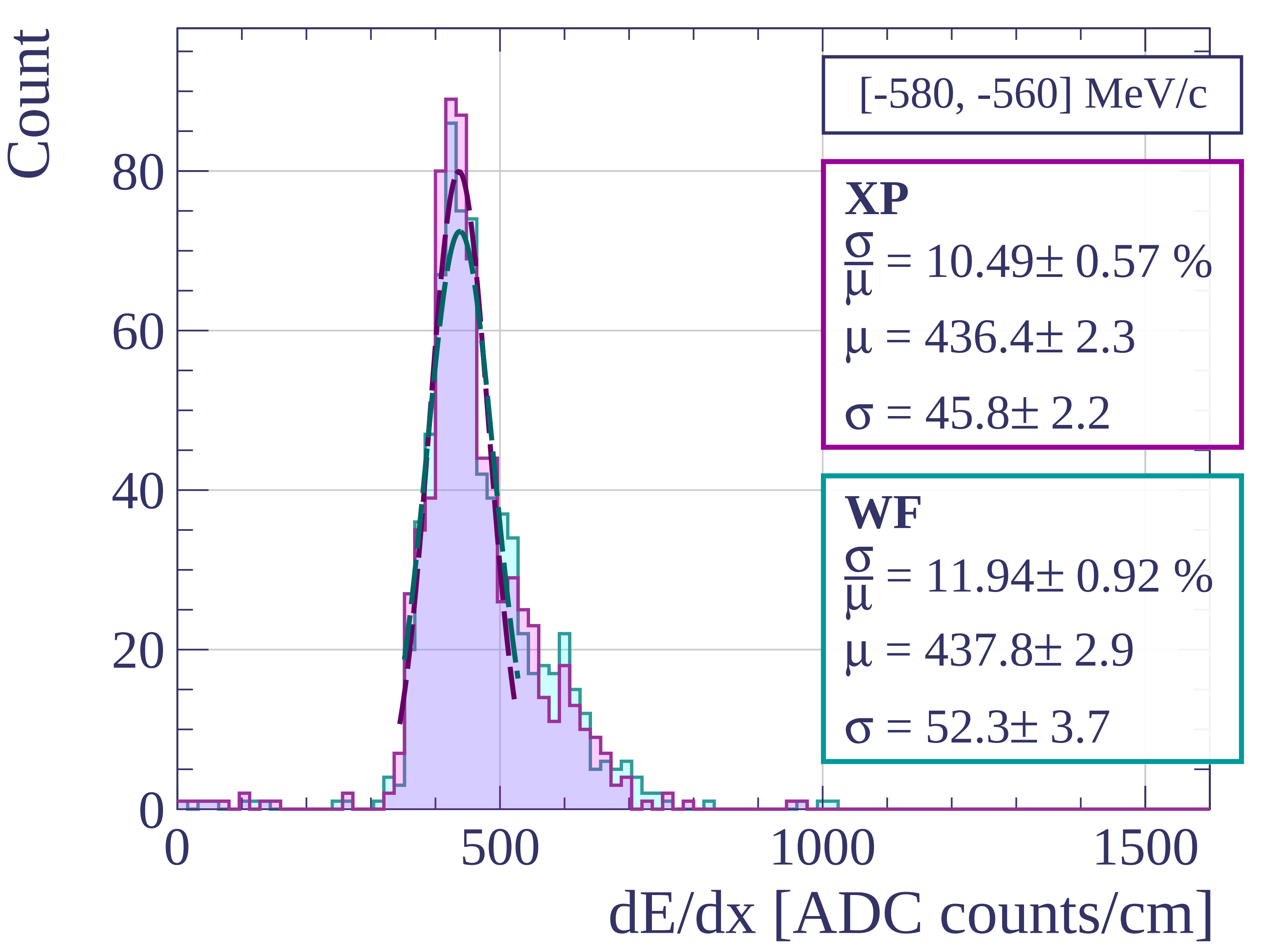


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.08± 0.51 %
μ = 438.1± 2.0
σ = 39.8± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.53 %
μ = 439.3± 2.0
σ = 38.1± 2.2

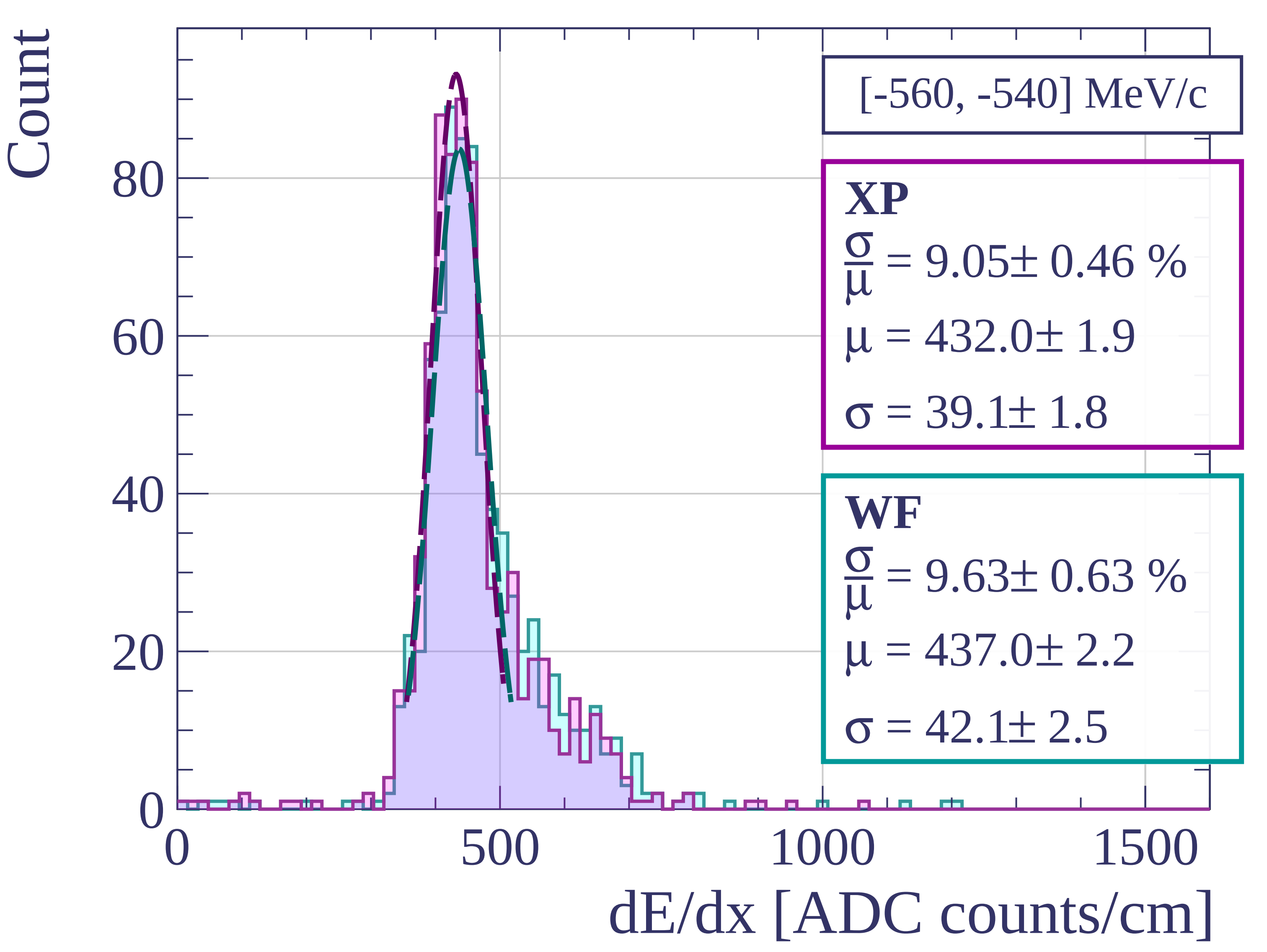


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-620, -600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.95± 0.50 %
μ = 438.1± 2.1
σ = 39.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 9.92± 0.57 %
μ = 443.9± 2.3
σ = 44.1± 2.3

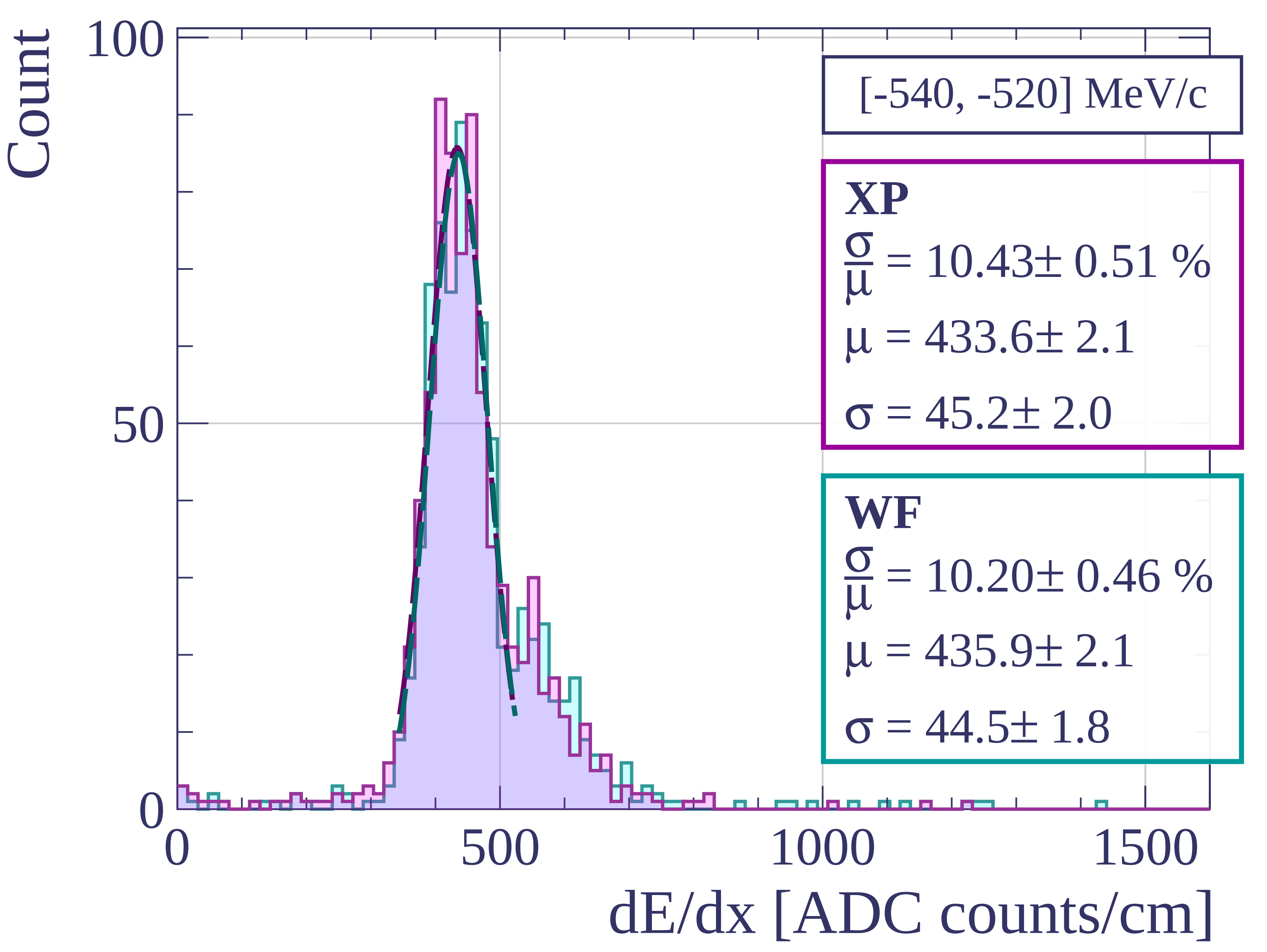

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.89± 0.57 %
μ = 444.6± 2.4
σ = 48.4± 2.3
WF
$\frac{\sigma}{\mu}$ = 11.81± 0.68 %
μ = 453.3± 2.7
σ = 53.5± 2.7



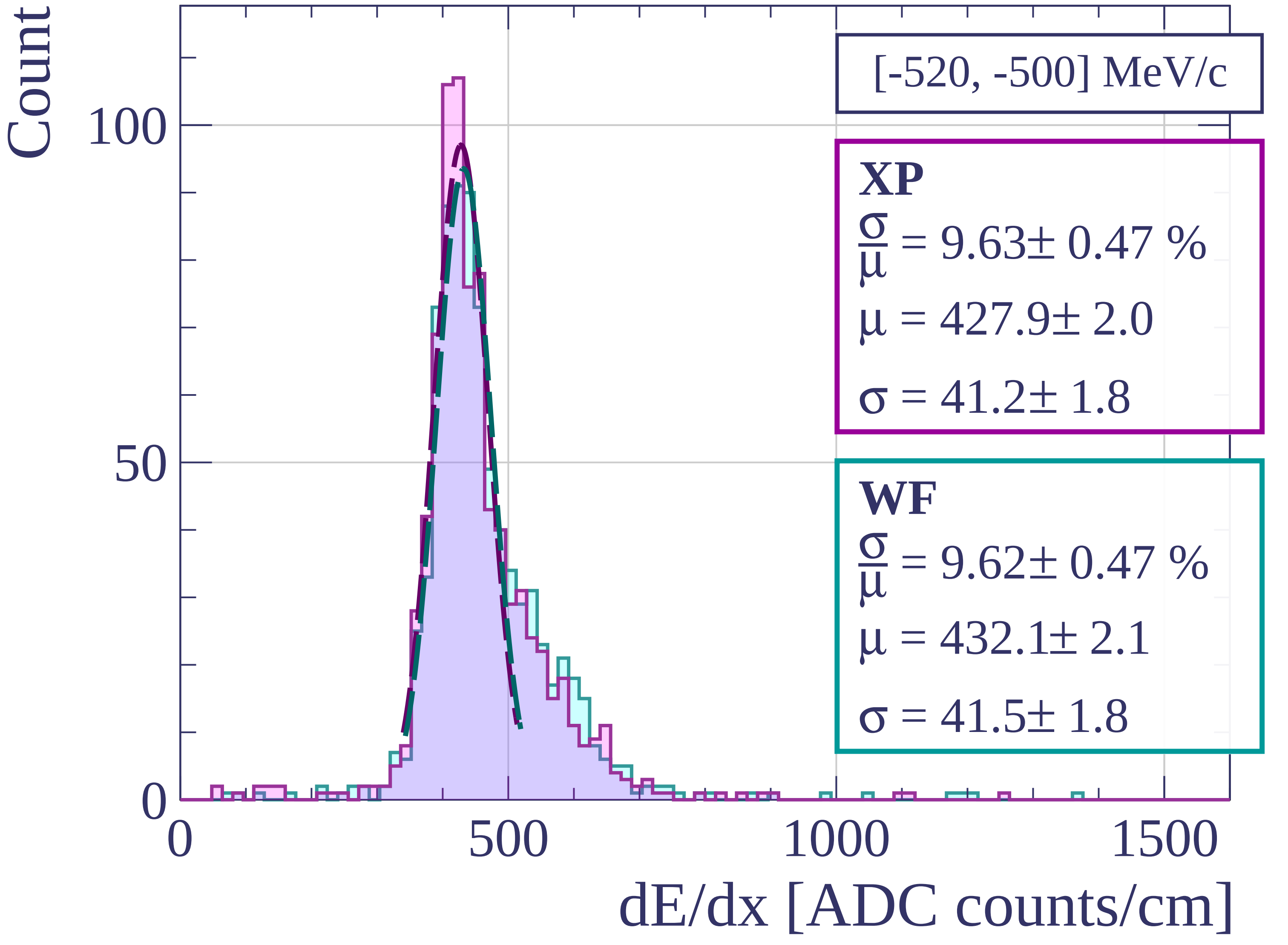
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-580, -560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.49± 0.57 %
μ = 436.4± 2.3
σ = 45.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 11.94± 0.92 %
μ = 437.8± 2.9
σ = 52.3± 3.7



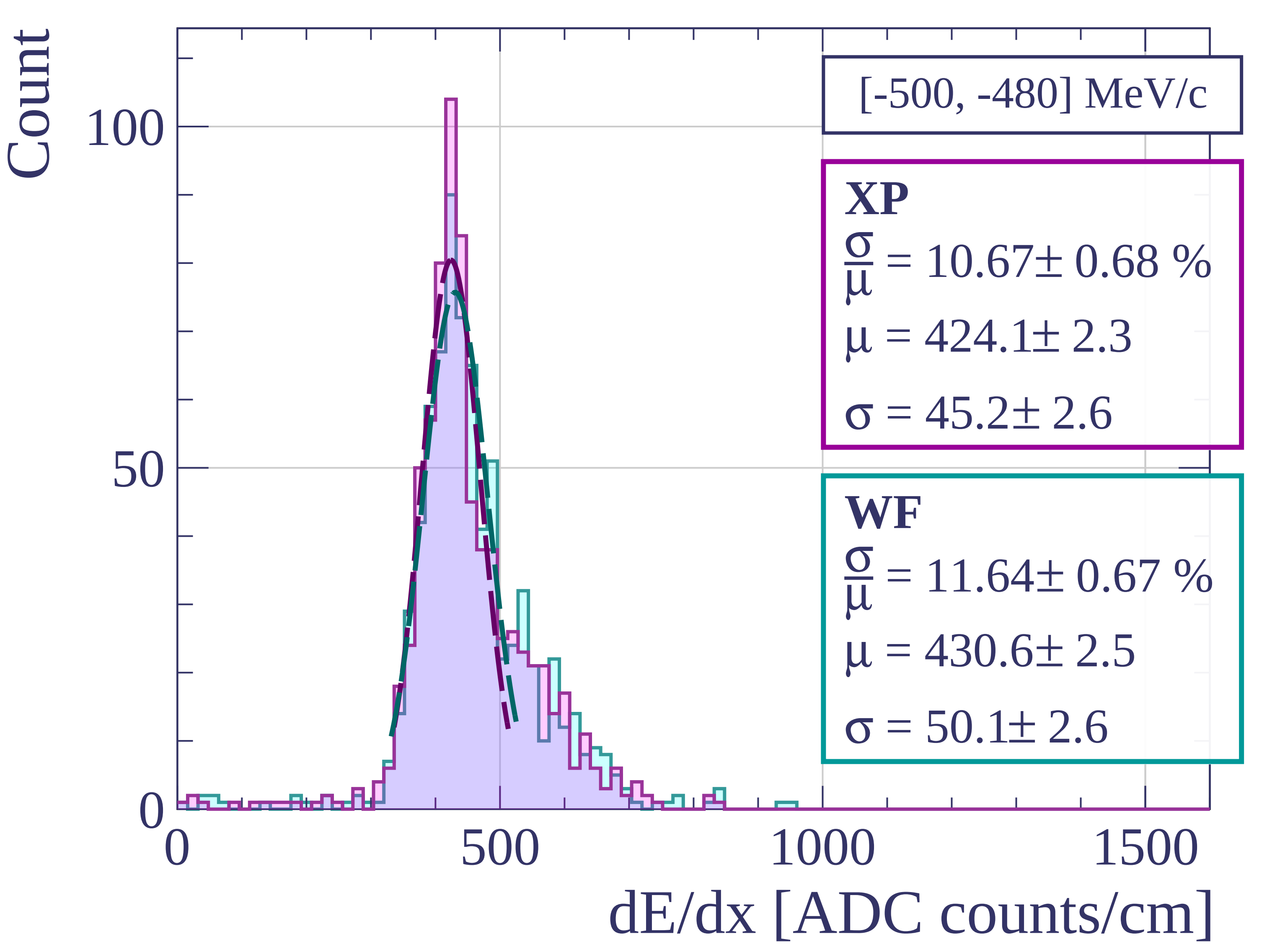
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
80
[-560, -540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.05± 0.46 %
μ = 432.0± 1.9
σ = 39.1± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.63± 0.63 %
μ = 437.0± 2.2
σ = 42.1± 2.5



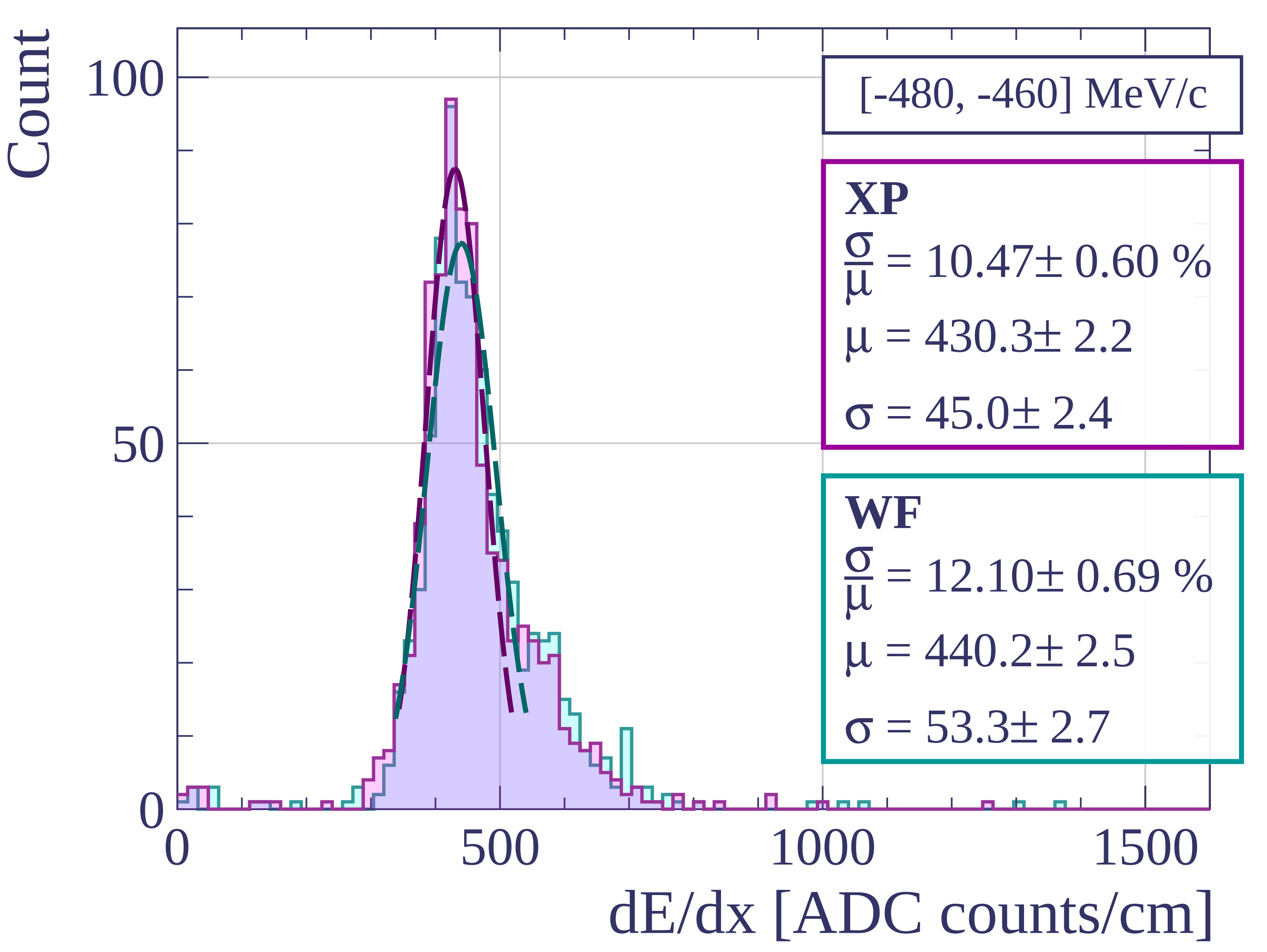
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
σ/μ = 10.43± 0.51 %
μ = 433.6± 2.1
σ = 45.2± 2.0
WF
σ/μ = 10.20± 0.46 %
μ = 435.9± 2.1
σ = 44.5± 1.8



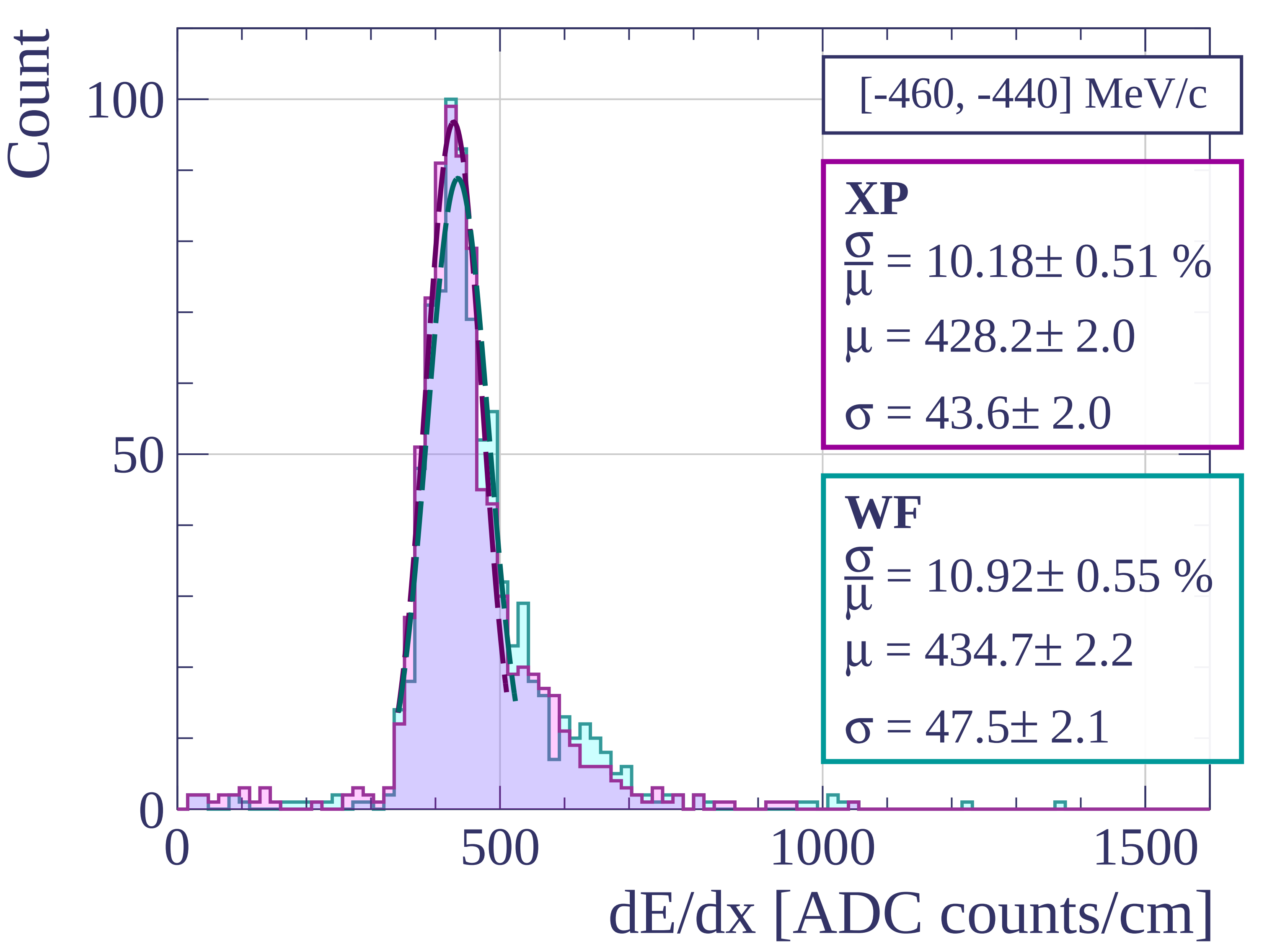
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-520, -500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.63± 0.47 %
μ = 427.9± 2.0
σ = 41.2± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.62± 0.47 %
μ = 432.1± 2.1
σ = 41.5± 1.8



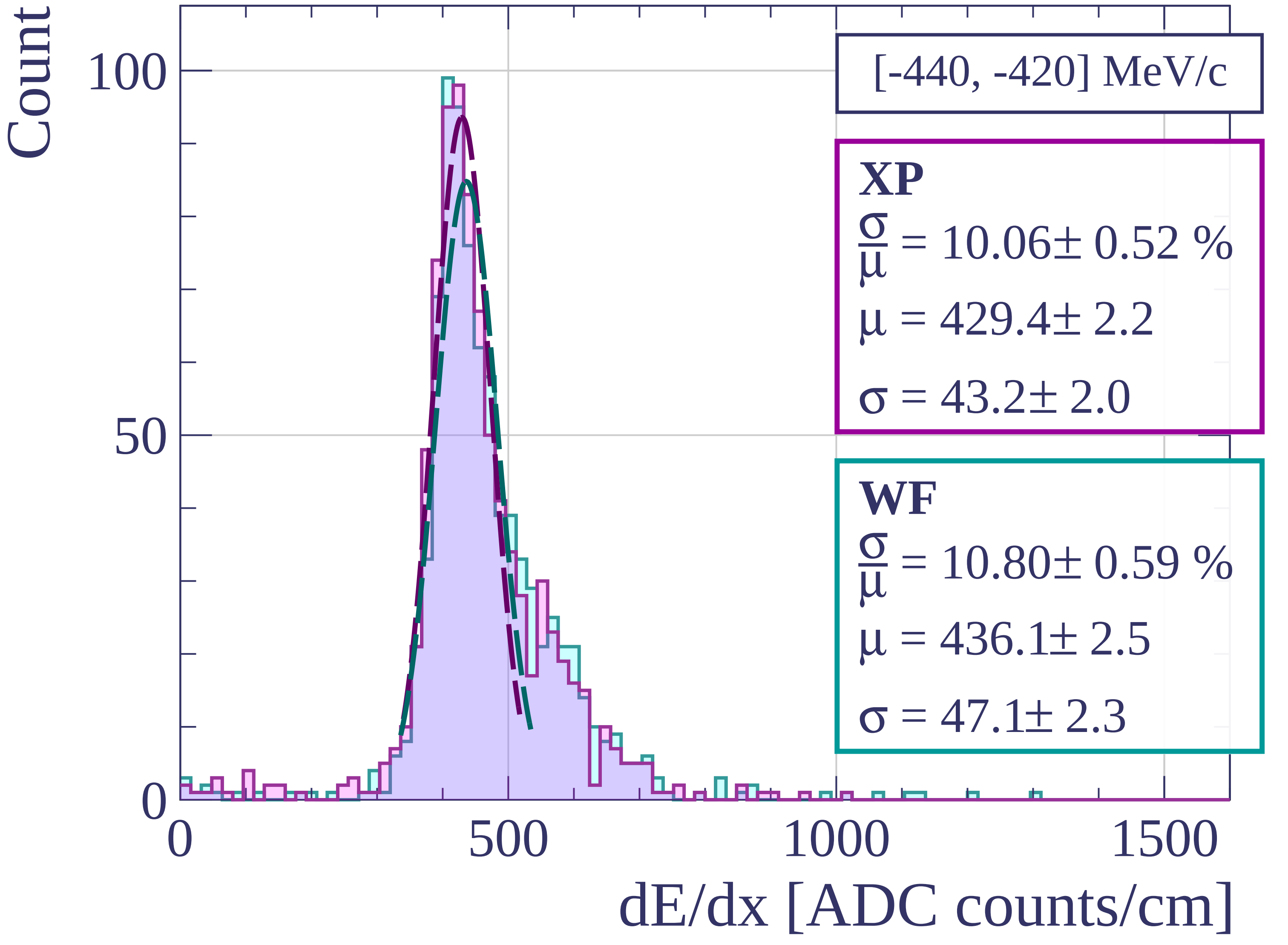
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-500, -480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.67± 0.68 %
μ = 424.1± 2.3
σ = 45.2± 2.6
WF
$\frac{\sigma}{\mu}$ = 11.64± 0.67 %
μ = 430.6± 2.5
σ = 50.1± 2.6

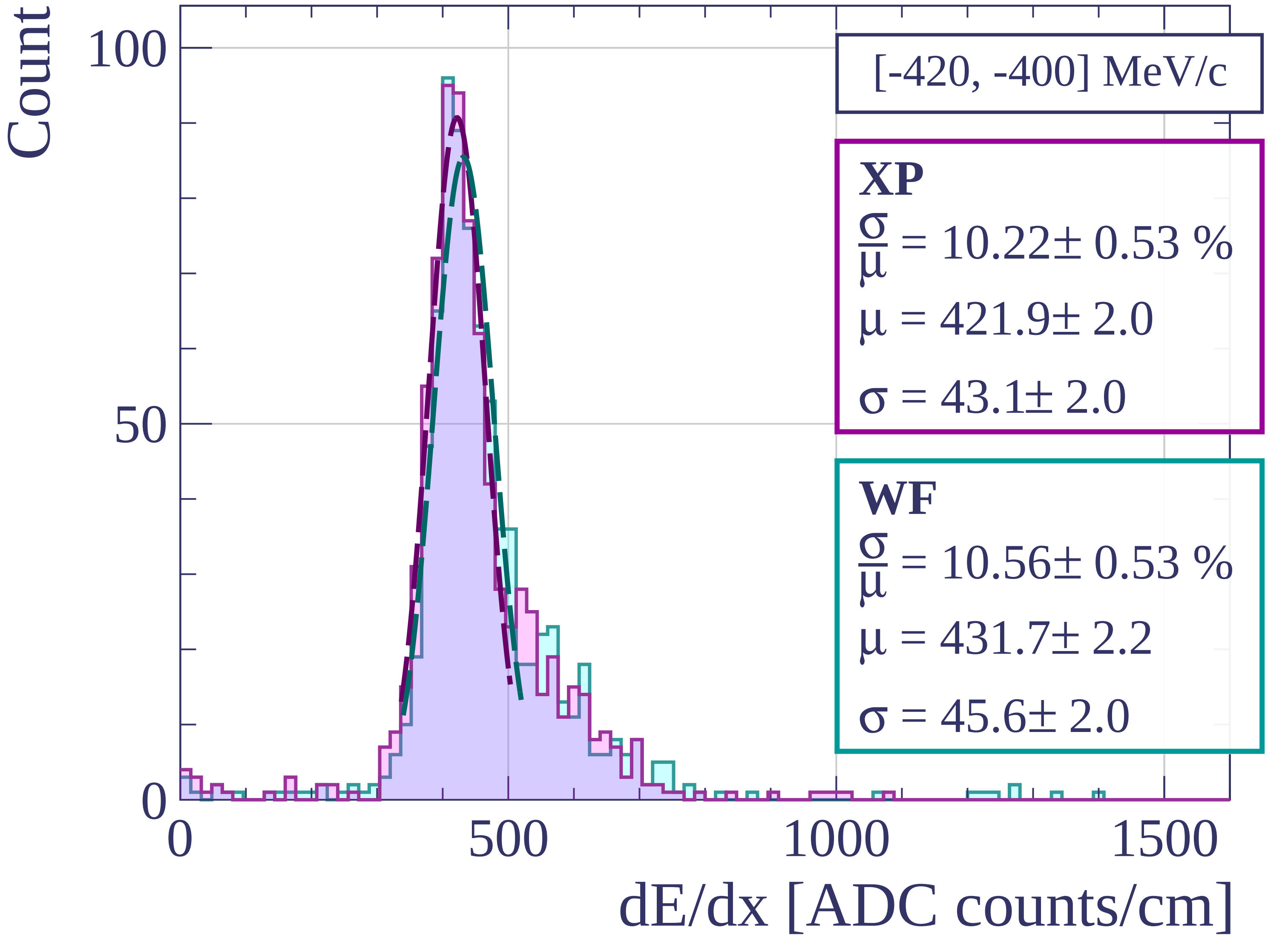


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-480, -460] MeV/c
XP
σ/μ = 10.47± 0.60 %
μ = 430.3± 2.2
σ = 45.0± 2.4
WF
σ/μ = 12.10± 0.69 %
μ = 440.2± 2.5
σ = 53.3± 2.7

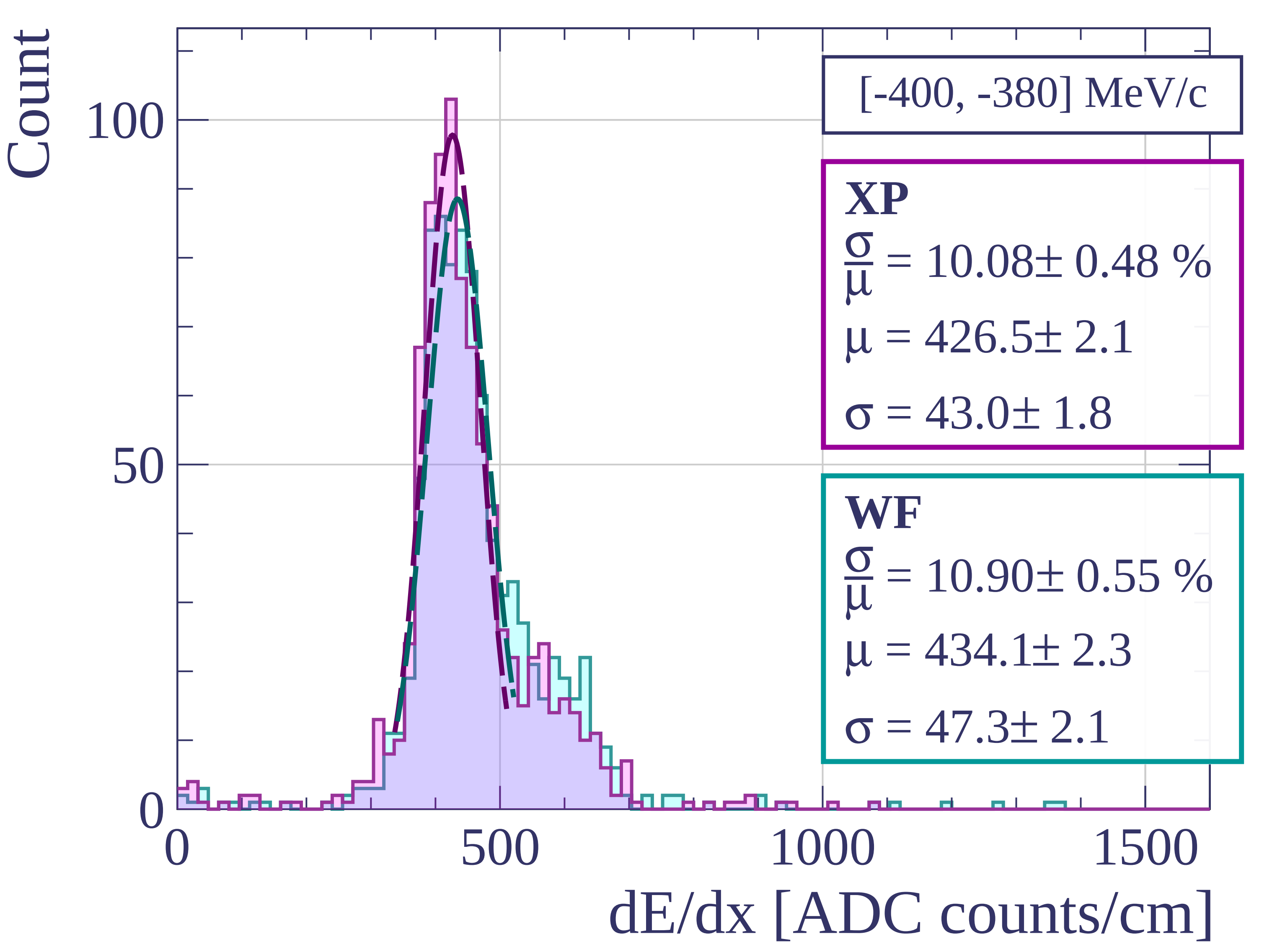


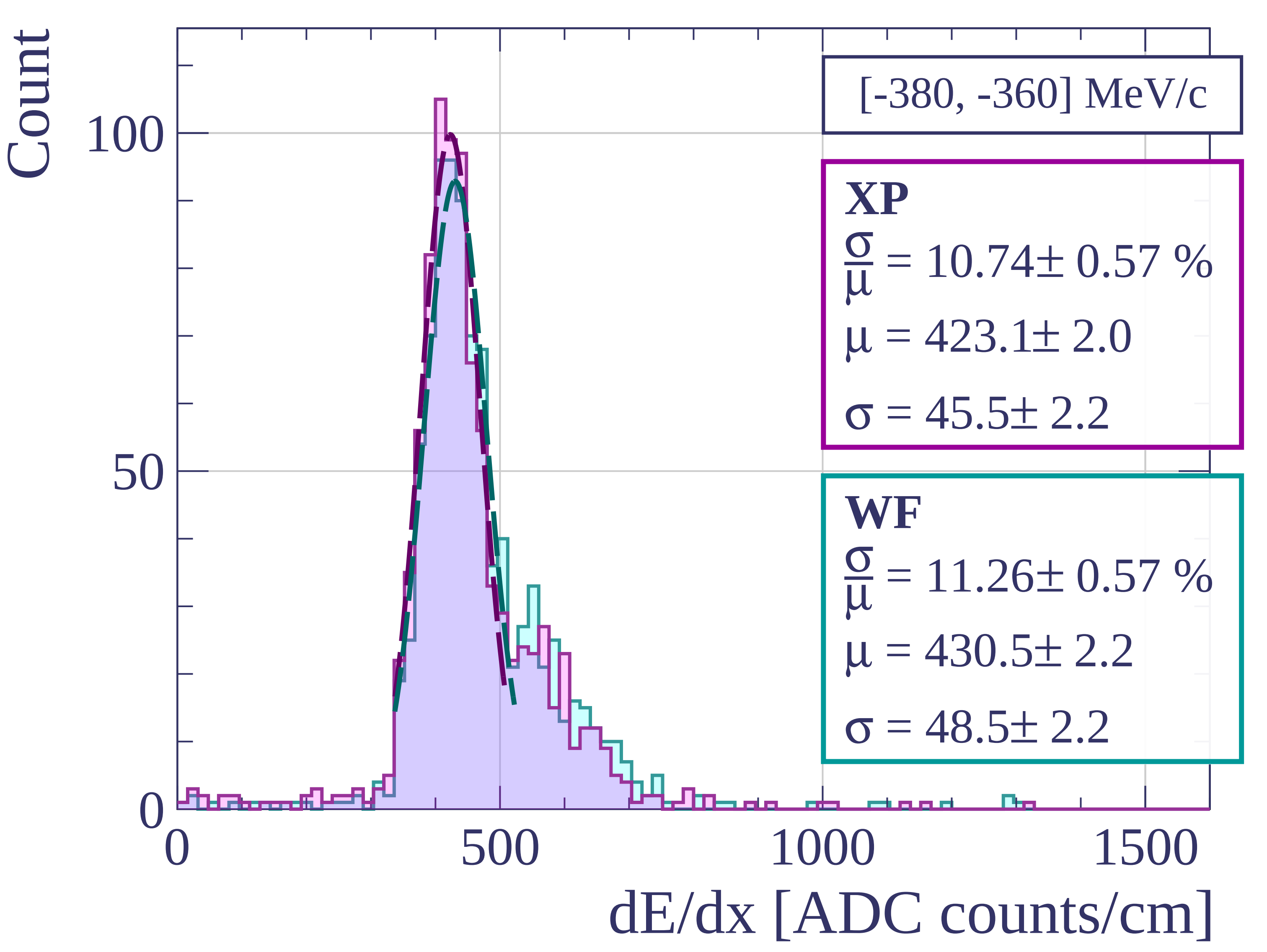
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.18± 0.51 %
μ = 428.2± 2.0
σ = 43.6± 2.0
WF
$\frac{\sigma}{\mu}$ = 10.92± 0.55 %
μ = 434.7± 2.2
σ = 47.5± 2.1


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-440, -420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.06± 0.52 %
μ = 429.4± 2.2
σ = 43.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 10.80± 0.59 %
μ = 436.1± 2.5
σ = 47.1± 2.3

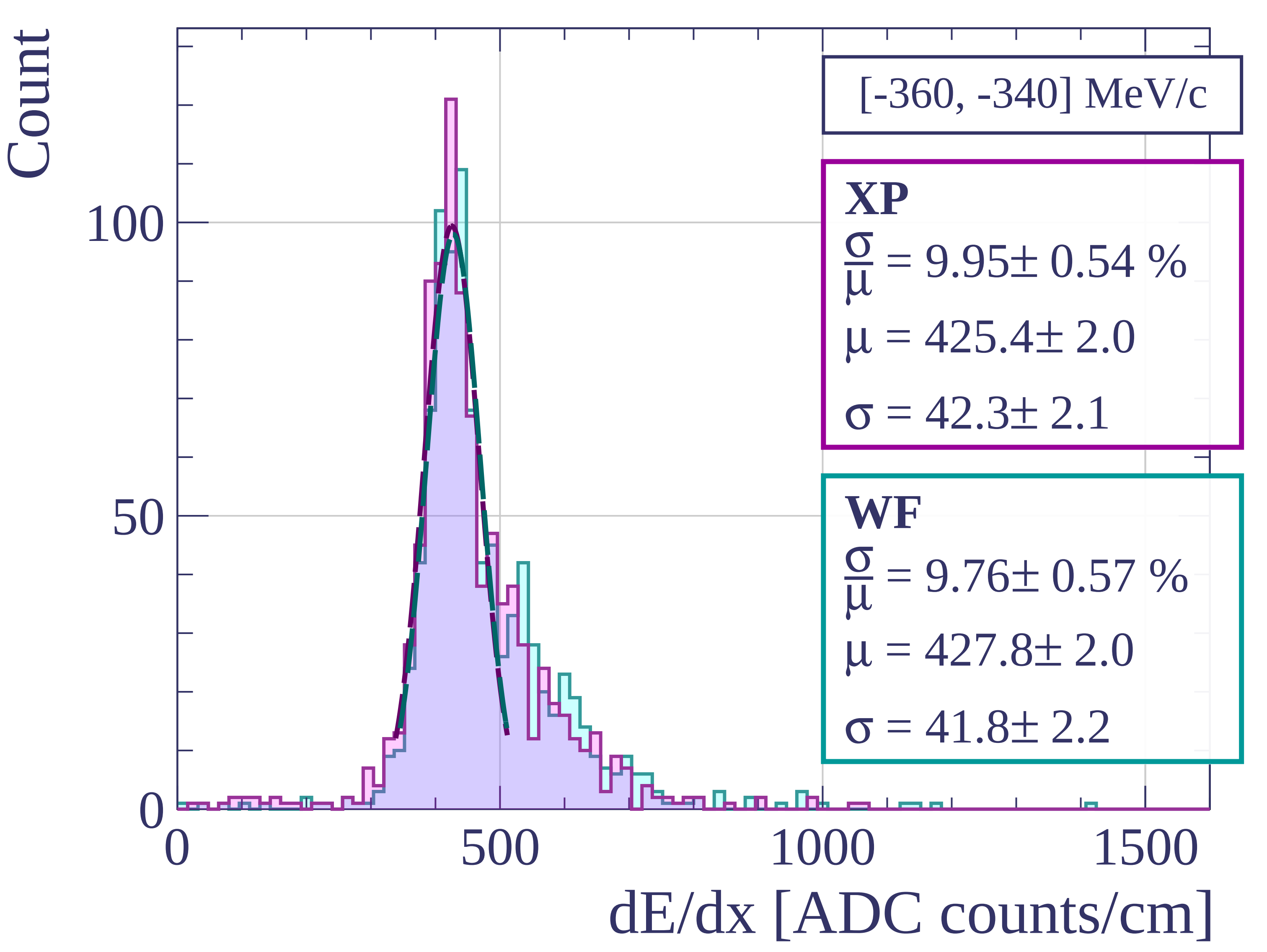


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.22± 0.53 %
μ = 421.9± 2.0
σ = 43.1± 2.0
WF
$\frac{\sigma}{\mu}$ = 10.56± 0.53 %
μ = 431.7± 2.2
σ = 45.6± 2.0

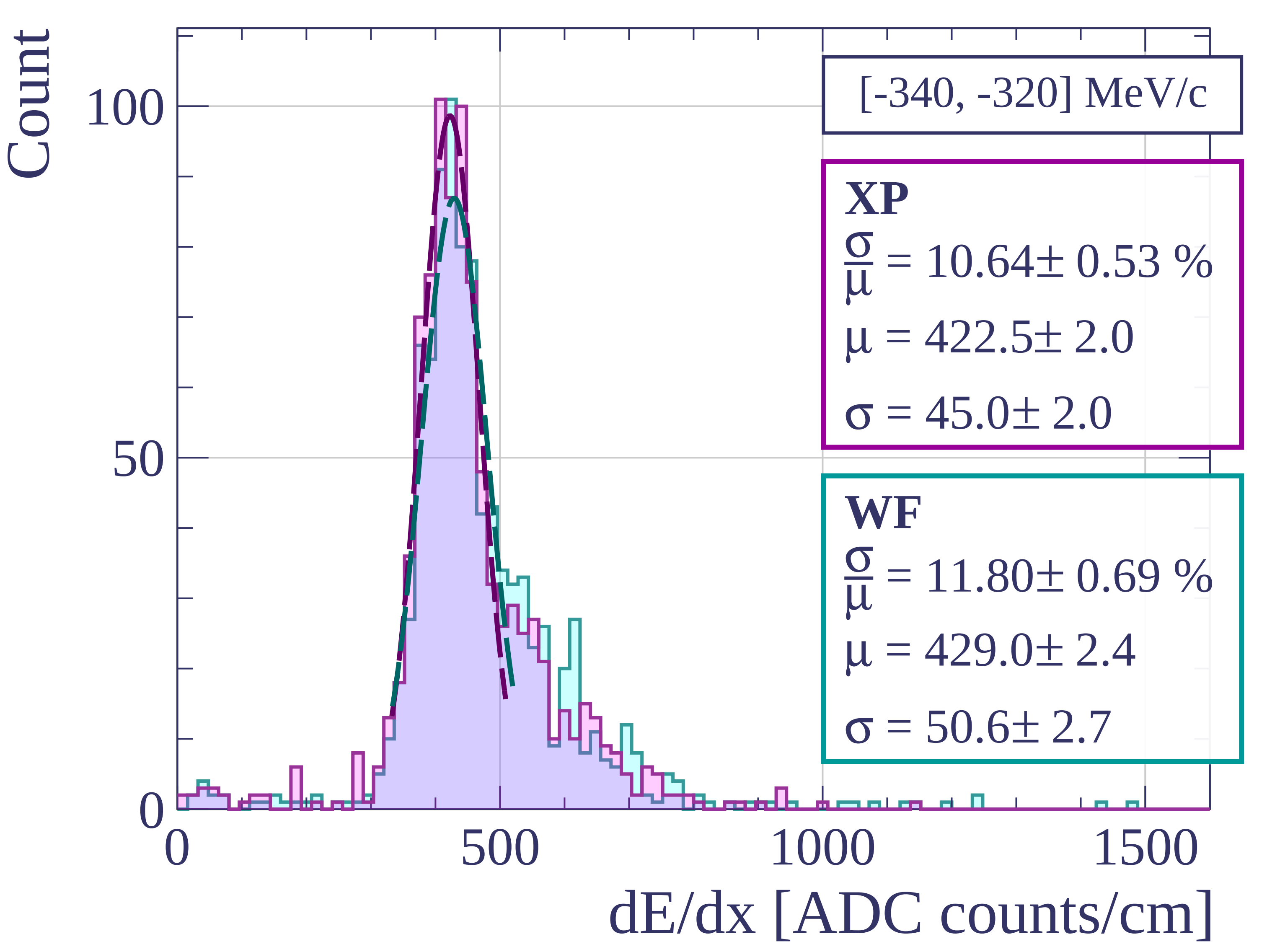

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.08± 0.48 %
μ = 426.5± 2.1
σ = 43.0± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.90± 0.55 %
μ = 434.1± 2.3
σ = 47.3± 2.1



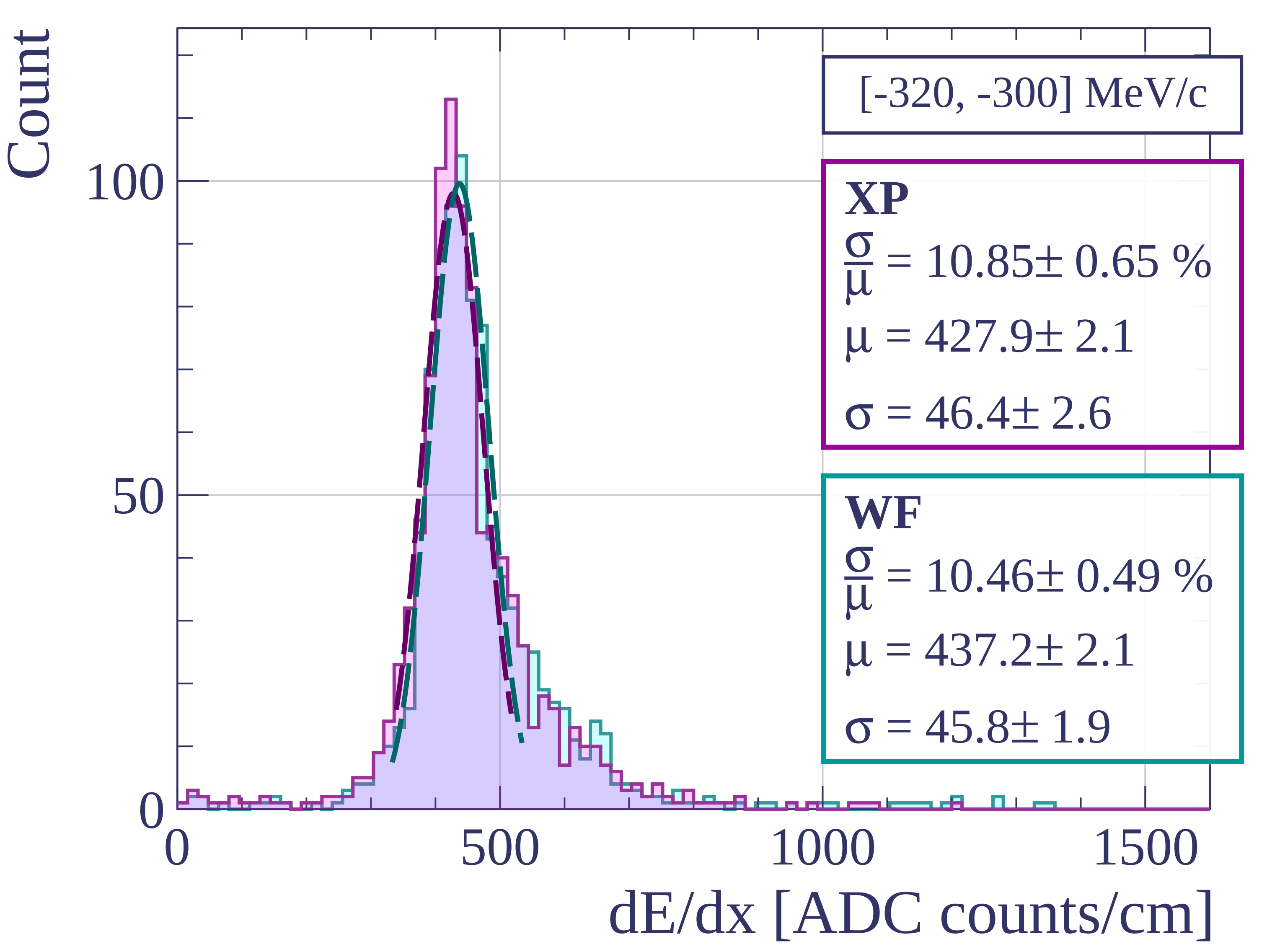
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-380, -360] MeV/c
XP
σ/μ = 10.74± 0.57 %
μ = 423.1± 2.0
σ = 45.5± 2.2
WF
σ/μ = 11.26± 0.57 %
μ = 430.5± 2.2
σ = 48.5± 2.2



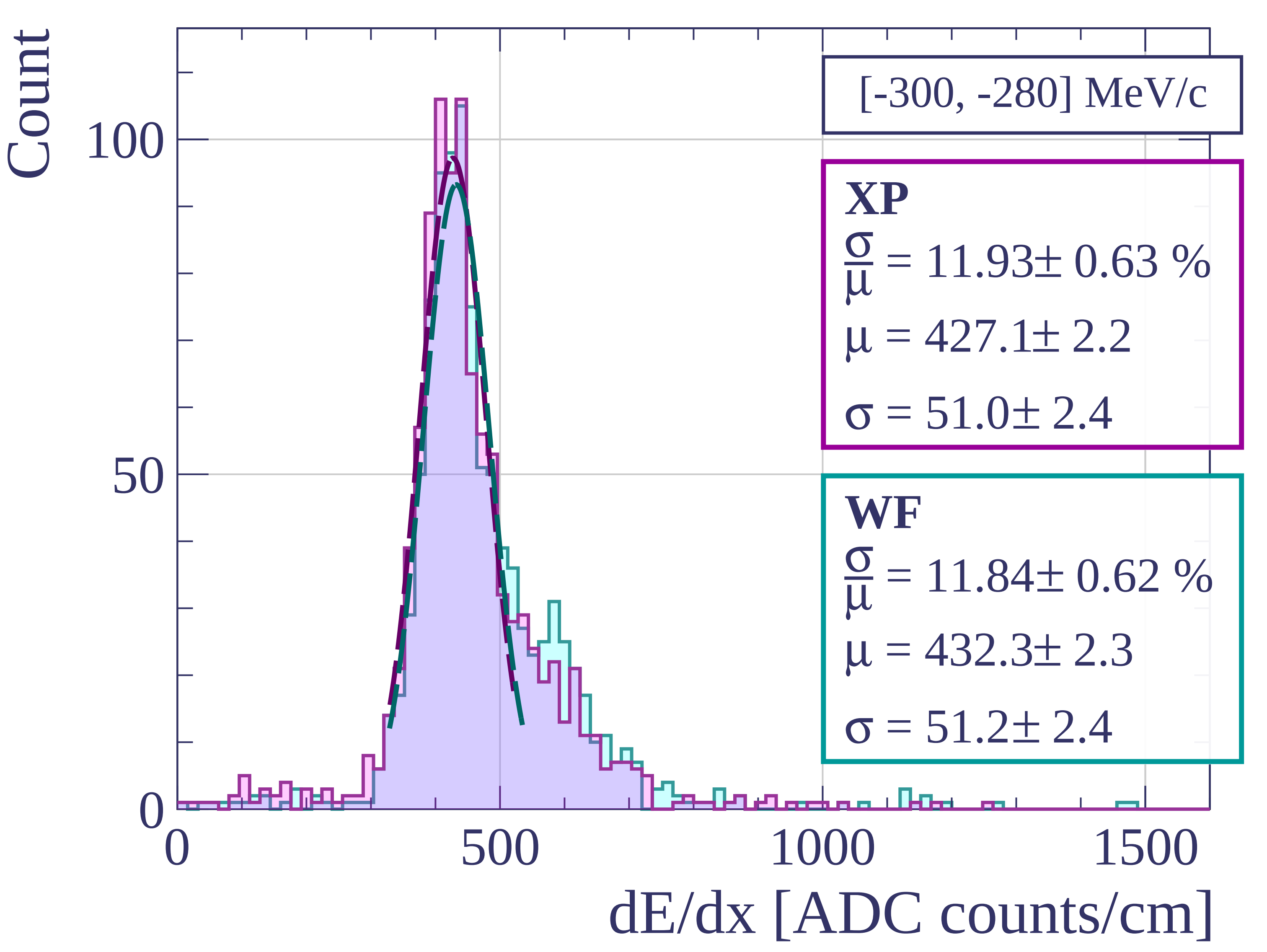
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[-360, -340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.95± 0.54 %
μ = 425.4± 2.0
σ = 42.3± 2.1
WF
$\frac{\sigma}{\mu}$ = 9.76± 0.57 %
μ = 427.8± 2.0
σ = 41.8± 2.2

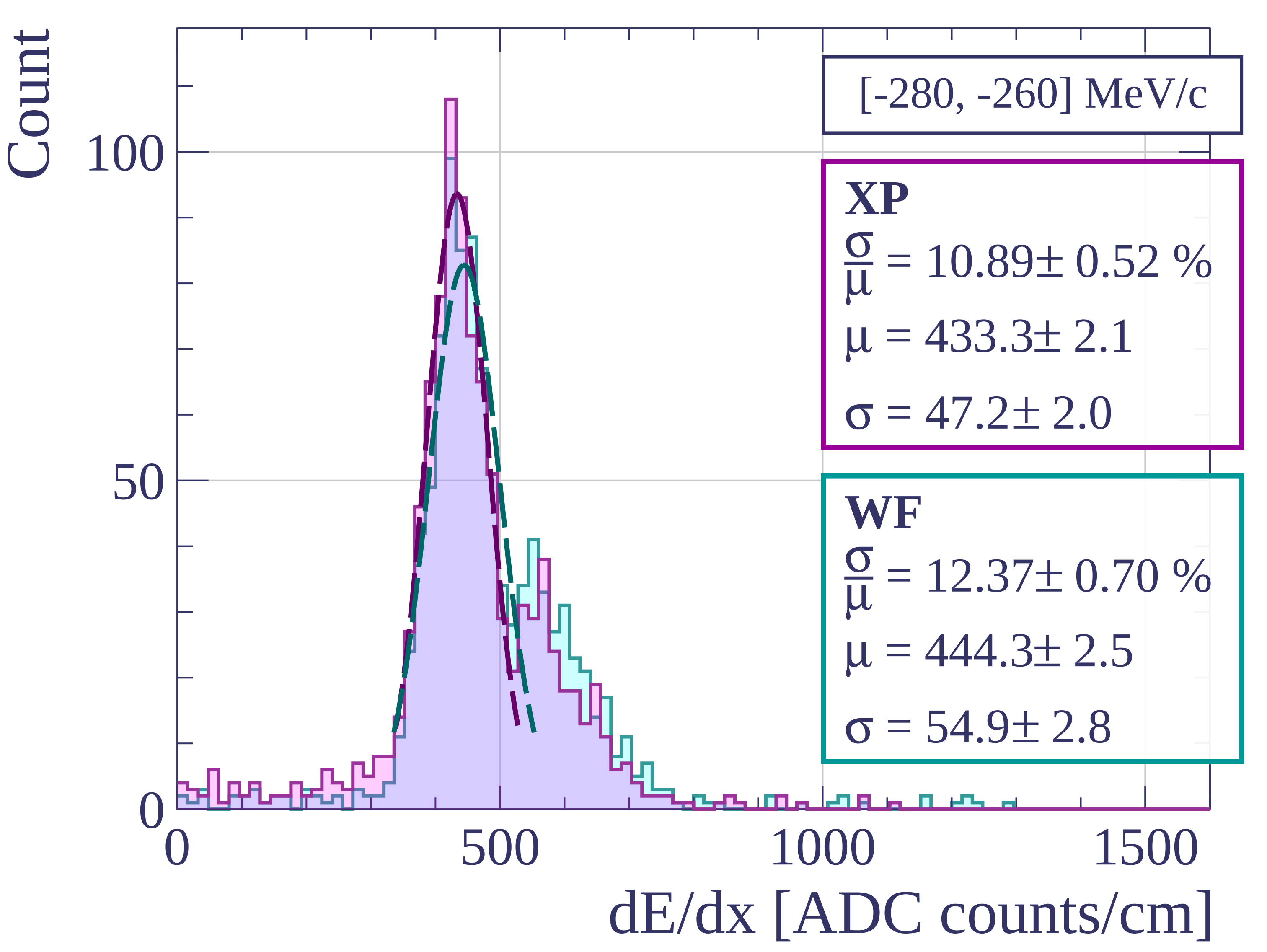


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-340, -320] MeV/c
XP
σ/μ = 10.64± 0.53 %
μ = 422.5± 2.0
σ = 45.0± 2.0
WF
σ/μ = 11.80± 0.69 %
μ = 429.0± 2.4
σ = 50.6± 2.7

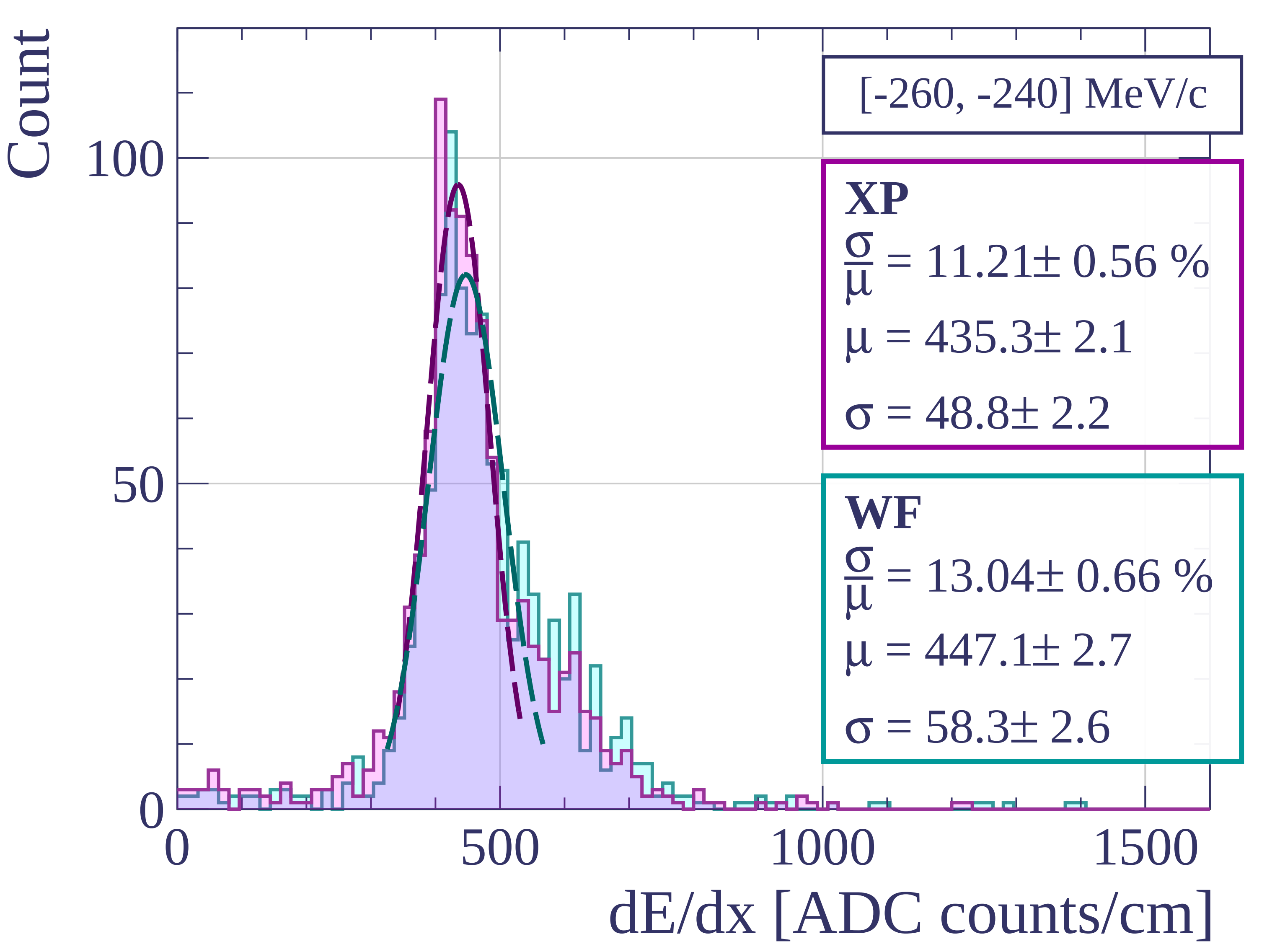


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-320, -300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.85± 0.65 %
μ = 427.9± 2.1
σ = 46.4± 2.6
WF
$\frac{\sigma}{\mu}$ = 10.46± 0.49 %
μ = 437.2± 2.1
σ = 45.8± 1.9

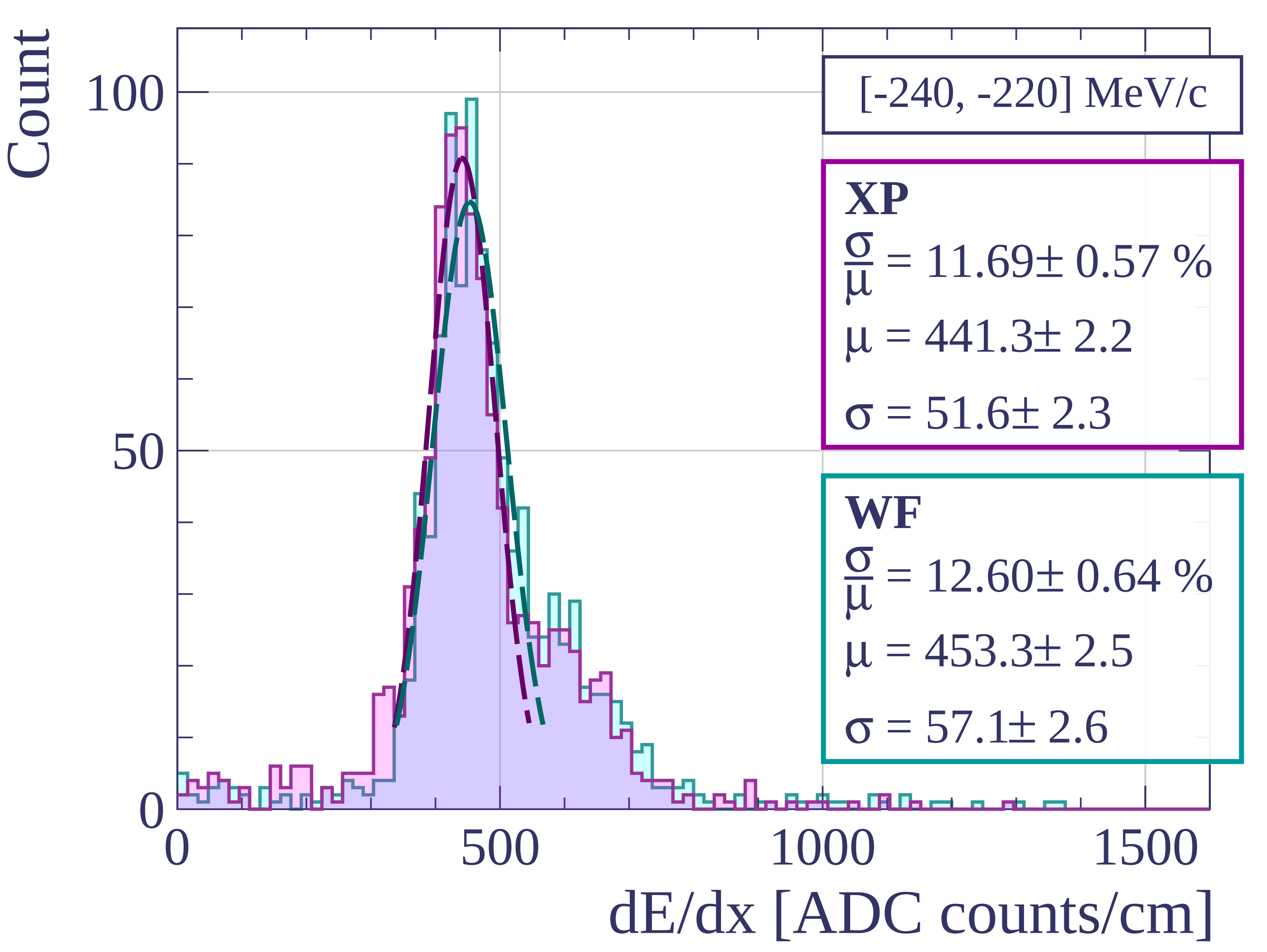

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-300, -280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.93± 0.63 %
μ = 427.1± 2.2
σ = 51.0± 2.4
WF
$\frac{\sigma}{\mu}$ = 11.84± 0.62 %
μ = 432.3± 2.3
σ = 51.2± 2.4

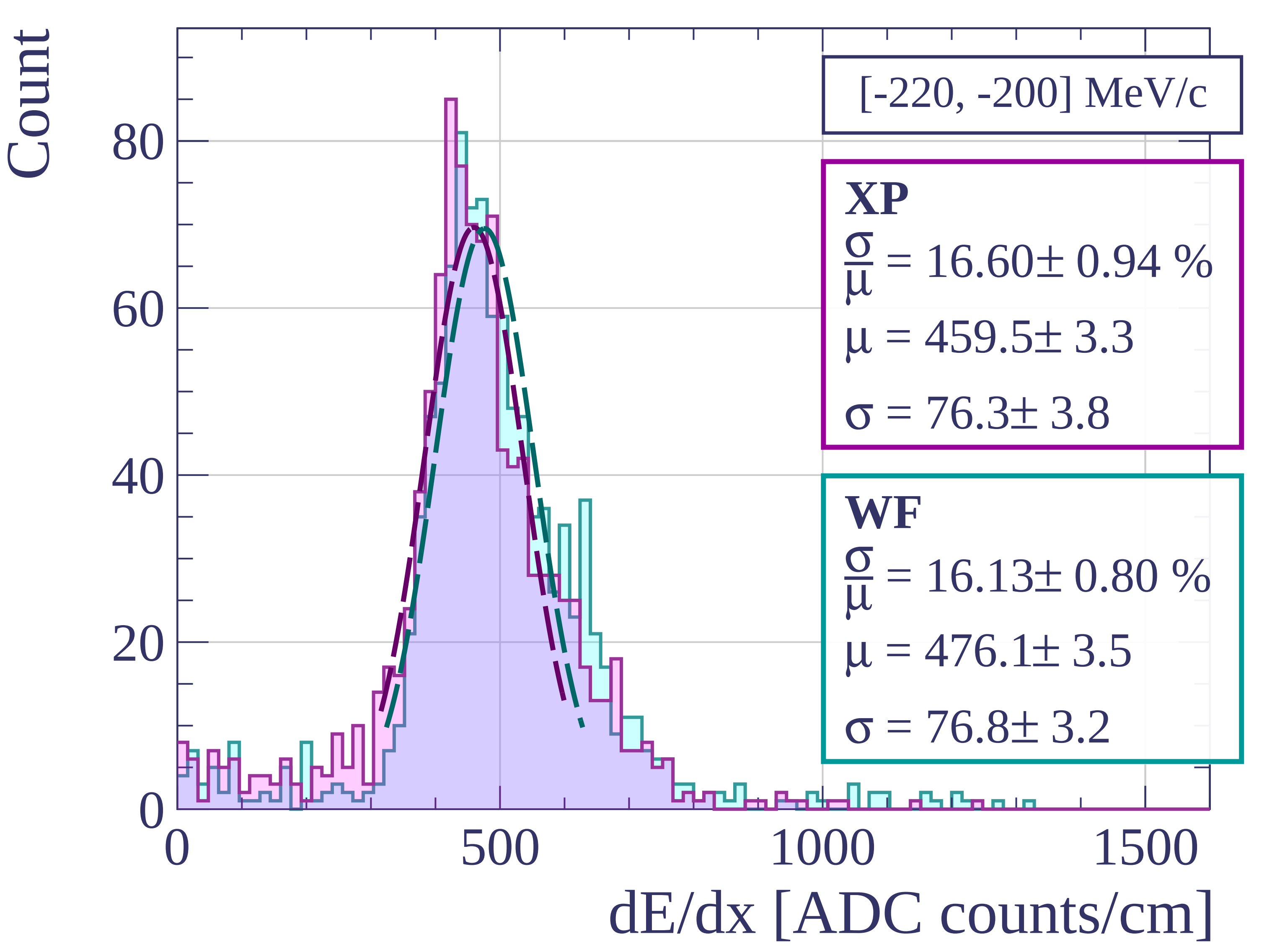


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-280, -260] MeV/c
XP
σ/μ = 10.89± 0.52 %
μ = 433.3± 2.1
σ = 47.2± 2.0
WF
σ/μ = 12.37± 0.70 %
μ = 444.3± 2.5
σ = 54.9± 2.8

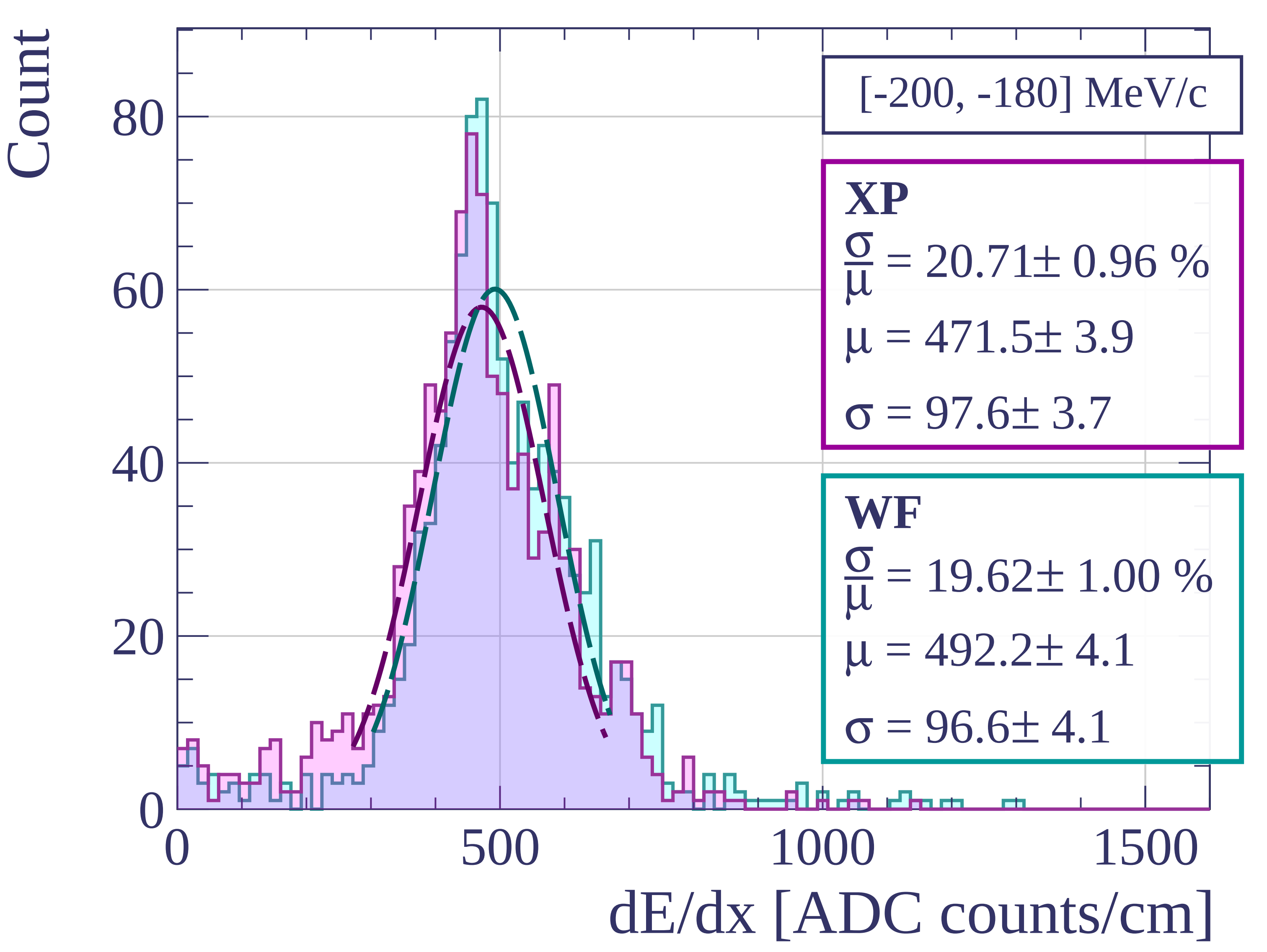


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-260, -240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.21± 0.56 %
μ = 435.3± 2.1
σ = 48.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 13.04± 0.66 %
μ = 447.1± 2.7
σ = 58.3± 2.6

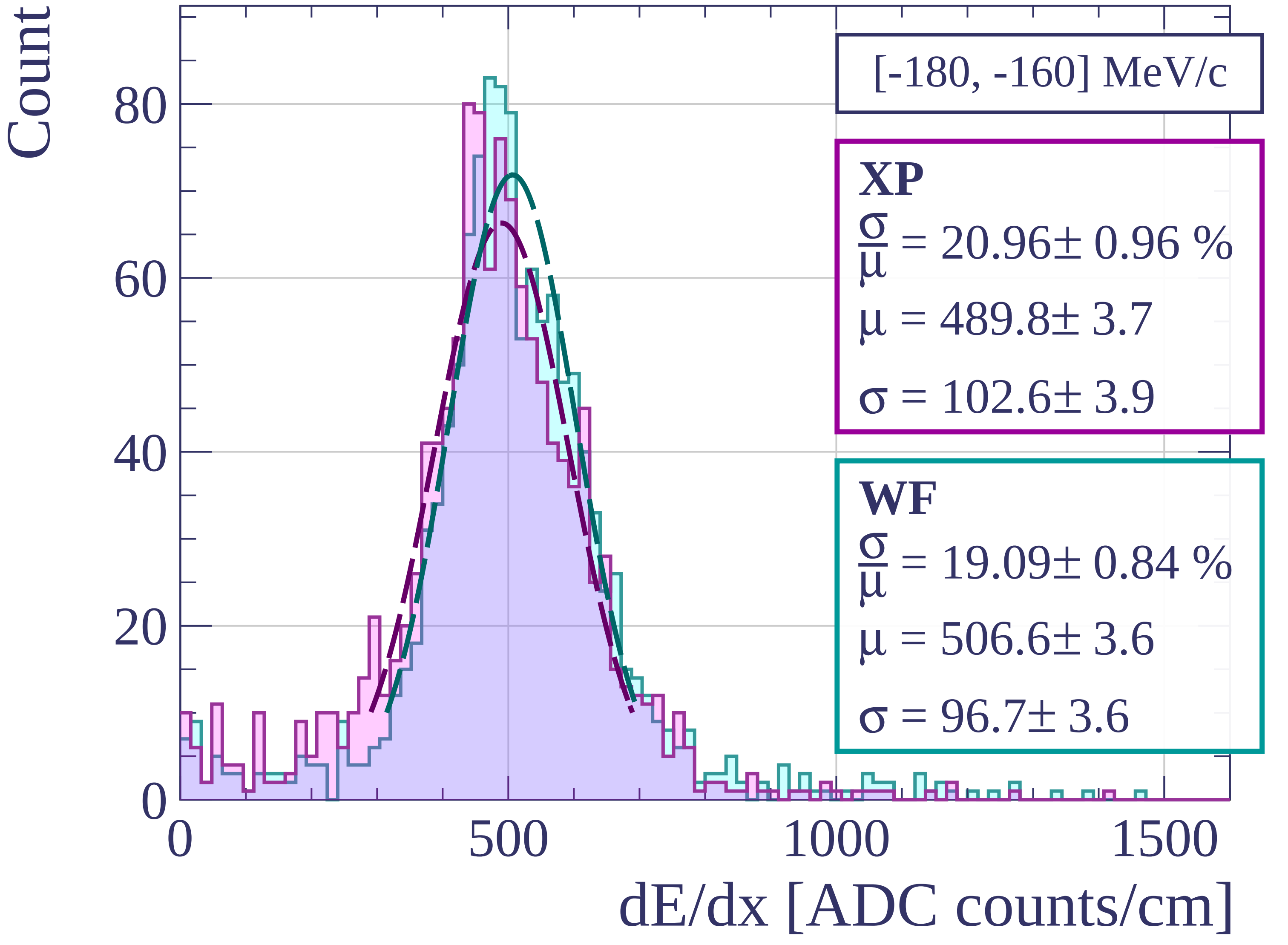

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-240, -220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.69± 0.57 %
μ = 441.3± 2.2
σ = 51.6± 2.3
WF
$\frac{\sigma}{\mu}$ = 12.60± 0.64 %
μ = 453.3± 2.5
σ = 57.1± 2.6



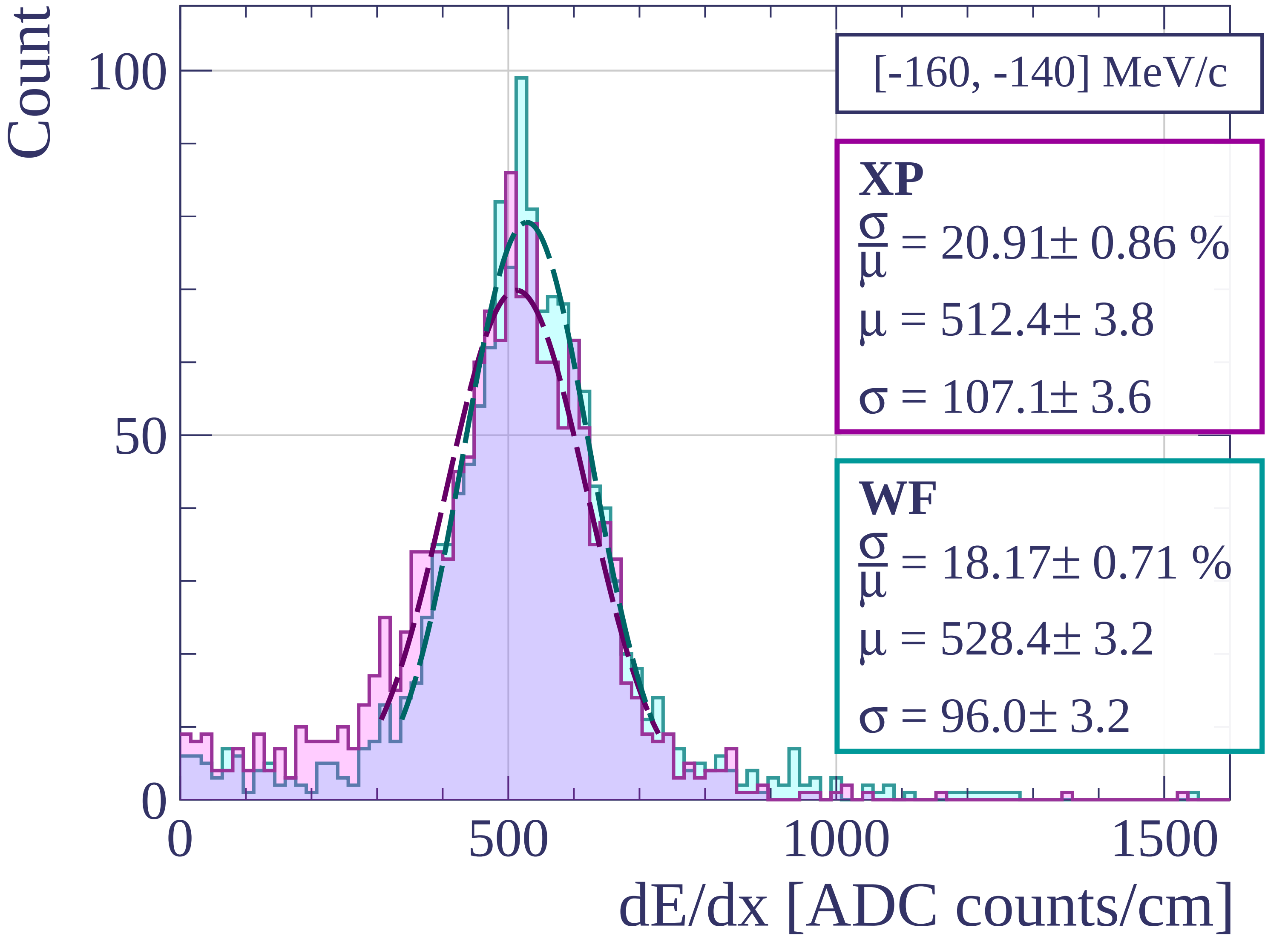
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 16.60± 0.94 %
μ = 459.5± 3.3
σ = 76.3± 3.8
WF
$\frac{\sigma}{\mu}$ = 16.13± 0.80 %
μ = 476.1± 3.5
σ = 76.8± 3.2

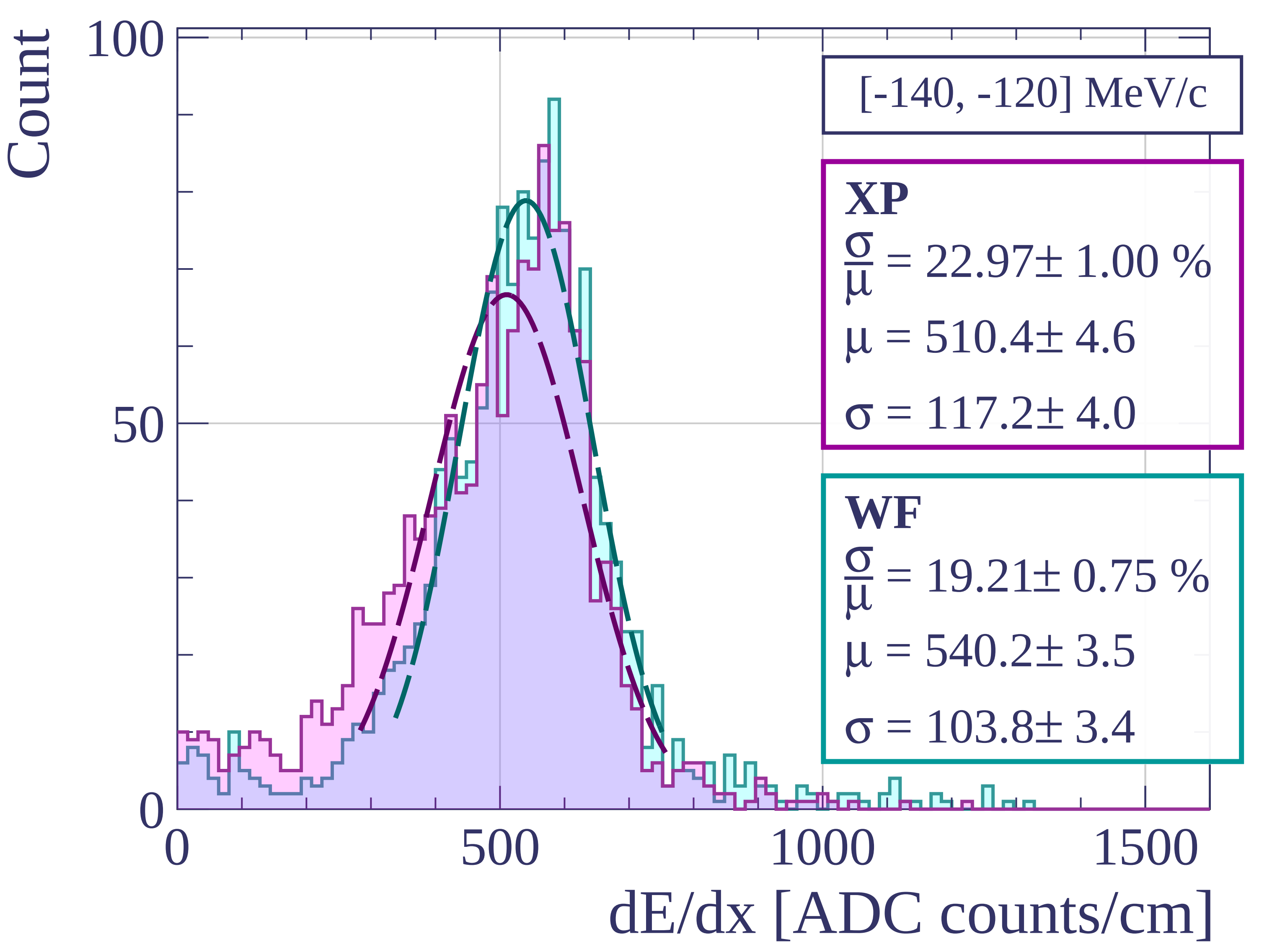


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
20
40
60
80
[-200, -180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.71± 0.96 %
μ = 471.5± 3.9
σ = 97.6± 3.7
WF
$\frac{\sigma}{\mu}$ = 19.62± 1.00 %
μ = 492.2± 4.1
σ = 96.6± 4.1

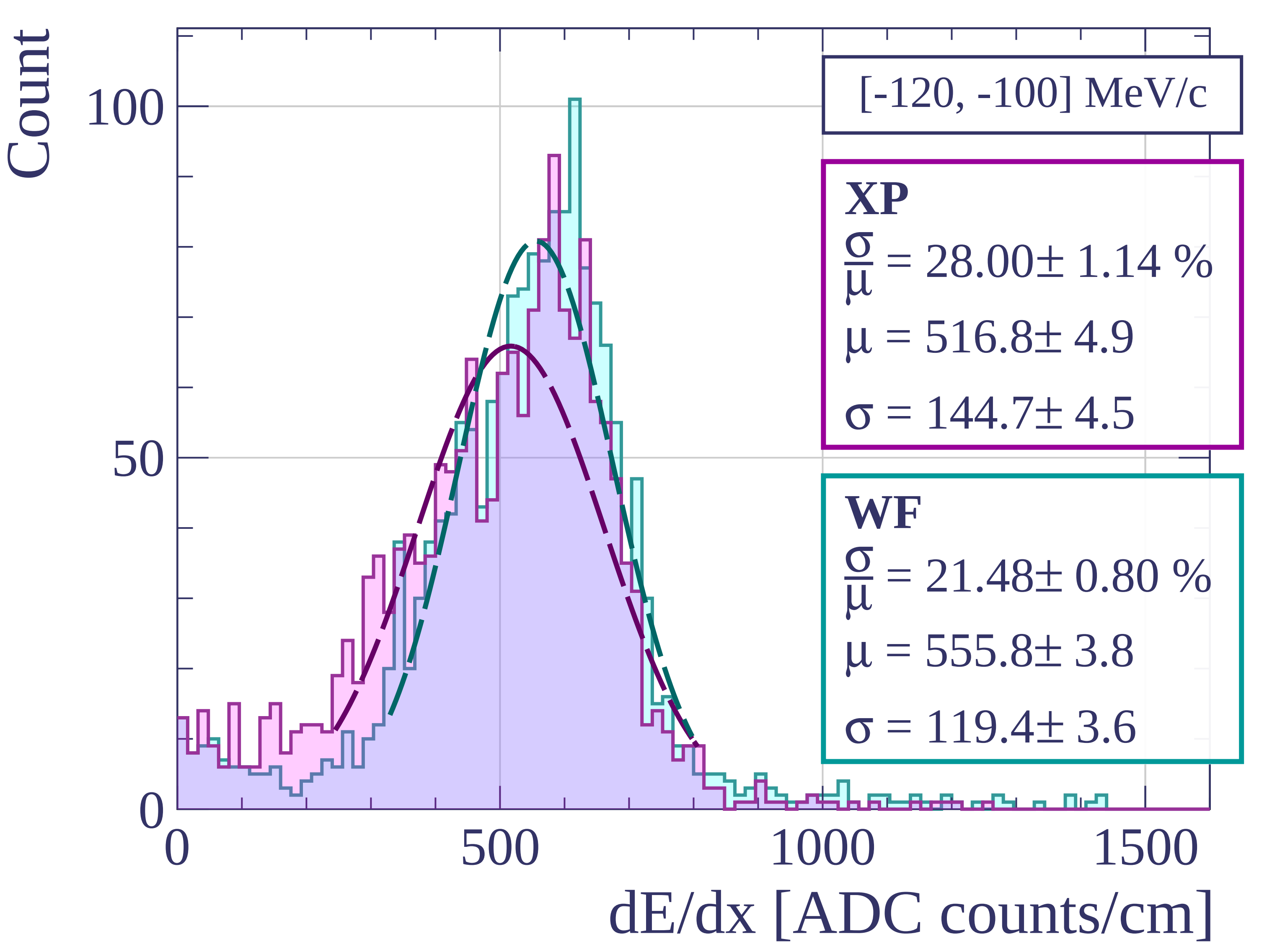


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.96± 0.96 %
μ = 489.8± 3.7
σ = 102.6± 3.9
WF
$\frac{\sigma}{\mu}$ = 19.09± 0.84 %
μ = 506.6± 3.6
σ = 96.7± 3.6

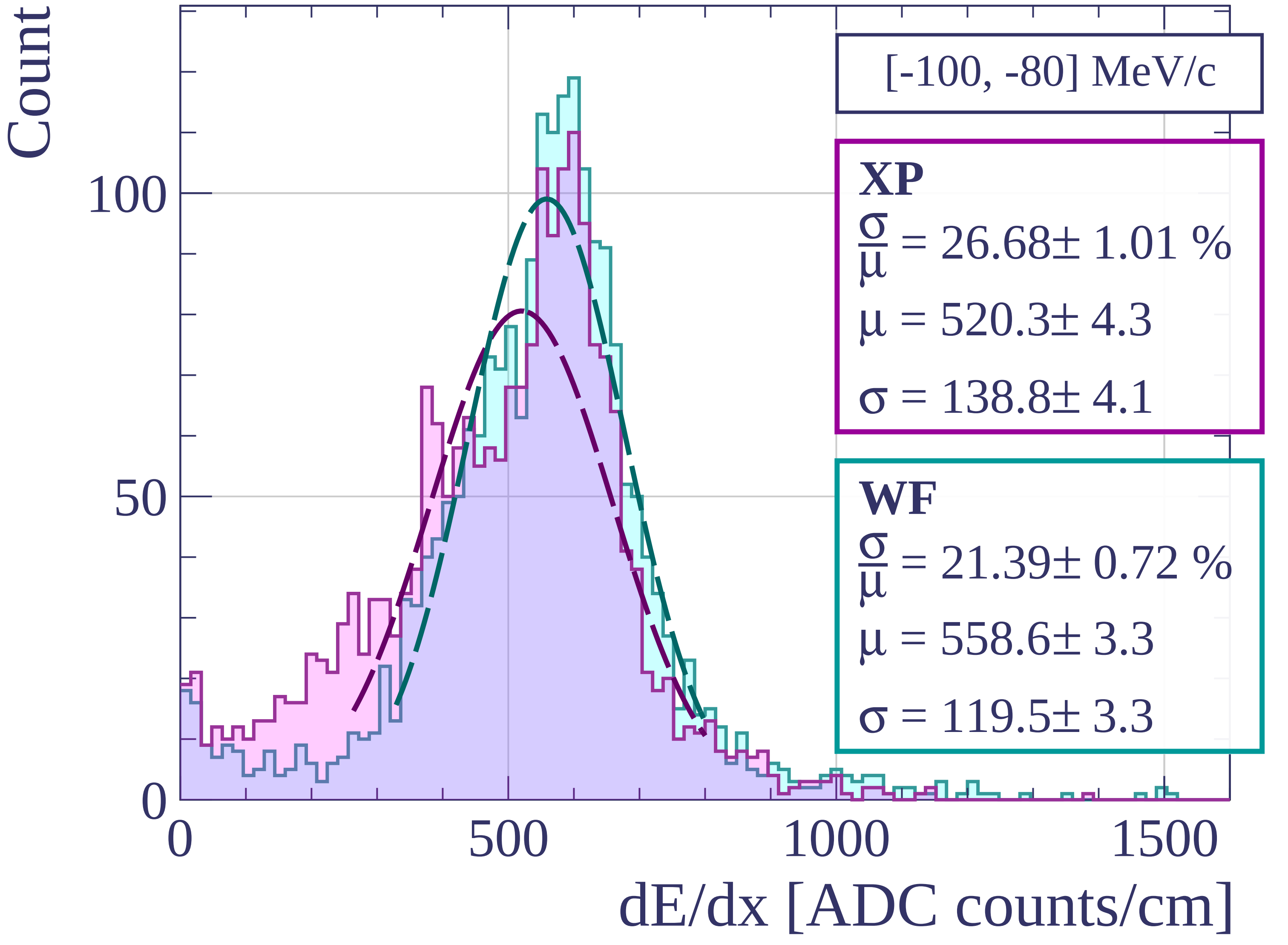

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
σ/μ = 20.91± 0.86 %
μ = 512.4± 3.8
σ = 107.1± 3.6
WF
σ/μ = 18.17± 0.71 %
μ = 528.4± 3.2
σ = 96.0± 3.2



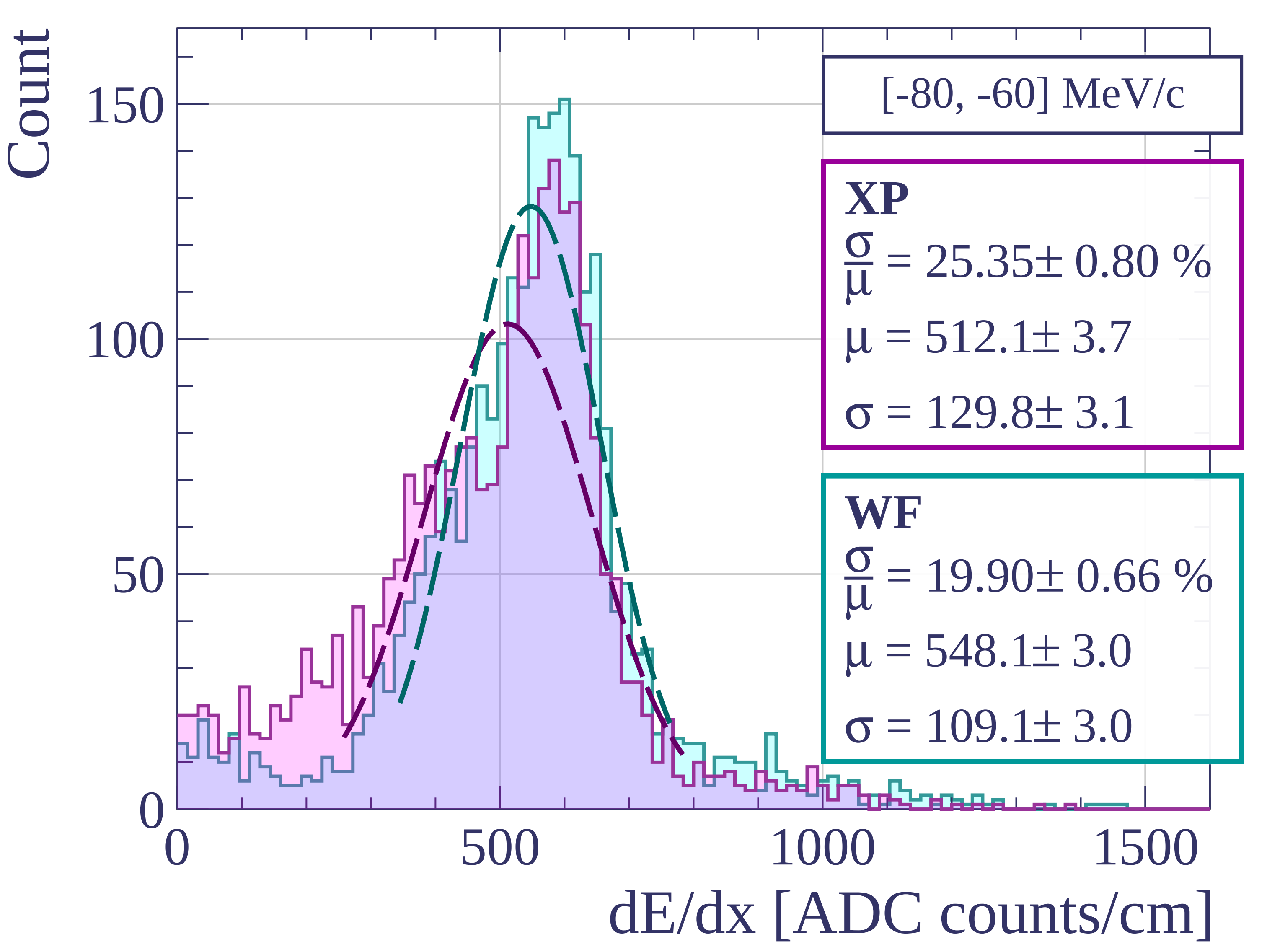
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 22.97± 1.00 %
μ = 510.4± 4.6
σ = 117.2± 4.0
WF
$\frac{\sigma}{\mu}$ = 19.21± 0.75 %
μ = 540.2± 3.5
σ = 103.8± 3.4



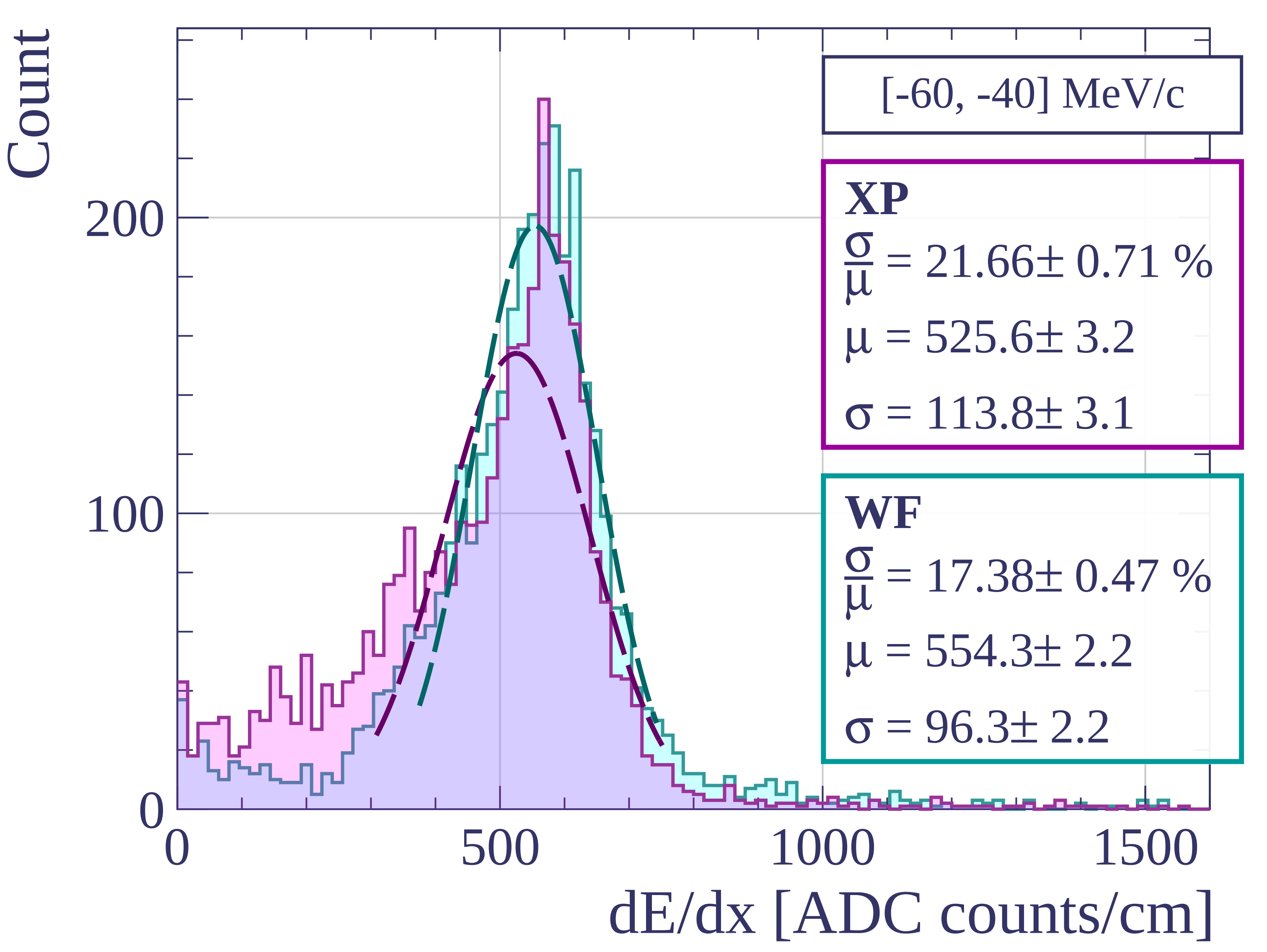
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-120, -100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 28.00± 1.14 %
μ = 516.8± 4.9
σ = 144.7± 4.5
WF
$\frac{\sigma}{\mu}$ = 21.48± 0.80 %
μ = 555.8± 3.8
σ = 119.4± 3.6



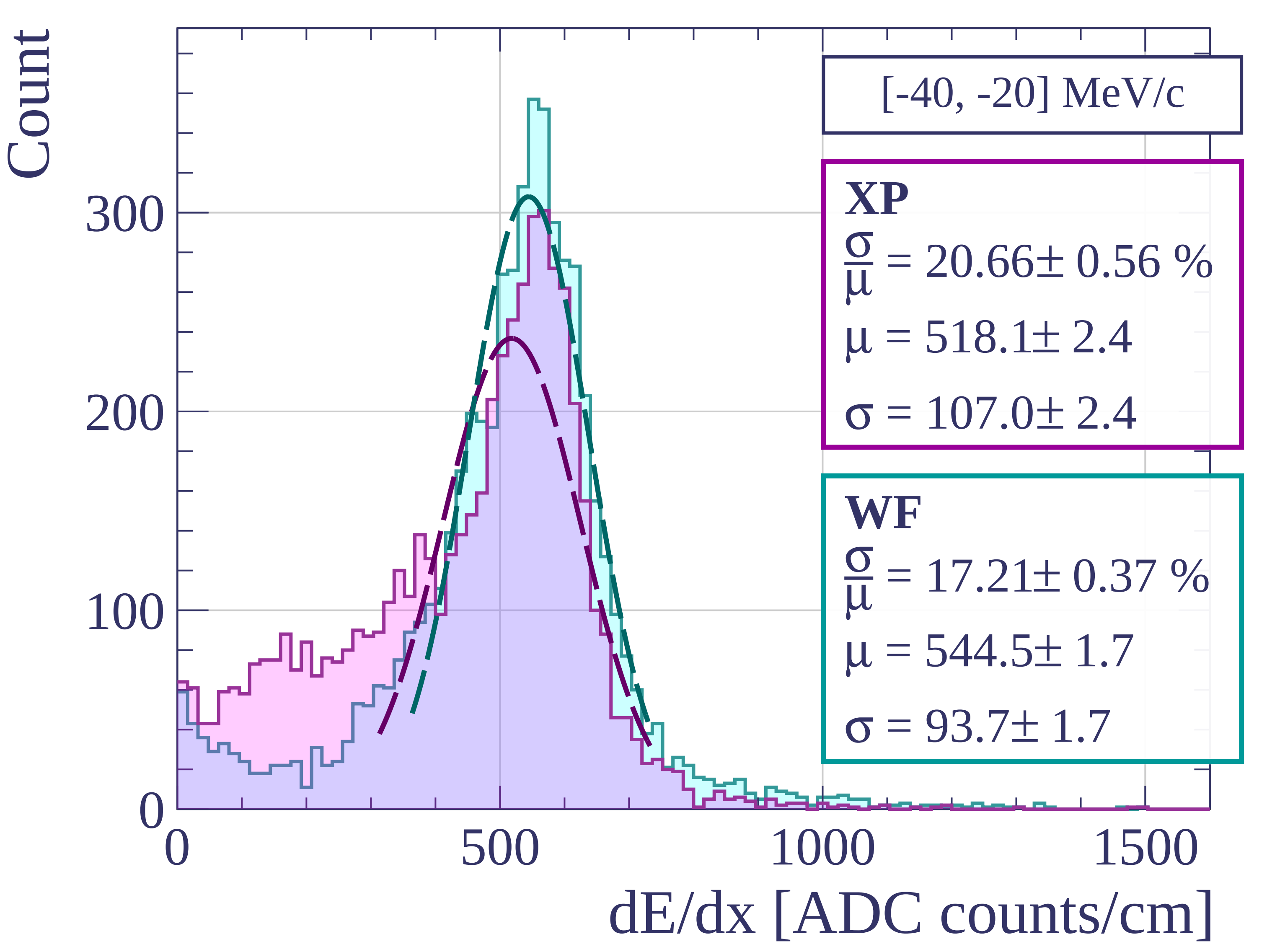
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[-100, -80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 26.68± 1.01 %
μ = 520.3± 4.3
σ = 138.8± 4.1
WF
$\frac{\sigma}{\mu}$ = 21.39± 0.72 %
μ = 558.6± 3.3
σ = 119.5± 3.3



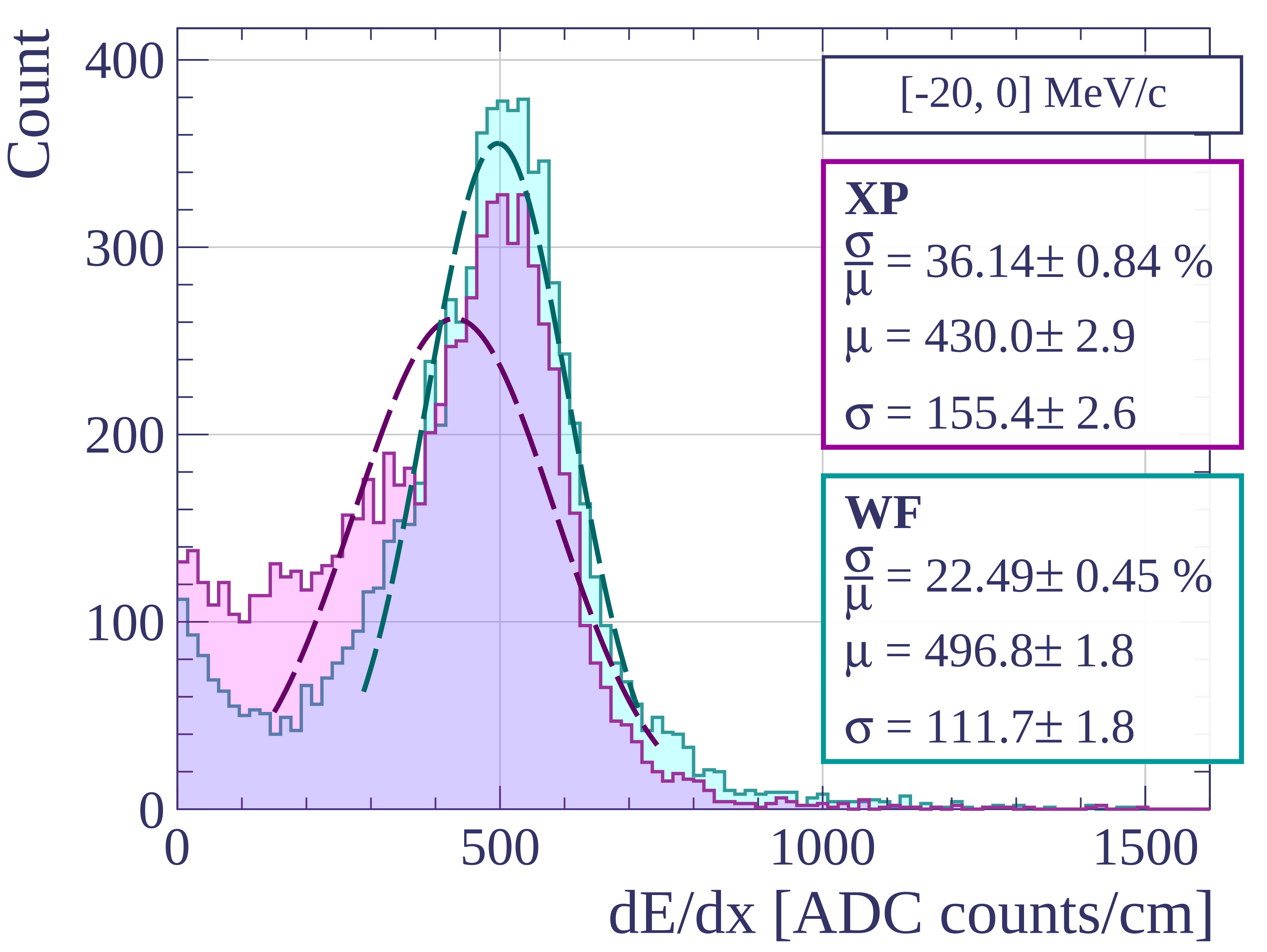
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-80, -60] MeV/c
XP
$\frac{\sigma}{\mu}$ = 25.35± 0.80 %
μ = 512.1± 3.7
σ = 129.8± 3.1
WF
$\frac{\sigma}{\mu}$ = 19.90± 0.66 %
μ = 548.1± 3.0
σ = 109.1± 3.0



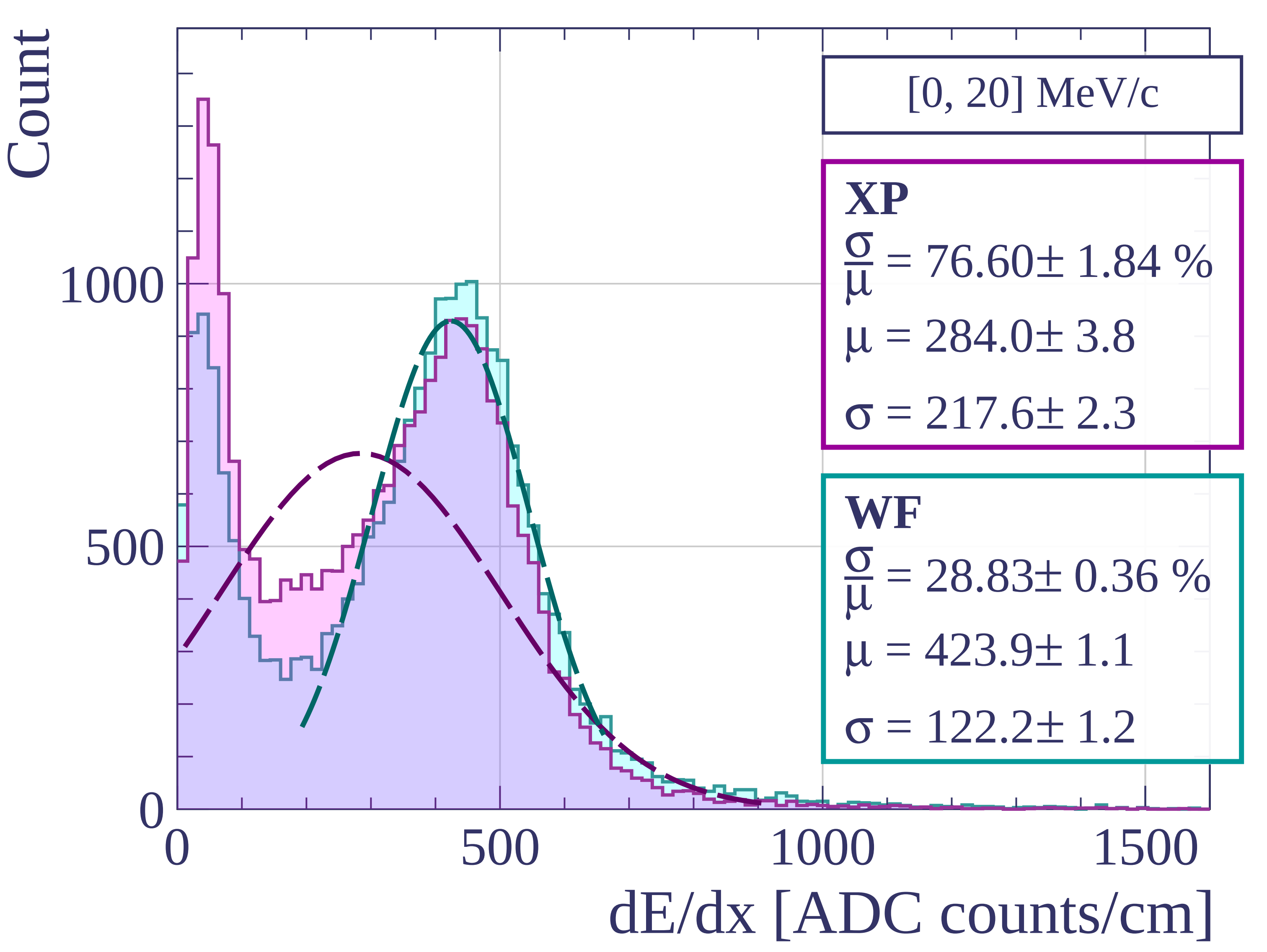
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-60, -40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 21.66± 0.71 %
μ = 525.6± 3.2
σ = 113.8± 3.1
WF
$\frac{\sigma}{\mu}$ = 17.38± 0.47 %
μ = 554.3± 2.2
σ = 96.3± 2.2



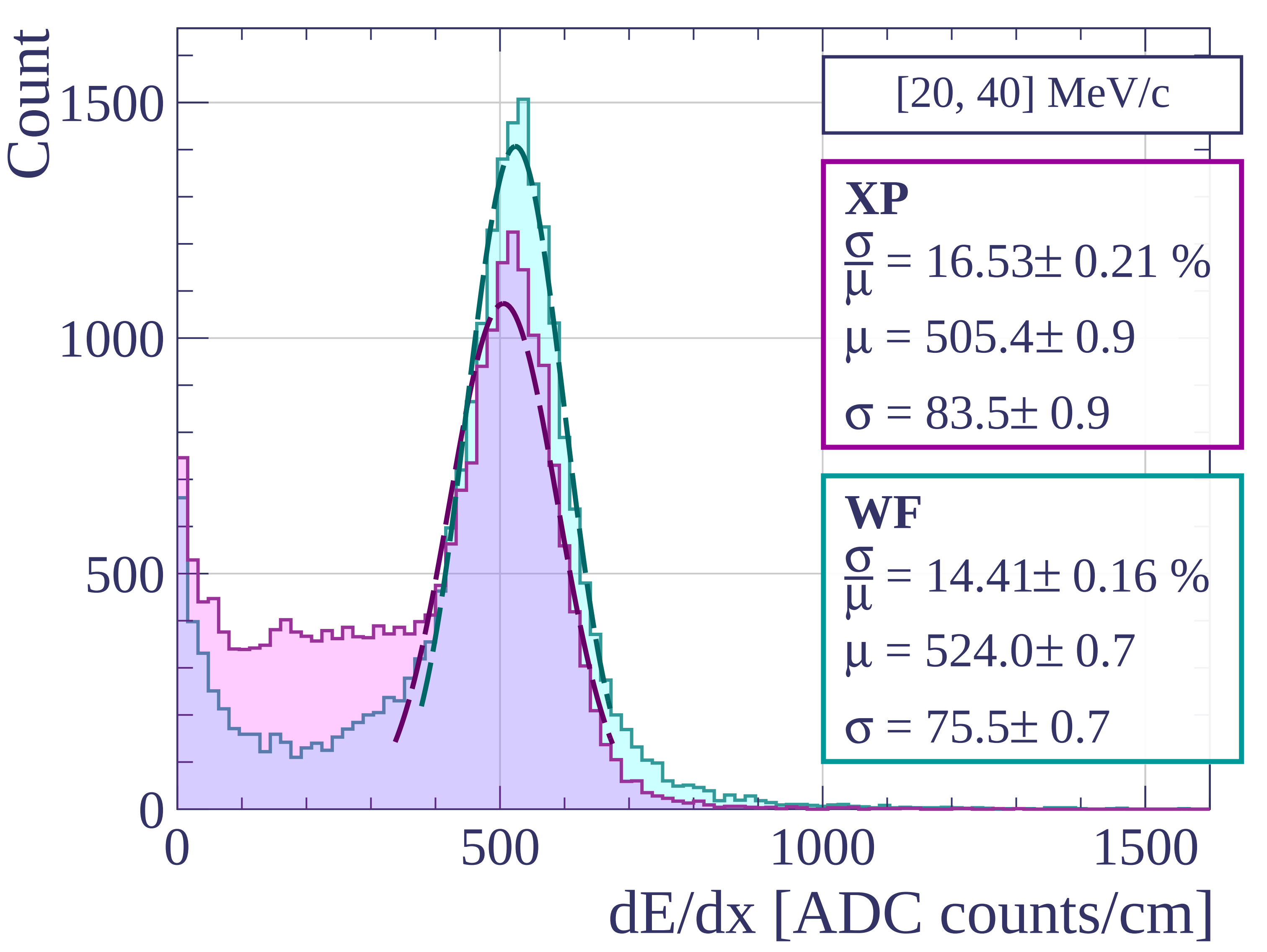
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-40, -20] MeV/c
XP
σ/μ = 20.66± 0.56 %
μ = 518.1± 2.4
σ = 107.0± 2.4
WF
σ/μ = 17.21± 0.37 %
μ = 544.5± 1.7
σ = 93.7± 1.7



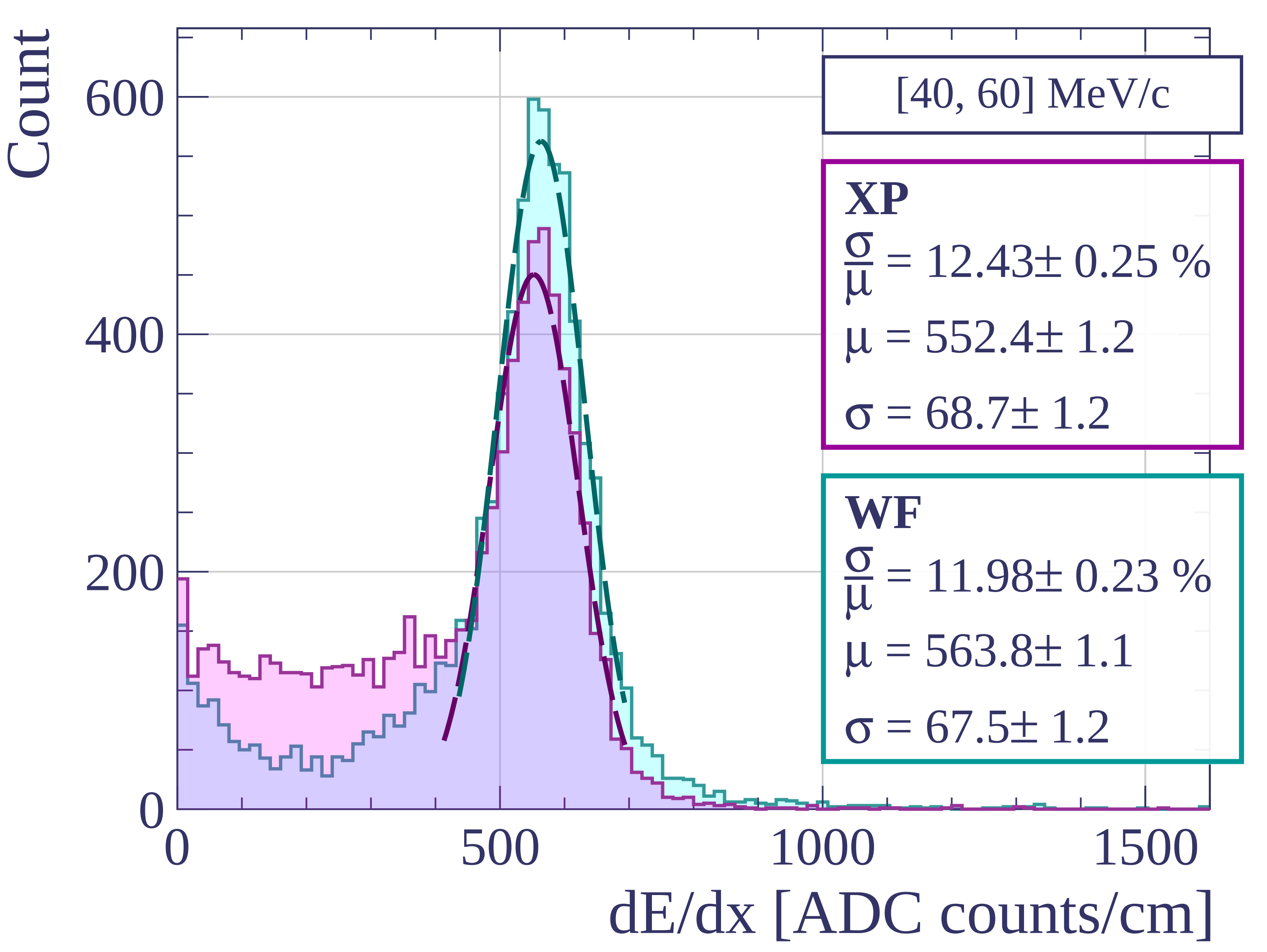
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
σ/μ = 36.14± 0.84 %
μ = 430.0± 2.9
σ = 155.4± 2.6
WF
σ/μ = 22.49± 0.45 %
μ = 496.8± 1.8
σ = 111.7± 1.8



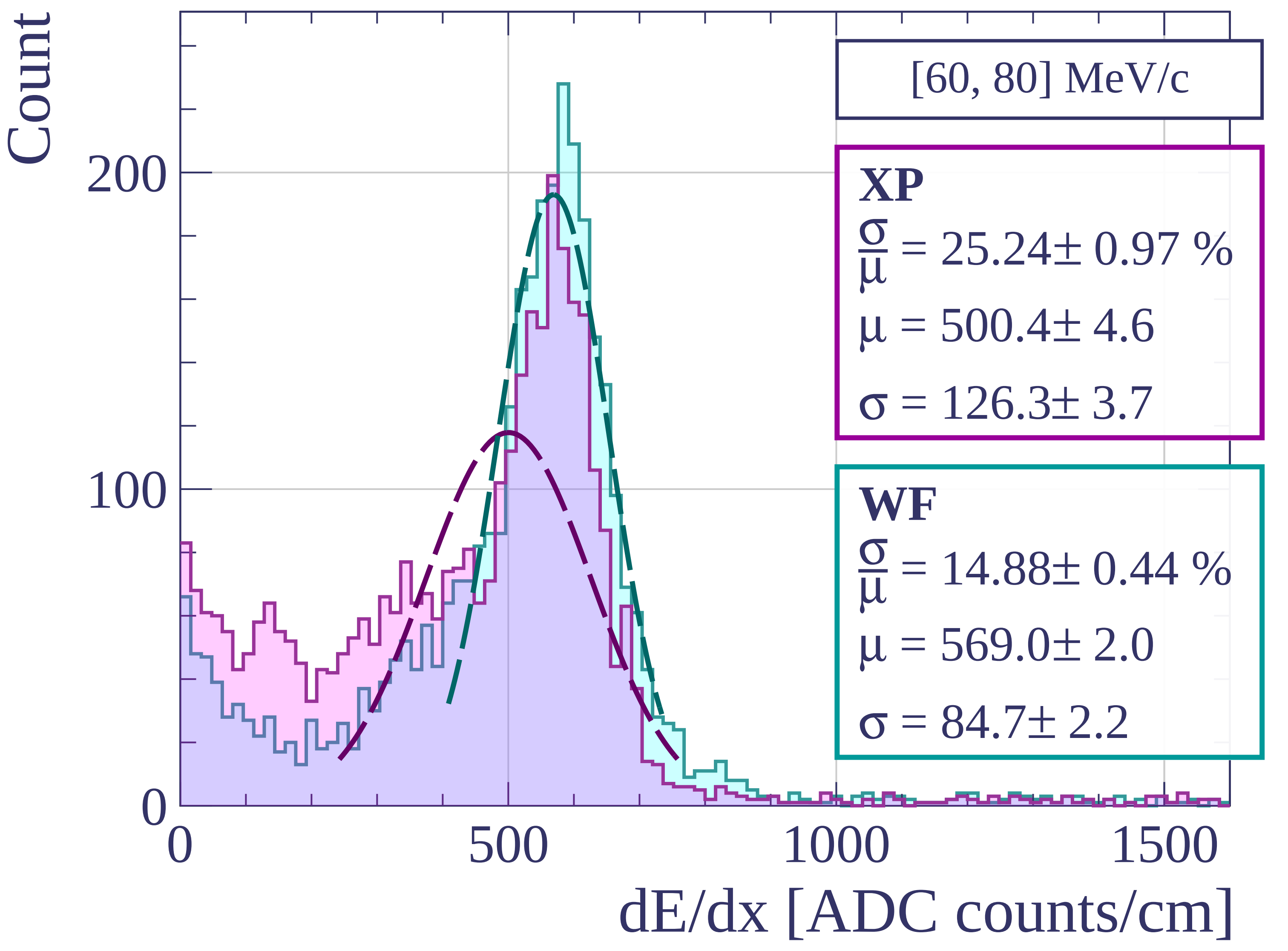
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
σ/μ = 76.60± 1.84 %
μ = 284.0± 3.8
σ = 217.6± 2.3
WF
σ/μ = 28.83± 0.36 %
μ = 423.9± 1.1
σ = 122.2± 1.2



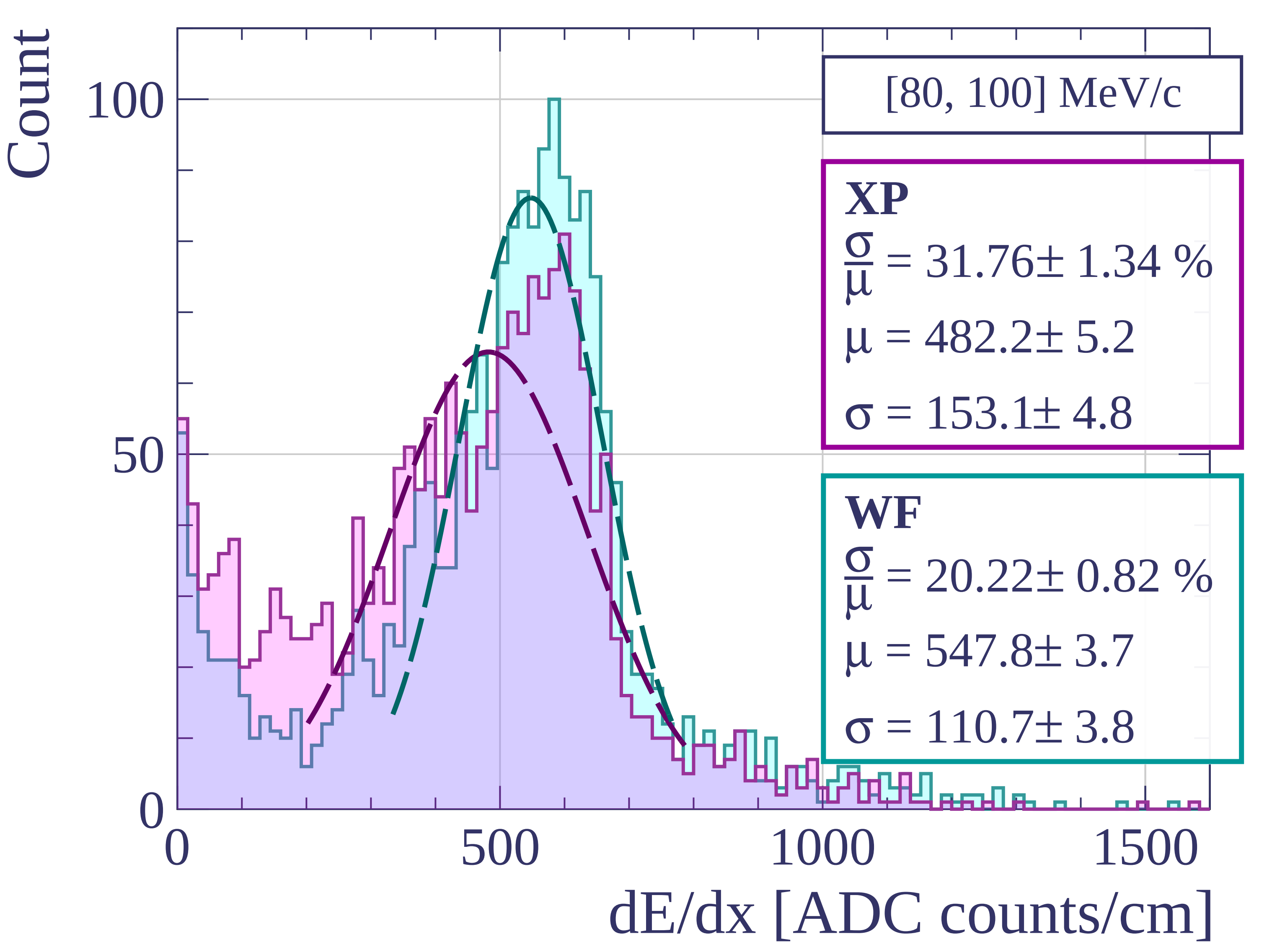
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[20, 40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 16.53± 0.21 %
μ = 505.4± 0.9
σ = 83.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 14.41± 0.16 %
μ = 524.0± 0.7
σ = 75.5± 0.7



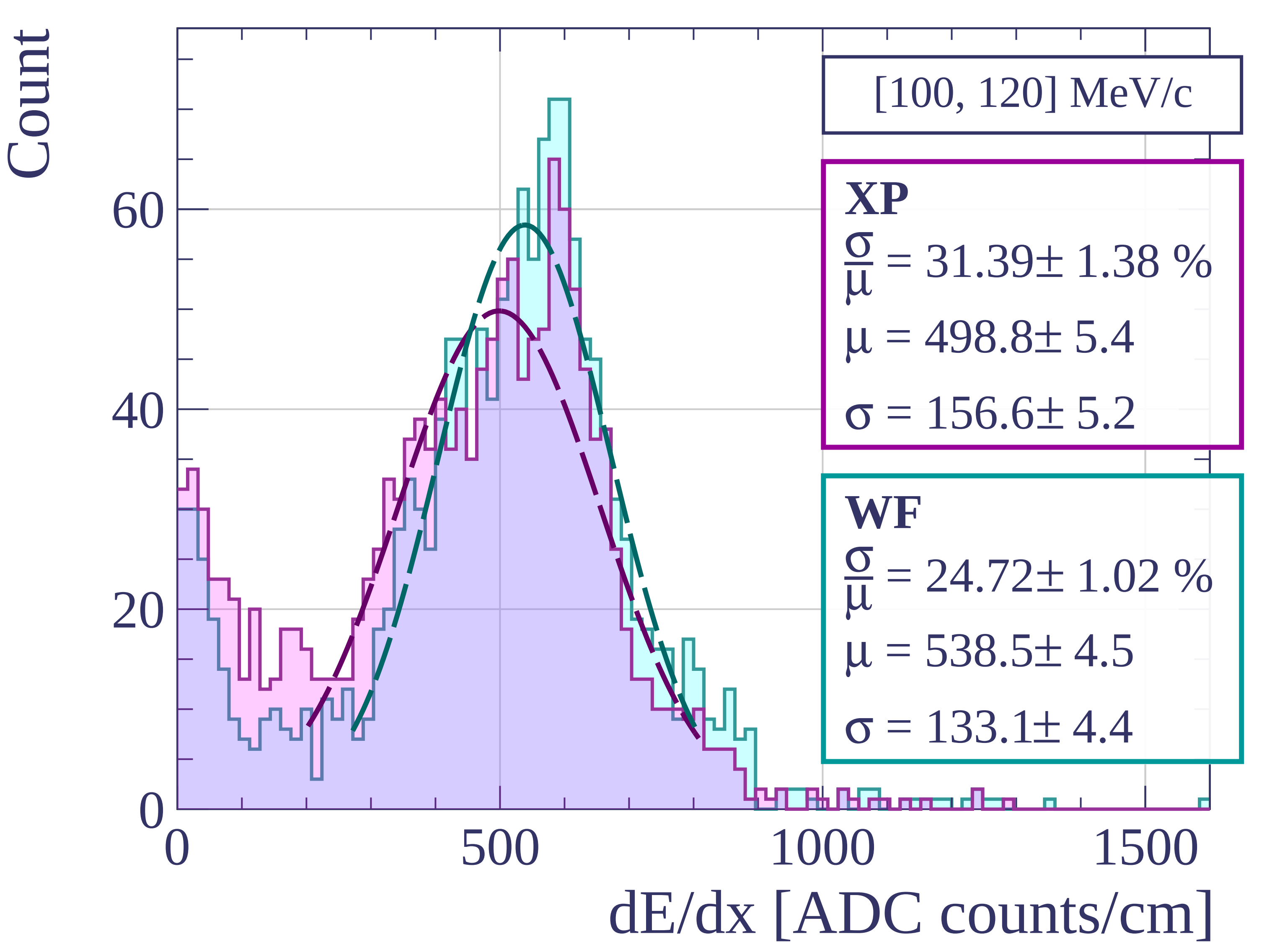
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
σ/μ = 12.43± 0.25 %
μ = 552.4± 1.2
σ = 68.7± 1.2
WF
σ/μ = 11.98± 0.23 %
μ = 563.8± 1.1
σ = 67.5± 1.2



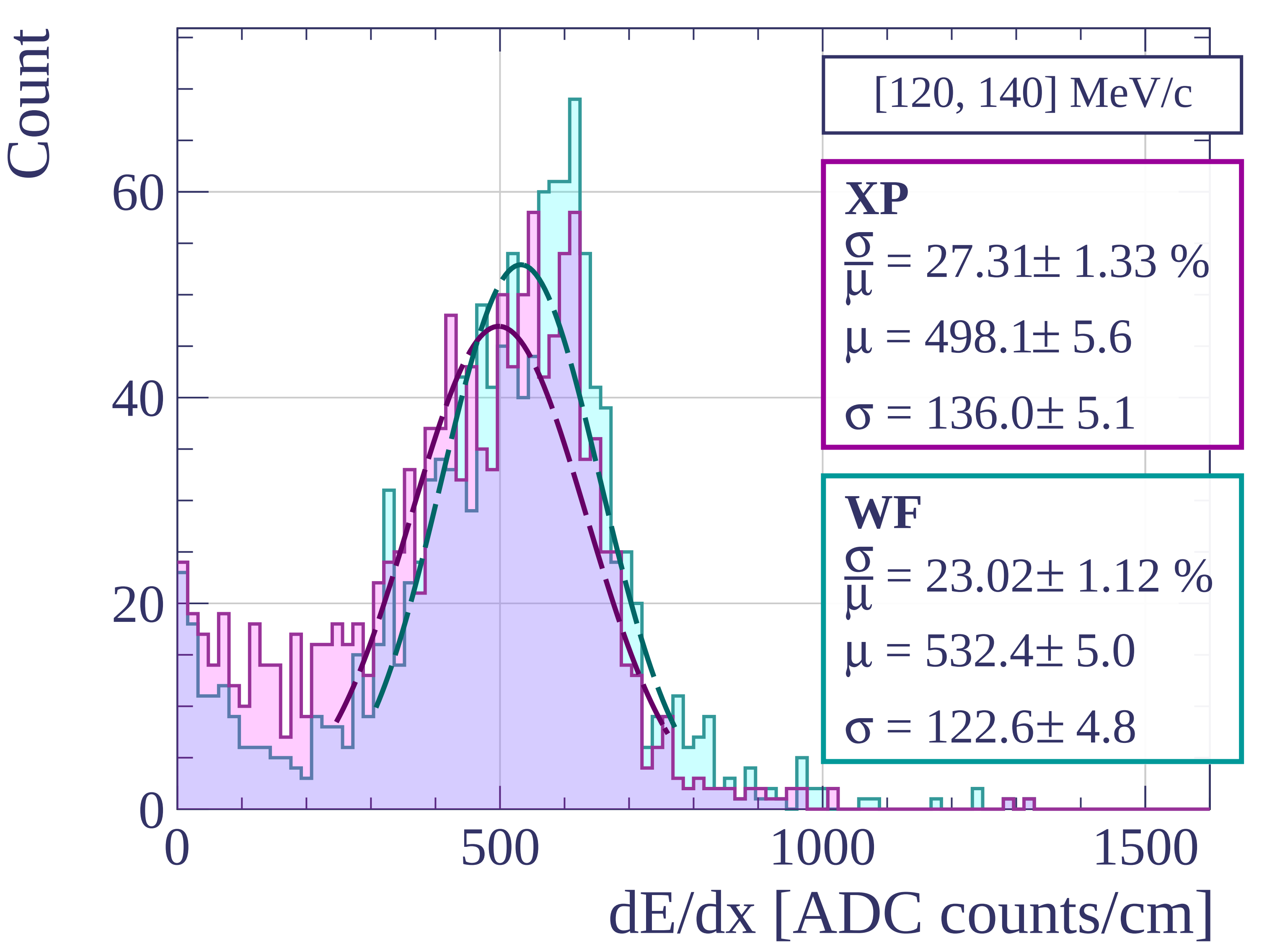
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[60, 80] MeV/c
XP
σ/μ = 25.24 ± 0.97 %
μ = 500.4 ± 4.6
σ = 126.3 ± 3.7
WF
σ/μ = 14.88 ± 0.44 %
μ = 569.0 ± 2.0
σ = 84.7 ± 2.2



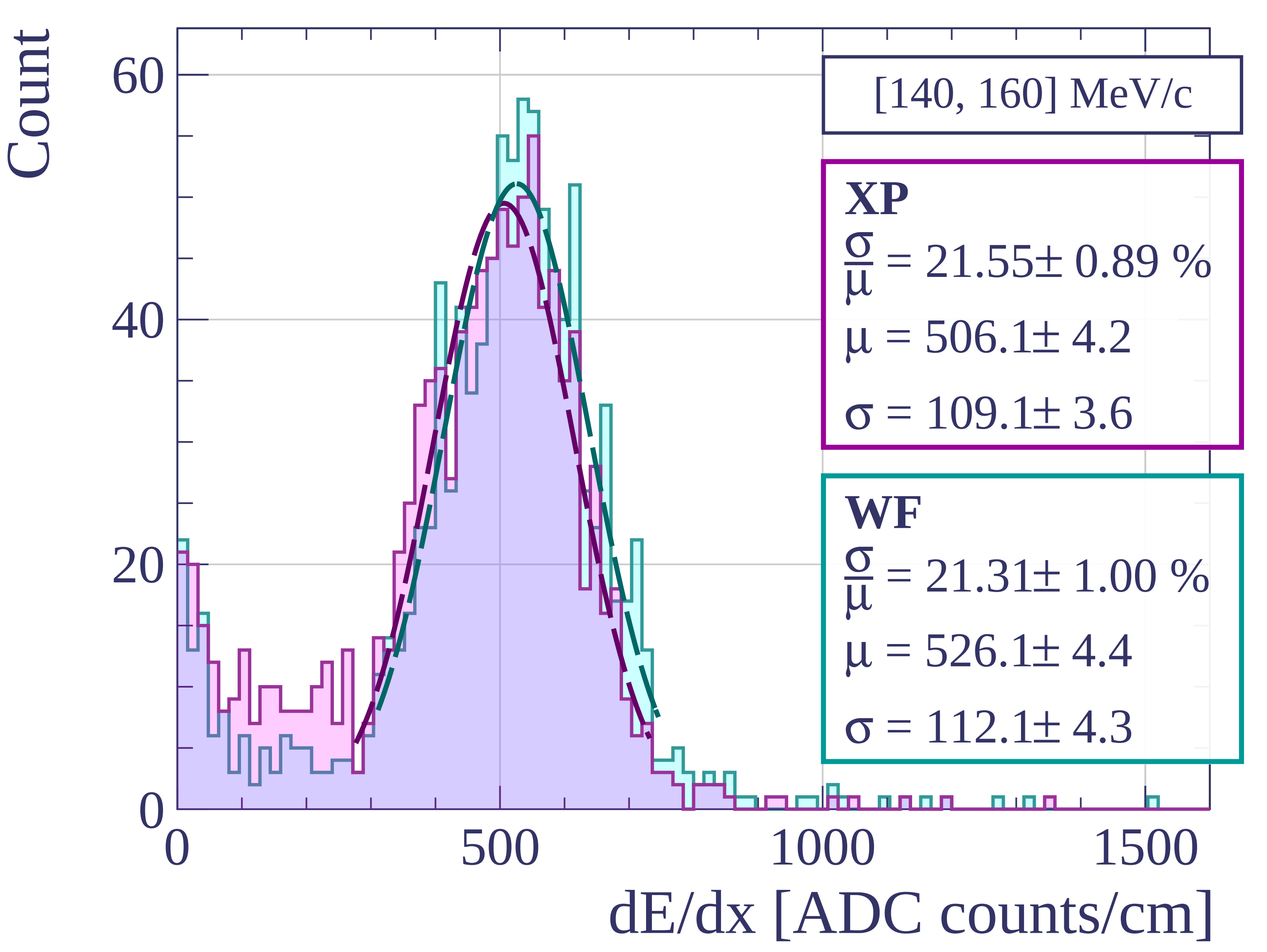
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 31.76± 1.34 %
μ = 482.2± 5.2
σ = 153.1± 4.8
WF
$\frac{\sigma}{\mu}$ = 20.22± 0.82 %
μ = 547.8± 3.7
σ = 110.7± 3.8



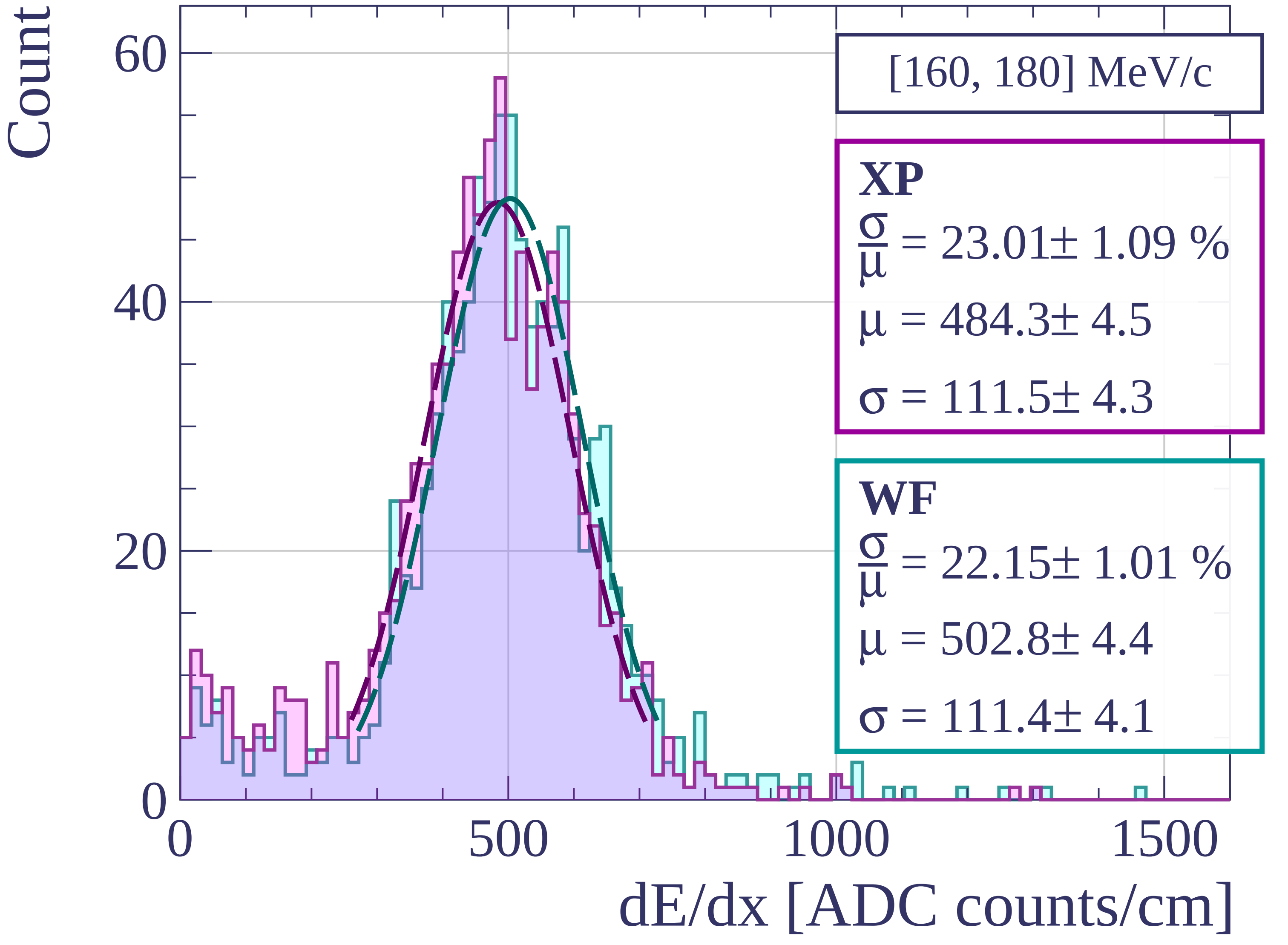
Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[100, 120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 31.39± 1.38 %
μ = 498.8± 5.4
σ = 156.6± 5.2
WF
$\frac{\sigma}{\mu}$ = 24.72± 1.02 %
μ = 538.5± 4.5
σ = 133.1± 4.4

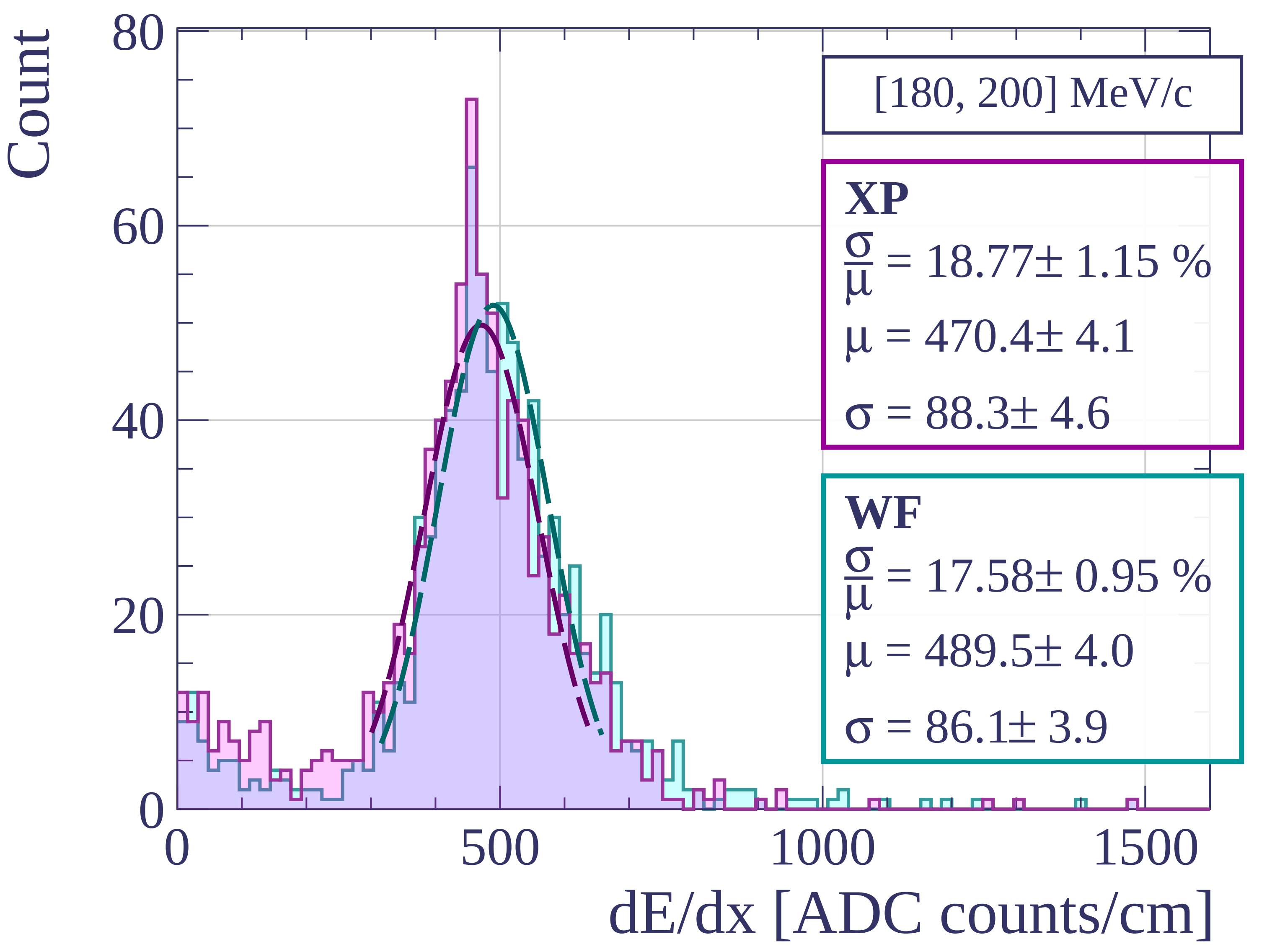


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
[120, 140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 27.31± 1.33 %
μ = 498.1± 5.6
σ = 136.0± 5.1
WF
$\frac{\sigma}{\mu}$ = 23.02± 1.12 %
μ = 532.4± 5.0
σ = 122.6± 4.8

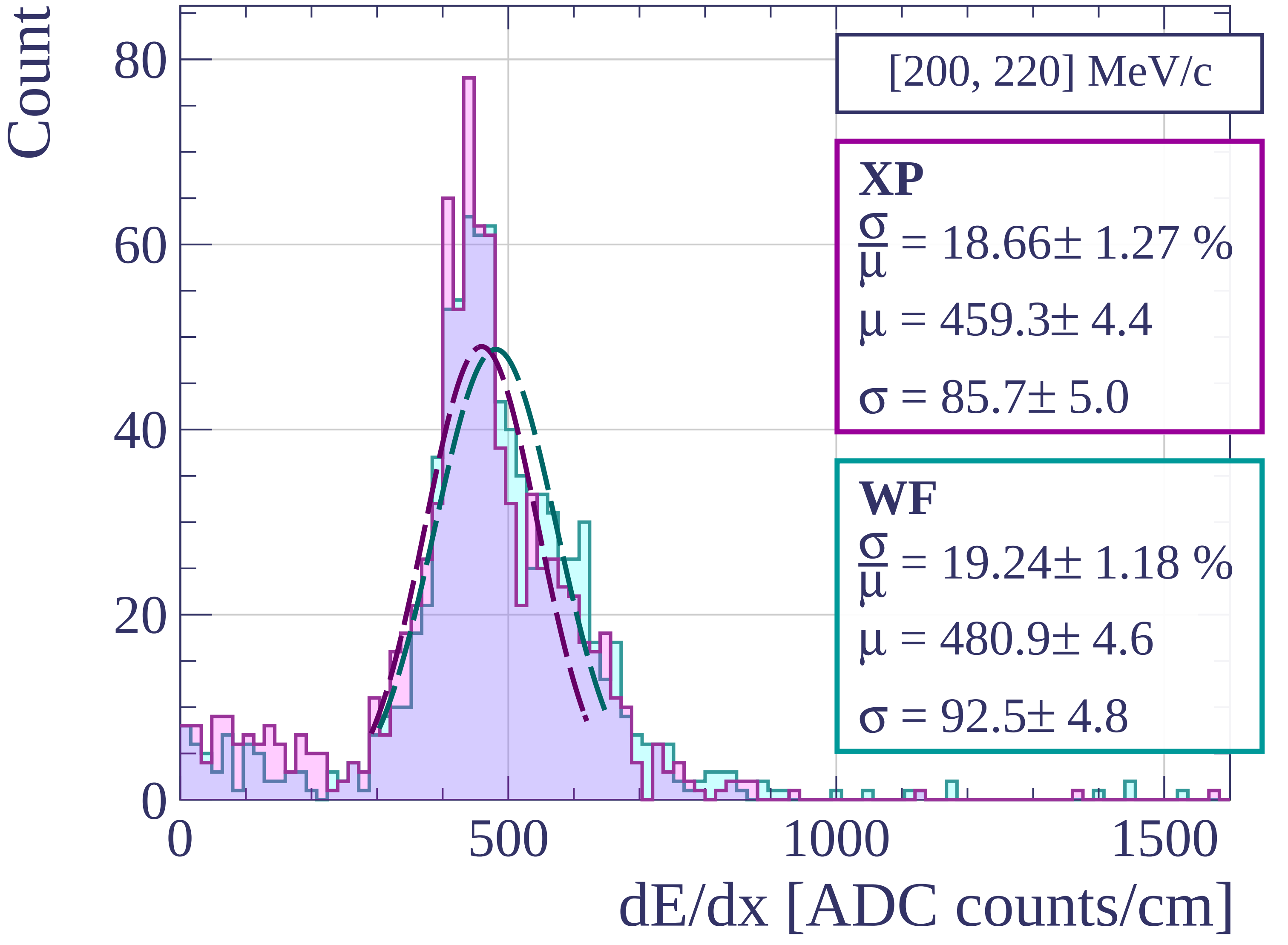


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
σ/μ = 21.55± 0.89 %
μ = 506.1± 4.2
σ = 109.1± 3.6
WF
σ/μ = 21.31± 1.00 %
μ = 526.1± 4.4
σ = 112.1± 4.3

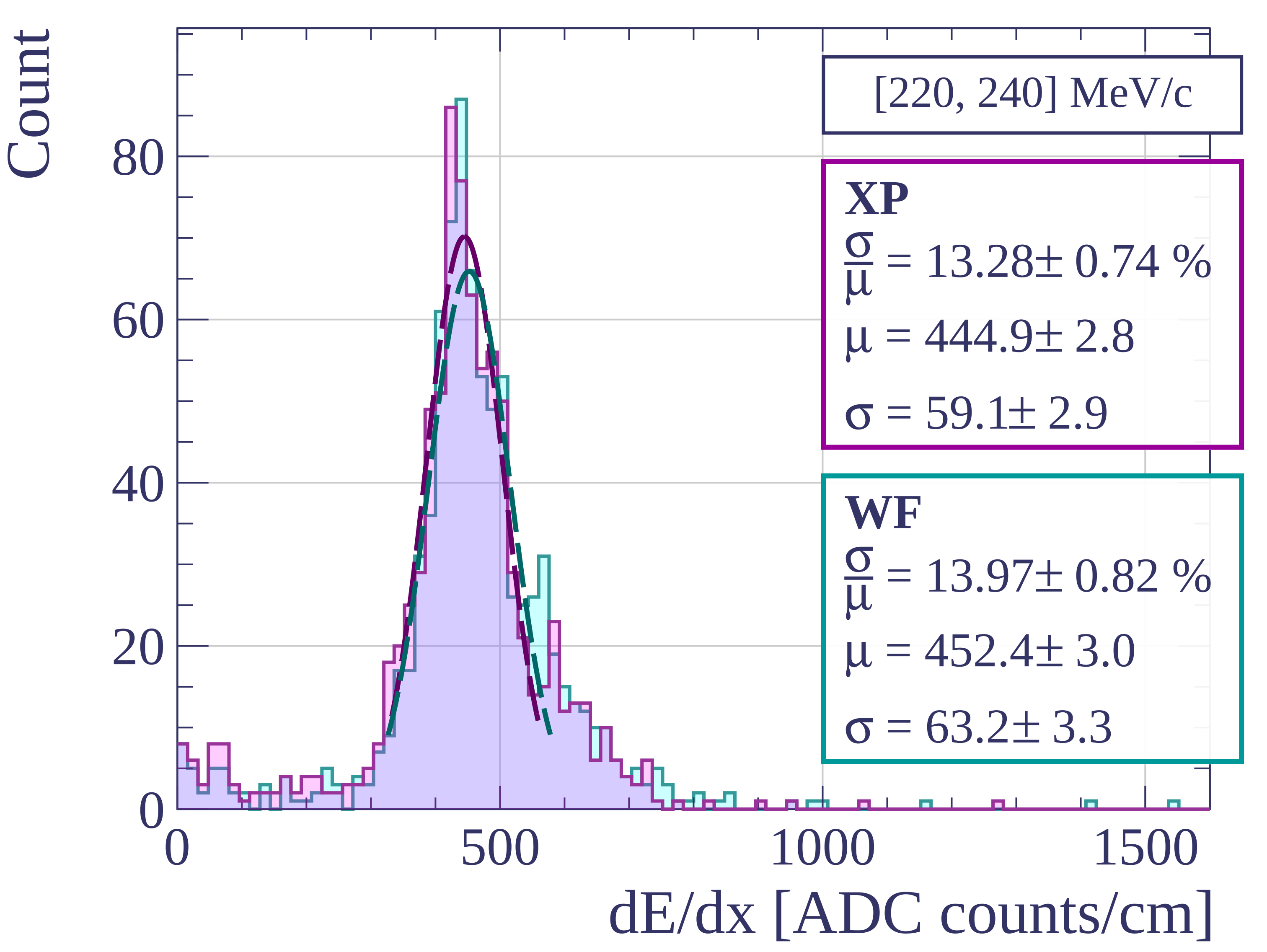

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.01± 1.09 %
μ = 484.3± 4.5
σ = 111.5± 4.3
WF
$\frac{\sigma}{\mu}$ = 22.15± 1.01 %
μ = 502.8± 4.4
σ = 111.4± 4.1



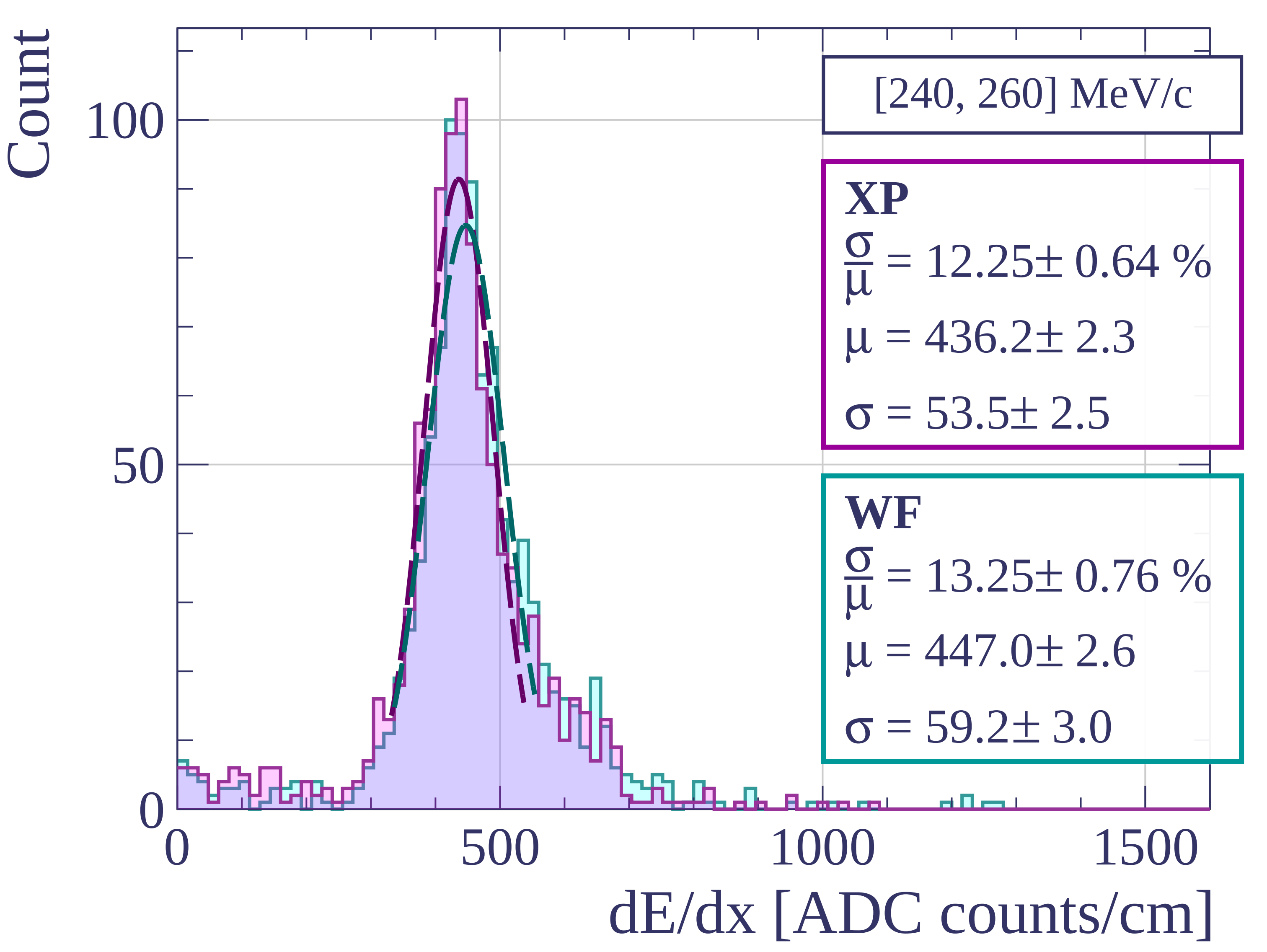
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 18.77± 1.15 %
μ = 470.4± 4.1
σ = 88.3± 4.6
WF
$\frac{\sigma}{\mu}$ = 17.58± 0.95 %
μ = 489.5± 4.0
σ = 86.1± 3.9



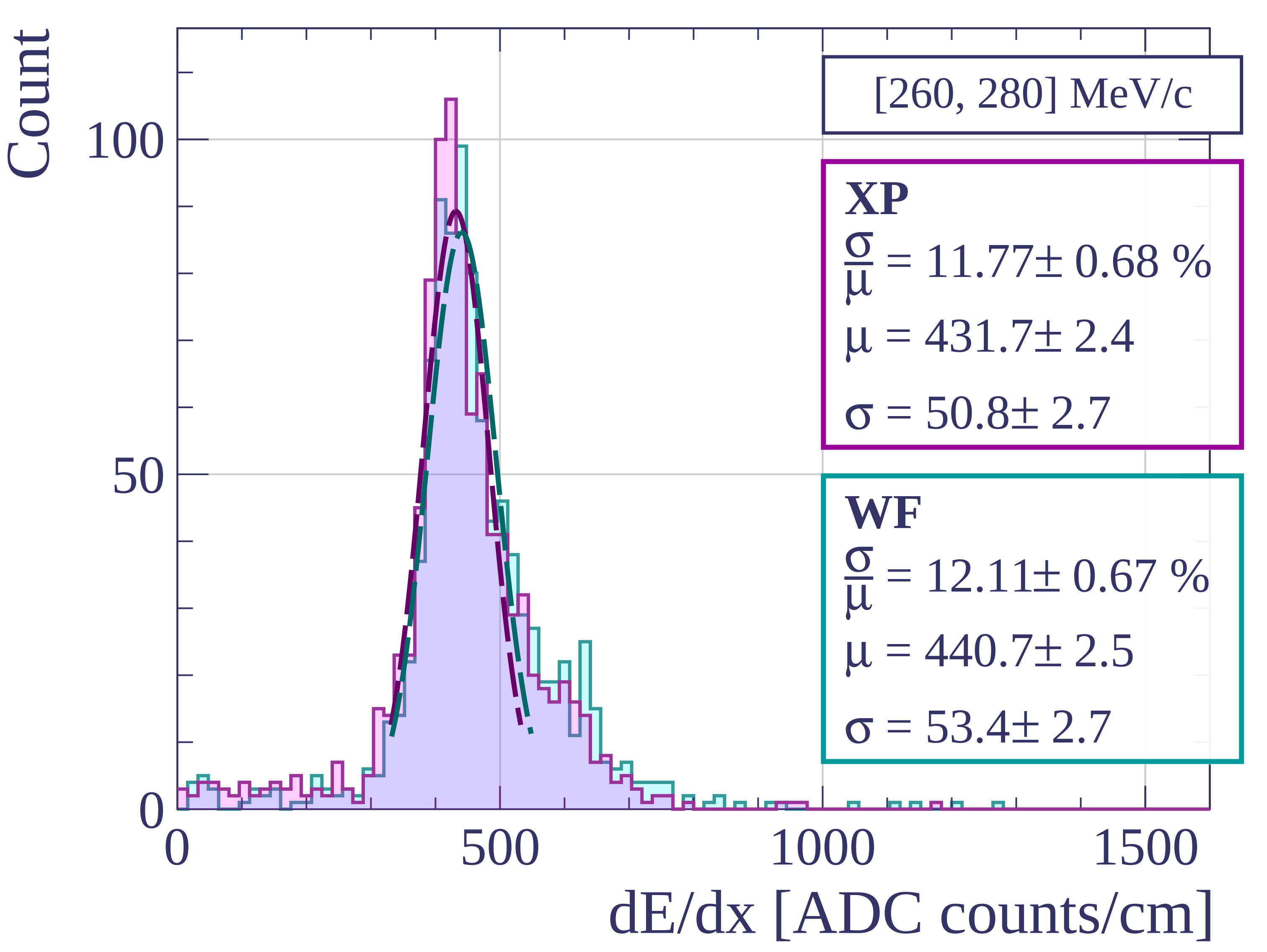
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[200, 220] MeV/c
XP
σ/μ = 18.66± 1.27 %
μ = 459.3± 4.4
σ = 85.7± 5.0
WF
σ/μ = 19.24± 1.18 %
μ = 480.9± 4.6
σ = 92.5± 4.8



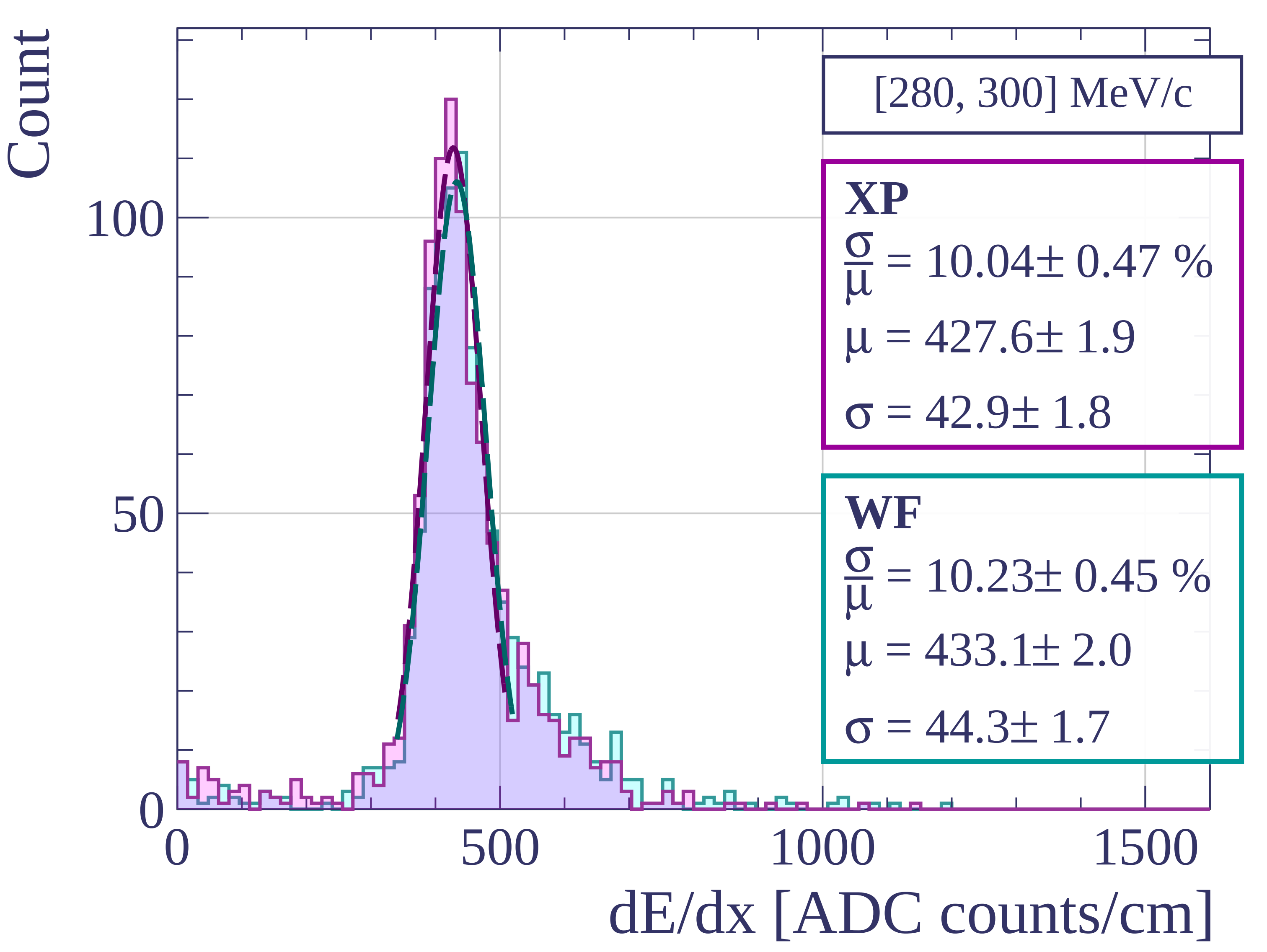
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[220, 240] MeV/c
XP
σ/μ = 13.28± 0.74 %
μ = 444.9± 2.8
σ = 59.1± 2.9
WF
σ/μ = 13.97± 0.82 %
μ = 452.4± 3.0
σ = 63.2± 3.3



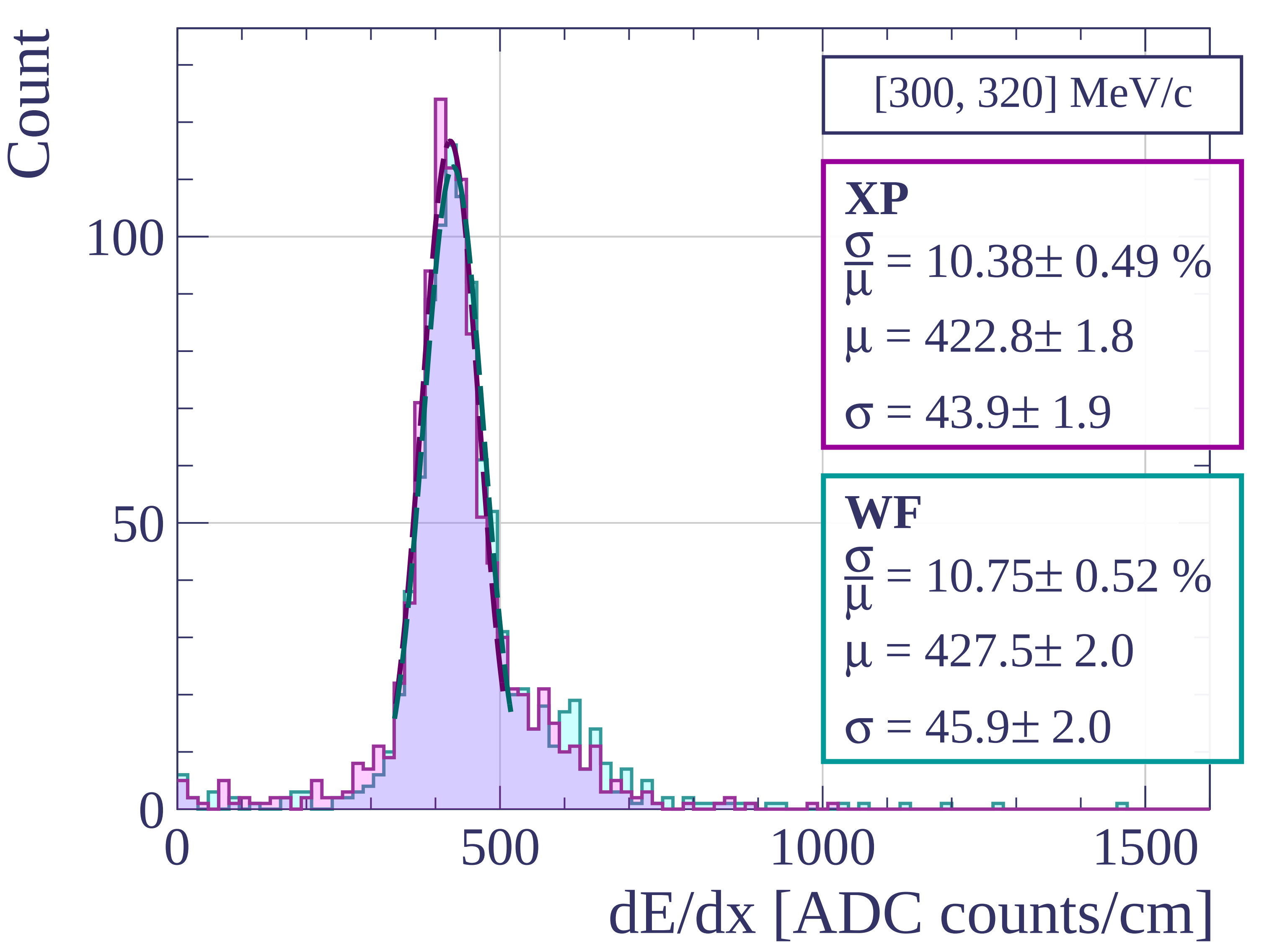
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
σ/μ = 12.25± 0.64 %
μ = 436.2± 2.3
σ = 53.5± 2.5
WF
σ/μ = 13.25± 0.76 %
μ = 447.0± 2.6
σ = 59.2± 3.0



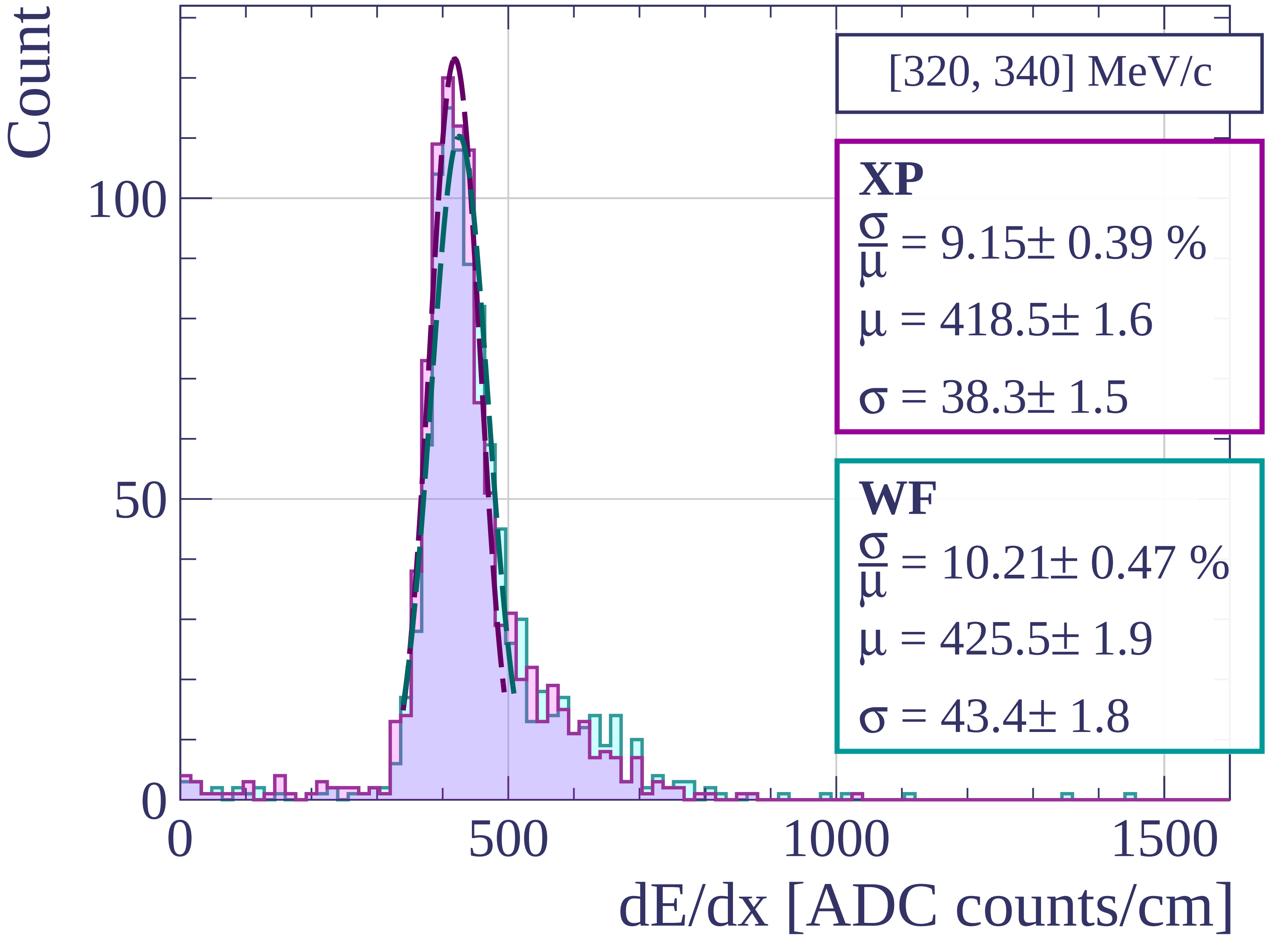
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[260, 280] MeV/c
XP
σ/μ = 11.77± 0.68 %
μ = 431.7± 2.4
σ = 50.8± 2.7
WF
σ/μ = 12.11± 0.67 %
μ = 440.7± 2.5
σ = 53.4± 2.7



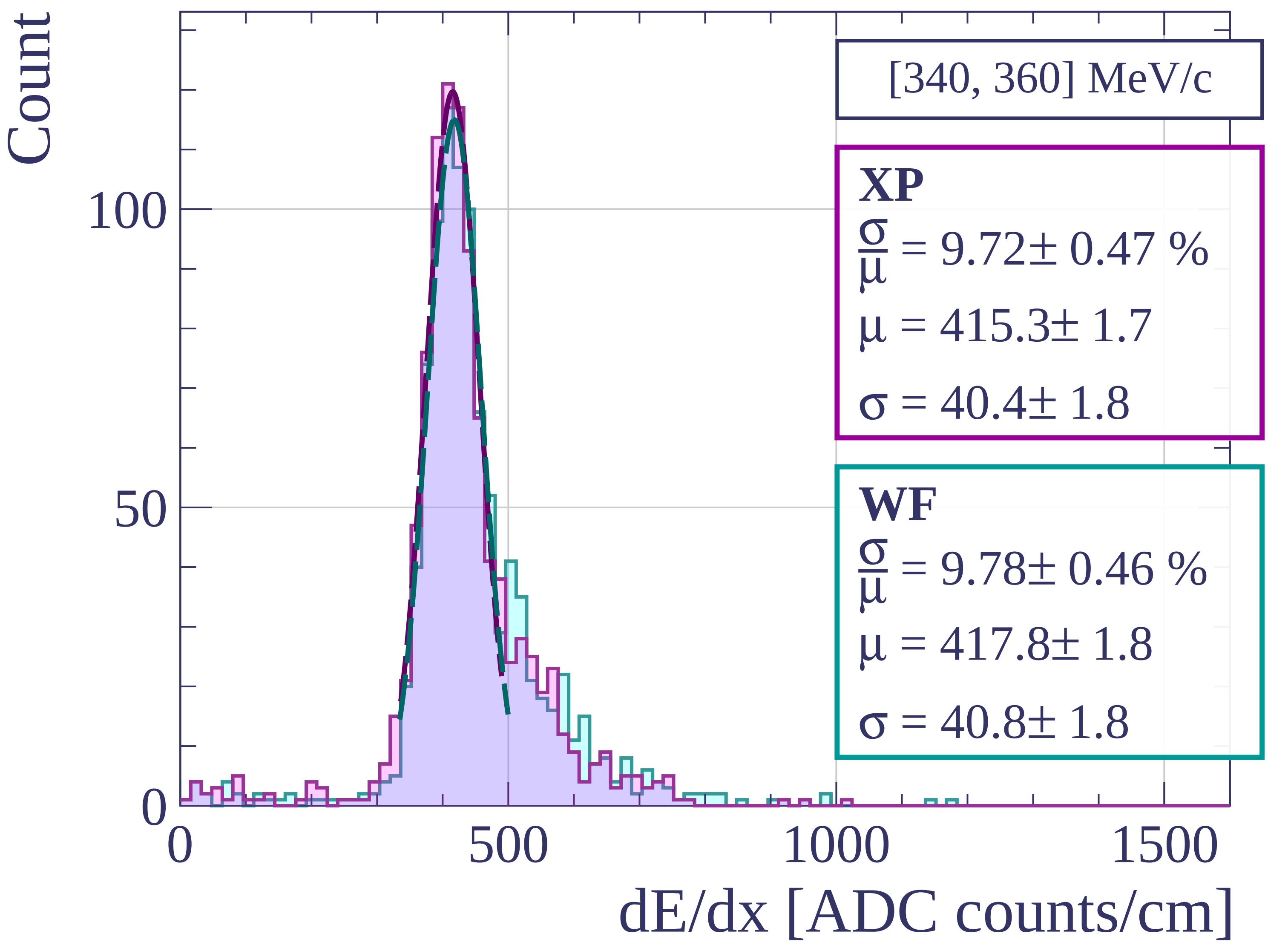
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
σ/μ = 10.04± 0.47 %
μ = 427.6± 1.9
σ = 42.9± 1.8
WF
σ/μ = 10.23± 0.45 %
μ = 433.1± 2.0
σ = 44.3± 1.7



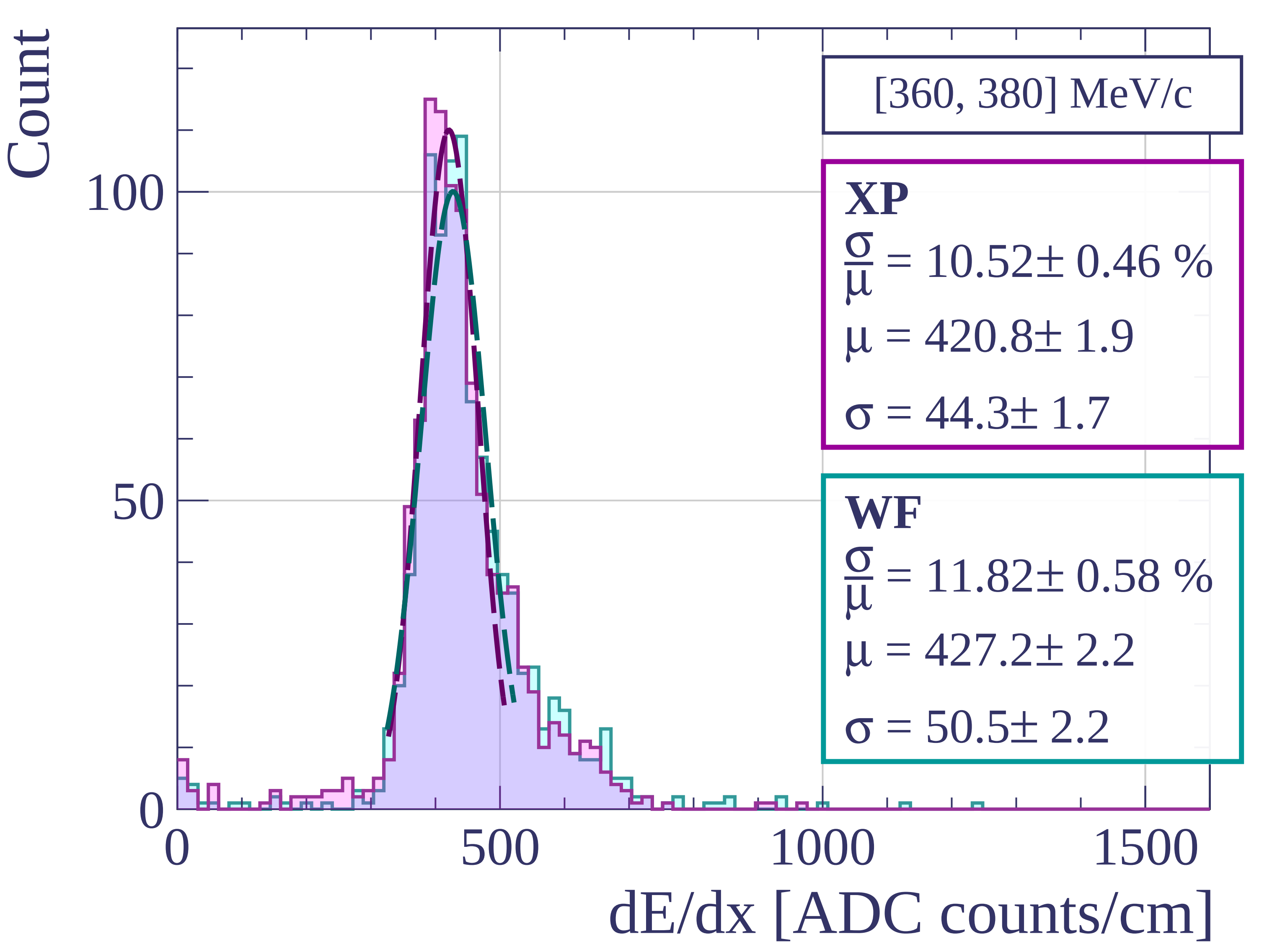
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[300, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.38± 0.49 %
μ = 422.8± 1.8
σ = 43.9± 1.9
WF
$\frac{\sigma}{\mu}$ = 10.75± 0.52 %
μ = 427.5± 2.0
σ = 45.9± 2.0



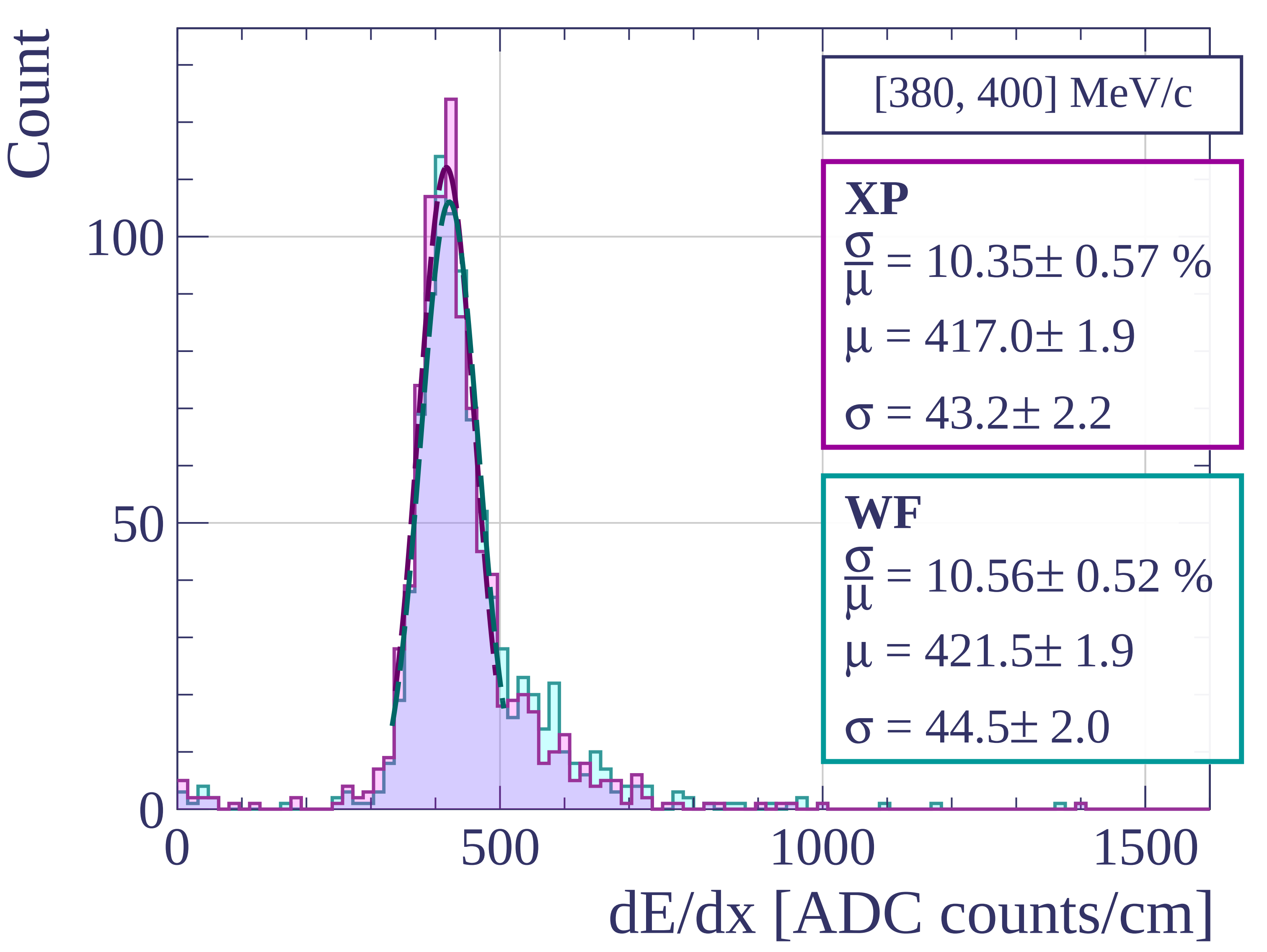
Count
dE/dx [ADC counts/cm]
[320, 340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.15± 0.39 %
μ = 418.5± 1.6
σ = 38.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 10.21± 0.47 %
μ = 425.5± 1.9
σ = 43.4± 1.8
0
50
100
0
500
1000
1500



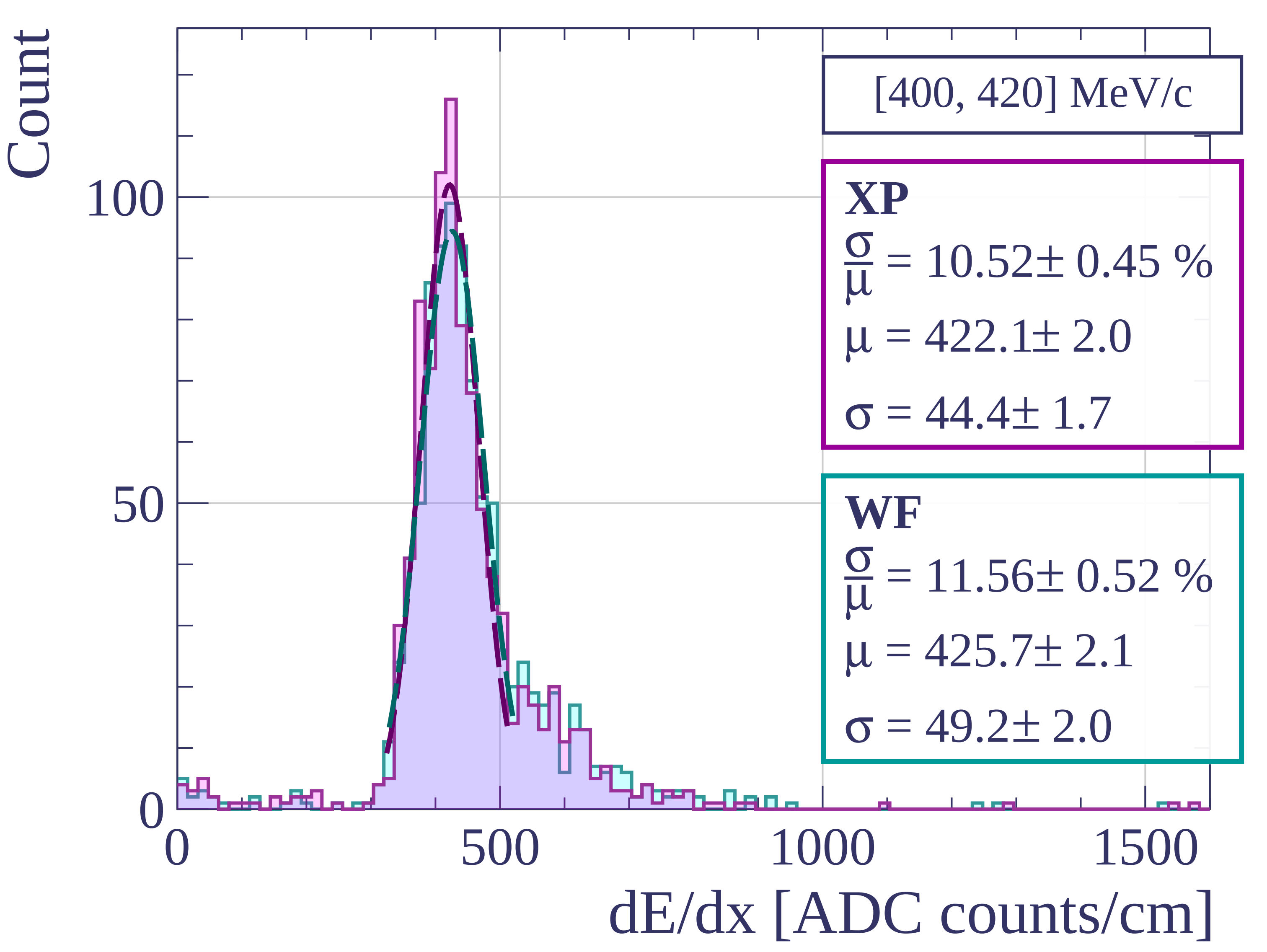
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.72± 0.47 %
μ = 415.3± 1.7
σ = 40.4± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.78± 0.46 %
μ = 417.8± 1.8
σ = 40.8± 1.8



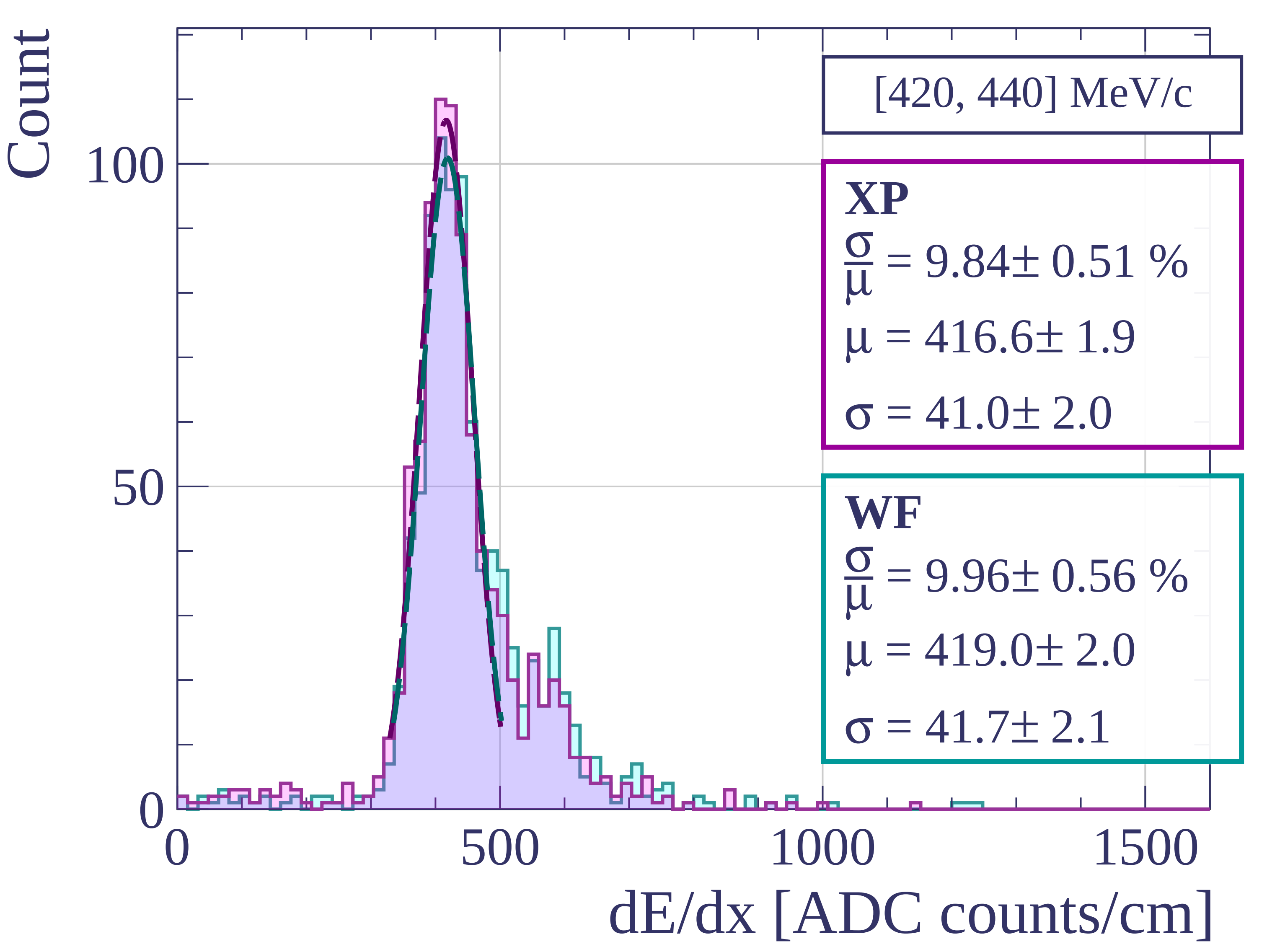
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
σ/μ = 10.52± 0.46 %
μ = 420.8± 1.9
σ = 44.3± 1.7
WF
σ/μ = 11.82± 0.58 %
μ = 427.2± 2.2
σ = 50.5± 2.2

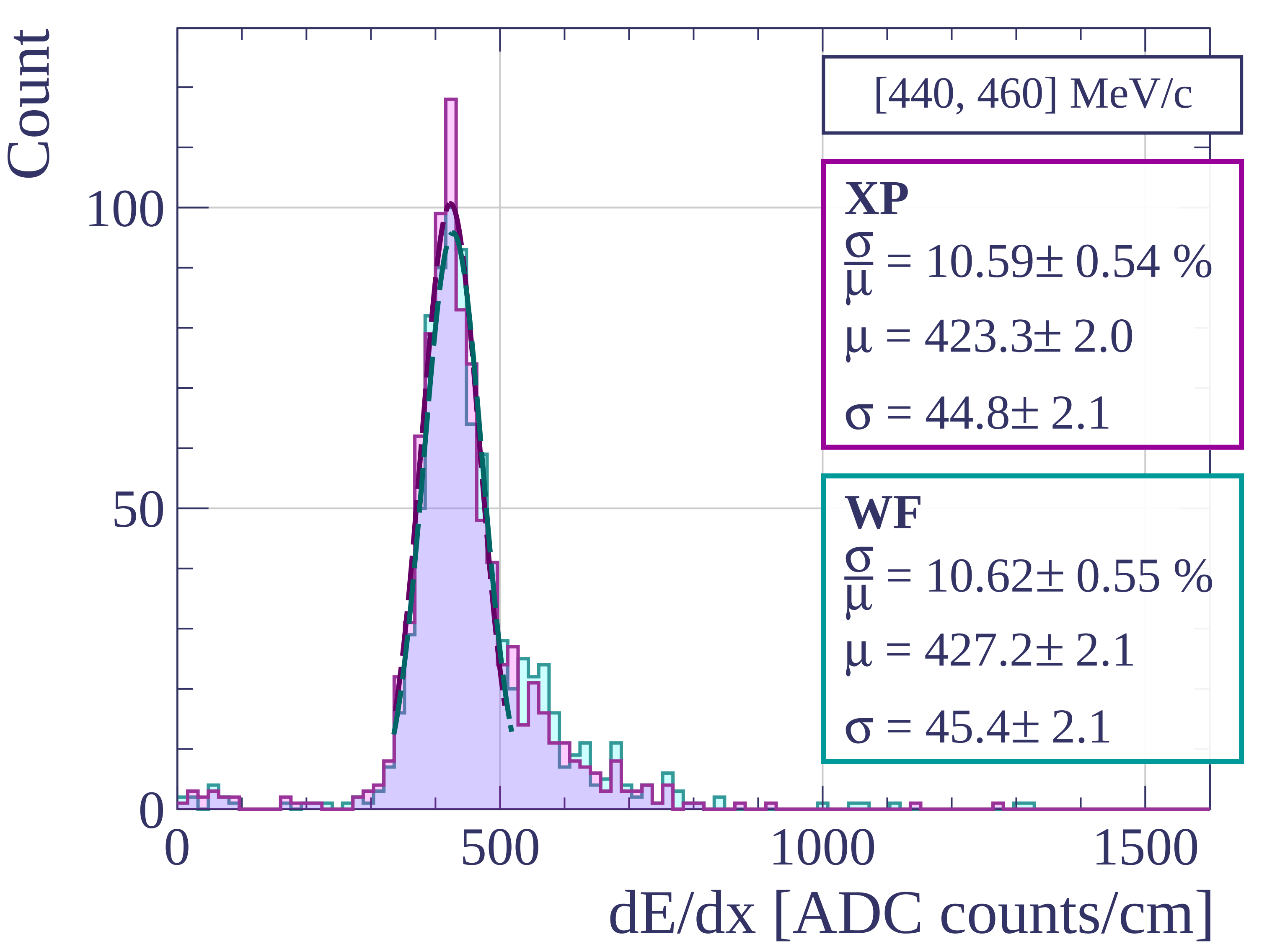


Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[380, 400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.35± 0.57 %
μ = 417.0± 1.9
σ = 43.2± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.56± 0.52 %
μ = 421.5± 1.9
σ = 44.5± 2.0

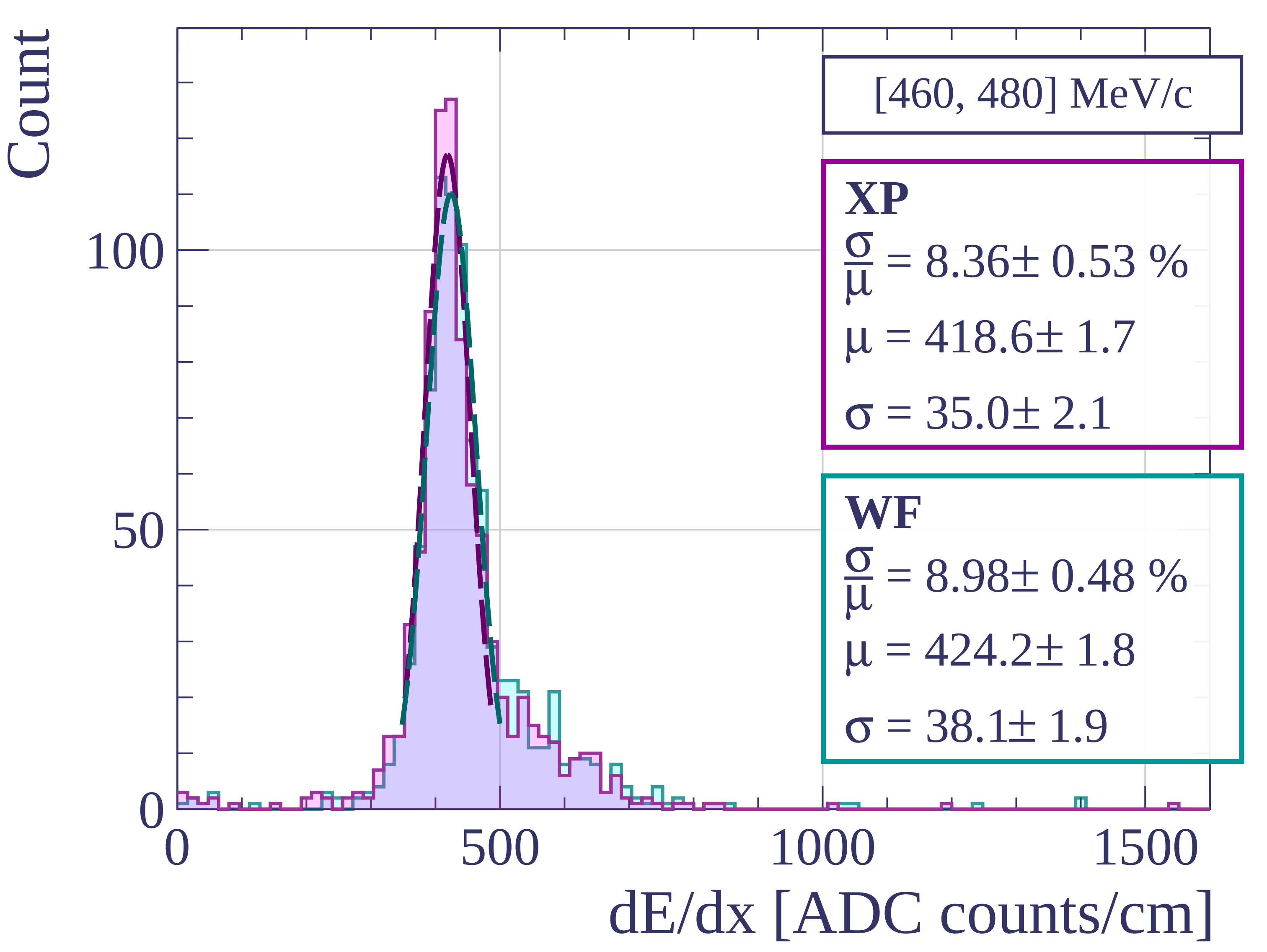


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
σ/μ = 10.52± 0.45 %
μ = 422.1± 2.0
σ = 44.4± 1.7
WF
σ/μ = 11.56± 0.52 %
μ = 425.7± 2.1
σ = 49.2± 2.0

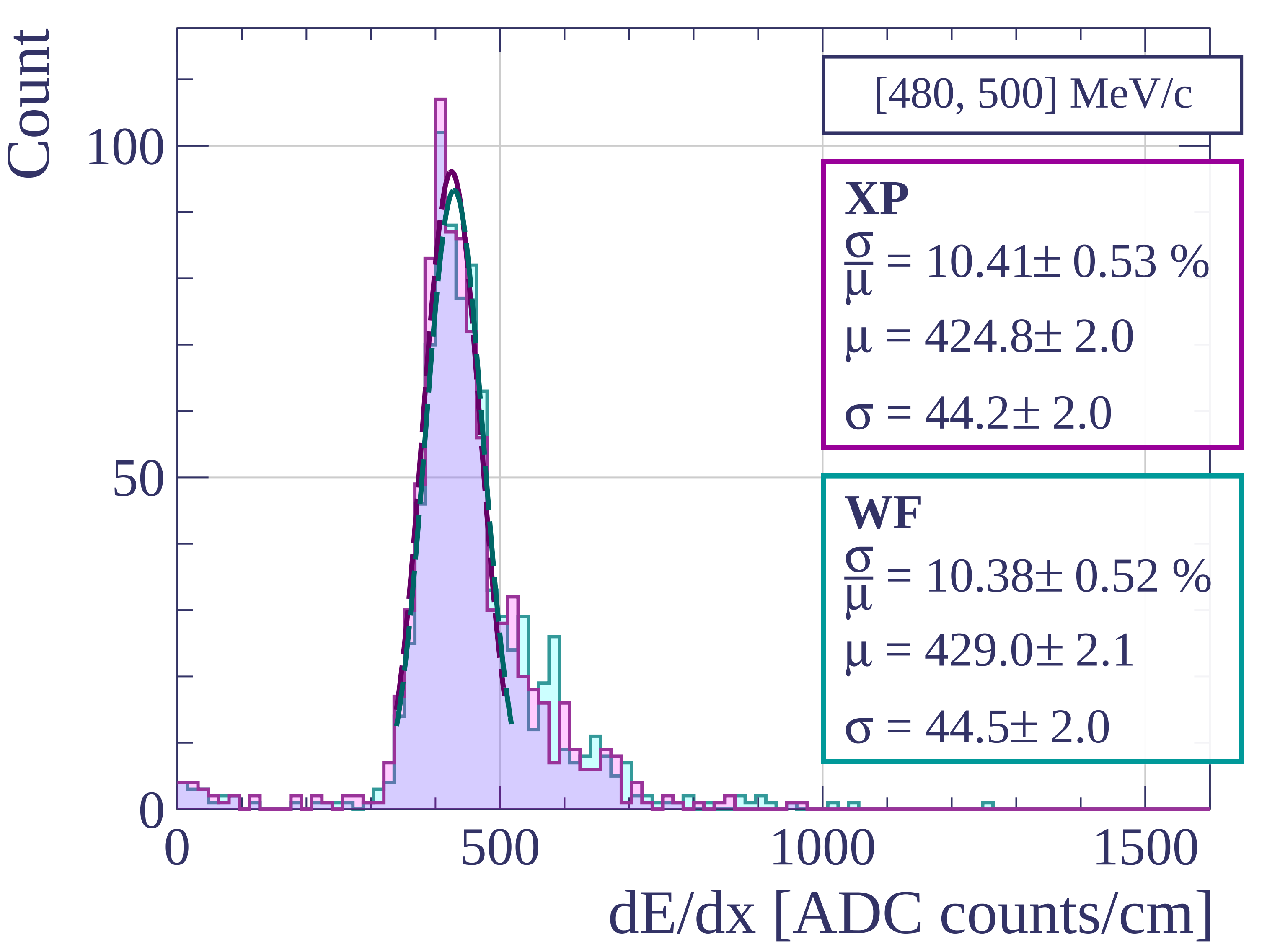

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[420, 440] MeV/c
XP
σ/μ = 9.84± 0.51 %
μ = 416.6± 1.9
σ = 41.0± 2.0
WF
σ/μ = 9.96± 0.56 %
μ = 419.0± 2.0
σ = 41.7± 2.1

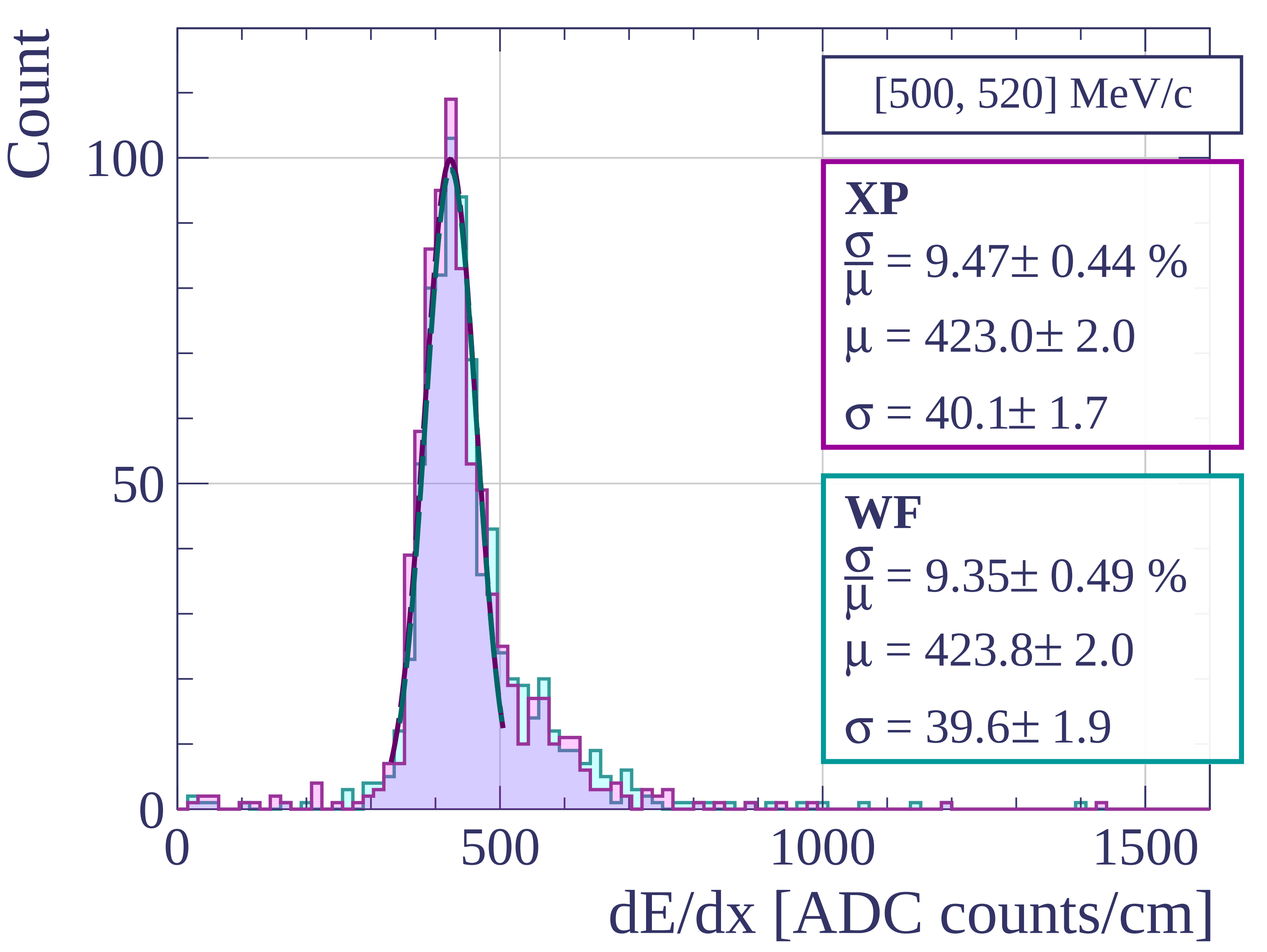


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[440, 460] MeV/c
XP
σ/μ = 10.59± 0.54 %
μ = 423.3± 2.0
σ = 44.8± 2.1
WF
σ/μ = 10.62± 0.55 %
μ = 427.2± 2.1
σ = 45.4± 2.1

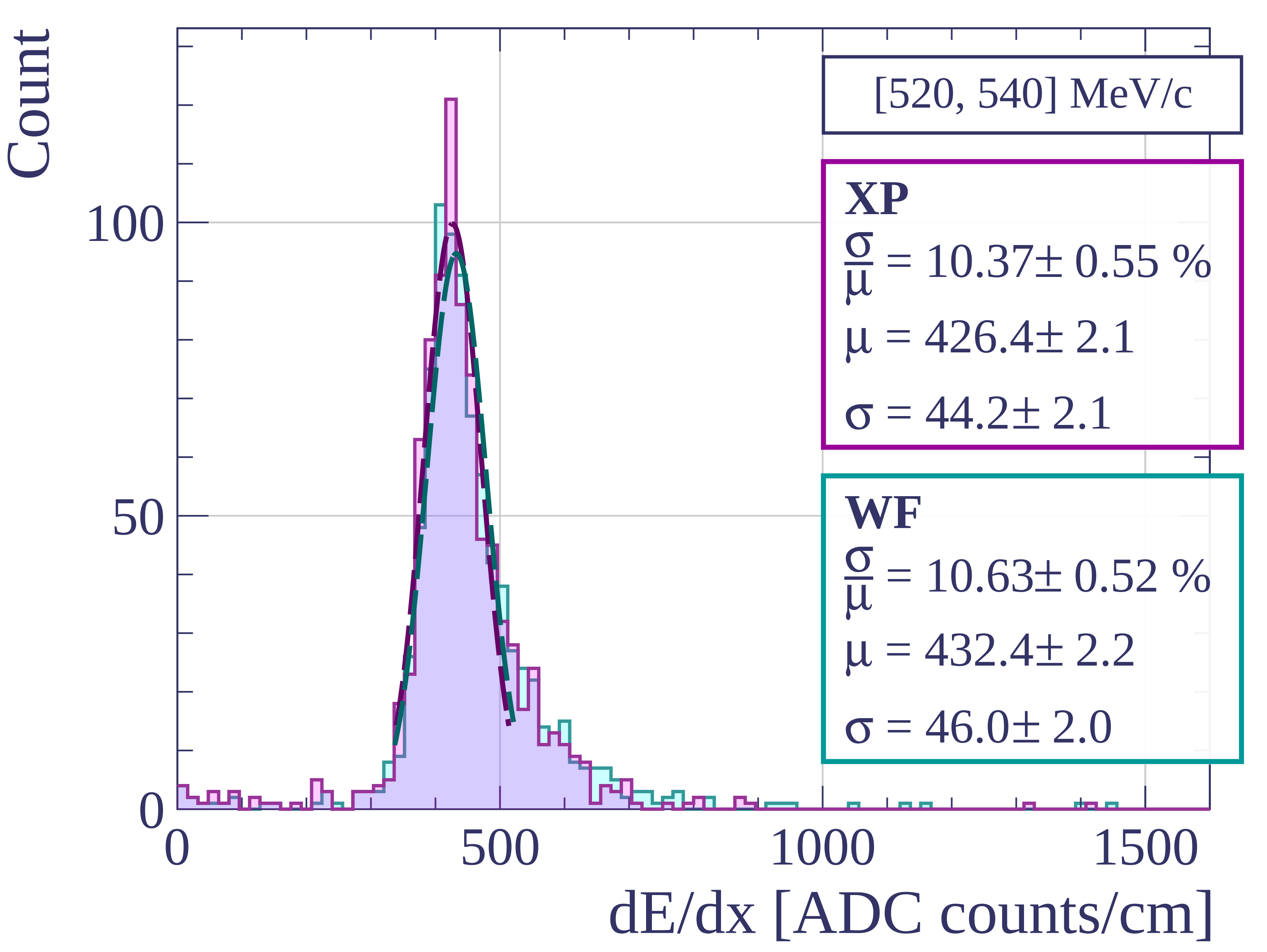


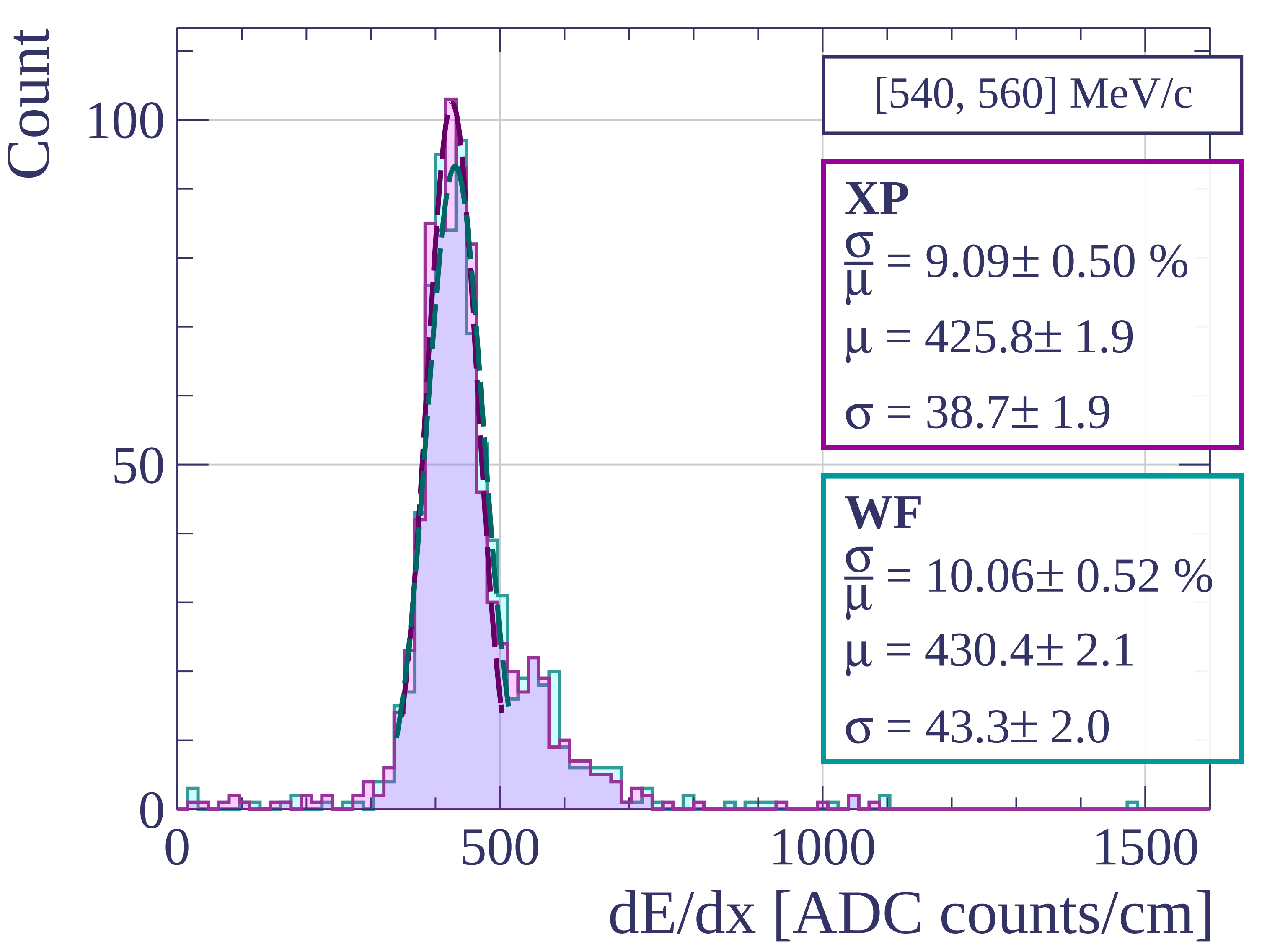
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
[460, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.36± 0.53 %
μ = 418.6± 1.7
σ = 35.0± 2.1
WF
$\frac{\sigma}{\mu}$ = 8.98± 0.48 %
μ = 424.2± 1.8
σ = 38.1± 1.9

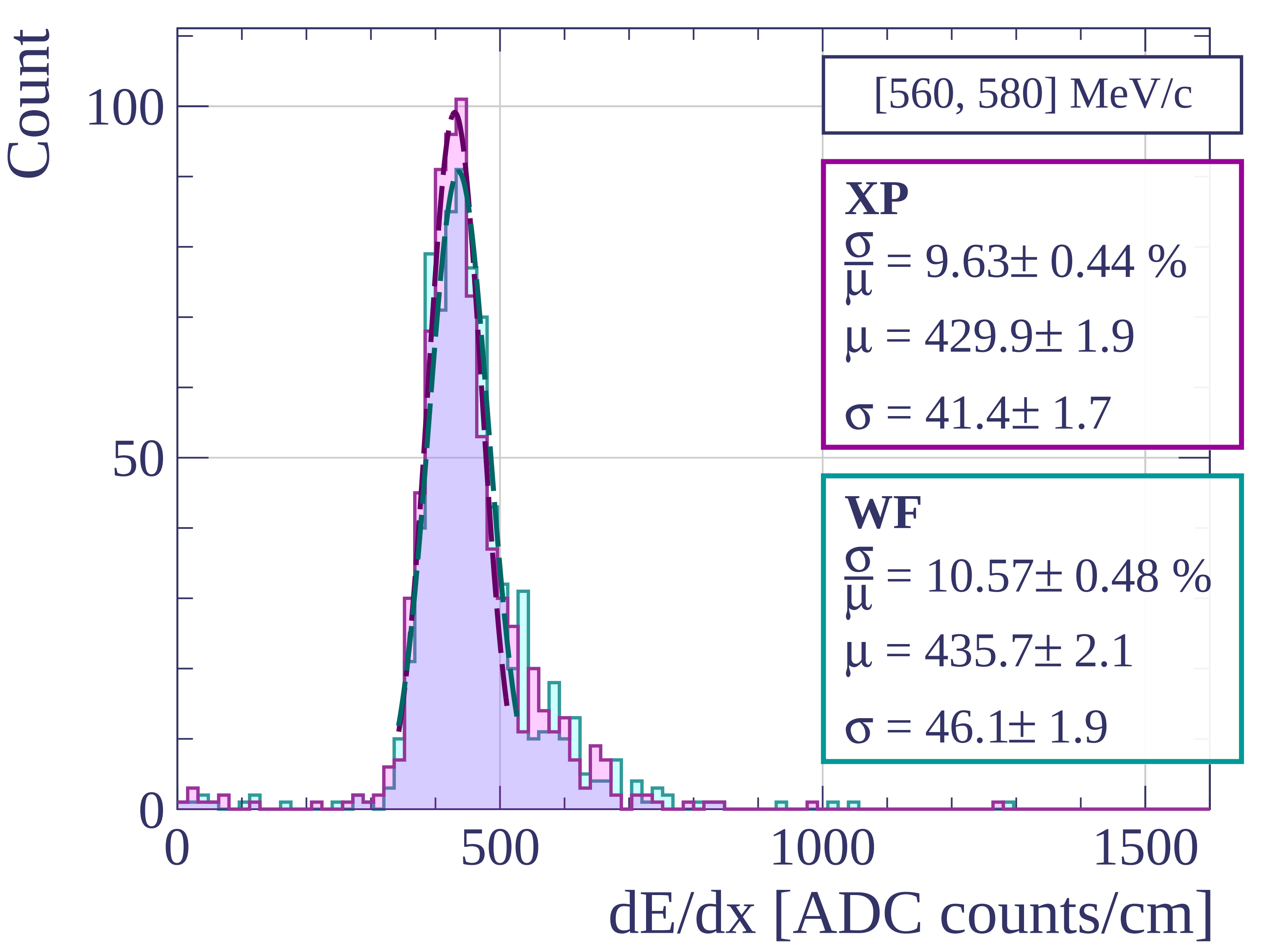

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.41± 0.53 %
μ = 424.8± 2.0
σ = 44.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 10.38± 0.52 %
μ = 429.0± 2.1
σ = 44.5± 2.0

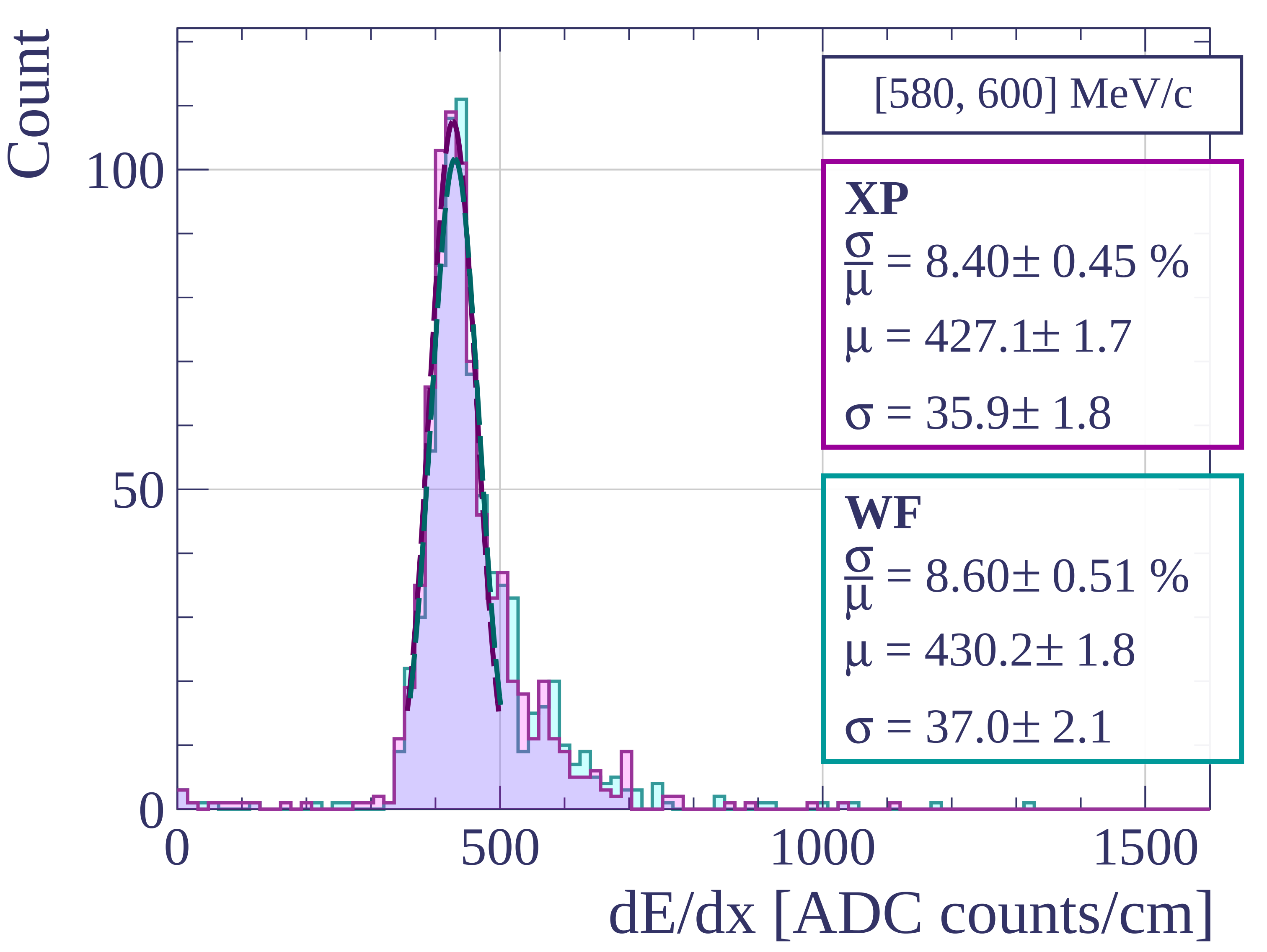


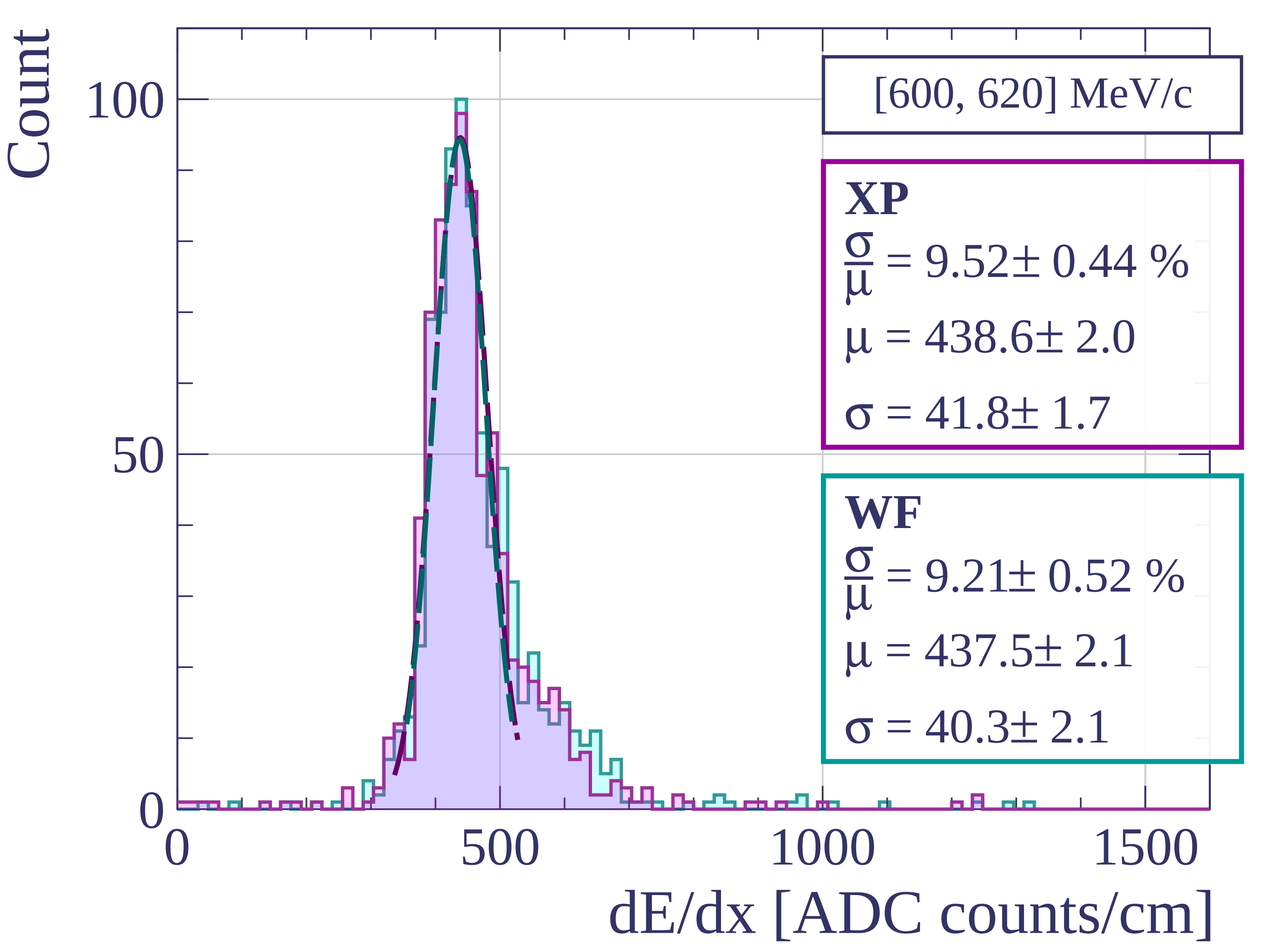
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[500, 520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.47± 0.44 %
μ = 423.0± 2.0
σ = 40.1± 1.7
WF
$\frac{\sigma}{\mu}$ = 9.35± 0.49 %
μ = 423.8± 2.0
σ = 39.6± 1.9


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.37± 0.55 %
μ = 426.4± 2.1
σ = 44.2± 2.1
WF
$\frac{\sigma}{\mu}$ = 10.63± 0.52 %
μ = 432.4± 2.2
σ = 46.0± 2.0

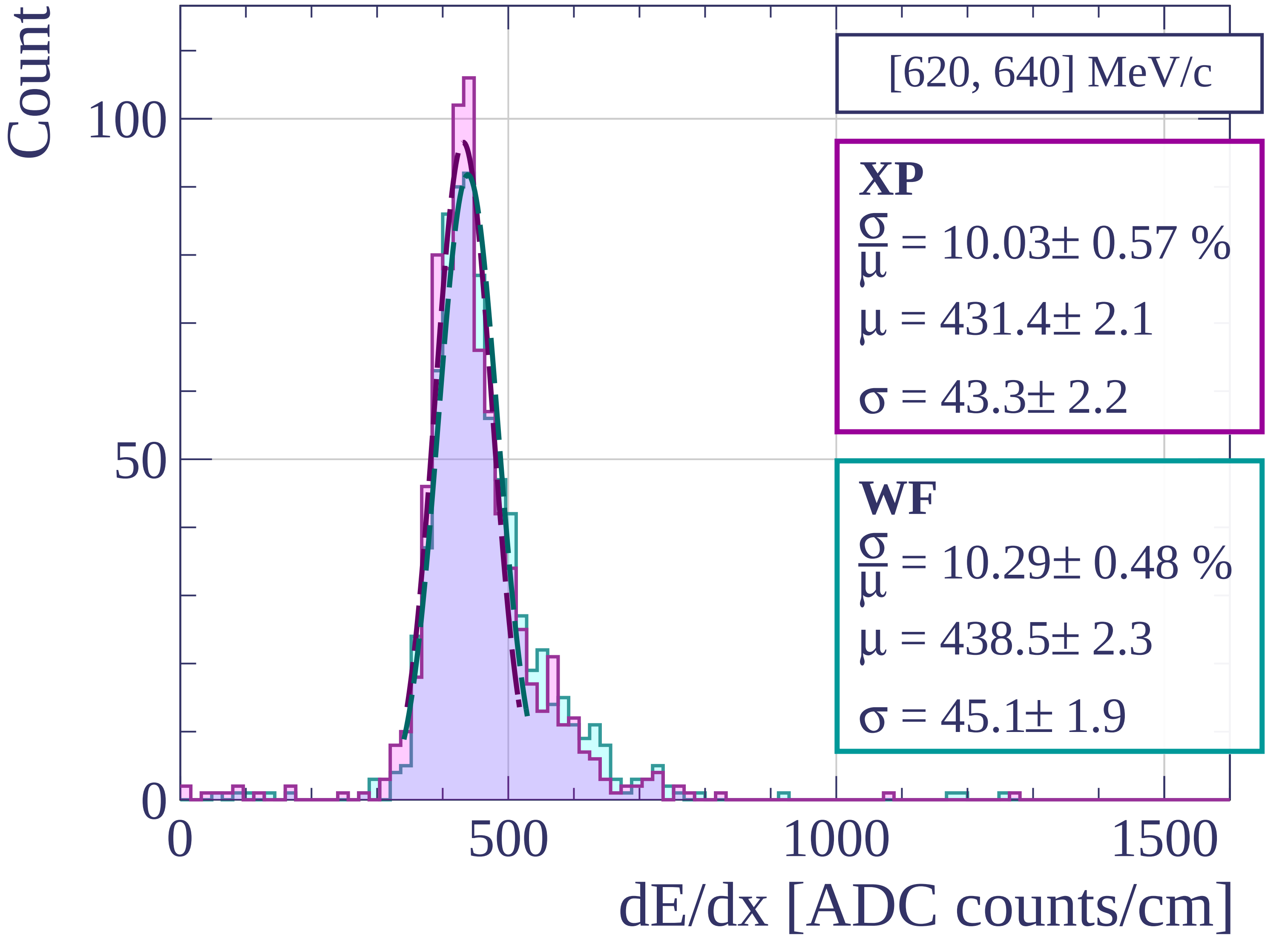

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[540, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.09± 0.50 %
μ = 425.8± 1.9
σ = 38.7± 1.9
WF
$\frac{\sigma}{\mu}$ = 10.06± 0.52 %
μ = 430.4± 2.1
σ = 43.3± 2.0


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.63± 0.44 %
μ = 429.9± 1.9
σ = 41.4± 1.7
WF
$\frac{\sigma}{\mu}$ = 10.57± 0.48 %
μ = 435.7± 2.1
σ = 46.1± 1.9

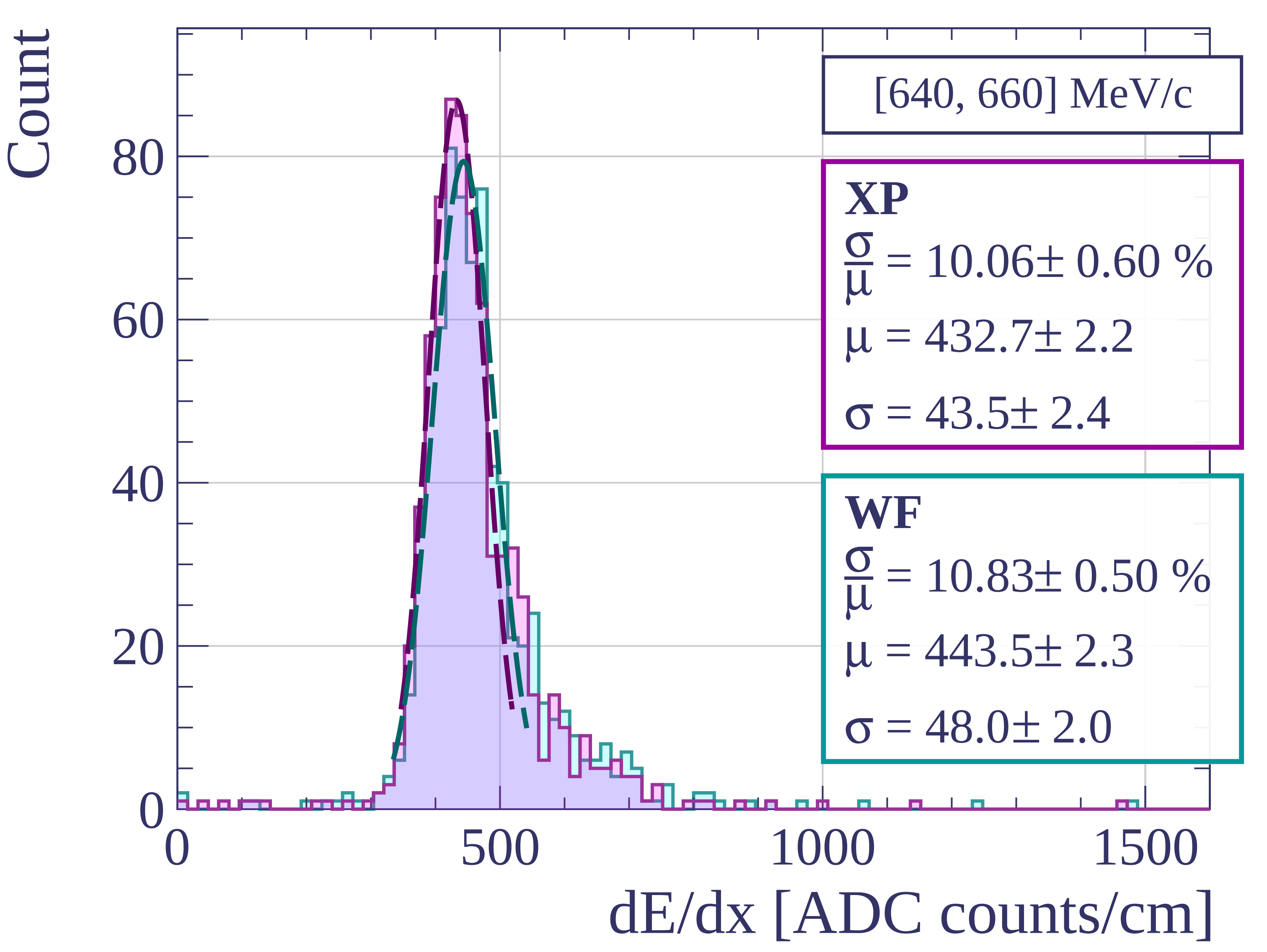

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[580, 600] MeV/c
XP
σ/μ = 8.40± 0.45 %
μ = 427.1± 1.7
σ = 35.9± 1.8
WF
σ/μ = 8.60± 0.51 %
μ = 430.2± 1.8
σ = 37.0± 2.1



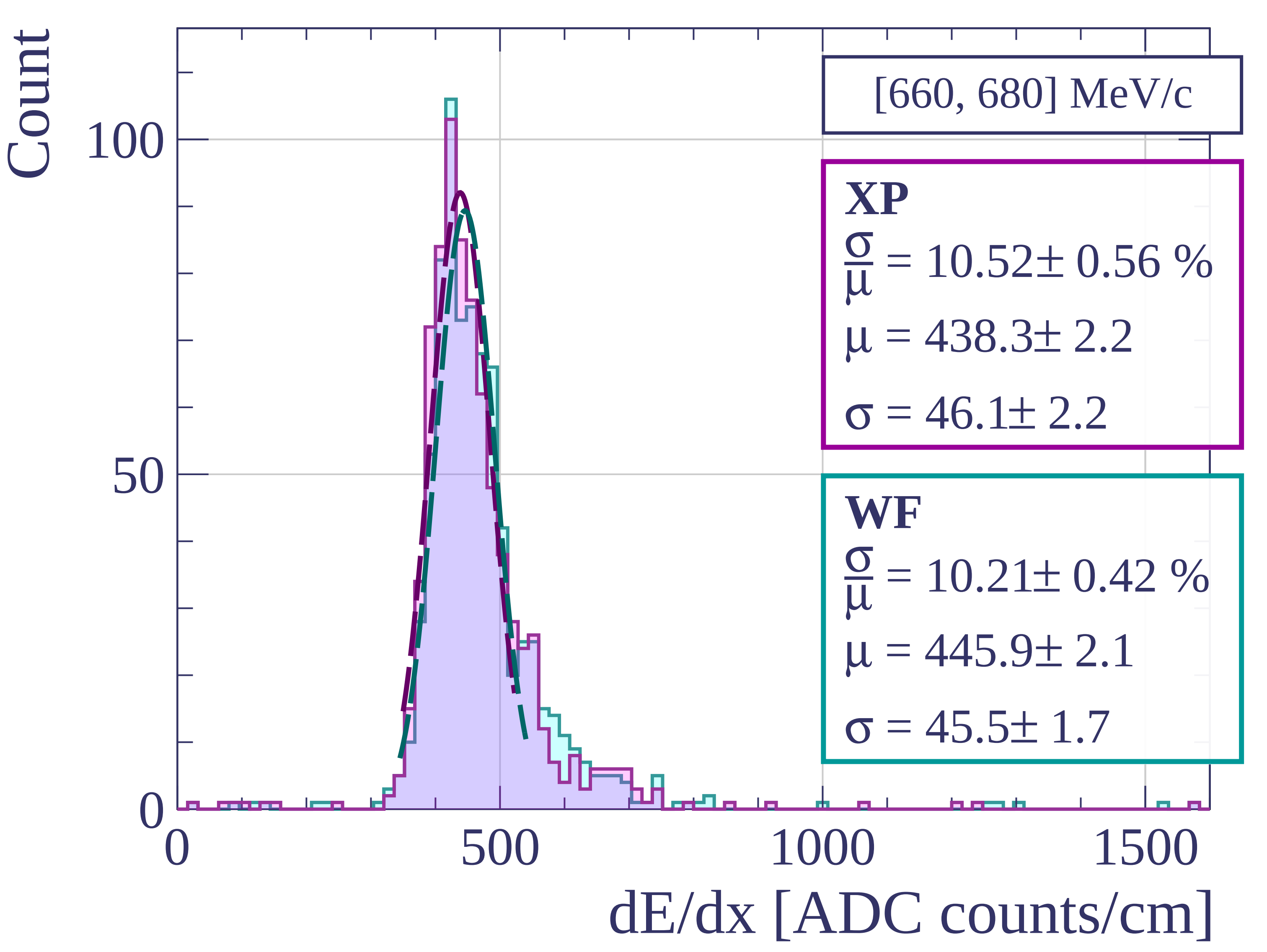
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.52± 0.44 %
μ = 438.6± 2.0
σ = 41.8± 1.7
WF
$\frac{\sigma}{\mu}$ = 9.21± 0.52 %
μ = 437.5± 2.1
σ = 40.3± 2.1

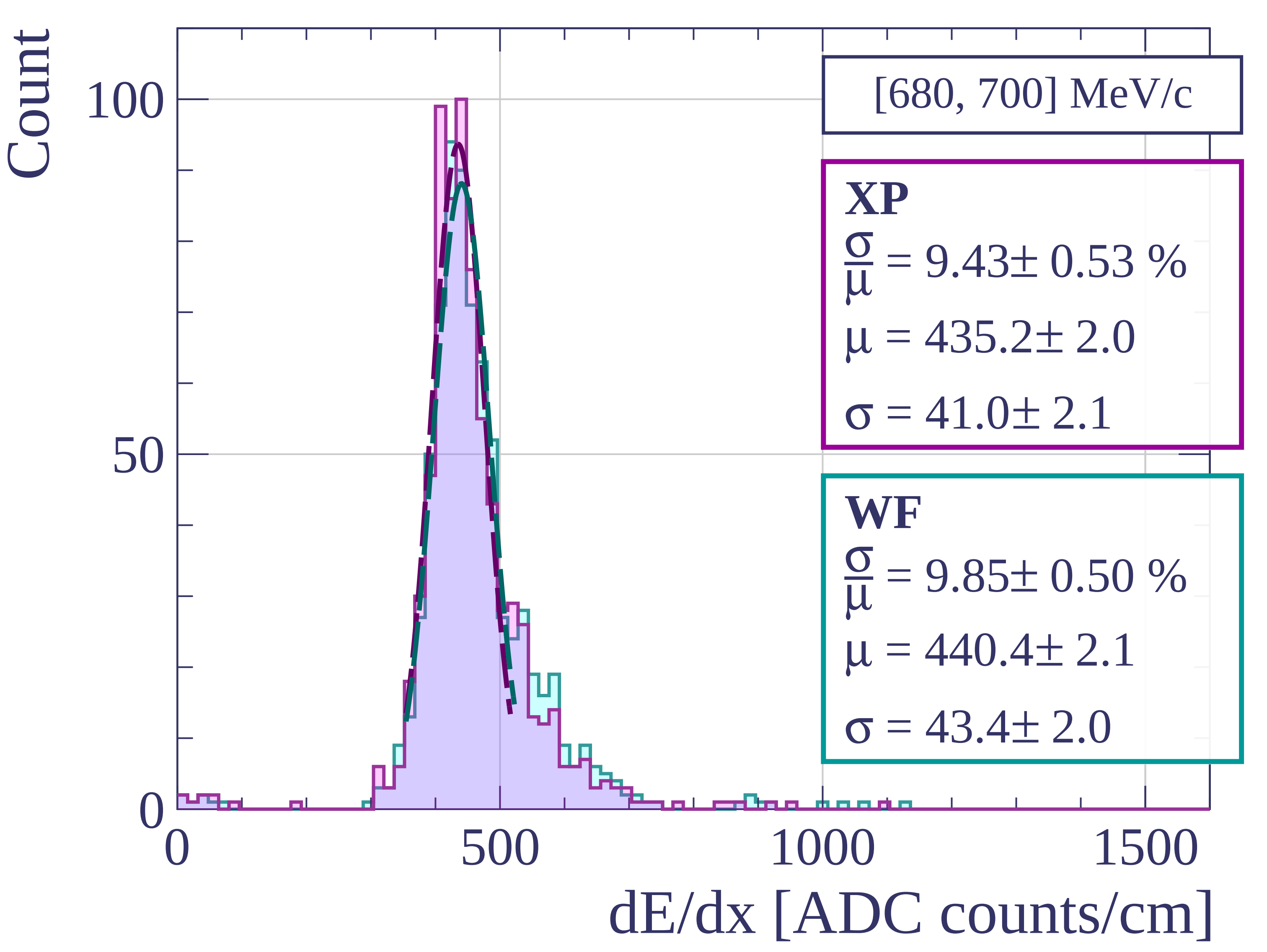


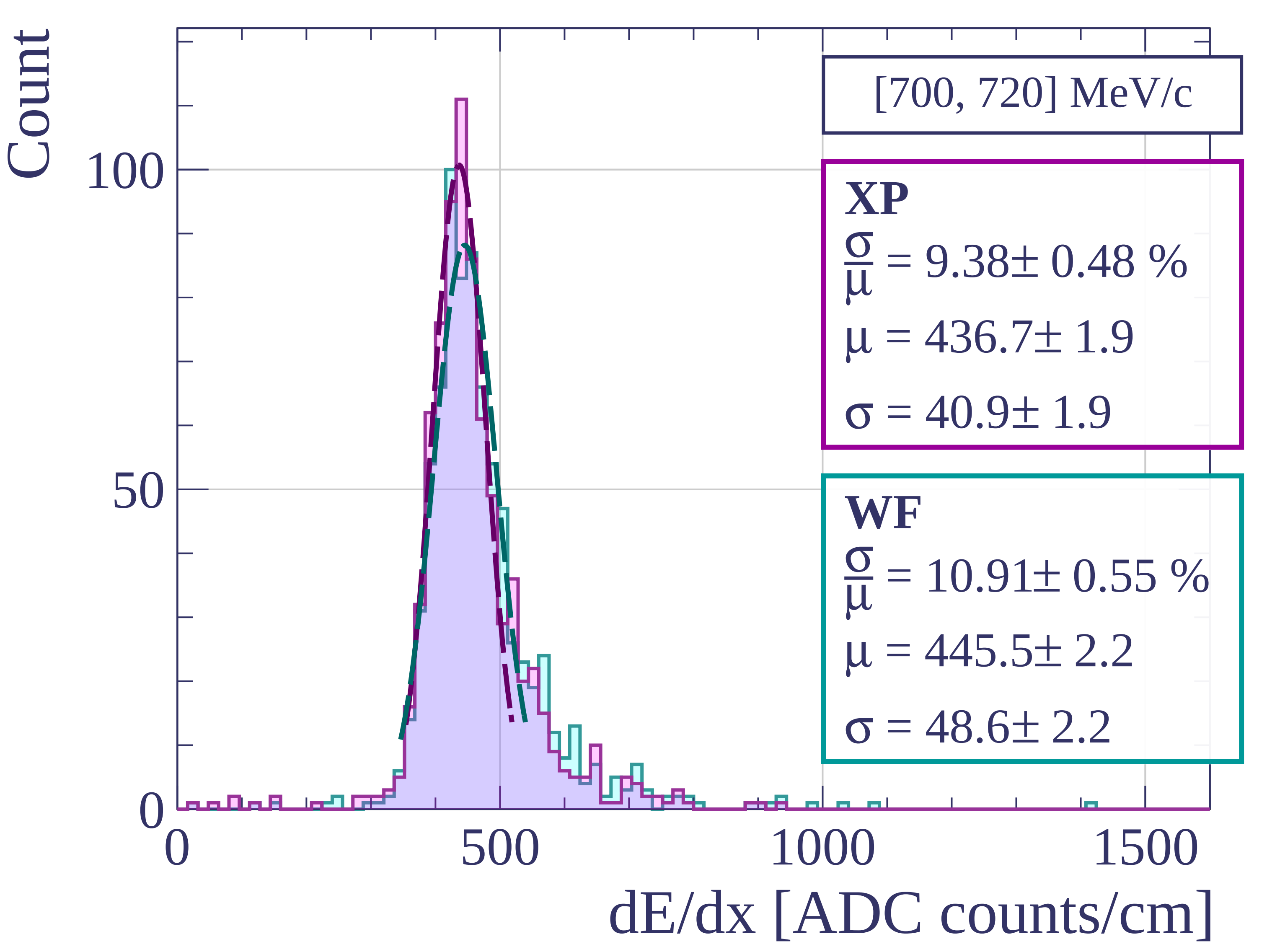
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[620, 640] MeV/c
XP
σ/μ = 10.03± 0.57 %
μ = 431.4± 2.1
σ = 43.3± 2.2
WF
σ/μ = 10.29± 0.48 %
μ = 438.5± 2.3
σ = 45.1± 1.9



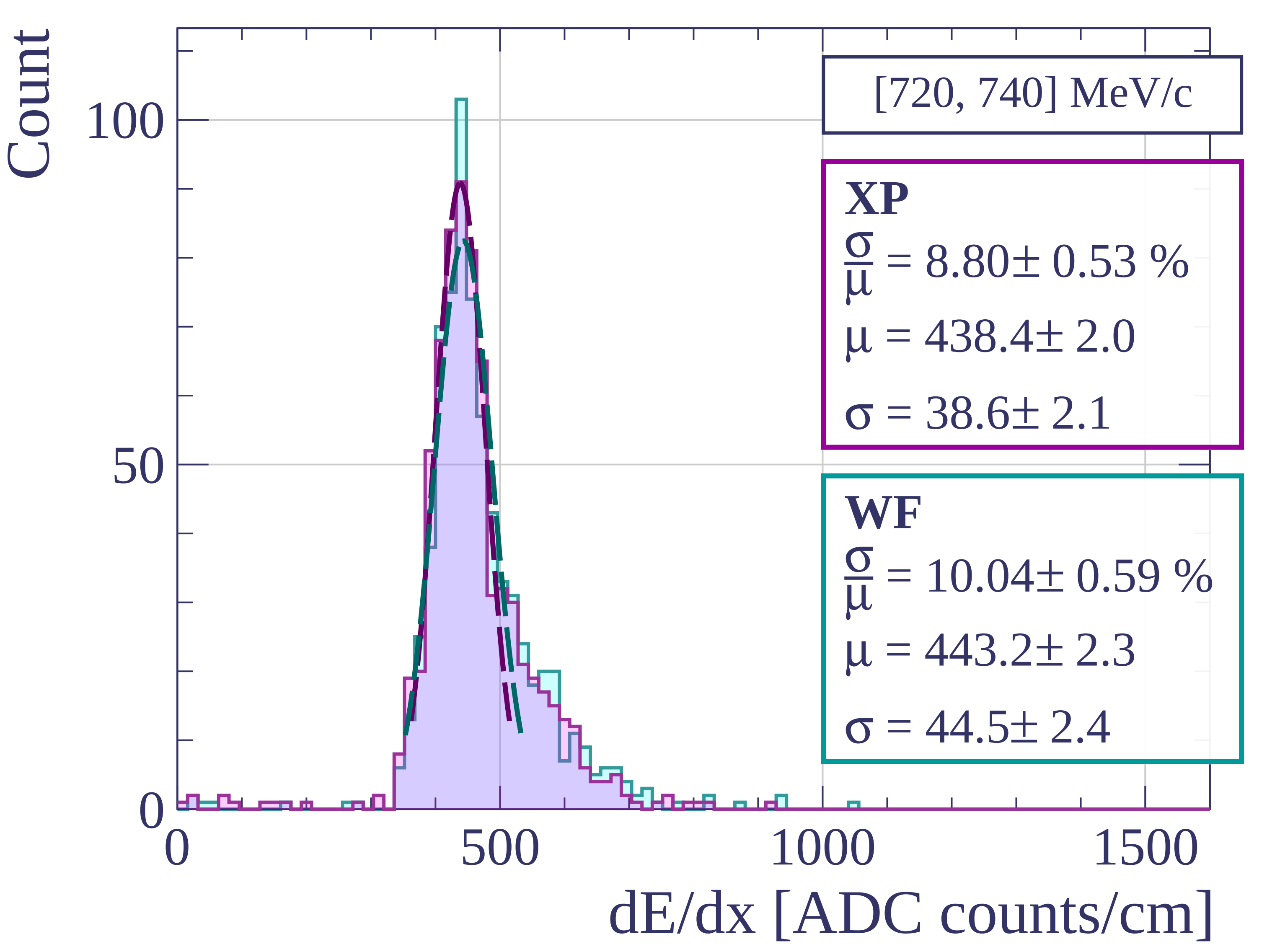
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.06± 0.60 %
μ = 432.7± 2.2
σ = 43.5± 2.4
WF
$\frac{\sigma}{\mu}$ = 10.83± 0.50 %
μ = 443.5± 2.3
σ = 48.0± 2.0


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.52± 0.56 %
μ = 438.3± 2.2
σ = 46.1± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.21± 0.42 %
μ = 445.9± 2.1
σ = 45.5± 1.7

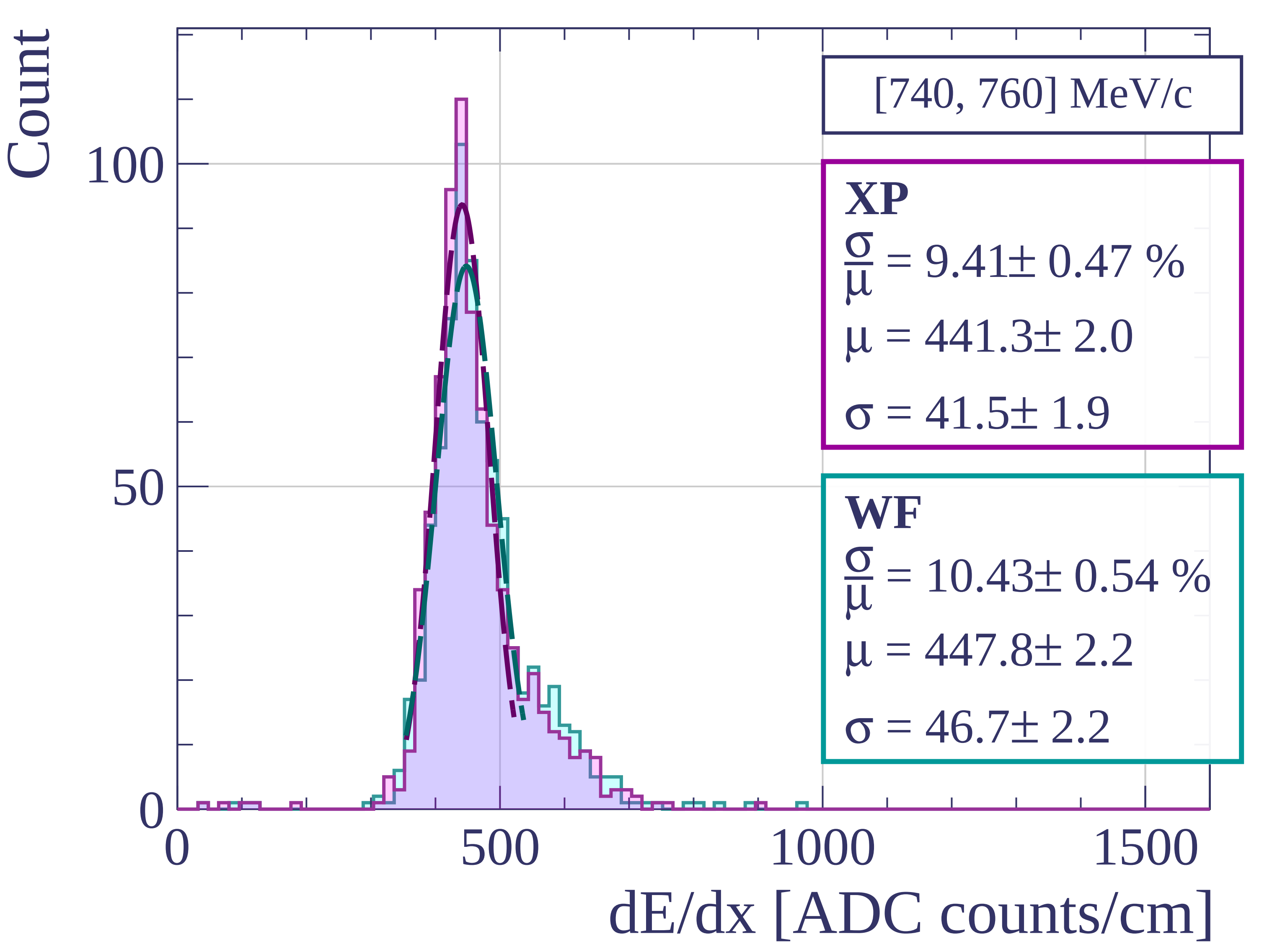

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[680, 700] MeV/c
XP
σ/μ = 9.43± 0.53 %
μ = 435.2± 2.0
σ = 41.0± 2.1
WF
σ/μ = 9.85± 0.50 %
μ = 440.4± 2.1
σ = 43.4± 2.0



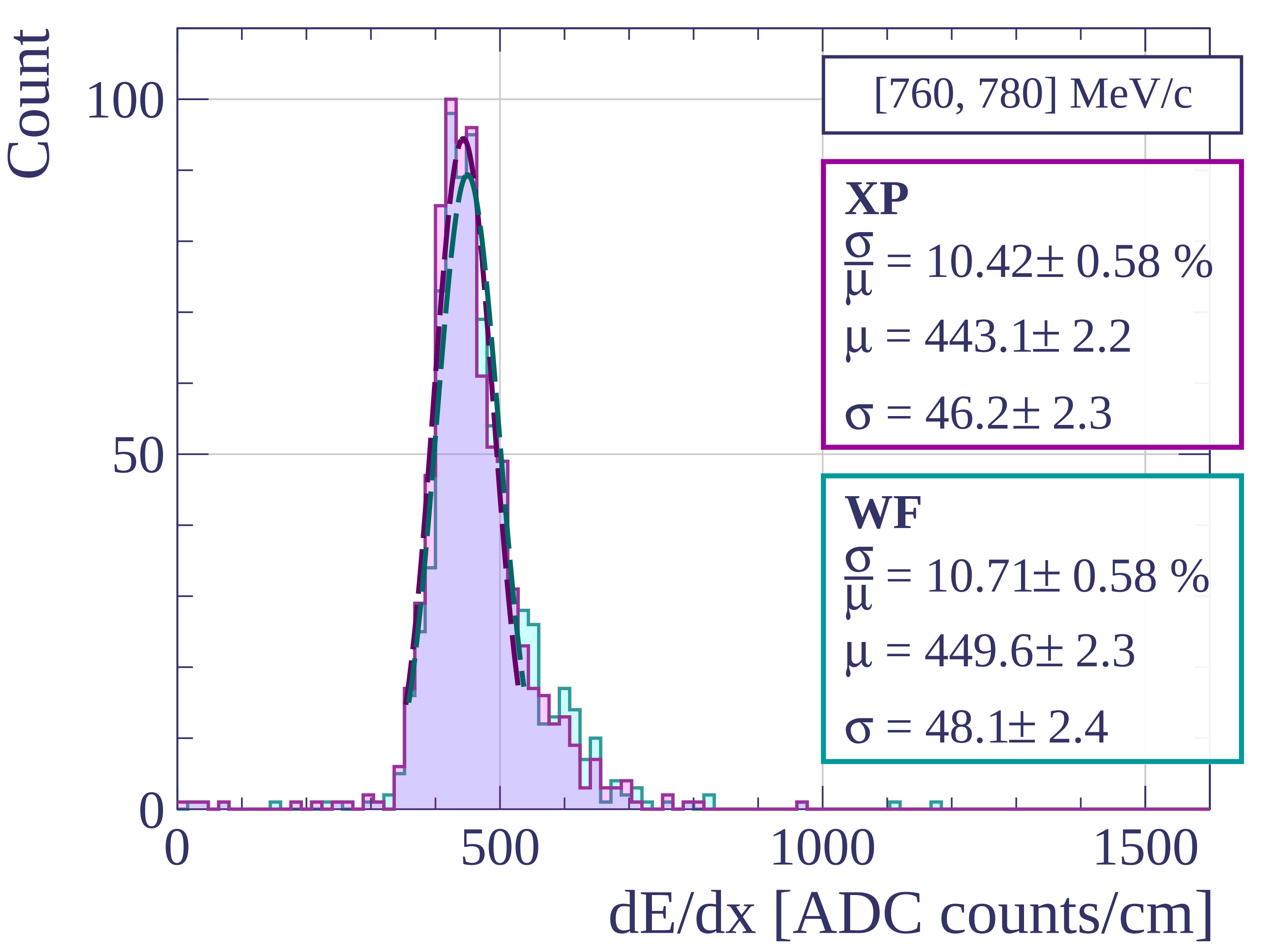
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.38± 0.48 %
μ = 436.7± 1.9
σ = 40.9± 1.9
WF
$\frac{\sigma}{\mu}$ = 10.91± 0.55 %
μ = 445.5± 2.2
σ = 48.6± 2.2



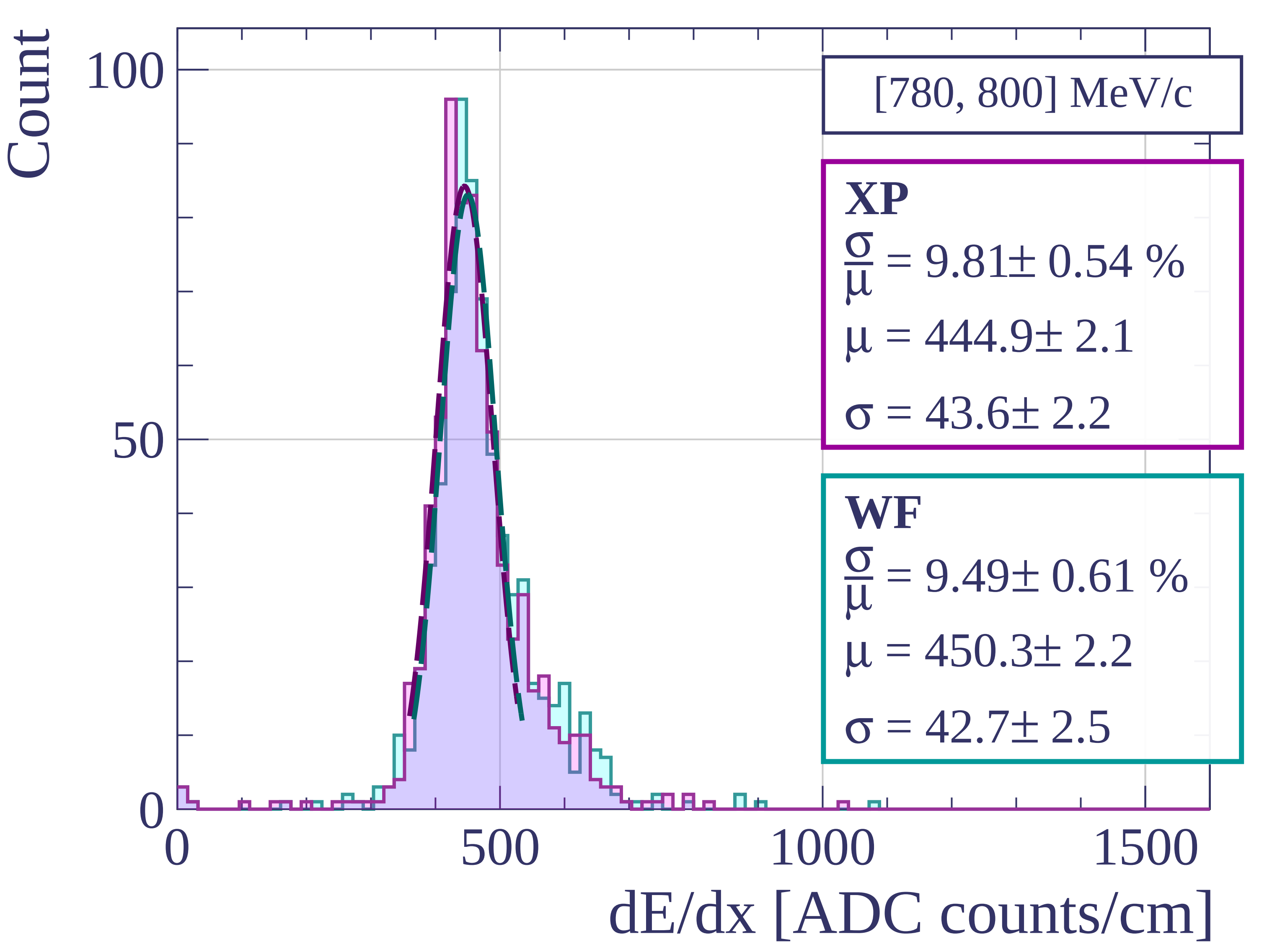
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 740] MeV/c
XP
σ/μ = 8.80± 0.53 %
μ = 438.4± 2.0
σ = 38.6± 2.1
WF
σ/μ = 10.04± 0.59 %
μ = 443.2± 2.3
σ = 44.5± 2.4



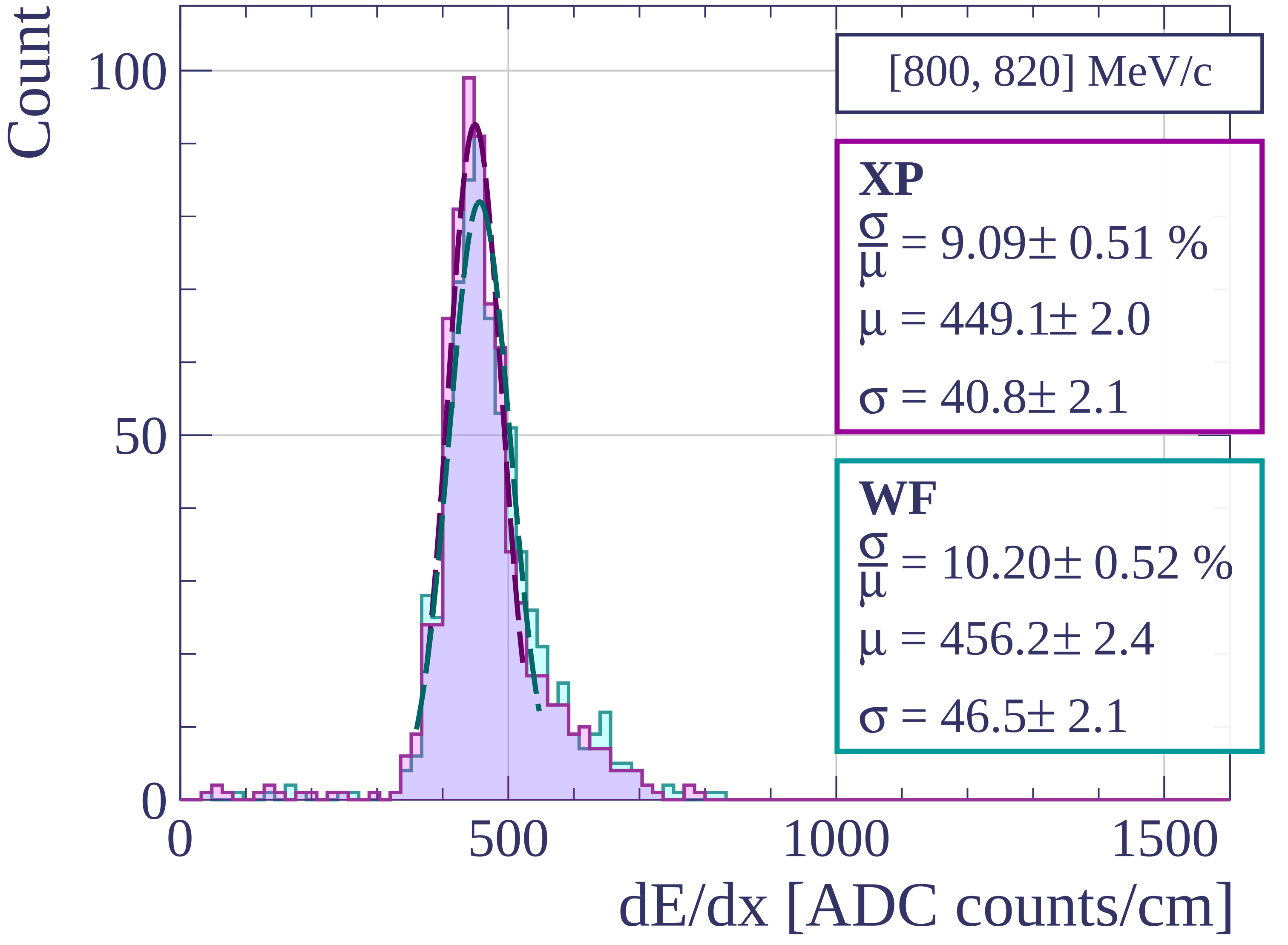
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[740, 760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.41± 0.47 %
μ = 441.3± 2.0
σ = 41.5± 1.9
WF
$\frac{\sigma}{\mu}$ = 10.43± 0.54 %
μ = 447.8± 2.2
σ = 46.7± 2.2



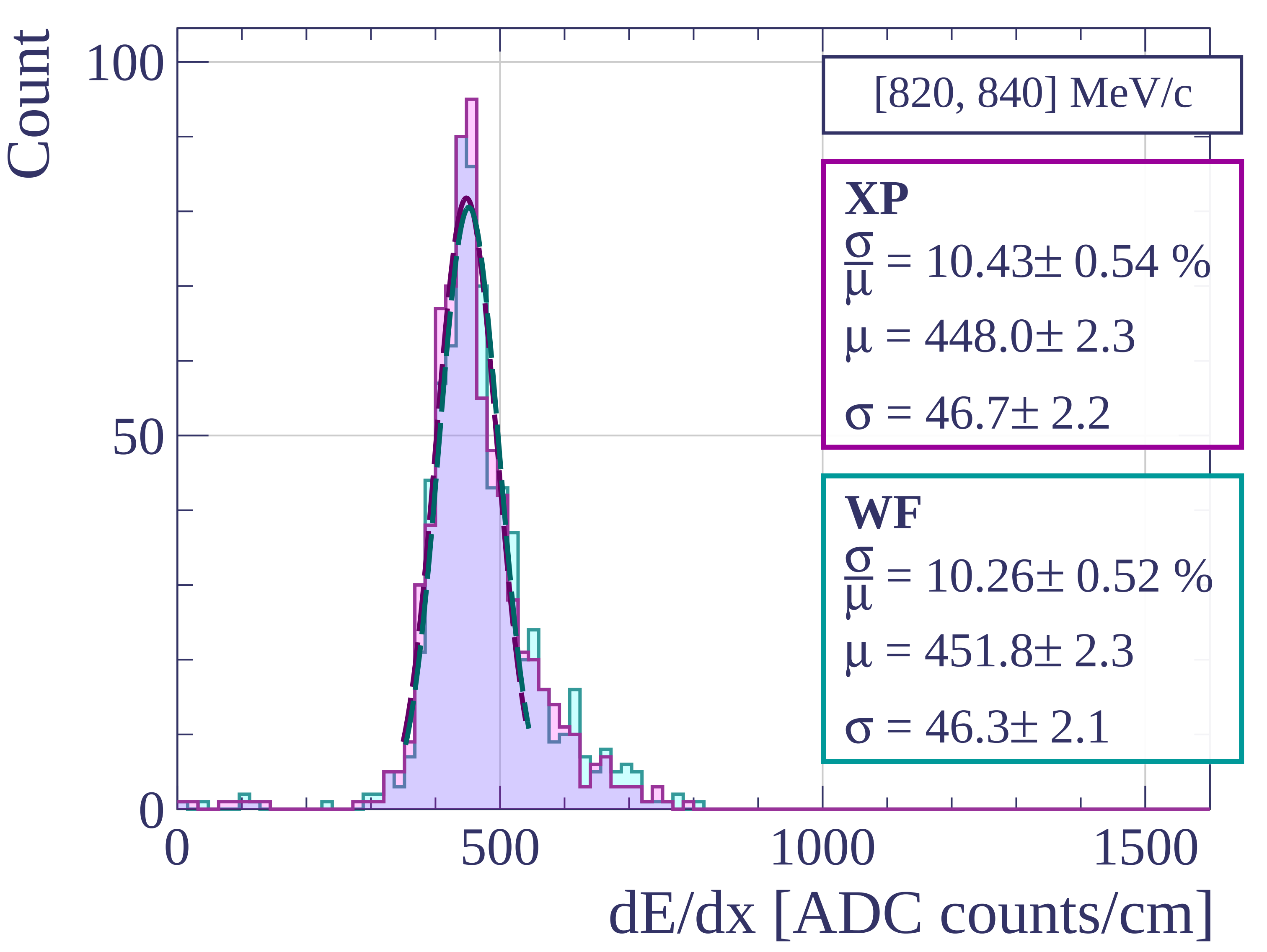
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[760, 780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.42± 0.58 %
μ = 443.1± 2.2
σ = 46.2± 2.3
WF
$\frac{\sigma}{\mu}$ = 10.71± 0.58 %
μ = 449.6± 2.3
σ = 48.1± 2.4

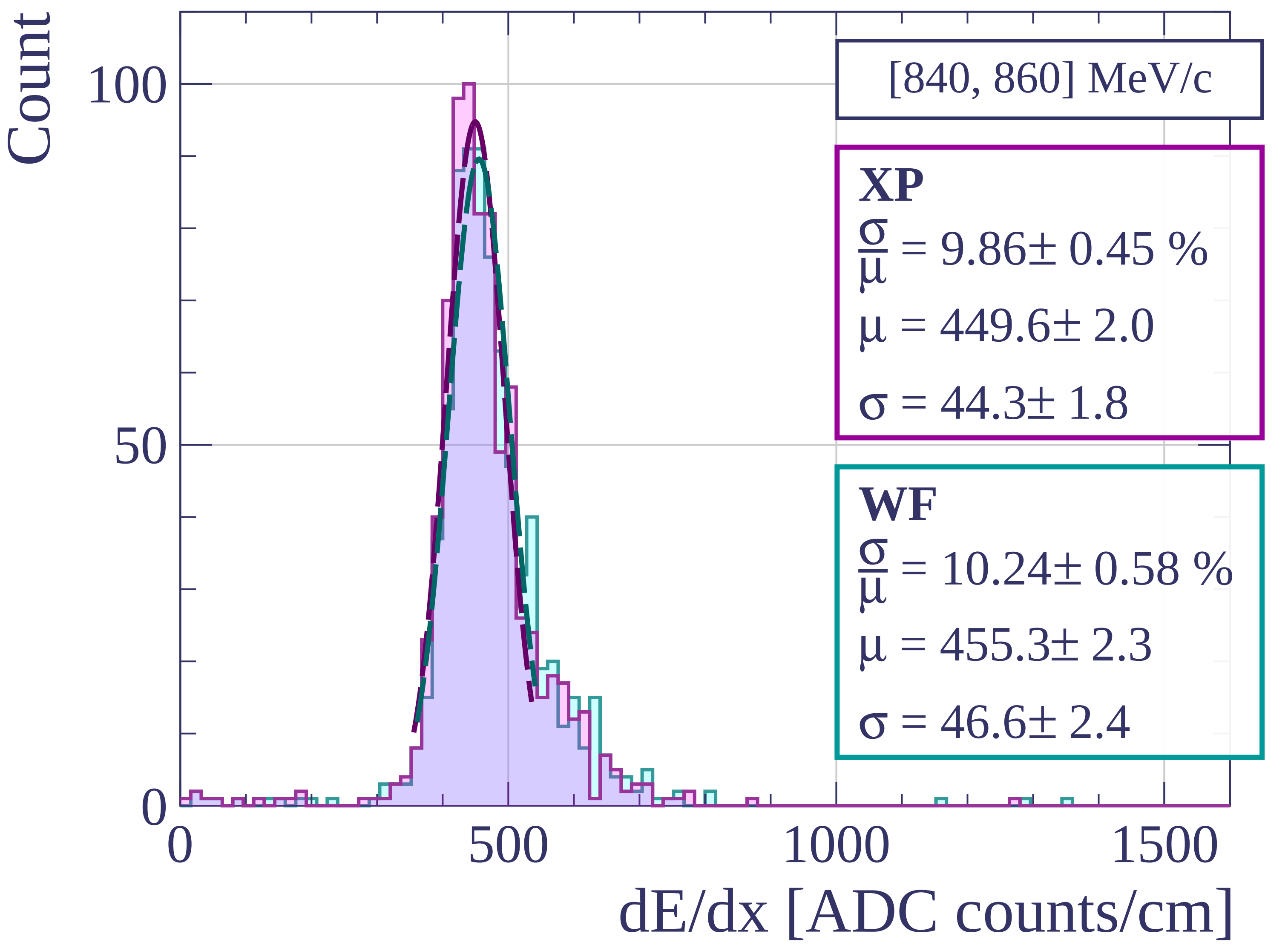


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[780, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.81± 0.54 %
μ = 444.9± 2.1
σ = 43.6± 2.2
WF
$\frac{\sigma}{\mu}$ = 9.49± 0.61 %
μ = 450.3± 2.2
σ = 42.7± 2.5

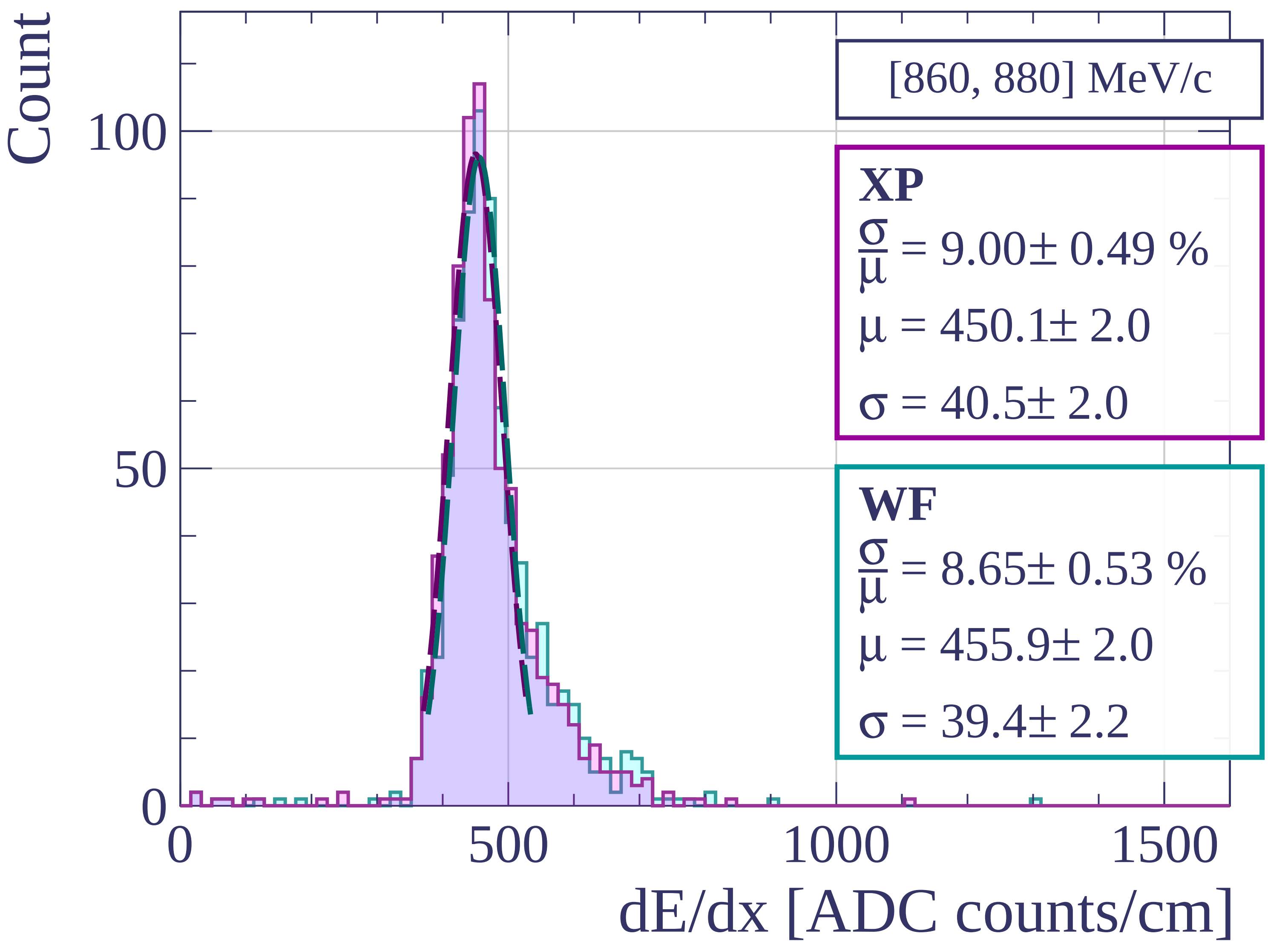


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.09± 0.51 %
μ = 449.1± 2.0
σ = 40.8± 2.1
WF
$\frac{\sigma}{\mu}$ = 10.20± 0.52 %
μ = 456.2± 2.4
σ = 46.5± 2.1

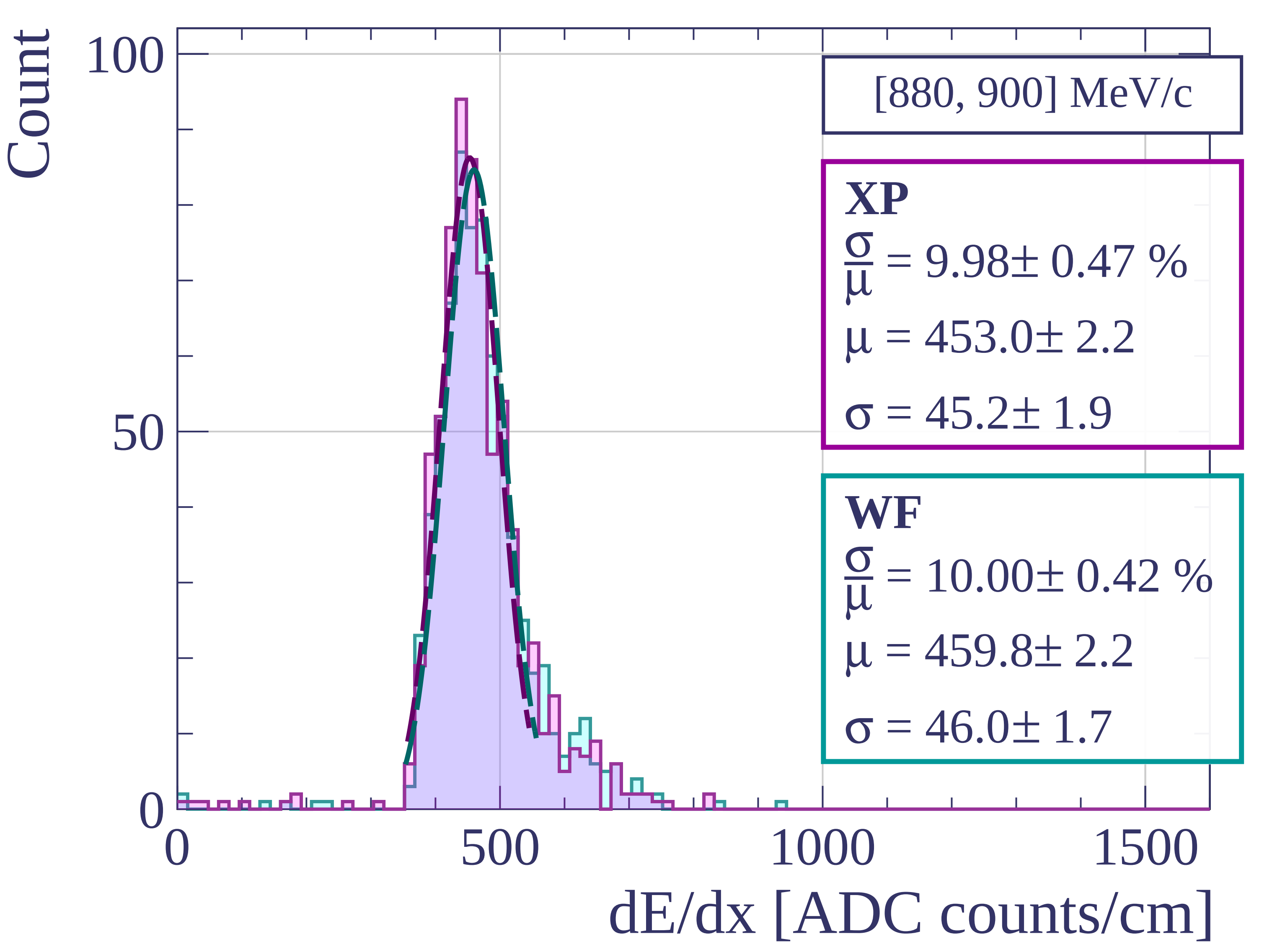

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[820, 840] MeV/c
XP
σ/μ = 10.43± 0.54 %
μ = 448.0± 2.3
σ = 46.7± 2.2
WF
σ/μ = 10.26± 0.52 %
μ = 451.8± 2.3
σ = 46.3± 2.1

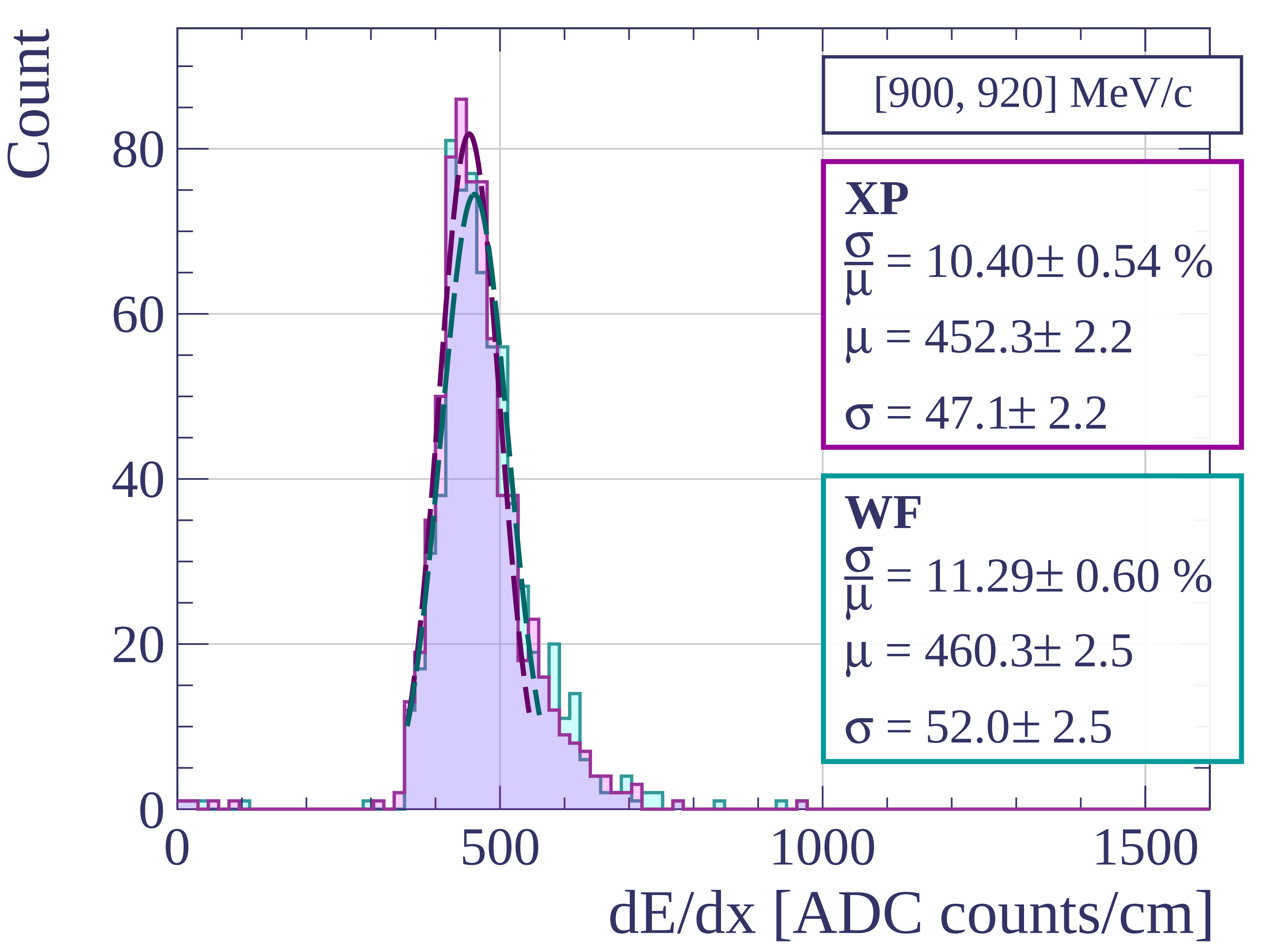


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[840, 860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.86± 0.45 %
μ = 449.6± 2.0
σ = 44.3± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.24± 0.58 %
μ = 455.3± 2.3
σ = 46.6± 2.4

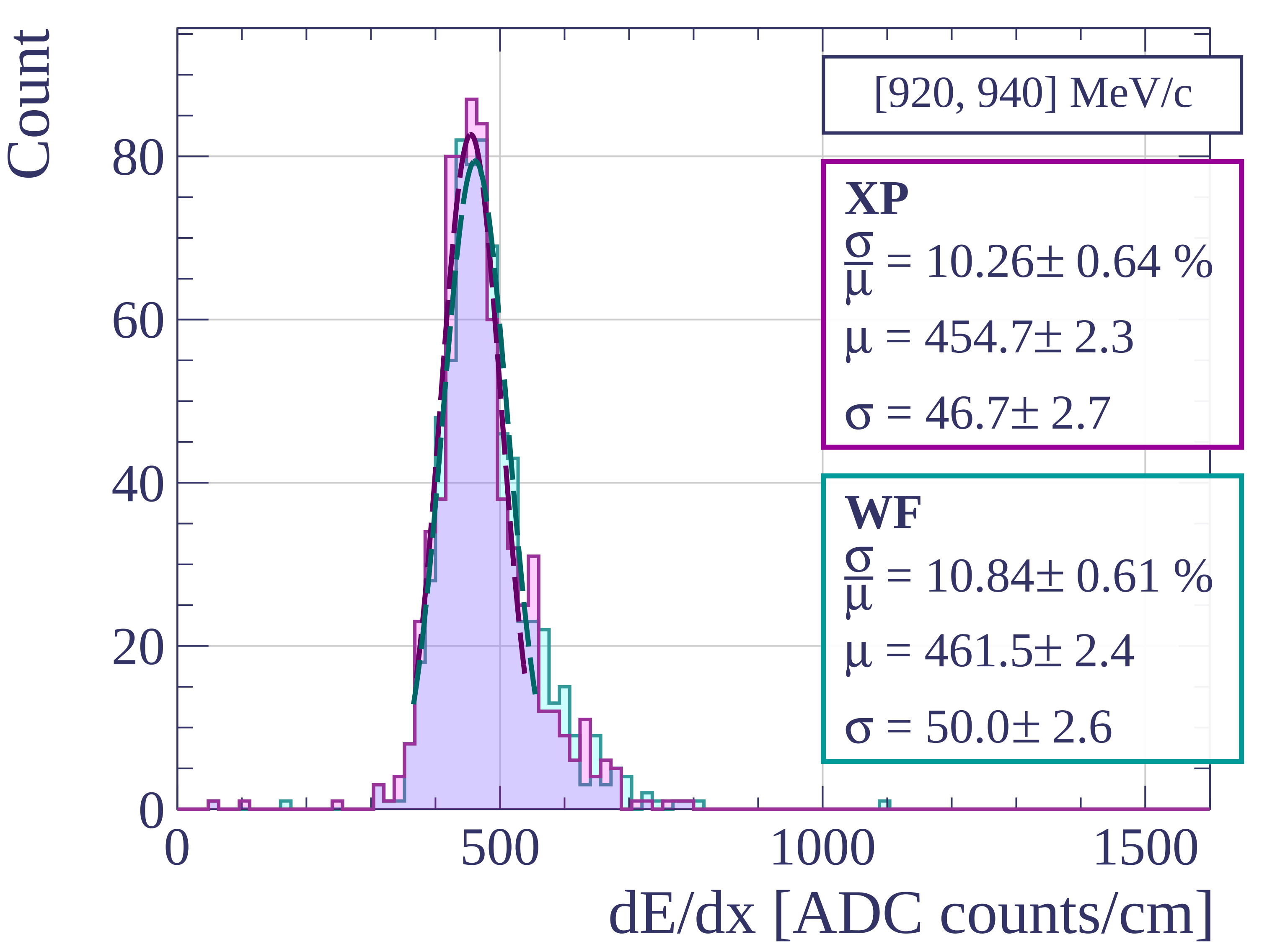


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[860, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.00± 0.49 %
μ = 450.1± 2.0
σ = 40.5± 2.0
WF
$\frac{\sigma}{\mu}$ = 8.65± 0.53 %
μ = 455.9± 2.0
σ = 39.4± 2.2

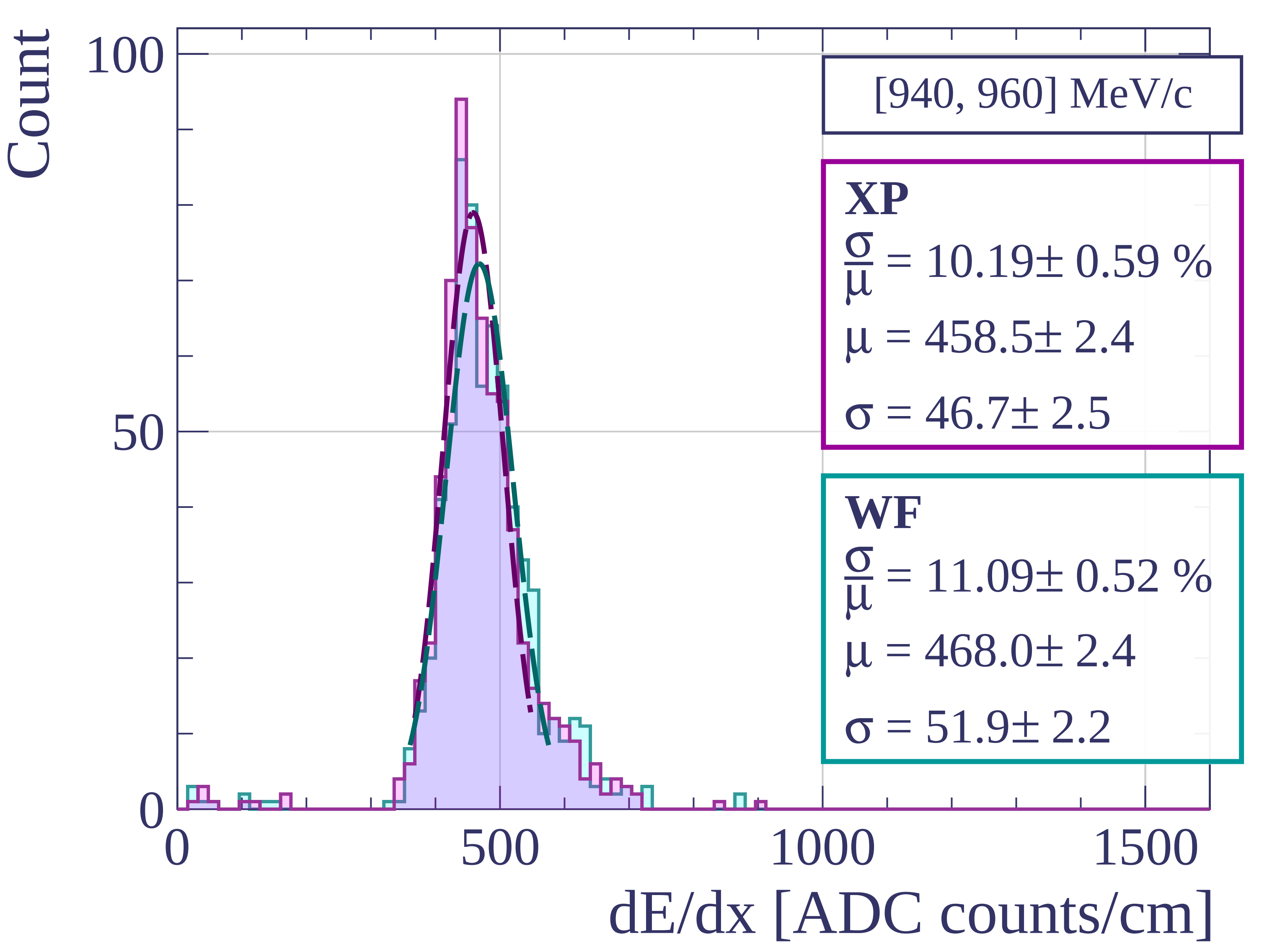

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 900] MeV/c
XP
σ/μ = 9.98± 0.47 %
μ = 453.0± 2.2
σ = 45.2± 1.9
WF
σ/μ = 10.00± 0.42 %
μ = 459.8± 2.2
σ = 46.0± 1.7



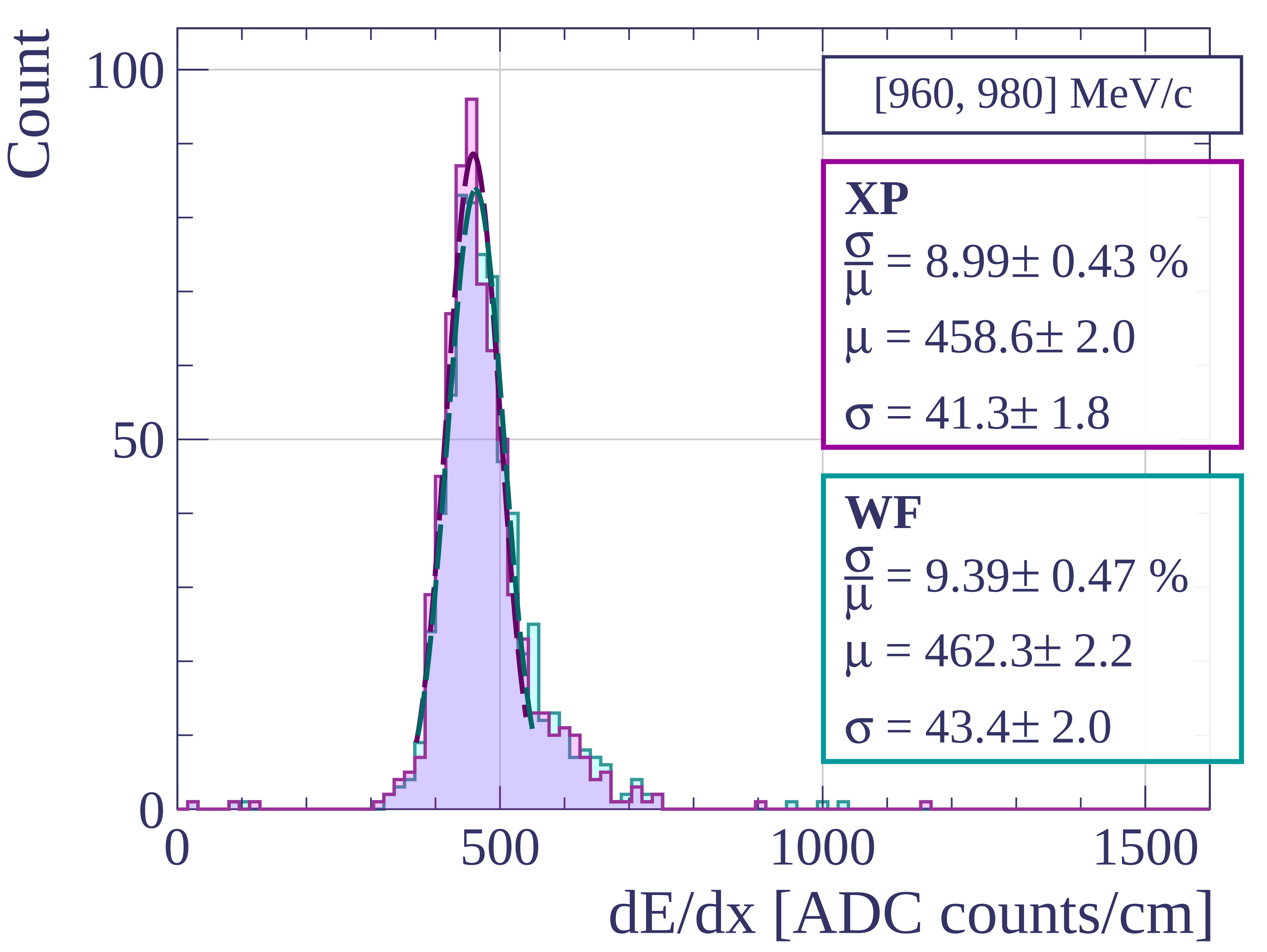
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[900, 920] MeV/c
XP
σ/μ = 10.40± 0.54 %
μ = 452.3± 2.2
σ = 47.1± 2.2
WF
σ/μ = 11.29± 0.60 %
μ = 460.3± 2.5
σ = 52.0± 2.5



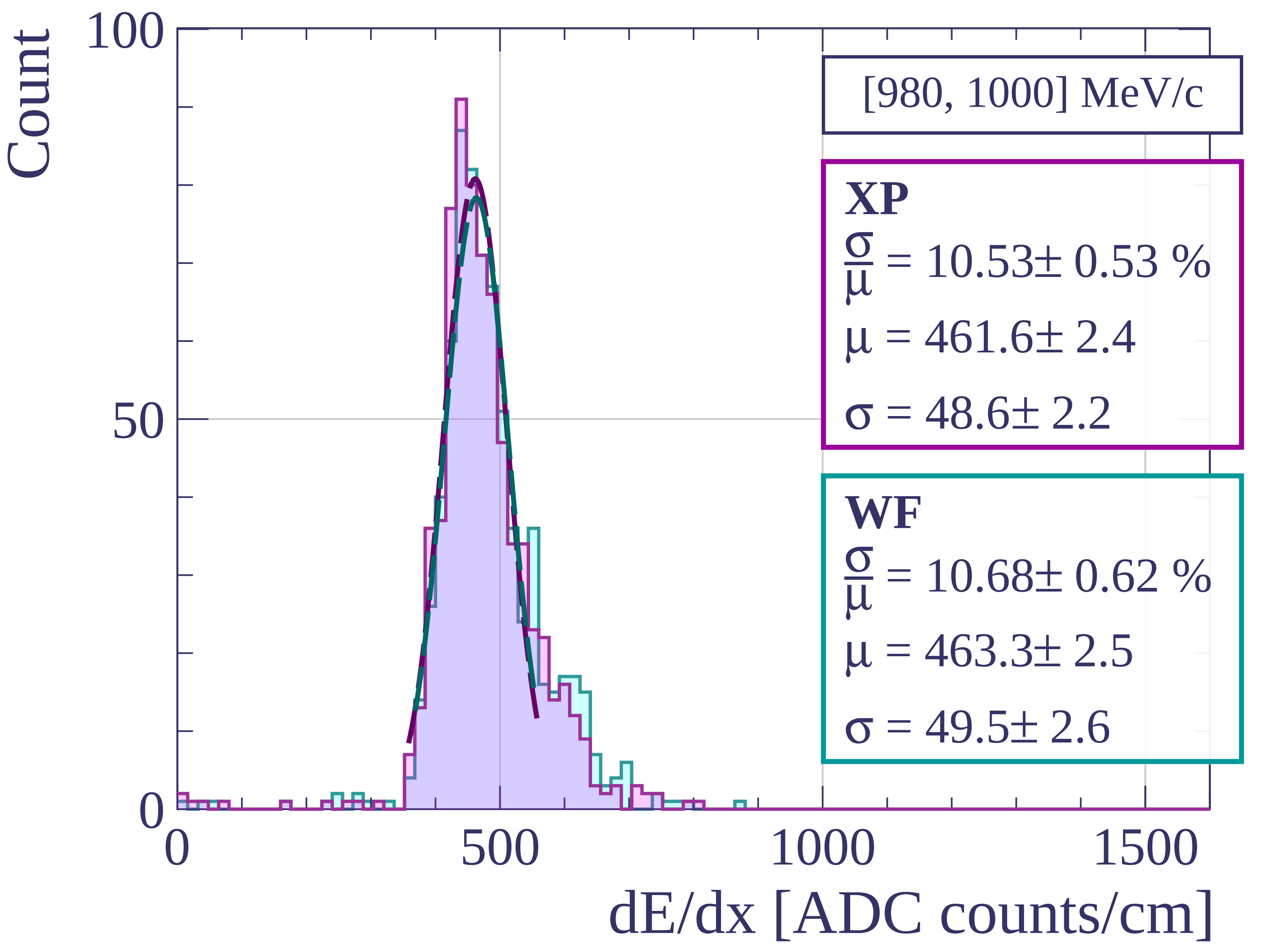
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[920, 940] MeV/c
XP
σ/μ = 10.26± 0.64 %
μ = 454.7± 2.3
σ = 46.7± 2.7
WF
σ/μ = 10.84± 0.61 %
μ = 461.5± 2.4
σ = 50.0± 2.6

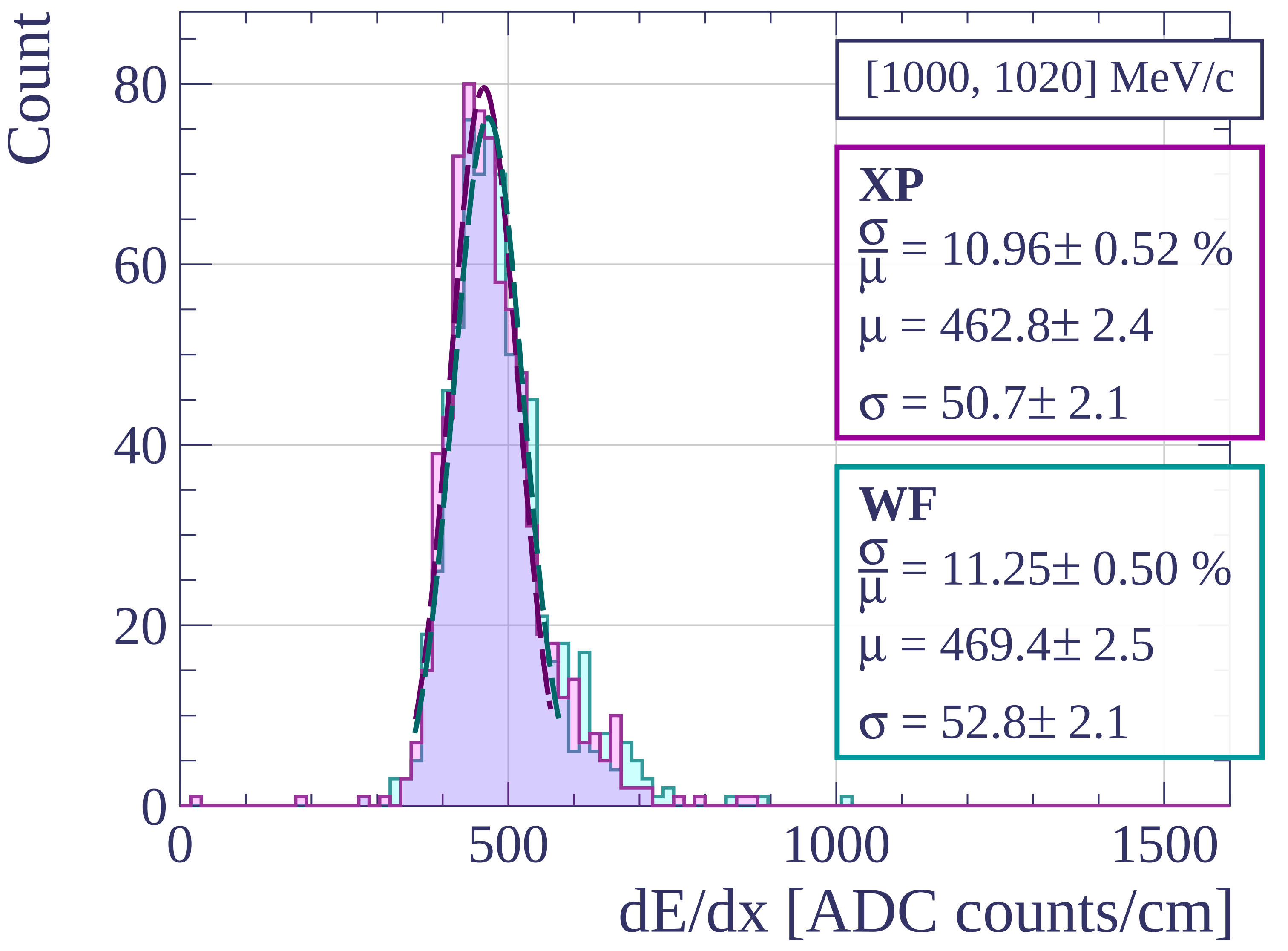


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.19± 0.59 %
μ = 458.5± 2.4
σ = 46.7± 2.5
WF
$\frac{\sigma}{\mu}$ = 11.09± 0.52 %
μ = 468.0± 2.4
σ = 51.9± 2.2

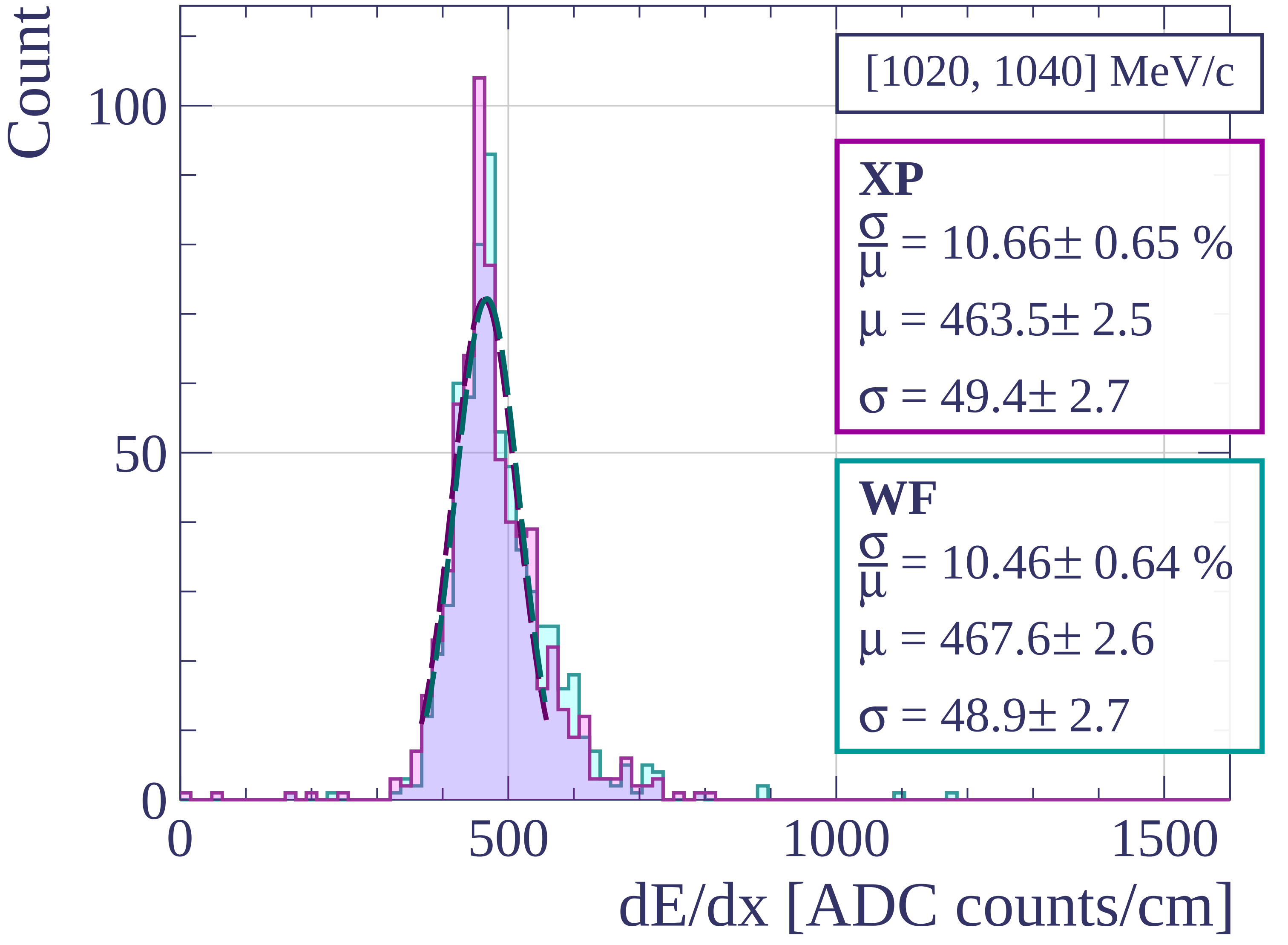


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.99± 0.43 %
μ = 458.6± 2.0
σ = 41.3± 1.8
WF
$\frac{\sigma}{\mu}$ = 9.39± 0.47 %
μ = 462.3± 2.2
σ = 43.4± 2.0

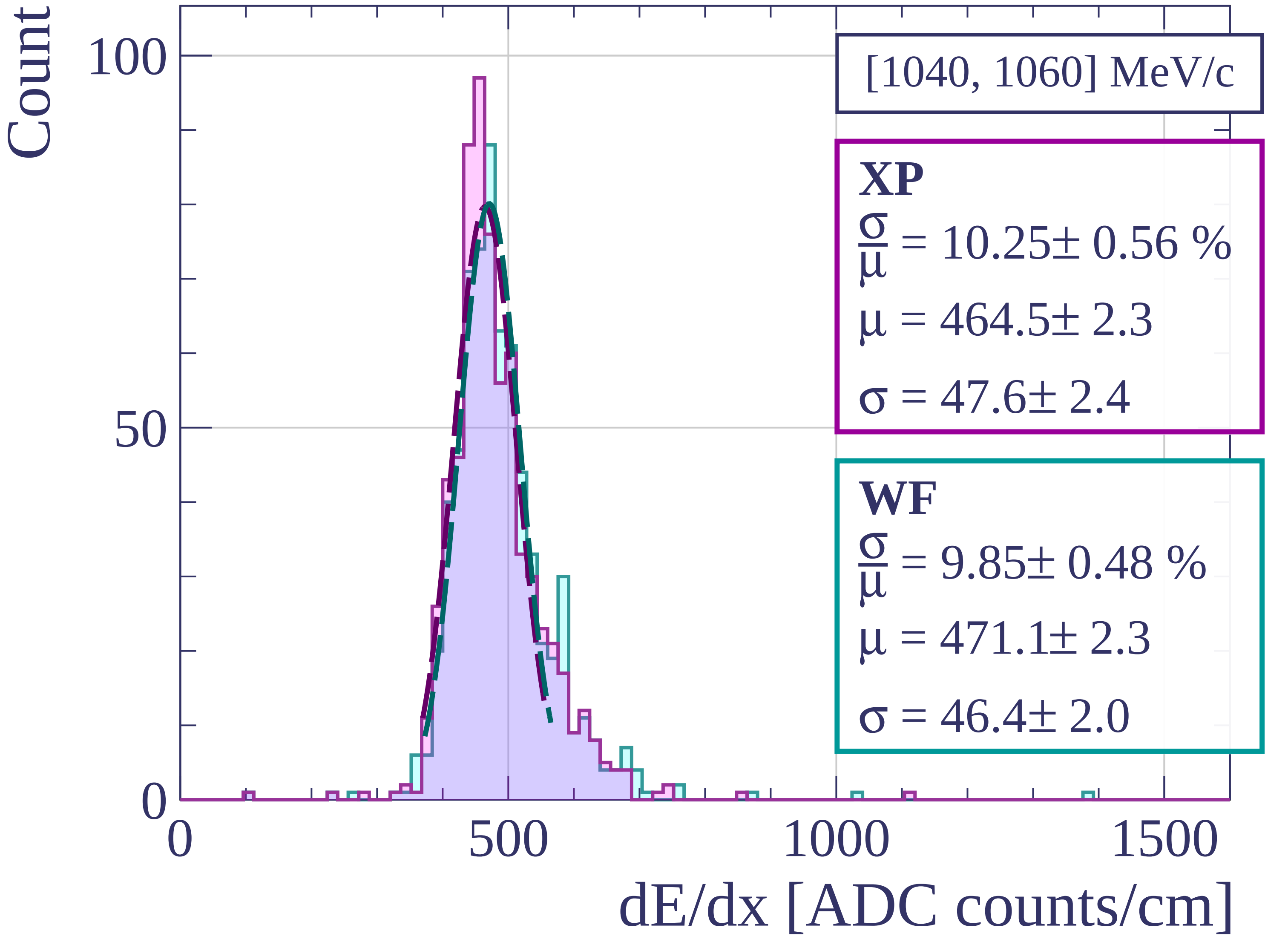

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
σ/μ = 10.53± 0.53 %
μ = 461.6± 2.4
σ = 48.6± 2.2
WF
σ/μ = 10.68± 0.62 %
μ = 463.3± 2.5
σ = 49.5± 2.6

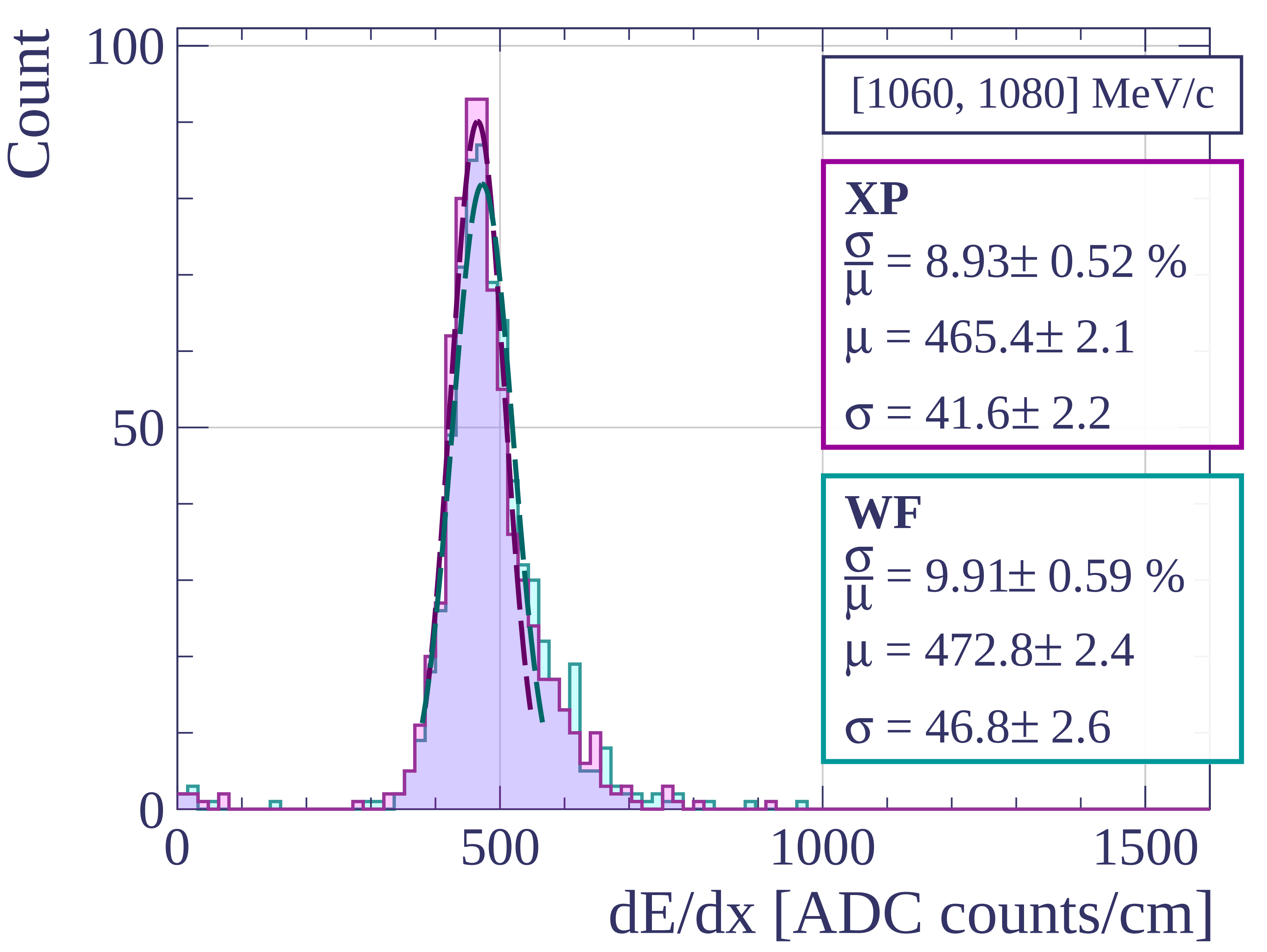


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1000, 1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.96± 0.52 %
μ = 462.8± 2.4
σ = 50.7± 2.1
WF
$\frac{\sigma}{\mu}$ = 11.25± 0.50 %
μ = 469.4± 2.5
σ = 52.8± 2.1

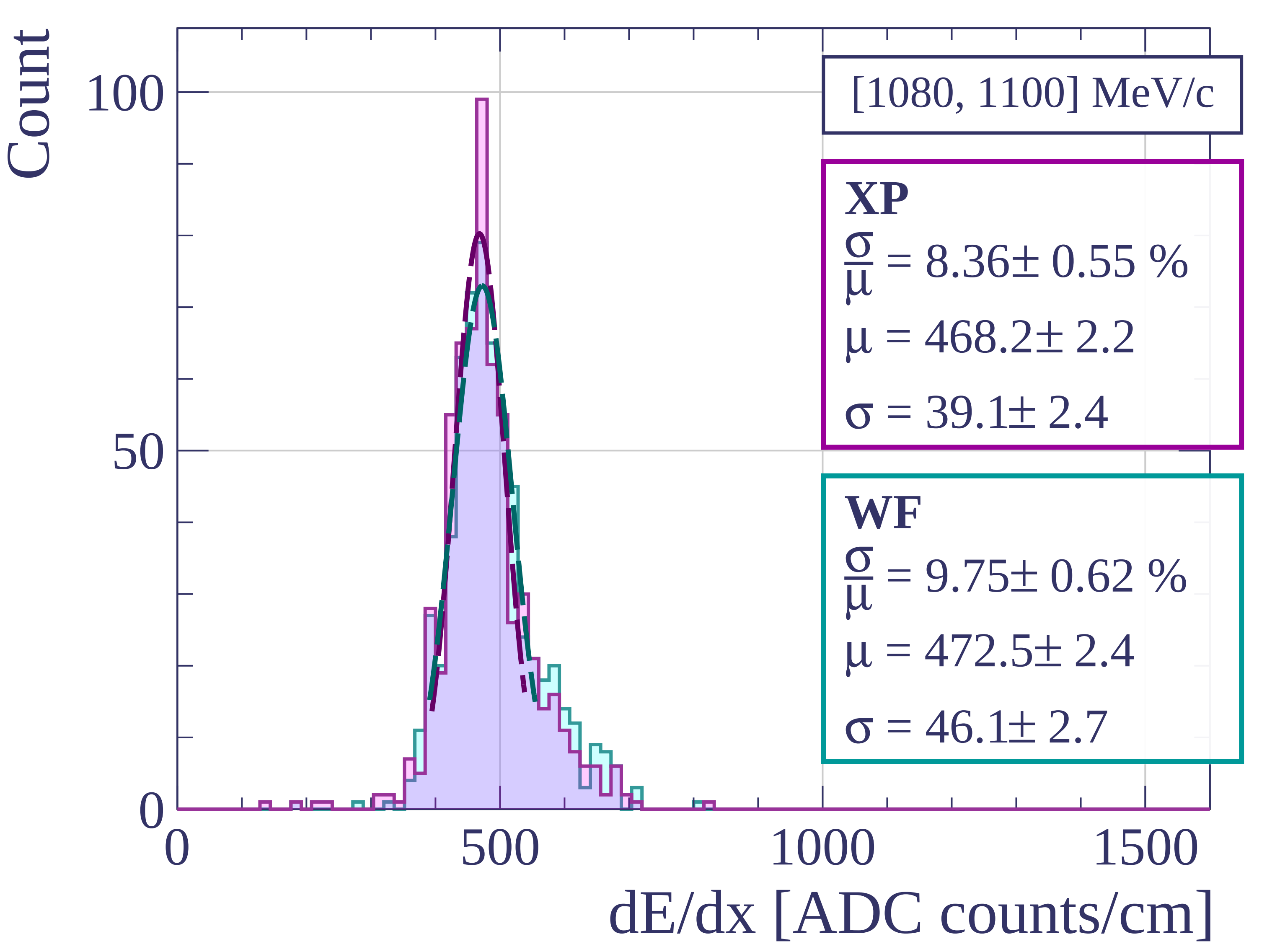


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.66± 0.65 %
μ = 463.5± 2.5
σ = 49.4± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.46± 0.64 %
μ = 467.6± 2.6
σ = 48.9± 2.7

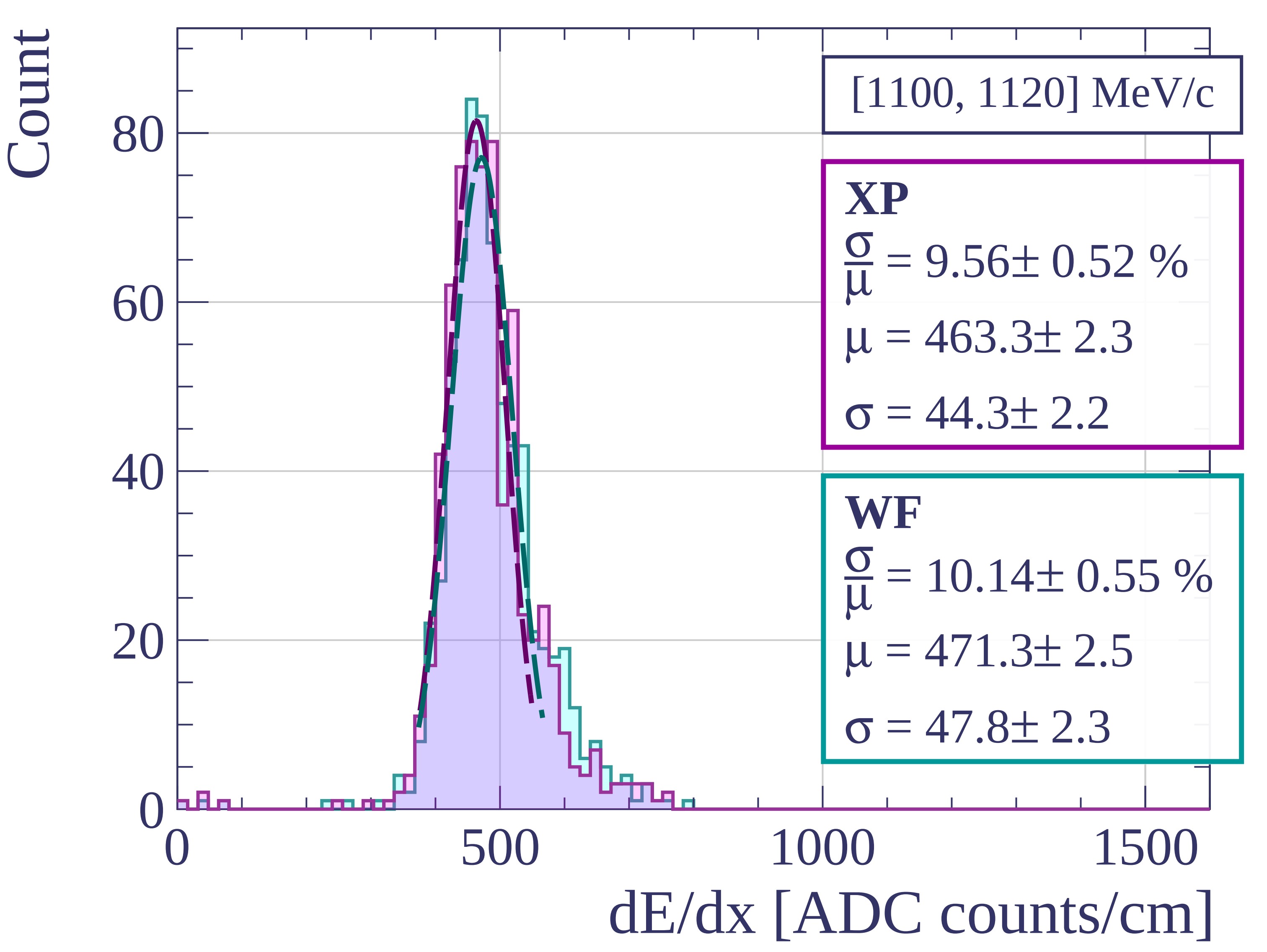

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.25± 0.56 %
μ = 464.5± 2.3
σ = 47.6± 2.4
WF
$\frac{\sigma}{\mu}$ = 9.85± 0.48 %
μ = 471.1± 2.3
σ = 46.4± 2.0



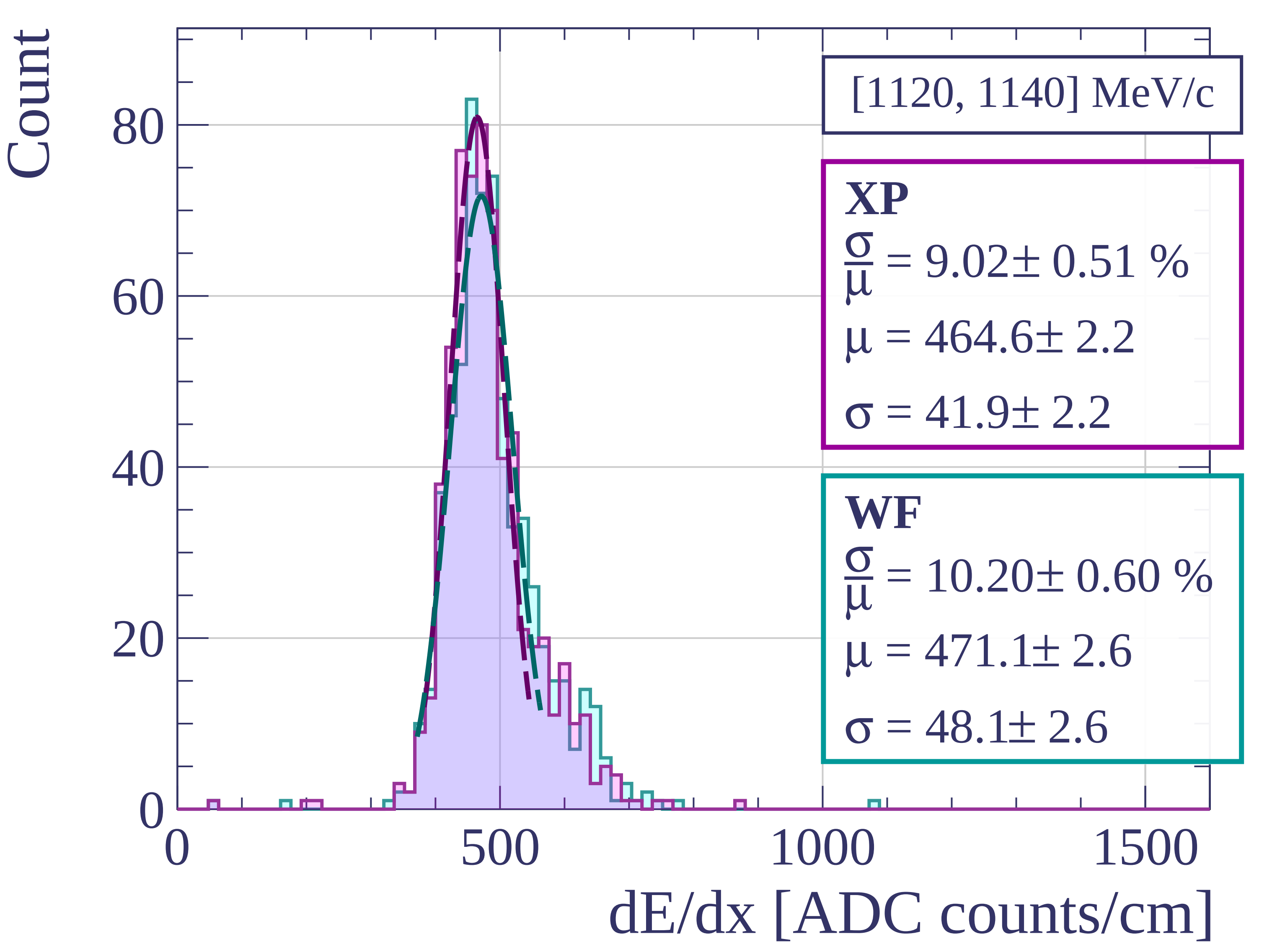
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
σ/μ = 8.93± 0.52 %
μ = 465.4± 2.1
σ = 41.6± 2.2
WF
σ/μ = 9.91± 0.59 %
μ = 472.8± 2.4
σ = 46.8± 2.6



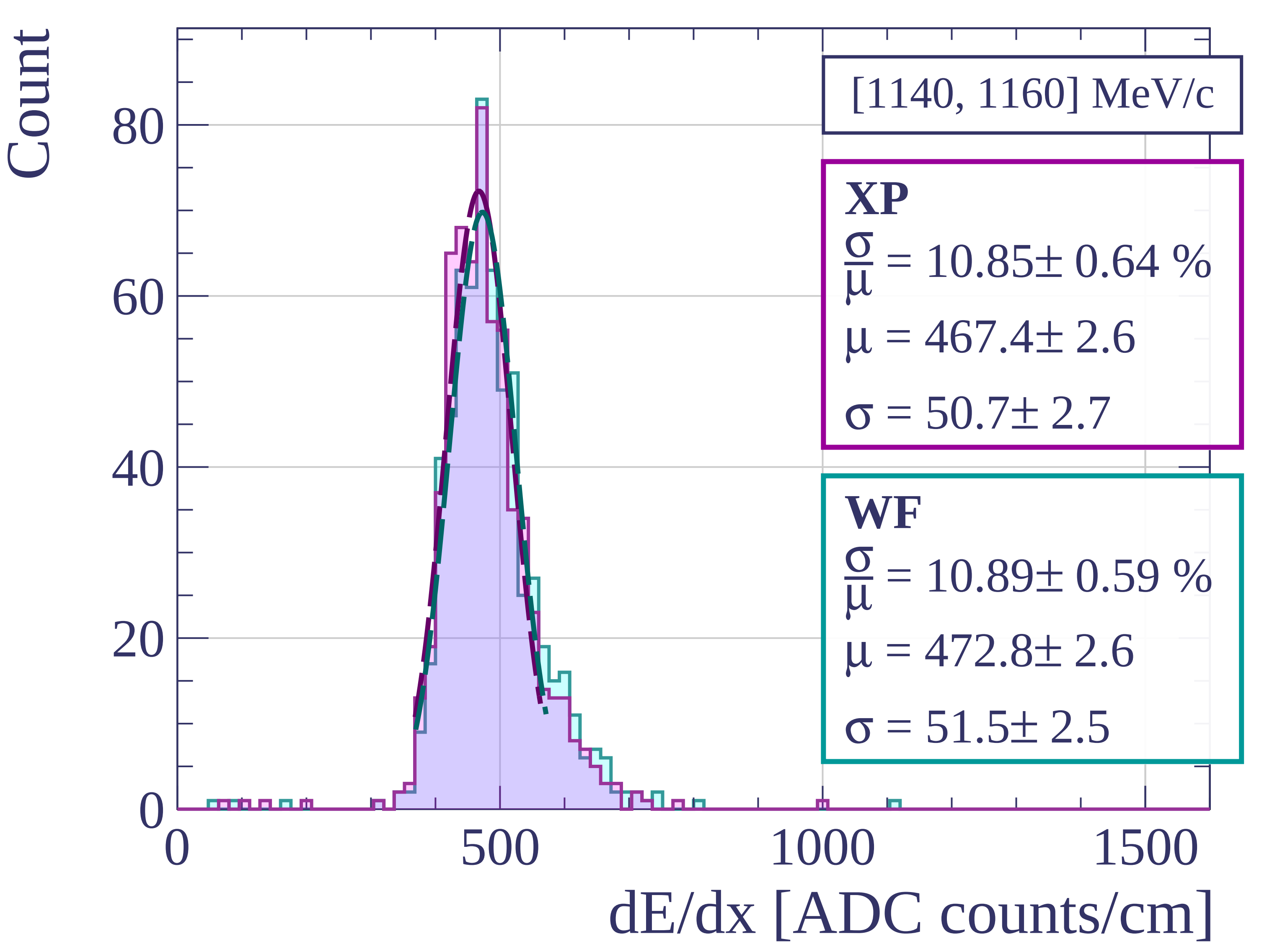
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1080, 1100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.36± 0.55 %
μ = 468.2± 2.2
σ = 39.1± 2.4
WF
$\frac{\sigma}{\mu}$ = 9.75± 0.62 %
μ = 472.5± 2.4
σ = 46.1± 2.7



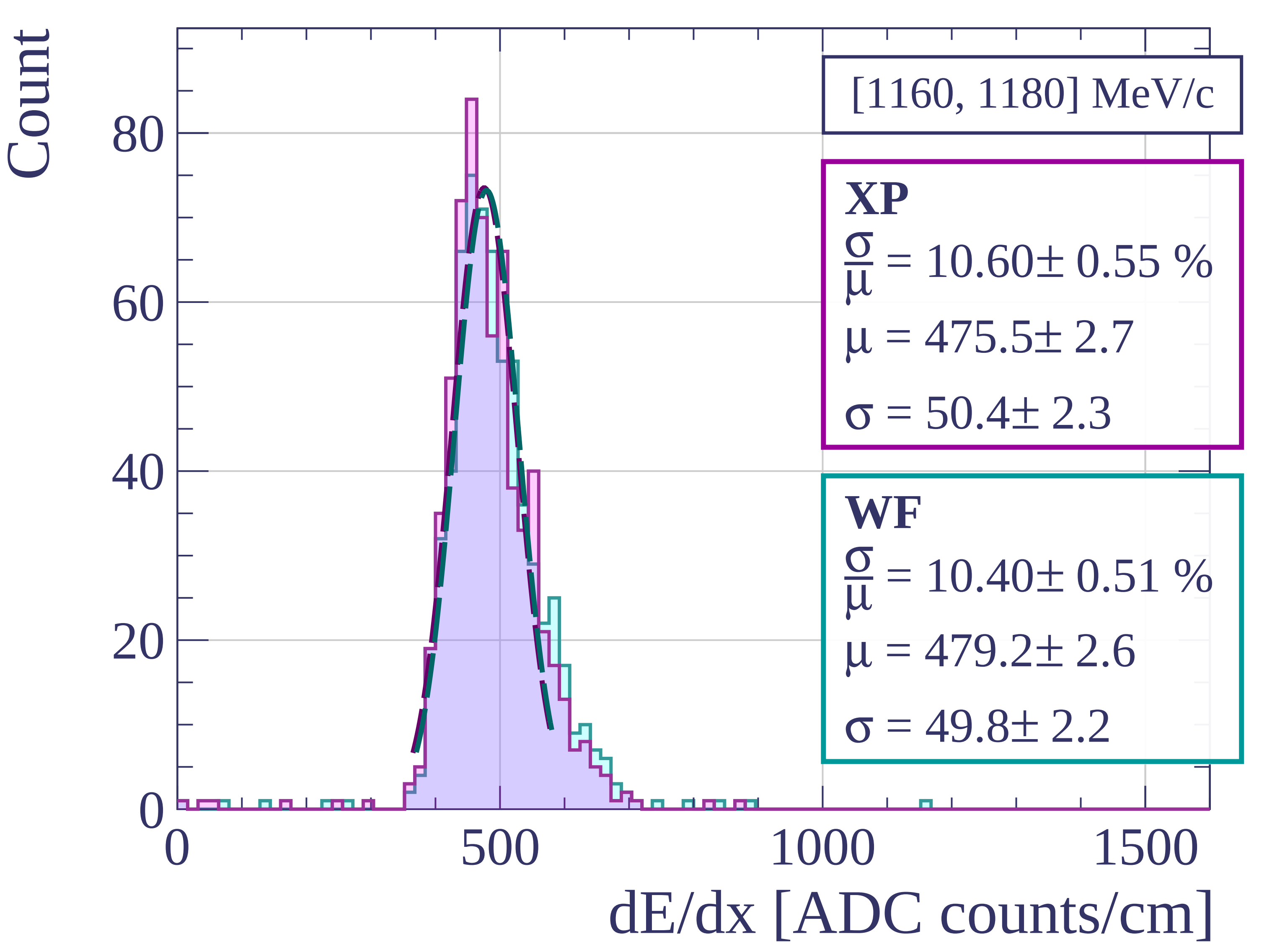
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.56± 0.52 %
μ = 463.3± 2.3
σ = 44.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.14± 0.55 %
μ = 471.3± 2.5
σ = 47.8± 2.3



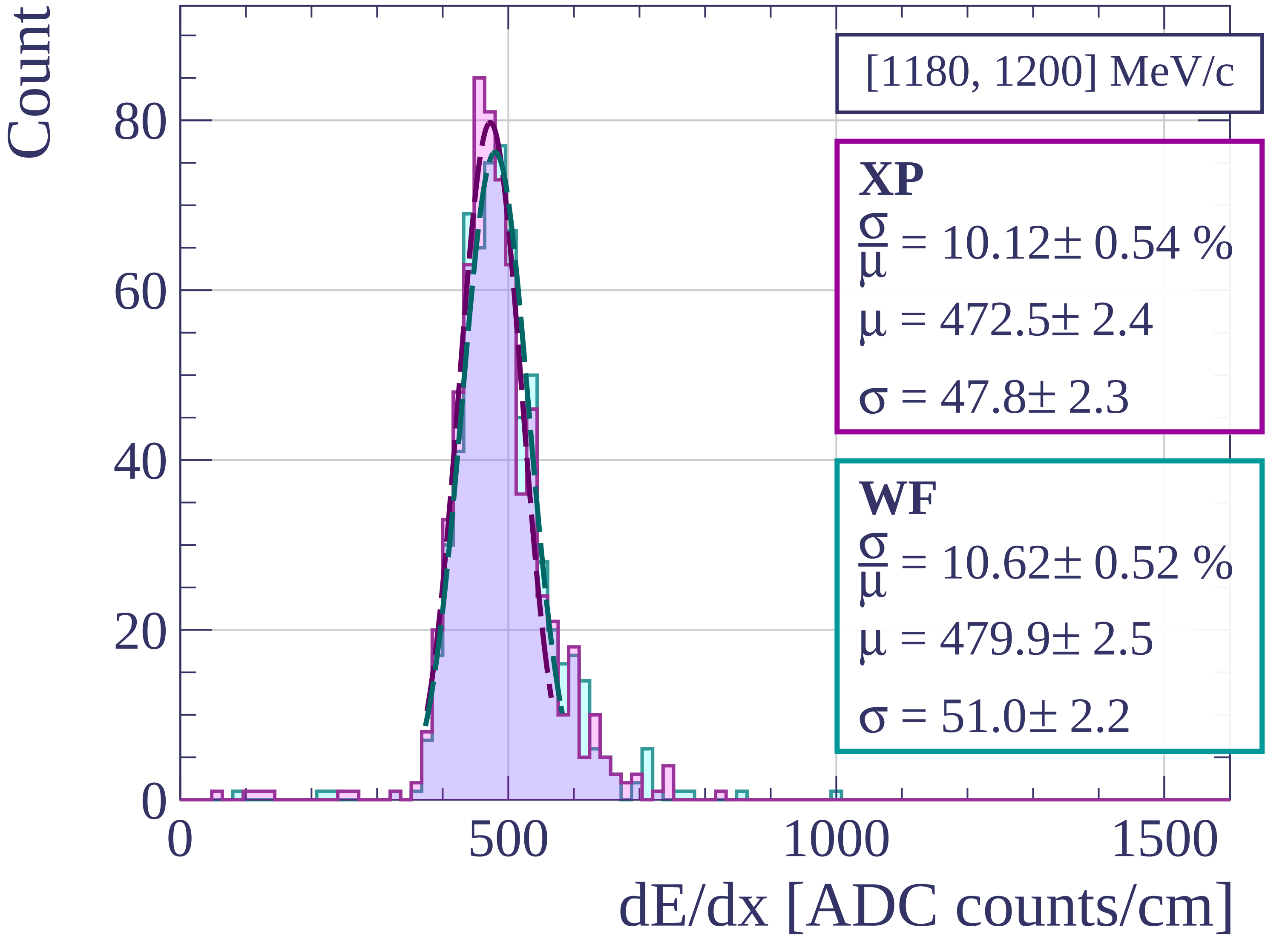
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
σ/μ = 9.02± 0.51 %
μ = 464.6± 2.2
σ = 41.9± 2.2
WF
σ/μ = 10.20± 0.60 %
μ = 471.1± 2.6
σ = 48.1± 2.6



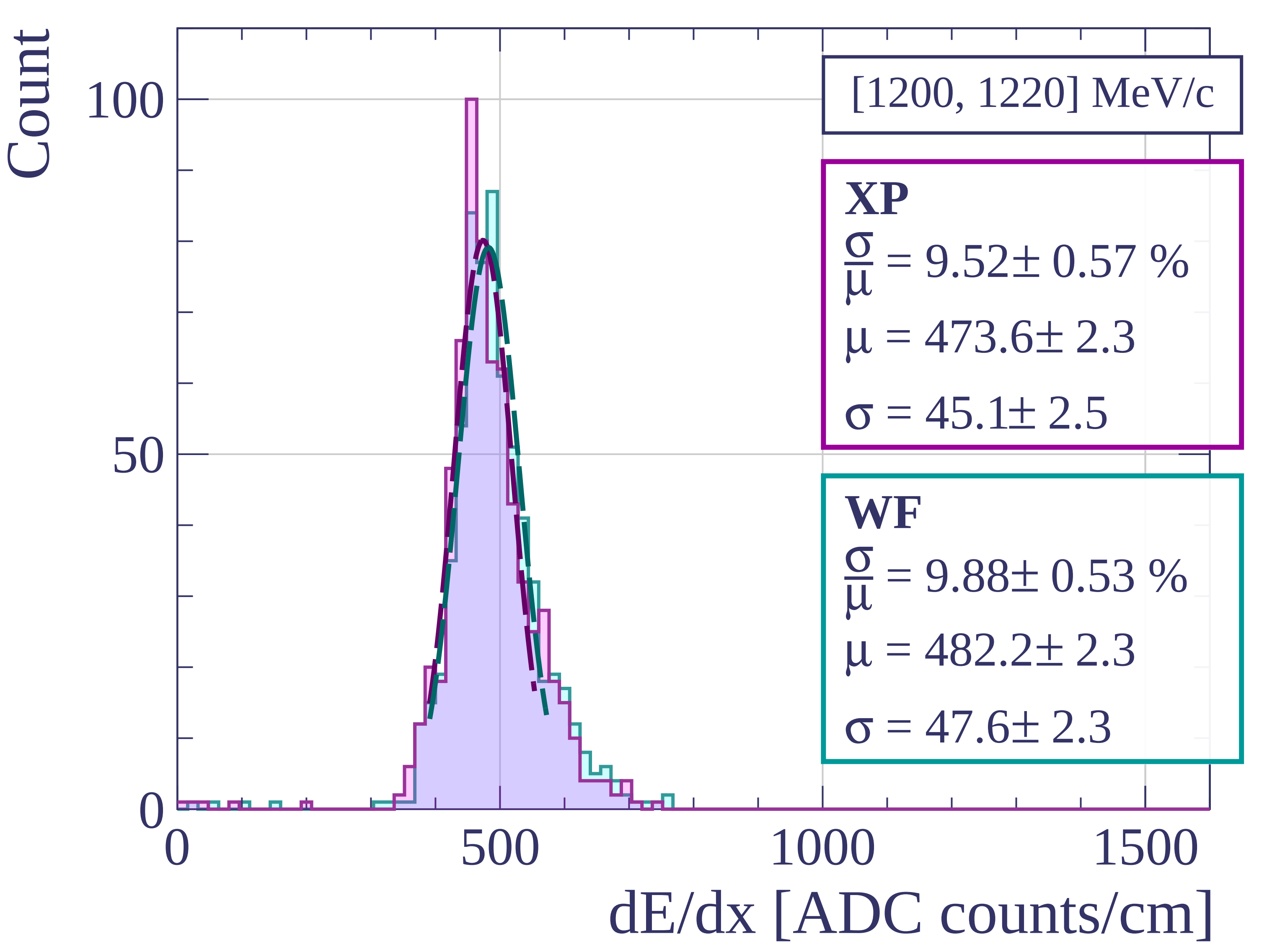
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1140, 1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.85± 0.64 %
μ = 467.4± 2.6
σ = 50.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.89± 0.59 %
μ = 472.8± 2.6
σ = 51.5± 2.5



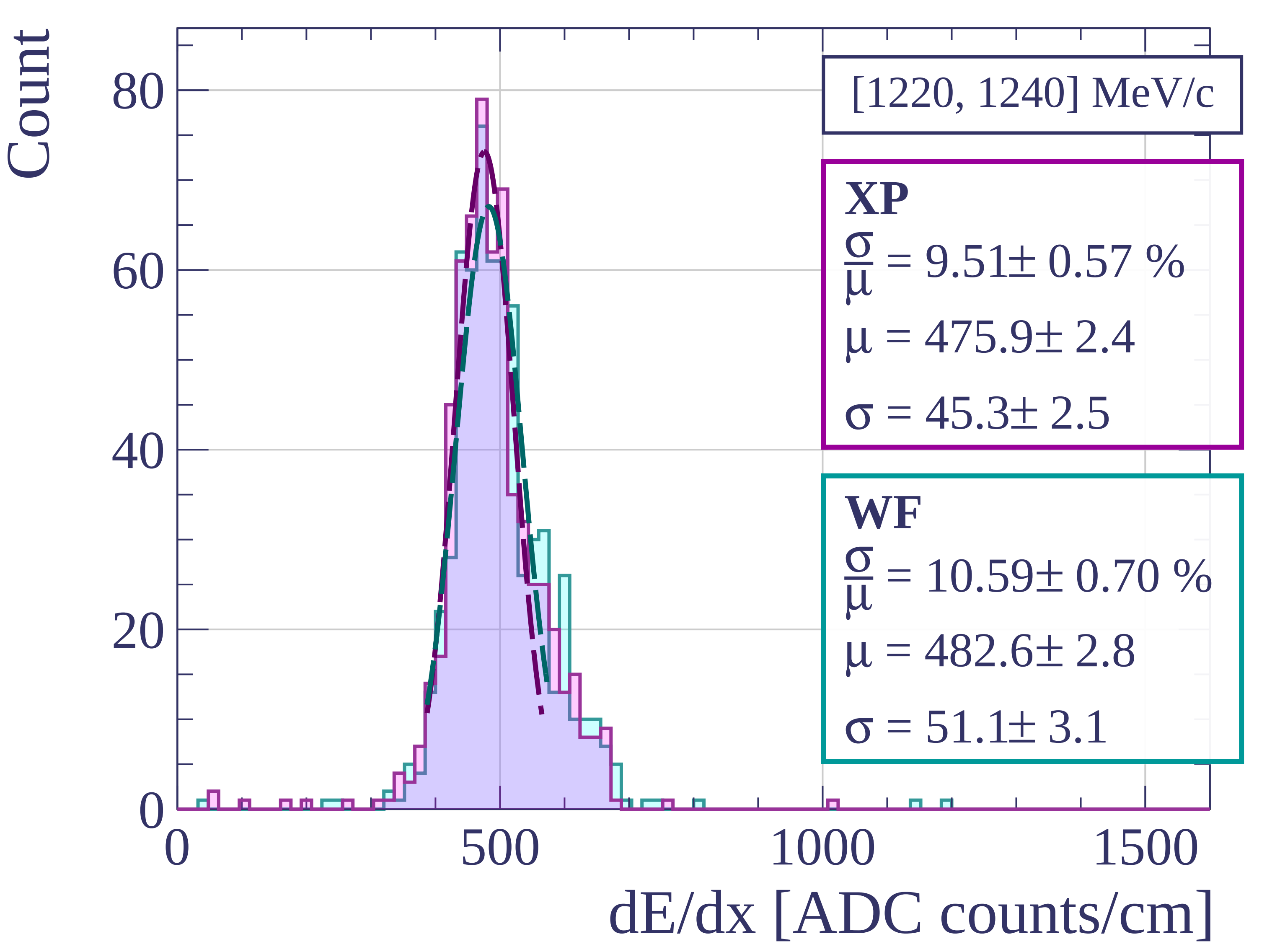
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[1160, 1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.60± 0.55 %
μ = 475.5± 2.7
σ = 50.4± 2.3
WF
$\frac{\sigma}{\mu}$ = 10.40± 0.51 %
μ = 479.2± 2.6
σ = 49.8± 2.2



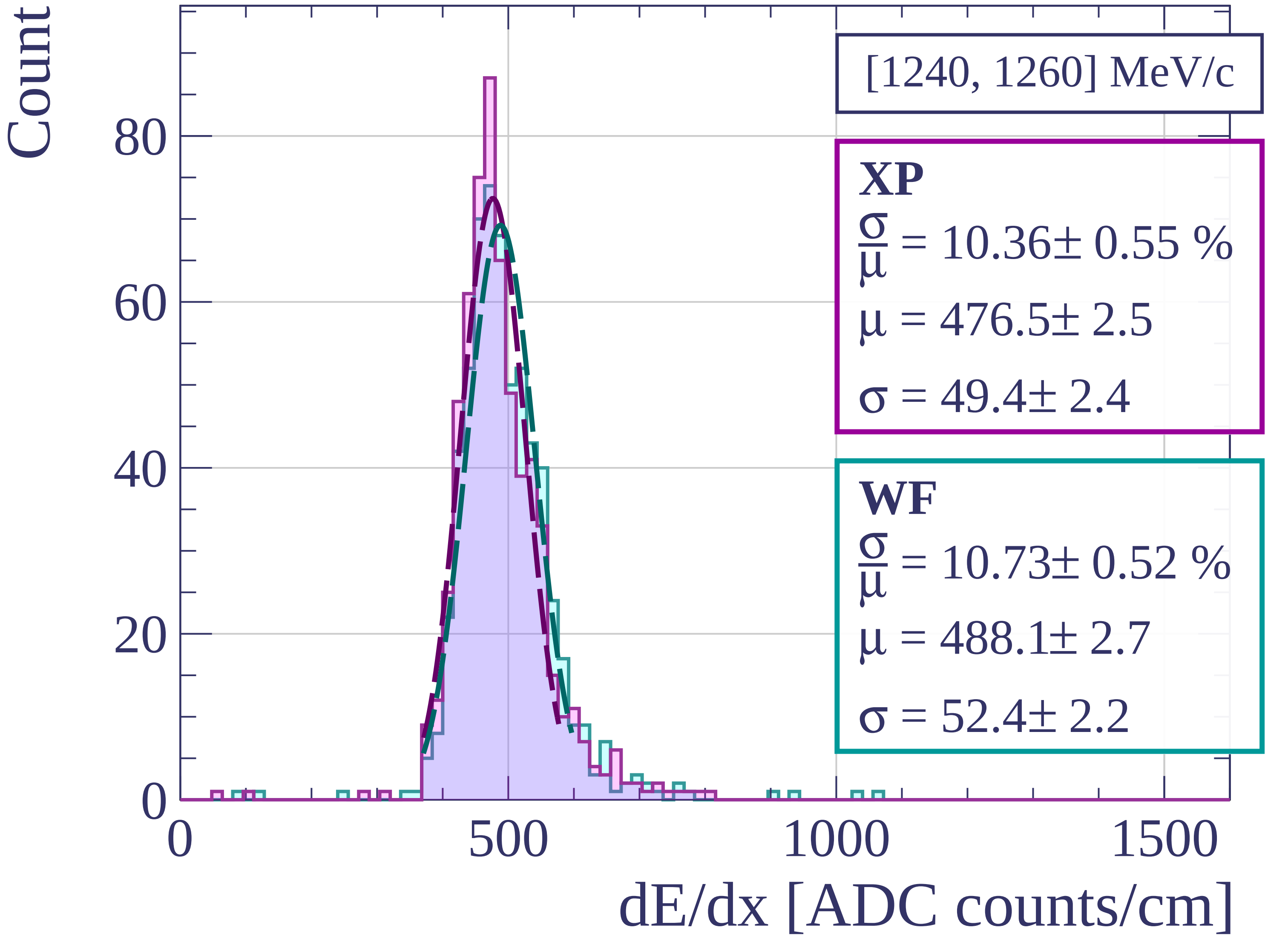
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1180, 1200] MeV/c
XP
σ/μ = 10.12± 0.54 %
μ = 472.5± 2.4
σ = 47.8± 2.3
WF
σ/μ = 10.62± 0.52 %
μ = 479.9± 2.5
σ = 51.0± 2.2



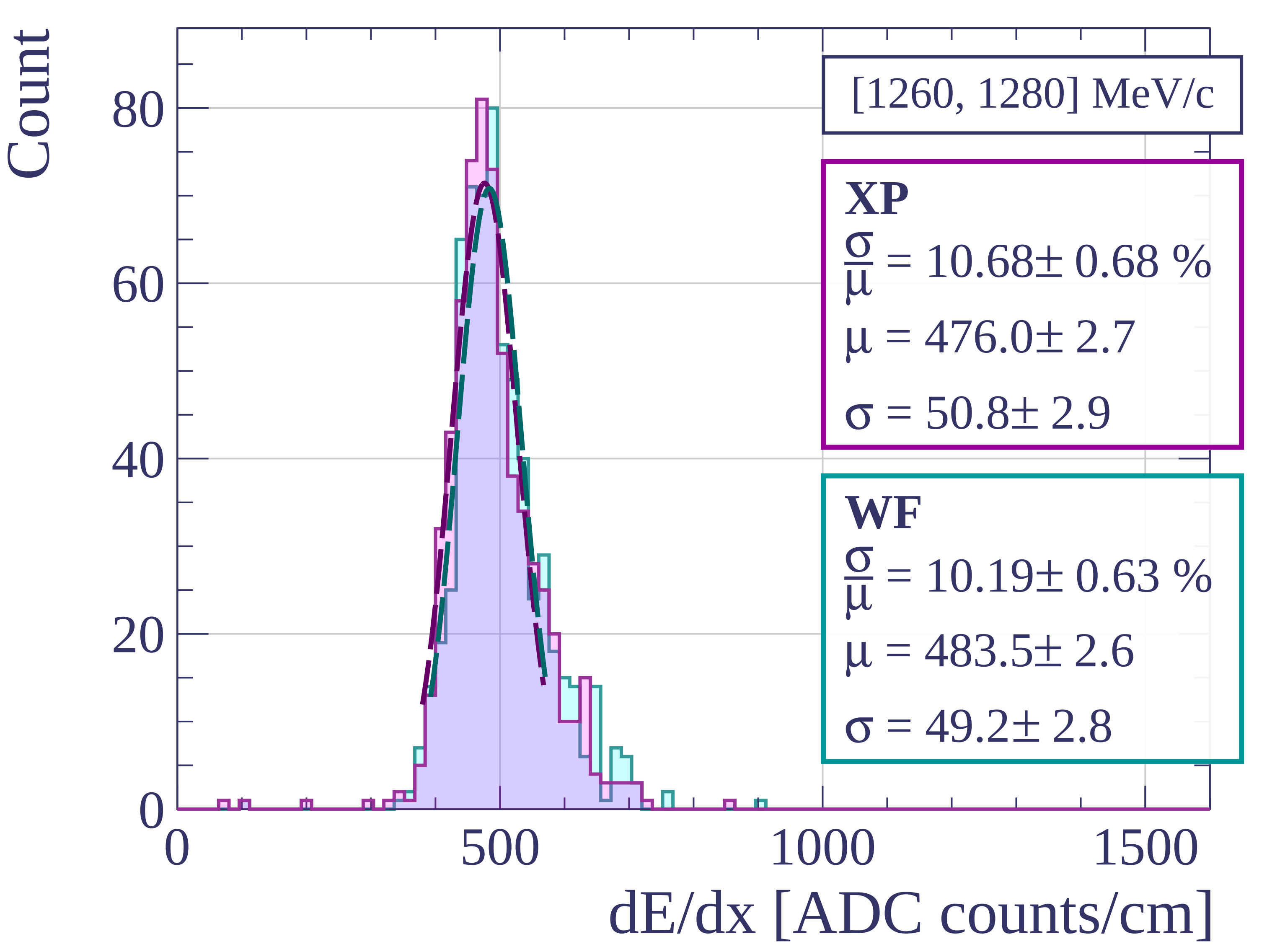
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.52± 0.57 %
μ = 473.6± 2.3
σ = 45.1± 2.5
WF
$\frac{\sigma}{\mu}$ = 9.88± 0.53 %
μ = 482.2± 2.3
σ = 47.6± 2.3

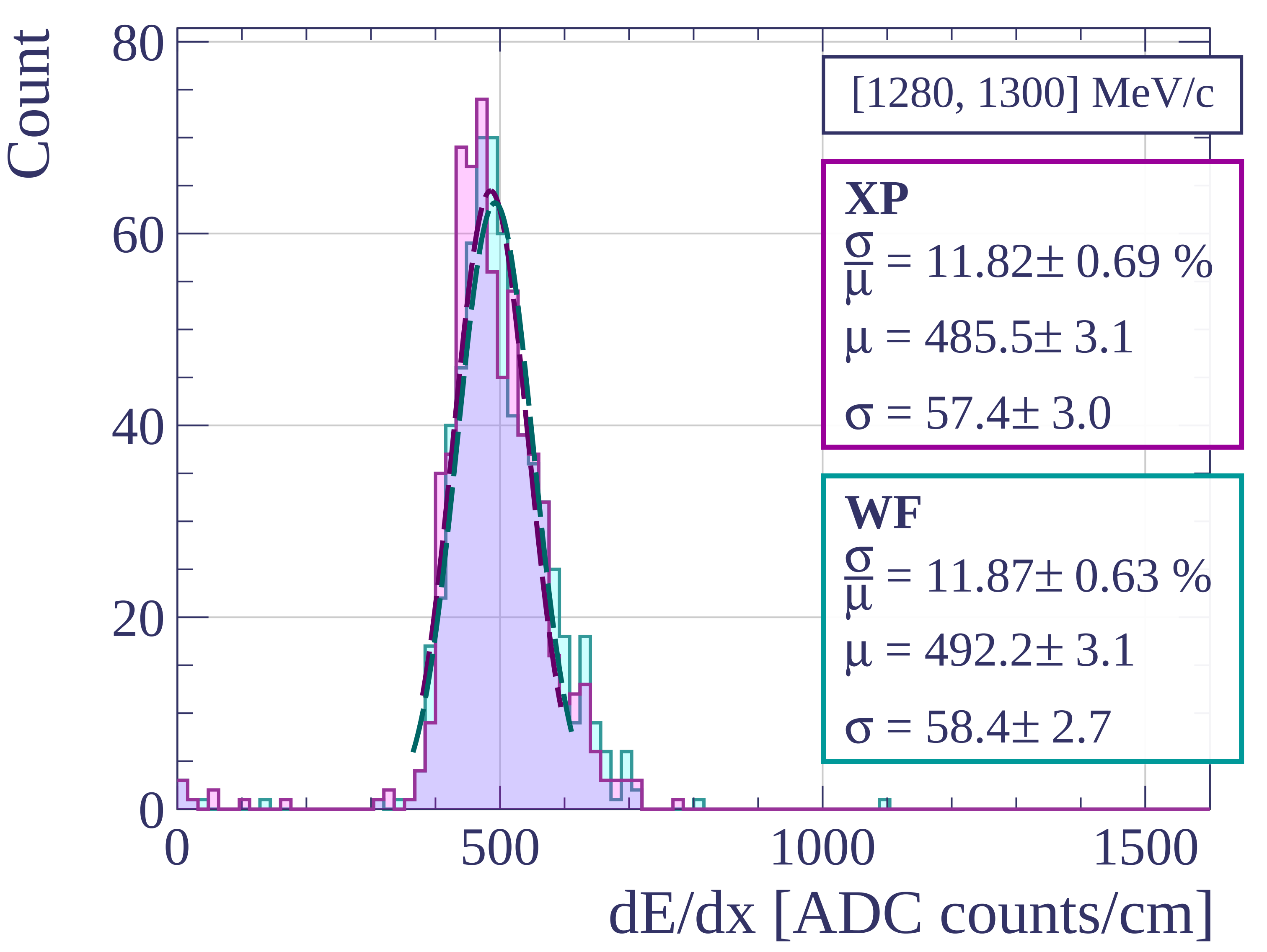


Count
dE/dx [ADC counts/cm]
[1220, 1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.51± 0.57 %
μ = 475.9± 2.4
σ = 45.3± 2.5
WF
$\frac{\sigma}{\mu}$ = 10.59± 0.70 %
μ = 482.6± 2.8
σ = 51.1± 3.1
0
20
40
60
80
0
500
1000
1500

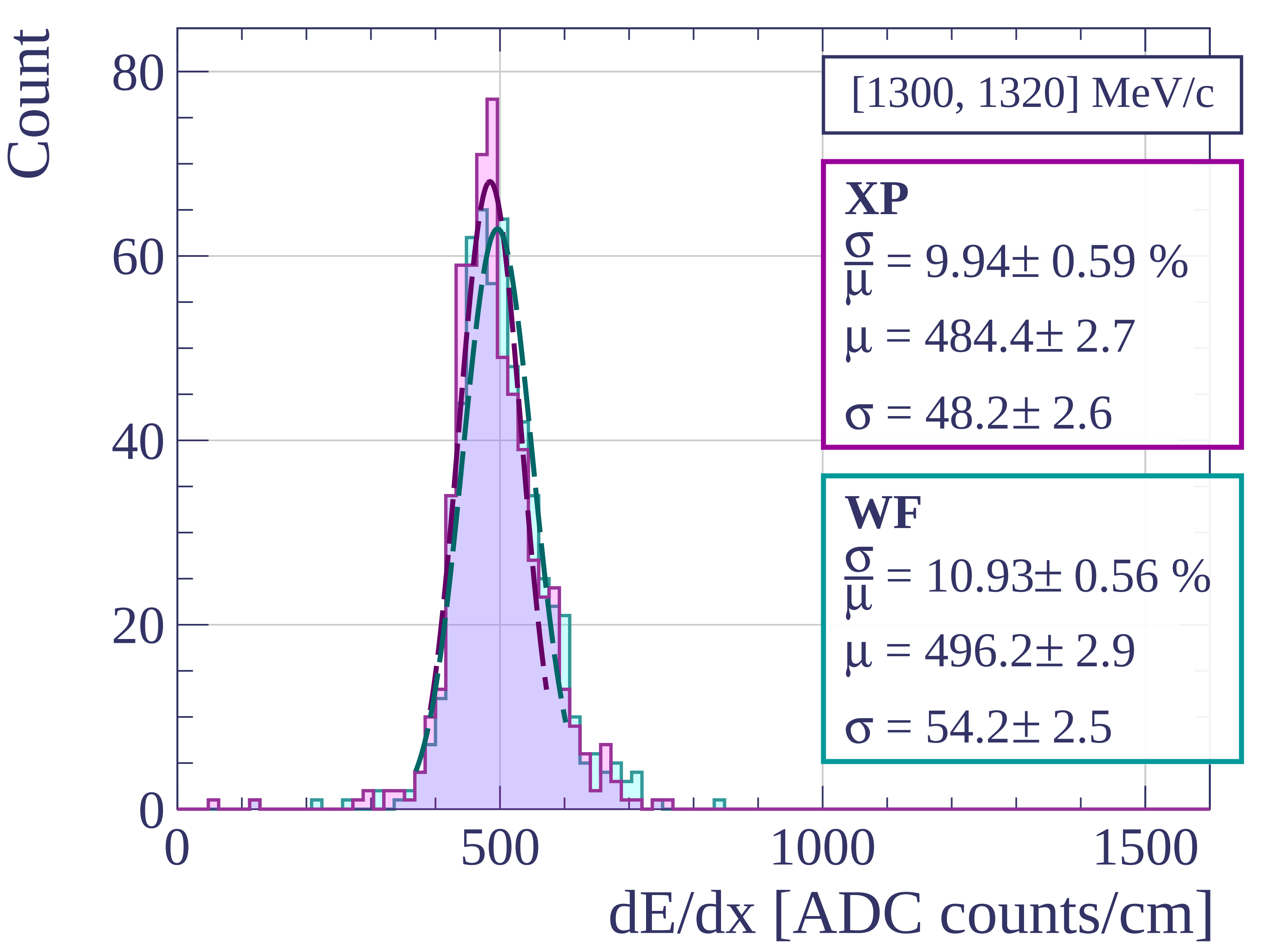


Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[1240, 1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.36± 0.55 %
μ = 476.5± 2.5
σ = 49.4± 2.4
WF
$\frac{\sigma}{\mu}$ = 10.73± 0.52 %
μ = 488.1± 2.7
σ = 52.4± 2.2

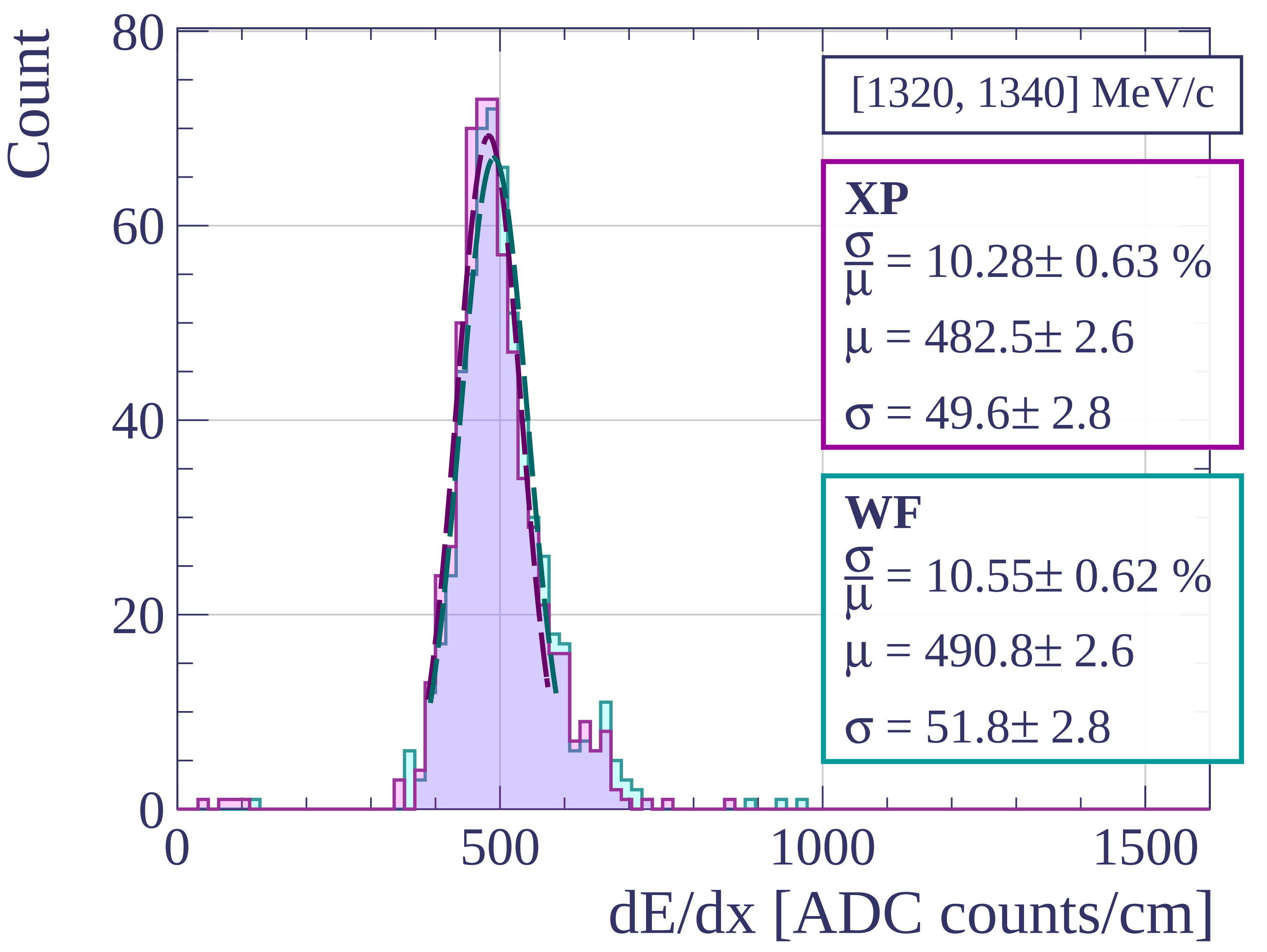

Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[1260, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.68± 0.68 %
μ = 476.0± 2.7
σ = 50.8± 2.9
WF
$\frac{\sigma}{\mu}$ = 10.19± 0.63 %
μ = 483.5± 2.6
σ = 49.2± 2.8



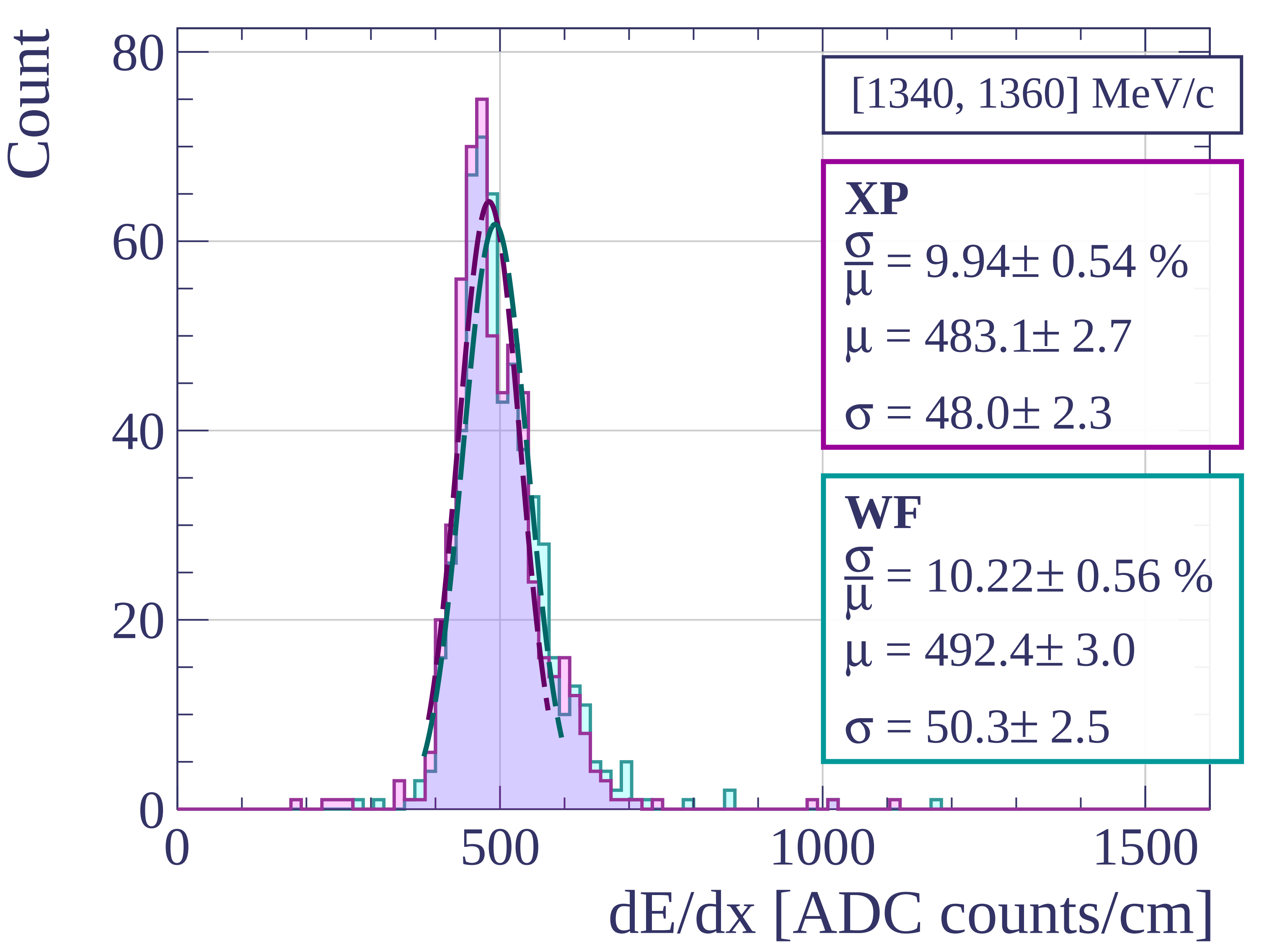
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
σ/μ = 11.82± 0.69 %
μ = 485.5± 3.1
σ = 57.4± 3.0
WF
σ/μ = 11.87± 0.63 %
μ = 492.2± 3.1
σ = 58.4± 2.7



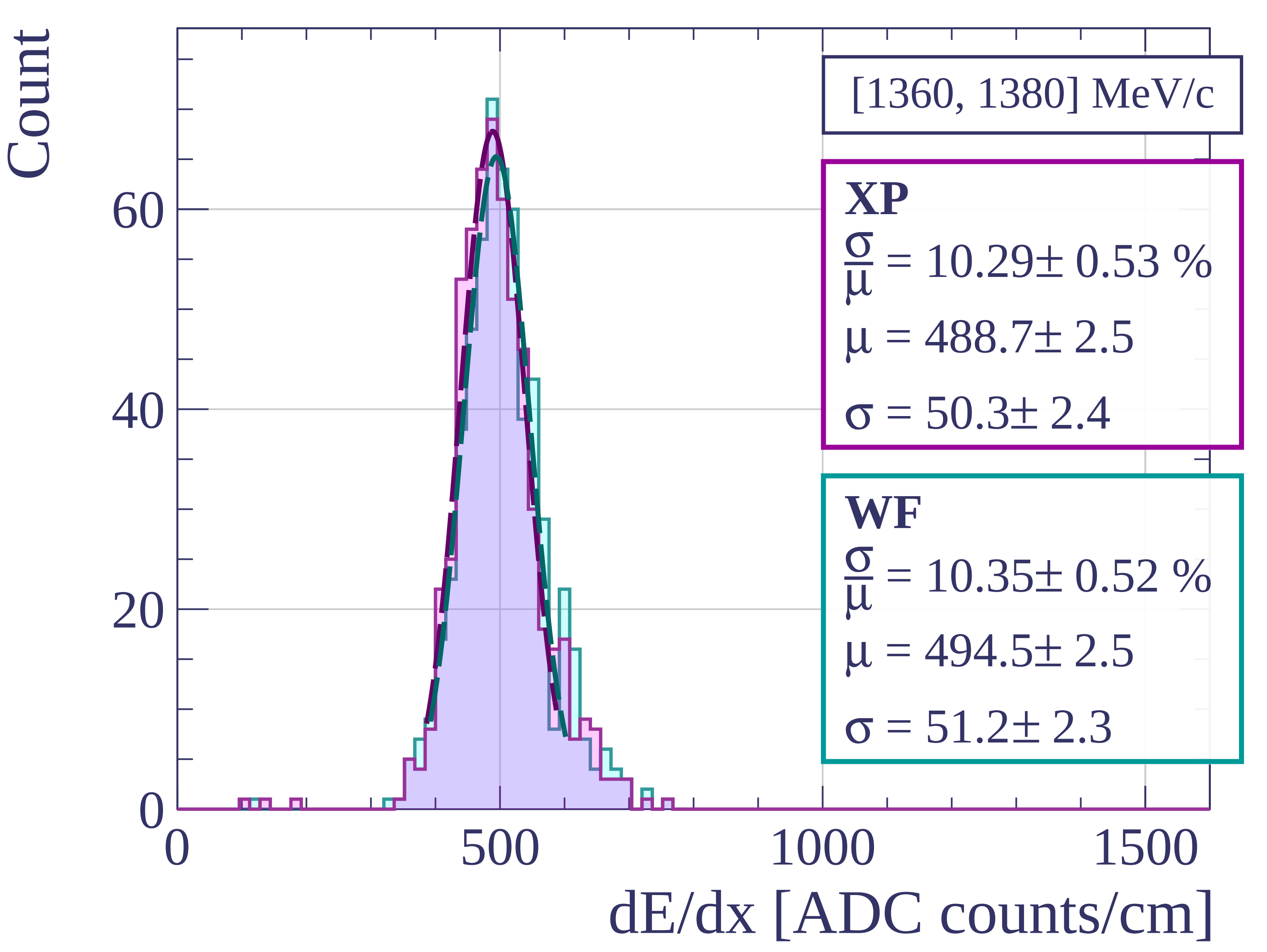
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1300, 1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.94± 0.59 %
μ = 484.4± 2.7
σ = 48.2± 2.6
WF
$\frac{\sigma}{\mu}$ = 10.93± 0.56 %
μ = 496.2± 2.9
σ = 54.2± 2.5



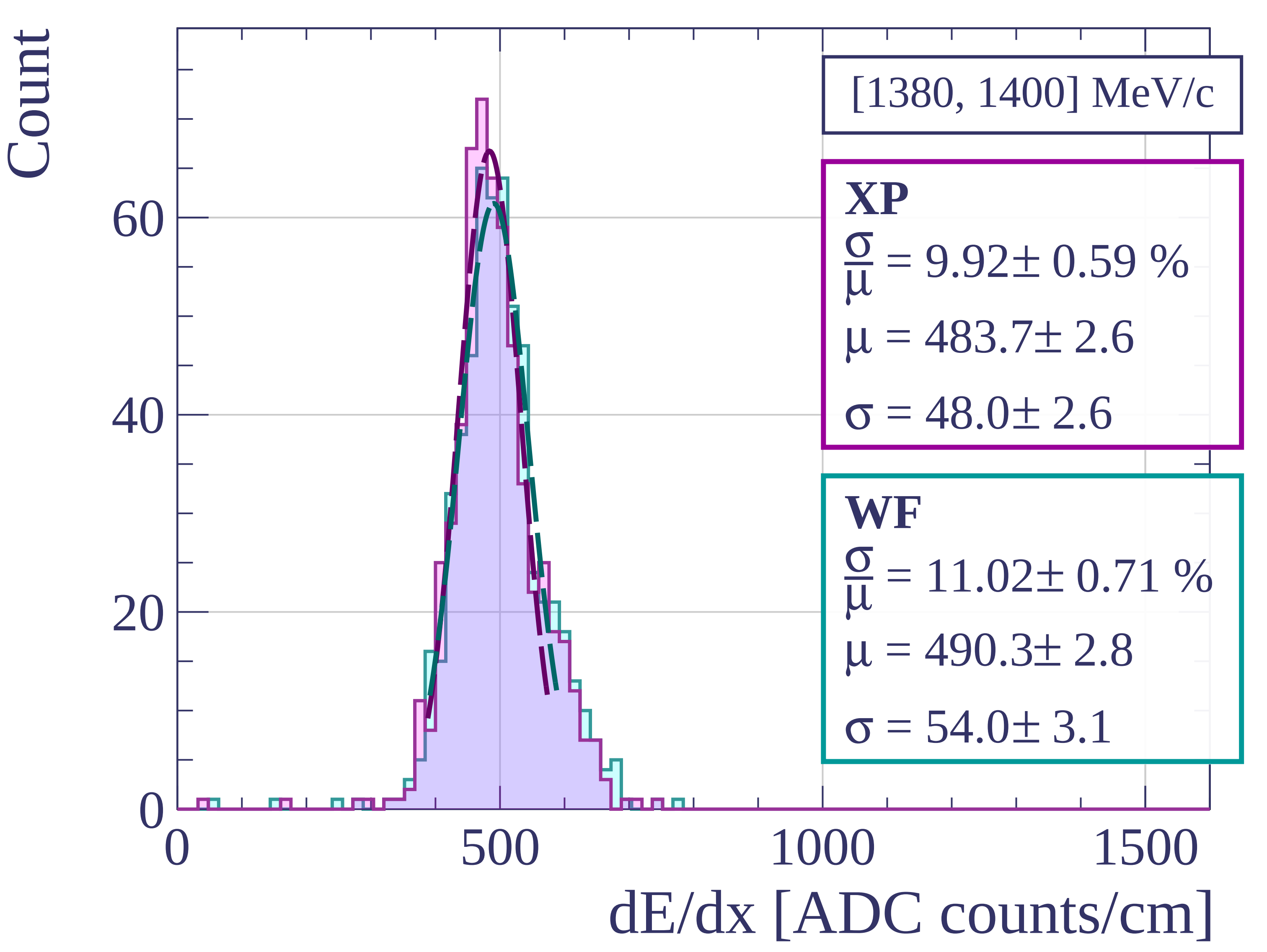
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[1320, 1340] MeV/c
XP
σ/μ = 10.28± 0.63 %
μ = 482.5± 2.6
σ = 49.6± 2.8
WF
σ/μ = 10.55± 0.62 %
μ = 490.8± 2.6
σ = 51.8± 2.8



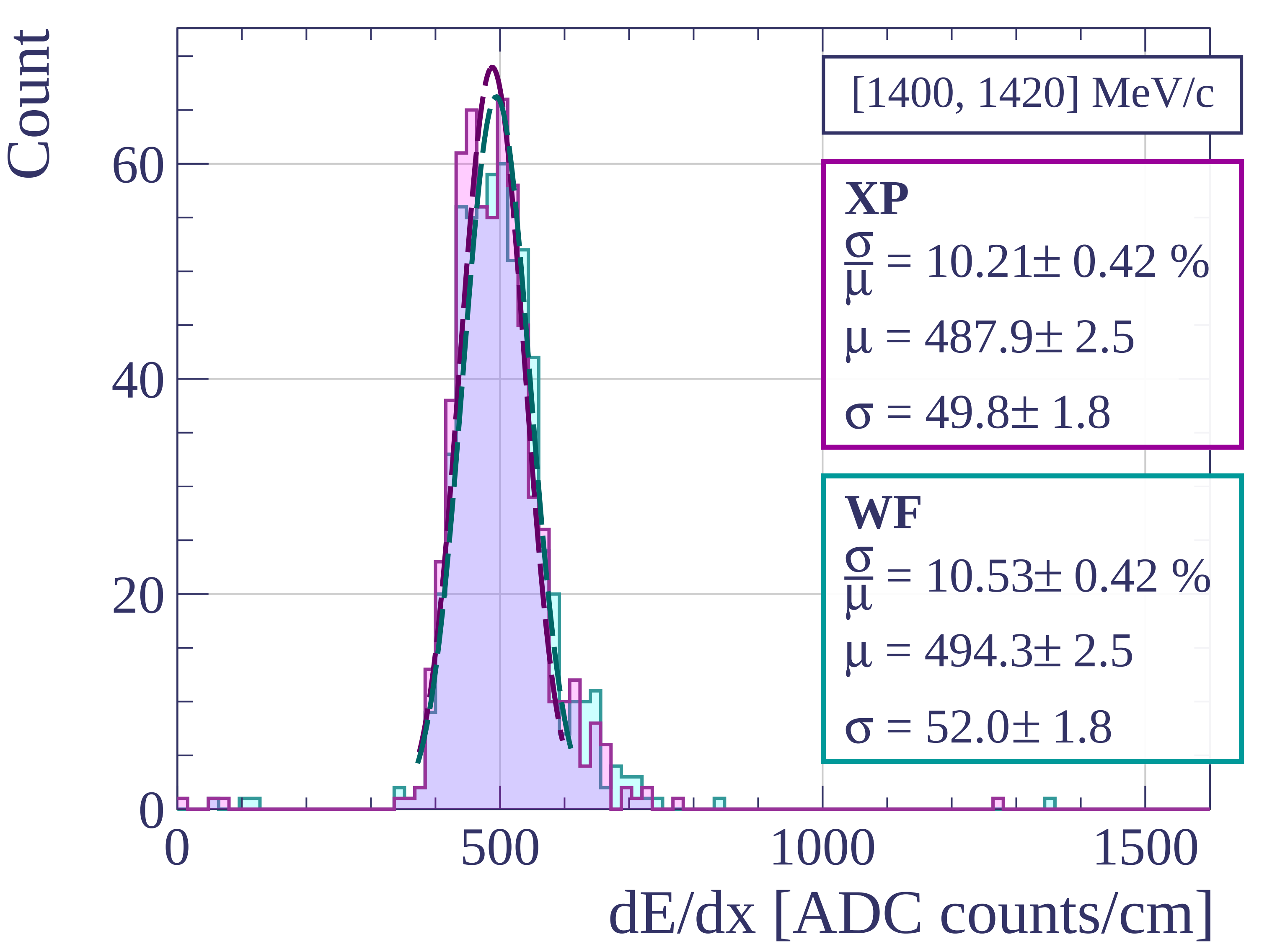
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1340, 1360] MeV/c
XP
σ/μ = 9.94± 0.54 %
μ = 483.1± 2.7
σ = 48.0± 2.3
WF
σ/μ = 10.22± 0.56 %
μ = 492.4± 3.0
σ = 50.3± 2.5



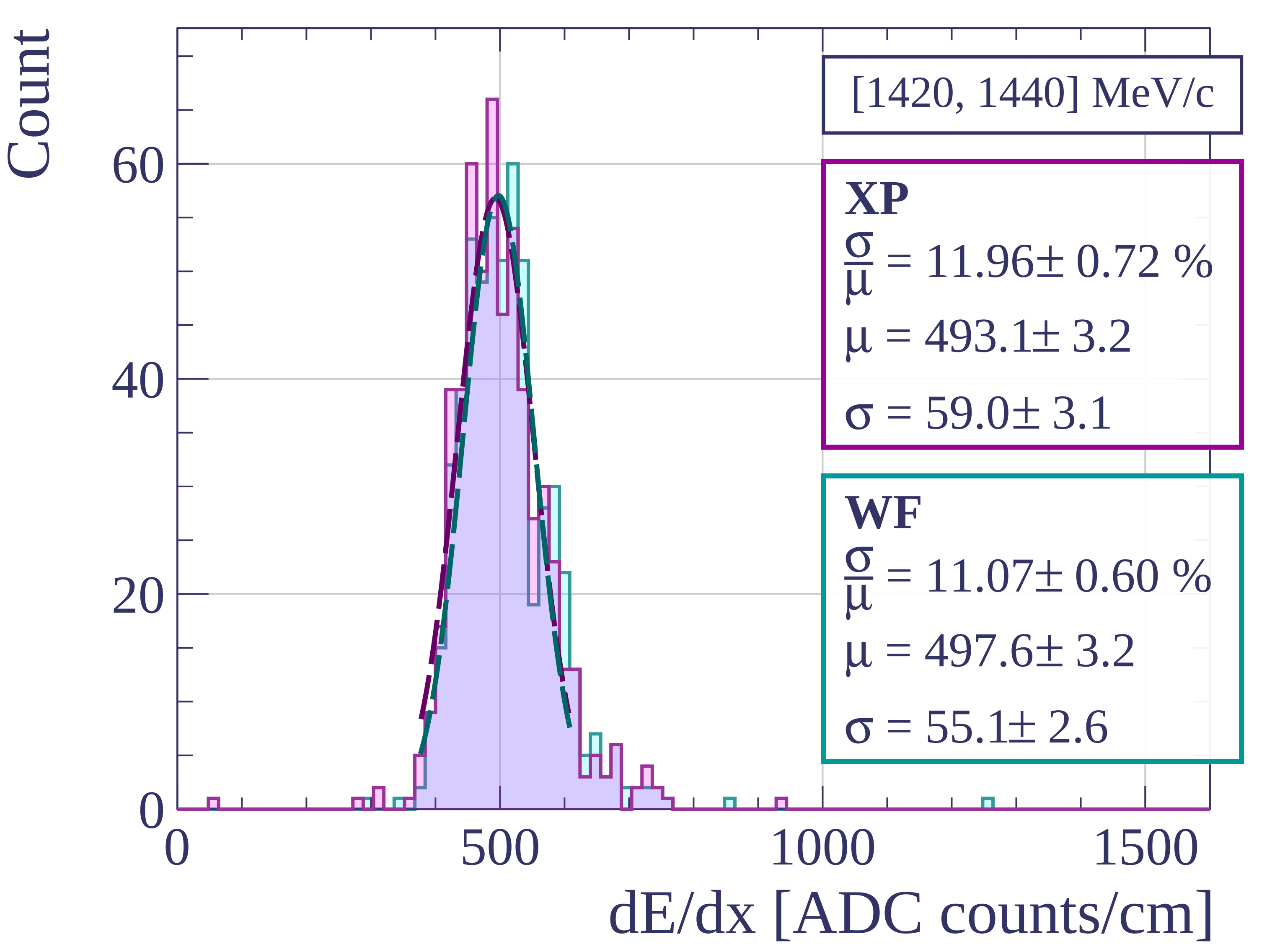
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.29± 0.53 %
μ = 488.7± 2.5
σ = 50.3± 2.4
WF
$\frac{\sigma}{\mu}$ = 10.35± 0.52 %
μ = 494.5± 2.5
σ = 51.2± 2.3



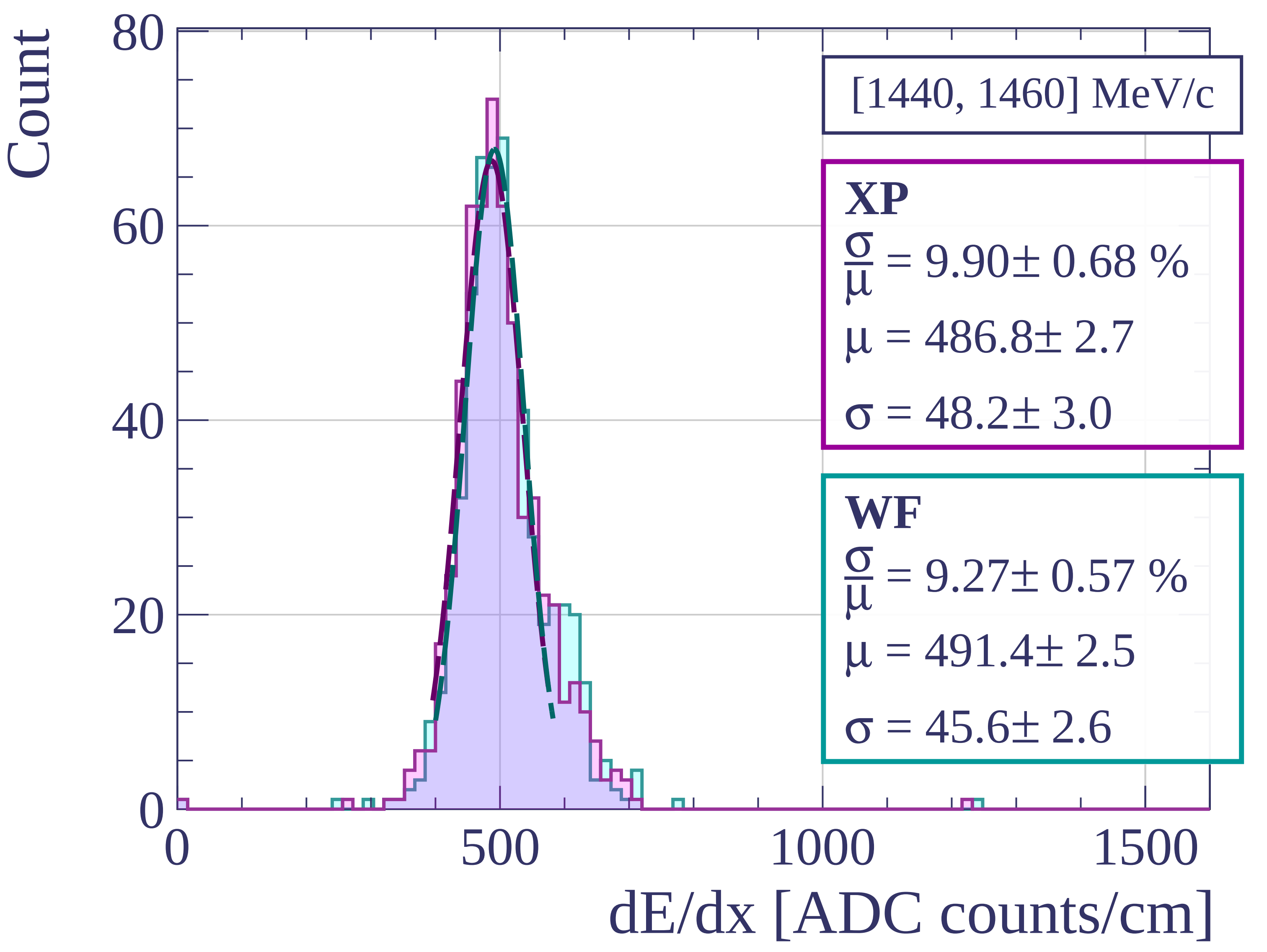
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1380, 1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.92± 0.59 %
μ = 483.7± 2.6
σ = 48.0± 2.6
WF
$\frac{\sigma}{\mu}$ = 11.02± 0.71 %
μ = 490.3± 2.8
σ = 54.0± 3.1



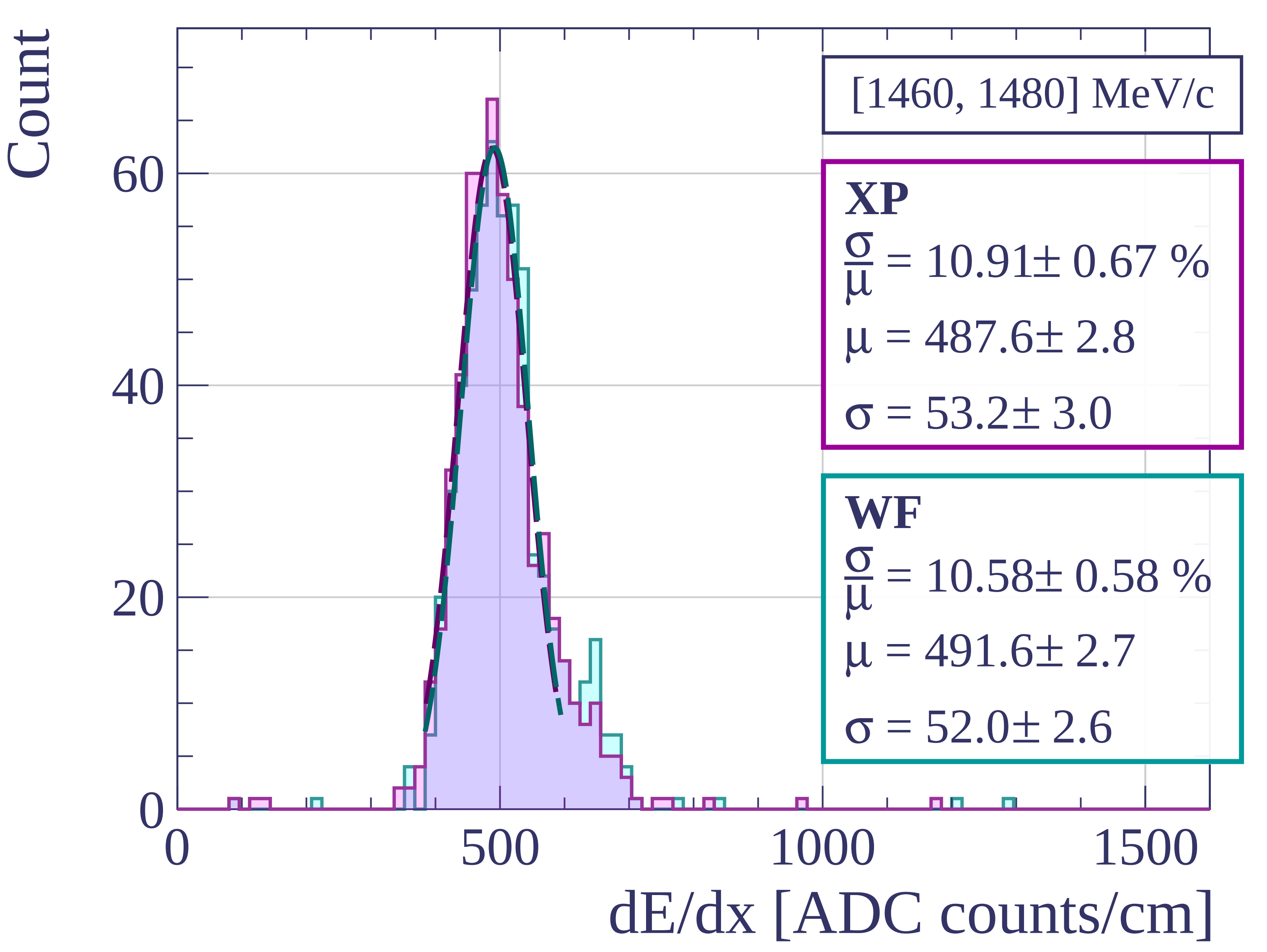
Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[1400, 1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.21± 0.42 %
μ = 487.9± 2.5
σ = 49.8± 1.8
WF
$\frac{\sigma}{\mu}$ = 10.53± 0.42 %
μ = 494.3± 2.5
σ = 52.0± 1.8



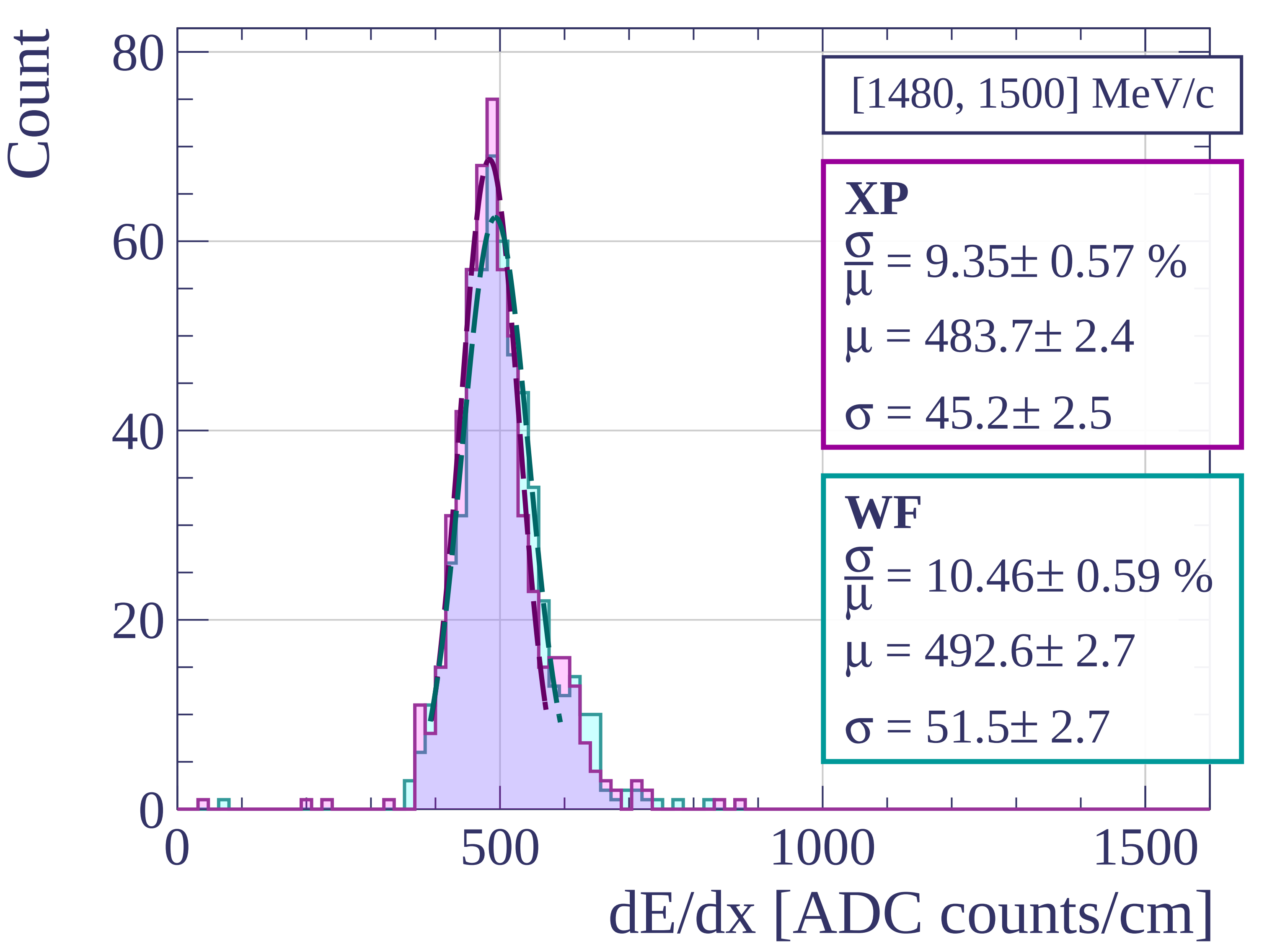
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1420, 1440] MeV/c
XP
σ/μ = 11.96± 0.72 %
μ = 493.1± 3.2
σ = 59.0± 3.1
WF
σ/μ = 11.07± 0.60 %
μ = 497.6± 3.2
σ = 55.1± 2.6



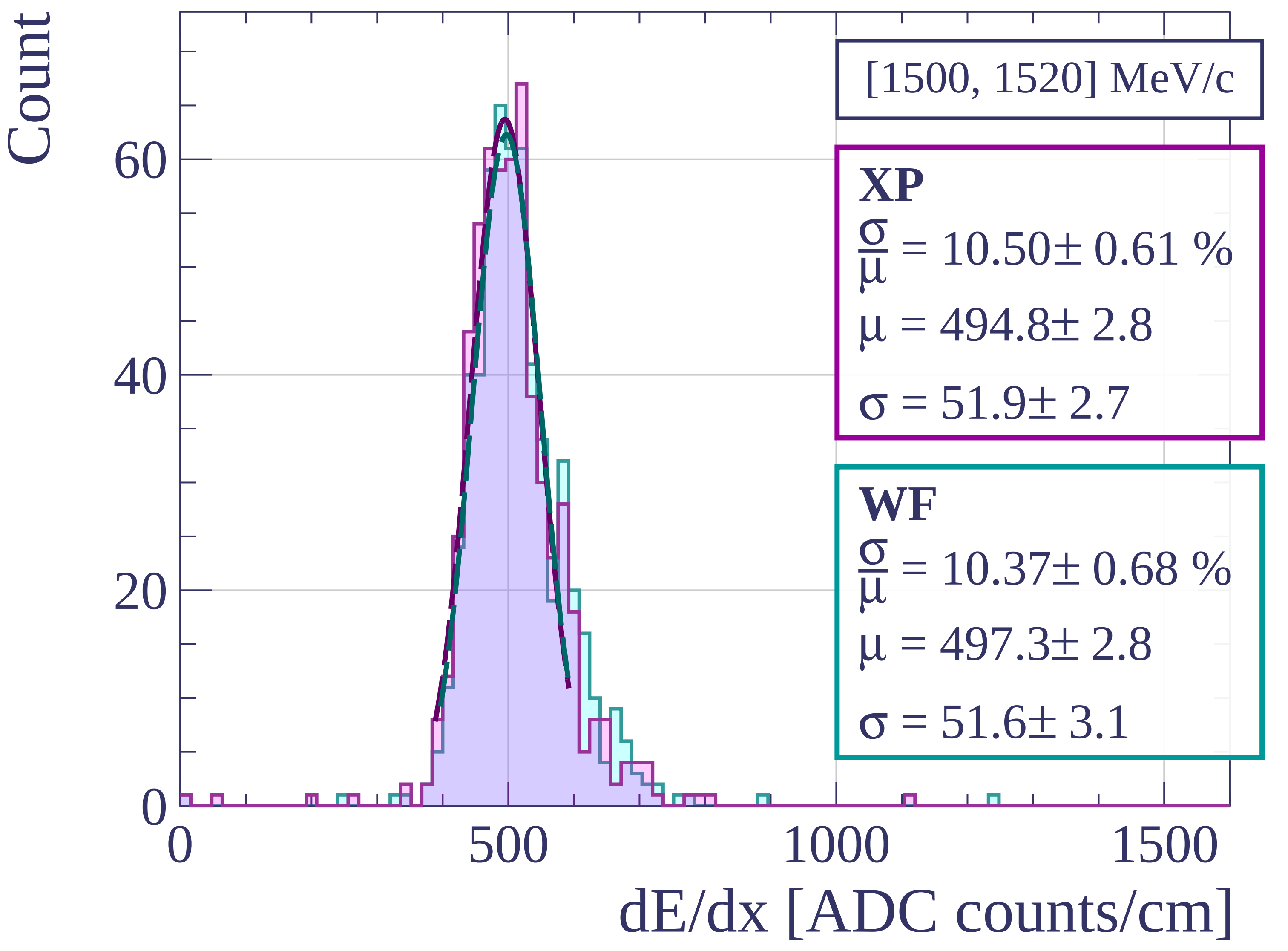
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.90± 0.68 %
μ = 486.8± 2.7
σ = 48.2± 3.0
WF
$\frac{\sigma}{\mu}$ = 9.27± 0.57 %
μ = 491.4± 2.5
σ = 45.6± 2.6



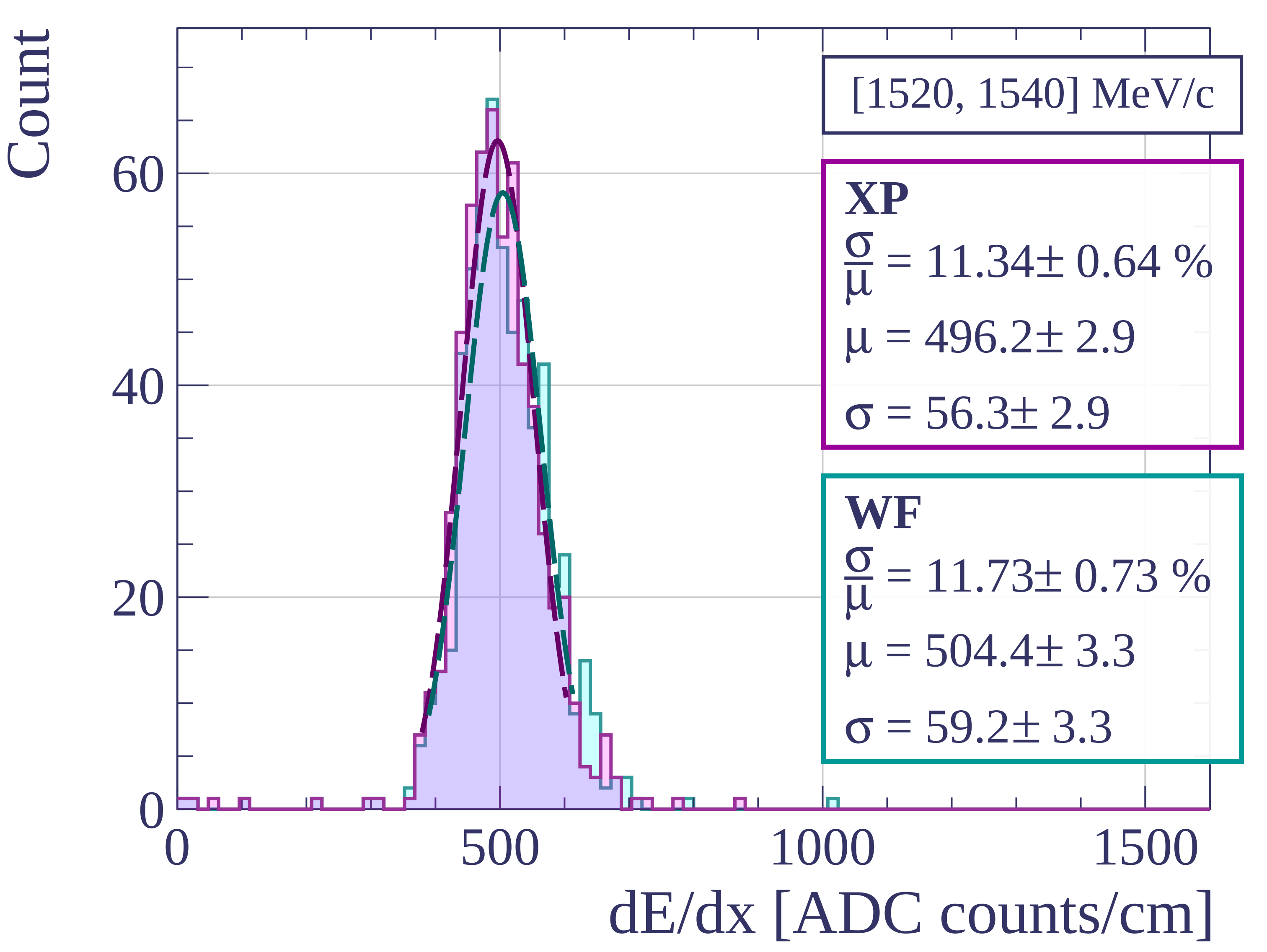
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.91± 0.67 %
μ = 487.6± 2.8
σ = 53.2± 3.0
WF
$\frac{\sigma}{\mu}$ = 10.58± 0.58 %
μ = 491.6± 2.7
σ = 52.0± 2.6

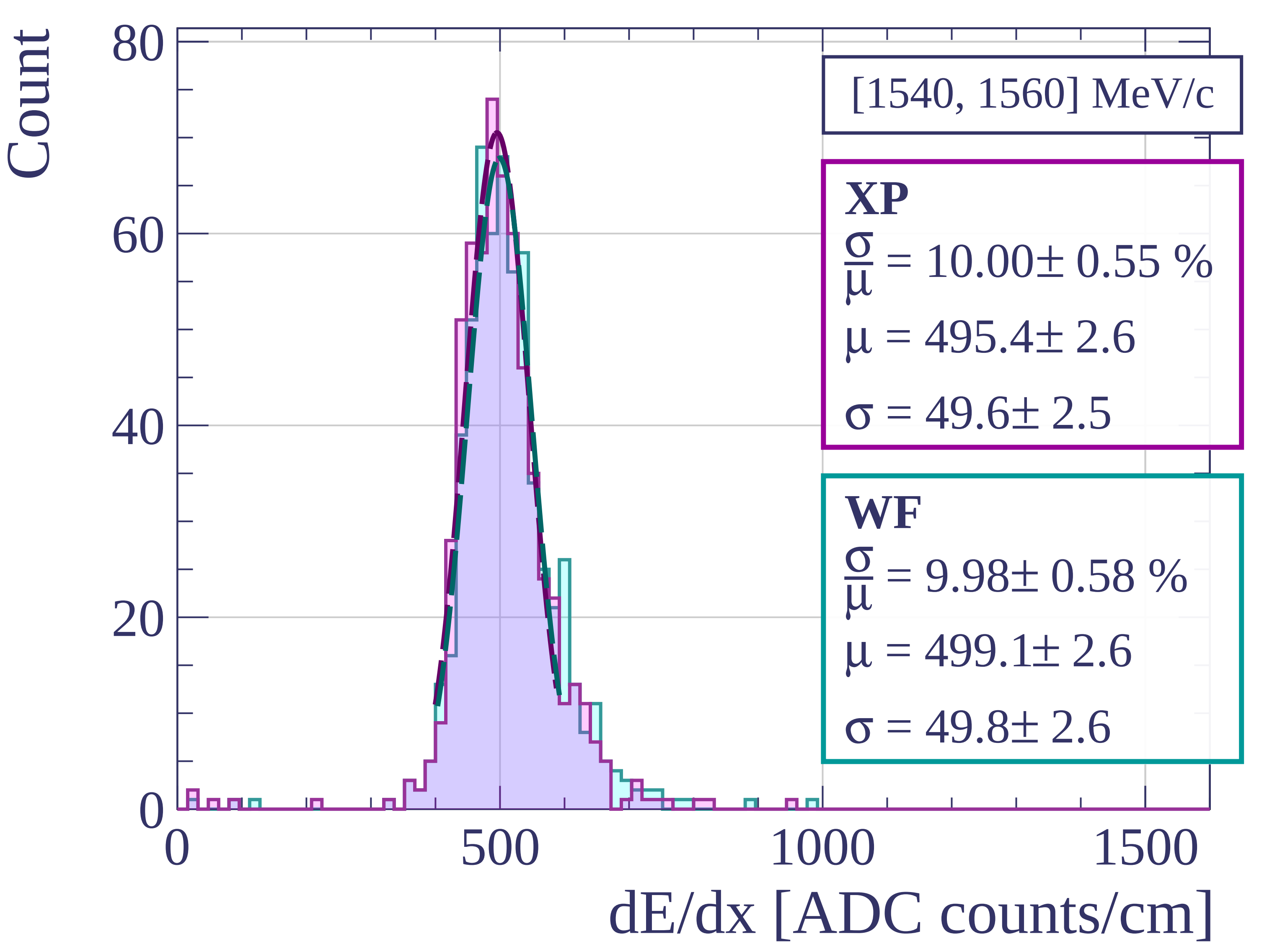


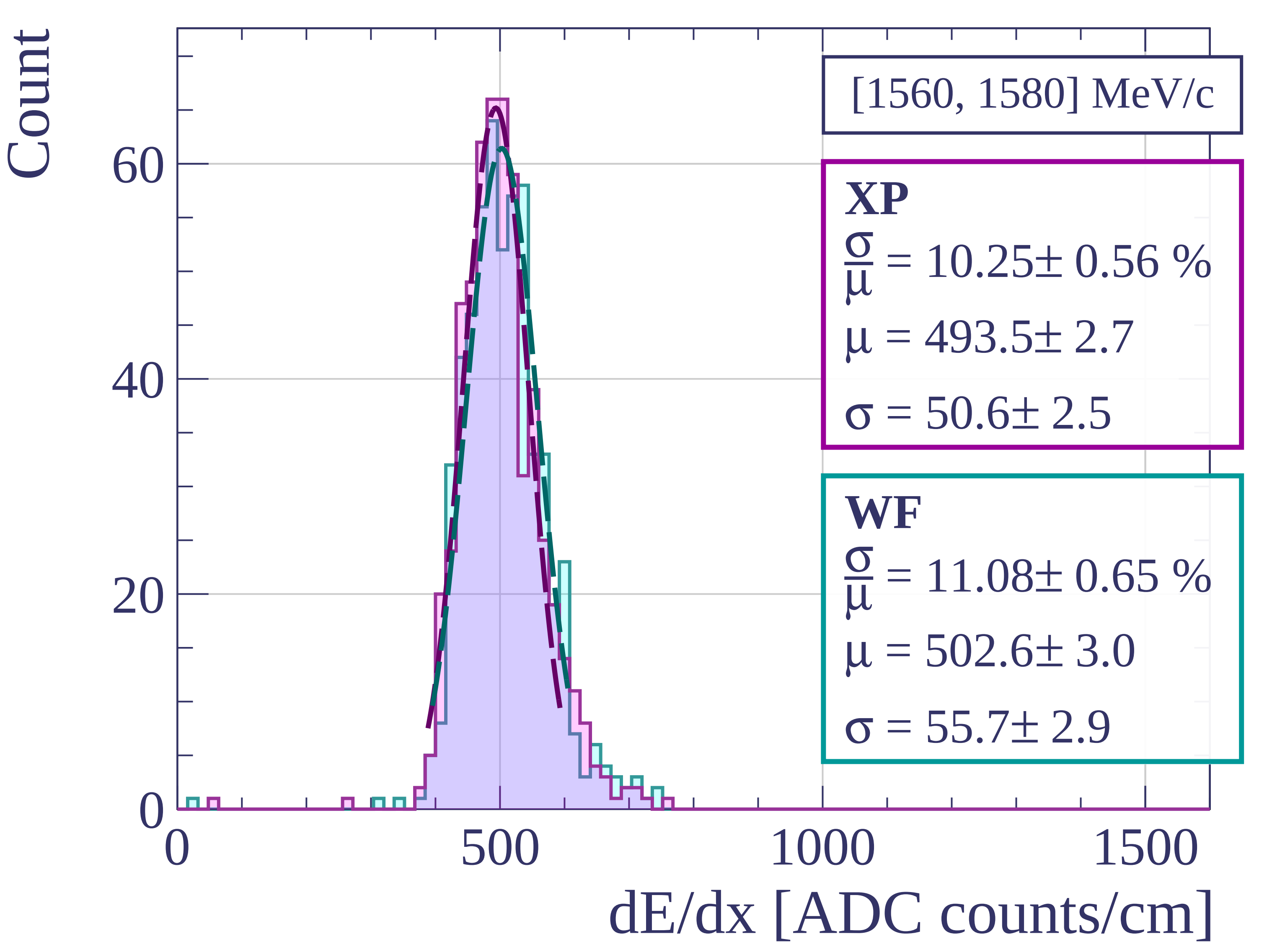
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.35± 0.57 %
μ = 483.7± 2.4
σ = 45.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 10.46± 0.59 %
μ = 492.6± 2.7
σ = 51.5± 2.7

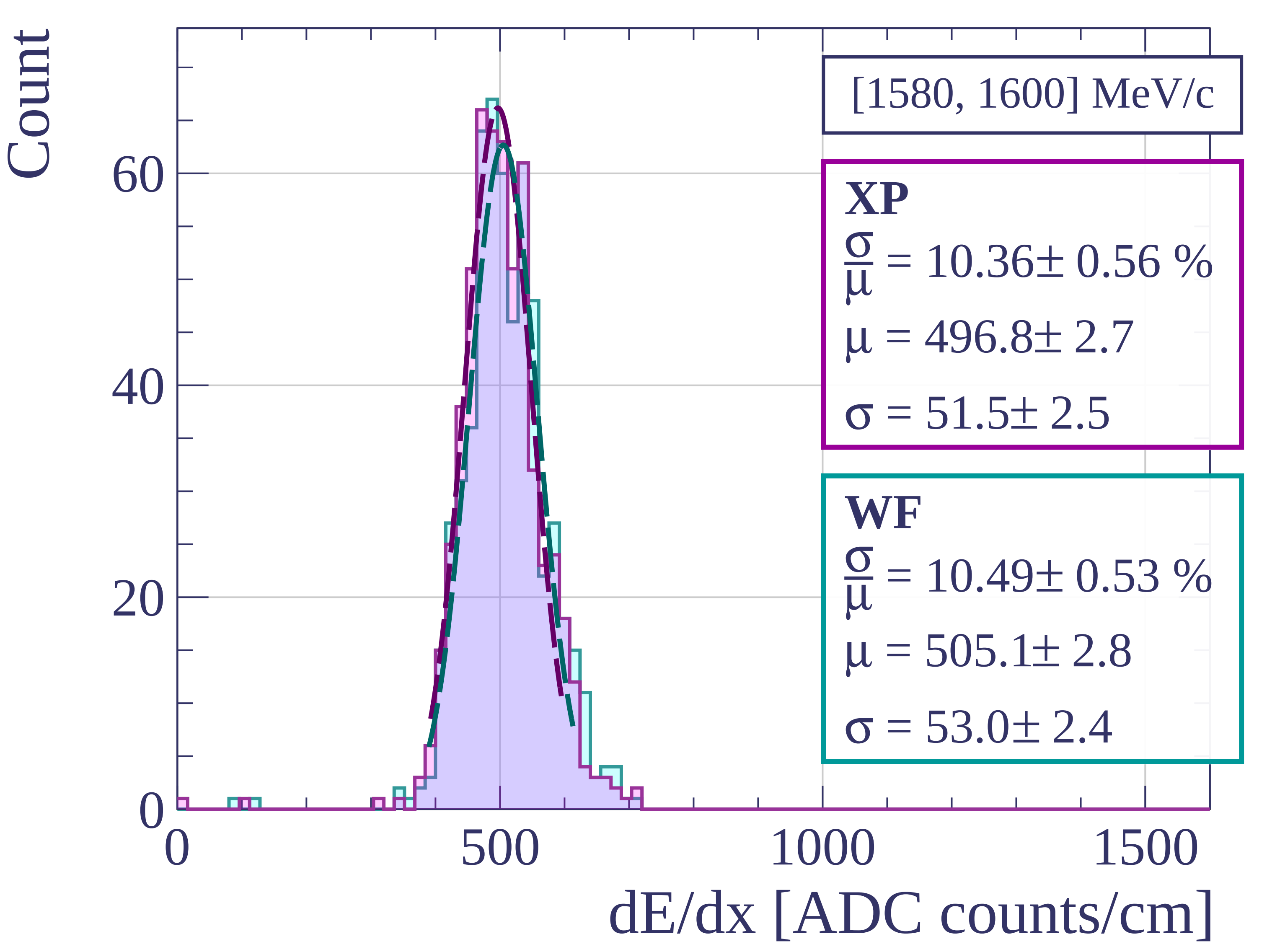


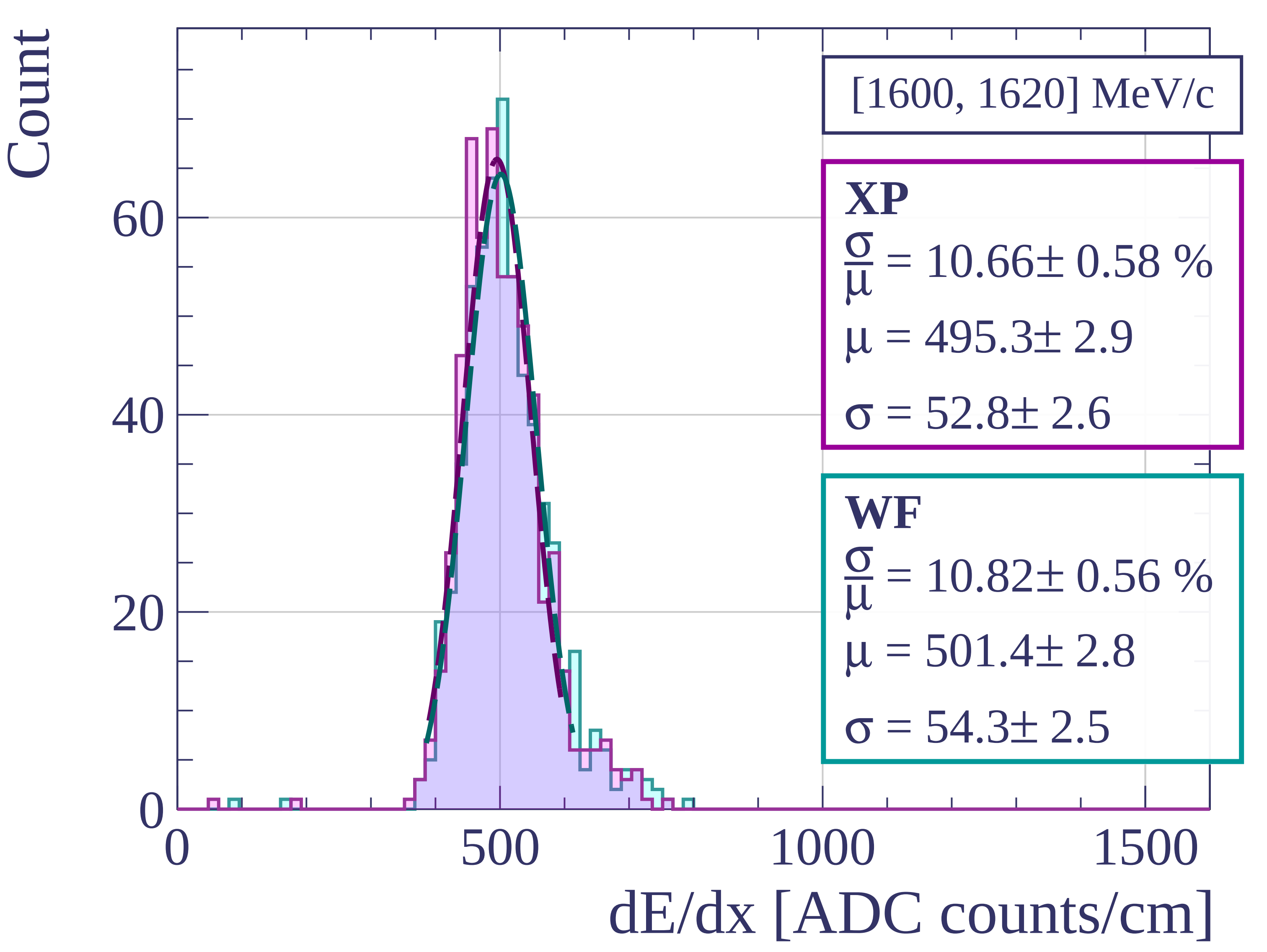
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1500, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.50± 0.61 %
μ = 494.8± 2.8
σ = 51.9± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.37± 0.68 %
μ = 497.3± 2.8
σ = 51.6± 3.1

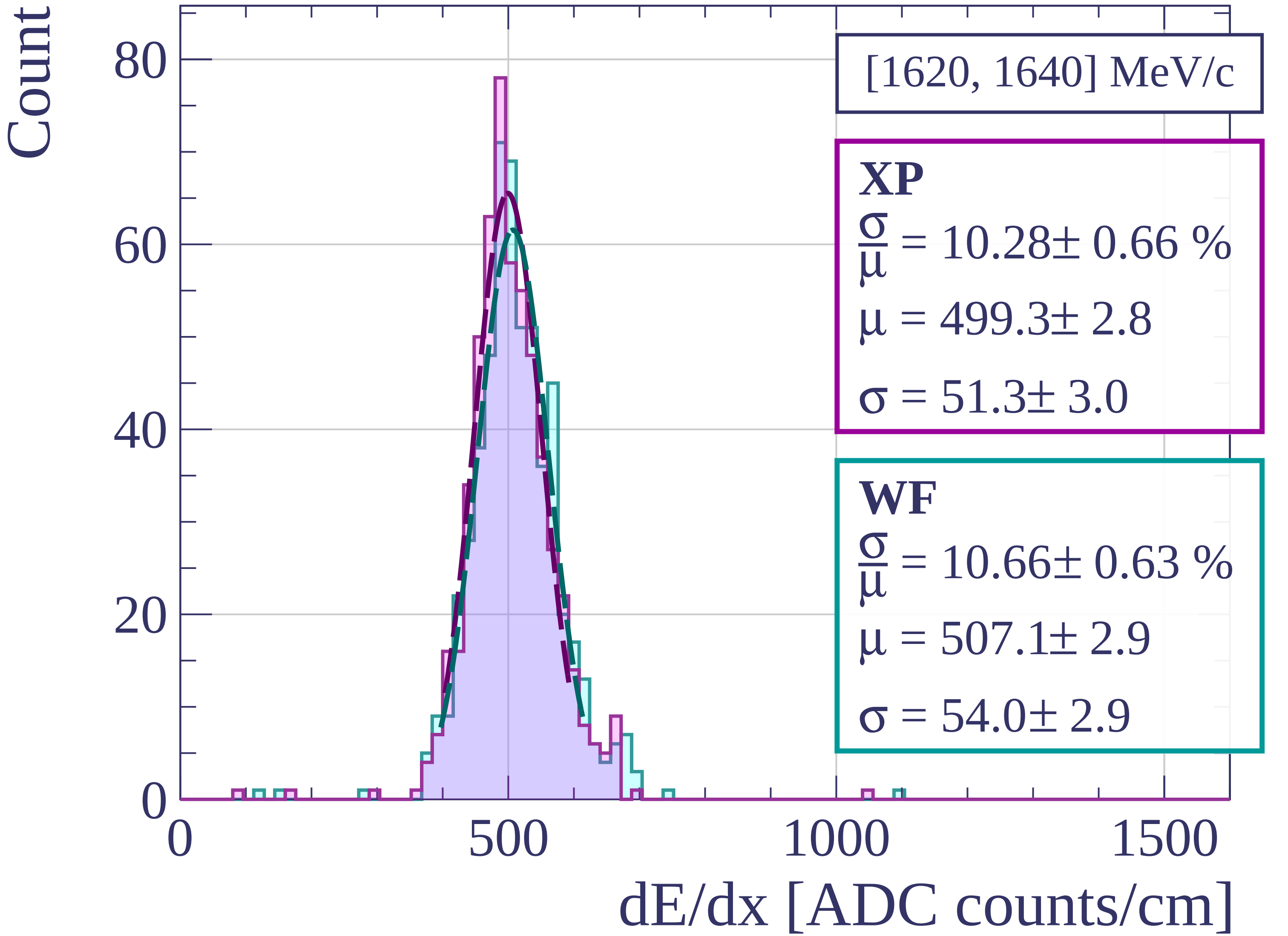

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.34± 0.64 %
μ = 496.2± 2.9
σ = 56.3± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.73± 0.73 %
μ = 504.4± 3.3
σ = 59.2± 3.3

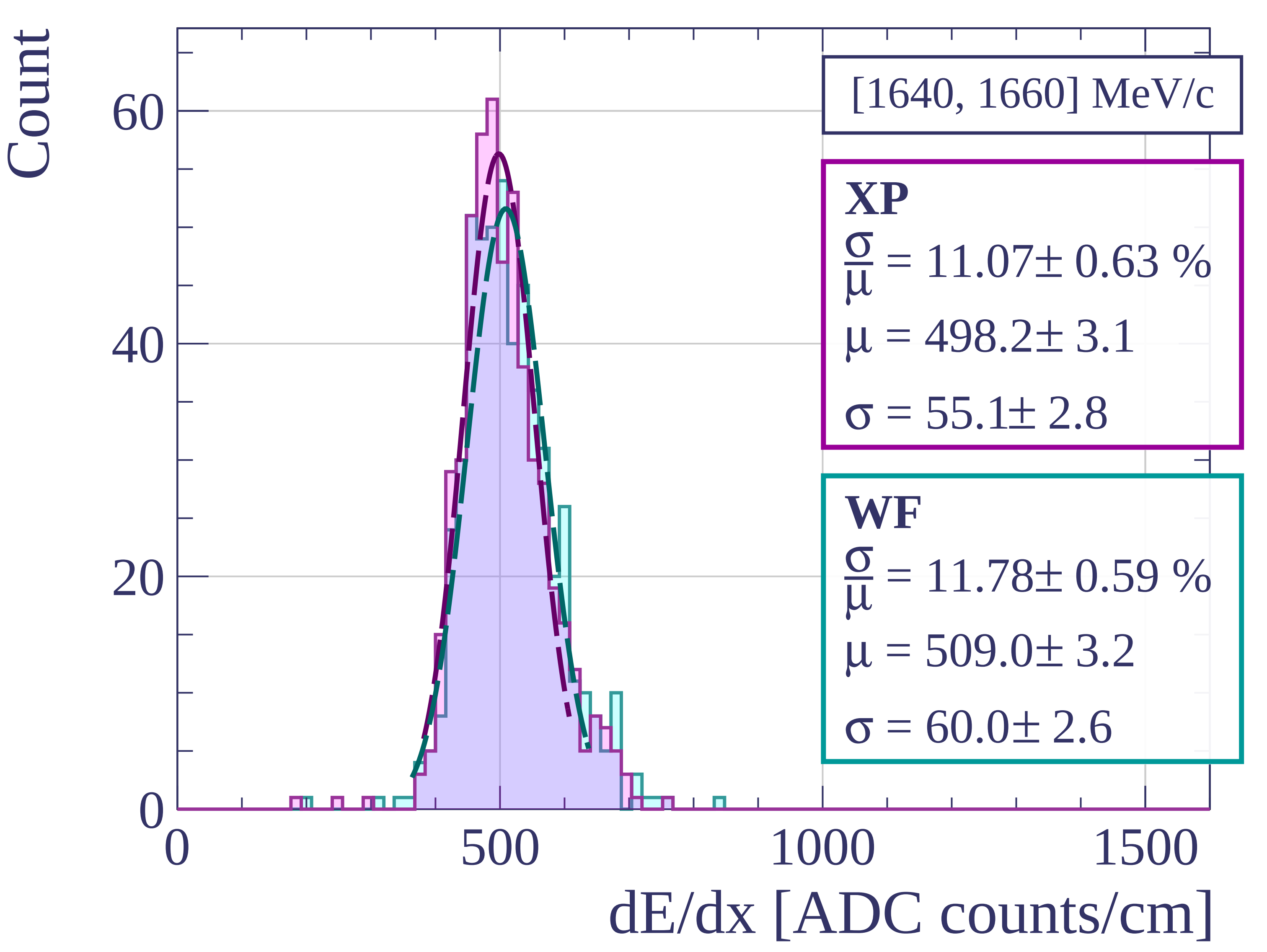

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
80
[1540, 1560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.00± 0.55 %
μ = 495.4± 2.6
σ = 49.6± 2.5
WF
$\frac{\sigma}{\mu}$ = 9.98± 0.58 %
μ = 499.1± 2.6
σ = 49.8± 2.6

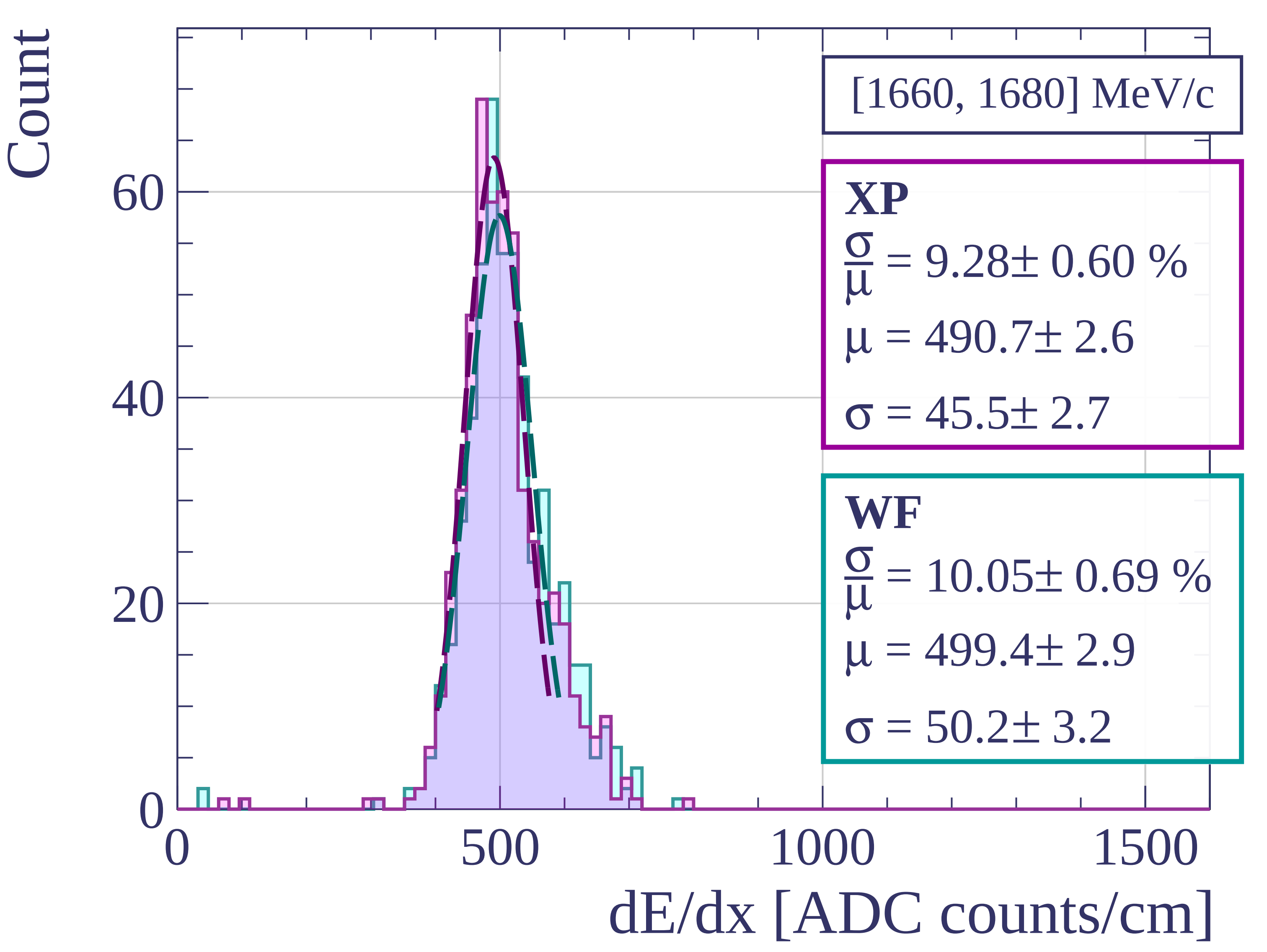

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1560, 1580] MeV/c
XP
σ/μ = 10.25± 0.56 %
μ = 493.5± 2.7
σ = 50.6± 2.5
WF
σ/μ = 11.08± 0.65 %
μ = 502.6± 3.0
σ = 55.7± 2.9


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1580, 1600] MeV/c
XP
σ/μ = 10.36± 0.56 %
μ = 496.8± 2.7
σ = 51.5± 2.5
WF
σ/μ = 10.49± 0.53 %
μ = 505.1± 2.8
σ = 53.0± 2.4

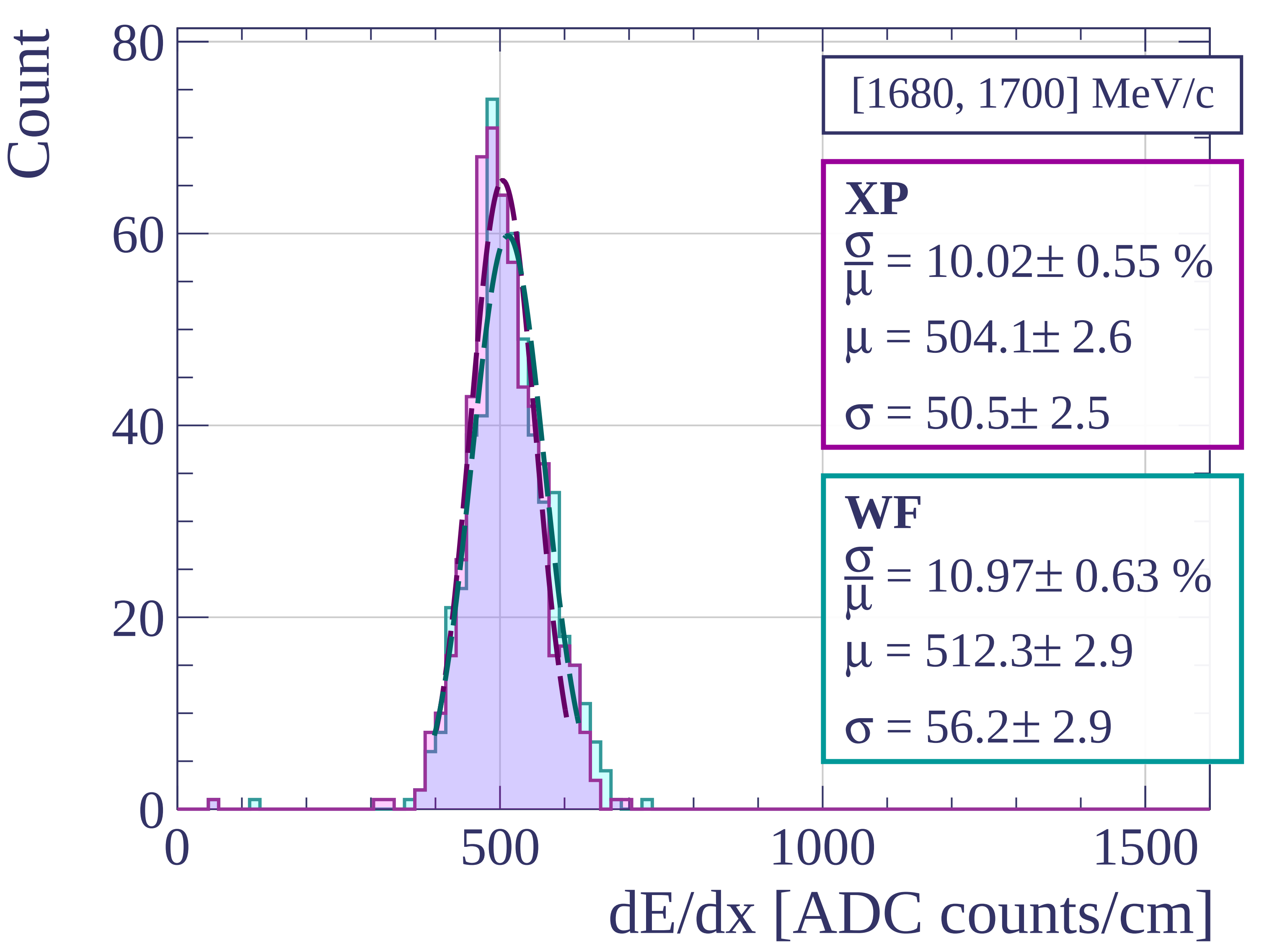

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.66± 0.58 %
μ = 495.3± 2.9
σ = 52.8± 2.6
WF
$\frac{\sigma}{\mu}$ = 10.82± 0.56 %
μ = 501.4± 2.8
σ = 54.3± 2.5


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1620, 1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.28± 0.66 %
μ = 499.3± 2.8
σ = 51.3± 3.0
WF
$\frac{\sigma}{\mu}$ = 10.66± 0.63 %
μ = 507.1± 2.9
σ = 54.0± 2.9

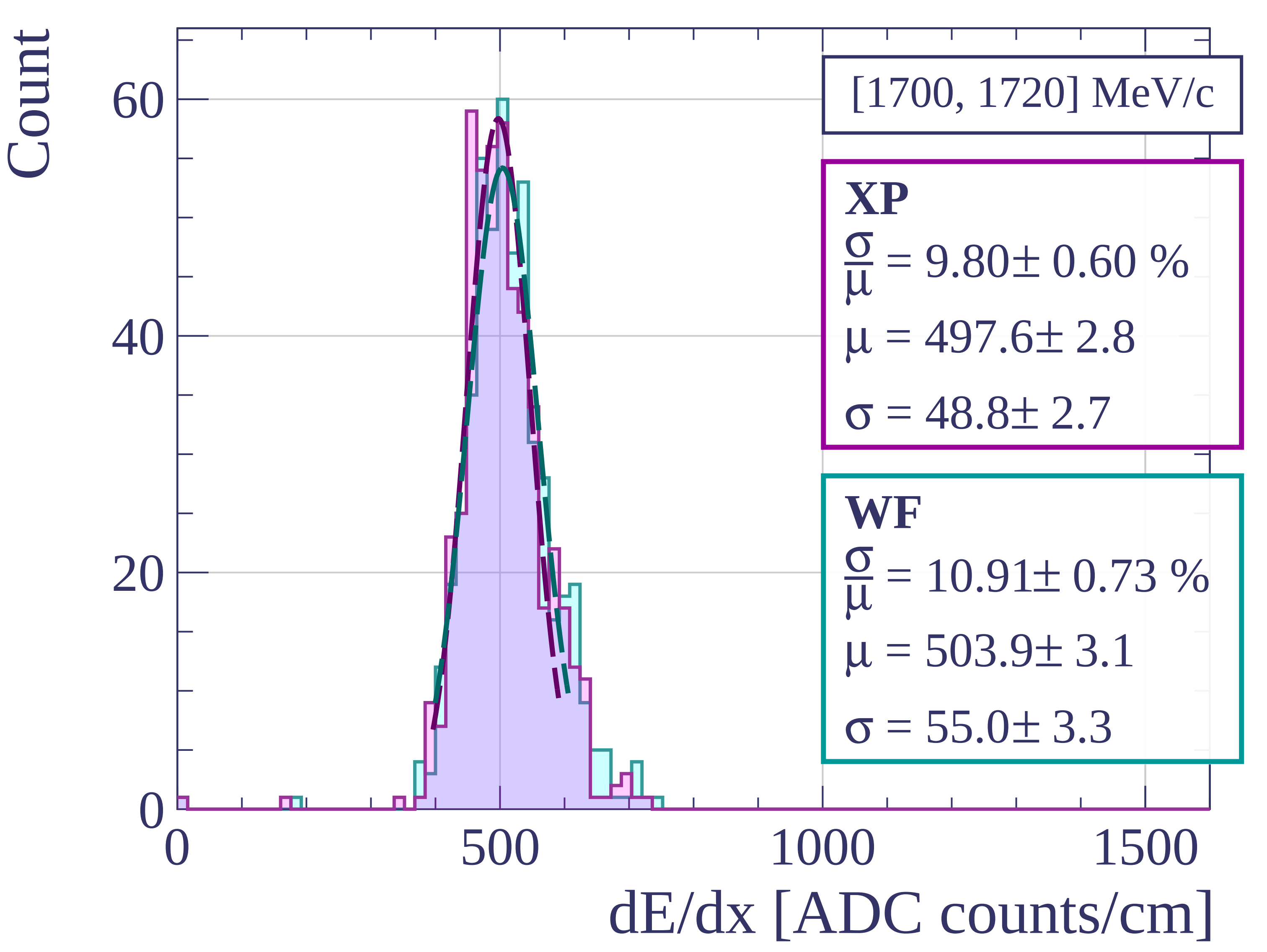

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1640, 1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.07± 0.63 %
μ = 498.2± 3.1
σ = 55.1± 2.8
WF
$\frac{\sigma}{\mu}$ = 11.78± 0.59 %
μ = 509.0± 3.2
σ = 60.0± 2.6

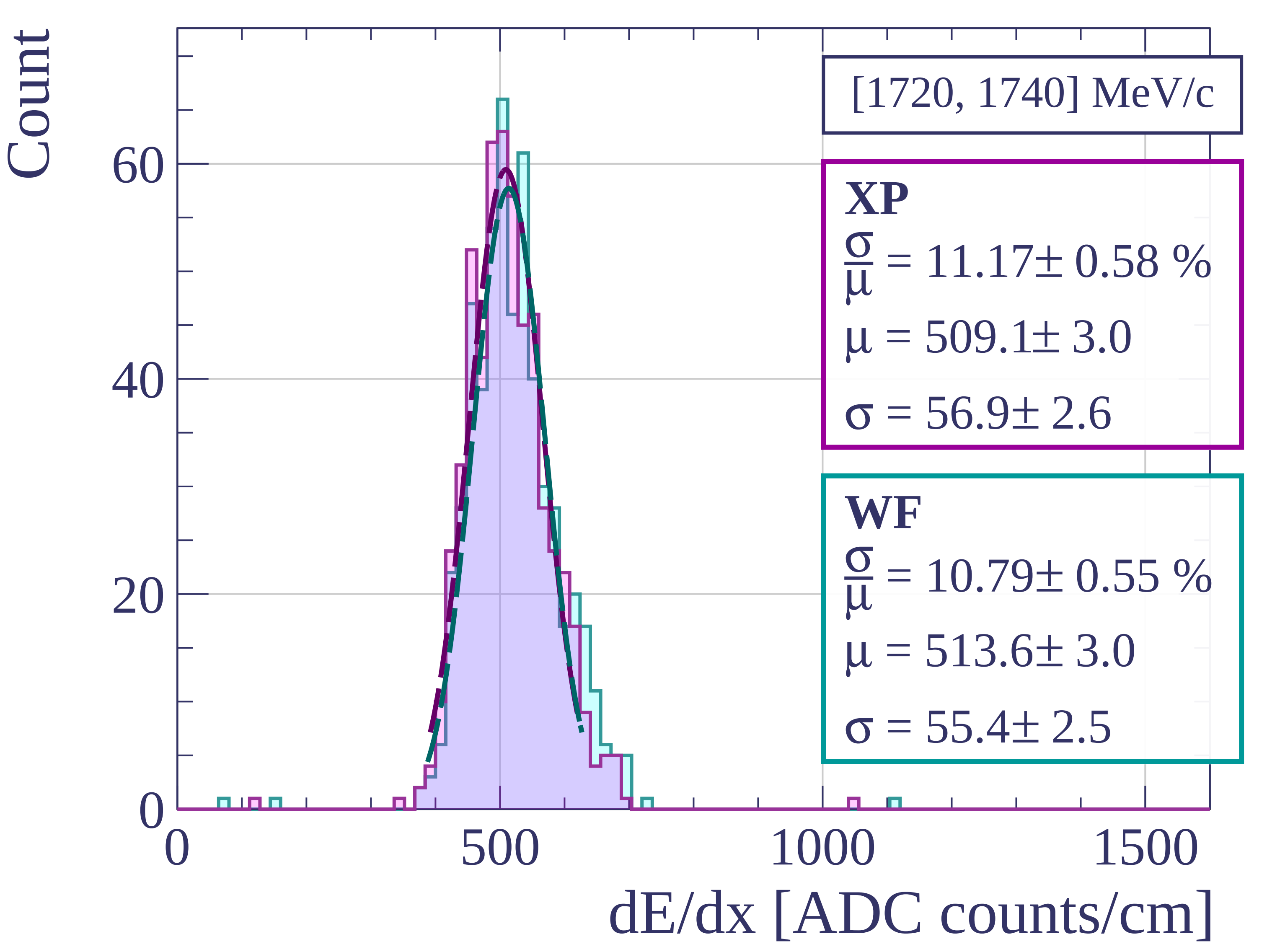


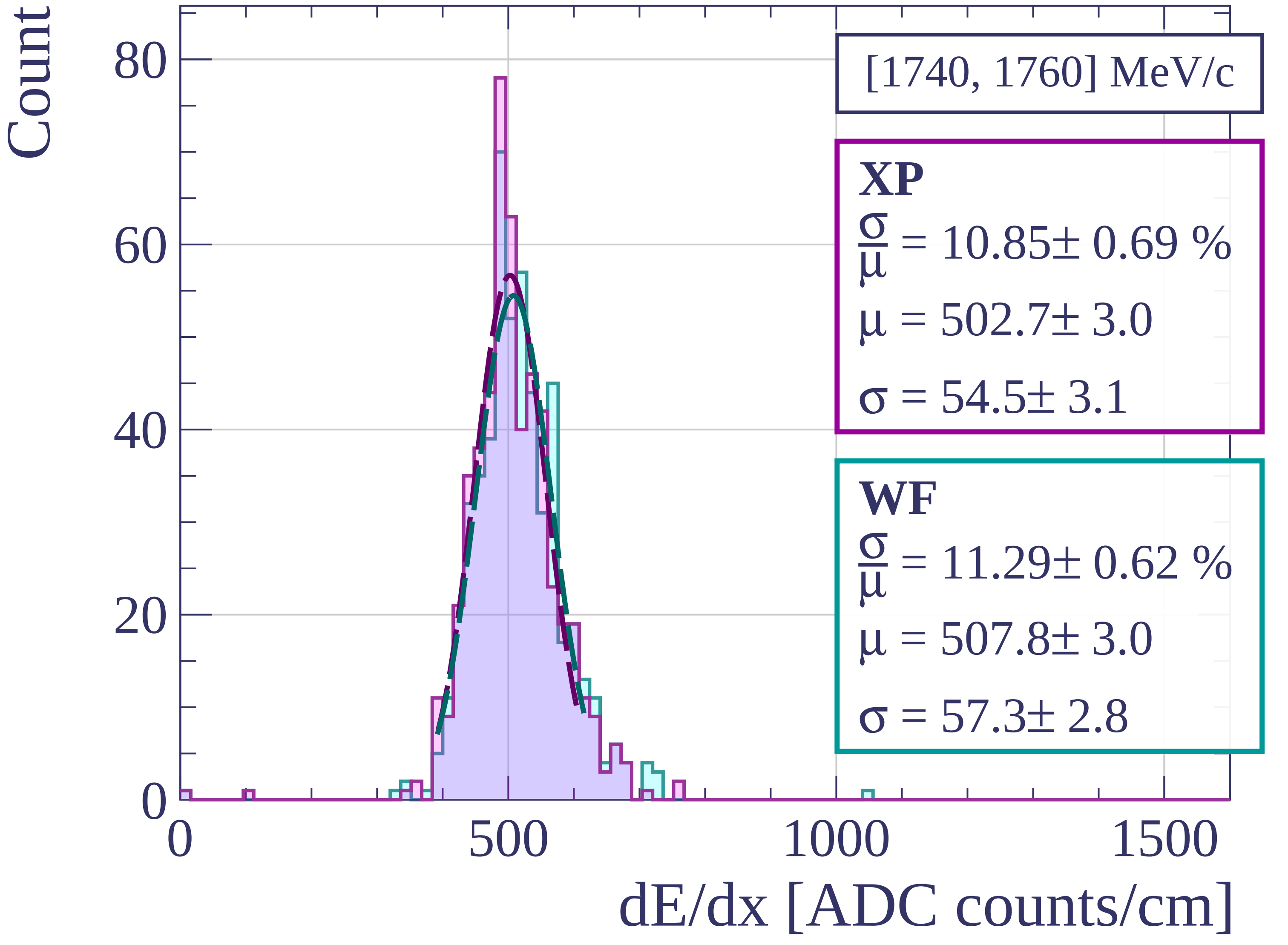
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1660, 1680] MeV/c
XP
$\frac{\sigma}{\mu} = 9.28 \pm 0.60$ %
$\mu = 490.7 \pm 2.6$
$\sigma = 45.5 \pm 2.7$
WF
$\frac{\sigma}{\mu} = 10.05 \pm 0.69$ %
$\mu = 499.4 \pm 2.9$
$\sigma = 50.2 \pm 3.2$



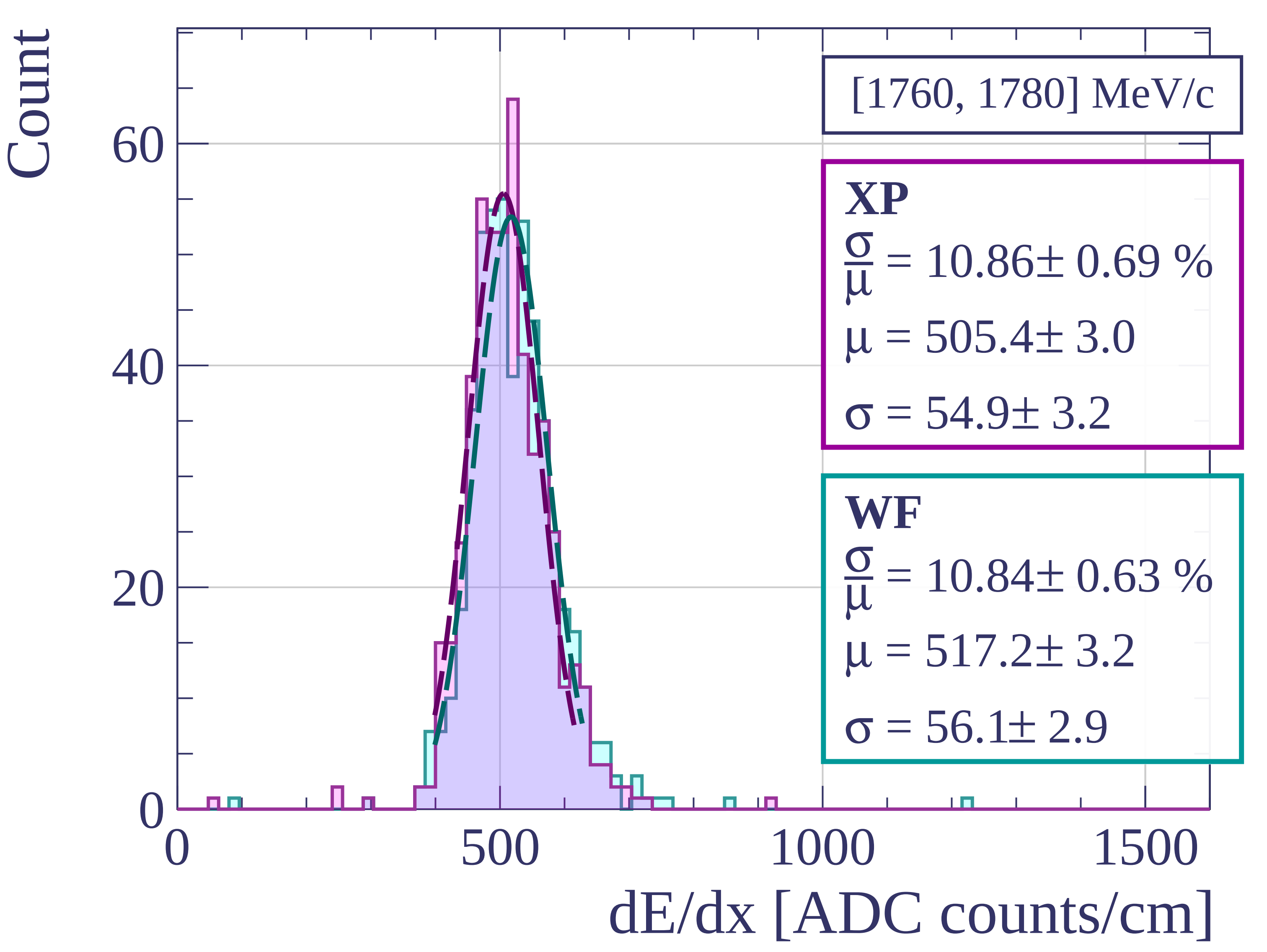
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1700] MeV/c
XP
σ/μ = 10.02± 0.55 %
μ = 504.1± 2.6
σ = 50.5± 2.5
WF
σ/μ = 10.97± 0.63 %
μ = 512.3± 2.9
σ = 56.2± 2.9

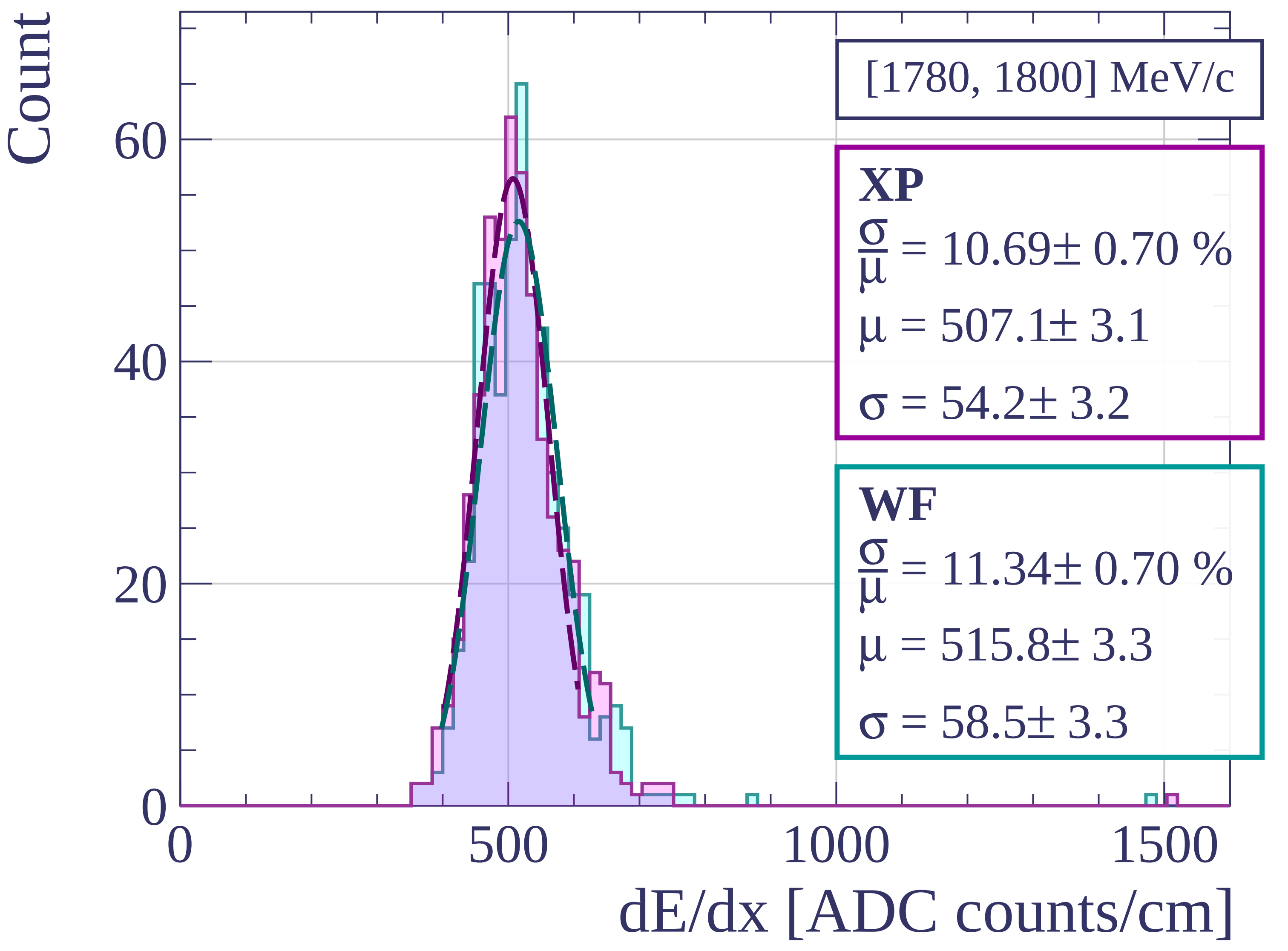

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1700, 1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.80± 0.60 %
μ = 497.6± 2.8
σ = 48.8± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.91± 0.73 %
μ = 503.9± 3.1
σ = 55.0± 3.3

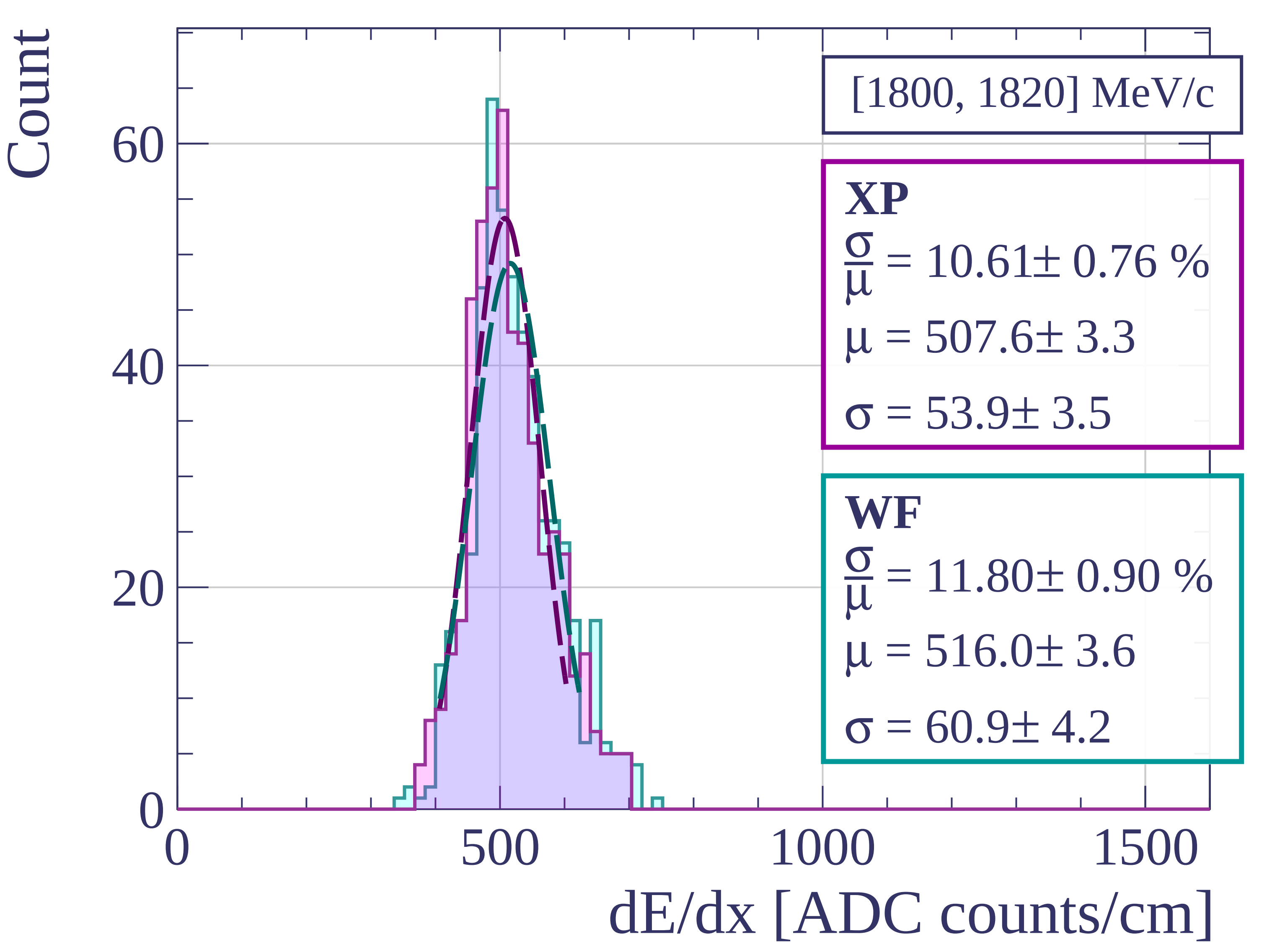

Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[1720, 1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.17± 0.58 %
μ = 509.1± 3.0
σ = 56.9± 2.6
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.55 %
μ = 513.6± 3.0
σ = 55.4± 2.5



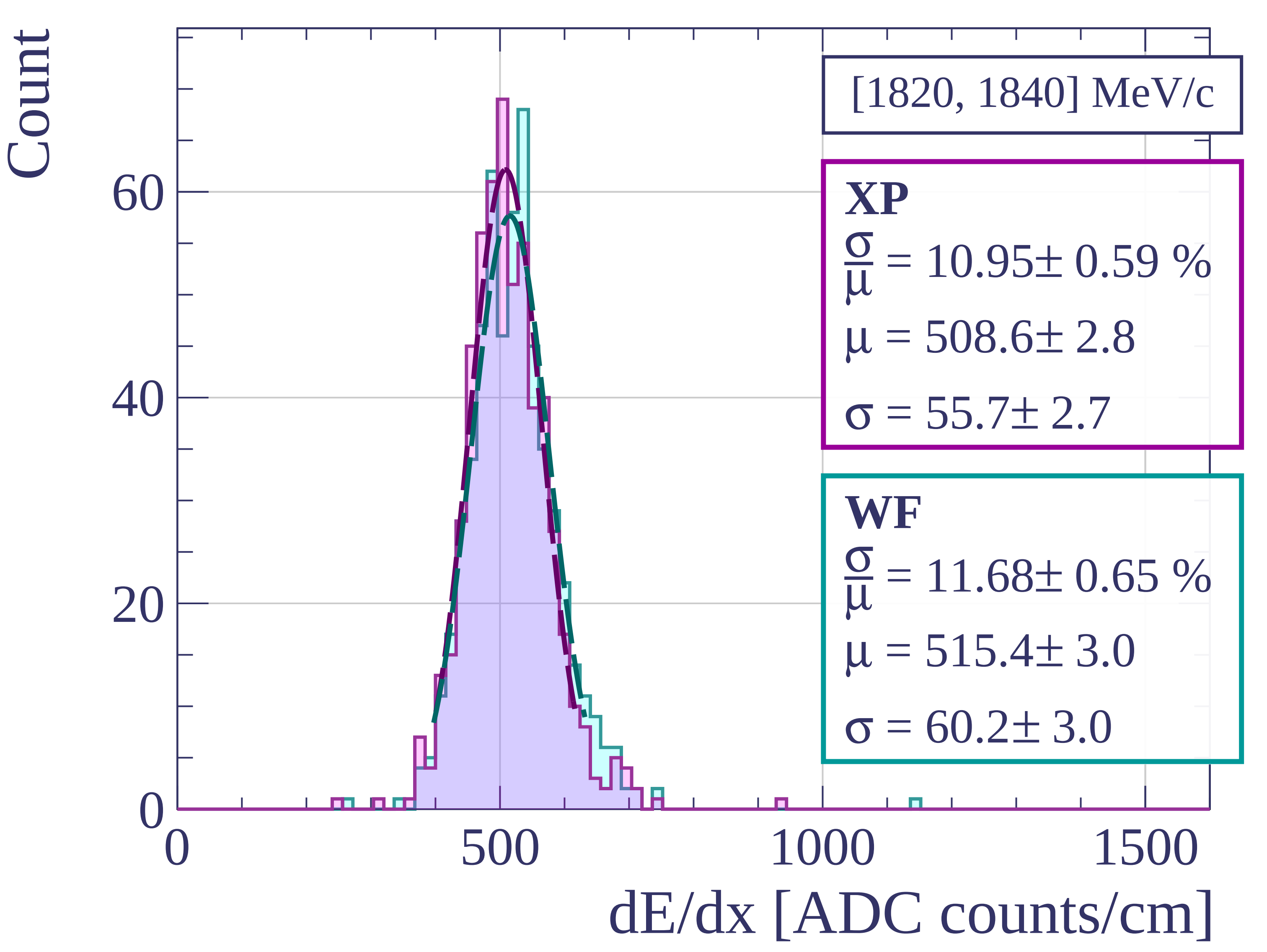
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1740, 1760] MeV/c
XP
σ/μ = 10.85± 0.69 %
μ = 502.7± 3.0
σ = 54.5± 3.1
WF
σ/μ = 11.29± 0.62 %
μ = 507.8± 3.0
σ = 57.3± 2.8


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.86± 0.69 %
μ = 505.4± 3.0
σ = 54.9± 3.2
WF
$\frac{\sigma}{\mu}$ = 10.84± 0.63 %
μ = 517.2± 3.2
σ = 56.1± 2.9

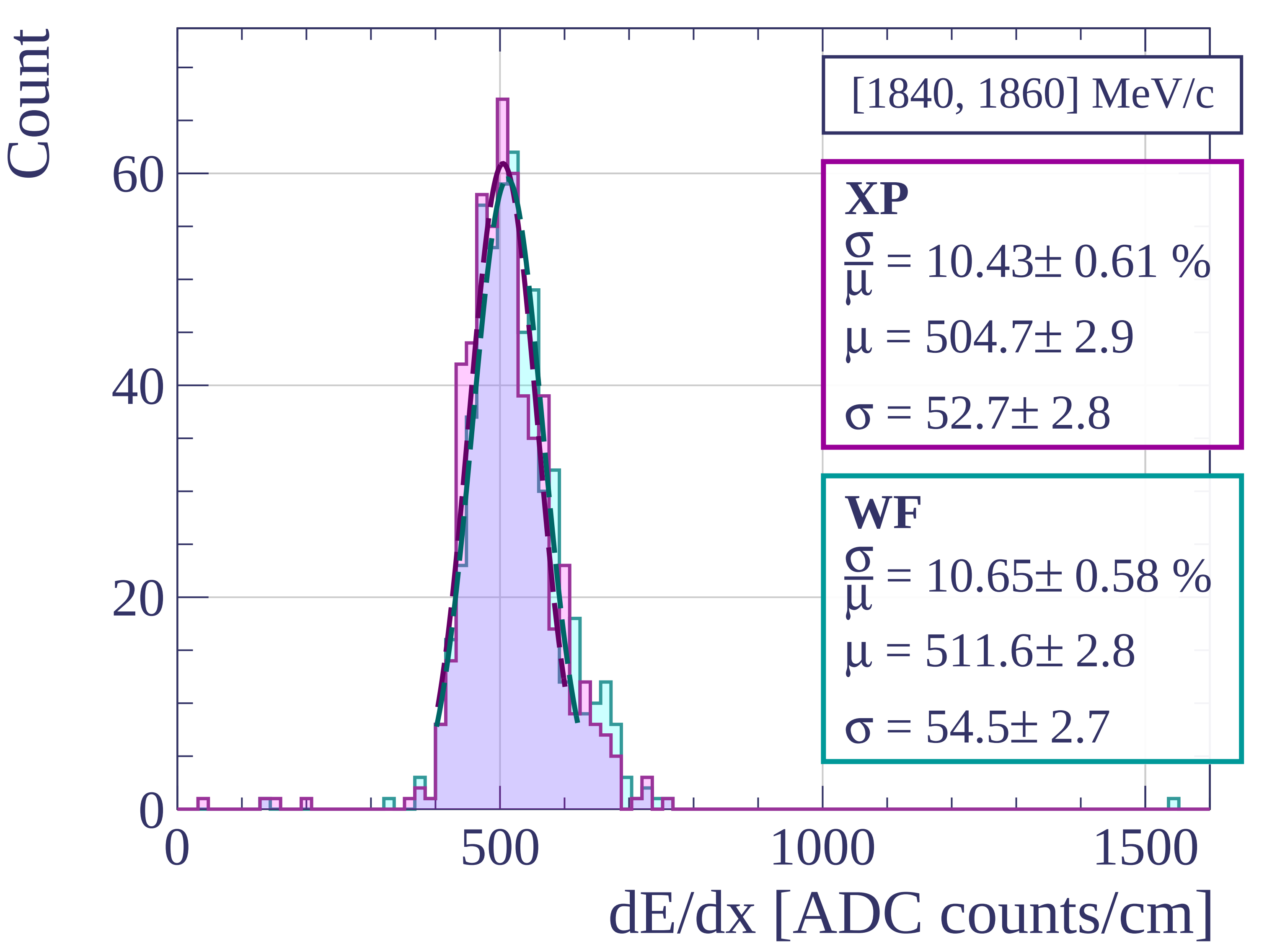

Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[1780, 1800] MeV/c
XP
σ/μ = 10.69± 0.70 %
μ = 507.1± 3.1
σ = 54.2± 3.2
WF
σ/μ = 11.34± 0.70 %
μ = 515.8± 3.3
σ = 58.5± 3.3



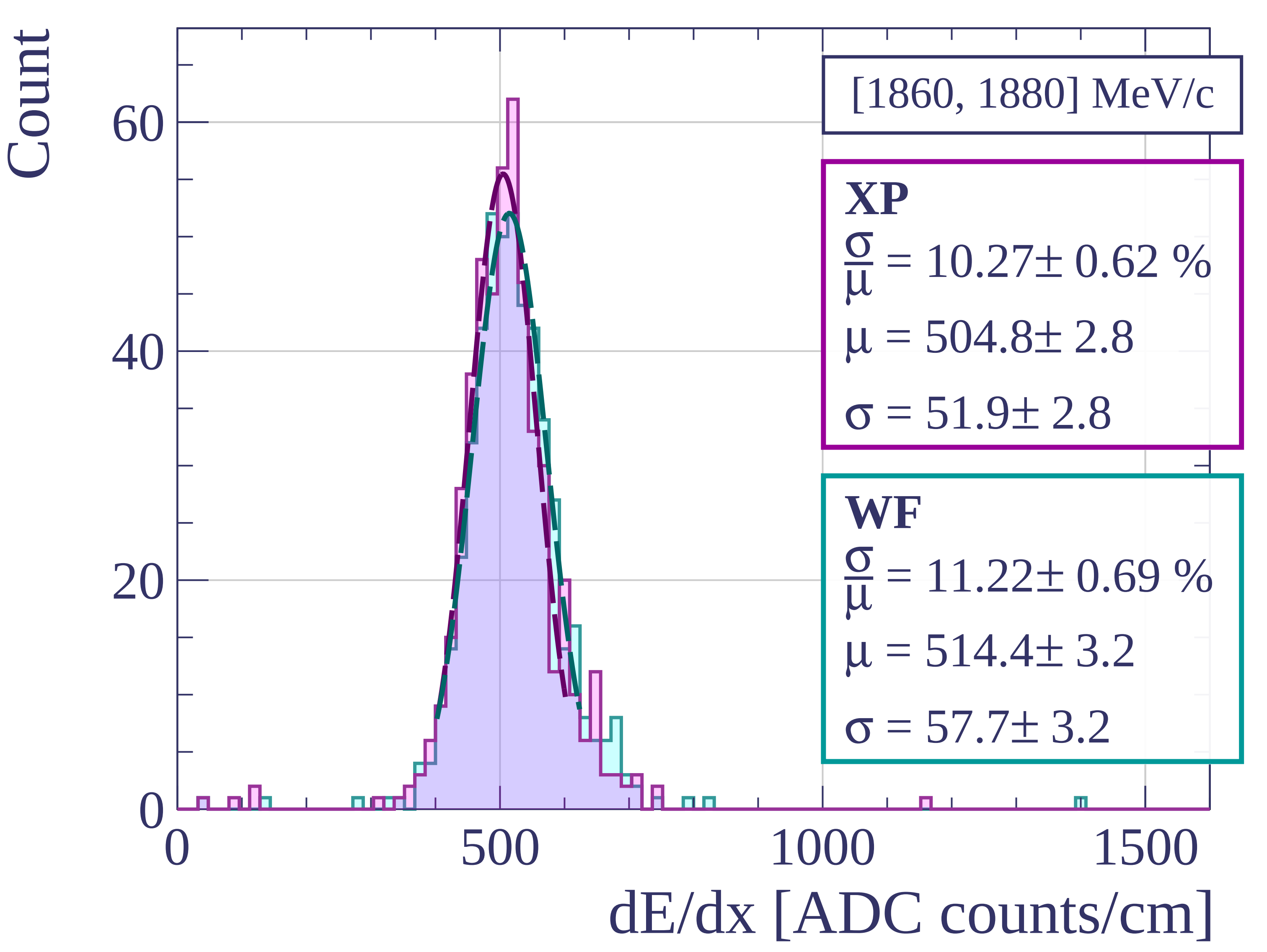
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1800, 1820] MeV/c
XP
σ/μ = 10.61± 0.76 %
μ = 507.6± 3.3
σ = 53.9± 3.5
WF
σ/μ = 11.80± 0.90 %
μ = 516.0± 3.6
σ = 60.9± 4.2



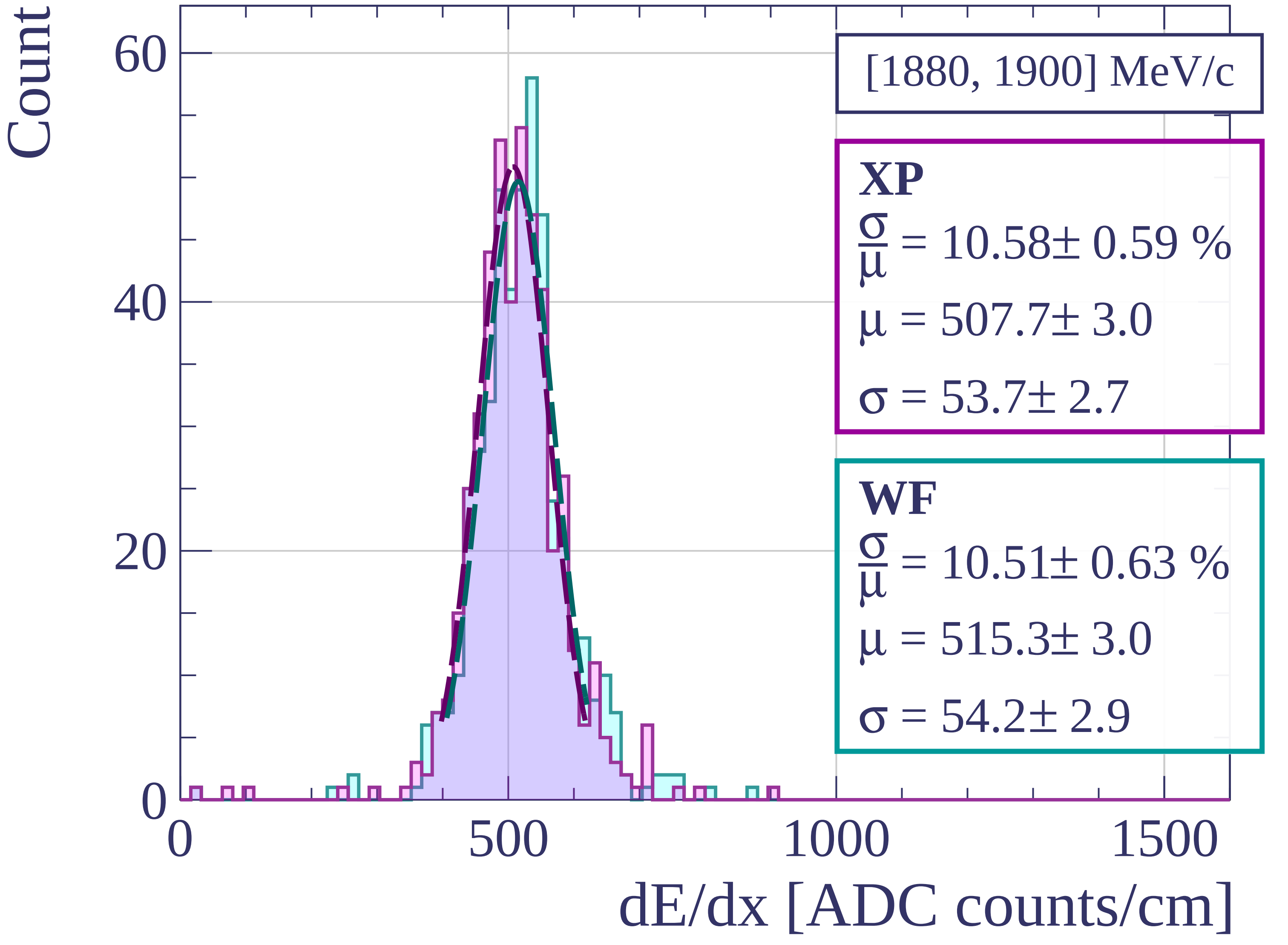
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1820, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.95± 0.59 %
μ = 508.6± 2.8
σ = 55.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.68± 0.65 %
μ = 515.4± 3.0
σ = 60.2± 3.0



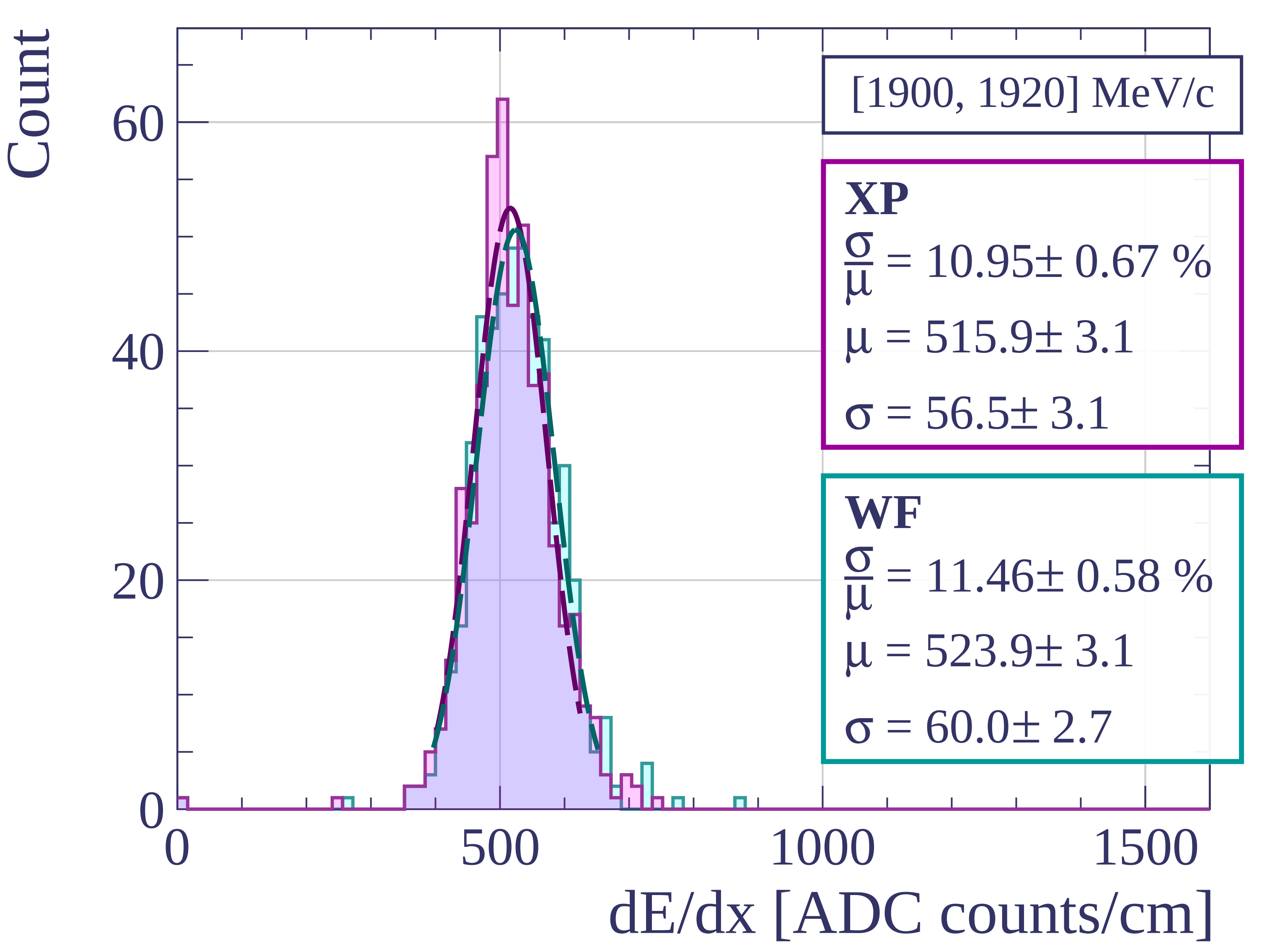
Count
dE/dx [ADC counts/cm]
[1840, 1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.43± 0.61 %
μ = 504.7± 2.9
σ = 52.7± 2.8
WF
$\frac{\sigma}{\mu}$ = 10.65± 0.58 %
μ = 511.6± 2.8
σ = 54.5± 2.7
0
20
40
60
0
500
1000
1500

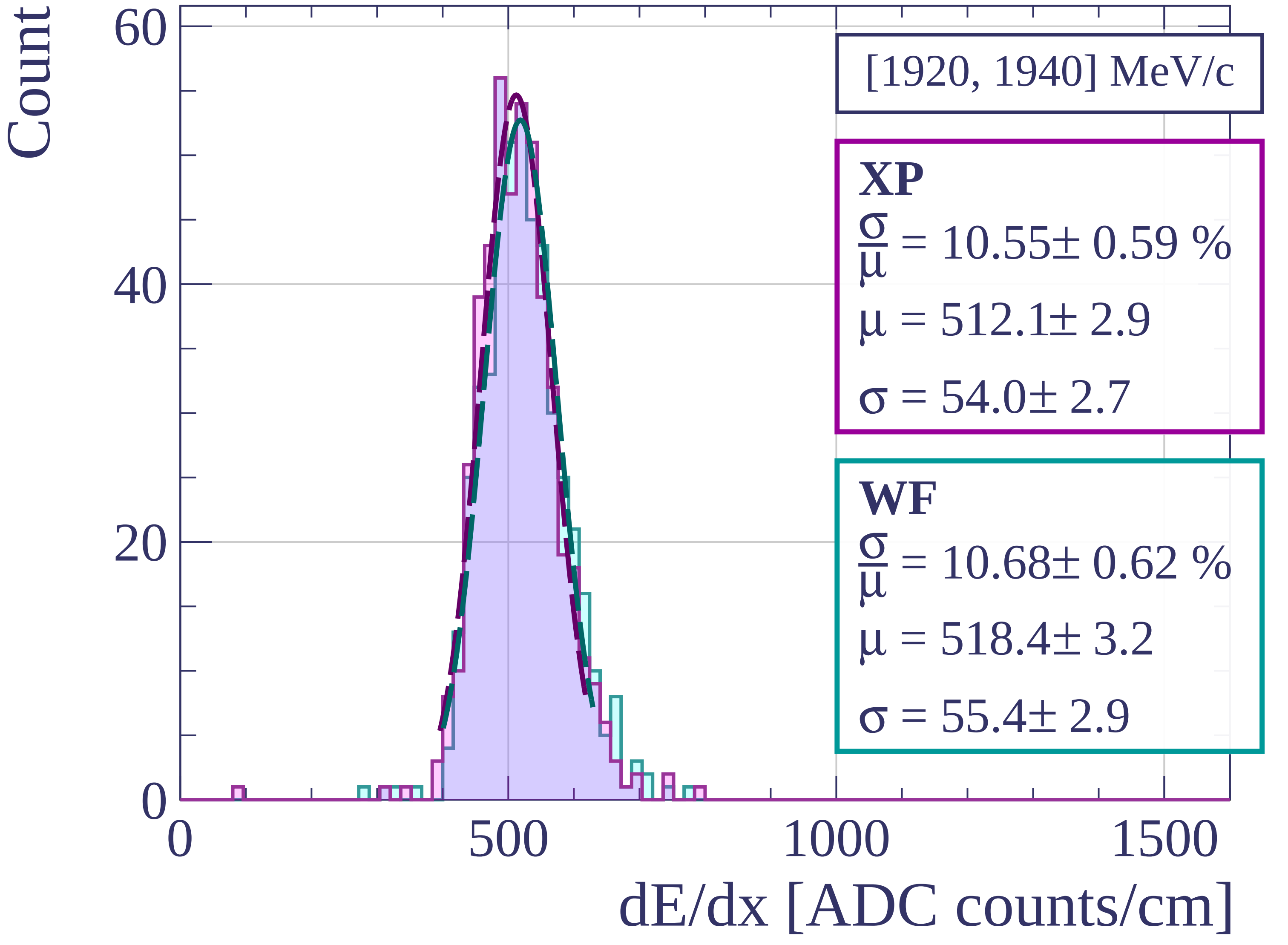


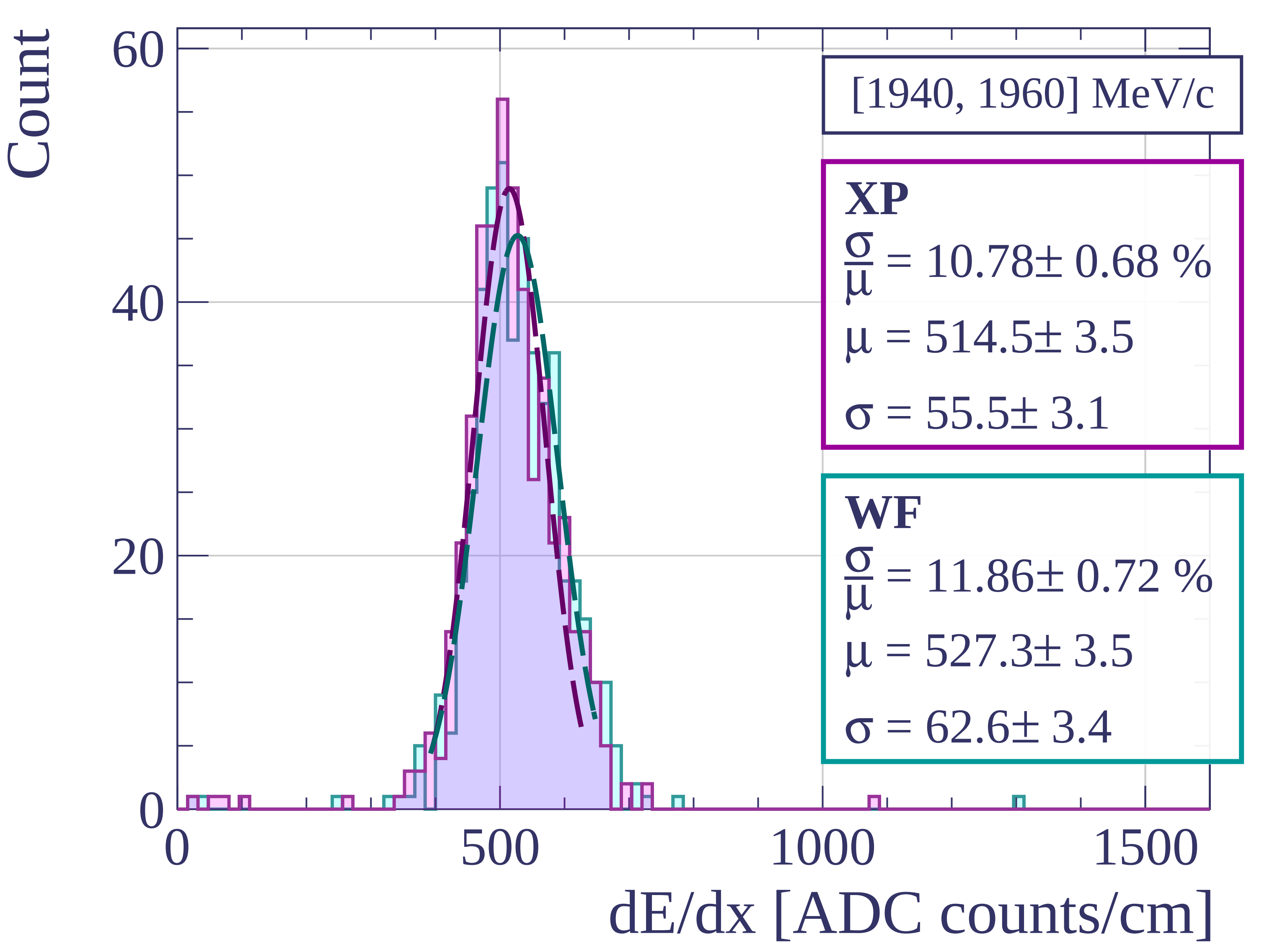
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
σ/μ = 10.27± 0.62 %
μ = 504.8± 2.8
σ = 51.9± 2.8
WF
σ/μ = 11.22± 0.69 %
μ = 514.4± 3.2
σ = 57.7± 3.2



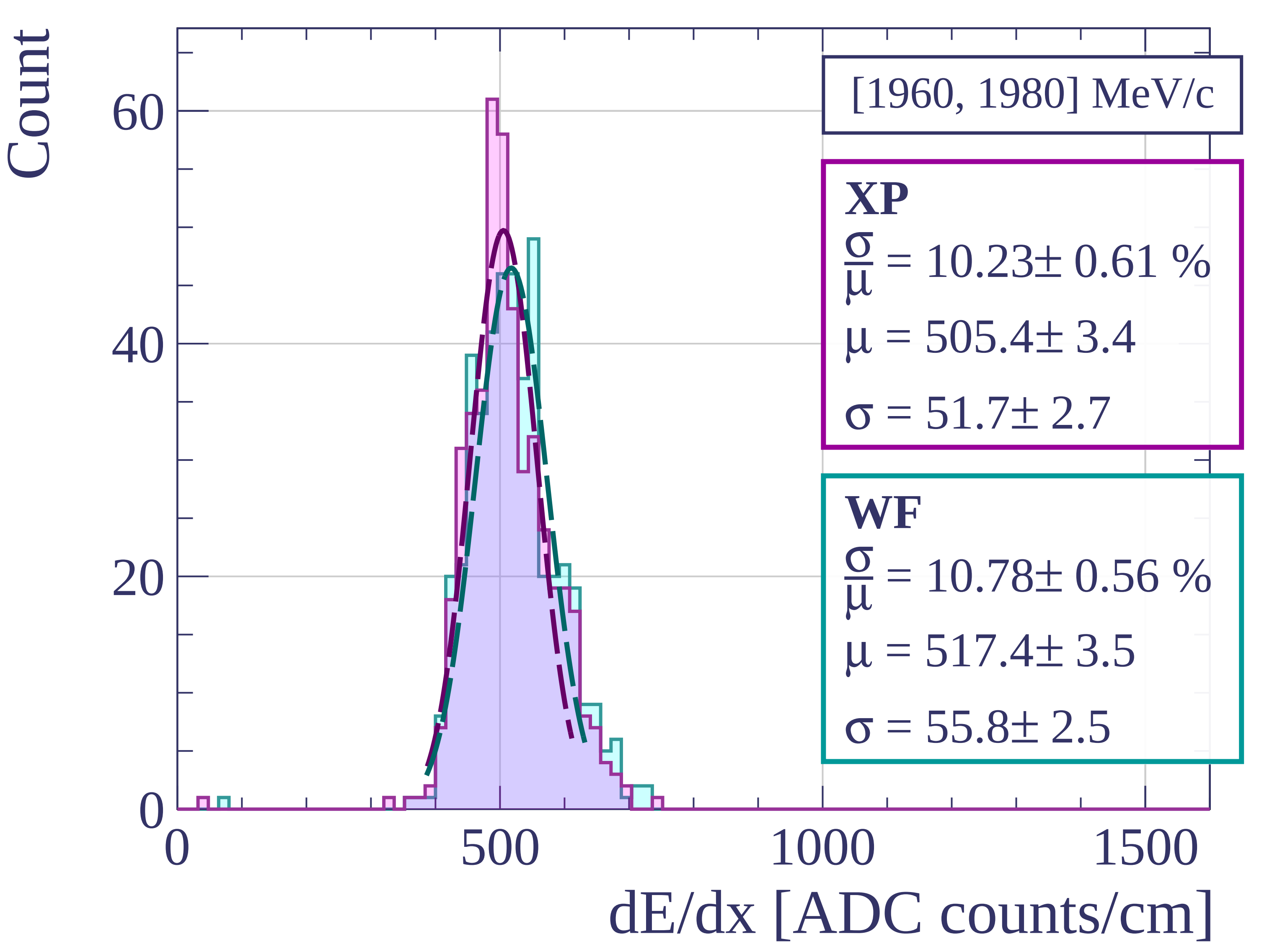
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1880, 1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.58± 0.59 %
μ = 507.7± 3.0
σ = 53.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.51± 0.63 %
μ = 515.3± 3.0
σ = 54.2± 2.9


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1900, 1920] MeV/c
XP
σ/μ = 10.95± 0.67 %
μ = 515.9± 3.1
σ = 56.5± 3.1
WF
σ/μ = 11.46± 0.58 %
μ = 523.9± 3.1
σ = 60.0± 2.7

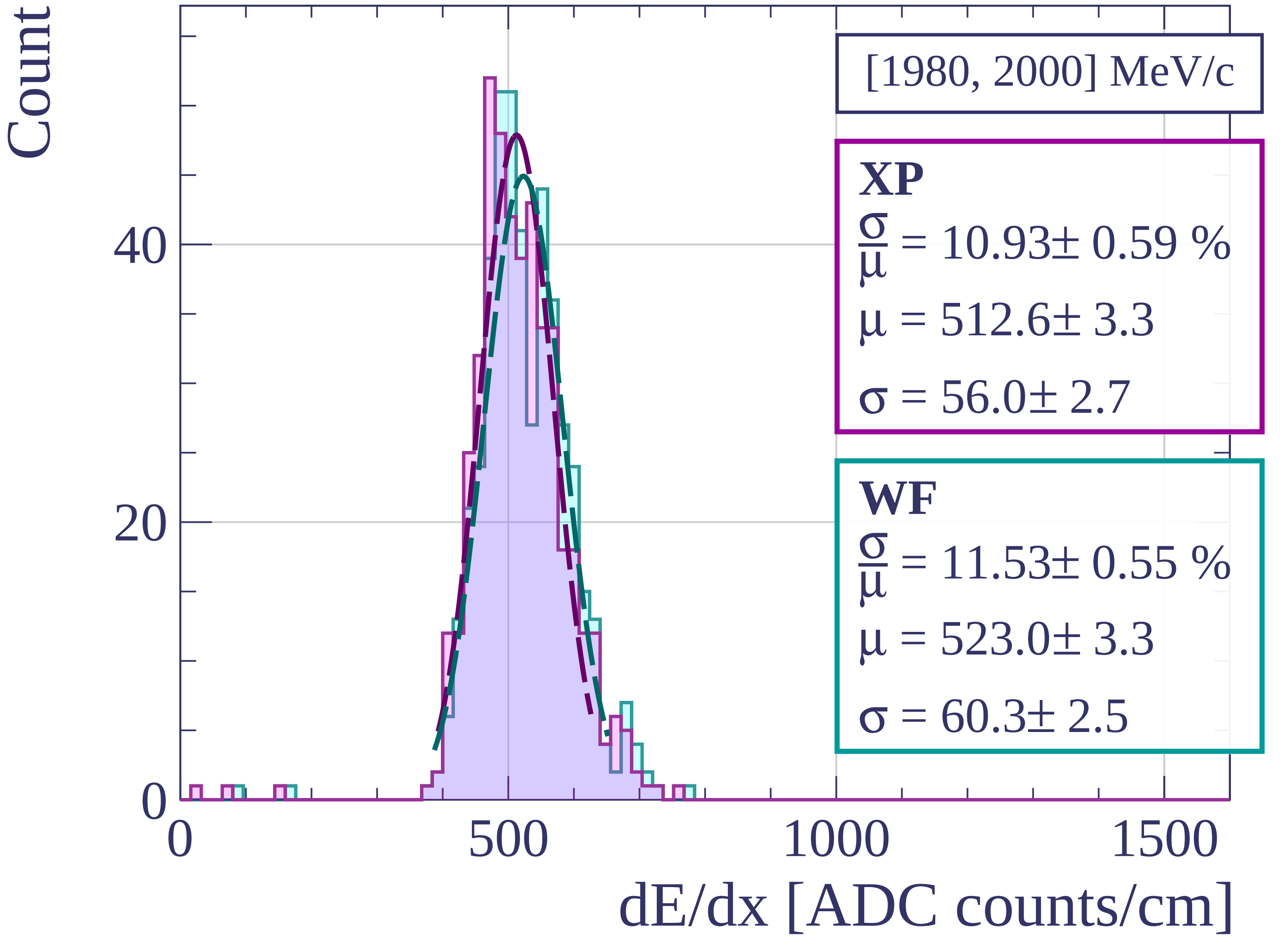

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.55± 0.59 %
μ = 512.1± 2.9
σ = 54.0± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.68± 0.62 %
μ = 518.4± 3.2
σ = 55.4± 2.9



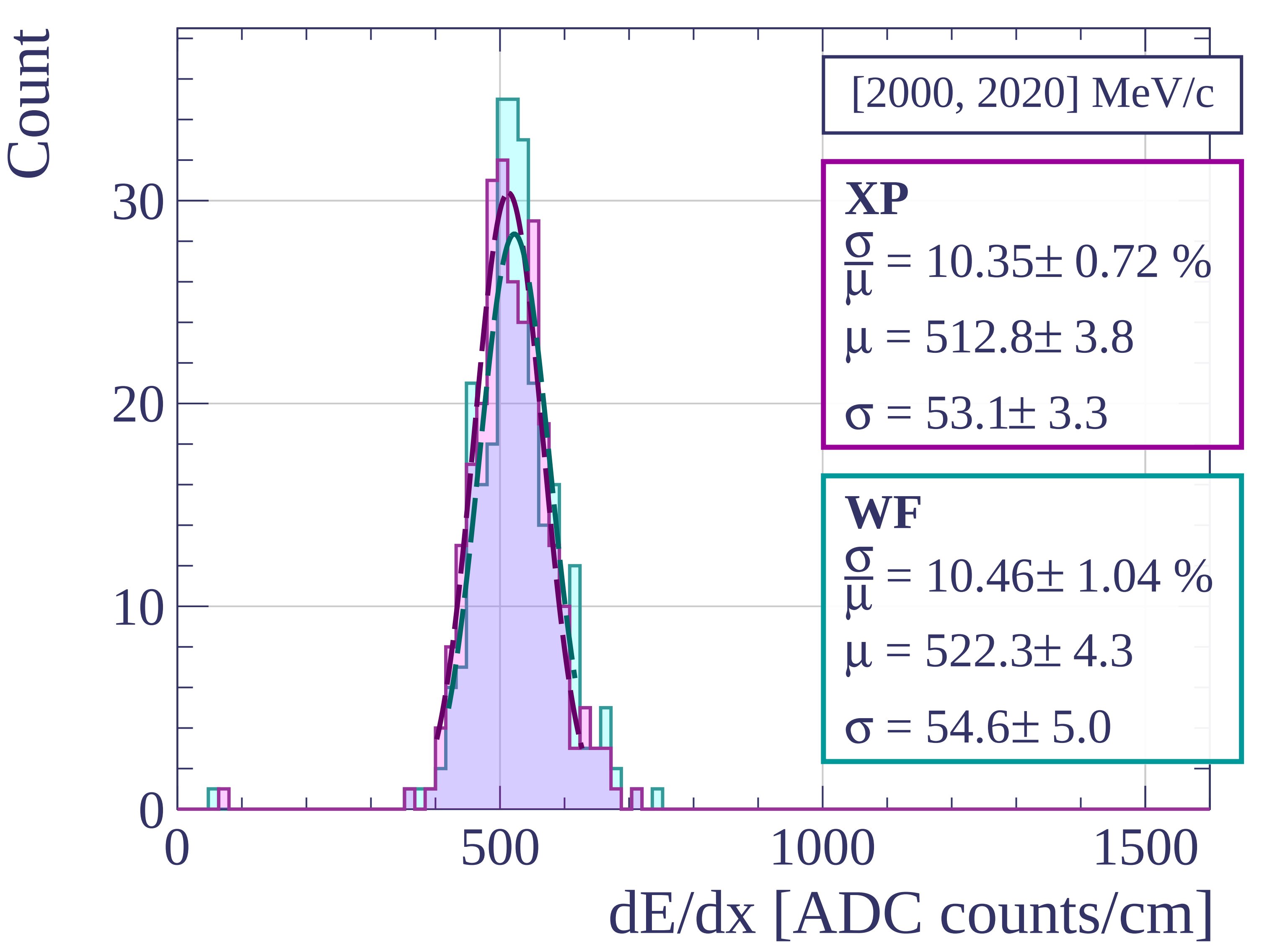
Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[1940, 1960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.78± 0.68 %
μ = 514.5± 3.5
σ = 55.5± 3.1
WF
$\frac{\sigma}{\mu}$ = 11.86± 0.72 %
μ = 527.3± 3.5
σ = 62.6± 3.4



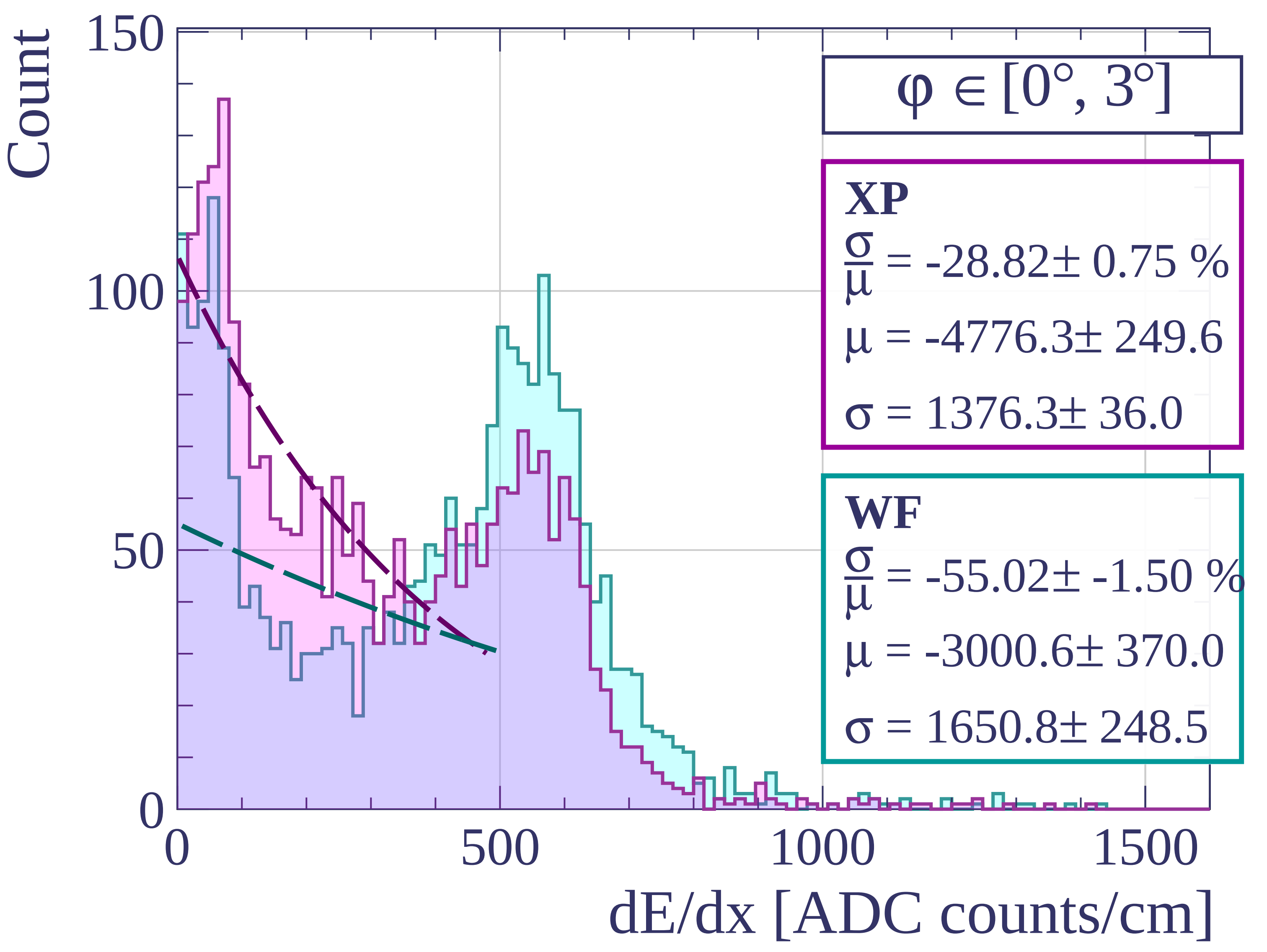
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.23± 0.61 %
μ = 505.4± 3.4
σ = 51.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 10.78± 0.56 %
μ = 517.4± 3.5
σ = 55.8± 2.5

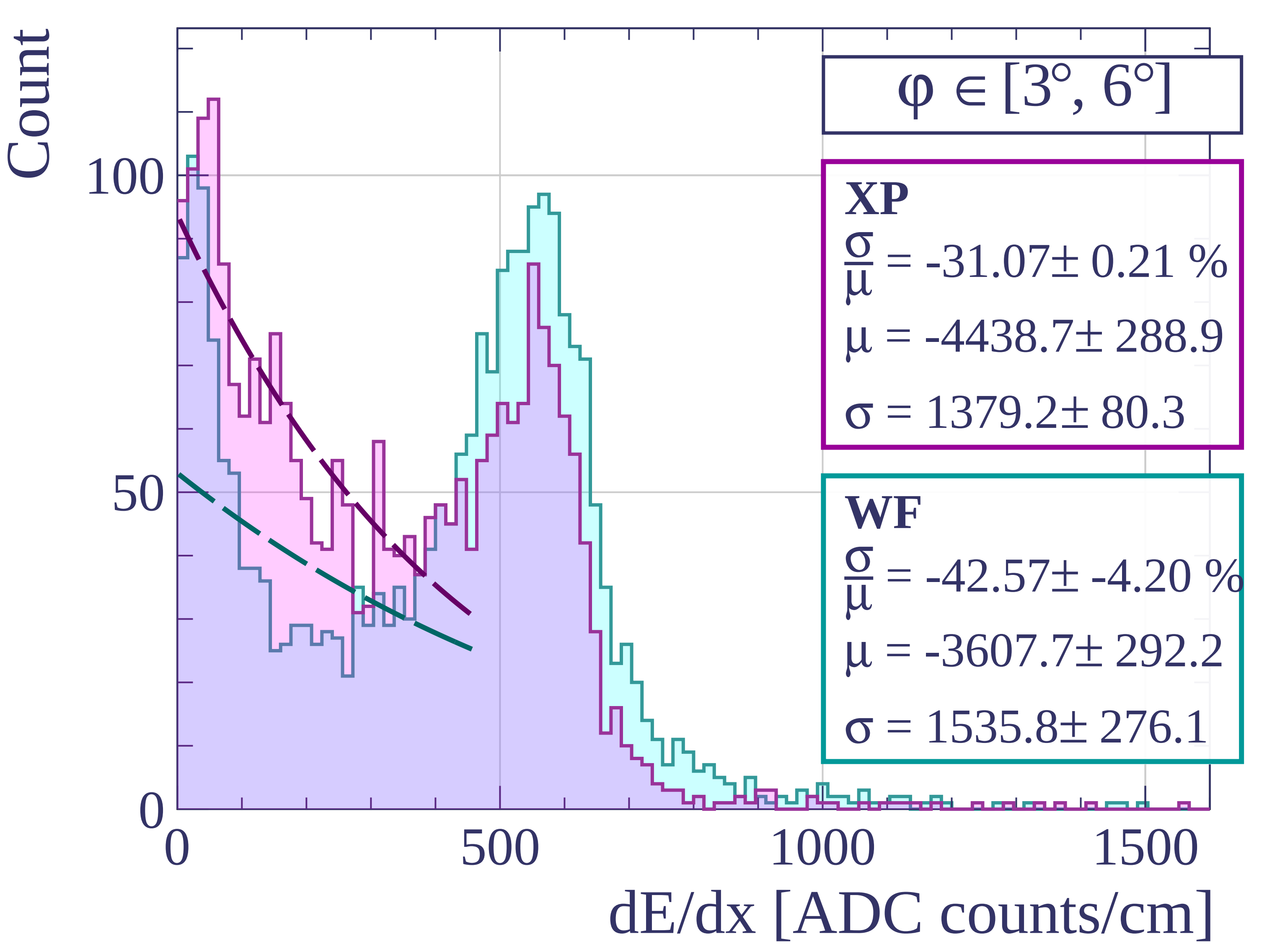


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1980, 2000] MeV/c
XP
σ/μ = 10.93± 0.59 %
μ = 512.6± 3.3
σ = 56.0± 2.7
WF
σ/μ = 11.53± 0.55 %
μ = 523.0± 3.3
σ = 60.3± 2.5

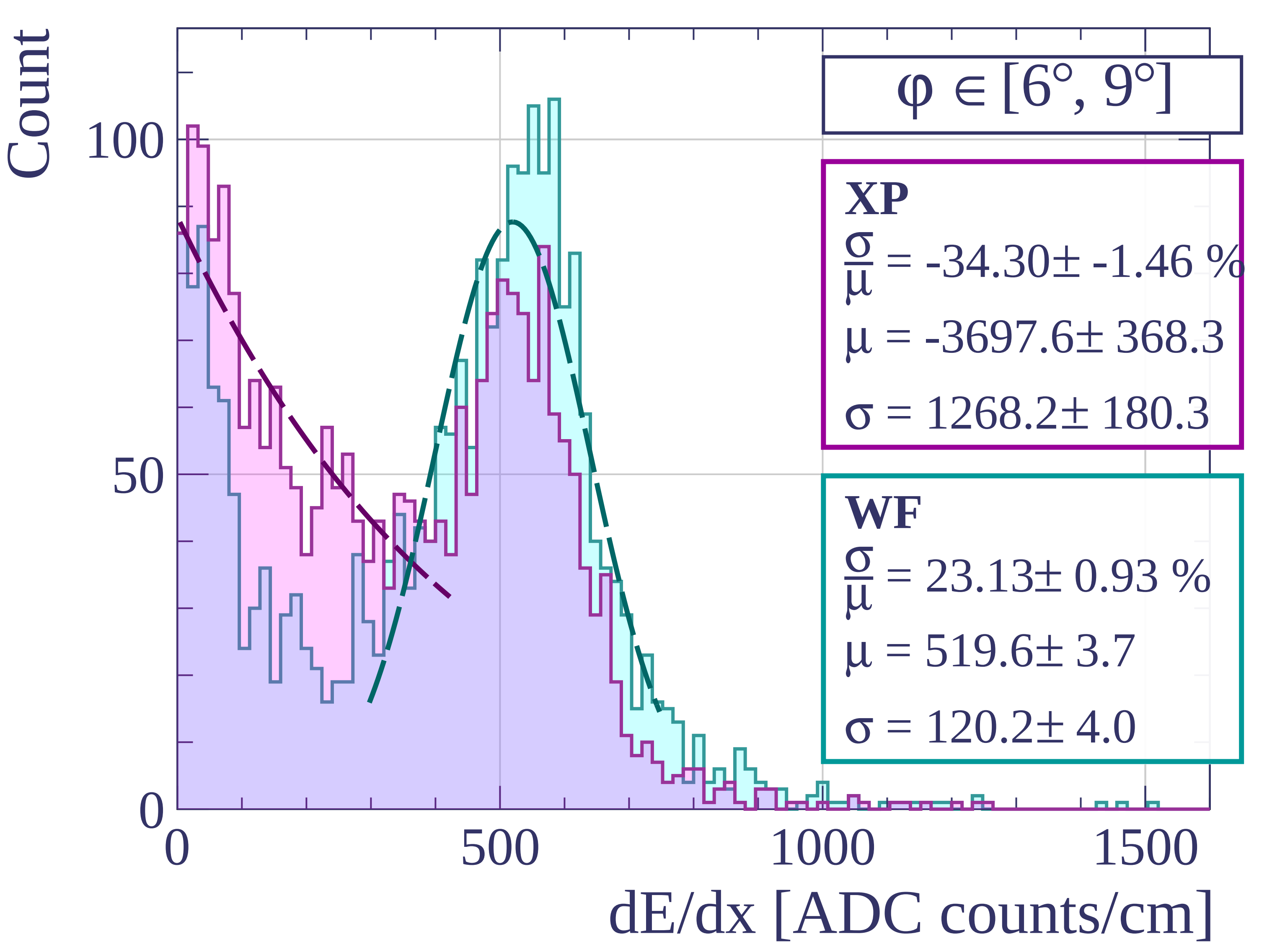


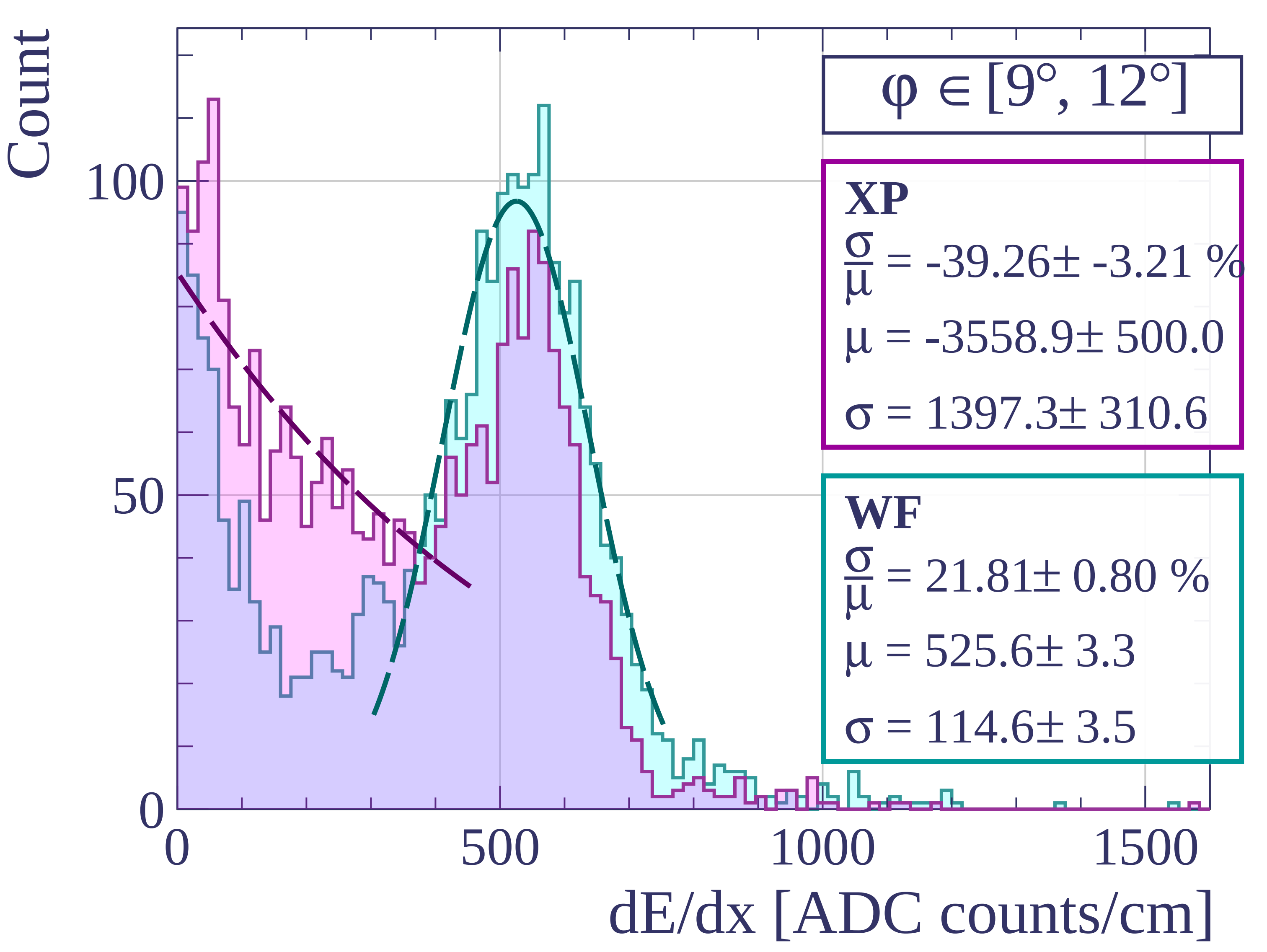
Count
dE/dx [ADC counts/cm]
[2000, 2020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.35± 0.72 %
μ = 512.8± 3.8
σ = 53.1± 3.3
WF
$\frac{\sigma}{\mu}$ = 10.46± 1.04 %
μ = 522.3± 4.3
σ = 54.6± 5.0


Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
φ ∈ [0°, 3°]
XP
σ/μ = -28.82± 0.75 %
μ = -4776.3± 249.6
σ = 1376.3± 36.0
WF
σ/μ = -55.02± -1.50 %
μ = -3000.6± 370.0
σ = 1650.8± 248.5

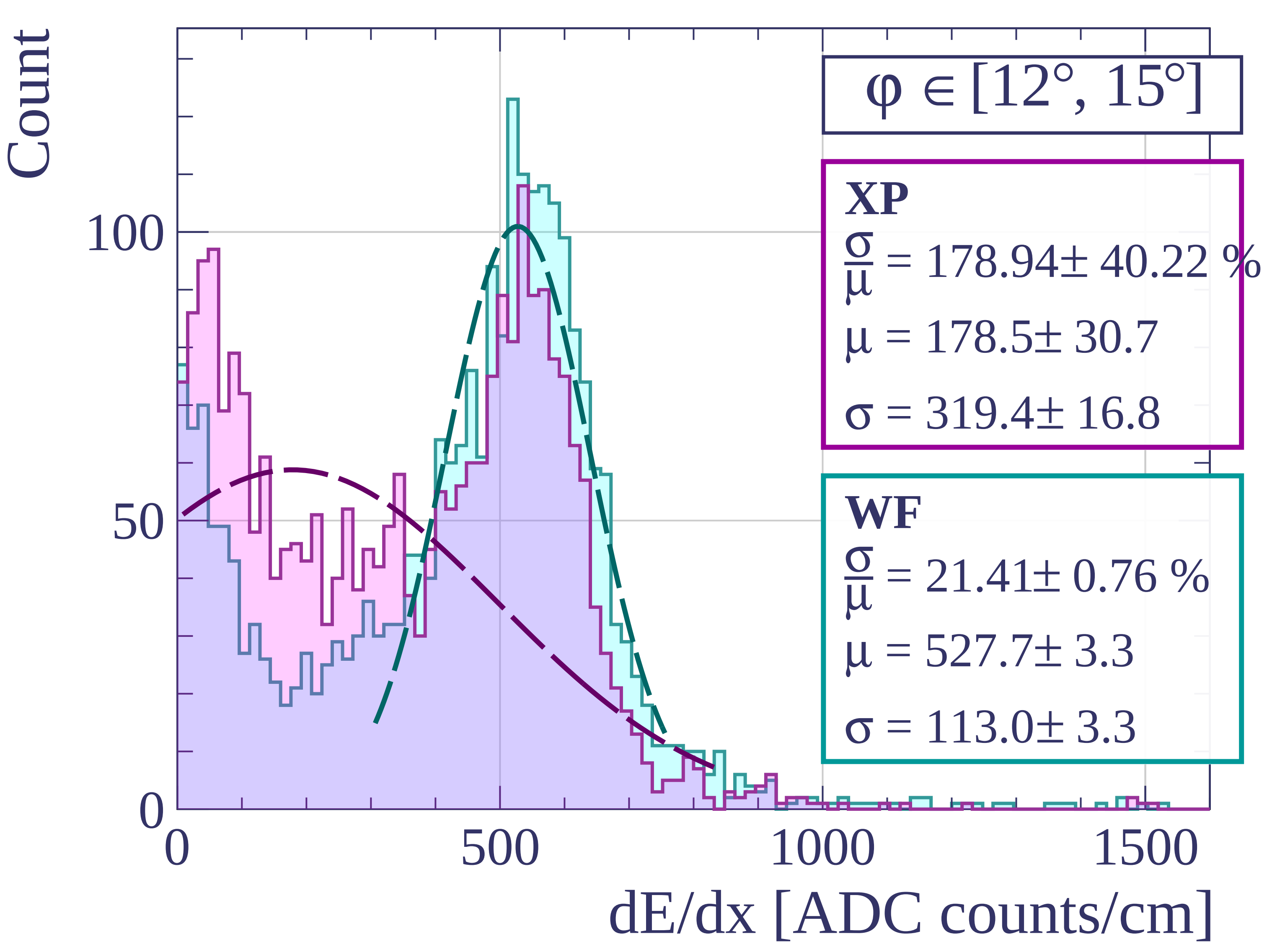


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [3°, 6°]
XP
σ/μ = -31.07± 0.21 %
μ = -4438.7± 288.9
σ = 1379.2± 80.3
WF
σ/μ = -42.57± -4.20 %
μ = -3607.7± 292.2
σ = 1535.8± 276.1

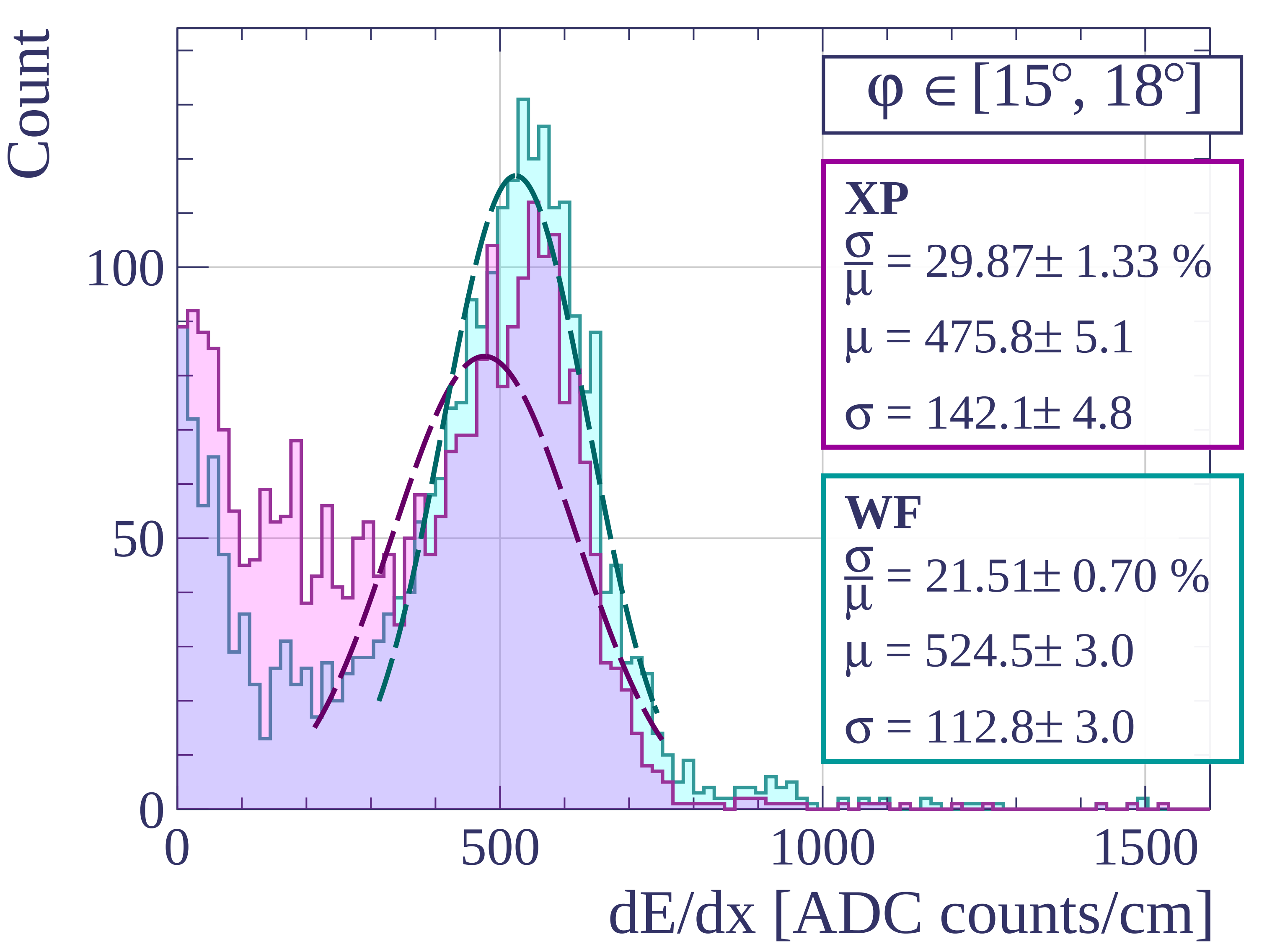

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = -34.30± -1.46 %
μ = -3697.6± 368.3
σ = 1268.2± 180.3
WF
σ/μ = 23.13± 0.93 %
μ = 519.6± 3.7
σ = 120.2± 4.0



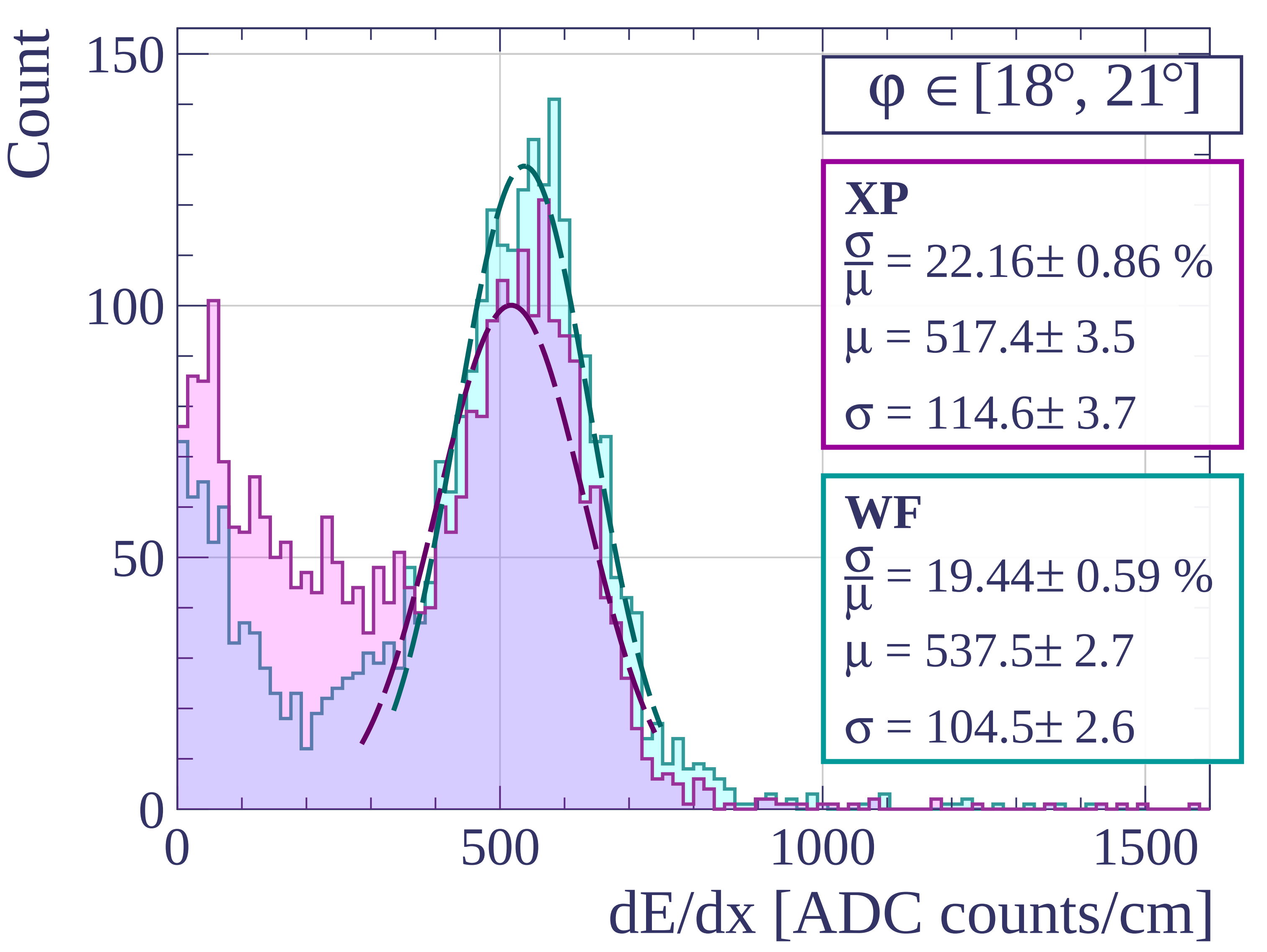
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = -39.26± -3.21 %
μ = -3558.9± 500.0
σ = 1397.3± 310.6
WF
σ/μ = 21.81± 0.80 %
μ = 525.6± 3.3
σ = 114.6± 3.5



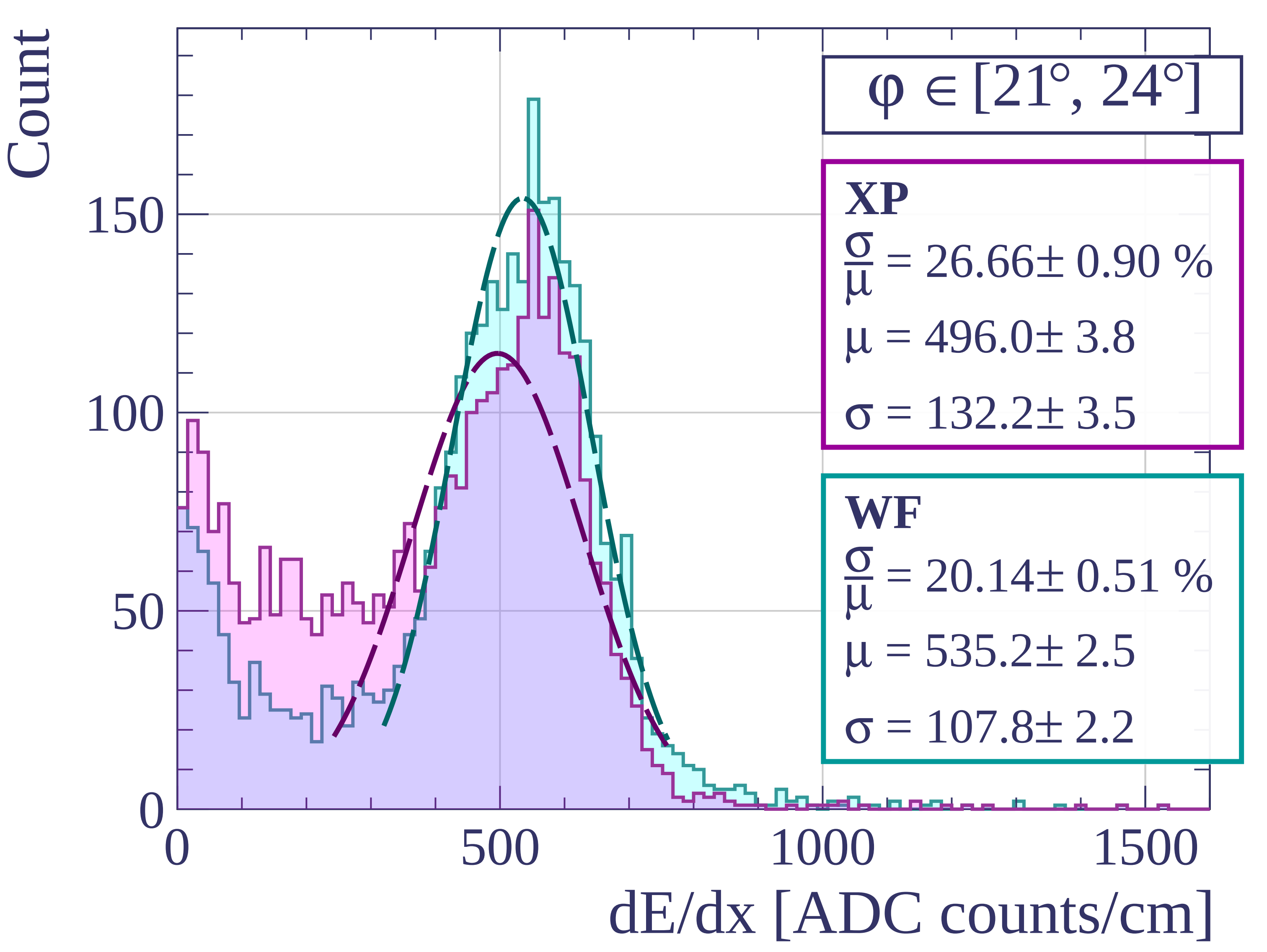
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 178.94± 40.22 %
μ = 178.5± 30.7
σ = 319.4± 16.8
WF
σ/μ = 21.41± 0.76 %
μ = 527.7± 3.3
σ = 113.0± 3.3



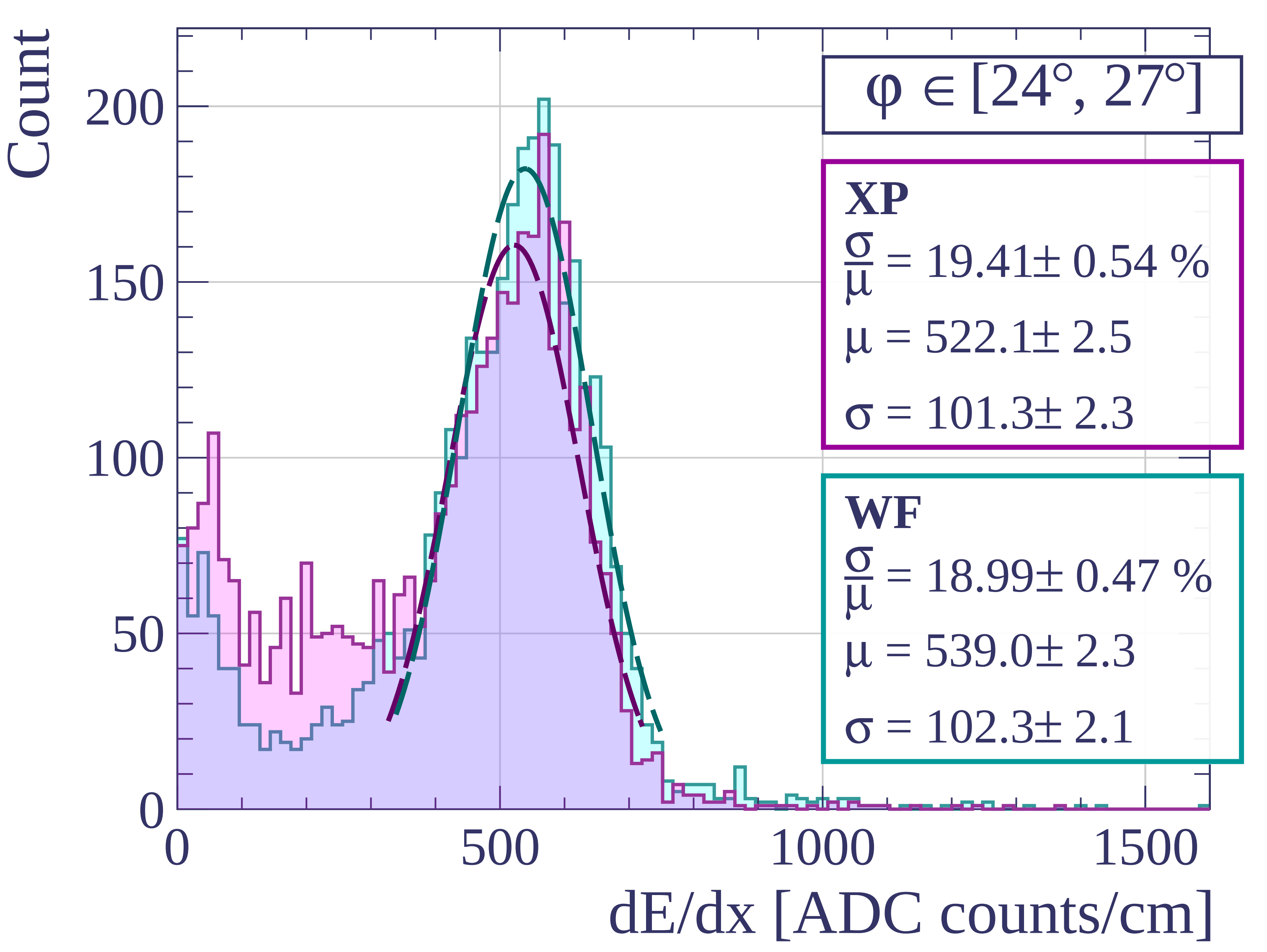
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 29.87± 1.33 %
μ = 475.8± 5.1
σ = 142.1± 4.8
WF
σ/μ = 21.51± 0.70 %
μ = 524.5± 3.0
σ = 112.8± 3.0



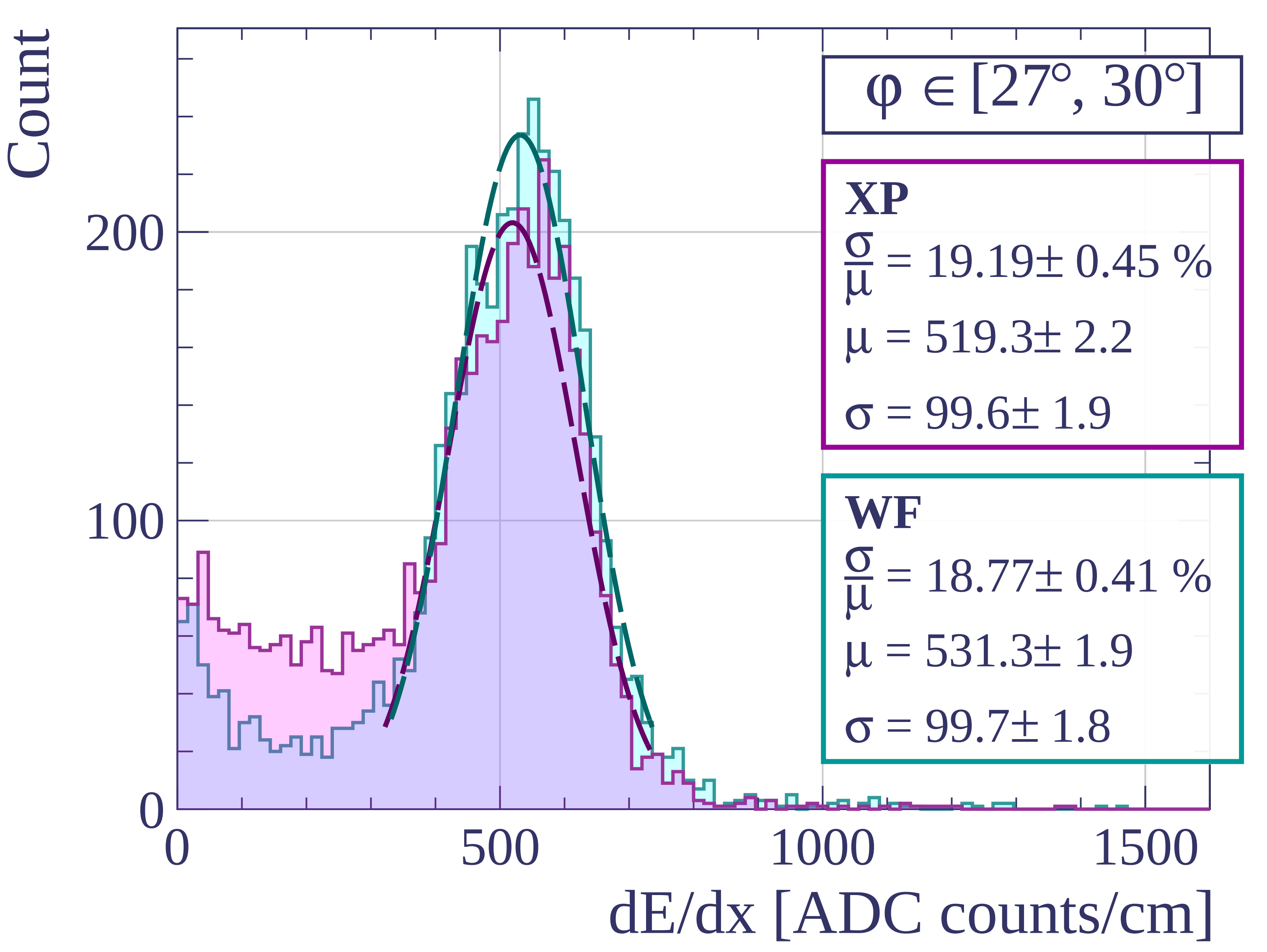
Count
dE/dx [ADC counts/cm]
φ ∈ [18°, 21°]
XP
σ/μ = 22.16± 0.86 %
μ = 517.4± 3.5
σ = 114.6± 3.7
WF
σ/μ = 19.44± 0.59 %
μ = 537.5± 2.7
σ = 104.5± 2.6
0
50
100
150
0
500
1000
1500



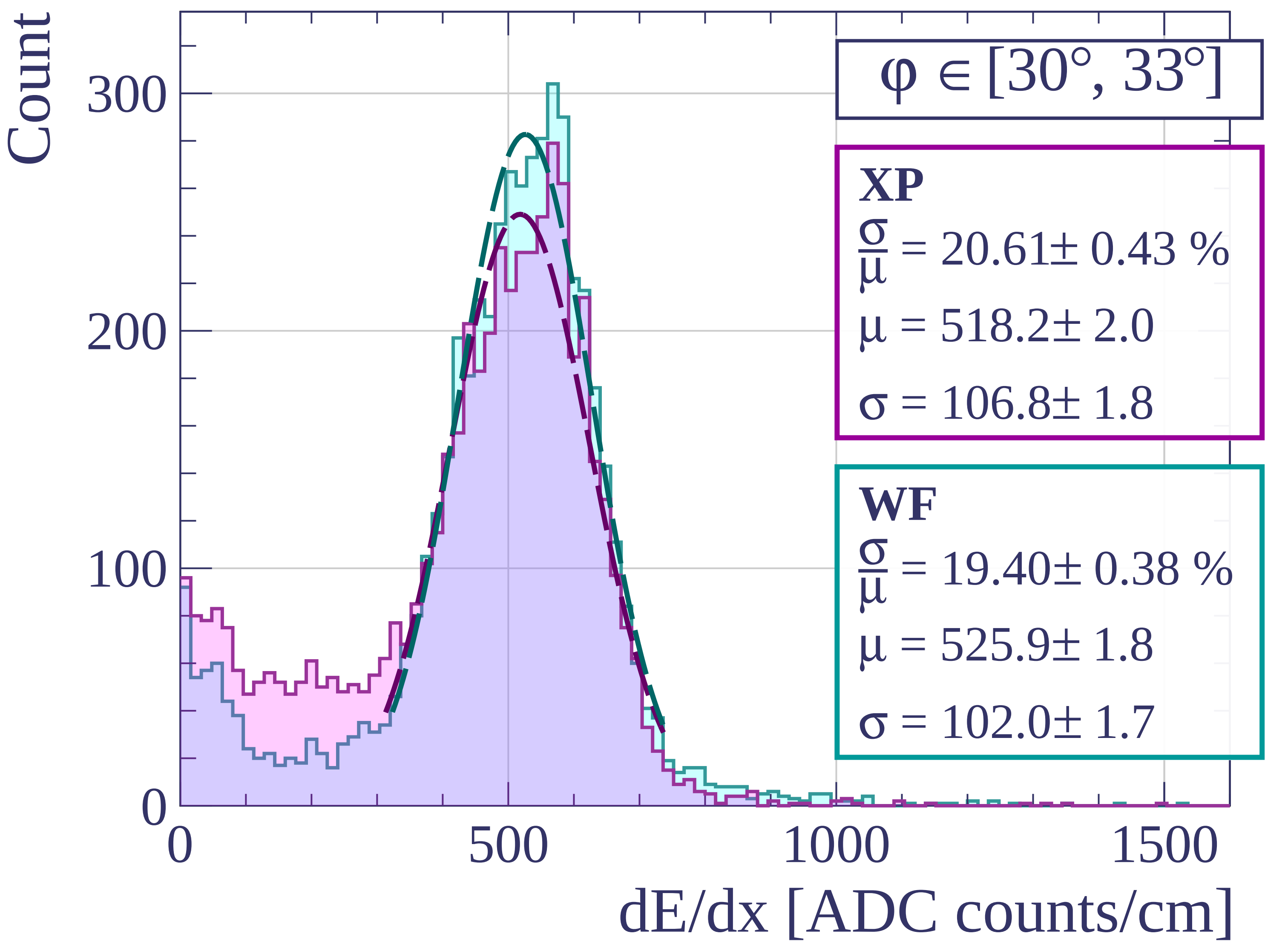
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
φ ∈ [21°, 24°]
XP
σ/μ = 26.66± 0.90 %
μ = 496.0± 3.8
σ = 132.2± 3.5
WF
σ/μ = 20.14± 0.51 %
μ = 535.2± 2.5
σ = 107.8± 2.2



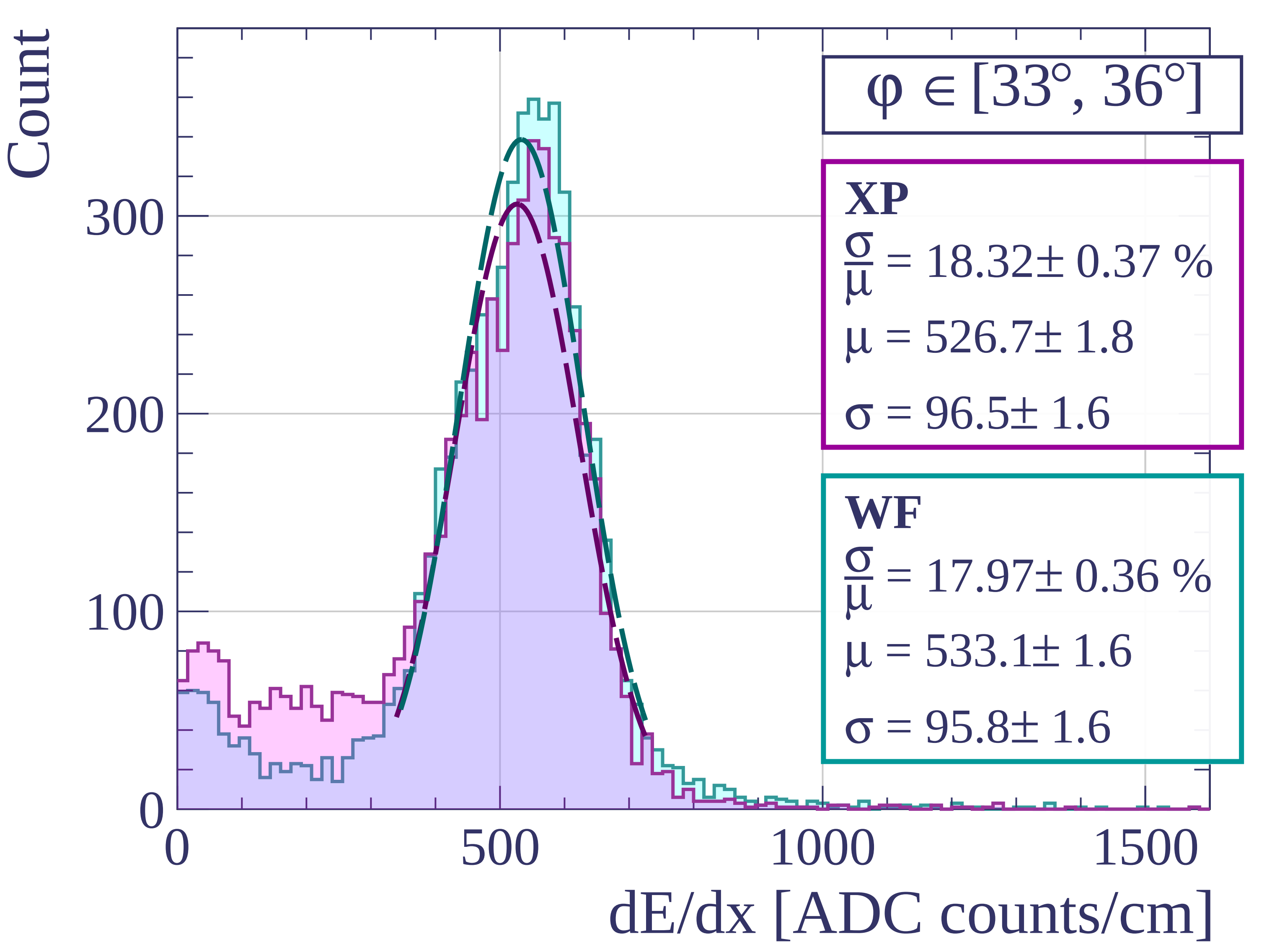
Count
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 19.41± 0.54 %
μ = 522.1± 2.5
σ = 101.3± 2.3
WF
σ/μ = 18.99± 0.47 %
μ = 539.0± 2.3
σ = 102.3± 2.1
0
50
100
150
200
0
500
1000
1500

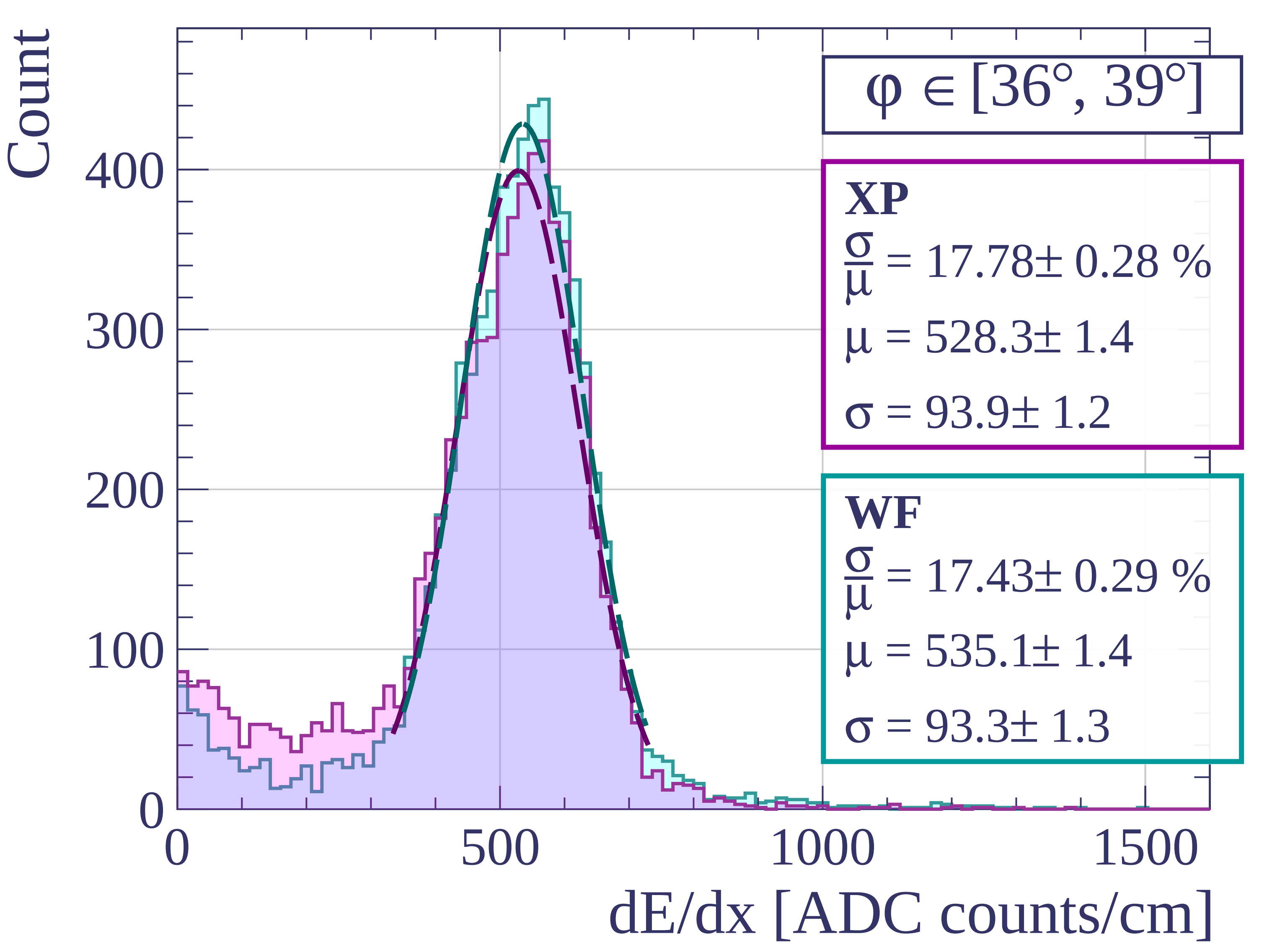


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [27°, 30°]
XP
σ/μ = 19.19± 0.45 %
μ = 519.3± 2.2
σ = 99.6± 1.9
WF
σ/μ = 18.77± 0.41 %
μ = 531.3± 1.9
σ = 99.7± 1.8

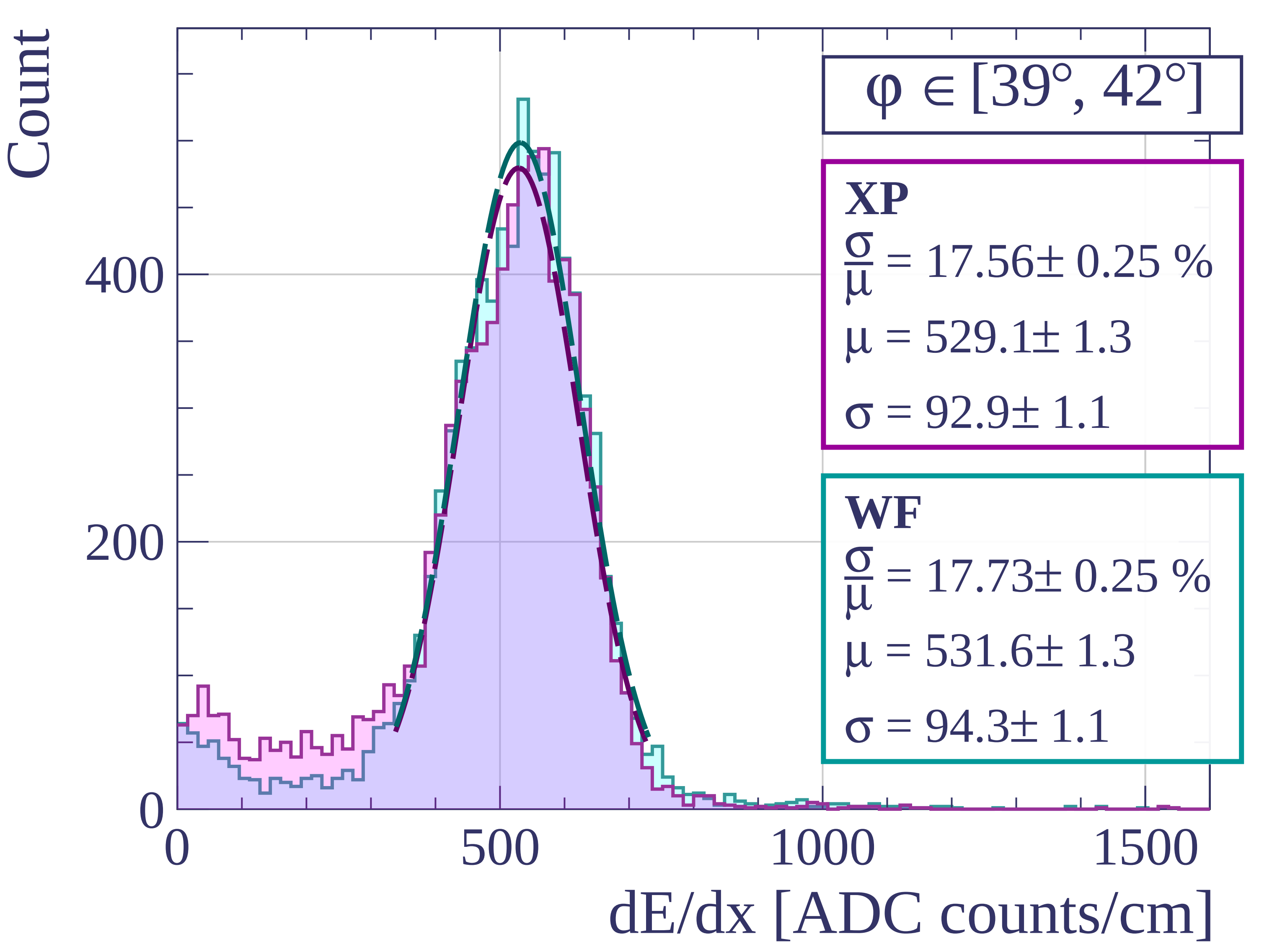


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [30°, 33°]
XP
$\frac{\sigma}{\mu}$ = 20.61± 0.43 %
μ = 518.2± 2.0
σ = 106.8± 1.8
WF
$\frac{\sigma}{\mu}$ = 19.40± 0.38 %
μ = 525.9± 1.8
σ = 102.0± 1.7

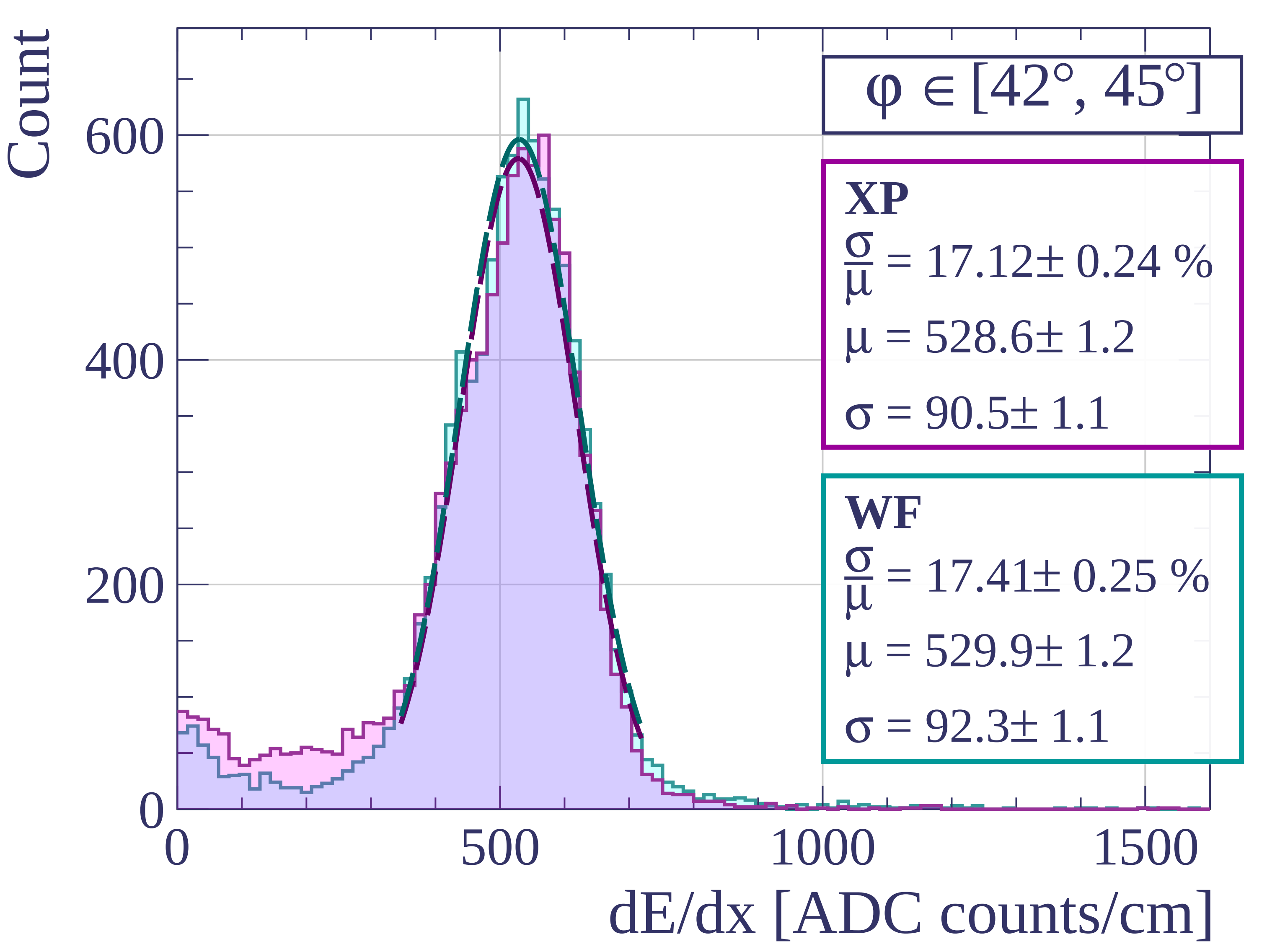

Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
φ ∈ [33°, 36°]
XP
σ/μ = 18.32± 0.37 %
μ = 526.7± 1.8
σ = 96.5± 1.6
WF
σ/μ = 17.97± 0.36 %
μ = 533.1± 1.6
σ = 95.8± 1.6

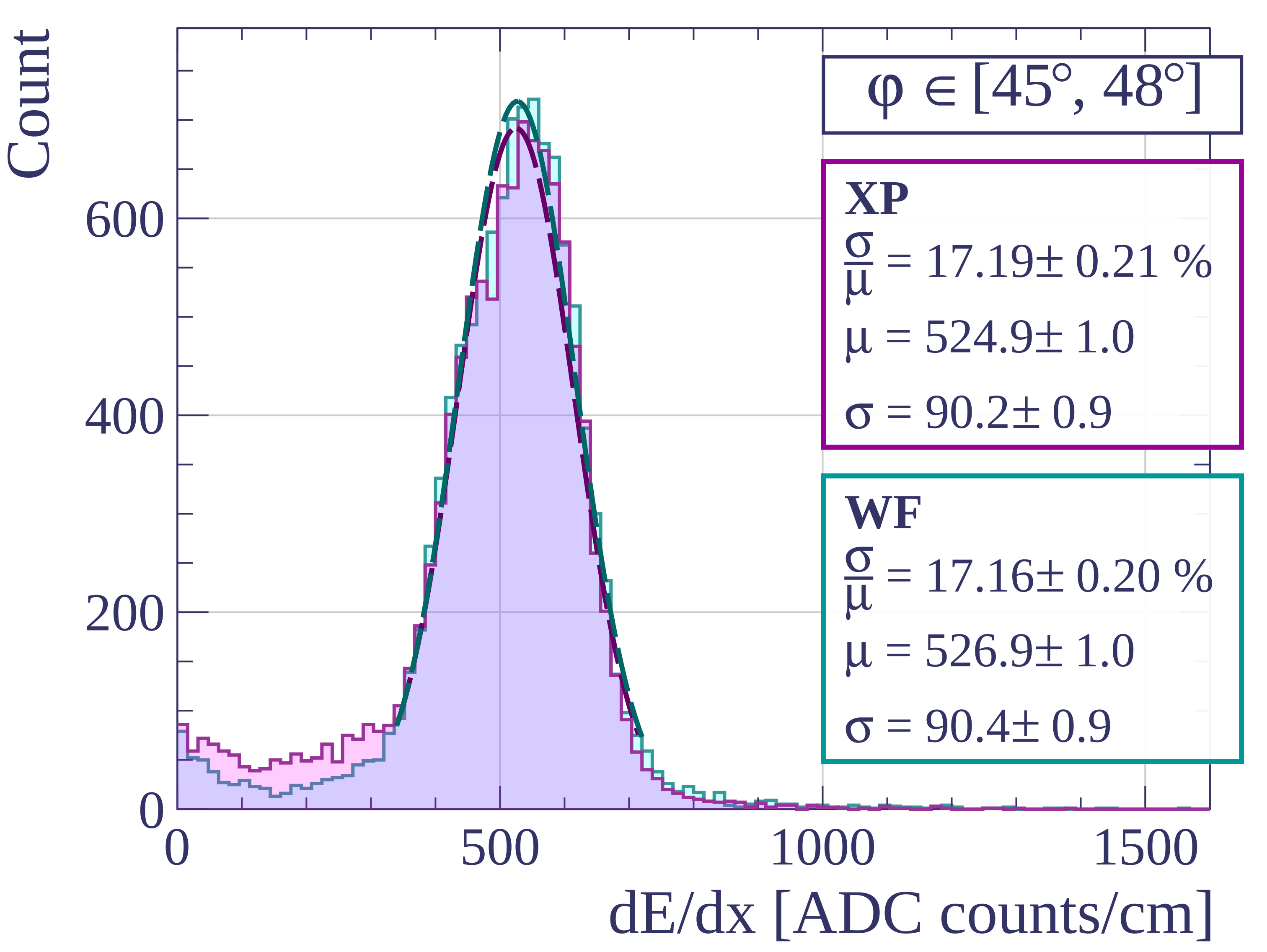


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 17.78± 0.28 %
μ = 528.3± 1.4
σ = 93.9± 1.2
WF
σ/μ = 17.43± 0.29 %
μ = 535.1± 1.4
σ = 93.3± 1.3

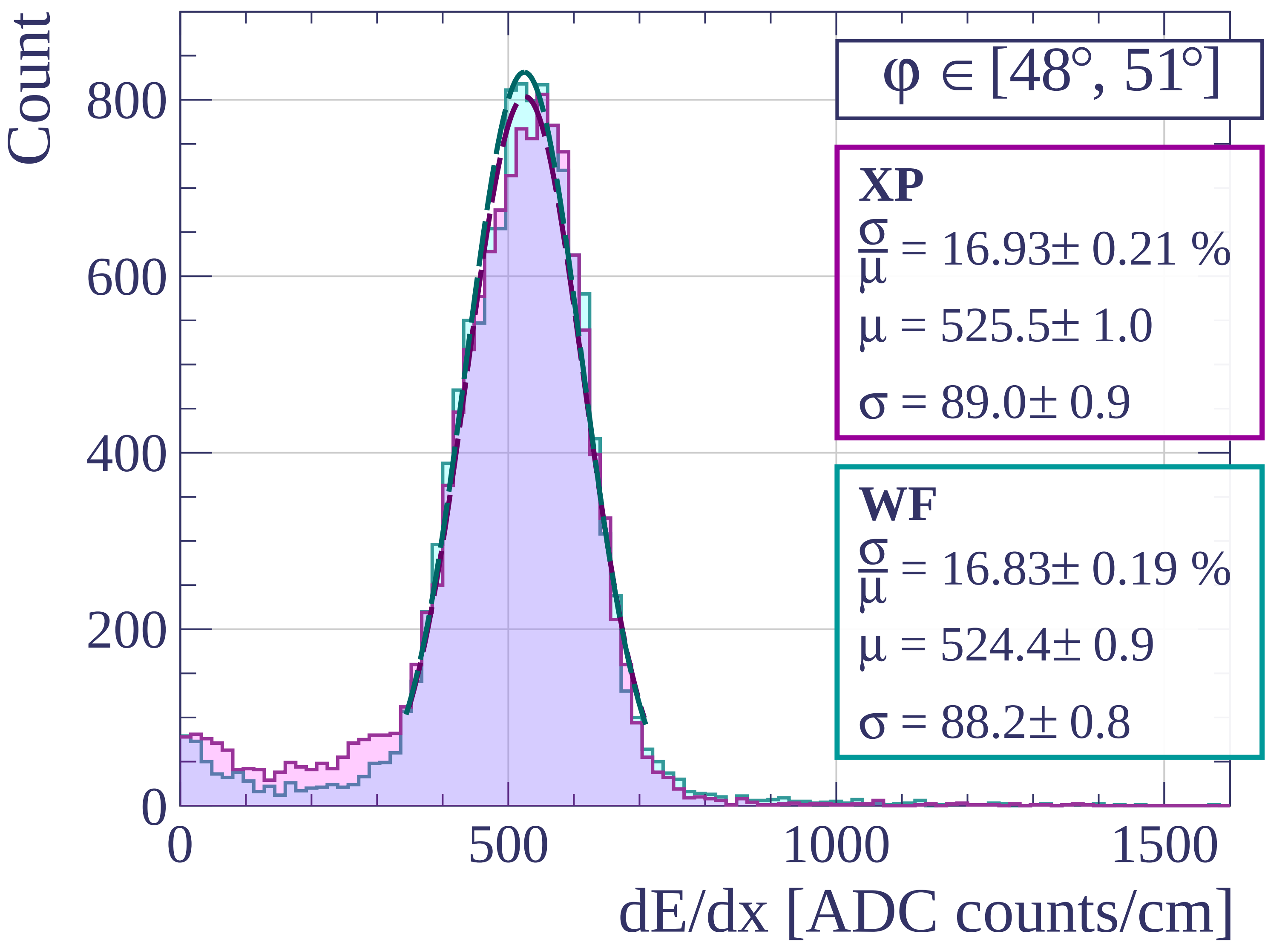


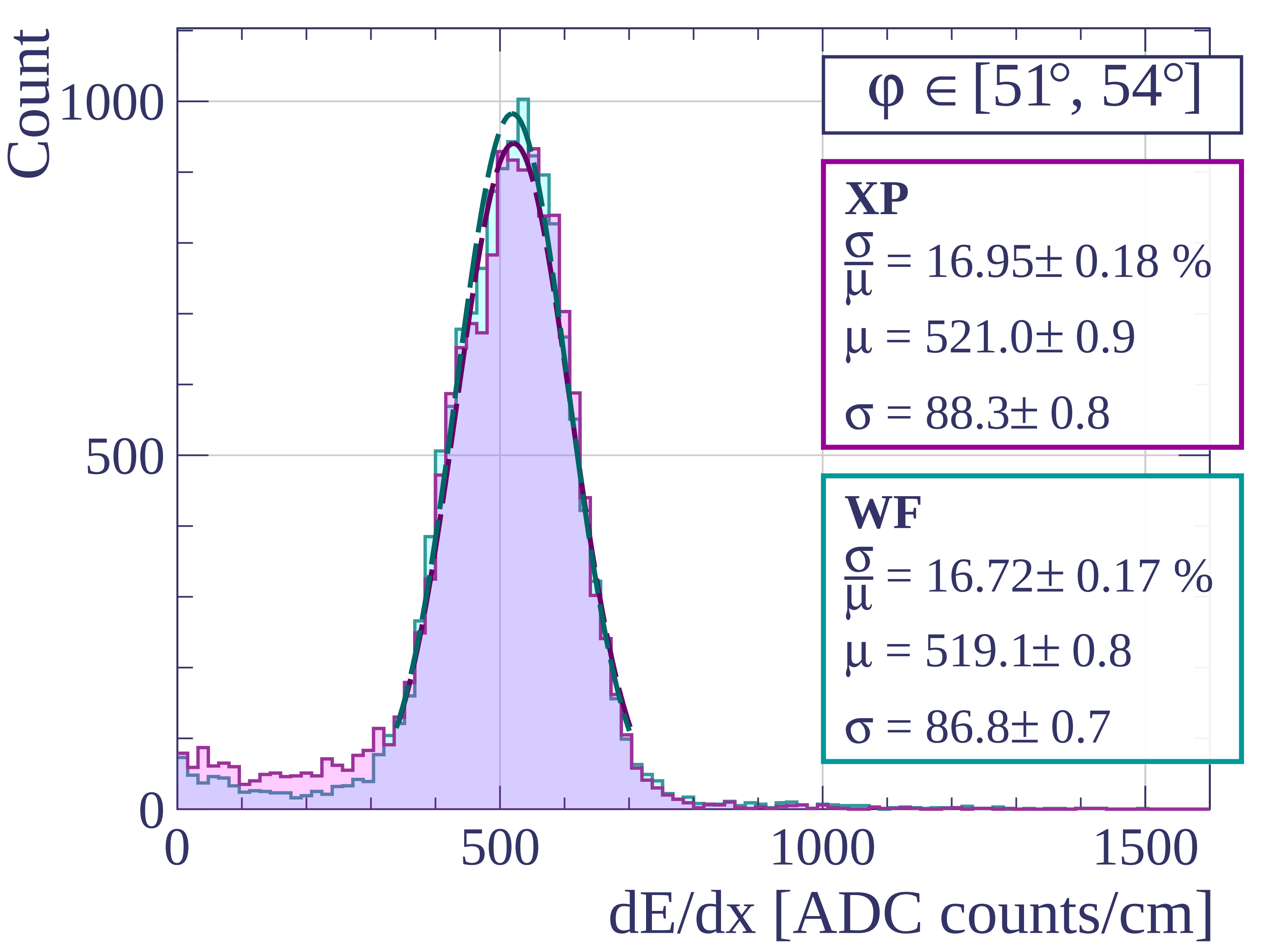
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
σ/μ = 17.56± 0.25 %
μ = 529.1± 1.3
σ = 92.9± 1.1
WF
σ/μ = 17.73± 0.25 %
μ = 531.6± 1.3
σ = 94.3± 1.1


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
$\frac{\sigma}{\mu}$ = 17.12± 0.24 %
μ = 528.6± 1.2
σ = 90.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 17.41± 0.25 %
μ = 529.9± 1.2
σ = 92.3± 1.1

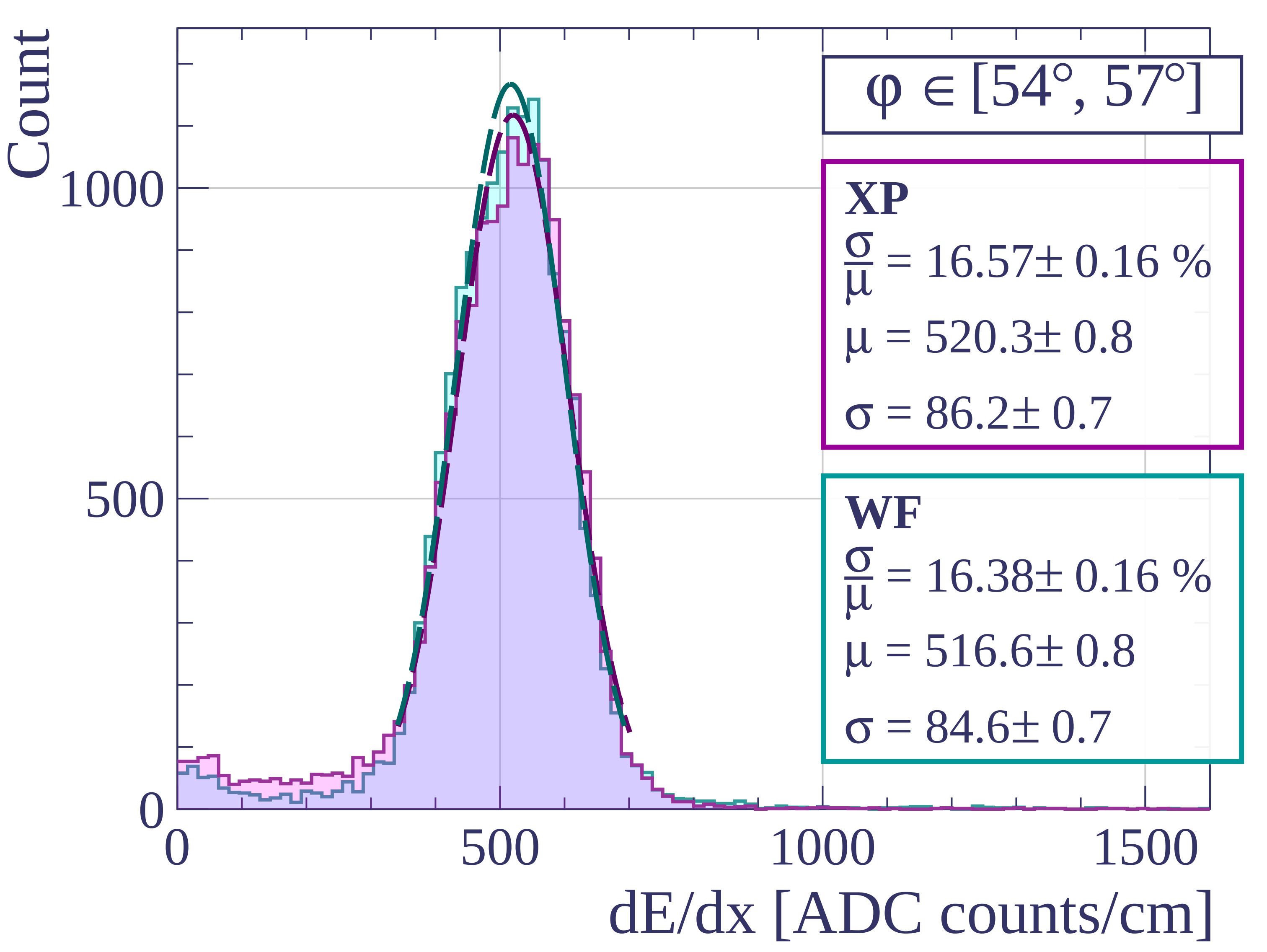


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [45°, 48°]
XP
σ/μ = 17.19± 0.21 %
μ = 524.9± 1.0
σ = 90.2± 0.9
WF
σ/μ = 17.16± 0.20 %
μ = 526.9± 1.0
σ = 90.4± 0.9

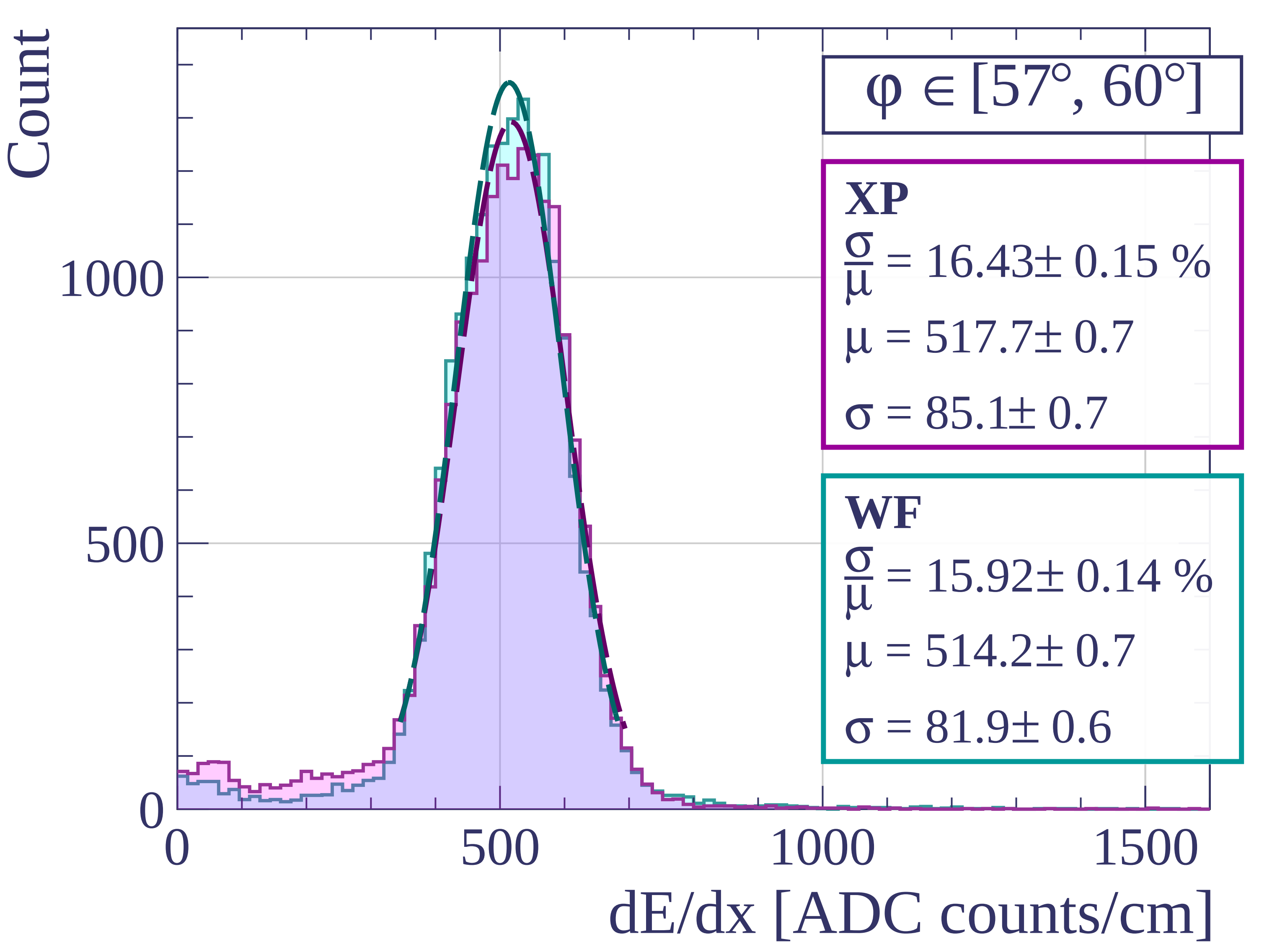

Count
dE/dx [ADC counts/cm]
φ ∈ [48°, 51°]
XP
σ/μ = 16.93± 0.21 %
μ = 525.5± 1.0
σ = 89.0± 0.9
WF
σ/μ = 16.83± 0.19 %
μ = 524.4± 0.9
σ = 88.2± 0.8
0
200
400
600
800
0
500
1000
1500



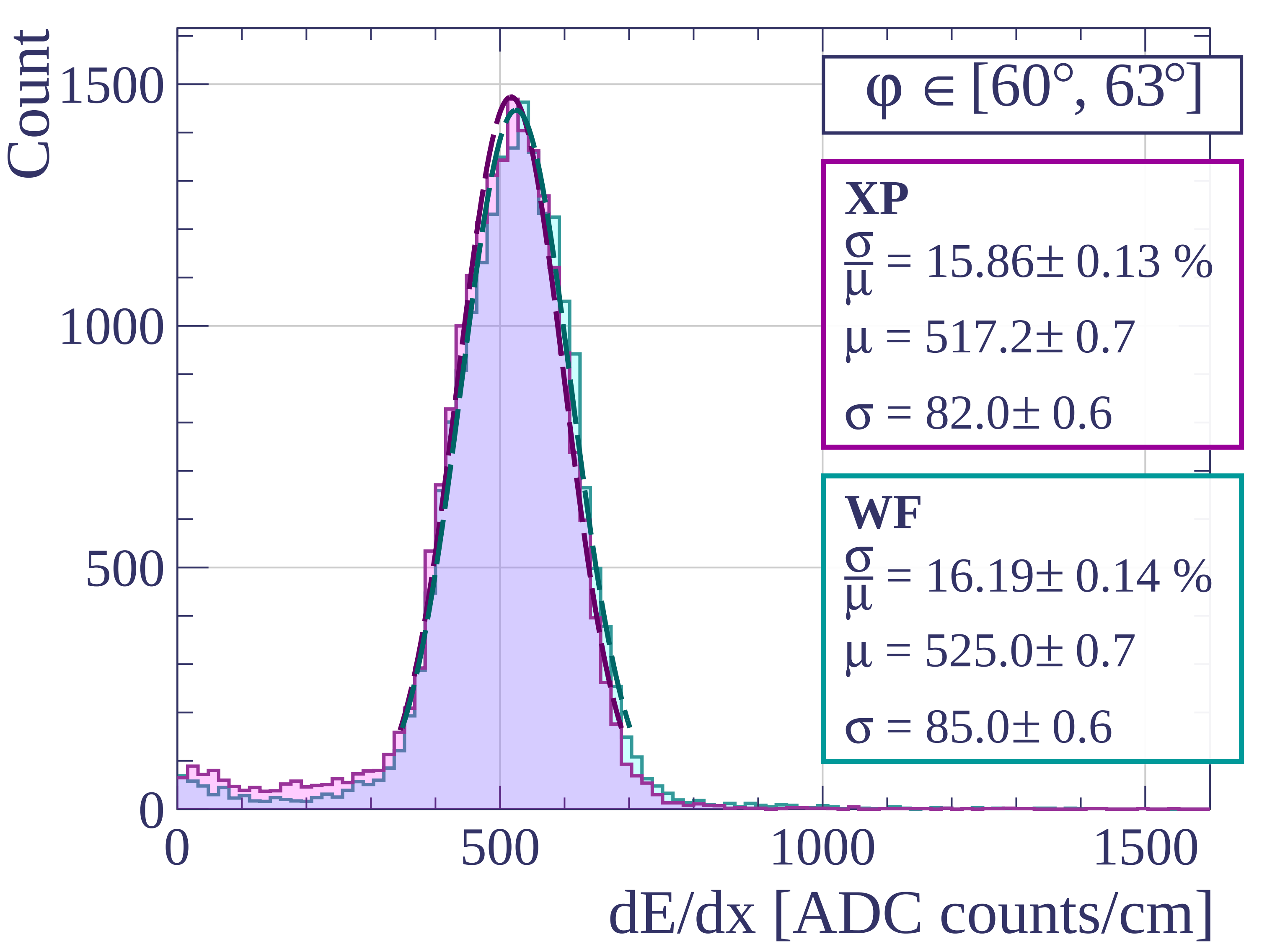
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [51°, 54°]
XP
σ/μ = 16.95± 0.18 %
μ = 521.0± 0.9
σ = 88.3± 0.8
WF
σ/μ = 16.72± 0.17 %
μ = 519.1± 0.8
σ = 86.8± 0.7



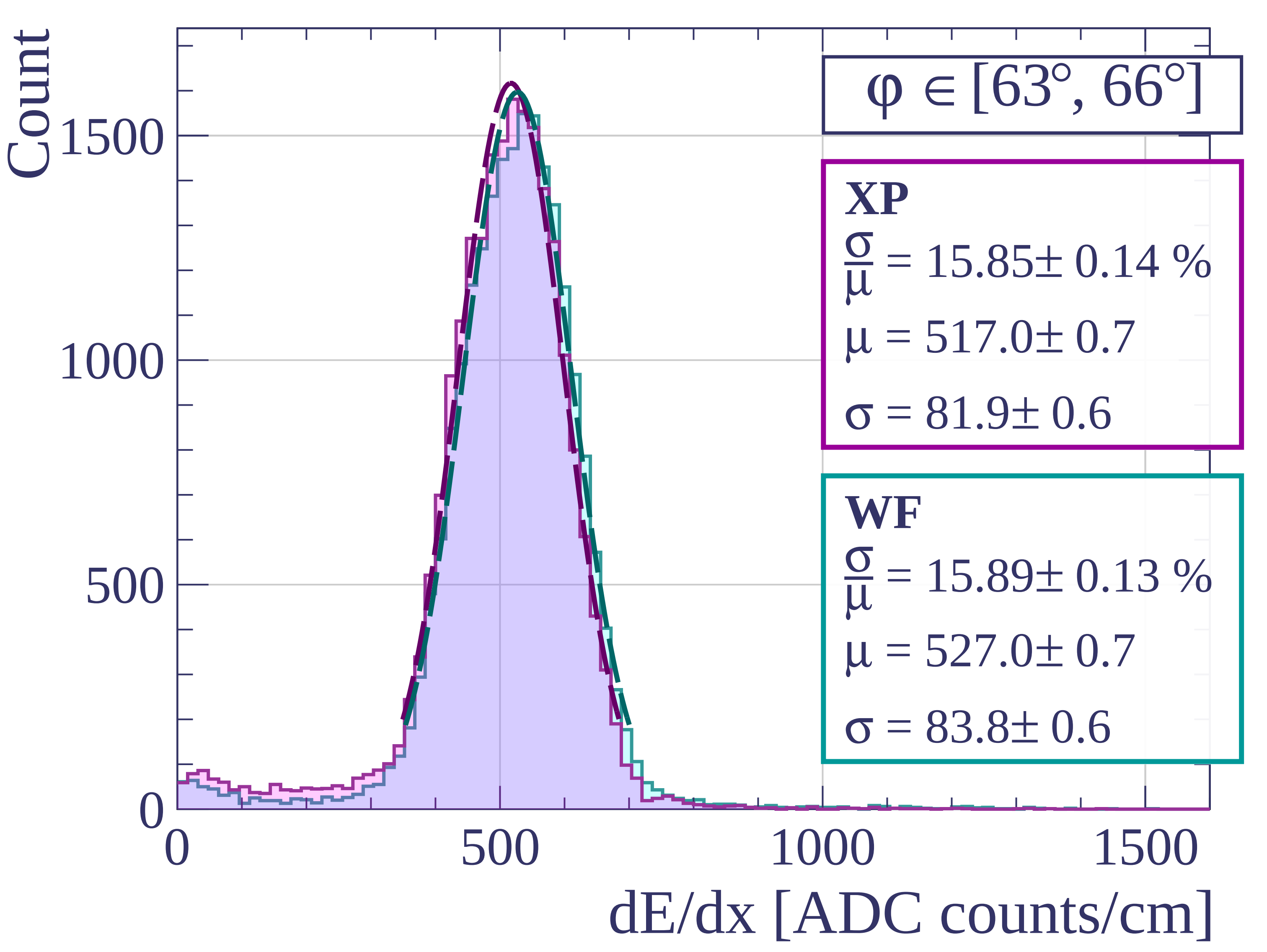
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [54°, 57°]
XP
σ/μ = 16.57± 0.16 %
μ = 520.3± 0.8
σ = 86.2± 0.7
WF
σ/μ = 16.38± 0.16 %
μ = 516.6± 0.8
σ = 84.6± 0.7



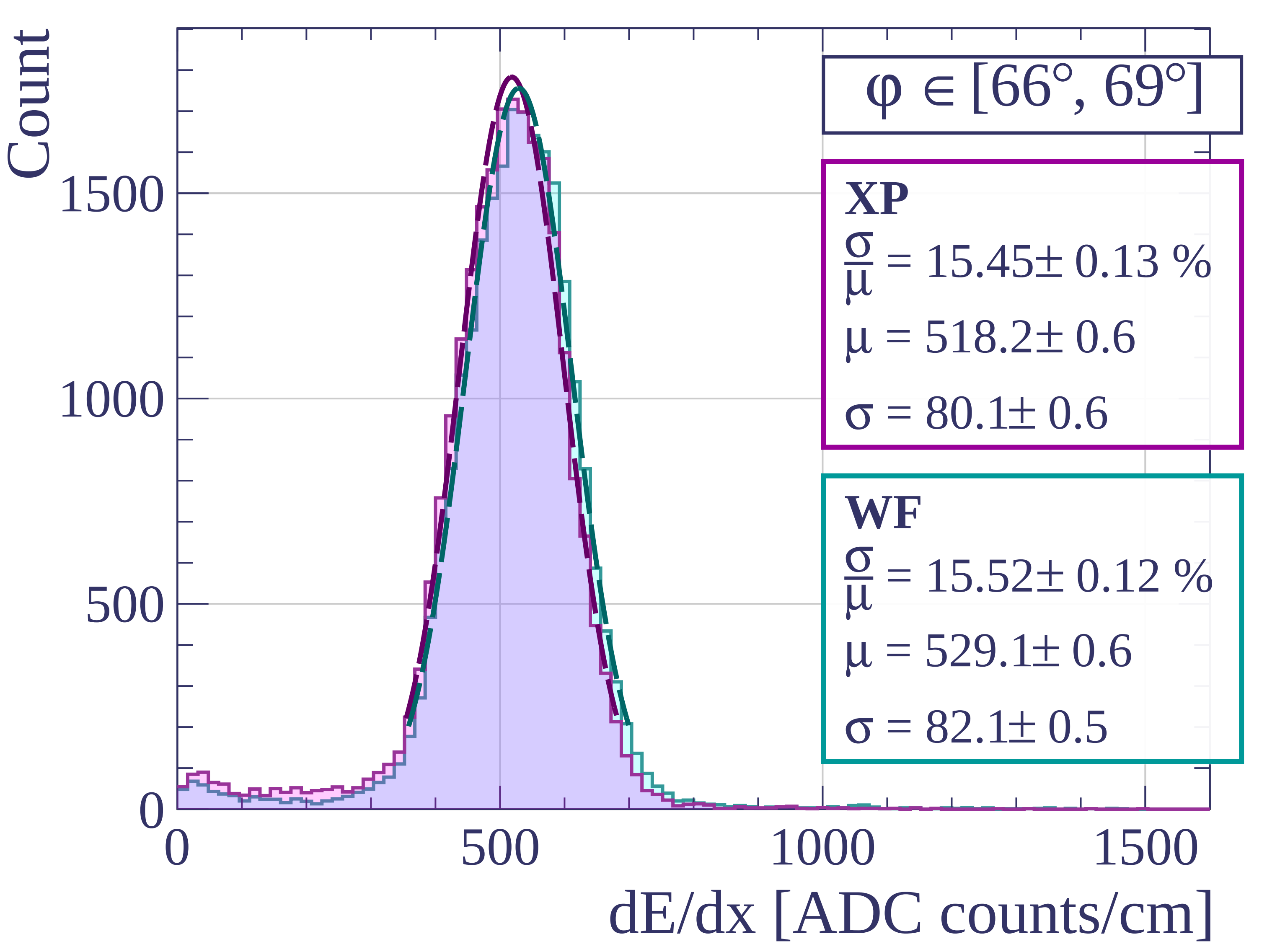
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [57°, 60°]
XP
σ/μ = 16.43± 0.15 %
μ = 517.7± 0.7
σ = 85.1± 0.7
WF
σ/μ = 15.92± 0.14 %
μ = 514.2± 0.7
σ = 81.9± 0.6



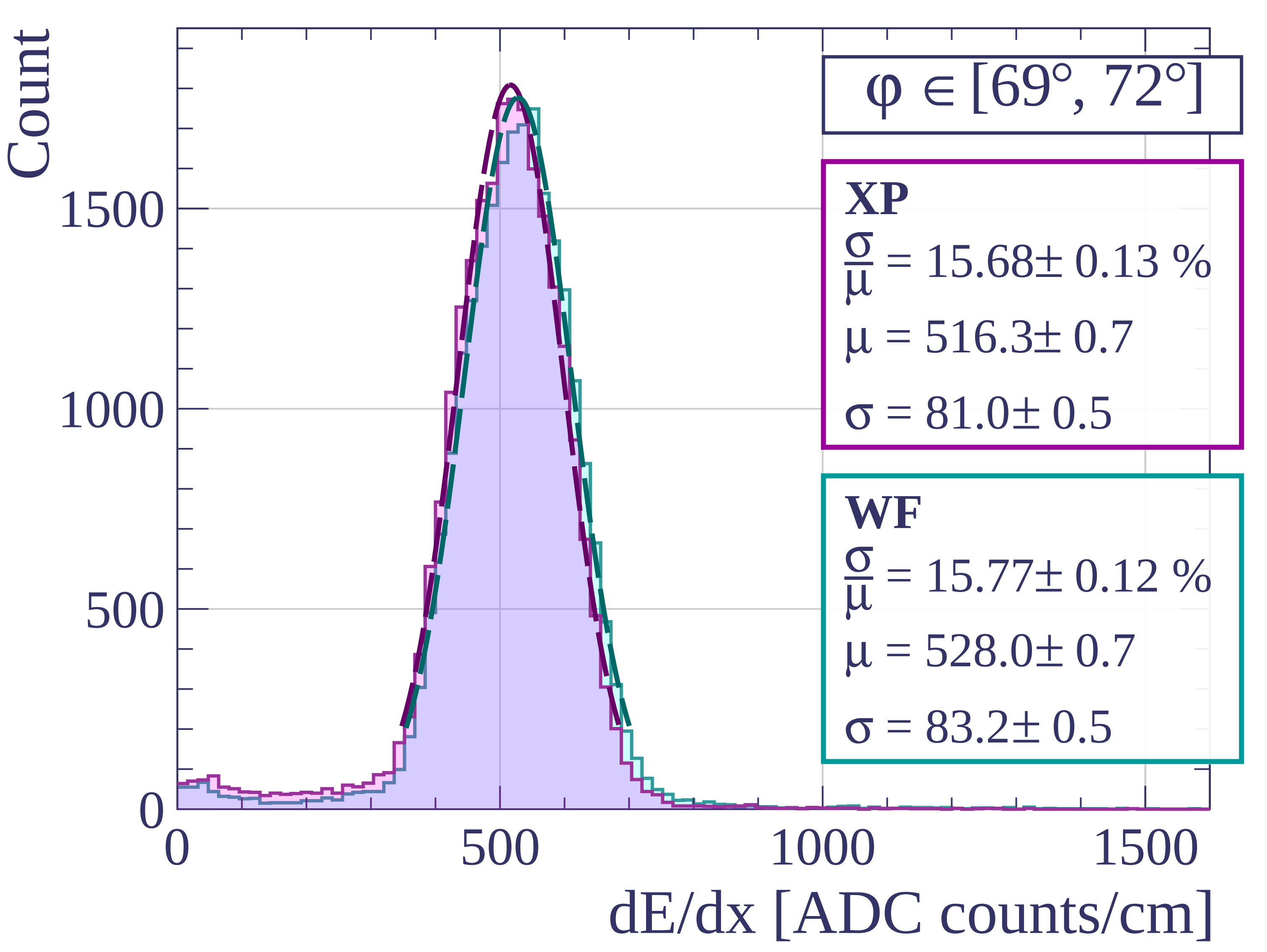
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
σ/μ = 15.86± 0.13 %
μ = 517.2± 0.7
σ = 82.0± 0.6
WF
σ/μ = 16.19± 0.14 %
μ = 525.0± 0.7
σ = 85.0± 0.6



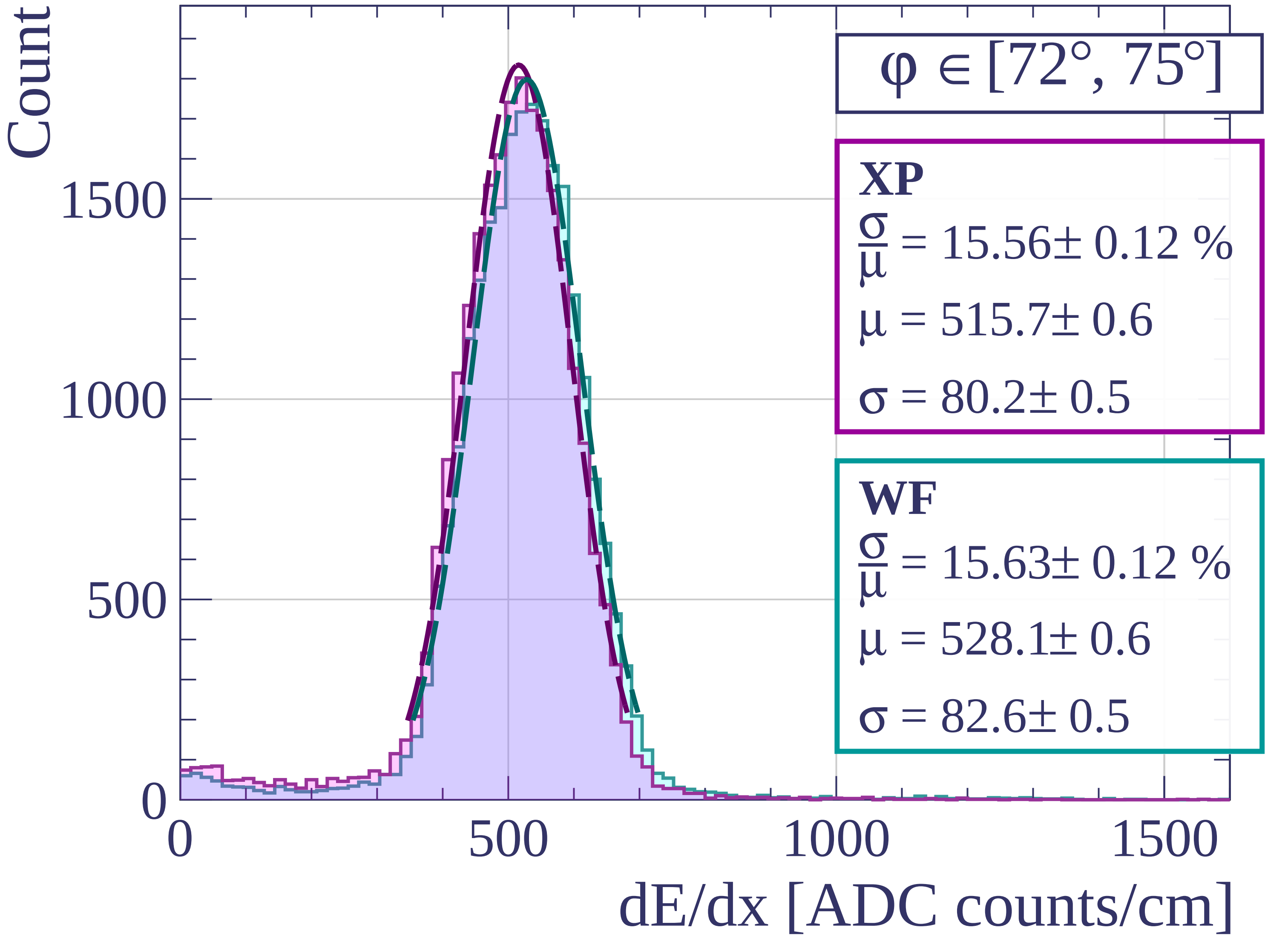
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [63°, 66°]
XP
σ/μ = 15.85± 0.14 %
μ = 517.0± 0.7
σ = 81.9± 0.6
WF
σ/μ = 15.89± 0.13 %
μ = 527.0± 0.7
σ = 83.8± 0.6

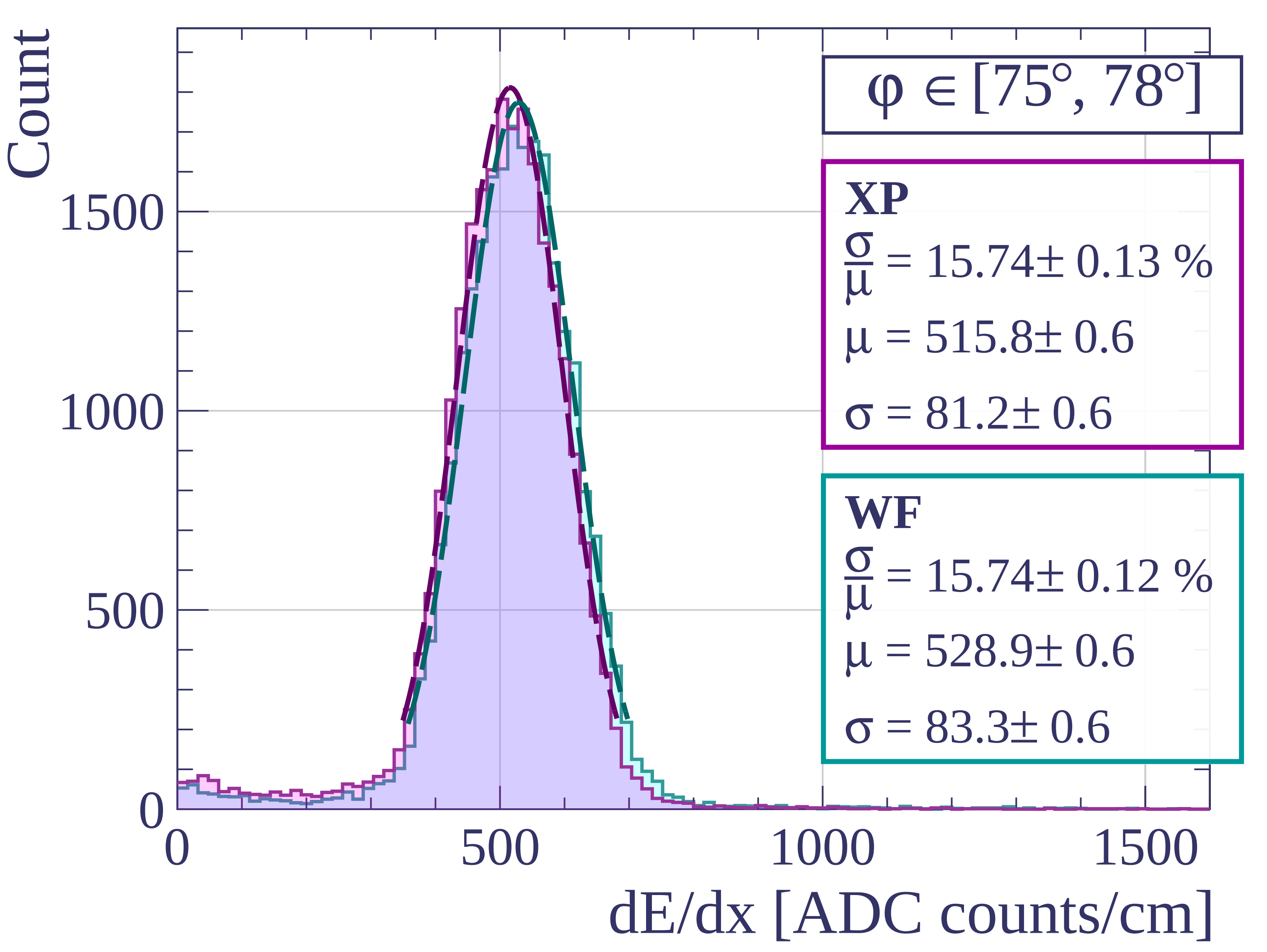


Count
dE/dx [ADC counts/cm]
φ ∈ [66°, 69°]
XP
σ/μ = 15.45± 0.13 %
μ = 518.2± 0.6
σ = 80.1± 0.6
WF
σ/μ = 15.52± 0.12 %
μ = 529.1± 0.6
σ = 82.1± 0.5
0
500
1000
1500

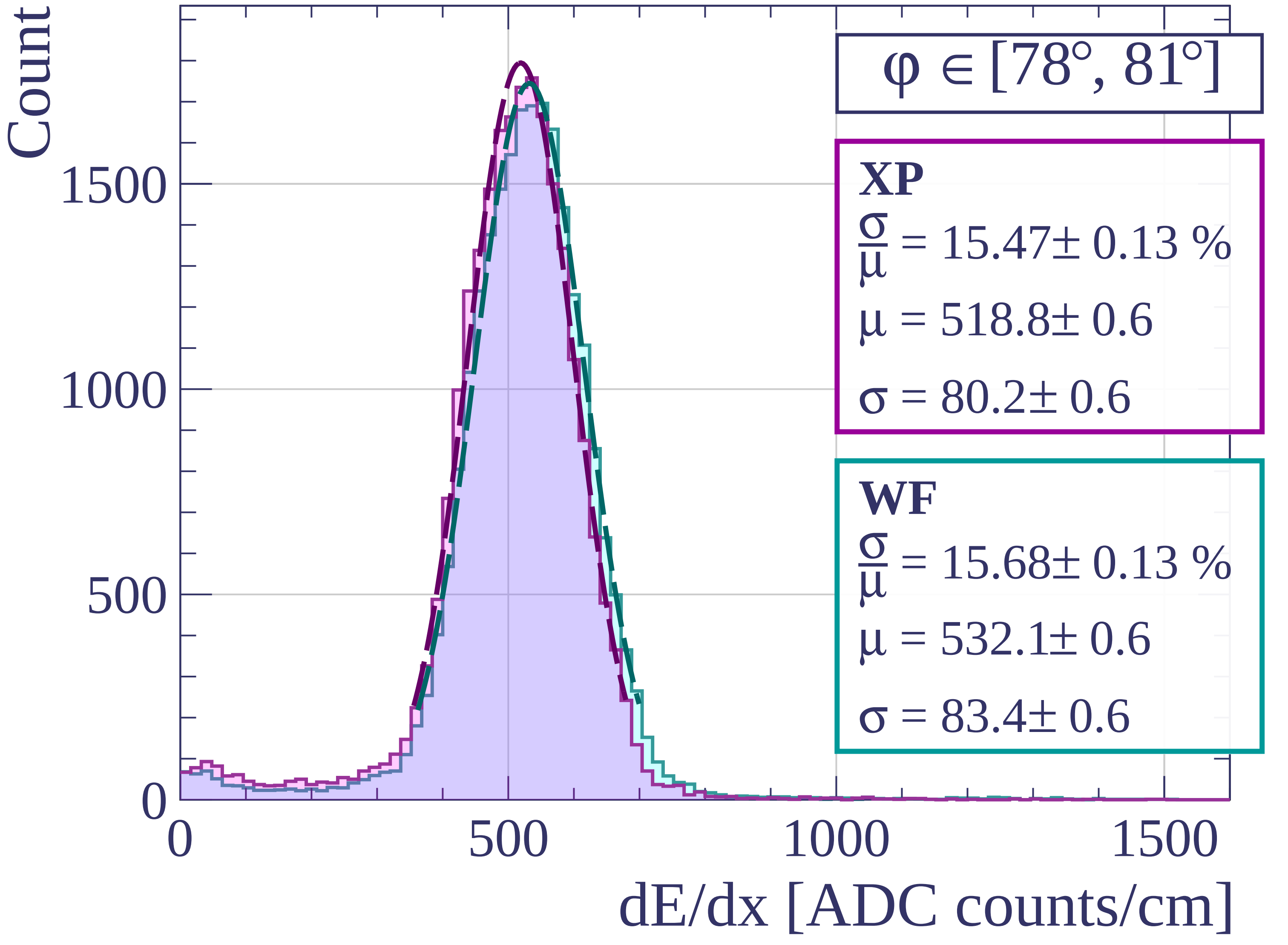


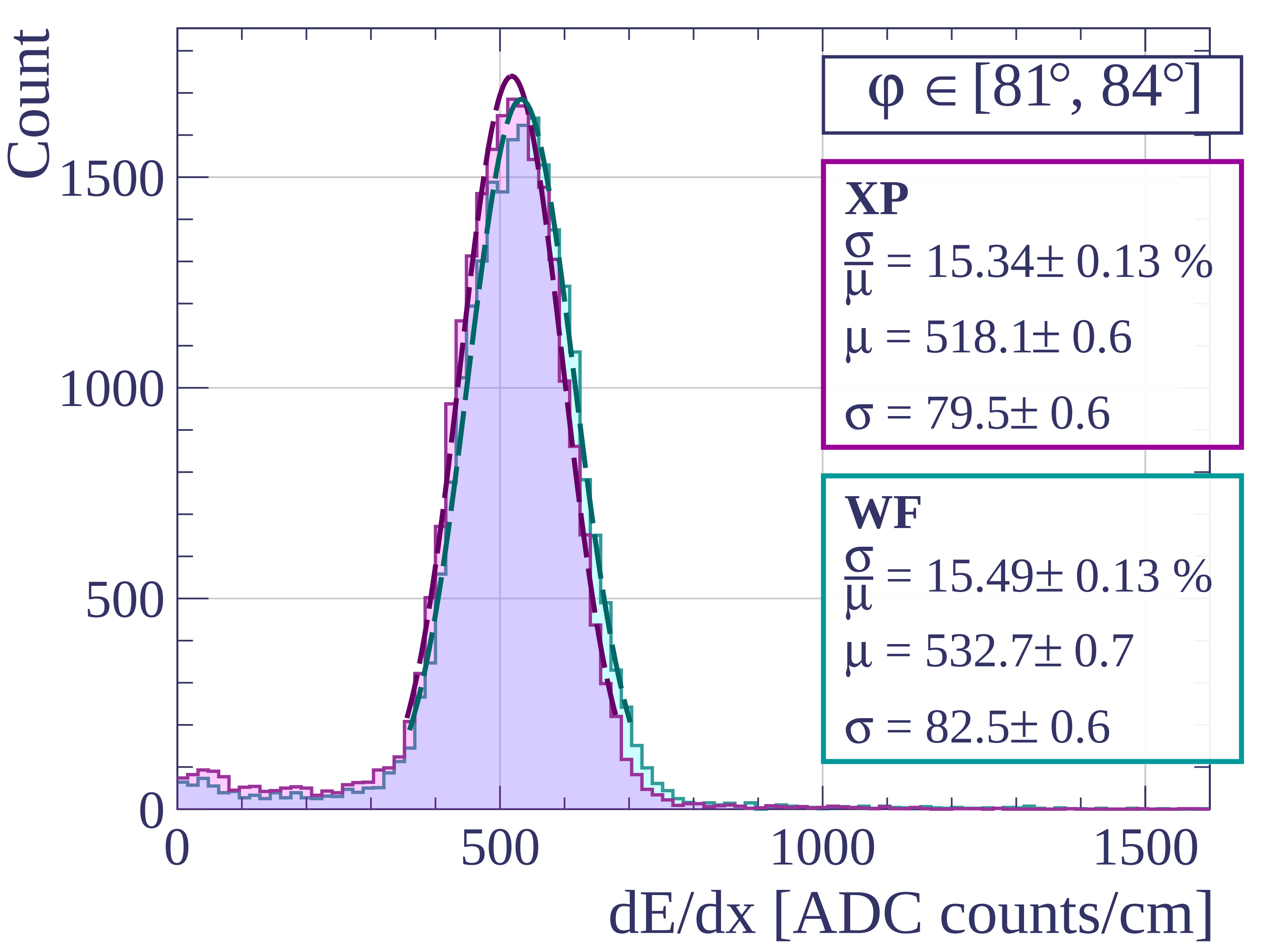
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
φ ∈ [69°, 72°]
XP
σ/μ = 15.68± 0.13 %
μ = 516.3± 0.7
σ = 81.0± 0.5
WF
σ/μ = 15.77± 0.12 %
μ = 528.0± 0.7
σ = 83.2± 0.5


Count
0
500
1000
1500
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [72°, 75°]
XP
σ/μ = 15.56± 0.12 %
μ = 515.7± 0.6
σ = 80.2± 0.5
WF
σ/μ = 15.63± 0.12 %
μ = 528.1± 0.6
σ = 82.6± 0.5

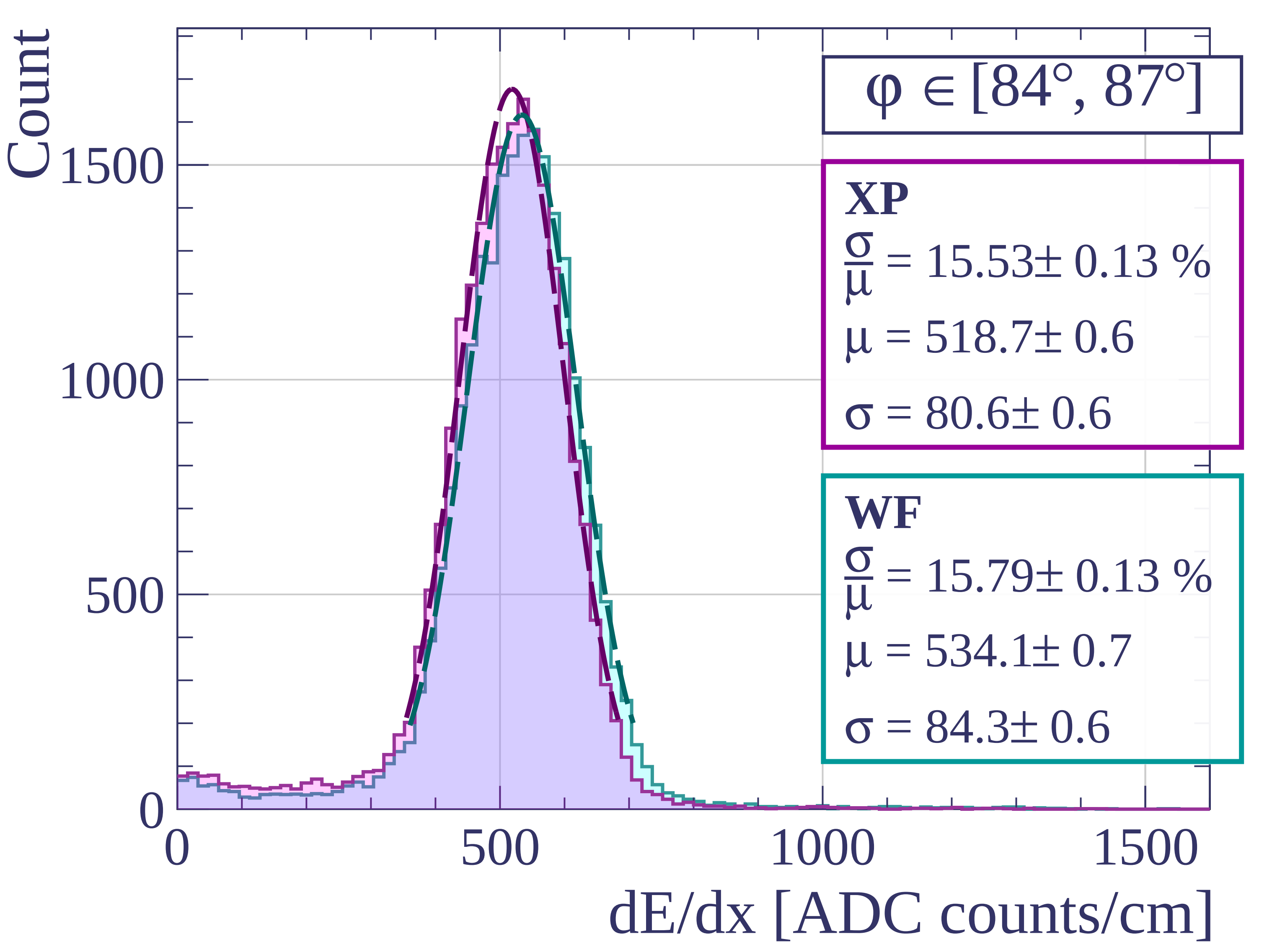


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
φ ∈ [75°, 78°]
XP
σ/μ = 15.74± 0.13 %
μ = 515.8± 0.6
σ = 81.2± 0.6
WF
σ/μ = 15.74± 0.12 %
μ = 528.9± 0.6
σ = 83.3± 0.6

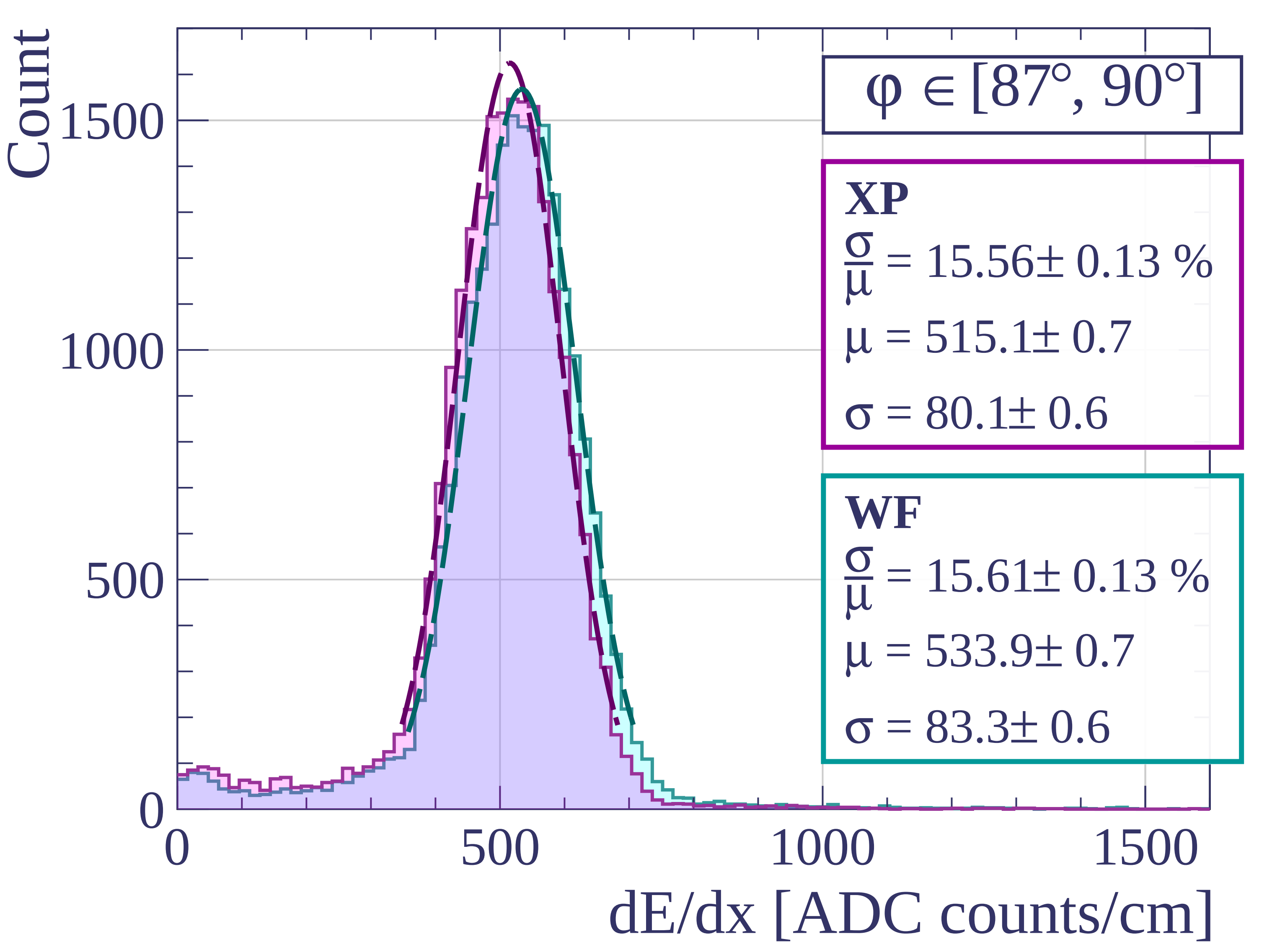

Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [78°, 81°]
XP
σ/μ = 15.47± 0.13 %
μ = 518.8± 0.6
σ = 80.2± 0.6
WF
σ/μ = 15.68± 0.13 %
μ = 532.1± 0.6
σ = 83.4± 0.6



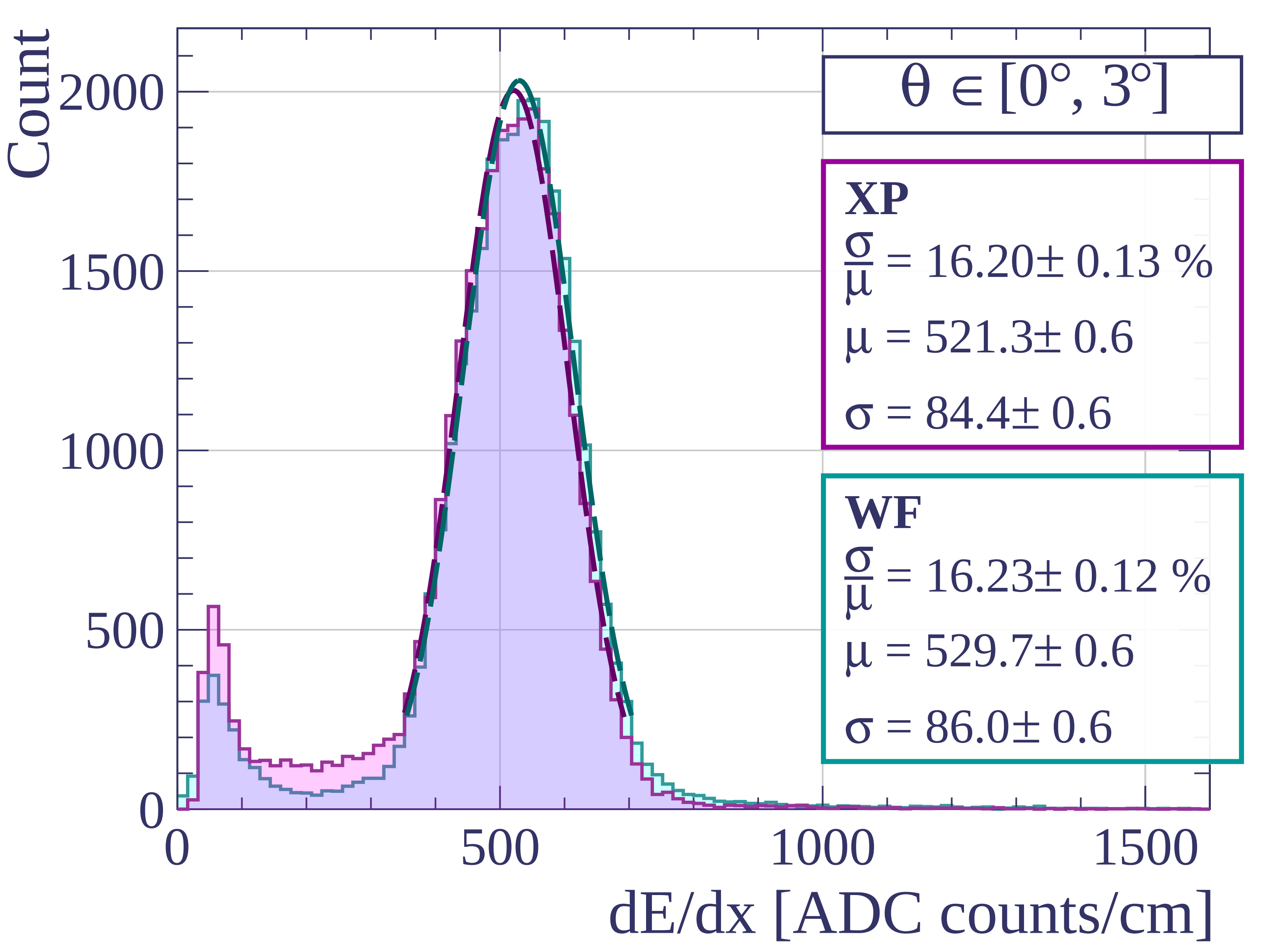
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [81°, 84°]
XP
σ/μ = 15.34± 0.13 %
μ = 518.1± 0.6
σ = 79.5± 0.6
WF
σ/μ = 15.49± 0.13 %
μ = 532.7± 0.7
σ = 82.5± 0.6



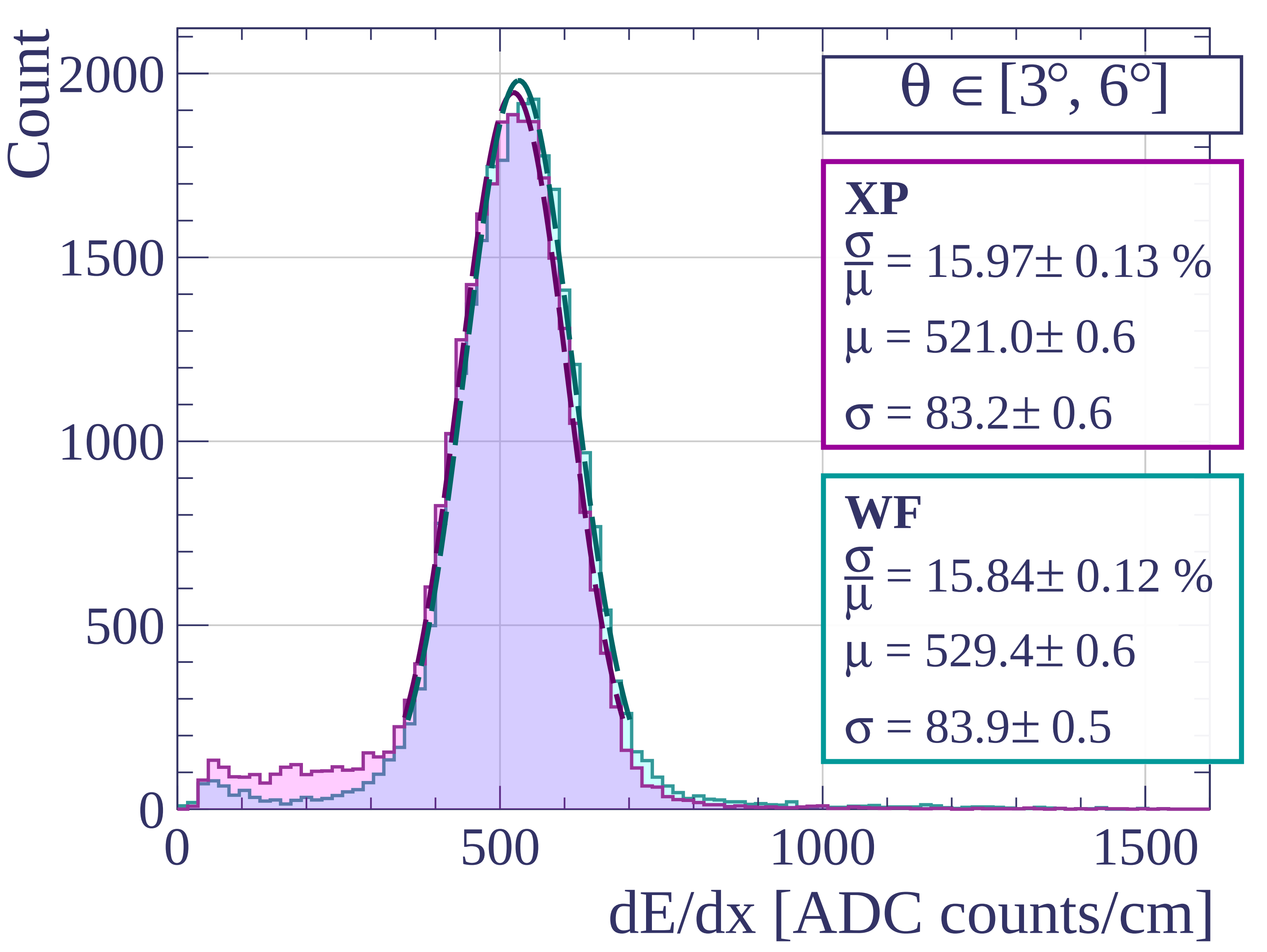
φ ∈ [84°, 87°]
XP
σ/μ = 15.53± 0.13 %
μ = 518.7± 0.6
σ = 80.6± 0.6
WF
σ/μ = 15.79± 0.13 %
μ = 534.1± 0.7
σ = 84.3± 0.6
Count
0
500
1000
1500
dE/dx [ADC counts/cm]



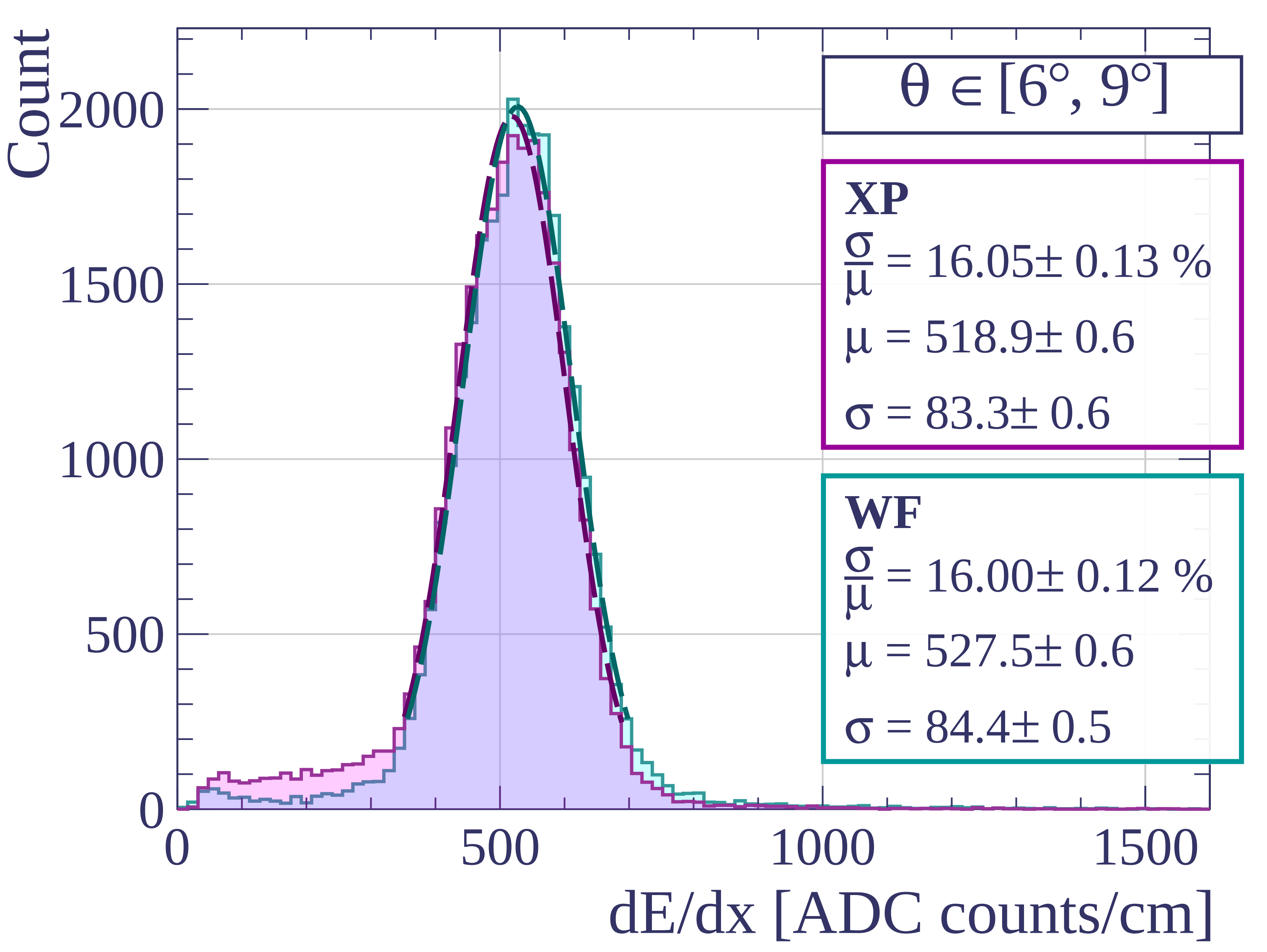
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
φ ∈ [87°, 90°]
XP
σ/μ = 15.56± 0.13 %
μ = 515.1± 0.7
σ = 80.1± 0.6
WF
σ/μ = 15.61± 0.13 %
μ = 533.9± 0.7
σ = 83.3± 0.6



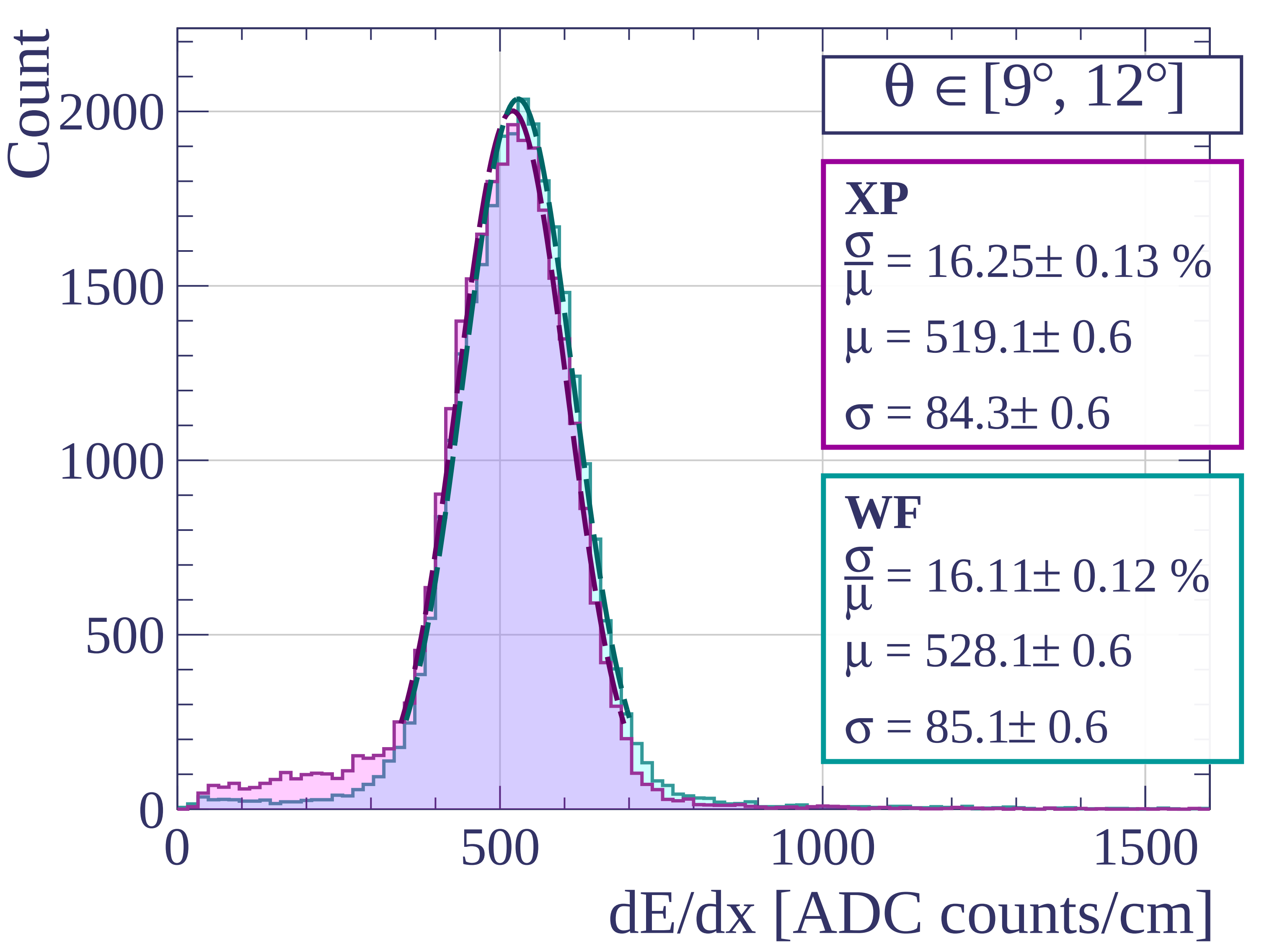
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [0°, 3°]
XP
σ/μ = 16.20± 0.13 %
μ = 521.3± 0.6
σ = 84.4± 0.6
WF
σ/μ = 16.23± 0.12 %
μ = 529.7± 0.6
σ = 86.0± 0.6



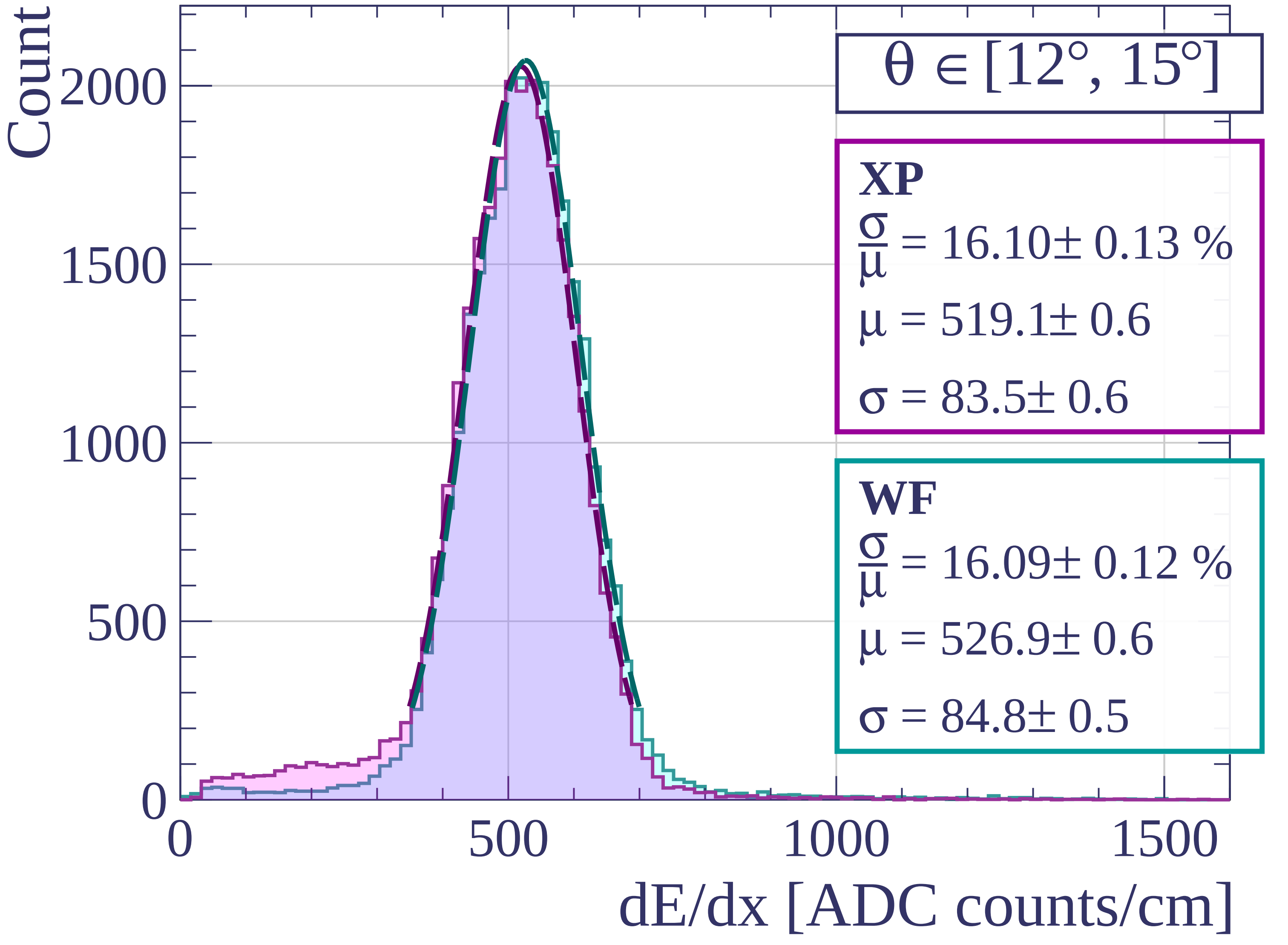
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [3°, 6°]
XP
σ/μ = 15.97± 0.13 %
μ = 521.0± 0.6
σ = 83.2± 0.6
WF
σ/μ = 15.84± 0.12 %
μ = 529.4± 0.6
σ = 83.9± 0.5



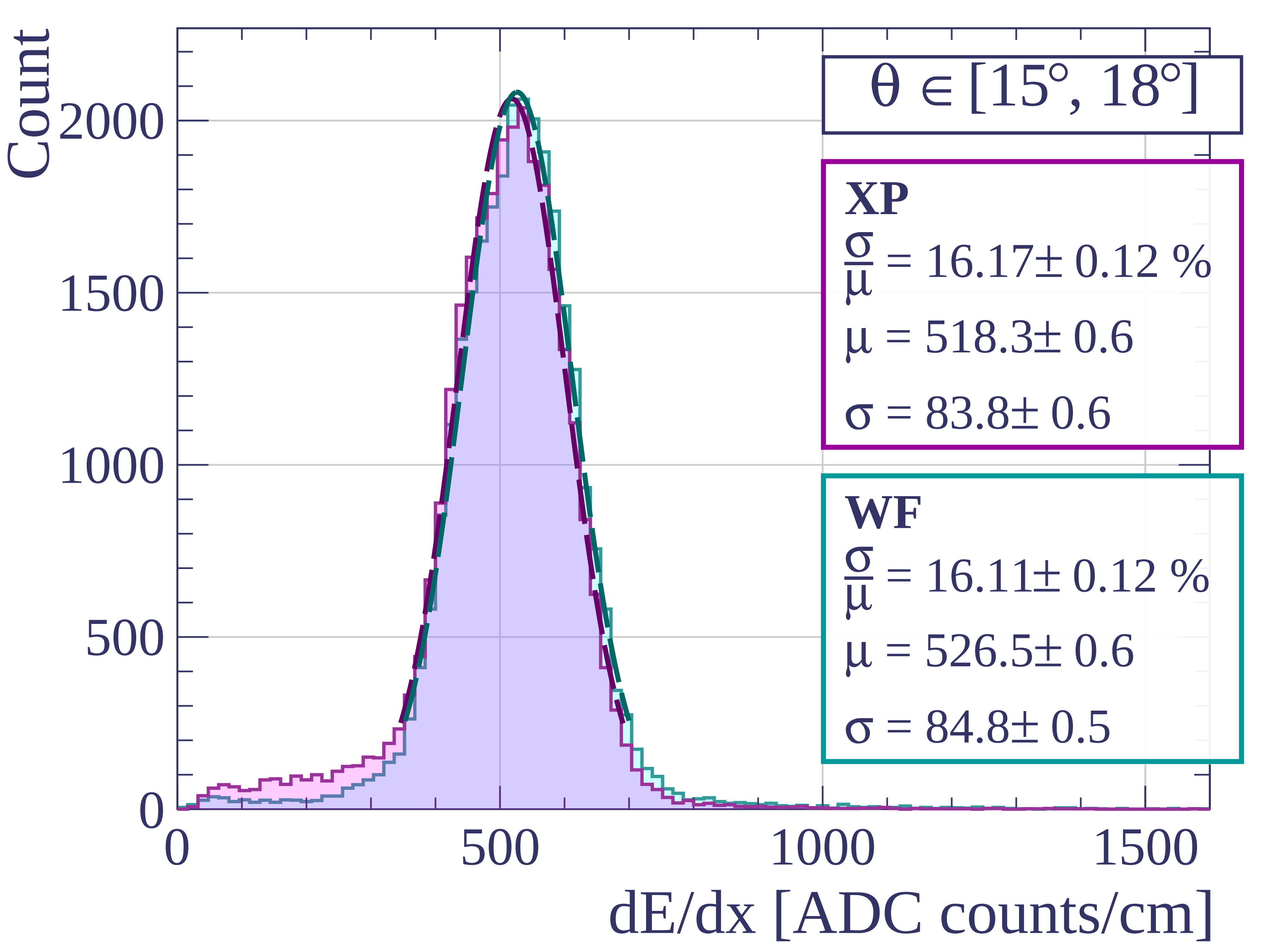
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [6°, 9°]
XP
σ/μ = 16.05± 0.13 %
μ = 518.9± 0.6
σ = 83.3± 0.6
WF
σ/μ = 16.00± 0.12 %
μ = 527.5± 0.6
σ = 84.4± 0.5



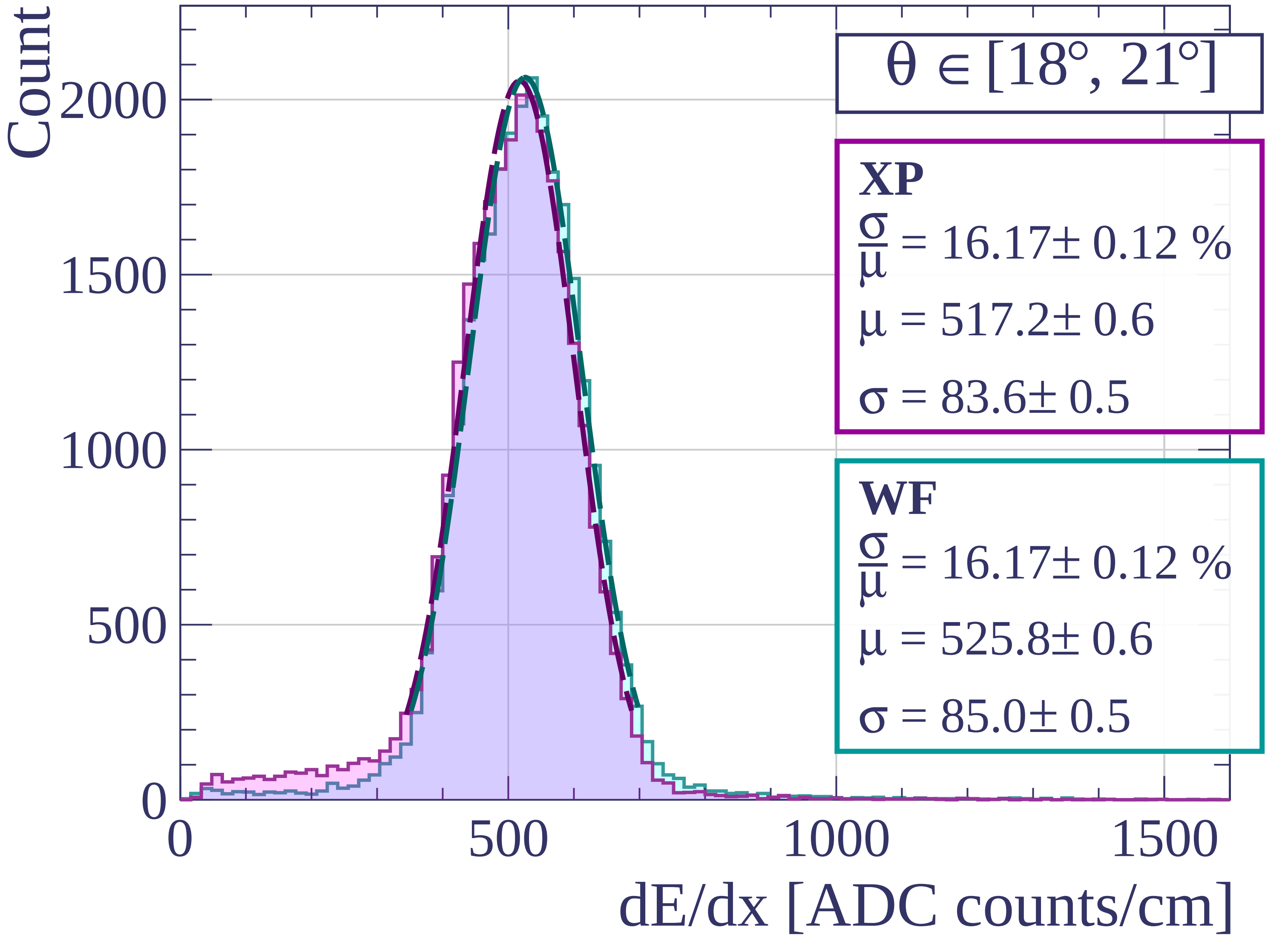
Count
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 16.25± 0.13 %
μ = 519.1± 0.6
σ = 84.3± 0.6
WF
σ/μ = 16.11± 0.12 %
μ = 528.1± 0.6
σ = 85.1± 0.6
0
500
1000
1500
2000

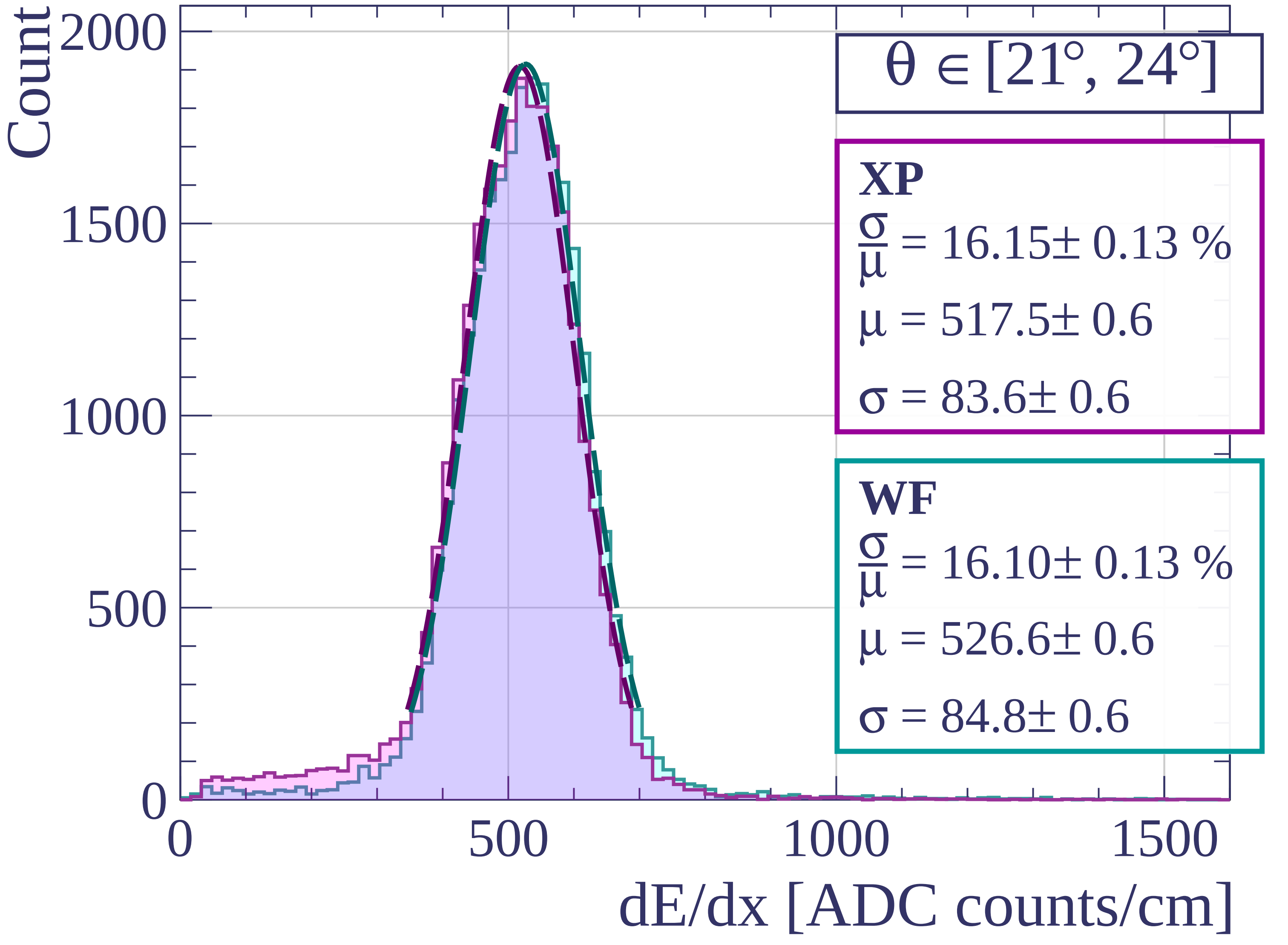


Count
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 16.10± 0.13 %
μ = 519.1± 0.6
σ = 83.5± 0.6
WF
σ/μ = 16.09± 0.12 %
μ = 526.9± 0.6
σ = 84.8± 0.5
0
500
1000
1500
2000

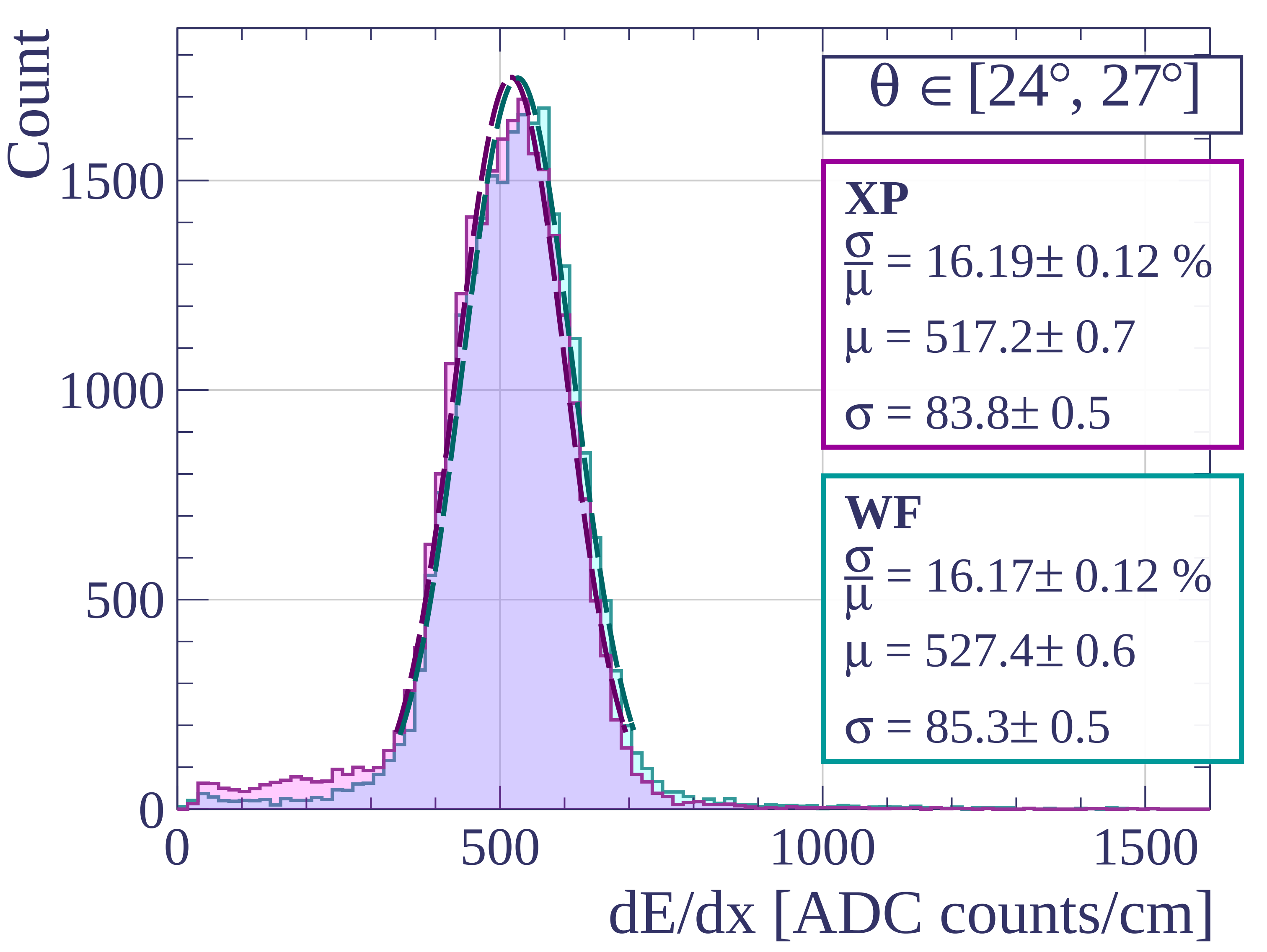


Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [15°, 18°]
XP
σ/μ = 16.17± 0.12 %
μ = 518.3± 0.6
σ = 83.8± 0.6
WF
σ/μ = 16.11± 0.12 %
μ = 526.5± 0.6
σ = 84.8± 0.5

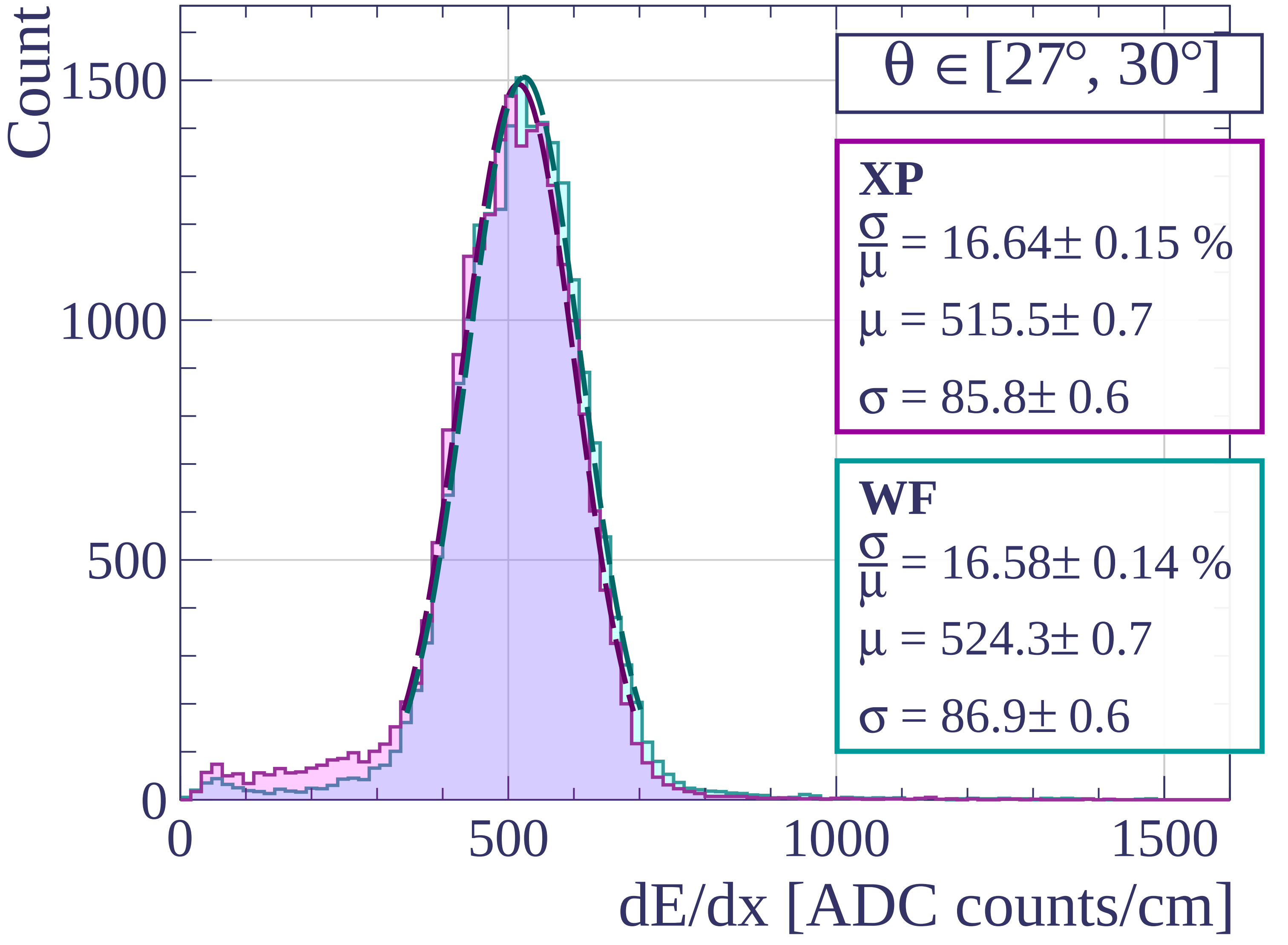

Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 16.17± 0.12 %
μ = 517.2± 0.6
σ = 83.6± 0.5
WF
σ/μ = 16.17± 0.12 %
μ = 525.8± 0.6
σ = 85.0± 0.5

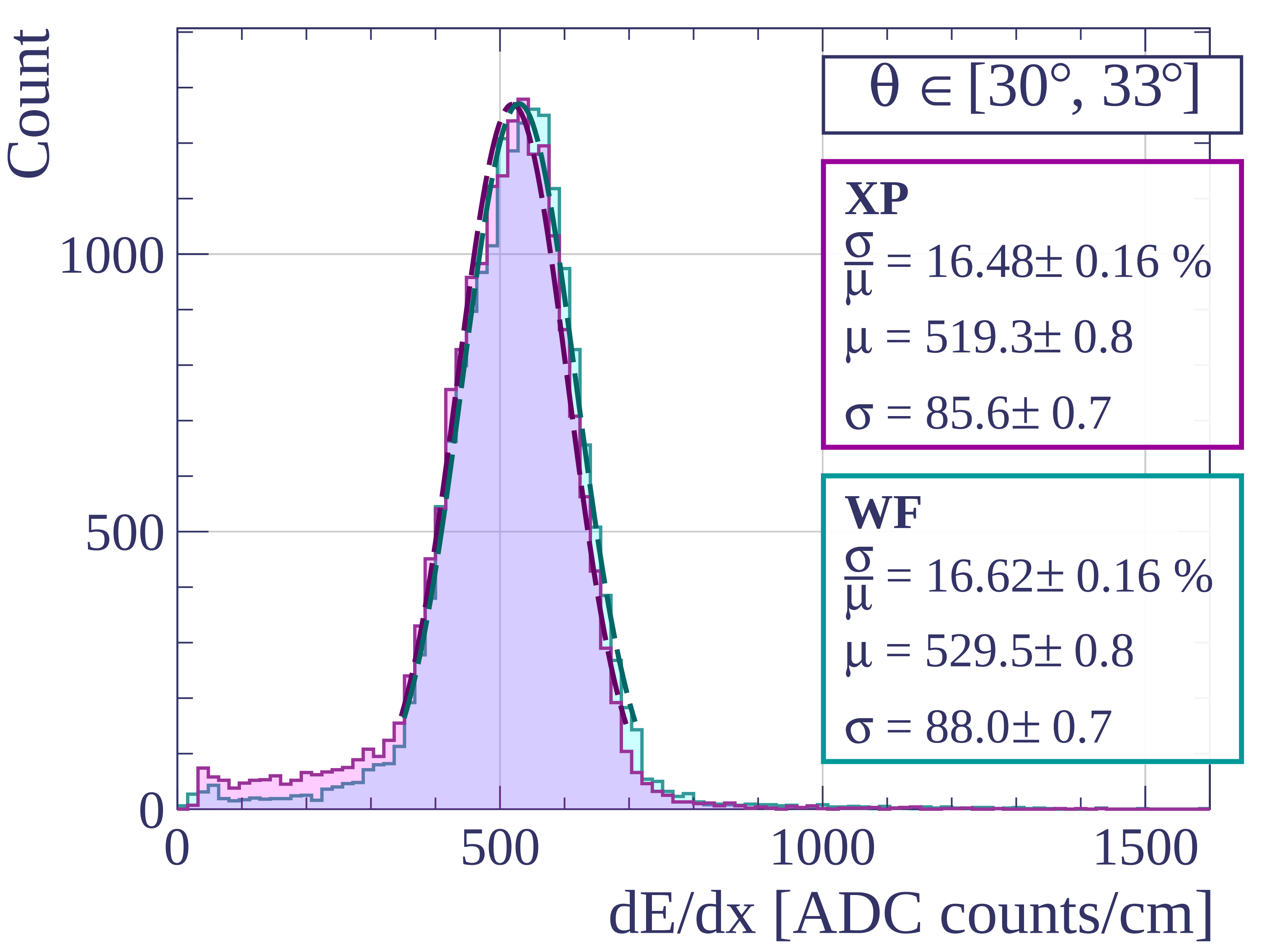


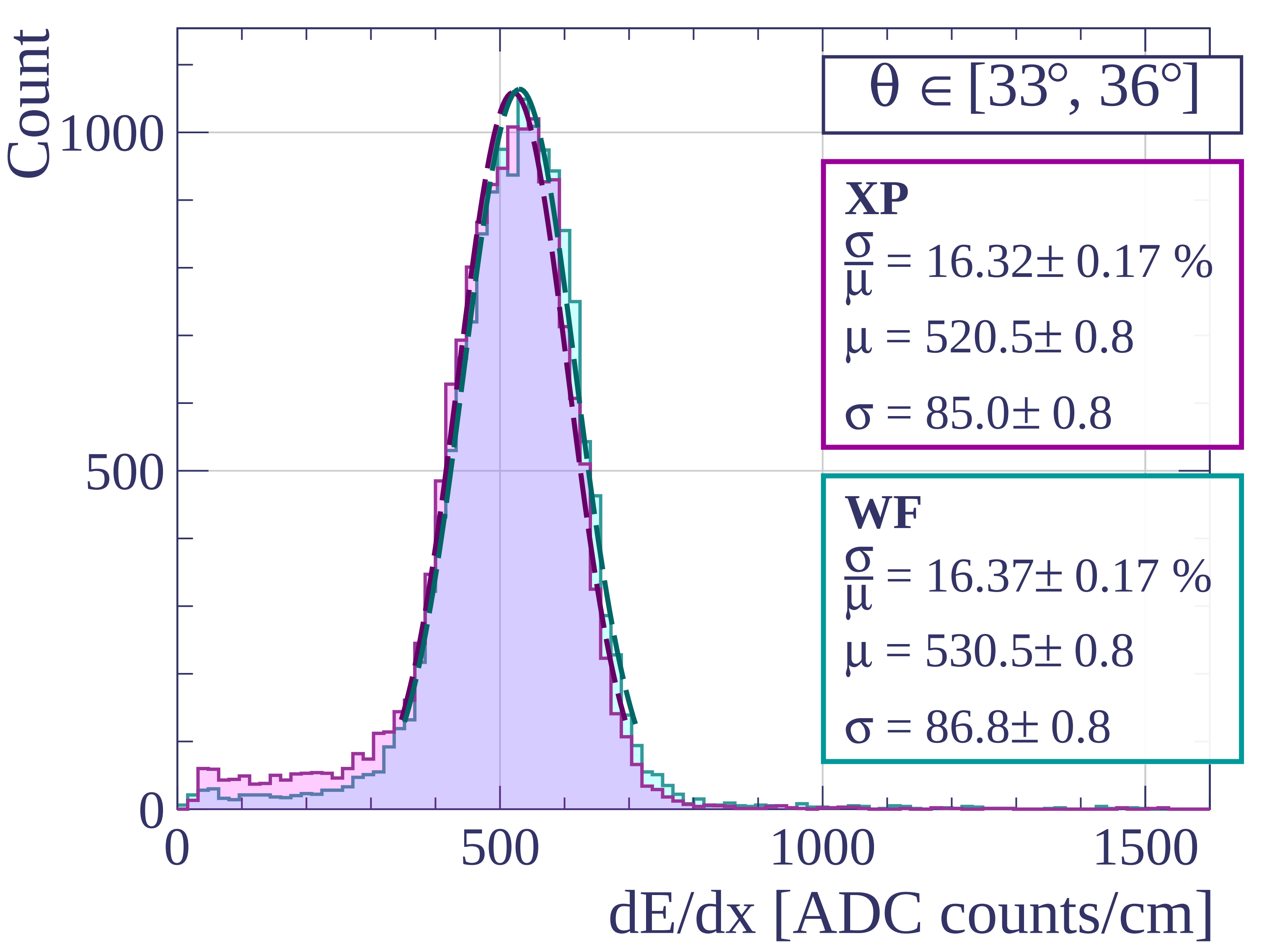
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [21°, 24°]
XP
σ/μ = 16.15± 0.13 %
μ = 517.5± 0.6
σ = 83.6± 0.6
WF
σ/μ = 16.10± 0.13 %
μ = 526.6± 0.6
σ = 84.8± 0.6

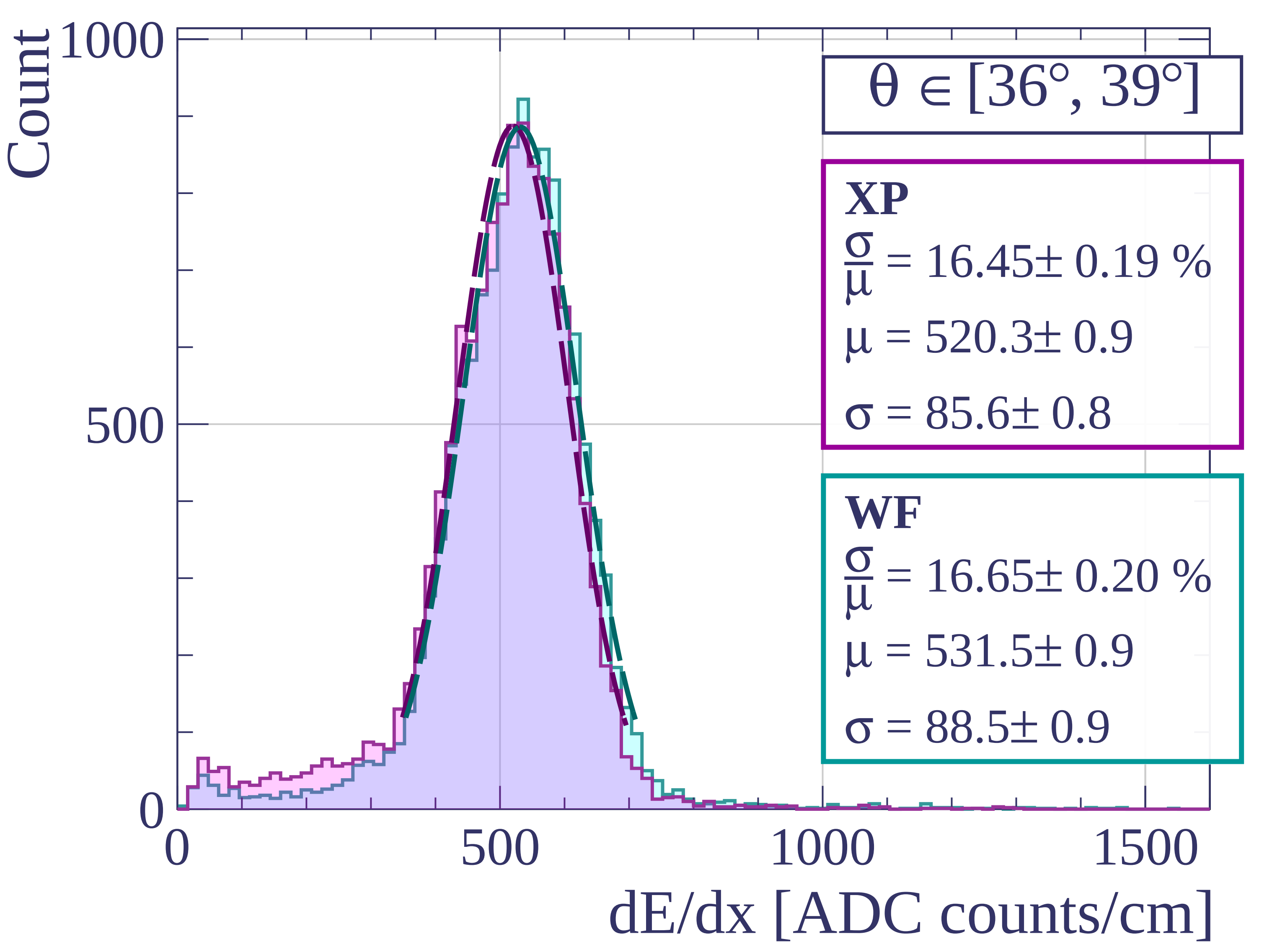


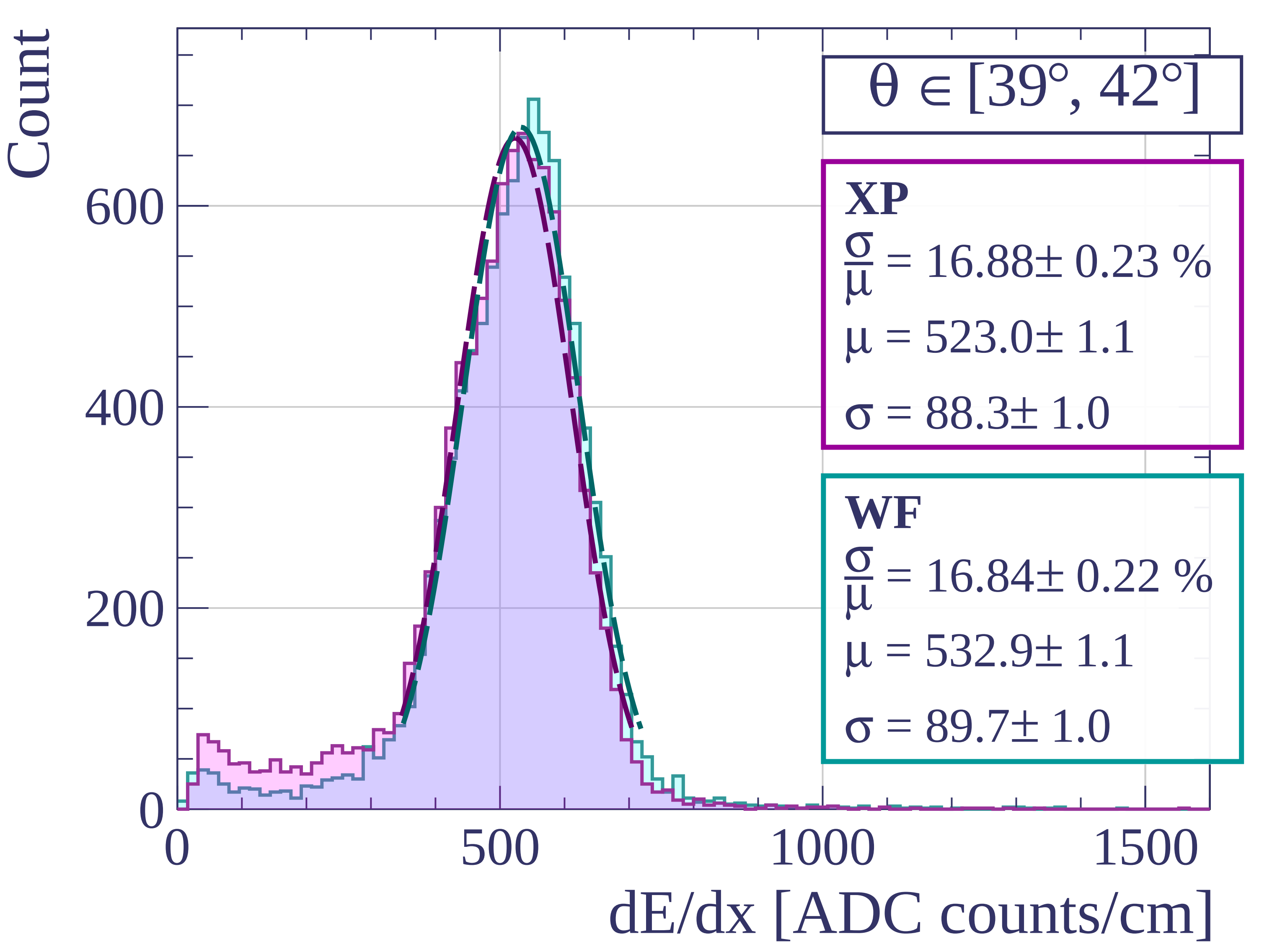
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
θ ∈ [24°, 27°]
XP
$\frac{\sigma}{\mu}$ = 16.19± 0.12 %
μ = 517.2± 0.7
σ = 83.8± 0.5
WF
$\frac{\sigma}{\mu}$ = 16.17± 0.12 %
μ = 527.4± 0.6
σ = 85.3± 0.5


Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [27°, 30°]
XP
σ/μ = 16.64± 0.15 %
μ = 515.5± 0.7
σ = 85.8± 0.6
WF
σ/μ = 16.58± 0.14 %
μ = 524.3± 0.7
σ = 86.9± 0.6

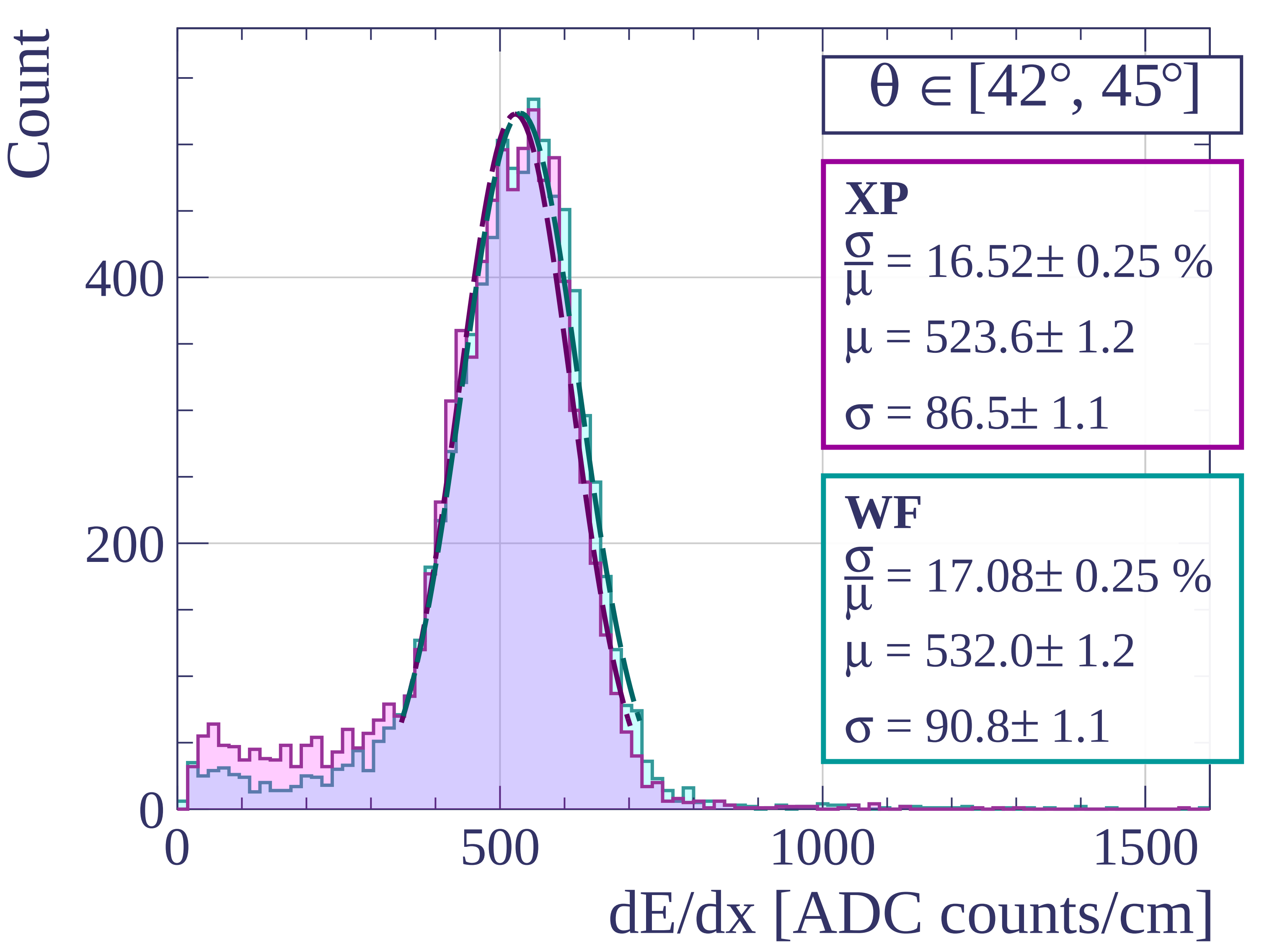

θ ∈ [30°, 33°]
XP
σ/μ = 16.48± 0.16 %
μ = 519.3± 0.8
σ = 85.6± 0.7
WF
σ/μ = 16.62± 0.16 %
μ = 529.5± 0.8
σ = 88.0± 0.7
Count
dE/dx [ADC counts/cm]
0
500
1000
1500

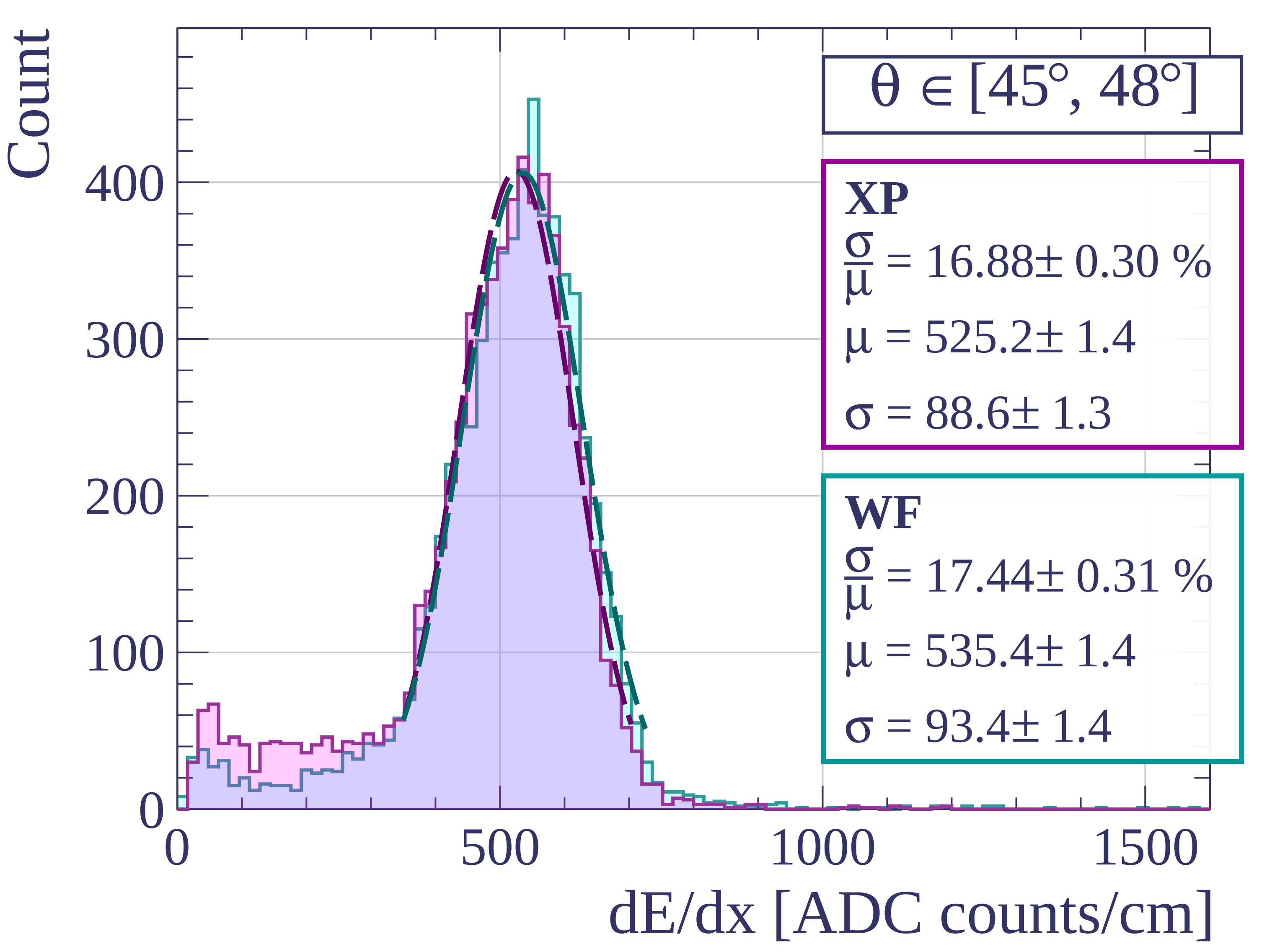

Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 16.32± 0.17 %
μ = 520.5± 0.8
σ = 85.0± 0.8
WF
σ/μ = 16.37± 0.17 %
μ = 530.5± 0.8
σ = 86.8± 0.8

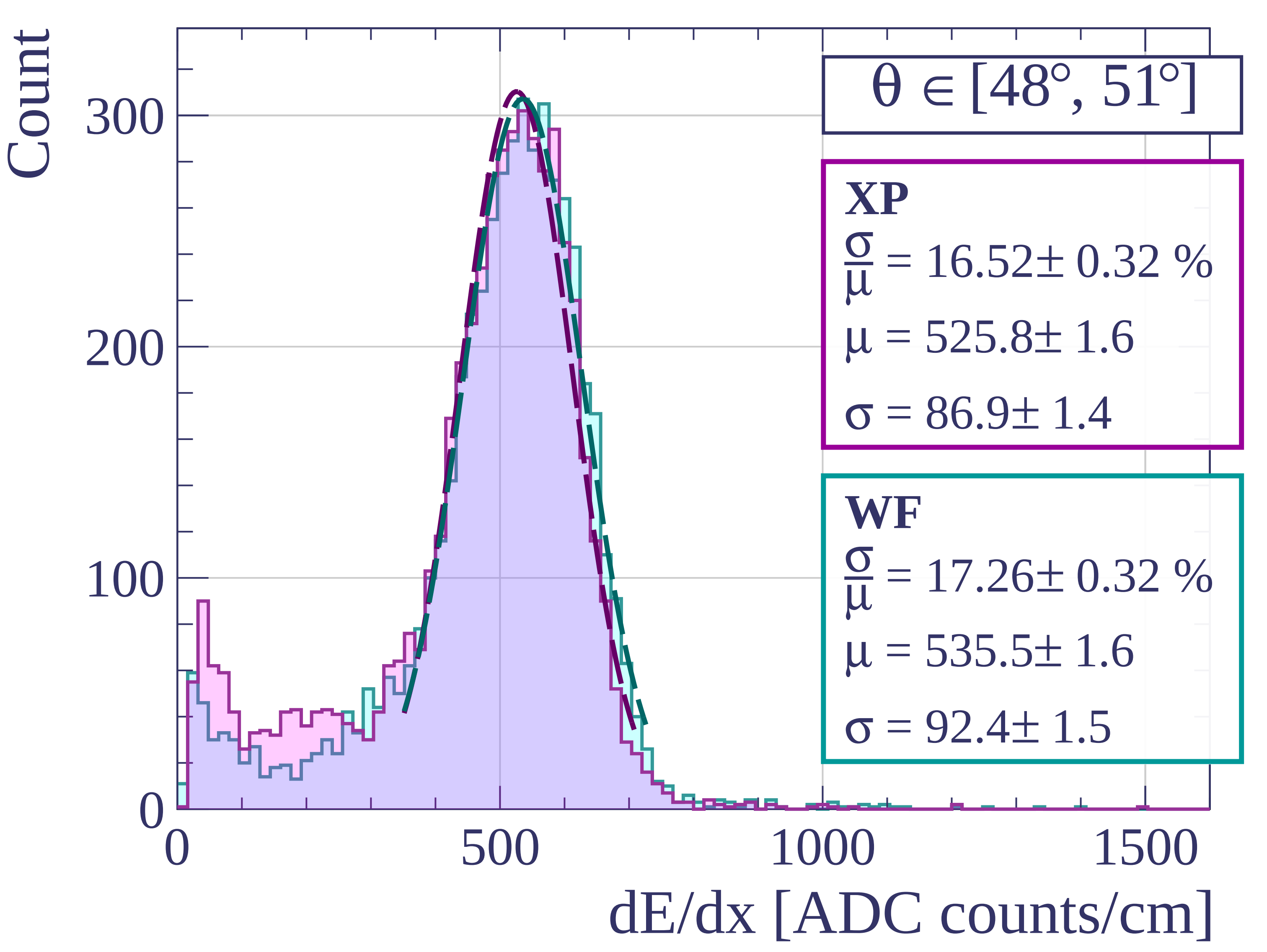

Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [36°, 39°]
XP
σ/μ = 16.45± 0.19 %
μ = 520.3± 0.9
σ = 85.6± 0.8
WF
σ/μ = 16.65± 0.20 %
μ = 531.5± 0.9
σ = 88.5± 0.9

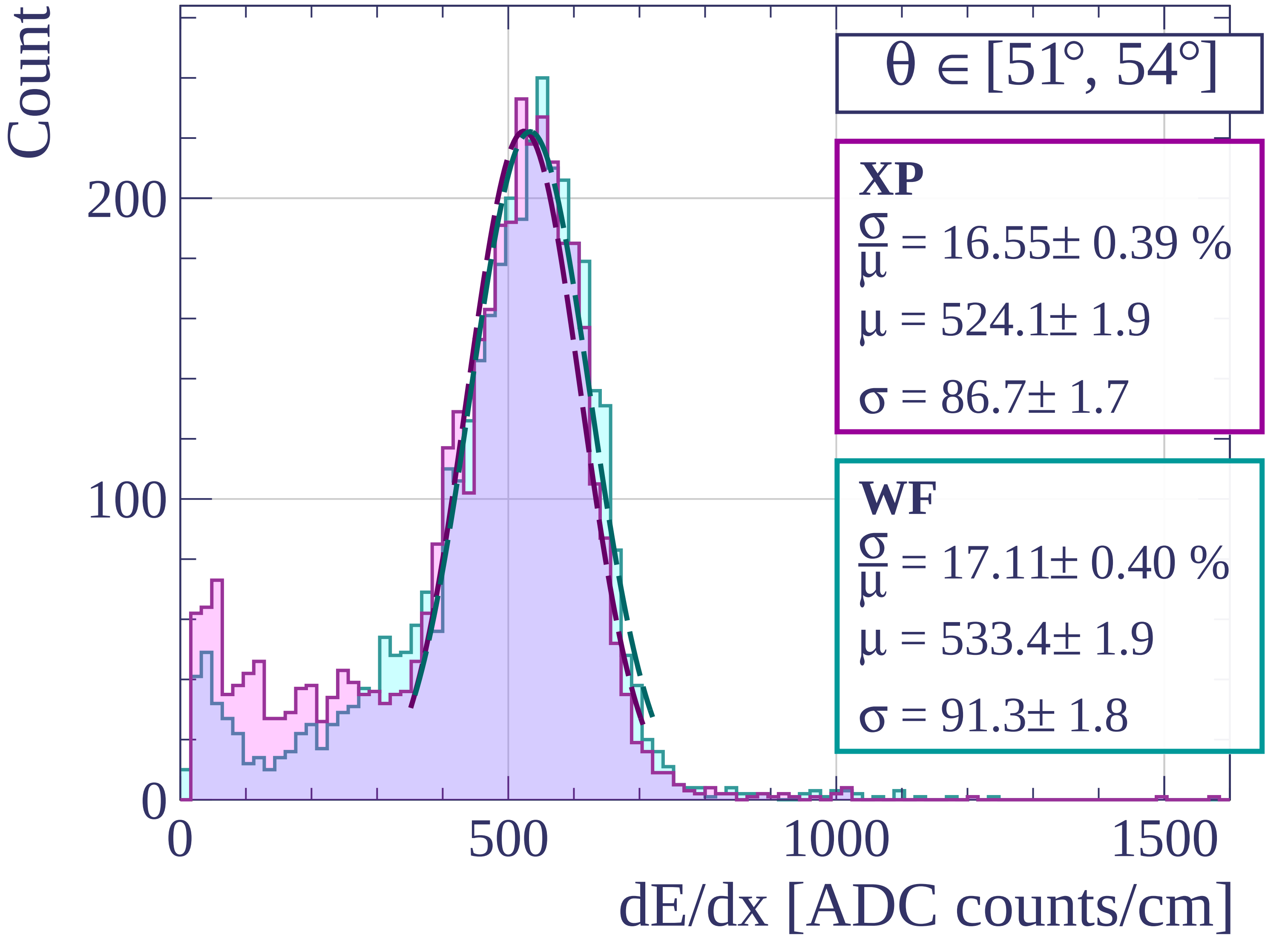


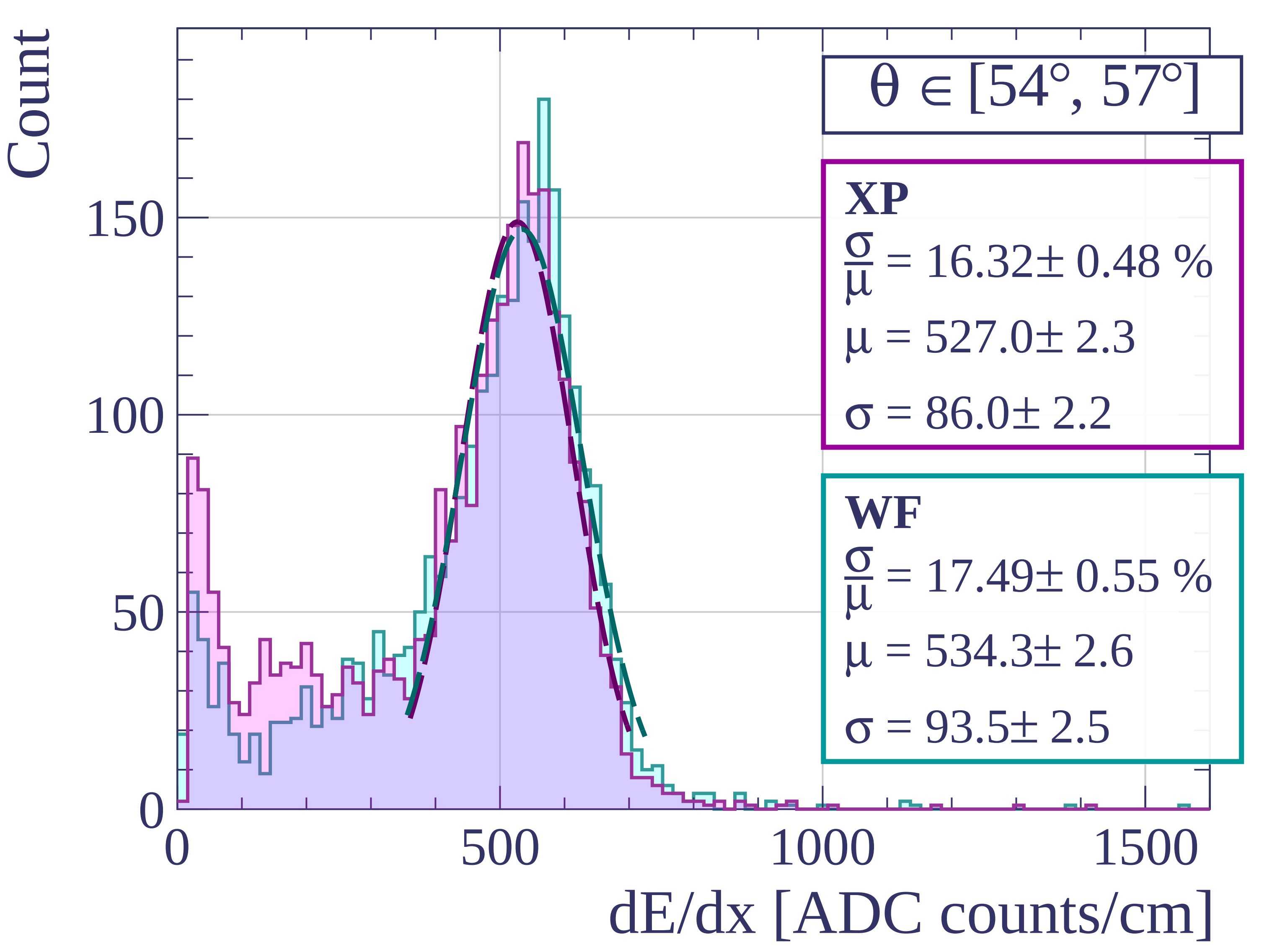
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [39°, 42°]
XP
σ/μ = 16.88± 0.23 %
μ = 523.0± 1.1
σ = 88.3± 1.0
WF
σ/μ = 16.84± 0.22 %
μ = 532.9± 1.1
σ = 89.7± 1.0


Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
θ ∈ [42°, 45°]
XP
σ/μ = 16.52± 0.25 %
μ = 523.6± 1.2
σ = 86.5± 1.1
WF
σ/μ = 17.08± 0.25 %
μ = 532.0± 1.2
σ = 90.8± 1.1

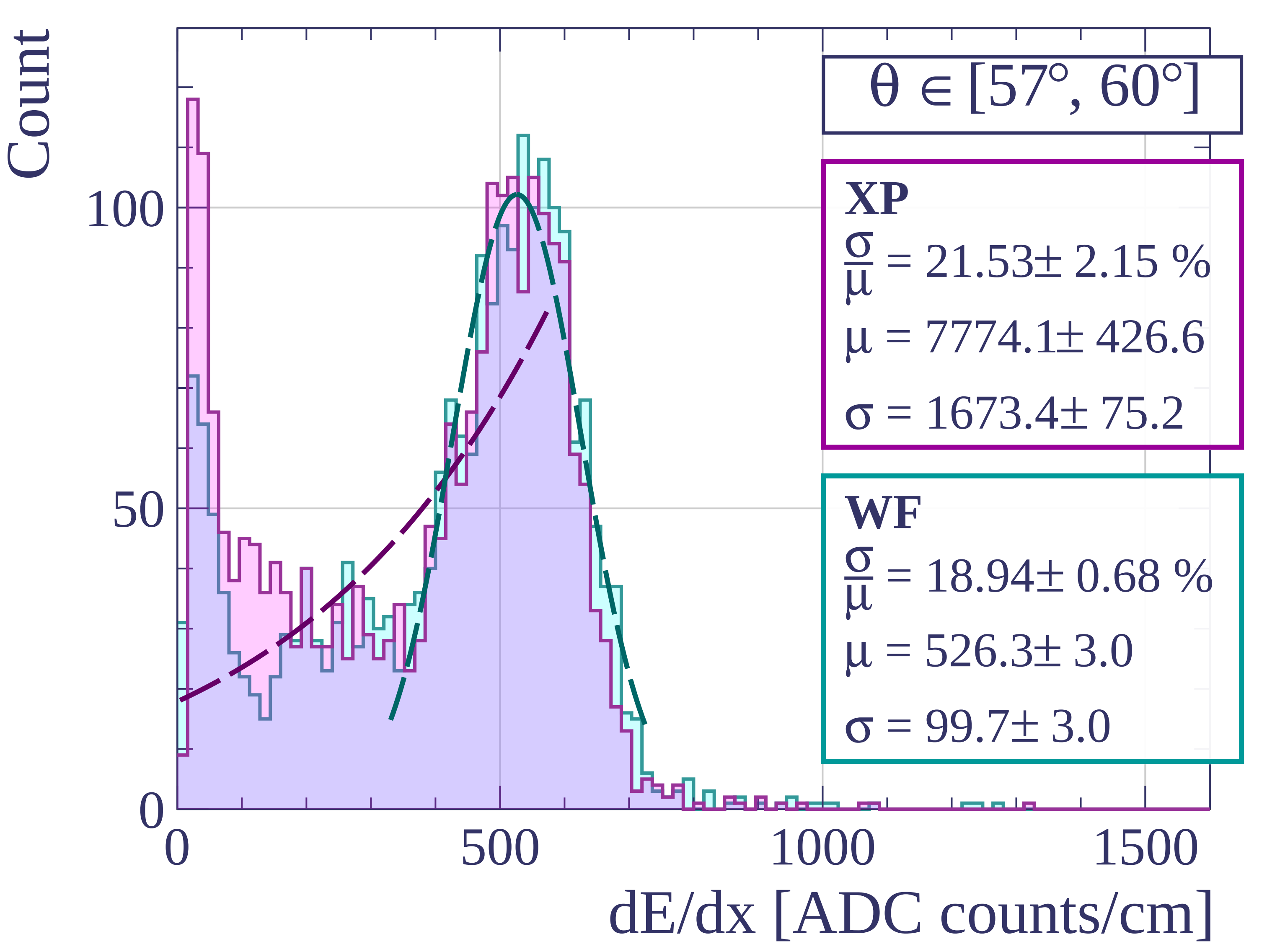

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [45°, 48°]
XP
σ/μ = 16.88± 0.30 %
μ = 525.2± 1.4
σ = 88.6± 1.3
WF
σ/μ = 17.44± 0.31 %
μ = 535.4± 1.4
σ = 93.4± 1.4

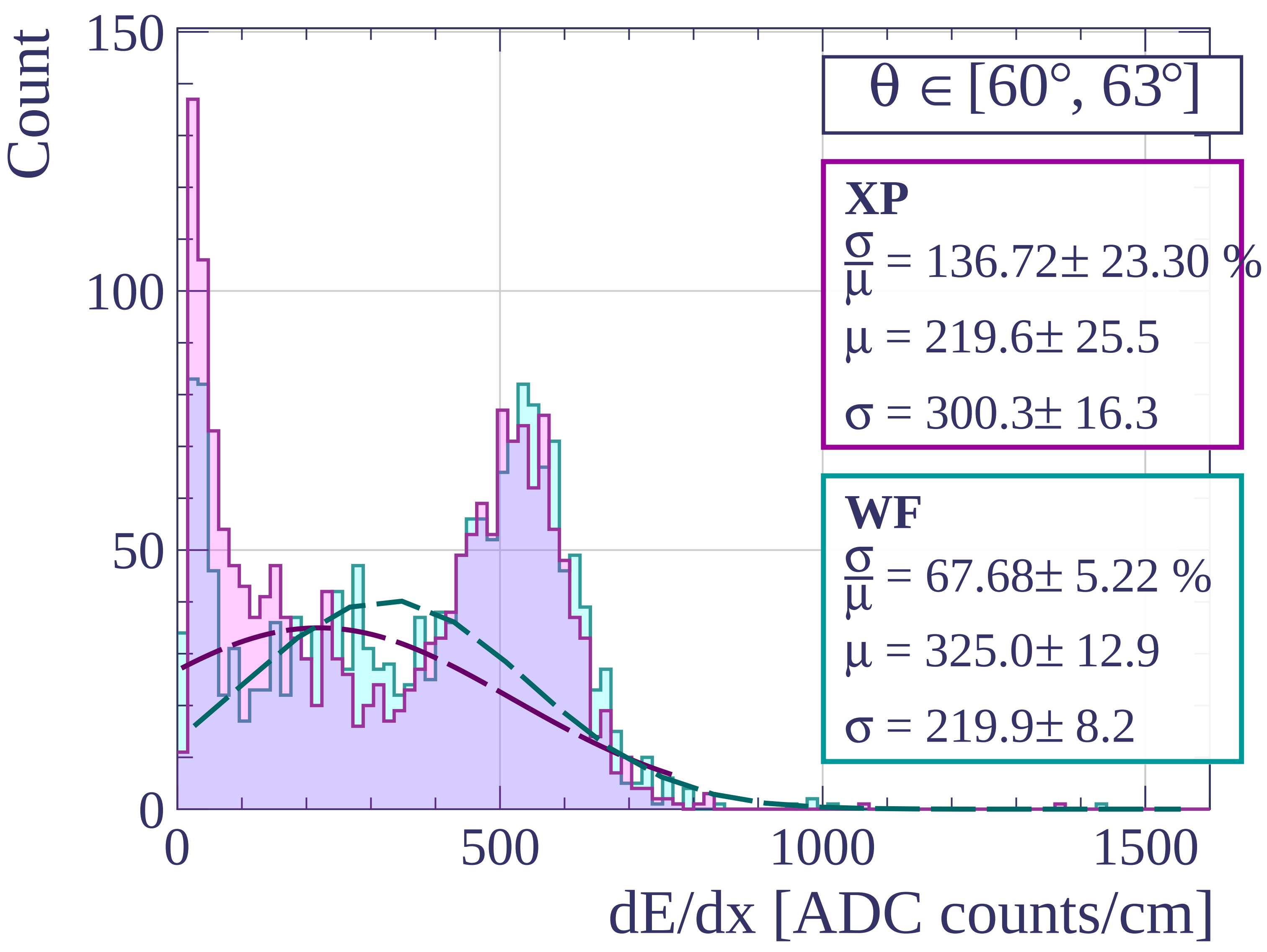

Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = 16.52± 0.32 %
μ = 525.8± 1.6
σ = 86.9± 1.4
WF
σ/μ = 17.26± 0.32 %
μ = 535.5± 1.6
σ = 92.4± 1.5


Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = 16.55± 0.39 %
μ = 524.1± 1.9
σ = 86.7± 1.7
WF
σ/μ = 17.11± 0.40 %
μ = 533.4± 1.9
σ = 91.3± 1.8

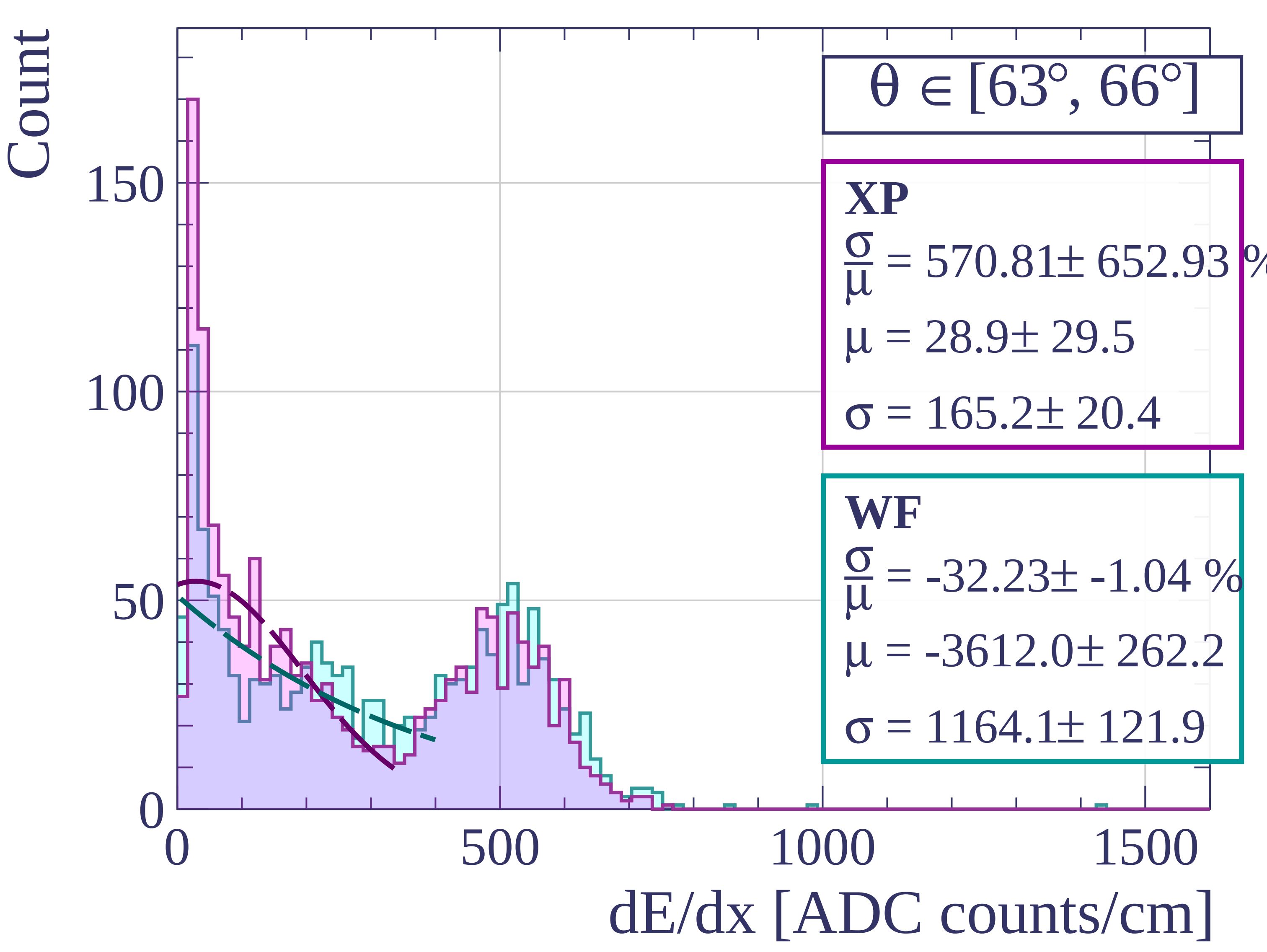


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
50
100
150
θ ∈ [54°, 57°]
XP
σ/μ = 16.32± 0.48 %
μ = 527.0± 2.3
σ = 86.0± 2.2
WF
σ/μ = 17.49± 0.55 %
μ = 534.3± 2.6
σ = 93.5± 2.5

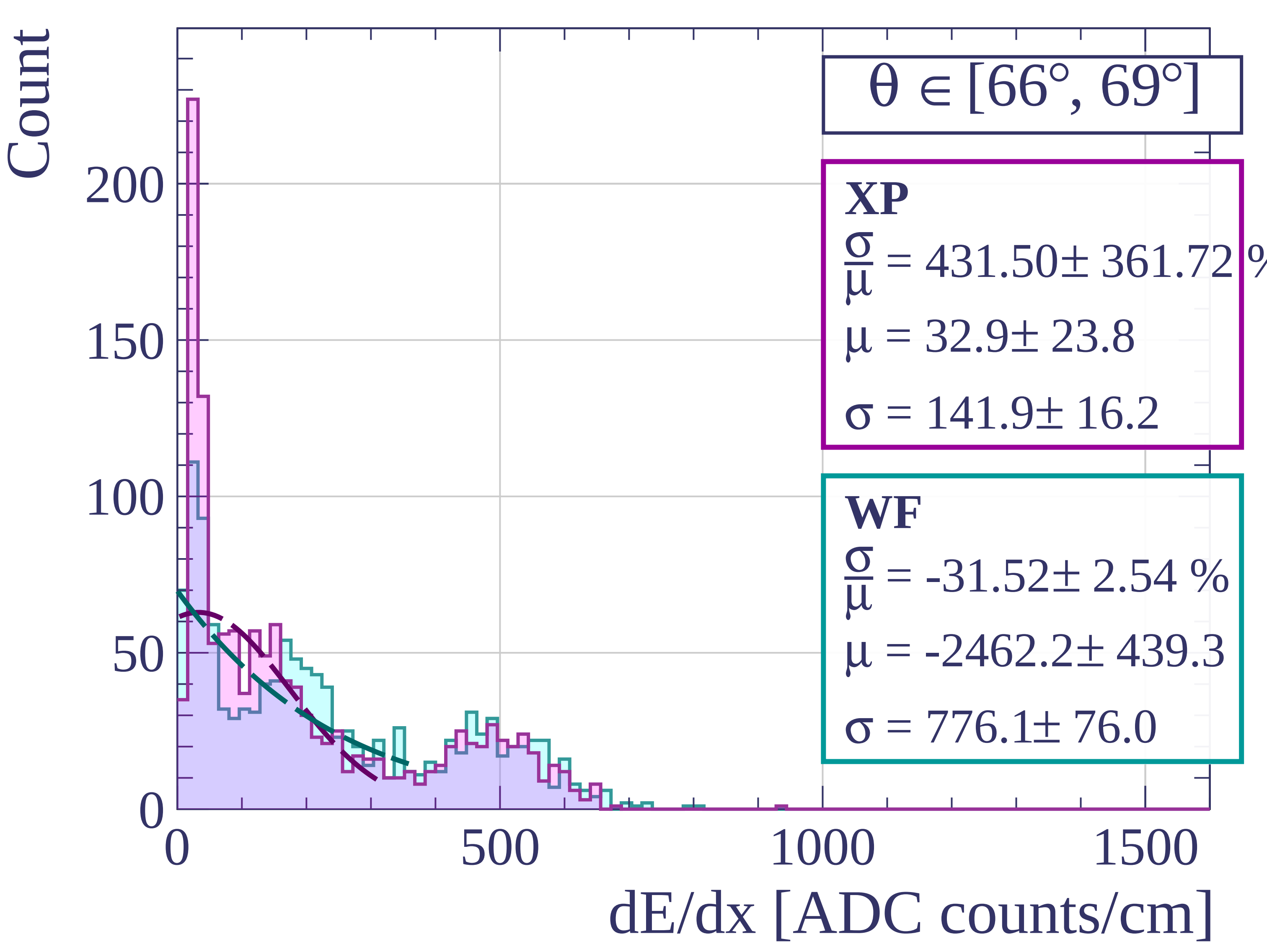

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = 21.53± 2.15 %
μ = 7774.1± 426.6
σ = 1673.4± 75.2
WF
σ/μ = 18.94± 0.68 %
μ = 526.3± 3.0
σ = 99.7± 3.0

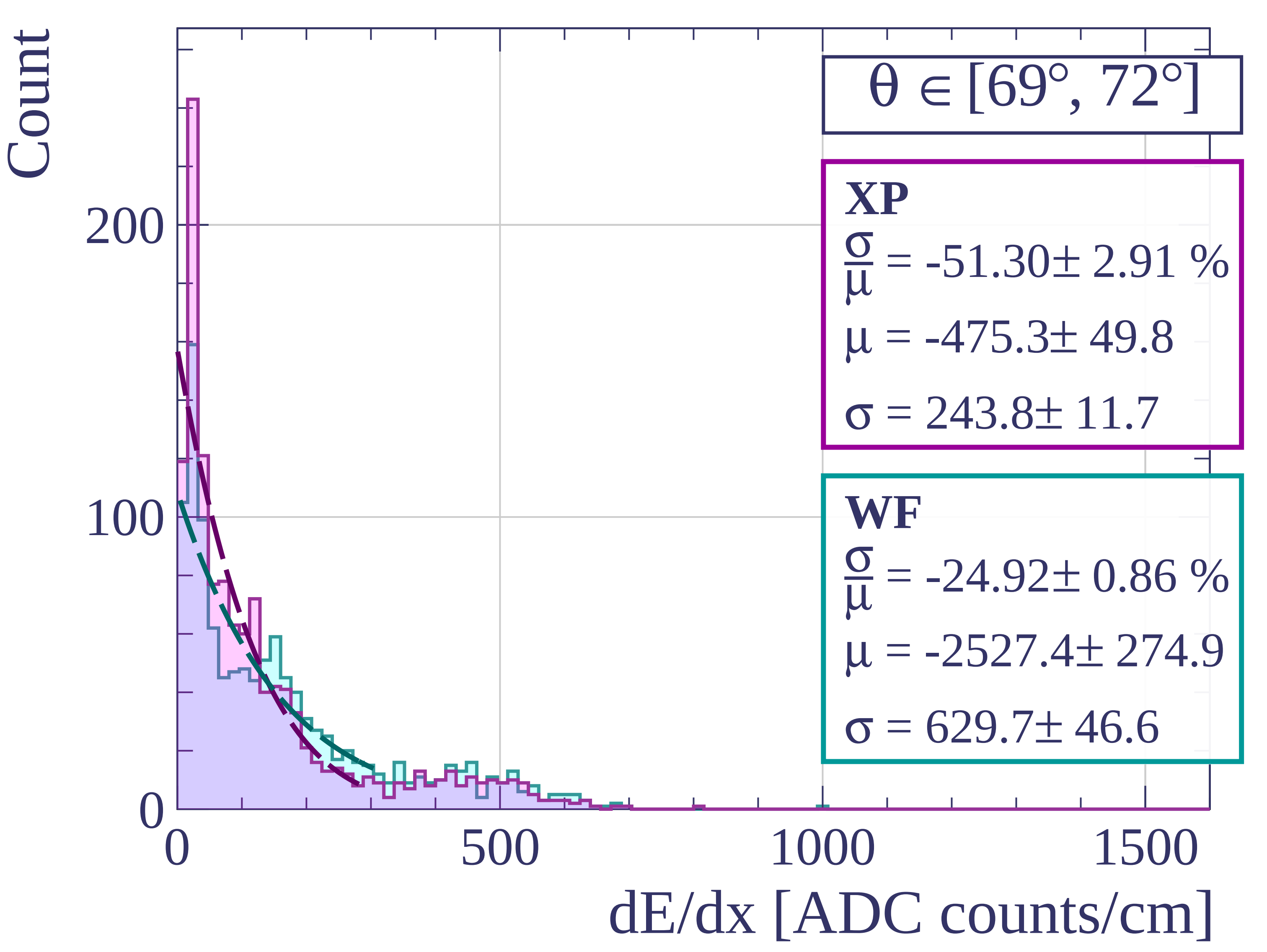


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
$\frac{\sigma}{\mu}$ = 136.72± 23.30 %
μ = 219.6± 25.5
σ = 300.3± 16.3
WF
$\frac{\sigma}{\mu}$ = 67.68± 5.22 %
μ = 325.0± 12.9
σ = 219.9± 8.2

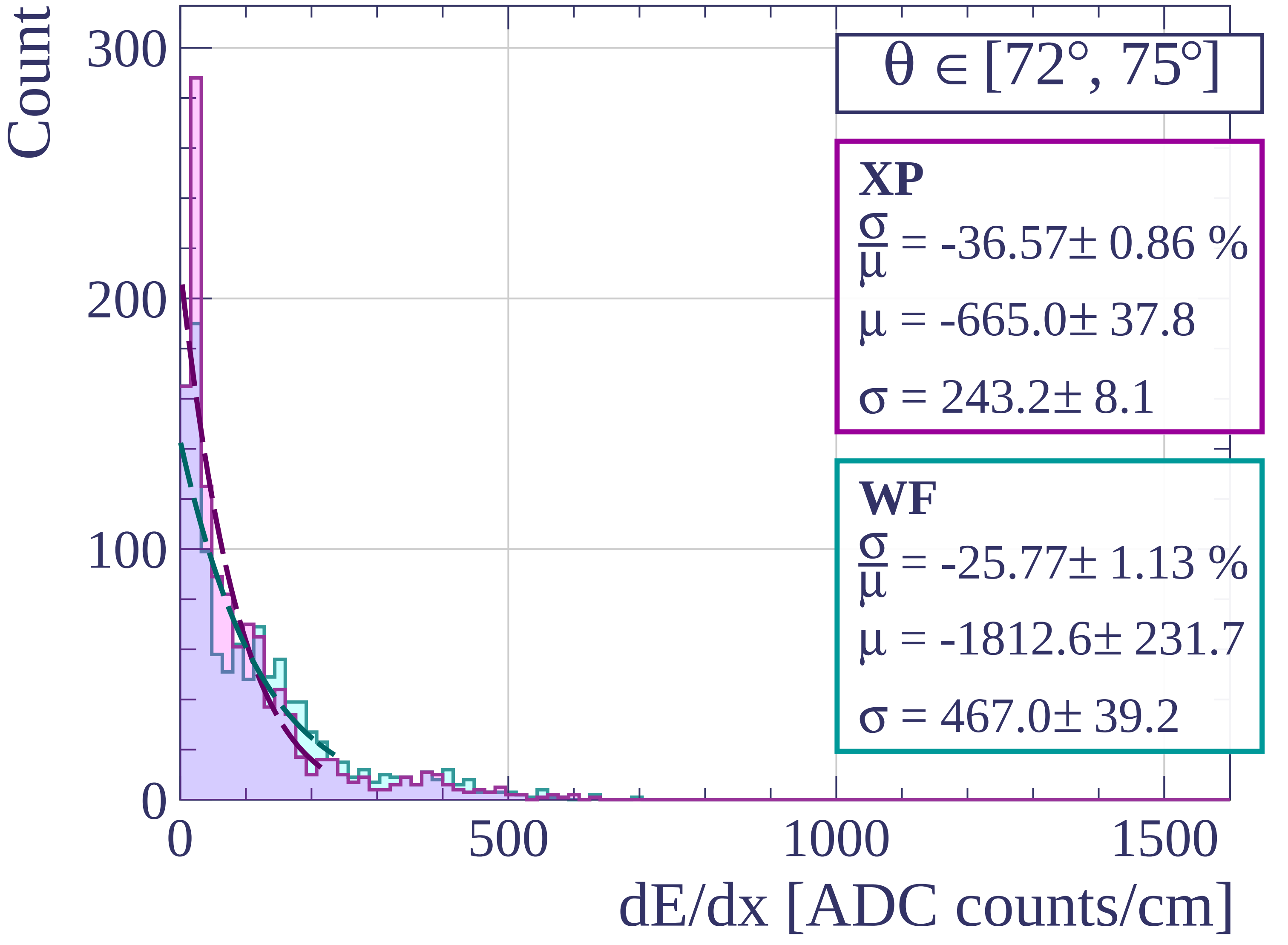


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [63°, 66°]
XP
σ/μ = 570.81± 652.93 %
μ = 28.9± 29.5
σ = 165.2± 20.4
WF
σ/μ = -32.23± -1.04 %
μ = -3612.0± 262.2
σ = 1164.1± 121.9

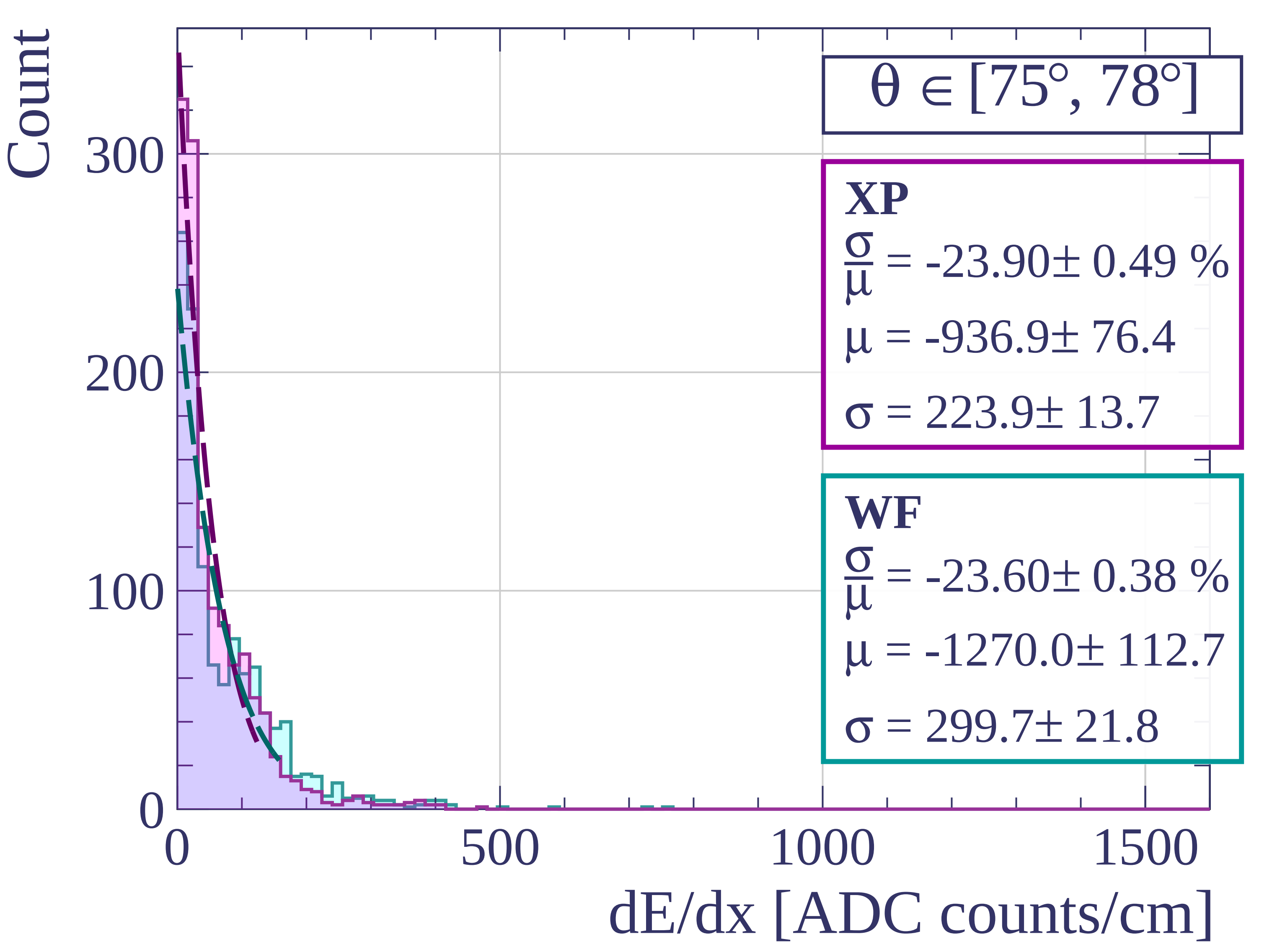

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [66°, 69°]
XP
σ/μ = 431.50± 361.72 %
μ = 32.9± 23.8
σ = 141.9± 16.2
WF
σ/μ = -31.52± 2.54 %
μ = -2462.2± 439.3
σ = 776.1± 76.0



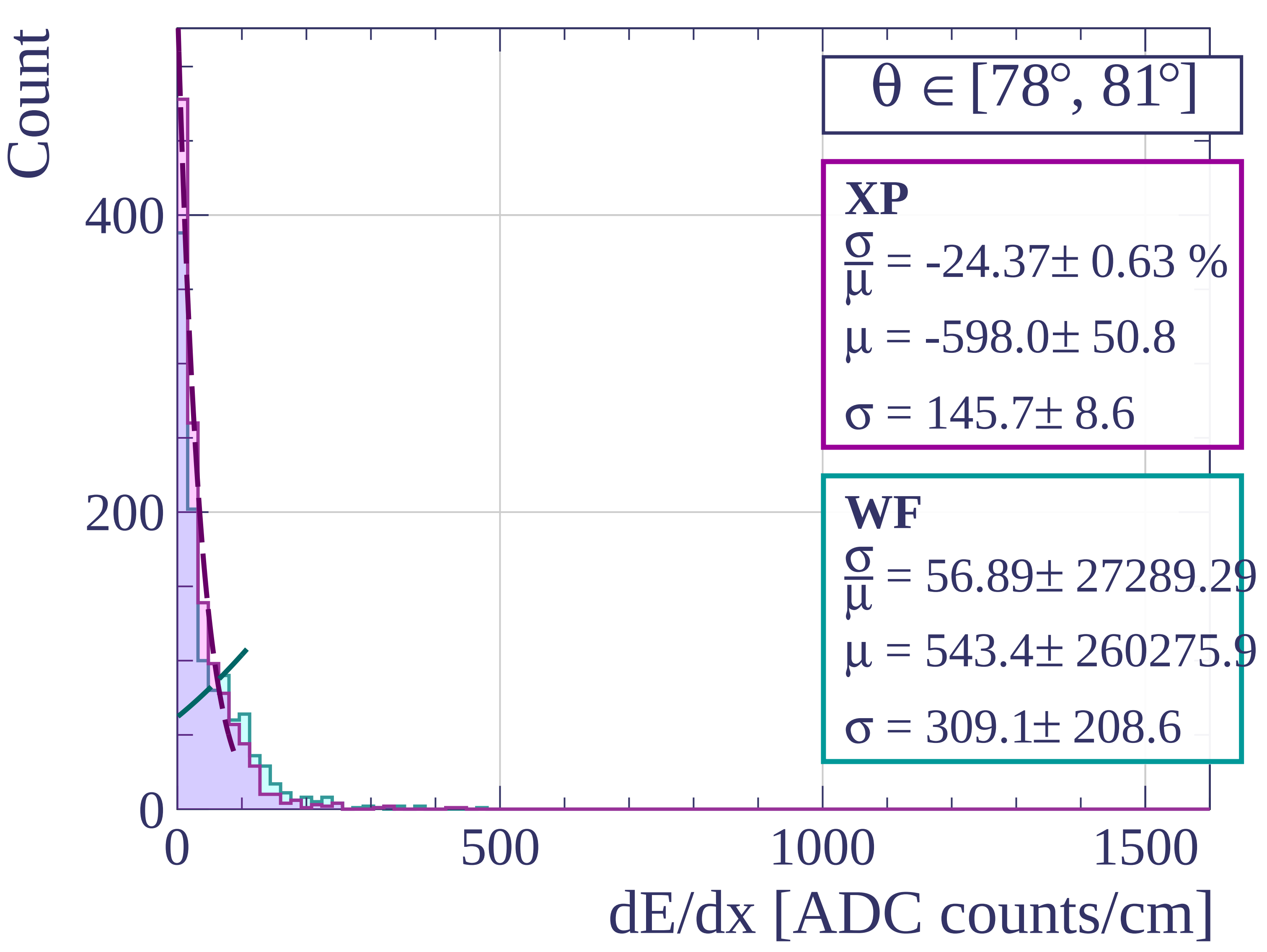
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [69°, 72°]
XP
σ/μ = -51.30± 2.91 %
μ = -475.3± 49.8
σ = 243.8± 11.7
WF
σ/μ = -24.92± 0.86 %
μ = -2527.4± 274.9
σ = 629.7± 46.6

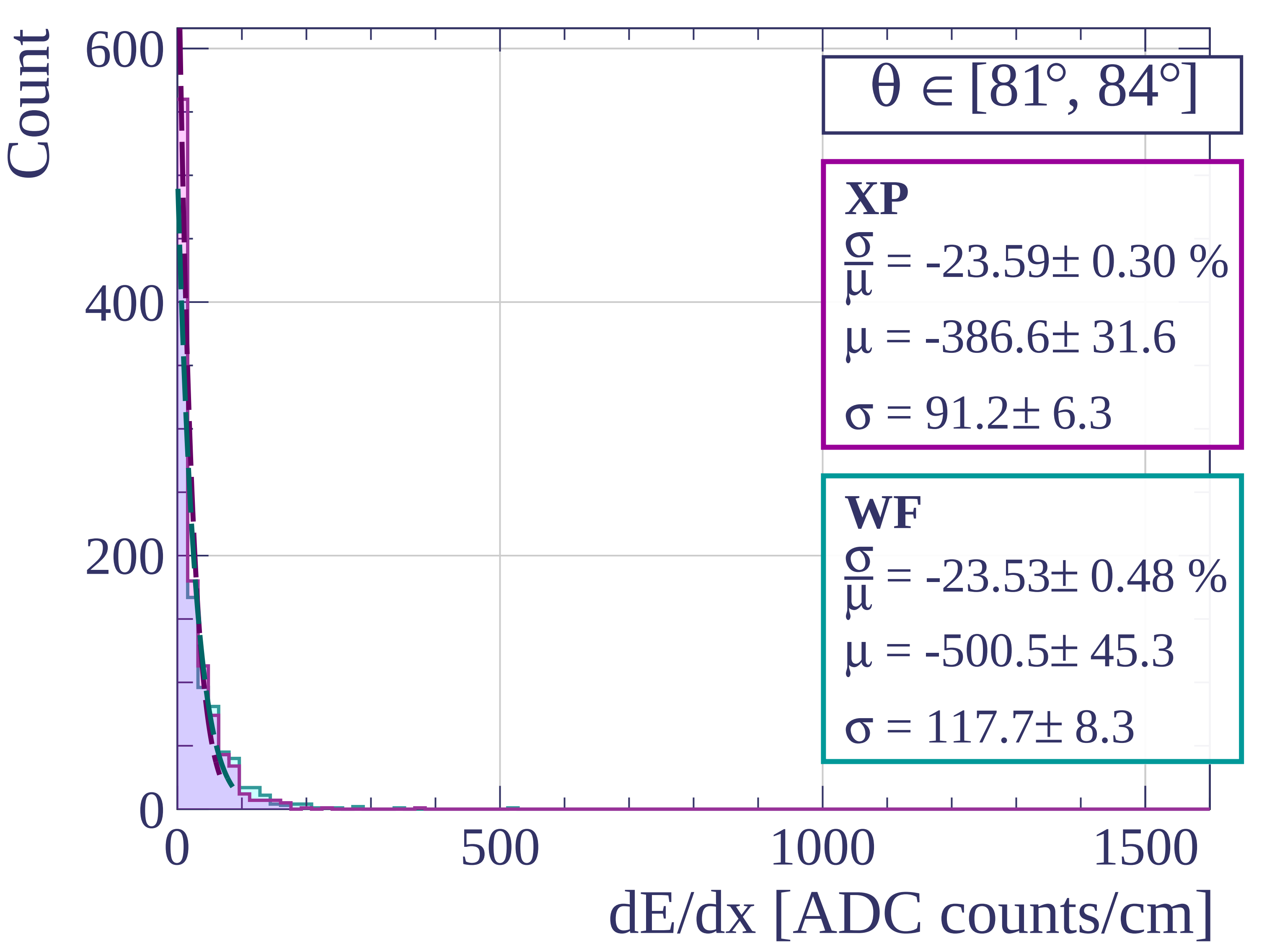


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = -36.57± 0.86 %
μ = -665.0± 37.8
σ = 243.2± 8.1
WF
σ/μ = -25.77± 1.13 %
μ = -1812.6± 231.7
σ = 467.0± 39.2

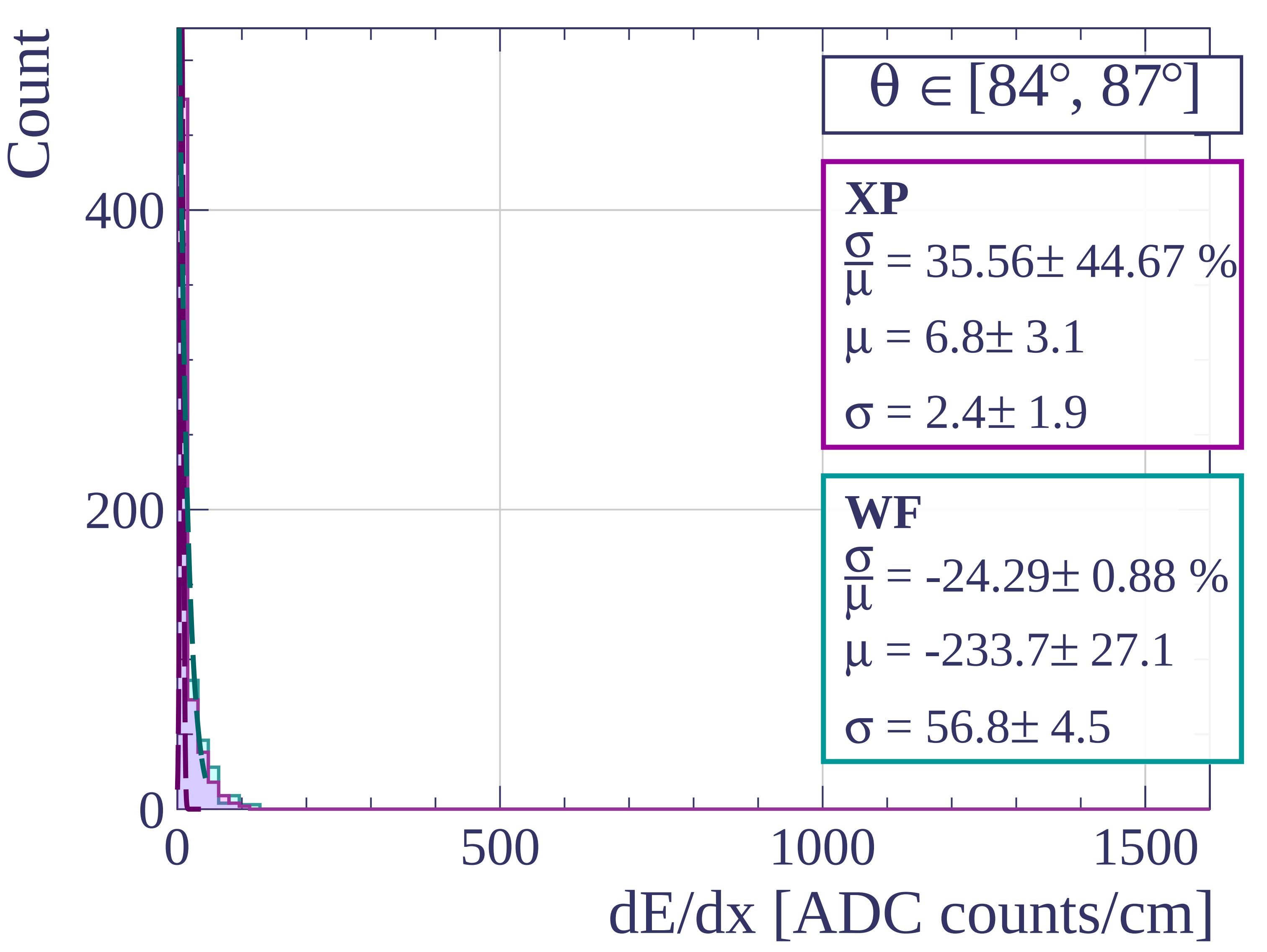


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
$\frac{\sigma}{\mu}$ = -23.90± 0.49 %
μ = -936.9± 76.4
σ = 223.9± 13.7
WF
$\frac{\sigma}{\mu}$ = -23.60± 0.38 %
μ = -1270.0± 112.7
σ = 299.7± 21.8

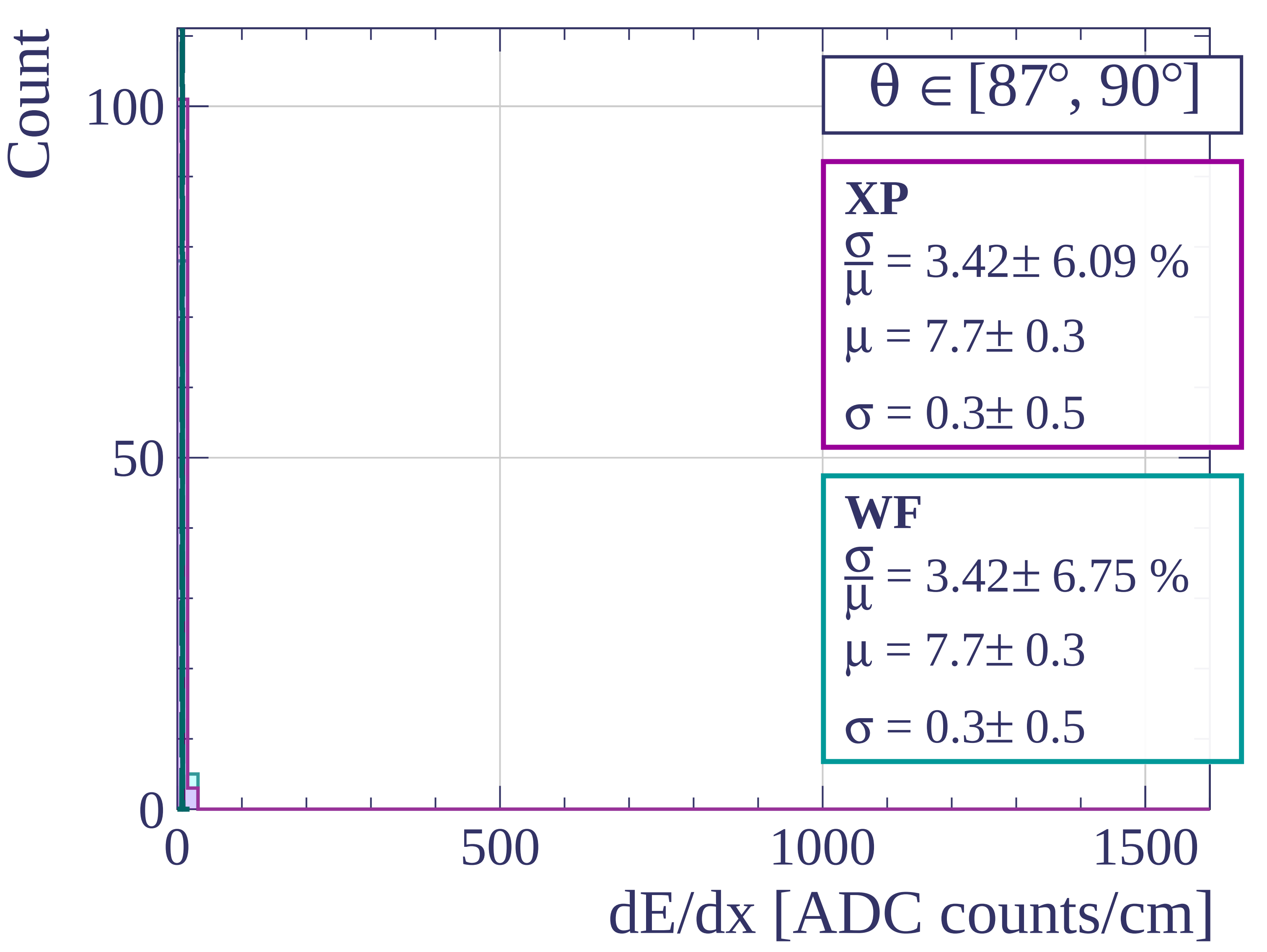

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
σ/μ = -24.37± 0.63 %
μ = -598.0± 50.8
σ = 145.7± 8.6
WF
σ/μ = 56.89± 27289.29
μ = 543.4± 260275.9
σ = 309.1± 208.6



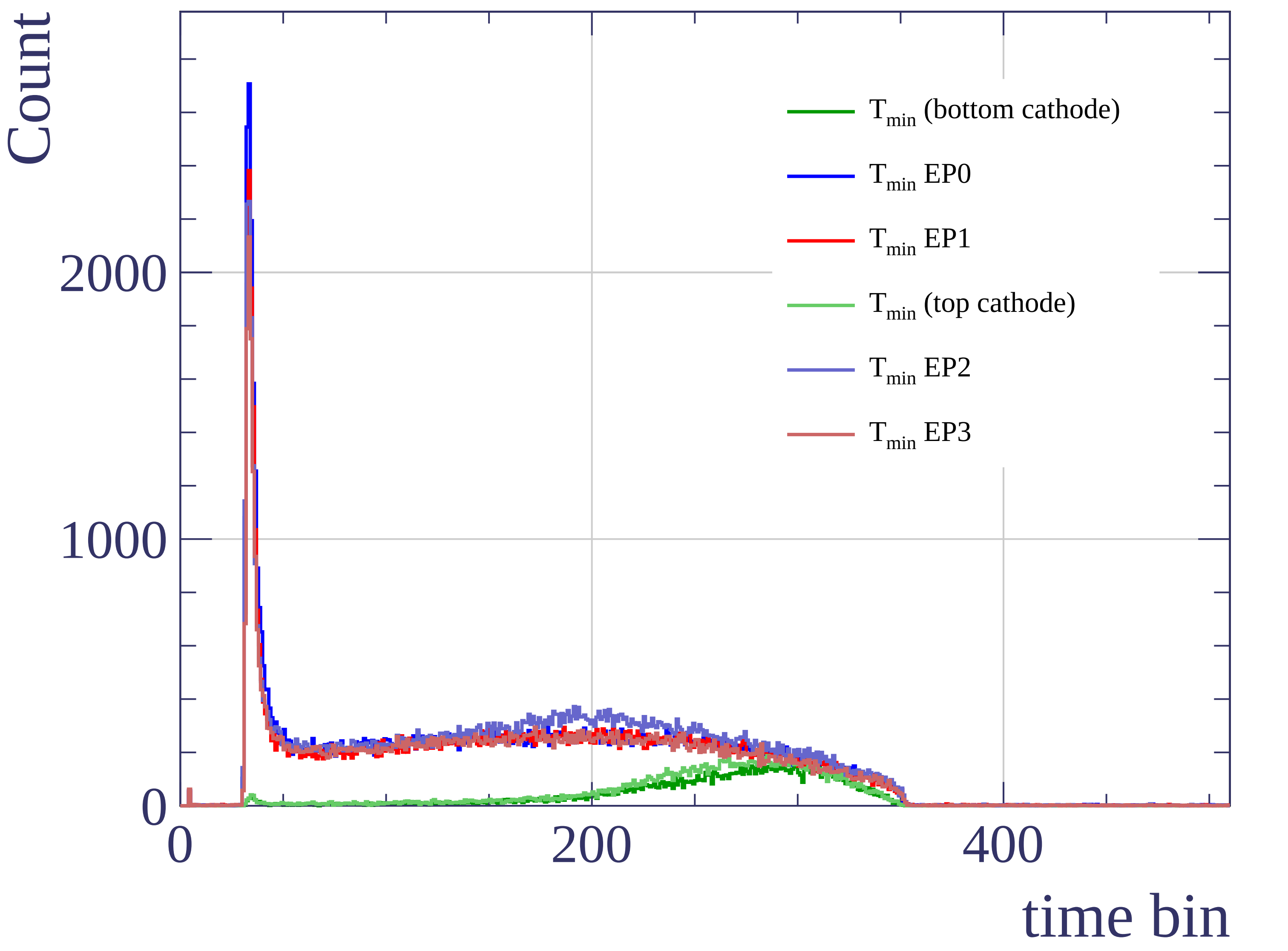
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -23.59± 0.30 %
μ = -386.6± 31.6
σ = 91.2± 6.3
WF
σ/μ = -23.53± 0.48 %
μ = -500.5± 45.3
σ = 117.7± 8.3



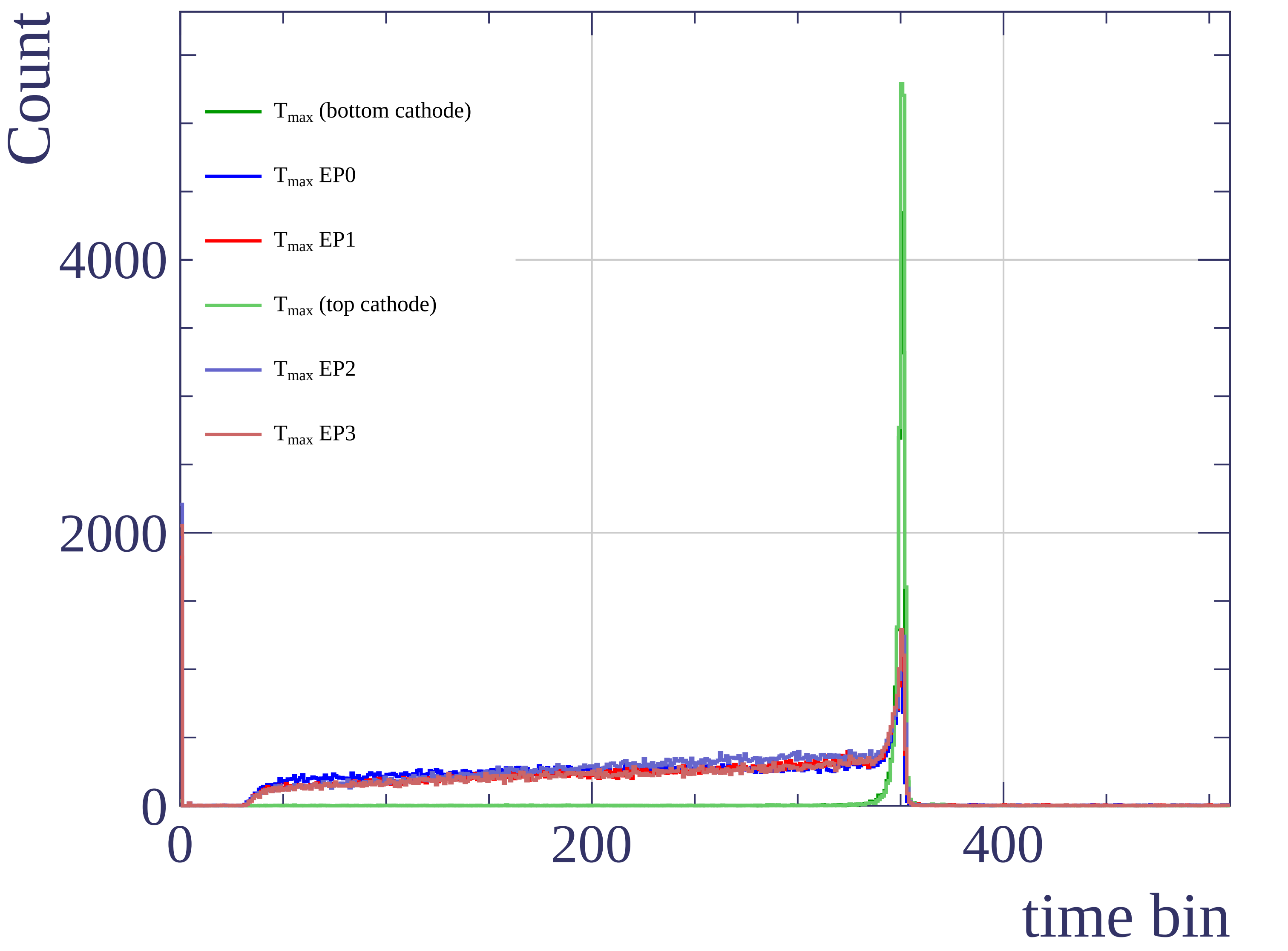
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = 35.56± 44.67 %
μ = 6.8± 3.1
σ = 2.4± 1.9
WF
σ/μ = -24.29± 0.88 %
μ = -233.7± 27.1
σ = 56.8± 4.5



Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
σ/μ = 3.42± 6.09 %
μ = 7.7± 0.3
σ = 0.3± 0.5
WF
σ/μ = 3.42± 6.75 %
μ = 7.7± 0.3
σ = 0.3± 0.5



Count
2000
1000
0
0
200
400
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
4000
2000
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3